

Set-Sized Quotients of the Omnific Integers

The universal constant-term quotient, exact cardinal thresholds,
support thresholds, and what survives in large quotients

Merged research report built from twenty manuscripts (03, 04, 06–11 of one batch,
12–16 of the next, 17–19 of a third, 20 of a fourth, 21 of a fifth, 22 of a sixth, 23 of a seventh)

22–23 September 2026; merged 23 September 2026

Abstract

Write an omnific integer uniquely as $n + s$ with $n \in \mathbb{Z}$ and s purely infinite, and let $\text{ct}(n + s) = n$. Every additive and multiplicative map from $\mathbf{Oz}$ to a set-sized ring, even a noncommutative, nonreduced or nonunital one, has the form $x \mapsto \text{ct}(x)e$ with e idempotent; every set-sized $\mathbf{Oz}$ -module is an abelian group with scalar action through ct . The same holds for $D + \Pi_K$ over any set field K and unital $D \subseteq K$, for the Gaussian ring $\mathbf{Oz}[i]$ (with $\mathbb{Z}[i]$), and, on the finite side, for standard part on the finite surreal and surcomplex rings. Countable supports already force this collapse while finite supports do not.

Over set-sized exponent groups the theorem becomes a detection threshold. For each of five explicit set-sized rings (four integer parts inside $\mathbf{Oz}$, of sizes $\kappa^{\aleph_0}$, $\kappa^{<\kappa}$, κ and λ , and a lexicographic Hahn ring of size $\kappa^{<\kappa}$), the least size of a ring or module detecting any nonzero purely infinite element equals the size of the ring; two of these values, and the exact thresholds of two examples in fixed exponent groups, are new to this merge. The report also keeps the large-quotient side (a copy of $\mathbf{No}$ and a Cantor algebra inside $\mathbf{Oz}/(1 + \omega^a)$), the integral and complete integral closures of $\mathbf{Oz}$, the arithmetic of integer-valued polynomials on $\mathbf{Oz}$, and a negative answer to a posted question on natural ordinal arithmetic. Five later manuscripts add the ordinary homological and tensor algebra of subrings of $\mathbf{Oz}$: for set-sized finite-support cores the arithmetic quotient has flat dimension d when the exponents form a coordinatewise cone of rank d and 1 when they form a totally ordered cone, and coefficient-lattice ideals have explicit tensor, Rees and duality algebra; and they add a fresh-scale image-gap theorem for polynomial and rational maps. Three manuscripts of a third batch add exact relation counts and Tor groups of these ideals, a sharp bound on the ordinary outputs of integer-valued polynomials, and the integral closure of $\mathbf{Oz}$ modulo the ideal generated by Π as a Boolean power of the algebraic integers. A fourth adds that the Boolean algebra of that power is atomless and a proper class, split uniformly over every set-sized subring, with integrality local to the exponent group of an element. A fifth shows that for the coordinatewise cone of rank d the projective dimension of the arithmetic quotient is $d + 1$. A sixth certifies rational functions of several variables by finitely many values at one fresh scale, removes set-sized exceptional sets in every number of variables, and transfers ordinary constant test sets exactly. A seventh shows that over one cyclic exponent group the extended ideal of the integral closure is not radical, while the Boolean algebra of the reduced fibre is still atomless. It is an AI-assisted merge of unrefereed manuscripts; nothing in it is verified in Lean.

Status. Proof-bearing research report assembled from twenty AI-assisted manuscripts written on 22 and 23 September 2026. The sources call their main results candidate original; priority is not certified, no named conjecture is claimed solved, the proofs have not been refereed, and no statement is formalized. Results added in the merge are marked [merge]; the source of every other statement is tagged [NN].

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1 Introduction

1.1 The question and the answer

The omnific integers $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$ form a discretely ordered subring of $\mathbf{No}$ whose fraction field is all of $\mathbf{No}$. Their purely infinite ideal Π is a proper class; each of its elements has a set-sized normal form, but below any positive exponent there is a proper class of smaller exponent scales. The eight manuscripts of the first merge (03, 04, 06–11) ask the same question: *what can an ordinary, set-sized ring or module see of an omnific integer?* All eight give the same answer, by different proofs, and two later manuscripts (13, 15) reprove it independently: exactly its ordinary integer constant term. This is much stronger than the observation that a proper-class field cannot embed in a set: the source is a ring, its homomorphisms have large kernels, and the targets may have zero divisors, nilpotents, noncommutative multiplication and any characteristic (06, 11). Nor is it a consequence of divisibility of Π by every integer, which only disposes of targets of positive characteristic (04, 07).

The sources then go in different directions: exact cardinal thresholds for set-sized integer parts (03, 06, 08, 09, 11), the finite side and the support threshold (04, 11), the large quotients in which a purely infinite element becomes a unit (07), integral closure (03), polynomial maps (10), and natural ordinal arithmetic (11). Three later manuscripts add the ordinary homological and tensor algebra of subrings of $\mathbf{Oz}$ with finite supports or with coefficient lattices (12, 14; [Section 13](#)), and fresh-scale arguments for polynomial and rational maps (16; [Sections 16.6](#) and [16.7](#)). Three manuscripts of a third batch extend three of these directions: relation modules, Tor groups and isomorphism classes of coefficient-lattice ideals (17; [Section 13.9](#)), the ordinary-output loci of integer-valued polynomials (18; [Section 16.8](#)), and the arithmetic fibre of the integral closure (19; [Section 15.8](#)). A manuscript of a fourth batch determines how that fibre branches: its Boolean algebra of idempotents is atomless and a proper class, and integrality is local to the exponent group of an element (20; [Section 15.9](#)). A manuscript of a fifth batch determines the projective dimensions of the coordinate-cone cores, which 12 had bounded, and computes the Tor algebra of lattice quotients, with arithmetic applications (21; [Section 13.10](#)). A manuscript of a sixth batch certifies rational functions of several variables by finitely many values at a single fresh scale, removes set-sized exceptional sets in every number of variables and lifts ordinary constant test sets exactly; much of its one-variable content re-derives 16's (22; [Section 16.10](#)). A manuscript of a seventh batch decides the fixed-workspace questions on that fibre for cyclic exponent groups, with opposite answers: the extended ideal is not radical, while the Boolean algebra of the reduced fibre is atomless, both by ramification over the place $\omega^a = 0$ of the rational functions in ω^a (23; [Section 15.10](#)). This report prints the common spine once and keeps every result of every source.

1.2 Main theorems

Theorem A (Universal set-sized quotient). *Let K be a set field and $D \subseteq K$ a unital subring. Every additive multiplicative map from $D + \Pi_K$ to a set-sized ring kills Π_K and factors uniquely through ct ; unital maps correspond bijectively to unital maps from D ; every set-sized module is annihilated by Π_K . For $\mathbf{Oz}$ the unique unital map to a set-sized ring S is $x \mapsto \text{ct}(x)1_S$, and unital maps $\mathbf{Oz}[i] \rightarrow S$ correspond to $j \in S$ with $j^2 = -1$. The kernels admitting set-sized quotients are exactly the $\text{ct}^{-1}(\mathbf{a})$; for $\mathbf{Oz}$ they are Π and $n\mathbf{Oz}$ ($n \geq 1$). ([Theorems 5.1](#) and [6.2](#) and [Corollaries 5.2](#) and [5.3](#); the eight sources of the first merge, manuscripts 13 and 15, and manuscript 05 of the sibling report.)*

Theorem B (The finite side and the support threshold). *Standard part is the universal set-sized quotient of the finite surreal and surcomplex rings, and more generally of $D + \mathfrak{m}_k$ (04). For*

every infinite λ , the rings of normal forms with supports of size at most λ (full surreal exponents) already have universal set-sized quotient D on both sides, while the finite-support rings admit the augmentation $\sum c_g \omega^g \mapsto \sum c_g$; a map on the finite-support ring extends to the countable-support ring if and only if it factors through ct (the failure for the augmentation is 04's and 11's, the criterion is 11's). (*Theorems 8.2 and 9.1.*)

Theorem C (Detection threshold equals size). *For each of five explicit set-sized rings A , namely four integer parts inside $\mathbf{Oz}$, of cardinality $\kappa^{\aleph_0}$ (cf $\kappa > \aleph_0$; 03), $\kappa^{<\kappa}$ (κ regular uncountable; 06), κ (every infinite κ ; 08) and λ ($\lambda^{\aleph_0} = \lambda$; 09), and 11's lexicographic Hahn ring of cardinality $\kappa^{<\kappa}$ (coefficient field with at most $\max(\kappa, 2^{\aleph_0})$ elements), the least cardinality of a ring target or module detecting any nonzero purely infinite element is exactly $|A|$, in ZFC. (*Theorem 10.10.*) The values for 06's and 11's rings are new here: 06 and 11 prove exactness only under $\kappa^{<\kappa} = \kappa$, and the unconditional value $\kappa^{<\kappa}$ follows from 06's own collision lemma (*Theorem 11.11 and Proposition 11.23*). The same thresholds hold for the first and second Ext groups of the homological package of 06, which extends to all five rings (*Theorems 12.2 and 12.3*).*

Theorem D (The field bound in a fixed exponent group). *In a set-sized Hahn integer part, a ring or module detecting an element divisible by X^{2^c} has at least as many elements as the field $L_c(F)$ of elements of the ambient field F supported in the scale subgroup H_c (03, generalized through 06's lemma). This bound is exact in 04's example $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$, where it equals $2^{\aleph_0}$, and in an example with rational coefficients it exceeds by far all the monomial and weighted counts of 04, 07 and 10, showing that 10's bound need not be attained. (*Proposition 10.3 and Examples 10.4 and 10.5.*)*

Theorem E (Large internal fields). *If a nonzero purely infinite element becomes a unit in a nonzero ring, the ring contains an explicitly embedded copy of $\mathbf{No}$. The nonzero quotient $\mathbf{Oz}/(1 + \omega^a)$, which has no nonzero set-sized image, contains an algebraically closed field containing $\mathbf{No}[i]$, exact cyclotomic algebras over it, and the Cantor algebra of locally constant functions on $\mathbb{Z}_2$ (07; one proposition of 06). (*Theorems 14.7, 14.15 and 14.19.*)*

Theorem F (Closures and invisible equations). *The complete integral closure of $\mathbf{Oz}$ is $\mathbf{No}$, while its integral closure is a proper, fine-dense subring with zero conductor that no set generates; its constant slice is the real algebraic integers, it has no finite quotients, and no nontrivial set-valued valuation of $\mathbf{No}$ contains $\mathbf{Oz}$ (03). For $\gamma > 0$ and $m \geq 2$ the equation $X^m = \omega^\gamma + 1$ has no root in $\mathbf{Oz}[i]$, yet its image in every set-sized ring has the root 1 (08). The extended ideal $\Pi\mathcal{N}$ is radical, $\mathcal{N}/\Pi\mathcal{N}$ is the finite-partition Boolean power $\overline{\mathbb{Z}}[\text{Idem}]$ of the ring of all complex algebraic integers, complexification doubles it integrally, and every set-sized domain image of $\mathcal{N}$ or $\mathcal{N}_{\mathbb{C}}$ is $\overline{\mathbb{Z}}$ or $\overline{\mathbb{F}}_p$ (19). Its Boolean algebra $\text{Idem}(\mathcal{N}/\Pi\mathcal{N})$ is atomless and a proper class, one idempotent splits every nonzero element of any set-sized subring of $\mathcal{N}/\Pi\mathcal{N}$, and no set-indexed family of maps to set-sized rings is jointly faithful on it; an element of $\mathcal{N}$ supported in a set-sized group G is integral over the omnific integers supported in G (20). In a cyclic workspace $\mathbb{Z} + \omega^a \mathbb{R}[\omega^a]$ the extended ideal of the integral closure is not radical, its fibre having nilpotents of every index, while the Boolean algebra of the reduced fibre is atomless of cardinality and order density $2^{\aleph_0}$, and the bounded-denominator Puiseux limit is reduced (23). (*Theorems 15.5, 15.12, 15.22, 15.25, 15.29, 15.38, 15.42, 15.46, 15.57, 15.66, 15.73, 15.79, 15.86, 15.100 and 15.111 and Corollary 15.110.*)*

Theorem G (Polynomial maps). *The ring of finite-degree surreal polynomials preserving $\mathbf{Oz}^r$ is $\Pi[\mathbf{X}] \oplus \text{Int}(\mathbb{Z}^r)$ with universal set-sized image $\text{Int}(\mathbb{Z}^r)$; a Newton least-common-multiple condition decides congruence preservation modulo every omnific integer; the p -adic completion is $C(\mathbb{Z}_p^r, \mathbb{Z}_p)$ and most of its characters are not evaluations; rational self-maps of $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ are polynomial and the bijections are affine (10). For every unital $D \subseteq k \in \{\mathbb{R}, \mathbb{C}\}$ and every polynomial f*

of degree at least two with coefficients in a set-sized Hahn field K_G , a fresh monomial t has $f(D + \Pi_k) \cap (f(t) + K_G) = \{f(t)\}$, every root of $f(Y) = f(t) + c$ ($c \neq 0$) having an explicit forbidden block of negative exponents; rational self-maps of $D + \Pi_k$ are polynomials and the surjective ones are the affine maps $uX + a$, $u \in D^\times$; $d+1$ points test degree d and no fewer do, while no set-sized sample tests all rational self-maps (16). A polynomial of degree $d \geq 1$ preserving $\mathbf{Oz}$ (or $\mathbf{Oz}[i]$) sends at most d inputs to $\mathbb{Z}$ (or $\mathbb{Z}[i]$) unless $f(X) = \text{ct}_*(f)(X - \eta)$ for a unique purely infinite η , which the two highest coefficients determine; in degree at least two at most one coset $b + \mathbb{Z}$ (or $b + \mathbb{Z}[i]$) has more than d preimages (18). For rational functions of m variables with coordinate degrees at most d , the values on an ordinary box and on m grids with one coordinate at a single fresh monomial, at most $(d+1)^m + m(2d^2 + 2d + 1)^{m-1}$ of them, certify preservation of $\mathbf{Oz}$ (similarly of $\mathbf{Oz}[i]$); set-sized exceptional sets are removable in every number of variables; $\mathcal{N}_m(\mathbf{Oz}) \cong \mathbf{Oz} \otimes_{\mathbb{Z}} \text{Int}(\mathbb{Z}^m)$ and $\mathcal{N}_m(\mathbf{Oz}[i]) \cong \mathbf{Oz}[i] \otimes_{\mathbb{Z}[i]} \text{Int}(\mathbb{Z}[i]^m)$; and a finite subset of $\mathbb{Z}^m$ (or $\mathbb{Z}[i]^m$) tests preservation of $\mathbf{Oz}$ (or $\mathbf{Oz}[i]$) exactly when it tests $\text{Int}(\mathbb{Z}^m)$ (or $\text{Int}(\mathbb{Z}[i]^m)$) (22). (Theorems 16.7, 16.14, 16.36, 16.43, 16.46, 16.51, 16.59, 16.61, 16.70, 16.75, 16.80 and 16.82 and Corollaries 16.20 and 16.29.)

Theorem H (Natural ordinal arithmetic). *The Grothendieck ring of the ordinals under natural operations is not a quotient of $\mathbf{Oz}$; every unital ring map from $\mathbf{Oz}$ to it has image $\mathbb{Z}$. This answers negatively the quotient question in MathOverflow question 188430 of Jesse Elliott (11; independently 13). (Theorem 17.1.)*

Theorem I (Homological and tensor algebra of omnific cores). *Let $D \subseteq K$ be $\mathbb{Z} \subset \mathbb{R}$ or $\mathbb{Z}[i] \subset \mathbb{C}$. The set-sized subring $A_d = D + \bigoplus_{0 \neq \alpha \in \mathbb{Q}_{\geq 0}^d} KX^\alpha$ of $\mathbf{Oz}$ (or $\mathbf{Oz}[i]$), whose exponents form a coordinatewise cone of rank d , has $\text{Tor}_{>0}^{A_d}(D, D) = 0$ but $\text{fd}_{A_d} D = d$, with a complete multigraded computation of $\text{Tor}^{A_d}(D, A_d/(X_i : i \in F))$, and its finite quotients are those of D (12). Enlarging the cone to all positive exponents of the same ordered group is a nonflat extension over which $\text{fd} D = 1$ and $\text{pd} D = 2$, as for the totally ordered cones of Section 12 (12, with the merge). Over A_d itself $\text{pd}_{A_d} D = d + 1$, one more than the flat dimension, with an explicit free telescope resolution and an embedding $D^{\mathbb{N}}/D^{(\mathbb{N})} \hookrightarrow \text{Ext}_{A_d}^{d+1}(D, A_d)$ (21). For a coefficient lattice $M \subset K$ of rank $r \geq 2$, the module $L(M) = M + \Pi$ over $\mathbf{Oz}$ or $\mathbf{Oz}[i]$ (or over the set-sized model A_1) has first syzygy Π^{r-1} , tensor square $\Pi \oplus (M \otimes M)$ whose torsion is $\ker(M \otimes M \rightarrow M^2)$, nonlinear Rees equations exactly the ordinary relations among the generators (the first one in degree d for $(\omega, 2^{1/d}\omega)$), isomorphism type determined by M up to a scalar, ideal classes realizing Pic of an order, and dual Π (14). Over $\mathbf{Oz}$ such a module has no presentation by a set of generators and a set of relations; over the lexicographic model $R_{\Gamma_\kappa, < \kappa}^{11} \subset \mathbf{Oz}$ every finite free presentation of it needs exactly κ relations, for every uncountable regular κ (17) and for $\kappa = \aleph_0$ (merge); there, for $J = X^g L(M)$ and $J' = X^h L(N)$, $\text{Tor}_1(J, A/J') \cong \text{Tor}_2(A/J, A/J') \cong \mathcal{T}(M, N) = \ker(M \otimes_D N \rightarrow K)$, and the ideals $(\omega, \alpha\omega)$, $\alpha \notin \text{Frac}(D)$, form $2^{\aleph_0}$ isomorphism classes, the orbits of $\text{GL}_2(D)$ acting by fractional linear maps (17). Over A_d the projective dimensions of $L(M)$ and $A_d/J_g(M)$ are $d+1$ and $d+2$; over $T_{\mathbb{Q}}$ and over the class ring read in two universes, the second Tor of the quotients by $J_g(\mathfrak{o}_1)$ and $J_h(\mathfrak{o}_2)$ for two orders vanishes exactly when their fields are linearly disjoint, the self- Tor product is the antisymmetrized boundary tensor, and the framed second Tor of an order recovers its multiplication (21). (Theorems 13.7, 13.9, 13.11, 13.14, 13.18, 13.21, 13.24, 13.25, 13.27, 13.29, 13.31, 13.34, 13.38, 13.48, 13.53, 13.54 and 13.56 and Corollaries 13.8 and 13.47.)*

1.3 Sources, provenance and where the merge chose

The report is built from twenty manuscripts. Eight (03, 04, 06–11), dated 22 September 2026 and placed in commit `be06fc8`, were merged first; their numbers are arrival numbers in their batch. Five

(12–16), dated 23 September 2026 and placed in commit `cf350b1`, are items 02–06 of the following batch; they are numbered here by the file prefixes under which their programs and data are stored in this directory. Three (17–19), also dated 23 September 2026 and placed in commit `f4c9504`, are items 01, 03 and 05 of a third batch, numbered in the same way; the other seven items of that batch are merged into the reports on omnific-preserving automorphisms and on omnific groups and lattices [47, 48]. One (20), also dated 23 September 2026 and placed in commit `c6359e4`, is item 01 of a fourth batch, numbered in the same way; the other eight items of that batch were placed in other reports. One (21), also dated 23 September 2026 and placed in commit `66d7e55`, is item 01 of a fifth batch, numbered in the same way; the other eight items of that batch were placed in other reports. One (22), also dated 23 September 2026 and placed in commit `9d28e28`, is item 09 of a sixth batch, numbered in the same way; the other eight items of that batch were placed in other reports. One (23), also dated 23 September 2026 and placed in commit `7d04483`, is item 01 of a seventh batch, numbered in the same way; the other eight items of that batch were placed in other reports. None has a reviewer, and none is shipped with this report.

	Manuscript (pages)	Pin	Contribution and placement
03	<i>Cardinal Visibility and Normalization of Omnific Integers</i> (25)	cd80e5e	field bound (Proposition 10.3); model A_κ^{03} with threshold $\kappa^{\aleph_0}$ (Section 11.1); flatness, Tor; integral and complete integral closure, valuations (Section 15)
04	<i>What Set-Sized Algebra Can See of Surreal Arithmetic</i> (24)	cd80e5e	$D + \Pi_k$ theorem via separation and telescope; finite side (Section 8); support threshold with negative side (Section 9); fixed-group bounds; localizations; modules, derivations, algebras
06	<i>The Universal Set-Sized Quotient of the Omnific Integers</i> (25)	a5c2a97	base : coefficient-ring theorem for any set field, scaled-field collision lemma, direct module argument; model A_κ^{06} ; homological package (Section 12); matrix representations; polynomial tests; prime-reservoir proposition
07	<i>Small Quotients and Large Internal Fields of the Omnific Integers</i> (27)	37eefca	theorem for any set field via weighted certificates; completions; compression $\mathbf{No} \hookrightarrow \mathcal{F}_a$, transfer, binomial kernel, Cantor algebra (Section 14)
08	<i>What Small Rings Can See of Omnific Integers</i> (26)	37eefca	Hartogs formulation, nonunital maps; closure formula; invisible equation; class-indexed derivations; criterion; model A_κ^{08} for every infinite κ ; cyclic-module lemma
09	<i>Set-Sized Shadows of Omnific Integers</i> (20)	37eefca	direct division of $D + \Pi_k$ elements; model A_λ^{09} , residue fields, non-arithmetic maximal ideals, $\aleph_1$ generators; Gaussian ramification
10	<i>Set-Sized Shadows of Omnific Arithmetic</i> (26)	a5c2a97	theorem for any set field incl. maps on Π_K ; quantitative bound; endomorphisms; numerical polynomials, p -adic completion and characters, lattice rigidity (Section 16)
11	<i>What Set-Sized Algebra Can See of the Omnific Integers</i> (20)	37eefca	pairwise-difference families; localization and presentation calculus; extension criterion; coinitality theorem; residual targets; ordinal ring (Section 17)

	Later manuscript (pages)	Pin	Contribution and placement
12	<i>Arithmetic Invisibility and Homological Dimension in Omnific Integer Cores</i> (21)	none	coordinate-cone cores A_d : Boolean flat resolution, multigraded Tor, fd $D = d$; noncoherence certificate; ordered-cone collapse to fd $D = 1$; projective bounds (Section 13; attained by 21's Corollary 13.47)
13	<i>Small-Target Rigidity and Arithmetic Specialization</i> (21)	71e9606	independent proofs of the universal theorem and of the ordinal answer (credited in Section 5.1 and Section 17); its fiber and norm theorems belong to the Diophantine report [46]
14	<i>Arithmetic Tensors in the Omnific Integers</i> (19)	71e9606	coefficient-lattice modules: syzygy, tensor normal form, symmetric and exterior powers, Rees equations, Hom, ideal classes, duals, matrix kernels, fd over A_0 (Section 13)
15	<i>Omnific Integers Have Only Ordinary Set-Sized Representations</i> (21)	71e9606	an independent proof of the universal theorem and its consequences (credited throughout); two one-line additions (Corollary 5.5 and Proposition 7.3)
16	<i>Fresh-Scale Image Gaps and Rational Rigidity for Omnific Integers</i> (23)	71e9606	image gap and forbidden layers, fresh fibers, rational collapse and surjective maps over any constant ring, optimal and Gaussian finite tests, rational probes, sampling obstructions (Sections 16.6 and 16.7)
	Third-batch manuscript (pages)	Pin	Contribution and placement
17	<i>Proper-Class Relations and Arithmetic Endomorphisms of Omnific Ideals</i> (19)	71e9606	independent proofs of 14's matrix, syzygy, Hom, tensor and power-growth theorems and of 12's intersection (credited in Section 13); no set-presentation, exactly κ relations, $\text{Tor}_1 \cong \text{Tor}_2 \cong \mathcal{T}(M, N)$ with the self-Tor range and field test, $\text{GL}_2(D)$ moduli, orders, an embedding of $R_{\Gamma, < \kappa}^{11}$ in $\mathbf{Oz}$ (Section 13.9)
18	<i>Polynomial Rigidity of Omnific Integer Lattices</i> (21)	befe739	ordinary-output dichotomy over any discretely ordered ring, canonical translation, rich targets, composition alignment, a smaller Gaussian test, matrix and jet reductions (Section 16.8); independent proofs of 10's and 16's classification, tests, rational collapse and surjections (credited in Section 16)
19	<i>Normalization and Arithmetic Fibres of the Omnific Integers</i> (23)	befe739	radical extended ideal, saturation, real-rooted lifting, the residue with $v^2 = -1$, the Boolean power $\mathbb{Z}[\text{Idem}]$, integral doubling, domain images and kernel detection, a finite-axiom criterion (Section 15.8); independent proofs of four results of 03 and of the universal theorem (credited)
	Fourth-batch manuscript (pages)	Pin	Contribution and placement
20	<i>Proper-Class Boolean Branching in the Normalization of the Omnific Integers</i> (24)	4cdeaec	monic radical certificates, support projection and exact descent, one-scale rational functions, independent quadratic branches, uniform splitting, rank extension of set-sized Hahn fibres, relative Boolean freeness, an explicit ordinal family, atomless proper-class Idem, reduced fibres of fixed Hahn fields, no jointly faithful set of small targets, 2^κ branches through prescribed data (Section 15.9); independent proofs of 19's fibre theory and of the universal theorem (credited)

	Fifth-batch manuscript (pages)	Pin	Contribution and placement
21	<i>Exact Homological Dimensions and Arithmetic Intersections in Omnific Integer Rings</i> (25)	9693b28	$\mathrm{pd}_{A_d} D = d + 1$ and the face quotients, by a free Koszul telescope and an embedded $D^{\mathbb{N}}/D^{(\mathbb{N})}$ in the top Ext; exact projective dimensions of lattice modules over A_d ; a flat DGA resolution, the first-Tor window, the self-Tor product, a linear-disjointness test, framed reconstruction of orders, the square-zero constant-term shadow (Section 13.10); independent proofs of 17's second Tor, self-Tor rank and Tor against tail-killed modules (credited)
	Sixth-batch manuscript (pages)	Pin	Contribution and placement
22	<i>Finite Tests at New Surreal Scales</i> (24)	bcac55a	single-scale certificates for rational functions of m variables over $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ by resultant specialization, common tests for a set of functions, removal of set-sized exceptions in m variables, a coordinate-degree Gaussian grid, tensor base change, exact lifting of constant test sets, image-ideal generators, birational rigidity, fresh scales in a fixed Hahn workspace (Section 16.10); independent proofs of 16's Laurent blocks, detector, finite tests, one-variable certificate, no-set-test theorem and one-variable removal of exceptions, and of 10's grid test and decomposition (credited)
	Seventh-batch manuscript (pages)	Pin	Contribution and placement
23	<i>Ramification in Omnific Normalizations</i> (25)	343dc2c	in the cyclic workspace $\mathcal{A}(\mathbb{Z}a)$: a nonradical extended ideal with nilpotents of every index and exact refinement thresholds, a residue criterion for $\mathcal{N}(\mathbb{Z}a)$ with $k\mathcal{N}(\mathbb{Z}a) = \mathcal{B}$, the ramification filtration, a reduced bounded-denominator Puiseux limit, Prüfer local structure with value group $\mathbb{Q}$, an idempotent nilradical, the associated graded ring and completion, splitting by fresh ramification with atomless Idem of size and density $2^{\aleph_0}$ and $2^{2^{\aleph_0}}$ branches, a periodic resolution over the coefficient fibre, a Cantor fragment (Section 15.10); independent proofs of the first equality of 20's rational slice and of 20's multiquadratic lemma (credited)

The pins are the repository commits the manuscripts inspected (cd80e5e0adc2, a5c2a97df4b5, 37eefca1a309, all of 22 September 2026, Pacific time; and 71e9606d297c, 23 September 2026, for 13–16). The commit 71e9606 precedes the placement of the first eight manuscripts, so 13–16 saw neither this report nor its sibling; 12 records no commit, only access to the repository through a GitHub connector on 23 September 2026. Of the third batch, 17 also pins 71e9606 and saw neither report, and 18 and 19 pin bfe739b56f4 (23 September 2026), which follows the placement be06fc8 but precedes the writing of this report: at that commit this directory held manuscript 06 as its text and the other first-batch manuscripts only as programs and audit files, so 18 and 19 saw 06's text but not 03's normalization, 10's polynomials or the second batch. 20 pins 4cdeaec43ff1 (23

September 2026), which also precedes the writing of this report: at that commit this directory again held 06’s text, so 20 saw 06’s small-image theorem but neither 16’s Laurent blocks nor 19; it read 19 as a saved draft (“the preceding normalization manuscript”) and does not assert that 19 was in the repository. 21 pins 9693b28c24e6 (23 September 2026), which follows the writing of the second merge and precedes that of the third (aae58bb): at that commit this report was built from thirteen manuscripts and held 12’s and 14’s section, including [Proposition 13.12](#) and the paragraph after it, but not 17’s [Section 13.9](#) or the material of 18–20. 22 pins bcac55ae6fd3 (23 September 2026), a commit that contains the writing of the third merge (aae58bb) but not those of the fourth and fifth: at that commit this report held 16’s and 18’s subsections, including [Theorems 16.51](#) and [16.52](#) and [Corollary 16.50](#) and [Question 18.14](#), but not 20’s or 21’s material. 22 read the repository and catalogue READMEs, the openings of the Diophantine and holonomic-rigidity reports and the fractions audit of the Diophantine report, not this report. 23 pins 343dc2c47121 (23 September 2026), which contains the writing of the fifth merge (a8f35b5) but not that of the sixth: at that commit this report held 20’s and 21’s material but not 22’s; 23 read its normalization, fibre and branching sections. Manuscript 05 of the same batch, *Omnific Integers and Diophantine Rigidity*, is merged into the sibling report [46]. It also proves the unital universal theorem for **Oz** (its “set-target rigidity”), the clearing of set supports and common monomial divisors, and it asks the “exact size boundary” question that 03, 06, 08, 09 and 11 answer; 03 credits the first three to 05 and 04 credits its finite-quotient background to manuscript 02 of the same sibling report. Those credits are kept.

Where the merge chose.

- *Base.* 06 is the base: its coefficient-ring theorem over any set field (also proved by 07 and 10) is the most general statement, and its scaled-field collision lemma (with 03’s field bound, a special case) is the mechanism that yields the sharp cardinal bounds where counting monomials does not suffice (03, 06, 11 and the fixed-group examples; 08 and 09 reach their thresholds by counting monomials). The proof is written in 08’s choice-free Hartogs form, with 08’s and 10’s nonunital forms and 06’s direct module argument. 04’s explicit telescope, 09’s and 10’s direct division, 11’s pairwise-difference families and 07’s weighted certificates are kept as second routes ([Section 5.1](#)).
- *Support threshold.* Printed once, in 11’s formulation (every infinite support bound, with the extension criterion) together with 04’s negative side ([Theorem 9.1](#)).
- *Thresholds.* Organized by one theorem ([Theorem 10.10](#)) with the five constructions of [Section 11](#); generator counts are listed per construction and no general law is stated.
- *Homological algebra.* 06’s package is proved under three hypotheses that all five constructions satisfy ([Theorem 12.2](#)).
- *Foundations.* NBG with choice for sets; global choice, stated by 04, 06, 09 (and 05) and by 17, 19 and 20, is used only for the class ultrafilters of 19 and 20, and the results depending on them say so ([Section 2.1](#) and [Remark 15.75](#)).
- *Later manuscripts.* 13 and 15 reprove the universal theorem and its consequences; they are credited where each result is printed, and only two one-line additions of 15 are printed ([Corollary 5.5](#) and [Proposition 7.3](#)); 13’s answer to Elliott’s question is credited in [Section 17](#). 12 and 14 form one new section ([Section 13](#)), placed after the homological package it is to be compared with; the ring $A_0 = A_1$ and the two-generator syzygy are printed once, the syzygy in 14’s rank generality over 12’s rings. 16’s results sit with 10’s polynomial maps; where they overlap (rational collapse, affine surjections, transfer, grids) the general- D statement is printed once with 16’s fresh-scale

proof and 10's statements keep their finite-difference and lattice proofs as a second route. 14's criterion for unimodular projective directions and 13's fiber and norm theorems are printed in the Diophantine report [46], not here.

- *Third batch.* Each of 17–19 joins the section it extends, with its new results in one new subsection. About half of 17 re-derives 14 (matrix kernels, lattice syzygy, Hom and isomorphism types, orders, tensor torsion, power growth, flat dimensions) and 12 (the intersection $X^g A \cap cX^g A$); these are printed once, under 14's and 12's labels with 17 added as a proving source, and 17's new statements are in [Section 13.9](#). 18's description of $\mathcal{N}_r$, its tests, its rational collapse and its classification of surjections repeat 10 and 16 and are credited there; its dichotomy package is [Section 16.8](#). 19 re-derives 03's constant slices, the absence of finite quotients, the complexification defect (with a different witness) and the universal theorem (with 04's telescope); these are credited, and its fibre theory is [Section 15.8](#). Genuinely different proofs are kept as short second routes.
- *Fourth batch.* 20 answers 19's question on the Boolean algebra and joins 19's section in one new subsection ([Section 15.9](#)), placed after 19's. About a third of 20 re-derives 19 (contraction, radicality, saturation, the residue v , the coefficient copy, spectral idempotents and the Boolean power, doubling and $w_{\pm}$, factorization of set-sized images, domain images, class ultrafilters and detection) with the same constructions; these are printed once, under 19's labels with 20 added as a proving source, and 20's proof of the universal theorem for **Oz** is credited at [Theorem 5.1](#). 20's dominant-scale lemma is 16's Laurent-block theorem, which 20 could not see; it is printed as 16's with 20's surjectivity step added ([Lemma 15.62](#)). 20's class-ultrafilter lemma, which starts from a set with the finite intersection property, is printed beside 19's ([Lemma 15.78](#)).
- *Fifth batch.* 21 answers the first clause of 12's and 14's question [Question 18.12](#) and joins their section in one new subsection ([Section 13.10](#)), placed at the end of [Section 13](#) so that no earlier statement is renumbered; the bounds it sharpens ([Proposition 13.12](#)) carry pointers to it. Its second part re-derives 17's second-Tor identification with its rank, the self-Tor rank, the pair $\mathbb{Z}[\sqrt{2}]$, $\mathbb{Z}[\sqrt{3}]$ and the Tor groups against modules killed by the tail, which 21 could not see; these are printed once, under 17's labels with 21 added as a proving source ([Theorems 13.34](#) and [13.42](#) and [Corollary 13.36](#)), and 21's extension of them to the class ring in a two-universe reading is stated in [Section 13.10](#). 21's flat face resolution, flat dimensions and coefficient syzygy repeat 12's and 14's ([Theorems 13.4](#), [13.9](#) and [13.11](#) and [Corollary 13.8](#)), as 21 says, and are cited, not reprinted. Its three questions are merged into [Questions 18.12](#), [18.13](#) and [18.15](#).
- *Sixth batch.* 22 joins [Section 16](#) in one new subsection ([Section 16.10](#)), placed after [Section 16.9](#) at the end of the section so that no earlier statement is renumbered. About half of 22 re-derives 10's and 16's results, which were in this report at 22's pin but which 22 did not read; these are printed once, under their labels with 22 added as a proving source ([Lemmas 16.5](#) and [16.30](#) and [Theorems 16.6](#), [16.7](#), [16.9](#), [16.31](#), [16.46](#) and [16.51](#)), and its one-variable certificate, detector, multivariate collapse and one-variable removal of exceptions are credited at [Corollary 16.50](#) and [Theorems 16.33](#), [16.49](#) and [16.52](#). 22's witness for the no-set-test theorem is kept as a second witness ([Remark 16.73](#)), and its multivariate certificate is kept beside 16's smaller probe certificate and compared with it ([Remark 16.71](#)). Its question on Gaussian constant tests is merged into [Question 18.14](#), and its other questions form [Questions 18.19](#) and [18.20](#).
- *Seventh batch.* 23 decides for cyclic exponent groups the fixed-workspace clause of [Question 18.17\(b\)](#), which 20 left open, and joins [Section 15](#) in one new subsection ([Section 15.10](#)), placed after 20's at the end of the section so that no earlier statement is renumbered. 20's multi-

quadratic lemma, which 23 reproves, is printed once under 20's label with 23 added as a proving source (Lemma 15.69); the first equality of 20's rational slice (Theorem 15.60) is contained in 23's Gaussian and fixed-ideal corollary (Corollary 15.94), and 23 credits it. The independent family of size $2^{\aleph_0}$ in the reduced cyclic fibre is 20's (Theorem 15.70 with $G = \{0\}$), which 23 did not cite; 23's own family is kept. Its Puiseux radicality is kept with 19's finite-axiom theorem as a second route (Remark 15.102). Two of its questions are merged into Question 18.17(b), (c), and its other seven form Questions 18.21 to 18.27.

Results added in the merge. Each carries the tag [merge] and a complete proof: the field bound for an arbitrary subfield of a fixed-group Hahn field (Proposition 10.3); the exact thresholds $2^{\aleph_0}$ in 04's example and in the rational-coefficient example (Examples 10.4 and 10.5), which answer 04's question for its own example and show that 10's bound need not be attained; the criterion (b') in Theorem 10.7; the exact value $\kappa^{<\kappa}$ of the sizes and thresholds of A_κ^{06} and of 11's example, in ZFC (Lemma 11.10, Theorem 11.11, and Proposition 11.23), which answers the first half of 06's question on cardinal arithmetic (08 answers the second half); the organizing theorem itself (Theorem 10.10) and the reconciliation of the realized cardinals (Remark 10.11); the homological package and the Ext thresholds for all five constructions (Theorems 12.2 and 12.3); the negative-side extension criterion (Theorem 9.1(iv)); the embedding $\mathbf{No} \hookrightarrow \text{Frac}(\mathbf{Oz}/P)$ for primes not containing Π (Remark 14.13); and the explicit elements of the normalization provided by 08's equation (Remark 15.27). From the later manuscripts: the syzygy, noncoherence and nonflatness theorems in the common generality of 12's and 14's rings (Theorem 13.9 and Proposition 13.10); the flat dimensions of lattice modules and cyclic quotients over every core, which removes a superfluous hypothesis of 12 (Theorem 13.11); exact projective dimensions where 12 gives bounds, and $\text{pd } D = 2$ for every ring of Section 12 with countably generated tail (Proposition 13.12); and the comparison of the two kinds of flat-dimension statements (Remark 13.16). From the third batch: 17's relation count, flat dimensions and Tor formula for $\kappa = \aleph_0$, with $\text{pd } \Pi = 1$ and $\text{pd } L(M) = 2$ there (Remark 13.32); 17's Tor formula over 12's cores and 14's A_0 , which gives $\text{rank Tor}_2(A/H, A/H) = 4 - \text{rank}(M^2)$ for 12's cyclic modules without the hypothesis $c^2 \in D$ (Theorem 13.34 and Corollary 13.35); the comparison of 18's and 16's Gaussian tests (Proposition 16.64); and the explicit candidates for idempotents of $\mathcal{N}/\Pi\mathcal{N}$ and the domain-image form of 08's invisible roots (Remarks 15.47 and 15.52). From the fourth batch: the descent $\mathcal{N} \cap F_\kappa^{03} = \mathcal{N}_\kappa$ asked by 03, with its radical and Gaussian forms, from 20's support projection (Corollary 15.59); the splitting by 03's unit, which decides both candidates of Remark 15.52 (Proposition 15.68); the observation that 20's splitting results need no global choice (Remark 15.75); and the remark that $\mathbf{B}_\mathbb{R}$ is residually set-sized although no set of small targets is jointly faithful (after Theorem 15.79). From the fifth batch: 21's projective dimensions of lattice modules over A_d for every unital $D \subseteq K$ (21 assumes a principal ideal domain only to have a basis; Theorem 13.48); and the observation that 21's cores A_d satisfy every hypothesis of Proposition 13.12(v) except (H2) and have $\text{pd } D = d + 1$, so (H2) cannot be dropped there (Remark 13.49). From the sixth batch: the comparison of 22's single-scale certificate with 16's probe certificate (Remark 16.71); the conclusion of 16's no-set-test theorem for $\mathcal{A}_{D,k}$ with every unital $D \neq k$ (Remark 16.88); and the observation that where 22's criterion gives no fresh exponent, 18's discrete-gap collapse still holds (after Proposition 16.86). From the seventh batch: the comparison with the full class, that $\Pi\mathcal{N} \cap K_{\mathbb{Z}a}$ is the radical of 23's extended ideal, so that the map of the cyclic arithmetic fibre to $\mathcal{N}/\Pi\mathcal{N}$ kills exactly its nilradical and the radical in Theorem 15.57 cannot be removed (Remark 15.119), with the explicit $\sqrt{\omega^{2a} + \omega^a} = \omega^{a/2} \sqrt{1 + \omega^a} \in \Pi\mathcal{N}$ (Example 15.87); the second route to 23's Puiseux radicality through 19's finite-axiom theorem, with the Boolean-power form, doubling and absolute integral closedness of the Puiseux fibre and the atomlessness of its

Boolean algebra (Remark 15.102); the Boolean-power form of the cyclic reduced fibre, with an atomless parameter of size and density $2^{\aleph_0}$, and for the real algebraic fragment the ring of locally constant $\mathbb{Z}$ -valued functions on Cantor space (Remark 15.114); and 19's v_T as a witness inside the cyclic workspace (Remark 15.104). Each follows from lemmas of the sources; none is a new method.

1.4 Relation to the repository

The foundations report defines $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$ (found:eq:omnific in [42]), and omnific integers appear as kernels and integer parts in the trigonometry, gamma-zeta, analysis and euclidean-three-space reports; the sibling report [46] treats their Diophantine geometry. The euclidean-three-space report proves the group analogue of the finite-side theorem for $\mathrm{SO}(3, \mathbf{No})$ (Section 8). The first-kappa report proves closedness of bounded-support Hahn fields, which contains the closure step of 06's construction (06 proves that step itself; Section 11.2). The κ -bounded coefficient rings $\mathbb{Z} + K((G^{<0}))_\kappa$ of L'Innocente-Mantova [23] are the published precedent for the bounded-support rings of 06 and 11. The Hahn-evaluation report [52], cited by 12, separates finite polynomial evaluation at ω from evaluation of unrestricted series; 12's embeddings use finite normal forms only. No statement of this report has a Lean counterpart; the repository has no omnific module.

Stale statements corrected. Three sources report a repository code search for “omnific” with no hits (06, 08, 11; 09 reports an explicitly incomplete result). At their pins the repository already defined $\mathbf{Oz}$ at found:eq:omnific and used it in the reports listed above; what it lacked was a report devoted to omnific arithmetic, and the collection now has two (this one and [46]). 03's companion draft and 04's “prior manuscript” were not in the repository at their pins; they are now manuscripts 05 and 02 of [46]. 07's description of the staging directory (“Gamma-zeta ZIP archives”) was true at its pin and is a snapshot statement. Of the later manuscripts, 13 and 15 (and 15's audit file) report that a repository search for “omnific” returned no matches; at their pin 71e9606 the foundations report already defined $\mathbf{Oz}$ at found:eq:omnific. 13 and 15 call the universal theorem a proposed, respectively candidate-original, contribution, and 13 calls its answer to Elliott's question one; the same theorem and the same answer are in the eight manuscripts of the first merge, dated a day earlier (Theorems 5.1 and 17.1), which 13 and 15 could not see, so both are independent re-derivations. 16 found no integer-valued-polynomial treatment in the repository; at its pin that was accurate, and the collection now has 10's (Section 16). 14's “companion manuscripts” are manuscripts 02 and 05 of the Diophantine report [46] and 03 of this one. Of the third batch, 17 proposes as additions the constant-matrix descent criterion, the coefficient-lattice Hom and isomorphism classification and the multiplication-kernel computation of Tor, with their consequences for orders and ideal powers; at its pin 71e9606 none of these was in the repository, but the collection now prints the descent criterion, the Hom classification, the multiplication kernel of the tensor square and the power growth as 14's, and the intersection $(\omega^\gamma) \cap (\alpha\omega^\gamma) = \omega^\gamma\Pi$ as 12's (Theorems 13.9, 13.18, 13.24 and 13.28 and Corollary 13.23), so 17 is credited there as an independent second source; its catalogue count of 49 main texts is a snapshot. 18 reports that targeted repository searches for integer-valued polynomial and rational terminology returned no matches; at its pin bfe739 manuscript 10 was present only as its check program code/10-omnific-arithmetic-checks.py, and the collection now prints 10's and 16's integer-valued polynomials, finite tests and rational self-maps (Section 16). 19 reports that the catalogue lists nine omnific manuscripts awaiting integration; that was true at its pin, and the first batch has since been written into this report and its sibling; 19's “repository quotient manuscript” was 06's text, now this report. 20 (pin 4cdeaec) reports that the catalogue separates the main reports from companions awaiting integration and cites this report under 06's title and

date; both were true at its pin, and 19, which 20 read only as a saved draft, has since been placed and written into this report (Section 15.8). 20 proposes support-local descent, the dominant-scale branch construction, relative freeness, the answers on atoms and size and the representation bounds as new; the identification of a fresh scale with Laurent blocks over the old Hahn field is 16's (Theorem 16.31 and Corollary 16.32), which was not in the repository at 20's pin, and the reprov'd antecedents are 19's, as 20 says. 21 (pin 9693b28) quotes Proposition 13.12(iii) as $d \leq \text{pd}_{A_d} D \leq d + 1$ with the next paragraph leaving the value undetermined for $d \geq 2$, and (iv) as bounds for lattice modules; both were accurate at its pin, and 21 itself now settles them (Corollary 13.47 and Theorem 13.48). 21 lists as its contributions, beside the dimensions, a uniform arbitrary-scale calculation of all Tor groups of two lattice quotients, the additive second-Tor identification (which it calls a dimension-shift consequence of the report's tensor calculation), the self-Tor ranks, second Tor as a linear-disjointness test and the Tor groups against constant-term modules. At its pin the report did not yet contain 17's Section 13.9, written in aae58bb after 21's pin; the collection now prints the second-Tor identification with its rank, the self-Tor range, the pair $\mathbb{Z}[\sqrt{2}]$, $\mathbb{Z}[\sqrt{3}]$ and the Tor groups against modules killed by the tail as 17's (Theorems 13.34 and 13.42 and Corollary 13.36), and 21 is credited there as a further source. The "quadratic self-Tor example" that 21 finds in the report is 12's $\text{Tor}_2(A/H, A/H) \cong D \oplus D$ (Theorem 13.11). 22 (pin bcac55a) states, in its introduction and in its audit (22-finite-tests-SOURCE_AUDIT.md, its Section 4), that the precise statements of its proposed package were not located in the selected repository material it inspected. That material did not include this report, which at 22's pin already contained the Laurent-block detector at one fresh scale (Theorems 16.31 and 16.33), workspace-uniform rational certificates with $d + 2$ values in one variable and fresh probes in several (Corollary 16.50), a rational function of numerator degree 0 and denominator degree 1 with no pole on $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ that deceives any set (Theorem 16.51; 16's $\omega^\beta/(X - \frac{1}{2})$ is $2\omega^\beta/(2X - 1)$, with coefficients in $\mathbf{Oz}$), the empty-or-proper-class alternative in one variable (Theorem 16.52) and Gaussian tests with a one-variable lifting of ordinary test sets (Propositions 16.47 and 16.64); so its novelty list is incomplete, and what is new in it is listed in Section 16.10. 23 (pin 343dc2c) read this report's normalization, fibre and branching sections and states accurately that Question 18.17(b) leaves fixed-workspace radicality and atomlessness open and that Theorem 15.57 keeps the radical; it credits Theorem 15.60. It presents its independent family of continuum size and the cardinality of the fibre as new. At its pin this report already stated, after Theorem 15.70, that one Laurent scale carries continuum many independent idempotents over $\mathbb{Z}_{\mathbb{R}}$, which with $G = \{0\}$ is an independent family of size $2^{\aleph_0}$ in $\text{Idem}(\mathbb{C}(\mathbb{Z}a))$; so the lower-bound half of 23's cardinality statement for the reduced fibre is 20's. Its Puiseux radicality also follows from 19's Theorem 15.51, which was in the report at its pin. What is new in 23 is listed in Section 15.10. The shipped audit files keep these statements verbatim; the README records the corrections.

Corrections to the sources. 03's source lost a relation symbol and prints " $C \text{ e } B[i]$ " for $C \neq B[i]$ (Proposition 15.18). 09 attributes the separated-scale division to Proposition 8.2.1 of [23]; the displayed separated-scale factorization is Fact 8.0.1 (Gonshor's Theorem 8.6) with its example on p. 46, as 11 cites, while Proposition 8.2.1 is the truncation criterion for divisibility cited by 08 (Section 4.2). 09's hypothesis $\lambda \geq 2^{\aleph_0}$ is implied by $\lambda^{\aleph_0} = \lambda$ (Section 11.4). 04's remark on its $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$ example is true but misleading (Remark 10.6). 06's remark that $|A_\kappa^{06}/xA_\kappa^{06}| \geq \kappa$ "under the cardinal-arithmetic hypothesis" needs no hypothesis and in fact gives $\kappa^{<\kappa}$ (Section 12). 06, 04 and 09 state global choice, which is never used (Section 2.1). 12 assumes $c^2 \in D$ for its flat dimensions $d + 1$ and 2 of cyclic modules, but needs it only for its Tor_2 formula (Theorem 13.11), and gives only bounds for projective dimensions that are exact in rank one (Proposition 13.12; 21 has since determined them for every d , Corollary 13.47 and Theorem 13.48). 14 credits the Diophantine

report’s example $[1 : \sqrt{2}]$ but not its rationality theorem for primitive directions, of which 14’s criterion is the Gaussian and principal-ideal extension (Section 13.8). 16’s rational collapse has the D-ring precedent that 16 itself credits (Theorem 16.42). 17 assumes κ uncountable for its exact relation count, but the proof uses only regularity, so $\kappa = \aleph_0$ is allowed (Remark 13.32). 19 did not see 03’s text and reproves four of its results without credit; they are credited here (Section 15.8). The real analogue of 18’s question on minimal tests with omnific nodes is answered by 16’s optimality theorem; the Gaussian question itself stays open (Question 18.16). 20’s dominant-scale lemma repeats 16’s Laurent blocks without credit (it could not see them); they are credited (Lemma 15.62). 21 repeats 17’s Tor results without credit (it could not see them); they are credited (Section 13.10); its proofs contain no error found in this merge. 22 repeats 10’s and 16’s results without credit (it did not read this report); they are credited (Section 16.10), and its proofs contain no error found in this merge. 23 does not cite 20’s continuum family or 19’s finite-axiom theorem (see above; credited in Section 15.10); its proofs contain no error found in this merge. No source contains a false theorem.

2 Conventions, notation, and the constant-term retraction

2.1 Foundations

We work in von Neumann–Bernays–Gödel class theory (NBG) with the axiom of choice for sets. Every surreal number, every normal-form support and every coefficient field or target ring called *set-sized* is a set; the carriers **No**, **Oz**, **No**[i] and most rings of Sections 3 and 5 are proper classes. A homomorphism with a class source is a class function obeying the usual identities, and its restriction to a set is a set.

The sources state different foundations. Manuscript 06 works in Gödel–Bernays set theory with global choice, 09 in GBC, and 04 in “GBC, with ordinary choice for sets” (GBC already contains global choice); 07, 08 and 11 work in NBG with choice for sets, 10 in NBG with choice, and 03 in NBG for the classes and ZFC for its set-sized constructions; 07 remarks that global choice is unnecessary. *No argument of the first two batches uses global choice.* Of the third batch, 17 and 19 state Gödel–Bernays class theory with global choice; 17’s uses of it (choosing preimages, a Schanuel splitting) are avoided below, and 19’s one essential use, a class ultrafilter on a Boolean algebra that may be a proper class, is stated with its hypothesis (Lemma 15.48); only the results built on it (Theorem 15.49 and Corollary 15.50) need global choice. 20 states the same foundation; its splitting results need only choice for sets (Remark 15.75), and its extension of branch maps (Theorem 15.80) uses class ultrafilters as 19’s detection does. 23 uses only set-sized rings and choice for sets. The collision step of the universal theorem (Theorem 5.1) is choice-free in the formulation of 08, which we adopt: for a set S , its Hartogs ordinal is an ordinal that does not inject into S ; it exists in ZF. The same device is used in the rotation-group theorem `e3:cut:thm:smallquotient` of [43]. Choice for sets enters only where it is visibly needed: maximal ideals of set-sized rings (Section 11), cardinal arithmetic such as $\kappa^{\aleph_0} = \max(2^{\aleph_0}, |[\kappa]^{\leq \aleph_0}|)$, and the existence of residue fields.

Definition 2.1 (Set-sized quotients and detection). A class ideal J of a class ring R *admits a set-sized quotient* if there is a surjective class ring homomorphism $R \rightarrow S$ onto a set-sized ring with kernel J . A ring homomorphism φ *detects* x if $\varphi(x) \neq 0$; a module M *detects* x if $xM \neq 0$.

This is the convention of 04 (“set-realizable quotient”), 08, 09 (“set-sized realization”), 10 (“set-sized quotient presentation”) and 11. Manuscripts 06 and 07 alternatively represent a class quotient by Scott codes (the members of a coset of least set-theoretic rank form a nonempty set); every quotient statement below may be read in either way, and no set or class whose members are arbitrary

proper classes is formed. Notation such as $\text{Hom}(\mathbf{Oz}, S)$ abbreviates a statement about individual class maps and their factorizations; after the universal theorem these maps are coded by their restrictions to a set-sized coefficient ring.

The universe-relative reading. Manuscripts 04, 10 and 11 add an equivalent reading: fix a lower universe containing the coefficient fields, targets and supports, and place the surreal carrier in a higher universe; “set-sized” then means small in the lower universe. The size restriction must be carried in the statement. A set-sized fragment of surreal arithmetic viewed in a larger universe has its own identity map as a set-sized target, so it does not satisfy the full-class theorem (11 insists that its main theorem is relative to one fixed universe, not to fragments viewed in a larger one).

2.2 Normal forms and the orientation of exponents

A nonzero surreal has a unique Conway normal form $x = \sum_{\gamma \in S} r_\gamma \omega^\gamma$ with nonzero real coefficients and a *reverse well-ordered set* S of exponents (every nonempty subset has a greatest element) [12, 20, 5]. We use *large monomials*: ω^γ is infinitely large when $\gamma > 0$, and $\omega^{\gamma+\delta} = \omega^\gamma \omega^\delta$ is Conway’s monomial law, not an exponential. For an abstract ordered abelian group G we write formal monomials X^g with the same orientation, and $k((X^G))$ for the Hahn series over a set field k with reverse well-ordered set supports in G (for $G = \mathbf{No}$ a proper class of series). The usual small-monomial notation is recovered by $t^g = X^{-g}$; sources that write t with well-ordered supports reverse all signs. A family is strongly summable if the union of its supports is reverse well ordered and each exponent receives finitely many contributions; strong sums are coefficientwise and are *not* limits of partial sums in the fine order topology.

Purely infinite refers to support, not to size or sign: $-\omega$ is purely infinite, while $\omega - 1$ is infinite but not purely infinite. This is the convention of the foundations report, which defines $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$ with Π the purely infinite normal forms (`found:eq:omnific` in [42]); the gamma-zeta report uses the same symbol [51]. The trigonometry report writes the same class Π as the normal forms supported at *negative t -exponents* (`trigonometry:eq:split` in [44]), because it uses $t = \omega^{-1}$; the two descriptions agree after the sign change $t^g = \omega^{-g}$.

2.3 The rings

Let k be a set field, G an ordered abelian group (a set, or the class $\mathbf{No}$), and $D \subseteq k$ a unital subring. Put

$$\Pi_k(G) = \{f \in k((X^G)) : \text{supp } f \subseteq G_{>0}\}, \quad \mathfrak{m}_k(G) = \{f \in k((X^G)) : \text{supp } f \subseteq G_{<0}\}, \quad (2.1)$$

$$\mathcal{A}_{D,k}(G) = D + \Pi_k(G), \quad \mathcal{B}_{D,k}(G) = D + \mathfrak{m}_k(G). \quad (2.2)$$

Both sums are direct as additive groups, and the zero series belongs to both ideals. We drop (G) when $G = \mathbf{No}$ and write $\Pi = \Pi_{\mathbb{R}}$. For $k = \mathbb{R}$ and $k = \mathbb{C}$ the identification $X^g \mapsto \omega^g$ gives $\mathbb{R}((X^{\mathbf{No}})) = \mathbf{No}$ and $\mathbb{C}((X^{\mathbf{No}})) = \mathbf{No}[i]$ (a union of two reverse well-ordered supports is reverse well ordered, so complexification is coefficientwise), and

$$\mathbf{Oz} = \mathcal{A}_{\mathbb{Z},\mathbb{R}} = \mathbb{Z} \oplus \Pi, \quad \mathbf{Oz}[i] = \mathbf{Oz} + i\mathbf{Oz} = \mathcal{A}_{\mathbb{Z}[i],\mathbb{C}} = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}, \quad \Pi_{\mathbb{C}} = \Pi + i\Pi. \quad (2.3)$$

We call $\mathbf{Oz}[i]$ the *Gaussian omnific ring*; it is a subring of $\mathbf{No}[i]$, not the surcomplex field, and not the integral closure of $\mathbb{Z}$ in $\mathbf{No}[i]$ (08, 09). The finite surreal and surcomplex rings are $\mathcal{O}_{\mathbb{R}} = \mathcal{B}_{\mathbb{R},\mathbb{R}}$ and $\mathcal{O}_{\mathbb{C}} = \mathcal{B}_{\mathbb{C},\mathbb{C}}$, and $\mathcal{A}_{\mathbb{R},\mathbb{R}} = \mathbb{R} + \Pi$ is the nonnegative-support ring (03’s $P_{\mathbb{R}}$). For a general set

field k (06, 07, 10) the class field $k((X^{\mathbf{No}}))$ need not be realized inside the surreals; nothing below requires such a realization.

Lemma 2.2 (Constant term [03, 04, 06, 07, 08, 09, 10, 11, 13, 14, 15, 16]). *The map $\text{ct} : \mathcal{A}_{D,k}(G) \rightarrow D$ extracting the coefficient of X^0 is a surjective unital ring homomorphism with kernel $\Pi_k(G)$, split by the inclusion of constants. On $\mathcal{B}_{D,k}(G)$ the same coefficient is a split surjective ring homomorphism with kernel $\mathfrak{m}_k(G)$; on $\mathcal{O}_{\mathbb{R}}$ and $\mathcal{O}_{\mathbb{C}}$ it is the standard part, written st .*

Proof. All exponents occurring in $\mathcal{A}_{D,k}(G)$ are nonnegative, and a sum of two nonnegative exponents is zero only when both are zero; so the constant coefficient of a product is the product of the constant coefficients. Additivity, the kernel and the section are immediate from uniqueness of the decomposition (2.2). The negative side is identical with nonpositive exponents. $\square$

Remark 2.3 (What ct is not). The map ct is defined on a ring of generally infinite elements; it is not the standard part, and coefficient extraction at 0 is not multiplicative on $\mathbf{No}$: $\omega \cdot \omega^{-1} = 1$ although both factors have zero constant coefficient. This is the warning recorded in [49] and `diff:warn:ct` of [53]. Nor is ct order preserving: $\omega - N > 0$ but $\text{ct}(\omega - N) = -N$ for every ordinary $N \geq 1$ (04, 08, 09, 11). On $\mathcal{A}_{D,k}$ the two maps ct and st receive different names because they have different domains (04), and (8.1) below shows that they cannot be combined.

Lemma 2.4 (Degree and units [03, 04, 07, 08, 09, 10, 11, 13, 14, 16]). *For $x \neq 0$ in $k((X^G))$ put $\deg_{\omega}(x) = \max \text{supp } x$. Then $\deg_{\omega}(xy) = \deg_{\omega}(x) + \deg_{\omega}(y)$ for $x, y \neq 0$, so $\mathcal{A}_{D,k}(G)$ and $\mathcal{B}_{D,k}(G)$ are domains, and*

$$\mathcal{A}_{D,k}(G)^{\times} = D^{\times}, \quad \mathbf{Oz}^{\times} = \{\pm 1\}, \quad \mathbf{Oz}[i]^{\times} = \{\pm 1, \pm i\}.$$

In particular every nonconstant element of $\mathcal{A}_{D,k}(G)$, such as $1 + \omega^a$ or $\omega^{\sqrt{2}} + \omega + 1$, is a nonunit.

Proof. The leading coefficient of a product is the product of the leading coefficients, which is nonzero because k is a field; the degree identity follows. A nonzero element of $\mathcal{A}_{D,k}(G)$ has nonnegative degree, so $xy = 1$ forces both degrees to be 0, i.e. both factors are constants in D with $xy = 1$ in D . The two unit lists are the units of $\mathbb{Z}$ and $\mathbb{Z}[i]$. $\square$

Here $\deg_{\omega}$ is the largest growth exponent (04); it is not the ordinal-valued degree of the order type of a support used in parts of the factorization literature [23].

2.4 Notation and renamings

The manuscripts use many symbols for the purely infinite ideal and several for the constant term. We use the repository's Π and the common ct , and resolve every collision by renaming in the new material. The renamings, and the tags distinguishing the different set-sized constructions, are listed below; Section 13 and Section 16.6 list the further renamings of 12, 14 and 16, Sections 13.9, 15.8 and 16.8 those of 17, 18 and 19, Section 15.9 those of 20, Section 13.10 those of 21, Section 16.10 those of 22, and Section 15.10 those of 23.

Object	In the sources	Here
purely infinite ideal	$\mathcal{I}$ (03, 06, 09, 14, 21), $\mathcal{J}_k$ (04), $\mathfrak{J}$ (07, 08, 11), $\mathfrak{P}$ (10), J_k (13), $\mathcal{I}_K$ (15), $\mathcal{P}_k$ (16), $\mathfrak{m}$ (17), $\mathcal{J}$ (18, 19, 20)	Π , Π_k , $\Pi_{\mathbb{C}}$
constant term	c_0 (06), π , π_D (10), ε_D (09), ε (12), ct (others)	ct
coefficientwise constant term on polynomials	Π (10), c_0 (06)	ct_*
$D + \Pi_k$	$\mathcal{A}_{D,k}$ (04), $A_{D,k}$ (06), $A(D, K)$ (07), R_D (08), $\mathcal{A}_D$ (09), $R(D, K)$ (10), $\mathcal{Z}_{K,D}$ (11), $A(D, k, G)$ (03), A_D (13), R_A (15), $\mathcal{A}_D$ (16)	$\mathcal{A}_{D,k}$, $\mathcal{A}_{D,k}(G)$
Gaussian omnific ring	GOz (06), $\mathbf{Og}$ (09)	$\mathbf{Oz}[i]$
surcomplex field	$\mathbf{No}(i)$ (04, 07, 08, 09, 12, 15, 16, 19)	$\mathbf{No}[i] = \mathbf{No}[i]$ [49]
Hahn field over H_a	L_a (06), $\mathcal{F}_a$ (07), L_h (03, set version)	$\mathcal{F}_a(k)$; L_h for 03's set field
localization $\mathbf{Oz}[\omega^{-a}]$	L_a (07)	$\mathbf{Oz}[\omega^{-a}]$ (never L_a)
binomial quotient $\mathbf{Oz}/(1 + \omega^a)$	B_a (07)	$\mathcal{Q}_a$
07's compression map $a\Psi$	E_a	Ψ_a
integral closure	$B, C, \text{Int}_F(A)$ (03), $N_{\mathbb{R}}, N_{\mathbb{C}}, \text{Int}_A(F)$ (19, 20)	$\mathcal{N}, \mathcal{N}_{\mathbb{C}}, \text{IC}_F(A)$
integer-valued polynomials	$\text{Int}(\mathbb{Z}^r)$, I_r (10)	$\text{Int}(\mathbb{Z}^r)$
set-sized integer parts	A_{κ} (03, 08, 17), R_{κ} (06), $A_{\lambda,D}$ (09), $R_{\Gamma, < \kappa}$ (11)	$A_{\kappa}^{03}, A_{\kappa}^{06}, A_{\kappa}^{08}, A_{\lambda,D}^{09}, R_{\Gamma, < \kappa}^{11}$ (17's is $R_{\Gamma_{\kappa}, < \kappa}^{11}$)
their ideals	$I_{\kappa}, \mathcal{J}_{\kappa}, I_{\lambda,k}$	$\Pi_{\kappa}^{03}, \Pi_{\kappa}^{06}, \Pi_{\kappa}^{08}, \Pi_{\lambda}^{09}$
small exponents below a support	$\text{Tiny}(S)$ (09), $U(S)$ (10), $G_{\ll h}$ (03)	$U_G(S), H_h(G)$
binomial series	$B_m(t)$ (08)	$\mathbf{b}_m(t)$

Throughout, for $a > 0$ in an ordered group G ,

$$H_a(G) = \{h \in G : n|h| < a \text{ for every ordinary } n \geq 1\}, \quad H_a = H_a(\mathbf{No}), \quad (2.4)$$

and $h \ll a$ means $h > 0$ and $h \in H_a(G)$. This is a comparison of *exponents*: $h \ll a$ does not say that ω^h is small (it is infinite whenever $h > 0$). For a set $S \subseteq G_{>0}$ we write $U_G(S) = \{a > 0 : na < s \text{ for all } s \in S, n \geq 1\}$ (09's $\text{Tiny}(S)$, 10's $U(S)$). Letters $\kappa, \lambda, \mu, \theta$ denote cardinals and are restated locally; $\mu = \text{cf } \kappa$ and $\theta = \kappa^{\aleph_0}$ in the construction of 03.

3 Support gaps and the purely infinite ideal

This section collects the facts that use the full surreal exponent class. All eight manuscripts of the first merge prove them, as do 13 and 15 and manuscript 05 of the sibling report [46], and 14 and 16 use them; of the third batch, 17 and 19 reprove the gaps, the units and the common divisors, and 18 uses the constant term. They are background for the new results, not claimed as new. Throughout, k is a set field and $D \subseteq k$ a unital subring.

Lemma 3.1 (Gaps below and above a set [03, 04, 06, 07, 08, 09, 10, 11, 13, 15, 16]).

(i) Every set of surreal numbers has a strict upper and a strict lower bound in $\mathbf{No}$.

(ii) For every set $S \subseteq \mathbf{No}_{>0}$ there is $h > 0$ with $nh < s$ for all $s \in S$ and all ordinary $n \geq 1$; in particular $0 < h < S$.

(iii) For such S the class $U_{\mathbf{No}}(S)$ contains the interval $(0, h)$, which is not a set, and for every ordinal Λ the exponents

$$a_\alpha = \frac{h}{\omega^{\alpha+1}} \quad (\alpha < \Lambda) \quad (3.1)$$

are distinct elements of $U_{\mathbf{No}}(S)$, each $\ll h$.

Proof. For a set T , the cuts $\{T \mid \emptyset\}$ and $\{\emptyset \mid T\}$ are set cuts and give (i). For (ii), apply the set cut $\{0 \mid T\}$ to the set $T = \{s/n : s \in S, n \geq 1\}$ of positive surreals; for $S = \emptyset$ take $h = 1$. For (iii), $(0, h)$ is not a set: a positive lower bound of it would have a positive half still in the interval. The denominators $\omega^{\alpha+1}$ are the Conway monomials of the ordinals $\alpha + 1$ in the *exponent* field; they strictly increase with α and exceed every ordinary integer, so the a_α are distinct, positive and satisfy $na_\alpha < h$. $\square$

Only the existence of a separator is used, not its simplicity or birthday (08). The sources use several equivalent families in place of (3.1): $a/\omega^{\alpha+1}$ (04, 05, 06, 08, 09, 03), $a/(\omega + \alpha + 1)$ with the field sum of embedded ordinals (07), $\delta\omega^{-(\alpha+1)}$ (11), $\epsilon\xi/\kappa$ (10), $\epsilon/(\bar{\alpha} + 2)$ with $\bar{\alpha}$ the surreal ordinal (13), and $g\omega^{-(\xi+1)}$ (15). Each is a set whenever α ranges over a set.

Example 3.2 (Surreal gaps where real gaps fail [03, 06, 08, 10, 11]). The series $f = \sum_{n \geq 1} \omega^{1/n}$ lies in Π ; its support has no positive *real* lower bound but has positive surreal lower bounds, for instance $1/\omega$. With $a = 2/\omega$ and $b = 1/\omega$ one has $na < 1/m$ for all ordinary m, n (10). In the fixed exponent group $\mathbb{R}$ the same series is purely infinite but is not divisible by any X^h , $h > 0$, inside the nonnegative-support ring (03); this is where the passage from monomials to the whole ideal fails in a fixed group (Section 10).

Proposition 3.3 (Common monomial divisors [03, 04, 06, 07, 08, 09, 10, 11, 13, 14, 15]). *Let $A = \mathcal{A}_{D,k}$ and Π_k its purely infinite ideal.*

- (i) *Every set-sized family $\mathcal{F} \subseteq \Pi_k$ satisfies $\mathcal{F} \subseteq X^a \Pi_k$ for some $a > 0$.*
- (ii) *Each $f \in \Pi_k$ is a single product of two elements of Π_k ; hence $\Pi_k^2 = \Pi_k$, and $\Pi_k = \bigcup_{a > 0} X^a A$ is a directed union of principal ideals.*
- (iii) *Π_k is not generated, as an ideal of A , by any set of its elements.*
- (iv) *$d\Pi_k = \Pi_k$ for every $0 \neq d \in D$; in particular $n\Pi_k = \Pi_k$ for every nonzero ordinary integer n .*

Proof. (i) The union of the supports of the members of $\mathcal{F}$ is a set of positive surreals; choose $0 < a <$ that union by Lemma 3.1. Translating each support by $-a$ leaves it positive and reverse well ordered, so $X^{-a}f \in \Pi_k$. (ii) Apply (i) to $\{f\}$: $f = X^a(X^{-a}f)$ with both factors in Π_k . If $0 < b < a$ then $X^a A \subseteq X^b A$, so the union is directed. (iii) If a set generated Π_k , (i) would place the generated ideal inside $X^a A$; but $X^{a/2} \in \Pi_k$ is not in $X^a A$, since its unique quotient by X^a in the ambient field is $X^{-a/2}$, which has a negative exponent. (iv) Divide every coefficient by d , which is invertible in k . $\square$

Idempotence and divisibility do not by themselves prove the universal theorem: an idempotent, rationally divisible ideal can act nontrivially on infinite set-sized modules, and a homomorphism to an infinite characteristic-zero ring need not kill an arbitrarily divisible additive subgroup (04, 06, 07, 08). In positive characteristic the last item already suffices (Remark 5.10).

$$n\mathbf{Oz} = n\mathbb{Z} + \Pi = \text{ct}^{-1}(n\mathbb{Z}) \quad (n \geq 1), \quad d\mathbf{Oz}[i] = d\mathbb{Z}[i] + \Pi_{\mathbb{C}} \quad (0 \neq d \in \mathbb{Z}[i]). \quad (3.2)$$

Indeed $n\Pi = \Pi$ and $d\Pi_{\mathbb{C}} = \Pi_{\mathbb{C}}$ by [Proposition 3.3\(iv\)](#). Consequently

$$\Pi = \bigcap_{n \geq 1} n\mathbf{Oz} = \bigcap_{p \text{ prime}} p\mathbf{Oz}, \quad \Pi_{\mathbb{C}} = \bigcap_{n \geq 1} n\mathbf{Oz}[i], \quad (3.3)$$

since a nonzero ordinary integer (or Gaussian integer) is not divisible by every n or every prime (08, 10, 11). For a general D the intersection formula can fail: for $D = \mathbb{Q}$ every element of $\mathcal{A}_{\mathbb{Q},\mathbb{R}}$ is divisible by every ordinary positive integer (11).

Proposition 3.4 (Clearing and fraction fields [03, 04, 06, 07, 08, 09, 11, 14, 15, 16]).

- (i) For every set $E \subseteq k((X^{\mathbf{No}}))$ there is $h > 0$ with $X^h E \subseteq \Pi_k$. For $E \subseteq \mathbf{No}$ or $E \subseteq \mathbf{No}[i]$ this reads $\omega^h E \subseteq \Pi$, respectively $\omega^h E \subseteq \Pi_{\mathbb{C}}$.
- (ii) $\text{Frac}(\mathcal{A}_{D,k}) = k((X^{\mathbf{No}}))$; in particular $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ and $\text{Frac}(\mathbf{Oz}[i]) = \mathbf{No}[i]$.
- (iii) Inverting the positive monomials gives the whole field: if $\mathcal{U} = \{X^a : a > 0\} \cup \{1\}$ then $\mathcal{U}^{-1}\mathcal{A}_{D,k} = k((X^{\mathbf{No}}))$.
- (iv) Neither $\mathbf{No}$ nor $\mathbf{No}[i]$, nor any localization of $\mathcal{A}_{D,k}$ in which some nonzero element of Π_k becomes a unit, admits a unital homomorphism to a nonzero set-sized ring or a nonzero set-sized unital module.

Proof. (i) The union of the supports of the members of E is a set; by [Lemma 3.1\(i\)](#) choose $h > 0$ larger than $-g$ for each exponent g in it. Then every exponent of $X^h x$, $x \in E$, is positive. (ii) and (iii) For $x \in k((X^{\mathbf{No}}))$ apply (i) to $\{x\}$: $x = (X^h x)/X^h$ with numerator and denominator in $\mathcal{A}_{D,k}$; the reverse inclusions hold because $\mathcal{A}_{D,k}$ is a subring of the field. (iv) is proved after the universal theorem ([Corollary 5.9](#)). $\square$

Part (i) is manuscript 05’s “Clearing supports” theorem, now in the sibling report [46]; 03 credits it to 05 and adds the algebra-wide formulation of [Section 15](#). Clearing one element is weaker than clearing all powers of it with one denominator, which [Section 15](#) also supplies (03).

4 The collision mechanism

Every proof of the universal theorem combines two finite multiplicative identities with a cardinal comparison. This section states the identities once, in the forms used by the different sources. The first ([Lemma 4.1](#), from 06, and in a special form from 03) gives a uniform proof of the sharp cardinal bounds of [Sections 10](#) and [11](#), including those that counting monomials cannot reach; the others are genuinely different routes and are kept as such.

4.1 Scaled-field collision

Lemma 4.1 (Scaled-field collision [06; 03]). *Let A be a commutative subring of a field F , let $L \subseteq F$ be a subfield and let $0 \neq m \in F$ satisfy $mL \subseteq A$.*

- (i) *If $\varphi : A \rightarrow S$ is a ring homomorphism, S any ring (not necessarily commutative, unital or reduced), and $\varphi(m^2) \neq 0$, then $x \mapsto \varphi(mx)$ is injective on L .*
 - (ii) *If M is a unital A -module, $v \in M$ and $m^2 v \neq 0$, then $x \mapsto (mx)v$ is injective on L .*
- Consequently, if $|L| > |S|$ then $\varphi(m^2) = 0$, and if $|L| > |M|$ then $m^2 M = 0$.*

Proof. Since $1 \in L$, $m \in A$. For distinct $x, y \in L$ the element $m/(x - y)$ lies in $mL \subseteq A$, and in A

$$(mx - my) \cdot \frac{m}{x - y} = m^2. \quad (4.1)$$

If $\varphi(mx) = \varphi(my)$, applying φ gives $\varphi(m^2) = 0$; no cancellation in S is used. For the module, apply $m/(x - y)$ to $(mx)v = (my)v$; commutativity of A turns the result into $m^2v = 0$. $\square$

The proof of (i) uses only that A is closed under addition and multiplication and contains mL ; so it applies verbatim when A is a nonunital subring such as Π_K , which is how it is used in [Theorem 5.1\(iii\)](#). The module version bounds $|M|$ itself. Deriving it through the ring $\text{End}_{\mathbb{Z}}(M)$ would only bound $|\text{End}_{\mathbb{Z}}(M)|$, which can be much larger than $|M|$ (06, 03, 09); this matters for every set-sized threshold below. In 03 the lemma appears as the identity $(X^{h/2}(f - f'))(X^{h/2}/(f - f')) = X^h$ for $f \neq f'$ in a smaller-scale field (03's “smaller-scale field obstruction”), i.e. the case $m = X^{h/2}$ of (4.1).

Lemma 4.2 (The scale subgroup and its Hahn field [03, 06, 07]). *Let G be an ordered abelian group (a set or $\mathbf{No}$) and $a \in G_{>0}$.*

- (i) $H_a(G)$ is a convex subgroup of G , divisible if G is, and $H_c(G) = H_{2c}(G)$ for every $c \in G_{>0}$.
- (ii) $k((X^{H_a(G)}))$ is a subfield of $k((X^G))$, and $X^c k((X^{H_a(G)})) \subseteq \Pi_k(G)$ for every $c \in G$ with $2c \geq a$; for instance $c = a$, or $c = a/2$ when $a/2 \in G$.
- (iii) For $G = \mathbf{No}$ the field $\mathcal{F}_a(k) = k((X^{H_a}))$ is a proper class: it contains the distinct monomials X^{a_α} of (3.1) for every ordinal α .

Proof. (i) If $h, h' \in H_a(G)$ then $2n|h| < a$ and $2n|h'| < a$ give $n|h + h'| < a$; negation and convexity are immediate, and if G is divisible then h/m satisfies the same inequalities. For every $c > 0$, $n|h| < c$ for all n is equivalent to $n|h| < 2c$ for all n . (ii) Supports of sums and products of series supported in a subgroup stay in it, and the Hahn inverse of $f = cX^g(1 + u)$, $\text{supp } u < 0$, is $c^{-1}X^{-g}\sum_{n \geq 0}(-u)^n$, whose support lies in the subgroup generated by $\text{supp } f$. If $2c \geq a$ and $h \in H_a(G)$ then $2|h| < a \leq 2c$, so $c + h > 0$; hence $X^c f$ has positive support for $f \in k((X^{H_a(G)}))$. (iii) Each $a_\alpha \ll a$ lies in H_a . $\square$

The field $\mathcal{F}_a = \mathcal{F}_a(\mathbb{R})$ is 07's $\mathcal{F}_a$ and 06's L_a with $k = \mathbb{R}$; it is real closed and contains a copy of all of $\mathbf{No}$ ([Proposition 14.1](#) and [Theorem 14.4](#)).

4.2 The geometric telescope

Lemma 4.3 (Bounded-exponent telescope [03, 04, 05, 06, 07, 08, 09, 10, 11, 15]). *Let G be an ordered abelian group, and let $0 < b < a$ and $a \ll c$ in G . Put $d = a - b$. Then*

$$q_{c;a,b} = \sum_{n \geq 0} X^{c-a-nd} \in \Pi_k(G), \quad (X^a - X^b) q_{c;a,b} = X^c. \quad (4.2)$$

The support of $q_{c;a,b}$ is countable and strictly decreasing. For every ordinary $N \geq 0$,

$$(X^a - X^b) \sum_{n=0}^N X^{c-a-nd} = X^c - X^{c-(N+1)d}, \quad (4.3)$$

and the remainder monomial $X^{c-(N+1)d}$ is still infinitely large.

Proof. For each ordinary $n \geq 0$, $a + nd \leq (n + 1)a < c$, so every exponent is positive; the exponents strictly decrease with n , so every nonempty set of them has a largest element. Multiplication by X^a gives the exponents $c - nd$ and multiplication by X^b gives $c - (n + 1)d$; at c the coefficient is 1 and at every $c - md$, $m \geq 1$, it is $1 - 1 = 0$. Each coefficient receives at most two contributions, so this is a Hahn identity. The finite identity is the same computation truncated; its remainder exponent $c - (N + 1)d \geq c - (N + 1)a > 0$. $\square$

Remark 4.4 (No limit is taken and no sum is transported). The remainders in (4.3) are positive monomials; they do not tend to 0 in the order, valuation or fine topology, not even at the ordinary scale 1. The infinite identity is a coefficient identity, not a limit of (4.3) (04, 06, 07, 08, 10, 03). A target homomorphism is never applied term by term to $q_{c;a,b}$: the series is first a single element of the source ring, and only the finite product identity is mapped (06, 08, 09, 11). This is why no continuity, valuation or strong-additivity hypothesis on homomorphisms appears anywhere. It is also why a ring of finite-support series behaves differently (Section 9): there $q_{c;a,b}$ is absent and the nonzero remainder of (4.3) cannot be removed.

Example 4.5 (A witness for ω [04, 10, 15]). With $c = 1$, $b = 1/\omega$ and $a = 2/\omega$, $(\omega^{2/\omega} - \omega^{1/\omega}) \sum_{n \geq 0} \omega^{1-(n+2)/\omega} = \omega$, while the finite partial products leave the nonzero remainders $\omega^{1-(N+1)/\omega}$ (04). In the lexicographic group $\mathbb{Z} \times \mathbb{Z}$ with $c = (1, 0)$, $a = (0, 2)$, $b = (0, 1)$ the quotient is $\sum_{n \geq 0} X^{(1, -n-2)}$ and the N -th remainder is $X^{(1, -N-1)}$ (10); under a compatible monomial realization each remainder is still infinite. 15 uses the same series $q = \sum_{n \geq 0} \omega^{1-(n+2)/\omega}$ to block the extension of the finite-support augmentation (Section 9).

Precedents. The countable geometric-series product with a monomial result is already in L’Innocente–Mantova [23, Remark 3.4.4], used there to show that $\deg \circ \text{ot}$ is not multiplicative when the value group does not embed in $\mathbb{R}$; 04 names it a direct precursor. The factorization of a series whose support meets two distinct nonzero Archimedean classes is [23, Fact 8.0.1] (Gonshor’s [20, Thm. 8.6]), with the geometric factorization displayed on p. 46 of the arXiv version; 11 cites it as the precedent of the division below. Manuscript 09 credited “Proposition 8.2.1 on separated-scale division”; that proposition is the truncation criterion (b divides c if and only if b divides a truncation $T_\Delta(c)$), whose proof divides the higher-scale part of c by b , and 08 cites it as the precedent for higher-scale divisibility. We credit all three places; none of the sources claims the geometric mechanism as new.

Lemma 4.6 (Weighted certificate [07]). *Let K be a set field, $r, s \in K^\times$, $a > 0$ and $\delta > \eta > 0$ with $\delta \ll a$. Then*

$$\frac{X^a}{rX^\delta - sX^\eta} = \frac{1}{r} \sum_{k \geq 0} \left(\frac{s}{r}\right)^k X^{a-\delta-k(\delta-\eta)} \in \Pi_K, \quad (4.4)$$

and if $\delta = \eta$ and $r \neq s$ the quotient is $(r - s)^{-1} X^{a-\delta}$.

Proof. The exponents are those of Lemma 4.3; multiplying by $rX^\delta - sX^\eta$ gives, at exponent $a - k(\delta - \eta)$ with $k \geq 1$, the two contributions $r \cdot r^{-1}(s/r)^k$ and $-s \cdot r^{-1}(s/r)^{k-1}$, which cancel. $\square$

4.3 Division of an arbitrary purely infinite element

The telescope divides a monomial. Manuscripts 09, 10 and 11 divide an arbitrary element directly, which removes the separate reduction to monomials.

Lemma 4.7 (Subordinate division [09, 10, 11, 13, 15]). *Let G be an ordered abelian group, $0 \neq s = \sum_{g \in S} c_g X^g \in \Pi_k(G)$ and $a > b > 0$ in $U_G(S)$. Put $d = a - b$. Then*

$$\frac{s}{X^a - X^b} = \sum_{g \in S} \sum_{n \geq 0} c_g X^{g-a-nd} \in \Pi_k(G), \quad (4.5)$$

so $s = (X^a - X^b)q$ with $q \in \Pi_k(G)$. The support of q lies in $\{g - a - nd : g \in S, n \in \mathbb{N}\}$, of cardinality at most $\max(|S|, \aleph_0)$; it is countable when S is.

Proof. In the ambient Hahn field $(X^a - X^b)^{-1} = X^{-a}(1 - X^{-d})^{-1} = X^{-a} \sum_{n \geq 0} X^{-nd}$. The product with s is a Hahn product: the sum of two reverse well-ordered sets is reverse well ordered and each exponent has finitely many representations (11 proves this two-support lemma in full; it is the standard Neumann support lemma). Every exponent that can occur satisfies $g - a - nd \geq g - (n+1)a > 0$ because $(n+1)a < g$; cancellation cannot create new exponents. Hence the quotient lies in $\Pi_k(G)$. $\square$

The inverse of $X^a - X^b$ is not in the ring; only its product with the larger-scale s is (11).

4.4 Separation principles

Lemma 4.8 (Separation [04, 09, 11; 03]). *Let R be a ring (possibly a proper class), $e \in R$, and $(x_t)_{t \in T}$ a set-indexed family with $e \in (x_s - x_t)R$ for all $s \neq t$.*

- (i) *If $\varphi : R \rightarrow S$ is a ring homomorphism with $\varphi(e) \neq 0$, then $t \mapsto \varphi(x_t)$ is injective; hence $|T| \leq |S|$ if S is a set.*
- (ii) *If R is commutative, M an R -module, $v \in M$ and $ev \neq 0$, then $t \mapsto x_tv$ is injective.*
- (iii) *If for every cardinal λ such a family with $|T| = \lambda$ exists, then every homomorphism from R to a set-sized ring kills e , and e annihilates every set-sized R -module.*

Proof. If $\varphi(x_s) = \varphi(x_t)$ with $e = (x_s - x_t)q$, then $\varphi(e) = (\varphi(x_s) - \varphi(x_t))\varphi(q) = 0$; neither commutativity of S nor preservation of 1 is used. For (ii), $ev = q(x_s - x_t)v = 0$. For (iii), take λ larger than the target, or than the Hartogs ordinal of the target, and apply the pigeonhole principle to a set. $\square$

Item (i) is 04's separation lemma, (ii) is 09's orbit separation and 03's exponent-family obstruction, and (iii) is 11's collision-divisibility principle.

Theorem 4.9 (Divisibility-separated families [11, 15]). *Let $\mathcal{S} \subseteq \Pi_k$ be a set and Λ an ordinal. There are distinct monomials $(u_\alpha)_{\alpha < \Lambda}$ in $\mathcal{A}_{D,k}$ with*

$$s \in (u_\alpha - u_\beta)\Pi_k \quad (s \in \mathcal{S}, \alpha \neq \beta).$$

Proof. Choose $\delta > 0$ strictly below the union of the supports of the members of $\mathcal{S}$ (Lemma 3.1), put $a_\alpha = \delta\omega^{-(\alpha+1)}$ and $u_\alpha = X^{a_\alpha}$. For $\alpha < \beta$, $0 < a_\beta < a_\alpha \leq \delta/\omega$, so $na_\alpha < \delta$ for every n , and Lemma 4.7 applies to each s ; reversing the indices only changes the sign of the quotient. $\square$

The coefficients of the u_α are 1; no independence of the coefficients of s is used, and cancellations inside s or the quotient are harmless (11). 15 proves the same with the family $u_\xi = \omega^{g\omega^{-(\xi+1)}}$, $\xi < \theta$, for every ordinal θ , and simultaneously for a set of elements s .

5 The universal set-sized quotient

5.1 The theorem

Theorem 5.1 (Universal set-sized quotient [04, 06, 07, 08, 09, 10, 11, 13, 15; 03, 05]). *Let K be a set field, $D \subseteq K$ a unital subring, and $A = \mathcal{A}_{D,K} = D + \Pi_K$ with the full surreal exponent class.*

(i) *Let S be a set-sized associative ring, not necessarily unital or commutative, and let $\varphi : A \rightarrow S$ be an additive and multiplicative class map, not necessarily preserving 1. Then $\varphi(\Pi_K) = 0$, and $\varphi = \psi \circ \text{ct}$ for a unique ring homomorphism $\psi : D \rightarrow S$, namely $\psi = \varphi|_D$.*

(ii) *For every set-sized unital ring S , restriction to D and composition with ct are mutually inverse bijections, natural in S ,*

$$\text{Hom}_1(A, S) \longleftrightarrow \text{Hom}_1(D, S).$$

(iii) *Every additive and multiplicative map from the nonunital ring Π_K to a set-sized ring is zero.*

(iv) *Every set-sized unital A -module M satisfies $\Pi_K M = 0$, and its action is $f \cdot v = \text{ct}(f)v$.*

Proof. We follow 06’s scaled-field argument in 08’s choice-free Hartogs form.

Positive monomials are killed. Let $\varphi : A \rightarrow S$ be as in (i), or an additive multiplicative map defined only on Π_K as in (iii). Fix $g > 0$, put $m = X^{g/2}$ and $L = \mathcal{F}_g(K) = K((X^{H_g}))$. By Lemma 4.2(ii) with $c = g/2$, $mL \subseteq \Pi_K$; also $m/(x - y) \in mL$ for distinct $x, y \in L$, so the identity (4.1) takes place inside Π_K . Let λ be the Hartogs ordinal of the set S , and put $b_\alpha = g/\omega^{\alpha+1}$ for $\alpha < \lambda$; these are distinct elements of H_g (Lemma 3.1(iii)), so $\alpha \mapsto X^{b_\alpha}$ is an injection $\lambda \rightarrow L$. If $\varphi(X^g) = \varphi(m^2) \neq 0$, Lemma 4.1(i) makes $x \mapsto \varphi(mx)$ injective on L , and composing gives an injection $\lambda \rightarrow S$, which is impossible. Hence $\varphi(X^g) = 0$ for every $g > 0$.

The whole ideal is killed. For $f \in \Pi_K$, Proposition 3.3 gives $f = X^a h$ with $a > 0$ and $h \in \Pi_K$, so $\varphi(f) = \varphi(X^a)\varphi(h) = 0$. This proves (iii), and for (i) it gives $\varphi(d + f) = \varphi(d)$ for $d \in D$, $f \in \Pi_K$. Thus $\varphi = \varphi|_D \circ \text{ct}$, and $\psi = \varphi|_D$ is forced because ct is surjective.

The correspondence. If φ is unital, so is ψ . Conversely, for a unital $\psi : D \rightarrow S$ the composite $\psi \circ \text{ct}$ is a unital homomorphism by Lemma 2.2, and its restriction to D is ψ . Post-composition with a map of targets preserves both constructions, which proves (ii).

Modules. Let M be a set-sized unital A -module and $v \in M$. If $X^g v \neq 0$, Lemma 4.1(ii) injects L , hence the Hartogs ordinal of M , into M ; so $X^g M = 0$ for every $g > 0$, and the factorization $f = X^a h$ gives $\Pi_K M = 0$. Writing $f = \text{ct}(f) + (f - \text{ct}(f))$ gives $f \cdot v = \text{ct}(f)v$. $\square$

Sources and routes. The statement is proved in the following generality by the sources: any set field K and unital $D \subseteq K$ (06, 07, 10); $k \in \{\mathbb{R}, \mathbb{C}\}$ and any unital $D \subseteq k$ (04, 09, 11); **Oz**, **Oz**[i] and complex coefficients with any unital $D \subseteq \mathbb{C}$ (08). Nonunital maps are treated by 08 (as a theorem), 06 (as a remark) and 10 (including maps defined on Π_K alone). Noncommutative targets are allowed in all eight sources. Manuscript 05 of the sibling report [46] proves the unital case for **Oz** with the telescope and a cardinal $\kappa > |A|$, and 03 reproves that case (credited to 05) through its exponent-family obstruction. Two later manuscripts prove the theorem again, independently and without seeing the others: 13 for any set field k and unital $D \subseteq k$ (unital maps, noncommutative targets), by direct division of an arbitrary purely infinite element with the Hartogs family $\epsilon/(\bar{\alpha} + 2)$ (the route of 09, 10 and 11 below), and 15 for $k \in \{\mathbb{R}, \mathbb{C}\}$ and unital $D \subseteq k$, with nonunital maps in a remark, by the separating families of Theorem 4.9 and a Hartogs ordinal, which is 08’s formulation; 15 also states the universe-relative reading of Section 2.1. Neither adds

a hypothesis-free generalization. Of the third batch, 17 proves the unital case for $D_0 + \Pi_k$ with $k \in \{\mathbb{R}, \mathbb{C}\}$ and any set-sized unital $D_0 \subseteq k$ by 06's scaled-field collision, choosing preimages by global choice, and declares it background; 19 proves the unital case for $\mathbf{Oz}$ by the explicit telescope below with a cardinal $\lambda > |S|$ and credits it to 06's text. Both are credited here as further proving sources; neither adds generality. So is 20, of the fourth batch, which proves the unital case for $\mathbf{Oz}$ by the same telescope with the exponents $a/\omega^{\alpha+1}$ ($\alpha < \lambda$, $\lambda > |S|$), crediting this report. The proofs differ, and the following are genuinely different routes, kept here in brief.

- *Explicit telescope* (04, 05, 06 appendix, 08). If $\varphi(X^c) \neq 0$, then for distinct $a_\alpha, a_\beta \ll c$ the telescope (4.2) puts X^c in $(X^{a_\alpha} - X^{a_\beta})A$, so Lemma 4.8(i) makes $\alpha \mapsto \varphi(X^{a_\alpha})$ injective, which is impossible for α ranging over a set larger than S ; Proposition 3.3 then kills all of Π_K . This is manuscript 05's proof, which the sibling report [46] quotes and refers here for; 05 chooses a cardinal $\kappa > |S|$ with global choice, where the Hartogs ordinal suffices. The route uses only countable supports, which is why it survives in the support-bounded rings of Section 9.
- *Direct division of an arbitrary element* (09, 10, 11). If $\varphi(s) \neq 0$ for $s \in \Pi_K$, Lemma 4.7 gives $s \in (X^a - X^b)\Pi_K$ for all $a > b$ in $U_{\mathbf{No}}(\text{supp } s)$, and Theorem 4.9 supplies such families of every set size, so Lemma 4.8(iii) applies to s itself; no separate reduction to monomials is needed. 10 phrases this as an injection $U(\text{supp } s) \rightarrow S$, $a \mapsto \varphi(X^a)$, valid also for nonunital maps on Π_K ; 11 phrases it as divisibility-separated families.
- *Weighted certificates* (07). With $\delta_\alpha = a/(\omega + \alpha + 1)$, $\alpha < \kappa^+$ for a cardinal $\kappa \geq |S|$, pairwise differences of the monomials X^{δ_α} divide X^a , and (4.4) even separates all rX^δ , $r \in K^\times$.
- *Modules through the endomorphism ring* (04, 07, 08, 10, 11). A set-sized module gives a unital ring map $A \rightarrow \text{End}_{\mathbb{Z}}(M)$ into a set-sized, generally noncommutative ring, and (ii) applies. This is valid for the full class but bounds only $|\text{End}_{\mathbb{Z}}(M)|$; the direct argument above (06, 09, 03) and 08's cyclic-module argument (Lemma 10.9) are the ones that give set-sized thresholds.

A group-level strengthening. The report on omnific groups and lattices [48], written concurrently, proves that for $n \geq 3$ the purely infinite congruence kernel of the elementary group $E_n(\mathbf{Oz})$ (and of $E_n(\mathbf{Oz}[i])$) has no nontrivial homomorphism to a set-sized group; applied to the elementary matrices $e_{12}(\psi(x))$ this implies (i) for unital maps.

5.2 Consequences for rings and representations

Corollary 5.2 (Omnific integers [03, 04, 05, 06, 07, 08, 09, 10, 11, 13, 15]). *For every set-sized unital ring S the only unital homomorphism $\mathbf{Oz} \rightarrow S$ is $x \mapsto \text{ct}(x)1_S$. Additive multiplicative maps $\mathbf{Oz} \rightarrow S$ into a set-sized ring correspond bijectively to idempotents $e \in S$, via $x \mapsto \text{ct}(x)e$.*

Proof. There is one unital map $\mathbb{Z} \rightarrow S$. A ring map $\psi : \mathbb{Z} \rightarrow S$ is determined by $e = \psi(1)$, which satisfies $e^2 = e$, and every idempotent defines one. Apply Theorem 5.1(i),(ii). $\square$

Corollary 5.3 (Gaussian omnific integers [04, 06, 07, 08, 09, 10, 11, 13, 15]). *For a set-sized unital ring S , unital maps $\mathbf{Oz}[i] \rightarrow S$ correspond bijectively to elements $j \in S$ with $j^2 = -1_S$, via $x + iy \mapsto \text{ct}(x)1_S + \text{ct}(y)j$ ($x, y \in \mathbf{Oz}$). Every such map discards all positive-exponent complex coefficients, not only their real parts.*

Proof. Apply Theorem 5.1 with $(D, K) = (\mathbb{Z}[i], \mathbb{C})$. A unital map from $\mathbb{Z}[i] = \mathbb{Z}[T]/(T^2 + 1)$ is a choice of a root of $T^2 + 1$; in a noncommutative S the subring generated by 1_S and j is commutative,

so evaluation is valid. $\square$

The image of i is part of the data: these are ring maps, not maps of complex algebras. For $(D, K) = (k, k)$ with $k \in \{\mathbb{R}, \mathbb{C}\}$, every unital map from $\mathcal{A}_{\mathbb{R}, \mathbb{R}}$ or $\mathcal{A}_{\mathbb{C}, \mathbb{C}}$ to a set-sized ring factors through the constant coefficient in k , and if the maps are required to fix k , the constant term is the unique such map to k (04). For any unital $D \subseteq \mathbb{C}$, the ring of complex-coefficient normal forms with positive support apart from a constant in D has universal set-sized quotient D (08); this is the case $K = \mathbb{C}$ of the theorem.

Corollary 5.4 (What all small observations identify [04]). *For $x, y \in \mathcal{A}_{D, k}$, $\text{ct}(x) = \text{ct}(y)$ if and only if $\varphi(x) = \varphi(y)$ for every unital homomorphism φ from $\mathcal{A}_{D, k}$ to a set-sized ring. Thus Π_k is the common kernel of all set-sized ring observations.*

Proof. The forward direction is the theorem; for the converse use the target D and the map ct . $\square$

This does not say that the elements of Π_k are zero: $\omega \neq 0$ in $\mathbf{Oz}$, yet every set-target ring homomorphism sends it to 0 (04).

Corollary 5.5 (A common monomial in a set of kernels [15]). *Let $(\varphi_j : \mathcal{A}_{D, k} \rightarrow B_j)_{j \in J}$ be a set-indexed family of ring homomorphisms, the targets possibly proper classes, together with specified nonzero $b_j \in \ker \varphi_j$. Then some nonzero monomial X^δ lies in every $\ker \varphi_j$; in particular the family is not jointly injective.*

Proof. Apply Proposition 3.4(i) to the set $\{b_j^{-1}\}$: some X^δ has $X^\delta/b_j \in \Pi_k$ for all j , so $X^\delta \in b_j \mathcal{A}_{D, k} \subseteq \ker \varphi_j$. $\square$

The witnesses are specified so that no choice over a proper class of families is needed; nothing is claimed about an intersection indexed by the class of all nonzero elements (15).

Corollary 5.6 (No faithful set-sized representation [04, 06, 07, 08, 11]). *No nonzero element of Π_k acts nontrivially on a set-sized unital $\mathcal{A}_{D, k}$ -module; in particular $\mathcal{A}_{D, k}$ has no faithful set-sized module. For every set-sized field F and set-sized F -vector space V (of any dimension), every unital ring map $\mathbf{Oz} \rightarrow \text{End}_F(V)$ is $x \mapsto \text{ct}(x) \text{id}_V$, and every unital ring map $\mathbf{Oz}[i] \rightarrow \text{End}_F(V)$ is $x + iy \mapsto \text{ct}(x) \text{id}_V + \text{ct}(y)J$ with $J^2 = -\text{id}_V$.*

Proposition 5.7 (Finite matrix representations [06, 11]). *Let $n \geq 1$.*

- (i) *The only unital map $\mathbf{Oz} \rightarrow M_n(F)$, F a set-sized field, is $x \mapsto \text{ct}(x)I_n$.*
- (ii) *Up to conjugacy in $\text{GL}_n(\mathbb{C})$, the unital maps $\mathbf{Oz}[i] \rightarrow M_n(\mathbb{C})$ are indexed by $p \in \{0, \dots, n\}$, with $i \mapsto \text{diag}(iI_p, -iI_{n-p})$.*
- (iii) *A unital map $\mathbf{Oz}[i] \rightarrow M_n(\mathbb{R})$ exists exactly when n is even, and then all such maps are conjugate to the one sending i to $n/2$ diagonal blocks $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.*

Proof. (i) is Corollary 5.2. By Corollary 5.3 a map in (ii) or (iii) is a matrix J with $J^2 = -I_n$. Over $\mathbb{C}$ its minimal polynomial divides $(T - i)(T + i)$, which has distinct roots, so J is diagonalizable with eigenvalues $\pm i$, and the multiplicity p of i determines the conjugacy class. Over $\mathbb{R}$, J makes $\mathbb{R}^n$ a complex vector space by $(u + iv)w = uw + vJw$, so n is even, and a complex basis gives the block form; conversely the blocks square to $-I_n$. $\square$

No norm bound, analytic hypothesis or finite-dimensionality is used to kill the purely infinite part; finite dimension enters only the classification of the possible images of i (06). For an infinite-dimensional set-sized space the reduction to one operator J with $J^2 = -1$ remains valid.

Algebras. A set-sized associative $\mathcal{A}_{D,k}$ -algebra is a set-sized ring with a unital map from $\mathcal{A}_{D,k}$ to its center; by the theorem this map factors through D , so set-sized $\mathcal{A}_{D,k}$ -algebras and D -algebras coincide, with the same underlying rings and morphisms; noncentral structure maps are covered by the noncommutative form (04). Enlarging the set-sized algebra in which a calculation takes place therefore never yields a faithful algebraic model of the full ring, although such calculations can be faithful on a chosen set-sized subring (04). Nor does the theorem forbid set-sized rings from embedding *into* $\mathbf{Oz}$: the inclusion of $\mathbb{Z}[\omega]$ detects ω ; it concerns maps defined on all of $\mathbf{Oz}$, and it makes no claim that $\mathbf{Oz}$ and $\mathbb{Z}$ have the same first-order theory (15).

Corollary 5.8 (No ordered specialization, no small retract, rigid endomorphisms [08, 10, 11; 05]).

- (i) *There is no order-preserving unital map from $\mathbf{Oz}$ to a nonzero ordered set-sized ring (11; 09 records the relevant value $\text{ct}(\omega - 1) = -1$).*
- (ii) *A set-sized unital subring $A \subseteq \mathbf{Oz}$ is the image of a ring retraction $\mathbf{Oz} \rightarrow A$ if and only if $A = \mathbb{Z}$ (08).*
- (iii) *Every unital class ring endomorphism σ of $\mathbf{Oz}$ satisfies $\text{ct} \circ \sigma = \text{ct}$; hence every automorphism preserves Π . Every automorphism of $\mathbf{Oz}[i]$ induces the identity or complex conjugation on $\mathbb{Z}[i]$ (10; the first assertion is also 05's, see [46]).*

Proof. (i) The unique map sends the positive element $\omega - 1$ to $-1 < 0$. (ii) A retraction is $f \mapsto \text{ct}(f)1_A$ by Corollary 5.2, whose image is $\mathbb{Z}$; and ct is a retraction onto $\mathbb{Z}$. (iii) $\text{ct} \circ \sigma$ is a unital map to the set $\mathbb{Z}$, hence equals ct . For an automorphism apply this to σ and σ^{-1} . In the Gaussian case σ and σ^{-1} induce mutually inverse ring automorphisms of $\mathbb{Z}[i]$ fixing $\mathbb{Z}$, and i goes to a root of $T^2 + 1$ in $\mathbb{Z}[i]$, i.e. to $\pm i$. $\square$

Passing to a small workspace is therefore not a homomorphic truncation of the full ring; the workspace may embed without being a retract (08). The endomorphism statement does not say that every endomorphism is the identity: $\mathbb{Z} \hookrightarrow \mathbf{Oz}$ composed with ct is a noninjective endomorphism.

Corollary 5.9 (Fields and unit-forcing localizations [04, 06, 09, 11]). *Let $0 \neq u \in \Pi_k$. The class rings $k((X^{\mathbf{No}}))$ (in particular $\mathbf{No}$ and $\mathbf{No}[i]$) and $\mathcal{A}_{D,k}[u^{-1}]$ admit no unital homomorphism to a nonzero set-sized ring and no nonzero set-sized unital module.*

Proof. A unital map from either ring to a set-sized S restricts to $\mathcal{A}_{D,k}$ and so kills u ; but u is invertible in the source, so $1_S = \varphi(u)\varphi(u^{-1}) = 0$. A nonzero set-sized module would give such a map to the nonzero set-sized ring $\text{End}_{\mathbb{Z}}(M)$. $\square$

This is not merely the observation that a proper-class field cannot inject into a set: the localizations need not be fields (04). For example $\mathbf{Oz}[\omega^{-1}]$ is a nonzero class domain, not a field (Proposition 6.11), with no nonzero set-sized module. This completes the proof of Proposition 3.4(iv).

Remark 5.10 (Positive characteristic and additive observations). If the target has positive characteristic n , the conclusion needs no cardinal argument: $f = n(f/n)$ with $f/n \in \Pi_k$ gives $\varphi(f) = n\varphi(f/n) = 0$ (04, 07); more generally an ideal of $\mathcal{A}_{D,k}$ containing a nonzero element of D contains Π_k (07, 15). The cardinal argument is needed exactly for characteristic-zero targets

such as $\mathbb{Q}$, $\mathbb{R}$ or $\text{End}_{\mathbb{Z}}(M)$ for infinite torsion-free M (11). The theorem concerns multiplicative maps: for a fixed $\gamma > 0$ the coefficient extraction $x \mapsto [\omega^\gamma]x$ is a nonzero additive map $\mathbf{Oz} \rightarrow \mathbb{R}$ detecting ω^γ , but it is not multiplicative, since $\omega^{\gamma/2}$ has coefficient 0 at γ while its square has coefficient 1 (04, 07, 09, 10, 11, 15). Valuations also detect nonconstant parts, but not by unital ring maps into set-sized rings (04).

Remark 5.11 (What the theorem does not say). The theorem is a universal property relative to set-sized targets, not an isomorphism $\mathbf{Oz} \cong \mathbb{Z}$. The ideal Π is nonzero, idempotent, not nilpotent and not generated by any set, and $\mathbf{Oz}$ is a domain (09, 08). The regular module $\mathbf{Oz}$ and the ideal Π are proper-class modules on which Π acts nontrivially, so the module correspondence is not a Morita equivalence of unrestricted module theories (06). The identity of $\mathbf{Oz}$ and the inclusion $\mathbf{Oz} \hookrightarrow \mathbf{No}$ are faithful class-valued maps outside the theorem (09). Individual monomials can be represented faithfully in small rings, for instance in polynomial rings or Hahn subfields, and every finite calculation lies in a set-sized subring with a faithful regular representation; the obstruction concerns one global homomorphism defined on *all* of the class ring, including the collision witnesses at all lower scales (04, 07, 08). In the language of 11, the set-sized reflection is $\text{Ref}_{\text{set}}(\mathbf{Oz}) = \mathbb{Z}$ and $\text{Ref}_{\text{set}}(\mathbf{Oz}[i]) = \mathbb{Z}[i]$: a surjection $q : R \rightarrow D$ onto a set-sized ring is a universal set-sized quotient if every map from R to a set-sized ring factors uniquely through q ; this refers to the explicit universal property, not to a category of all class rings with set-sized hom-sets.

Theorem 5.12 (Residually set-sized targets [11, 13]). *Call a class ring B residually set-sized if for each nonzero $b \in B$ there are a set-sized ring A_b and a unital map $\pi_b : B \rightarrow A_b$ with $\pi_b(b) \neq 0$ (the quantifier is elementwise; no set of detecting maps is formed). Every unital map $h : \mathcal{A}_{D,k} \rightarrow B$ into a residually set-sized ring kills Π_k and factors uniquely through D . For a set-sized commutative ring E and any class X of indeterminates, the ring $E[X]$ of finite polynomials is residually set-sized.*

Proof. If $h(s) \neq 0$ for some $s \in \Pi_k$, then $\pi \circ h$ detects s for a detecting map π with set-sized target, contrary to [Theorem 5.1](#). A nonzero polynomial $f \in E[X]$ uses a finite set $Y \subseteq X$ of variables; sending the other variables to 0 gives a map $E[X] \rightarrow E[Y]$ fixing f , with set-sized target. $\square$

6 Quotients, closures, completions and localizations

Throughout, $A = \mathcal{A}_{D,k}$ with k a set field and $D \subseteq k$ a unital subring, and $\Pi_k = \ker \text{ct}$.

6.1 Every quotient and its small image

Theorem 6.1 (Small reflection of an arbitrary quotient [06, 07, 09, 11, 15]). *Let J be a class ideal of A . Then $\text{ct}(J)$ is an ideal of D , and for every set-sized unital ring S*

$$\text{Hom}_1(A/J, S) \cong \text{Hom}_1(D/\text{ct}(J), S),$$

the universal map being $[x] \mapsto \text{ct}(x) \bmod \text{ct}(J)$. Its kernel is $(\Pi_k + J)/J$, and $\text{ct}^{-1}(\text{ct}(J)) = \Pi_k + J$.

Proof. ct is surjective, so the image of an ideal is an ideal. A map from A/J is a map $\varphi : A \rightarrow S$ vanishing on J ; by [Theorem 5.1](#), $\varphi = \psi \circ \text{ct}$ for a unique ψ , and $\varphi(J) = 0$ if and only if $\psi(\text{ct}(J)) = 0$. If $\text{ct}(a) = \text{ct}(b)$ with $b \in J$, then $a - b \in \Pi_k$, which gives the kernel and the preimage. $\square$

Thus a quotient can be a nonzero ring while all of its set-sized images are zero; a quotient presentation of a proper-class ring is not automatically a set-sized ring (09). “Reflection” is only a name for the displayed mapping property; no category of all proper classes is used (07).

Theorem 6.2 (Ideals with set-sized quotient [04, 06, 07, 08, 09, 10, 11, 13, 15]). *For a class ideal J of A the following are equivalent:*

- (i) J admits a set-sized quotient (Definition 2.1);
- (ii) $\Pi_k \subseteq J$;
- (iii) $J = \text{ct}^{-1}(\mathfrak{a}) = \Pi_k + \mathfrak{a}$ for an ideal $\mathfrak{a}$ of D .

Then $\mathfrak{a} = J \cap D = \text{ct}(J)$ is unique and $A/J \cong D/\mathfrak{a}$. For $\mathbf{Oz}$ the kernels are exactly

$$\Pi \text{ (quotient } \mathbb{Z}) \quad \text{and} \quad n\mathbf{Oz} = n\mathbb{Z} + \Pi \text{ (quotient } \mathbb{Z}/n\mathbb{Z}, n \geq 1),$$

the case $n = 1$ giving the zero ring. For $\mathbf{Oz}[i]$ they are $\Pi_{\mathbb{C}}$ (quotient $\mathbb{Z}[i]$) and $d\mathbf{Oz}[i] = d\mathbb{Z}[i] + \Pi_{\mathbb{C}}$ for $0 \neq d \in \mathbb{Z}[i]$, with finite quotient $\mathbb{Z}[i]/(d)$ of $|d|^2$ elements; associates give the same ideal.

Proof. (i) $\Rightarrow$ (ii) by Theorem 5.1 applied to the quotient map. (ii) $\Rightarrow$ (iii): if $\Pi_k \subseteq J$ then $d + f \in J$ ($f \in \Pi_k$) if and only if $d \in J \cap D$. (iii) $\Rightarrow$ (i): $A \rightarrow D \rightarrow D/\mathfrak{a}$ realizes the quotient. For $\mathbf{Oz}$ the ideals of $\mathbb{Z}$ are $n\mathbb{Z}$, $n \geq 0$; $n = 0$ gives Π and $n \geq 1$ gives $n\mathbf{Oz}$ by (3.2). For $\mathbf{Oz}[i]$, $\mathbb{Z}[i]$ is a principal ideal domain (division with remainder by rounding the real and imaginary parts of a complex quotient), and (3.2) gives $d\mathbf{Oz}[i] = d\mathbb{Z}[i] + \Pi_{\mathbb{C}}$. Multiplication by $d = b + ci$ on $\mathbb{Z}[i] \cong \mathbb{Z}^2$ has matrix $\begin{pmatrix} b & -c \\ c & b \end{pmatrix}$ of determinant $b^2 + c^2 = |d|^2$, the index of its image; more generally a nonzero ideal containing d contains the positive integer $d\bar{d}$, so it has finite index without any appeal to unique factorization (07, 06, 08). $\square$

The distinction between $n = 0$ and $n \geq 1$ matters: the characteristic-zero kernel is Π , not $0\mathbf{Oz}$; one must not set $n = 0$ in $n\mathbf{Oz} = n\mathbb{Z} + \Pi$ (04, 06, 07, 08, 09, 10, 11). For $D = k$, the only proper kernel with a set-sized quotient is Π_k , with quotient k (04).

Corollary 6.3 (Small primes, maximal ideals and fields [06, 07, 08, 10, 11, 13, 15]). *The prime class ideals of $\mathbf{Oz}$ with set-sized quotient are Π and $p\mathbf{Oz}$ (p an ordinary prime), with quotient domains $\mathbb{Z}$ and $\mathbb{F}_p$ and fraction fields $\mathbb{Q}$ and $\mathbb{F}_p$; the maximal ones are the $p\mathbf{Oz}$. A unital map from $\mathbf{Oz}$ to a set-sized field has kernel Π in characteristic 0 (its image is $\mathbb{Z}$, not a field) and $p\mathbf{Oz}$ in characteristic p ; the only nonzero set-sized field quotients are the $\mathbb{F}_p$. For $\mathbf{Oz}[i]$ replace these by $\Pi_{\mathbb{C}}$ and the ideals generated by Gaussian primes, with quotient domains $\mathbb{Z}[i]$ and the corresponding finite fields.*

In particular Π is prime but not maximal (it lies in $p\mathbf{Oz}$). These results classify only ideals with set-sized quotients: (0) is prime in $\mathbf{Oz}$ with a proper-class quotient, and no Zorn argument for a proper class of ideals is used to assert that proper class ideals lie in maximal class ideals; a maximal ideal outside this list, if one is given, has a proper-class residue field (07, 08, 11).

Proposition 6.4 (Principal quotients [04]). *Let $f \in A$ be nonconstant and $\beta = \deg_{\omega}(f) > 0$. For distinct $\gamma, \delta \in (0, \beta)$, $X^{\gamma} - X^{\delta} \notin fA$; hence $A/(f)$ has sets of pairwise distinct residues of every set cardinality and admits no set-sized quotient. Consequently (f) admits a set-sized quotient exactly when $f \in D \setminus \{0\}$, and then $fA = fD + \Pi_k$ and $A/(f) \cong D/fD$. Among the ideals with set-sized quotient, Π_k is the only one not generated by a set; every other one is generated by a subset of D , and for $\mathbf{Oz}$ these are the principal ideals $n\mathbf{Oz}$, $n \geq 1$.*

Proof. The difference has degree $\max(\gamma, \delta) < \beta$, while a nonzero multiple fq , $q \in A$, has degree $\beta + \deg_{\omega}(q) \geq \beta$ (Lemma 2.4); the interval $(0, \beta)$ contains sets of every cardinality (Lemma 3.1). For $0 \neq f \in D$, $f\Pi_k = \Pi_k$ (Proposition 3.3); the zero ideal would inject the class A into a set. For the last assertion write the ideal as $\Pi_k + \mathfrak{a}$; if $\mathfrak{a} = 0$ use Proposition 3.3(iii); otherwise some $0 \neq d \in \mathfrak{a}$ gives $d\Pi_k = \Pi_k$, so the ideal generated by the set $\mathfrak{a}$ already contains Π_k . $\square$

Theorem 6.5 (Quotient-size alternative [08, 13]). *Let J be a class ideal of $\mathbf{Oz}$ with $\Pi \not\subseteq J$. For every set cardinal μ there is a set of at least μ elements of $\mathbf{Oz}$ pairwise incongruent modulo J .*

Proof. Choose $f \in \Pi \setminus J$ and write $f = \omega^c h$ with $c > 0$, $h \in \Pi$; then $\omega^c \notin J$. For an ordinal λ of cardinality at least μ use the exponents $c/\omega^{\beta+1}$, $\beta < \lambda$: if two of the monomials were congruent modulo J , the telescope (4.2) would put ω^c in J . $\square$

A statement that a quotient “is not small” always has this concrete meaning: for every set cardinal it contains a larger set of distinct elements; no “cardinality of $\mathbf{No}$ ” is used (07).

6.2 Finite congruences, closures and invisible primes

For a class ideal J of $\mathbf{Oz}$ define its finite-congruence closure $\bar{J}^{\text{fin}} = \bigcap_{n \geq 1} (J + n\mathbf{Oz})$; the witness in J may depend on n (08).

Theorem 6.6 (Closure of every ideal [08, 09]). *If $\text{ct}(J) = d\mathbb{Z}$ with $d \geq 0$, then $\bar{J}^{\text{fin}} = J + \Pi = \text{ct}^{-1}(d\mathbb{Z})$. In particular J is dense for finite congruences exactly when $\text{ct}(J) = \mathbb{Z}$, and the closure of 0 is Π .*

Proof. $\text{ct}(J)$ is an ideal of $\mathbb{Z}$, and $J + n\mathbf{Oz} = \text{ct}^{-1}(d\mathbb{Z} + n\mathbb{Z}) = \Pi + \gcd(d, n)\mathbb{Z}$ by (3.2). If $d > 0$ the term $n = d$ is $d\mathbb{Z}$ and all terms contain it; if $d = 0$ the intersection is $\bigcap_n n\mathbb{Z} = 0$. $\square$

Corollary 6.7 (Dense proper principal ideals [08, 09]). *If $u \in \mathbf{Oz}$ is nonconstant with $\text{ct}(u) = \pm 1$, then (u) is proper and dense for finite congruences, and every unital map from $\mathbf{Oz}$ to a nonzero set-sized ring sends u to ± 1 . Every nonconstant irreducible $p \in \mathbf{Oz}$ has $\text{ct}(p) = \pm 1$; hence (p) is proper, profinitely dense, and $\mathbf{Oz}/(p)$ has no nonzero set-sized ring image.*

Proof. u is a nonunit by Lemma 2.4; apply Theorem 6.6 and Corollary 5.2. For an irreducible $p = c + f$, $0 \neq f \in \Pi$: if $c = 0$ then $p = 2(p/2)$, and if $|c| > 1$ then $p = c(p/c)$ with $p/c = 1 + f/c$ nonconstant, both factorizations into nonunits; so $c = \pm 1$. A map killing p would kill ± 1 . $\square$

The converse of the second statement is false: $\omega^2 - 1 = (\omega - 1)(\omega + 1)$ has constant term -1 and is reducible. The constant term is a complete invariant for set-sized ring observations, not for factorization inside $\mathbf{Oz}$ (09).

Corollary 6.8 (A prime quotient invisible to all set-sized rings [06, 08, 09, 11, 15]). *Let $q = \omega^{\sqrt{2}} + \omega + 1$. Then $\mathbf{Oz}/(q)$ is a nonzero proper-class integral domain of characteristic 0; it admits no unital map to a nonzero set-sized ring and no nonzero set-sized unital module.*

Proof. L’Innocente and Mantova proved that q is prime in $\mathbf{Oz}$ [23, Theorem B], answering Gonshor’s question [20, p. 117]; this primality is imported, not reproved. So (q) is a proper prime ideal and $\mathbf{Oz}/(q)$ is a nonzero domain. Since $\text{ct}(q) = 1$, Theorem 6.1 gives the zero ring as its universal set-sized image. On a set-sized module, inflated to $\mathbf{Oz}$, q acts both as $\text{ct}(q) = 1$ and as 0, so the module is zero. If $\mathbf{Oz}/(q)$ were a set, its identity map would contradict the first assertion. No nonzero integer n lies in (q) : $n = qa$ would force $a = n/q$, a nonzero infinitesimal, which is not omnific; so $\mathbb{Z} \hookrightarrow \mathbf{Oz}/(q)$. $\square$

This is no contradiction with the maximal-ideal theorem for set-sized rings: the source quotient is not a set, and the corollary does not say that a particular maximal class ideal fails to exist (06, 09). The same obstruction needs no primality.

Proposition 6.9 (Unit-constant ideals [06, 07]). *If a class ideal J of $\mathbf{Oz}$ contains an element with constant term 1, then $\mathbf{Oz}/J$ has no nonzero set-sized unital module and no unital map to a nonzero set-sized ring; if J is proper, $\mathbf{Oz}/J$ is a nonzero proper-class ring. More generally, if $\text{ct}(f) = c$, the small reflection of $\mathbf{Oz}/(f)$ is $\mathbb{Z}/c\mathbb{Z}$.*

Proof. The element acts as 1 by Theorem 5.1 and as 0 by the quotient relation. For a principal ideal $\text{ct}((f)) = c\mathbb{Z}$; apply Theorem 6.1. $\square$

For example $\omega + 1$ and $1 + \omega^a$ ($a > 0$) are nonunits, so $\mathbf{Oz}/(1 + \omega^a)$ is a nonzero class ring with no nonzero set-sized image; Section 14 shows how much it contains. The benefit of q in Corollary 6.8 is that the quotient is a domain (06). Under any map to a set-sized ring, ω^a becomes 0 and $1 + \omega^a$ becomes 1, which is the arithmetic reason (07).

Theorem 6.10 (Arithmetic completions [04, 06, 07, 08, 09, 10, 11, 13, 15]). *All finite quotient rings of $\mathbf{Oz}$ are the $\mathbb{Z}/n\mathbb{Z}$, and*

$$\widehat{\mathbf{Oz}} := \varprojlim_{n \geq 1} \mathbf{Oz}/n\mathbf{Oz} \cong \widehat{\mathbb{Z}}, \quad \varprojlim_{r \geq 1} \mathbf{Oz}/p^r \mathbf{Oz} \cong \mathbb{Z}_p \quad (p \text{ prime}).$$

The canonical maps are ct followed by the usual injections; their kernel is Π and their image is the diagonal copy of $\mathbb{Z}$. For the Gaussian ring, $\widehat{\mathbf{Oz}}[i] \cong \widehat{\mathbb{Z}}[i] = \widehat{\mathbb{Z}}[T]/(T^2 + 1)$ and $\varprojlim_n \mathbf{Oz}[i]/\ell^n \mathbf{Oz}[i] \cong \mathbb{Z}_\ell[i] = \mathbb{Z}_\ell[T]/(T^2 + 1)$ for an ordinary prime ℓ (not necessarily a domain), with kernel $\Pi_{\mathbb{C}}$.

Proof. By Theorem 6.2 the finite quotients and their transition maps are those of $\mathbb{Z}$ (respectively $\mathbb{Z}[i]$), identified by ct ; a nonzero integer is not divisible by every n or every p^r . For $\mathbf{Oz}[i]$, every finite quotient has some positive characteristic n , so its kernel contains $n\mathbf{Oz}[i] = n\mathbb{Z}[i] + \Pi_{\mathbb{C}}$; the ideals $n\mathbb{Z}[i]$ are cofinal among finite-index ideals, and the limit is free of rank two over $\widehat{\mathbb{Z}}$ (respectively $\mathbb{Z}_\ell$) with $i^2 = -1$. $\square$

The finite quotient system is an ordinary set-indexed system; no inverse limit over a proper class of targets is formed, and $\mathbf{Oz}$ is not claimed to be an object of the category of small topological rings (08, 09). The finite-congruence topology on $\mathbf{Oz}$ is not Hausdorff: Π is the closure of 0. It is unrelated to the fine order topology and to Hahn valuation topologies (07, 10, 11). Every $g \in \Pi$ is divisible in $\mathbf{Oz}$ by every power of p , but this does not represent g as a nonzero p -adic integer; for example $\omega^\omega + \pi\omega + 17$ maps to 17 in every finite quotient and in every $\mathbb{Z}_p$ (07). The finite-quotient and completion statements were already in the Diophantine manuscripts of the sibling report [46] (04 cites them as prior); the universal theorem explains why passing from finite targets to arbitrary set-sized targets recovers nothing more.

6.3 Localizations and presentations

Proposition 6.11 (Adjoining one monomial inverse [04, 07]). *For $a > 0$,*

$$\mathcal{A}_{D,k}[X^{-a}] = \{x \in k((X^{\mathbf{No}})) : \text{supp } x \subseteq [-Na, \infty) \text{ for some ordinary } N\},$$

independently of D . This is a proper subring of $k((X^{\mathbf{No}}))$ and not a field; it has no nonzero set-sized ring target (Corollary 5.9). In particular $\mathbf{Oz}[\omega^{-a}]$ is a nonzero class domain, not a field, with no nonzero set-sized unital module.

Proof. An element of the localization is fX^{-Na} with $f \in \mathcal{A}_{D,k}$, whose support is bounded below by $-Na$. Conversely, if $\text{supp } x \subseteq [-Na, \infty)$, then $X^{(N+1)a}x$ has strictly positive support, so $x = X^{(N+1)a}x/X^{(N+1)a}$ lies in the localization; the extra factor X^a removes the restriction on the coefficient at the lower endpoint, which explains the independence from D . The exponent $a\omega$ exceeds every Na , so $X^{-a\omega}$ is not in the ring although $X^{a\omega}$ is; the ring is proper and not a field. $\square$

The same argument shows $\omega^{-\omega} \notin \mathbf{Oz}[1/\omega]$: $\omega^{-\omega} = a/\omega^n$ would force $a = \omega^{n-\omega} \notin \mathbf{Oz}$ (11).

Theorem 6.12 (Quotient-localization calculus [08, 09, 11, 15]). *Let $R = \mathcal{A}_{D,k}$, J a class ideal and $S \subseteq R$ a multiplicative class containing 1. Put $D_0 = D/\text{ct}(J)$ and $T = \{\text{ct}(s) + \text{ct}(J) : s \in S\} \subseteq D_0$. For every set-sized commutative ring B ,*

$$\text{Hom}((R/J)[S^{-1}], B) \cong \text{Hom}(D_0[T^{-1}], B);$$

if $0 \in T$ the right side is empty for $B \neq 0$. For $R = \mathbf{Oz}$, $J = 0$ and S a set-sized multiplicative subset, the same holds for noncommutative B , with universal formula $f/s \mapsto \text{ct}(f)1_B(\text{ct}(s)1_B)^{-1}$.

Proof. A map from the left ring is a map $h : R \rightarrow B$ killing J and sending each $s \in S$ to a unit. By Theorem 5.1, $h = g \circ \text{ct}$ uniquely, and the conditions become $g(\text{ct}(J)) = 0$ and $g(\text{ct}(s)) \in B^\times$, which are exactly the conditions for a map $D_0[T^{-1}] \rightarrow B$; the sets $\text{ct}(J)$ and T are sets even when J and S are classes. The natural map to the reflection is surjective on fractions of constants. For the noncommutative statement, the denominators $\text{ct}(s)1_B$ are central, so the ordinary universal property of localization applies (08). $\square$

Example 6.13 (Three effects [08, 11, 15]). $\text{Ref}_{\text{set}}(\mathbf{Oz}[1/(\omega + 2)]) = \mathbb{Z}[1/2]$; $\text{Ref}_{\text{set}}(\mathbf{Oz}[1/\omega]) = 0$; $\text{Ref}_{\text{set}}(\mathbf{Oz}/(\omega + 1)) = 0$; and $\text{Ref}_{\text{set}}(\mathbf{Oz}[1/n]) = \mathbb{Z}[1/n]$ for a nonzero ordinary integer n . The second ring is a nonzero domain inside $\mathbf{No}$ and the third is nonzero because $\omega + 1$ is not a unit; both have no nonzero set-sized representation. A nonzero class ring can therefore have no nonzero set-sized unital module, so set-sized existence arguments for modules cannot be applied to a proper class of class ideals (11). If $\text{ct}(s) = 0$ for some $s \in S$, the displayed localization of $\mathbb{Z}$ is the zero ring (08). In the language of 09: no unital map to a nonzero set-sized ring can make a nonzero $f \in \Pi_k$ invertible or make an element with unit constant term zero.

Theorem 6.14 (Algebraic points forget infinite coefficients [06, 09, 11, 15]). *Let X be a set of commuting variables, $(f_j)_{j \in E}$ a set-indexed family in $\mathcal{A}_{D,k}[X]$, $B = \mathcal{A}_{D,k}[X]/(f_j)$ and $B_0 = D[X]/(\text{ct}_* f_j)$, where ct_* applies ct to coefficients. For every set-sized commutative ring S the natural map $B \rightarrow B_0$ induces $\text{Hom}(B_0, S) \cong \text{Hom}(B, S)$; thus $\text{Ref}_{\text{set}}(B) = B_0$.*

Proof. A map $\mathcal{A}_{D,k}[X] \rightarrow S$ is a map on $\mathcal{A}_{D,k}$, which factors through ct , together with images of the variables; for such data f_j vanishes exactly when $\text{ct}_* f_j$ does. B_0 is a set because D , X and the relations are sets, and $B \rightarrow B_0$ is surjective. $\square$

06 states this for finitely many variables and 11 for a set of variables; 09 states the corresponding reduction of localization and relation problems. The relation $\omega T = 1$ reduces to $0 = 1$, so it has no solution in a nonzero set-sized ring under any representation of its coefficients, although $T = \omega^{-1}$ solves it in $\mathbf{No}$ (06). For $B = \mathbf{Oz}[T]/(T^2 - \omega)$ one gets $\text{Ref}_{\text{set}}(B) = \mathbb{Z}[T]/(T^2)$: a set-sized field sends T to 0 and a set-sized ring may send it to any square-zero element, while the actual root $\omega^{1/2} \in \mathbf{Oz}$ gives a map to the class ring $\mathbf{Oz}$ (11). These are universal-property statements, not quantifier elimination, not a reconstruction of class-valued solutions, and not an assertion that a scheme with a proper-class coordinate ring is an ordinary scheme (06, 11).

6.4 Diophantine observation

If $P \in \mathbb{Z}[X_1, \dots, X_m]$, $x \in \mathbf{Oz}^m$ and $\varphi : \mathbf{Oz} \rightarrow S$ is a unital set-target map, then $\varphi(P(x)) = P(\text{ct}(x_1)1_S, \dots, \text{ct}(x_m)1_S)$, even for noncommutative S , since the coordinates are integer scalars (04). This does *not* say that $P(x) = 0$ whenever $P(\text{ct}(x)) = 0$ (take $P(X) = X$ and $x = \omega$), and it does not classify infinite points of curves; inequalities, supports, valuations and nonmultiplicative coefficient observations remain available (04).

Proposition 6.15 (Conservativity for integer coefficients [08, 13]). *A finite system of polynomial equations with coefficients in $\mathbb{Z}$ has a solution in $\mathbf{Oz}$ if and only if it has one in $\mathbb{Z}$; more generally, positive existential ring formulas with parameters in $\mathbb{Z}$ have the same truth value in $\mathbf{Oz}$ and $\mathbb{Z}$. With an inequation this fails: $\exists x \exists y (x^2 = 2y^2 \wedge y \neq 0)$ holds in $\mathbf{Oz}$ (take $x = \sqrt{2}\omega$, $y = \omega$) but not in $\mathbb{Z}$.*

Proof. Apply the homomorphisms ct and $\mathbb{Z} \hookrightarrow \mathbf{Oz}$, which preserve positive existential formulas (induction on their construction). A nonzero integer solution of $x^2 = 2y^2$ would give a rational square root of 2; ct maps the omnific witnesses to $x = y = 0$, preserving the equation but not the inequation. $\square$

The parameter restriction is essential (the coefficient $\omega^\gamma + 1$ of Theorem 15.25 is not an integer), and so is positivity (08). The sibling report [46] proves the same constant-term transfer for equations in its Diophantine setting.

7 Modules and derivations over the full class ring

Theorem 7.1 (Set-sized modules [04, 06, 07, 08, 09, 10, 11, 13, 15]). *Let $A = \mathcal{A}_{D,k}$. Restriction along $D \hookrightarrow A$ and inflation along ct are mutually inverse equivalences between set-sized unital A -modules and set-sized unital D -modules; the action of A on a set-sized module is $(d + f)m = dm$ ($d \in D$, $f \in \Pi_k$). The equivalence is the identity on underlying abelian groups and on morphisms, and it is exact. In particular set-sized $\mathbf{Oz}$ -modules are exactly abelian groups with $xm = \text{ct}(x)m$, and set-sized $\mathbf{Oz}[i]$ -modules are exactly $\mathbb{Z}[i]$ -modules inflated along ct .*

Proof. Theorem 5.1(iv) gives $\Pi_k M = 0$ and the action formula. Conversely, a D -module inflates along ct , and restriction recovers the D -action because $\text{ct}|_D = \text{id}$. An additive map between two such modules is A -linear exactly when it is D -linear. Kernels, cokernels and exact sequences are computed on underlying groups with the induced actions. $\square$

A $\mathbf{Oz}[i]$ -module structure on an abelian group is the choice of an endomorphism J with $J^2 = -\text{id}$, automatically invertible (08). Category language abbreviates the correspondence of individual objects and morphisms: each action is coded by its restriction to the set $D \times M$, and no class of arbitrary class actions is formed (04, 08, 11). The regular module A is a proper class on which Π_k acts nontrivially, so there is no conflict with recovering a ring from its regular module (04). The equivalence can transport extension problems between the two small-module categories, but it should not be rephrased as a statement about homological dimensions computed with arbitrary class-sized modules: the rank-one free module A is not an object of the small-module category (07). Section 12 treats the homological algebra over set-sized integer parts.

Proposition 7.2 (An additive splitting that is not a module splitting [08, 15]). *The exact sequence of class $\mathbf{Oz}$ -modules $0 \rightarrow \Pi \rightarrow \mathbf{Oz} \xrightarrow{\text{ct}} \mathbb{Z} \rightarrow 0$, with $\mathbb{Z}$ carrying the action $x \cdot n = \text{ct}(x)n$, does not split as a sequence of $\mathbf{Oz}$ -modules, although it splits as a sequence of additive groups and of rings.*

Proof. A module section s gives $e = s(1)$ with $\text{ct}(e) = 1$ and $ue = s(u \cdot 1) = s(0) = 0$ for every $u \in \Pi$. With $u = \omega$ and $\mathbf{Oz}$ a domain, $e = 0$, a contradiction. $\square$

Proposition 7.3 (Projectivity depends on the size convention [15]). *With the constant-term action, $\mathbb{Z}$ is projective in the category of set-sized $\mathbf{Oz}$ -modules, although $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$ has no $\mathbf{Oz}$ -linear section (Proposition 7.2). For every set-sized $\mathbf{Oz}$ -module M , $\text{Hom}_{\mathbf{Oz}}(M, \mathbf{Oz}) = 0$.*

Proof. By Theorem 7.1 the set-sized modules are abelian groups, and $\mathbb{Z}$ is free on one generator. If $f : M \rightarrow \mathbf{Oz}$ is linear and $m \in M$, then $\omega f(m) = f(\omega m) = f(0) = 0$, so $f(m) = 0$ in the domain $\mathbf{Oz}$. $\square$

No nonzero free $\mathbf{Oz}$ -module is set-sized, yet the set-sized module category has many projectives; this is a warning against importing the size convention of ordinary module theory, not a Morita statement (15).

Theorem 7.4 (Derivations into set-sized modules [04, 06, 08, 09, 10, 11, 13]). *Let $A = \mathcal{A}_{D,k}$ and M a set-sized A -module. Every derivation $\partial : A \rightarrow M$ (an additive class map with $\partial(xy) = x\partial(y) + y\partial(x)$) vanishes on Π_k , and restriction gives a natural bijection $\text{Der}_{\mathbb{Z}}(A, M) \cong \text{Der}_{\mathbb{Z}}(D, M)$ with inverse $\delta \mapsto \delta \circ \text{ct}$. Consequently:*

- (i) every derivation $\mathbf{Oz} \rightarrow M$ is zero;
- (ii) $\text{Der}_{\mathbb{Z}}(\mathbf{Oz}[i], M) \cong M[2] = \{m \in M : 2m = 0\}$ via $\partial \mapsto \partial(i)$, and $\text{Der}_{\mathbb{Z}[i]}(\mathbf{Oz}[i], M) = 0$;
- (iii) every k -linear derivation of $\mathcal{A}_{k,k}$ or of $\mathcal{O}_k$ ($k = \mathbb{R}, \mathbb{C}$) into a set-sized module is zero, and $\text{Der}_D(A, M) = 0$.

Proof. $\Pi_k M = 0$ by Theorem 7.1, and each $x \in \Pi_k$ is a product uv with $u, v \in \Pi_k$ (Proposition 3.3), so $\partial(x) = u\partial(v) + v\partial(u) = 0$. Thus $\partial(d+x) = \partial(d)$. Conversely, for a derivation δ of D , $\delta \circ \text{ct}$ satisfies the Leibniz rule because A acts on M through ct . A derivation of $\mathbb{Z}$ vanishes ($\partial(1) = 2\partial(1)$). A derivation of $\mathbb{Z}[i]$ is determined by $m = \partial(i)$; differentiating $i^2 + 1 = 0$ gives $2im = 0$, equivalently $2m = 0$ since i acts invertibly; conversely $\partial(a+bi) = bm$ satisfies the Leibniz rule, the discrepancy for a product being $2b\dim = 0$. It kills $\mathbb{Z}[i]$ exactly when $m = 0$. A k -linear derivation of k vanishes; the finite side uses Theorem 8.2 and $\mathfrak{m}_k^2 = \mathfrak{m}_k$ (Lemma 8.4). $\square$

A second route (09). Since M is a D -module, the square-zero extension $D \ltimes M$ with $(d, m)(e, n) = (de, dn + em)$ is a set-sized ring, and $a \mapsto (\text{ct}(a), \partial(a))$ is a unital ring map $A \rightarrow D \ltimes M$; the universal theorem forces it to factor through D .

Example 7.5 (The ramification at 2 survives [09, 11]). Give $\mathbb{F}_2$ the $\mathbb{Z}[i]$ -module structure $i \mapsto 1$. Then $\partial(a+bi+f) = b \bmod 2$ is a nonzero $\mathbb{Z}$ -derivation $\mathbf{Oz}[i] \rightarrow \mathbb{F}_2$. The statement “all set-valued derivations of the Gaussian omnific ring vanish” is false; only the positive-support directions vanish, and the surviving derivation reflects ordinary ramification of $\mathbb{Z}[i]$ at 2.

The coefficient-linearity qualification in (iii) cannot be dropped: the theorem reduces arbitrary derivations of $\mathcal{A}_{\mathbb{R}, \mathbb{R}}$ or $\mathcal{O}_{\mathbb{R}}$ to derivations of $\mathbb{R}$ over $\mathbb{Q}$, which are not zero (04). For a set-sized commutative D -algebra B , every D -algebra map $A \rightarrow B[\varepsilon]/(\varepsilon^2)$ kills Π_k , so relative first-order deformations with set-sized coefficients cannot detect Π_k ; no assertion about an absolute proper-class module of Kähler differentials is made (11).

Proposition 7.6 (Class-valued derivations [06, 08, 10]). *For $\eta \in \mathbf{No}$ let $\ell_\eta : \mathbf{No} \rightarrow \mathbb{R}$ extract the coefficient of ω^η from the normal form of its argument; it is additive. The formula*

$$D_\eta\left(\sum_x r_x \omega^x\right) = \sum_x \ell_\eta(x) r_x \omega^x$$

defines a derivation $\mathbf{No} \rightarrow \mathbf{No}$ which restricts to a derivation $\mathbf{Oz} \rightarrow \Pi$ into the class module Π , preserves strongly summable families, and satisfies $D_\eta(\omega^{\omega^\theta}) = \omega^{\omega^\theta}$ if $\eta = \theta$ and 0 otherwise. Every finite family of distinct D_η is linearly independent over $\mathbb{R}$, and $D_0(\omega) = \omega$. Coefficientwise complexification gives the analogous derivations of $\mathbf{No}[i]$ and $\mathbf{Oz}[i]$.

Proof. The output support is a subset of the input support, so the formula defines a surreal; the exponent 0 is killed because $\ell_\eta(0) = 0$, and no negative exponent appears for an omnific input. Additivity and preservation of strong sums are coefficientwise. At a fixed exponent z the coefficient of a product is a finite sum over pairs $x + y = z$, and $\ell_\eta(z) = \ell_\eta(x) + \ell_\eta(y)$; multiplying and summing gives the Leibniz rule. The exponent ω^θ has coefficient 1 at θ and 0 elsewhere, which gives the displayed values and, by evaluation at the monomials ω^{ω^θ} , linear independence. Finally $\ell_0(1) = 1$. $\square$

The proposition combines three sources, none of which states all of it: 08 introduces the family D_η for every η , as derivations $\mathbf{Oz} \rightarrow \Pi$, with the diagonal values and linear independence; the case $\eta = 0$ is 06’s Euler-type derivation $\mathcal{D}$ (06 defines it on $\mathbf{No}$, with the complex extension) and 10’s exponent-weighted derivation D_h ($h = \ell_0$; 10 records preservation of strongly summable families). The general- η statements on $\mathbf{No}$ and on strong sums follow by the same coefficientwise proof. There is no conflict with Theorem 7.4: the target Π is a proper-class module with its internal multiplication. Berarducci and Mantova’s surreal derivation with $\partial\omega = 1$ [6], restricted to $\mathbf{Oz}$ with values in $\mathbf{No}$, is another nonzero class-valued boundary example (09). None of the D_η is asserted to be the Berarducci–Mantova derivation, a fine derivative, or compatible with surreal exponentiation (06, 08, 10), and no uniqueness is claimed for the established derivation (09). The word “small” in Theorem 7.4 cannot be dropped.

8 The finite side: standard part is universal

This section is manuscript 04’s; no other source treats the negative-support side. Its proof is shorter because the ambient valuation produces the needed quotients directly, and it is a different route rather than a sign reversal of the positive side (04). Let $k \in \{\mathbb{R}, \mathbb{C}\}$, $D \subseteq k$ a unital subring, and for $x \neq 0$ let $v(x) = -\deg_\omega(x)$, so that $v(x/y) = v(x) - v(y)$; a nonzero normal form lies in $\mathfrak{m}_k$ exactly when $v(x) > 0$.

Lemma 8.1 (Division of an infinitesimal by small differences [04]). *Let $0 \neq \varepsilon \in \mathfrak{m}_k$ and $a = v(\varepsilon) > 0$. For $0 < \gamma < \delta < a$, $\varepsilon/(\omega^{-\gamma} - \omega^{-\delta}) \in \mathfrak{m}_k$.*

Proof. The denominator has largest exponent $-\gamma$, so valuation γ ; the quotient has valuation $a - \gamma > 0$, i.e. all of its exponents are negative. $\square$

Theorem 8.2 (Universal set-sized residue on the finite side [04]). *For every set-sized unital ring S , restriction gives a bijection $\text{Hom}_1(\mathcal{B}_{D,k}, S) \rightarrow \text{Hom}_1(D, S)$ with inverse $\psi \mapsto \psi \circ \text{st}$. Every such homomorphism annihilates $\mathfrak{m}_k$. In particular every unital map from $\mathcal{O}_{\mathbb{R}}$ or $\mathcal{O}_{\mathbb{C}}$ to a set-sized ring factors through $\text{st} : \mathcal{O}_{\mathbb{R}} \rightarrow \mathbb{R}$, respectively $\text{st} : \mathcal{O}_{\mathbb{C}} \rightarrow \mathbb{C}$.*

Proof. Suppose $0 \neq \varepsilon \in \mathfrak{m}_k$ and $\varphi(\varepsilon) \neq 0$, put $a = v(\varepsilon)$ and choose a set of exponents in $(0, a)$ of cardinality larger than $|S|$ (Lemma 3.1). By Lemma 8.1, ε is a multiple, by an element of $\mathfrak{m}_k \subseteq \mathcal{B}_{D,k}$, of the difference of any two of the monomials $\omega^{-\gamma}$, so Lemma 4.8(i) injects the chosen set into S , a contradiction. Hence $\varphi(\mathfrak{m}_k) = 0$ and $\varphi(d + \varepsilon) = \varphi(d)$; conversely $\psi \circ \text{st}$ extends every ψ . $\square$

For a target without a specified coefficient-field structure, the induced map on $\mathbb{R}$ or $\mathbb{C}$ is not presumed to be the identity: the assertion is factorization through the residue field, not uniqueness of maps from $\mathcal{O}_{\mathbb{C}}$ to an arbitrary ring (04). The statement involves no topology on S and so is stronger than a statement about continuous maps into topological rings; it must also be distinguished from the existence of standard part, which is classical and part of the repository's formal infrastructure [50].

Corollary 8.3 (Quotients, modules and derivations on the finite side [04]). *The kernels of $\mathcal{B}_{D,k}$ admitting a set-sized quotient are exactly $\mathfrak{m}_k + \mathfrak{a}$ for ideals $\mathfrak{a}$ of D ; the only nonzero set-sized quotients of $\mathcal{O}_{\mathbb{R}}$ and $\mathcal{O}_{\mathbb{C}}$ are $\mathbb{R}$ and $\mathbb{C}$, up to the residue identification. Set-sized $\mathcal{B}_{D,k}$ -modules are exactly D -modules inflated along st (set-sized $\mathcal{O}_k$ -modules are k -vector spaces), no faithful set-sized module exists, and $\text{Der}_{\mathbb{Z}}(\mathcal{B}_{D,k}, M) \cong \text{Der}_{\mathbb{Z}}(D, M)$ for set-sized M . For $0 \neq \varepsilon \in \mathfrak{m}_k$, the localization $\mathcal{B}_{D,k}[\varepsilon^{-1}]$ has no nonzero set-sized ring target and no nonzero set-sized module.*

Proof. Repeat the proofs of Theorems 6.2, 7.1 and 7.4 and Corollary 5.9 with Theorem 8.2 and st in place of the positive-side theorem and ct ; for $D = k$ the only ideals are 0 and k . Derivations use Lemma 8.4. $\square$

Lemma 8.4 (Idempotence on the finite side [04]). $\mathfrak{m}_k^2 = \mathfrak{m}_k$ as an ideal of $\mathcal{B}_{D,k}$.

Proof. For $0 \neq \varepsilon \in \mathfrak{m}_k$ with $a = v(\varepsilon)$, $\varepsilon = \omega^{-a/2}(\omega^{a/2}\varepsilon)$ with both factors of valuation $a/2 > 0$. $\square$

Two residues that cannot be combined (04). Every element of $k((X^{\mathbf{No}}))$ has a unique decomposition along $\Pi_k \oplus k \oplus \mathfrak{m}_k$, obtained by splitting its support at 0; so $\mathcal{A}_{k,k} \cap \mathcal{O}_k = k$ and $\mathcal{A}_{k,k} + \mathcal{O}_k$ is the whole field additively. The two constant-coefficient maps agree on the intersection, yet no unital ring map on the field extends both:

$$\text{ct}(\omega) = 0, \quad \text{st}(\omega^{-1}) = 0, \quad \omega \cdot \omega^{-1} = 1. \quad (8.1)$$

The obstruction is multiplicative, not an ambiguity of the additive decomposition, and Corollary 5.9 sharpens it: adjoining one inverse to the positive-support ring already destroys every nonzero set-sized target.

Relation to the rotation groups. The euclidean-three-space report proves that every class homomorphism from $\text{SO}(3, \mathbf{No})$ to a set-sized group factors through standard part (`e3:cut:thm:smallquotient`), that every nontrivial set-sized image is $\text{SO}(3, \mathbb{R})$ (`e3:cut:cor:onlyimage`), and the analogues for $\text{SO}(n, \mathbf{No})$, $n \geq 3$, and $\text{SU}(n, \mathbf{No}[i])$, $n \geq 2$ [43]. Theorem 8.2 is the ring analogue on the finite side; neither result implies the other. The group proof uses exact commutators and normal generation before a choice-free Hartogs contradiction; the ring proof uses divisibility by differences of monomials. Manuscripts 06, 07, 08, 09, 10 and 11 acknowledge the rotation theorem as the repository's conceptual predecessor for universal set-sized quotients, 04 refers to related size obstructions in other algebraic settings, and none of them uses the rotation theorem as a lemma.

9 The support threshold: finite supports fail, countable supports suffice

The universal theorem does not need normal forms of every set cardinality. Keep the *full* exponent class **No** and bound only the size of each individual support. For $k \in \{\mathbb{R}, \mathbb{C}\}$, $D \subseteq k$ unital, and an infinite cardinal λ , put

$$\mathcal{A}_{D,k}^{<\lambda} = D + \{f \in \Pi_k : |\text{supp } f| < \lambda\}, \quad \mathcal{B}_{D,k}^{<\lambda} = D + \{f \in \mathfrak{m}_k : |\text{supp } f| < \lambda\},$$

and, in 11's notation, $R_{\leq \lambda} = \mathcal{A}_{D,k}^{<\lambda+}$ and $R_{\text{fin}} = \mathcal{A}_{D,k}^{<\aleph_0}$. These are rings: the support of a sum lies in a finite union and that of a product in the image of the product of the two supports under addition, and for an infinite λ finite unions and products of sets of size $< \lambda$ (or $\leq \lambda$) keep that bound; regularity of λ is not needed. They remain proper classes, since a one-term series may have any positive exponent (04, 11).

Theorem 9.1 (Support threshold [04, 11; 03, 07, 09]).

(i) For every uncountable cardinal λ and every set-sized unital ring S , restriction induces bijections

$$\text{Hom}_1(\mathcal{A}_{D,k}^{<\lambda}, S) \cong \text{Hom}_1(D, S), \quad \text{Hom}_1(\mathcal{B}_{D,k}^{<\lambda}, S) \cong \text{Hom}_1(D, S),$$

with inverses given by *ct* and *st*. In particular countably supported normal forms ($\lambda = \aleph_1$, i.e. $R_{\leq \aleph_0}$) already have universal set-sized quotient D on both sides, and so does $R_{\leq \mu}$ for every infinite μ .

(ii) For finite supports both assertions fail: the augmentation

$$E\left(d + \sum_{g \in F} c_g \omega^g\right) = d + \sum_{g \in F} c_g \quad (F \text{ finite})$$

is a unital ring homomorphism from each finite-support ring to k , sending every monomial, including ω and ω^{-1} , to 1; it does not factor through *ct* or *st*.

(iii) (Extension criterion.) For a set-sized unital ring S and a unital map $h : R_{\text{fin}} \rightarrow S$ the following are equivalent: (a) h extends to $R_{\leq \aleph_0}$; (b) h extends to $R_{\leq \mu}$ for some infinite μ ; (c) $h = g \circ \text{ct}$ for a map $g : D \rightarrow S$; (d) h extends to $\mathcal{A}_{D,k}$. Whenever an extension exists it is unique at each of these support levels. In particular E does not extend to $R_{\leq \aleph_0}$.

(iv) [merge] The same equivalence holds on the finite side, with $\mathcal{B}_{D,k}^{<\aleph_0}$, $\mathcal{B}_{D,k}^{<\aleph_1}$, $\mathcal{B}_{D,k}^{<\lambda}$, $\mathcal{B}_{D,k}$ and *st* in place of R_{fin} , $R_{\leq \aleph_0}$, $R_{\leq \mu}$, $\mathcal{A}_{D,k}$ and *ct*.

Proof. (i), positive side. Fix $s \in \Pi_k$ with $|\text{supp } s| < \lambda$. For $a > b$ in $U_{\text{No}}(\text{supp } s)$, the quotient of Lemma 4.7 has support of cardinality at most $\max(|\text{supp } s|, \aleph_0) < \lambda$, and the monomials ω^a are in the ring; so the families of Theorem 4.9 and all their quotients lie in $\mathcal{A}_{D,k}^{<\lambda}$, and Lemma 4.8(iii) gives $\varphi(s) = 0$ (11). Alternatively (04), every monomial and every telescope $q_{c;a,b}$ has at most countable support, so the explicit-telescope route (Lemma 4.3 with Lemma 4.8(i); see the second routes after Theorem 5.1) kills all positive monomials inside the bounded ring (the field-collision proof of Theorem 5.1 does not apply here, since the Hahn field over H_g is not contained in the bounded ring), and the factorization $s = \omega^a(\omega^{-a}s)$ of Proposition 3.3 only translates the support, keeping its cardinality.

(i), finite side (04). For $0 \neq \varepsilon$ with $|\text{supp } \varepsilon| < \lambda$, $a = v(\varepsilon)$, $0 < \gamma < \delta < a$ and $d = \delta - \gamma$,

$$\frac{\varepsilon}{\omega^{-\gamma} - \omega^{-\delta}} = \varepsilon \omega^{\gamma} \sum_{n \geq 0} \omega^{-nd},$$

the geometric series being the inverse of $1 - \omega^{-d}$. Its support lies in $\{\beta + \gamma - nd : \beta \in \text{supp } \varepsilon, n \in \mathbb{N}\}$, of cardinality at most $|\text{supp } \varepsilon| \cdot \aleph_0 < \lambda$, and its largest exponent is $-a + \gamma < 0$. So the quotient of [Lemma 8.1](#) stays in the bounded ring, and the proof of [Theorem 8.2](#) applies. A valuation inequality alone would not control the support size; this check is needed.

(ii) Additivity and $E(1) = 1$ are clear; for finite supports $E(xy) = \sum_{\alpha, \beta} c_{\alpha} d_{\beta} = E(x)E(y)$ by expanding the finite double sum. $E(\omega) = 1 \neq 0 = \text{ct}(\omega)$ and $E(\omega^{-1}) = 1 \neq 0 = \text{st}(\omega^{-1})$.

(iii) An extension to some $R_{\leq \mu}$ factors through ct by (i), giving (a) $\Rightarrow$ (c) and (b) $\Rightarrow$ (c); an extension to $\mathcal{A}_{D,k}$ factors by [Theorem 5.1](#), giving (d) $\Rightarrow$ (c). Conversely $g \circ \text{ct}$ is defined on every ring in the list and extends h , and two extensions to one ring both factor through ct and agree on D . Since $E(\omega) = 1 \neq \text{ct}(\omega)$, E violates (c).

(iv) The same argument with (i) on the finite side, [Theorem 8.2](#) and st ; the extension $g \circ \text{st}$ exists on every ring of the list, and uniqueness follows as before. $\square$

Remark 9.2 (Why the threshold is about closure, not continuity). No axiom that target maps preserve countable sums is added: certain countable sums lie *inside* the source ring, where they satisfy finite identities. An extension of E to the countable-support ring would send both monomials in [Example 4.5](#) to 1 and the product to 1, forcing $(1 - 1)E(q) = 1$ (04, 11). A map that assigns different values to some monomials is handled by the full theorem: a large family forces a collision somewhere (11). The separating families may have cardinality far above λ ; their members are distinct elements, not terms of one series, and each witness separately has countable support (04, 07, 11).

The finite-support augmentation appears independently in 03 (“two necessary warnings”), 04, 09, 11 and 15 (which blocks its extension with the single series of [Example 4.5](#)), and 12’s cores carry it too ([Proposition 13.2](#)); 07 describes the same boundary in words (on a finite-support fragment the certificate (4.2) need not exist, and a faithful interpretation in a set-sized target can exist), and manuscript 05 of the sibling report [46] observes that $\mathbb{Z} + \omega\mathbb{R}[\omega]$ has many real evaluation maps that do not extend to **Oz**. Proper-class size alone therefore does not prove the universal theorem: the finite-support rings have a proper class of monomials and arbitrarily small positive exponents, yet E sees every monomial as 1 (04). Neither size nor additive divisibility alone suffices; the proof needs many subordinate scales together with enough support closure for geometric division (11). For $D = \mathbb{Z}$, finite targets still kill the positive-support ideal of R_{fin} ($f = n(f/n)$ in characteristic n), so its finite quotient theory is ordinary congruence theory; having profinite completion $\hat{\mathbb{Z}}$ is much weaker than having $\mathbb{Z}$ universal for all set-sized targets (09).

The quotient and module classifications extend to the bounded rings of (i), and so does the derivation statement, because the monomial factorizations proving idempotence preserve support cardinality (04). 04 does *not* infer that these smaller rings have fraction field $k((X^{\mathbf{No}}))$: arbitrary normal forms need not have the required support size. The result is also different from completion or omitted-type questions for bounded-support series over a *fixed* set-sized exponent group (compare [45]): here the exponent class is still large enough to supply a new separating family for every proposed target (04). The fixed-group question is the subject of the next section.

10 Fixed exponent groups: cardinal lower bounds and exact thresholds

Over a fixed set-sized exponent group the full-class theorem cannot hold verbatim: the ring is itself a set, and its identity map is a set-sized target that retains every element (03, 04, 07, 08, 10, 11). What survives is a lower bound on the size of a target that detects a given element. This section compares the bounds proved by the sources, proves a field bound that dominates them, and states the organizing theorem of the report: for each of the set-sized constructions of Section 11, the exact detection threshold is the size of the ring.

Throughout, Γ is an ordered abelian set group, k a set field, $D \subseteq k$ a unital subring, $F \subseteq k((X^\Gamma))$ a subfield containing k and every monomial X^g , $\Pi(F) = \{f \in F : \text{supp } f \subseteq \Gamma_{>0}\}$ and $A = D + \Pi(F)$. The case $F = k((X^\Gamma))$ is the ring $\mathcal{A}_{D,k}(\Gamma)$ of (2.2); the constructions of Section 11 take F to be a countable-support, a $< \kappa$ -support, or a controlled subfield.

10.1 The bounds of the sources

Proposition 10.1 (Monomial-count bounds [03, 04, 07, 10, 13, 15]). *Let $A = \mathcal{A}_{D,k}(\Gamma)$ and $\varphi : A \rightarrow S$ a ring homomorphism.*

- (i) *If $\varphi(X^a) \neq 0$ for some $a > 0$, then $\gamma \mapsto \varphi(X^\gamma)$ is injective on $H_a(\Gamma)_{>0}$ (04); more generally $rX^\delta \mapsto \varphi(rX^\delta)$ is injective on $\{rX^\delta : r \in k^\times, 0 < \delta \in H_a(\Gamma)\}$, and the same holds if X^a divides some g with $\varphi(g) \neq 0$ (07).*
- (ii) *If $0 \neq x \in \Pi_k(\Gamma)$ and $\varphi(x) \neq 0$, then $a \mapsto \varphi(X^a)$ is injective on $U_\Gamma(\text{supp } x)$; this also holds for a nonunital map defined on $\Pi_k(\Gamma)$ alone (10; 15's local criterion is this bound, and 13's criterion is the same injection on any set $T \subseteq \Gamma_{>0}$ with $s > a + n(a - b)$ for $a > b$ in T , $s \in \text{supp } x$).*
- (iii) *On the negative side, if $0 \neq \varepsilon \in \mathfrak{m}_k(\Gamma)$, $a = v(\varepsilon)$ and $\varphi(\varepsilon) \neq 0$ for a map φ on $\mathcal{B}_{D,k}(\Gamma)$, then $\gamma \mapsto \varphi(X^{-\gamma})$ is injective on $(0, a)_\Gamma$ (04).*
- (iv) *For a module M and $v \in M$ with $X^h v \neq 0$, $a \mapsto X^a v$ is injective on $\{a \in \Gamma : a \ll h\}$ (03).*

No divisibility of Γ is needed.

Proof. The telescope, the weighted certificate, the division lemma and the finite-side division (Lemmas 4.3, 4.6, 4.7 and 8.1) use only a , γ , δ , subtraction and ordinary integer multiples, so their witnesses lie in $\mathcal{A}_{D,k}(\Gamma)$ or $\mathcal{B}_{D,k}(\Gamma)$; apply Lemma 4.8. $\square$

Example 10.2 (Where these bounds are vacuous [04, 08, 10, 13]). If Γ is Archimedean, $H_a(\Gamma) = \{0\}$ and the telescopes do not exist. For $\Gamma = \mathbb{Z}$ a reverse well-ordered positive support is finite, $\mathcal{A}_{\mathbb{Z},\mathbb{R}}(\mathbb{Z}) = \mathbb{Z} + X\mathbb{R}[X]$, and evaluation at $X = 1$ is a map to the set field $\mathbb{R}$ detecting X (10). An arbitrary fixed Hahn workspace cannot inherit the full theorem merely because its notation resembles **No** (08).

10.2 The field bound

The monomial counts are far from optimal. The following bound is 03's “smaller-scale field obstruction”, obtained there for countable-support fields over divisible groups; the general form below is a direct application of 06's scaled-field collision lemma and is recorded as a merge statement.

Proposition 10.3 (Field bound [03, 06] [merge]). *Let $c \in \Gamma_{>0}$ and $L_c(F) = F \cap k((X^{H_c(\Gamma)}))$.*

- (i) $L_c(F)$ is a subfield of F and $X^c L_c(F) \subseteq \Pi(F) \subseteq A$.
- (ii) If $\varphi : A \rightarrow S$ is a ring homomorphism with $\varphi(X^{2c}) \neq 0$, then $f \mapsto \varphi(X^c f)$ is injective on $L_c(F)$; if M is an A -module and $X^{2c}v \neq 0$, then $f \mapsto (X^c f)v$ is injective on $L_c(F)$.
- (iii) Consequently, if $x \in X^{2c}A$ (for instance $x = X^{2c}$, or $0 \neq x \in \Pi(F)$ with $2c$ a strict lower bound of $\text{supp } x$), every ring homomorphism and every module detecting x has cardinality at least $|L_c(F)|$.

Proof. (i) $k((X^{H_c(\Gamma)}))$ is a subfield by Lemma 4.2(ii), so $L_c(F)$ is an intersection of two subfields. For $h \in H_c(\Gamma)$, $|h| < c$, so $c + h > 0$ and $X^c f \in F$ has positive support. (ii) is Lemma 4.1 with $m = X^c$ and $L = L_c(F)$, inside the field $k((X^\Gamma))$. (iii) If $x = X^{2c}y$ with $y \in A$ and $\varphi(x) \neq 0$, then $\varphi(X^{2c}) \neq 0$; if $xv \neq 0$ then $X^{2c}(yv) \neq 0$. If $2c < \text{supp } x$ then $X^{-2c}x \in F$ has positive support, so it lies in $\Pi(F) \subseteq A$. $\square$

For $F = H_\omega(k, \Gamma)$ and $c = h/2$ this is exactly 03's theorem ($H_{h/2} = H_h$). The bound contains 04's and 10's monomial counts whenever those monomials lie in $H_c(\Gamma)$, and 07's weighted count, because $L_c(F)$ contains every rX^δ with $r \in k$ and $\delta \in H_c(\Gamma)$; it also contains k itself, so with real coefficients every detecting target has at least $2^{\aleph_0}$ elements (03). The following examples show that it can be strictly and decisively larger.

Example 10.4 (The exact threshold in 04's example [merge]). Let $\Gamma = \mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$ (first coordinate dominant), $k \in \{\mathbb{R}, \mathbb{C}\}$, $A = \mathcal{A}_{D,k}(\Gamma)$ with full Hahn supports, and $a = (1, 0)$. 04 computes the positive part $H_a(\Gamma)_{>0} = \{(0, q) : q \in \mathbb{Q}_{>0}\}$, so $H_a(\Gamma) = \{0\} \oplus \mathbb{Q}$, and notes that retaining X^a requires only a countable family of distinct images. With $c = (1/2, 0)$, $H_c(\Gamma) = \{0\} \oplus \mathbb{Q}$ and $L_c = k((X^{\{0\} \oplus \mathbb{Q}})) \cong k((t^\mathbb{Q}))$ has cardinality $2^{\aleph_0}$; and $|A| \leq |k((X^\Gamma))| \leq 2^{\aleph_0} \cdot (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$, since Γ is countable. Hence the least cardinality of a ring target, and of a module, detecting any nonzero $x \in X^{(1,0)}A$ is exactly $2^{\aleph_0}$, attained by the identity map and the regular module. The same holds on the negative side: if $\varepsilon \in \mathfrak{m}_k(\Gamma)$ has $v(\varepsilon) = (1, 0)$, then with $c = (1/4, 0)$ the element X^{-2c} divides ε in $\mathcal{B}_{D,k}(\Gamma)$ (the cofactor $X^{2c}\varepsilon$ has largest exponent $(-1/2, 0) < 0$), $X^{-c}L_c \subseteq \mathfrak{m}_k(\Gamma)$, and Lemma 4.1 gives the bound $2^{\aleph_0}$, which is again attained.

In this example 07's weighted bound ($|k^\times| \cdot \aleph_0 = 2^{\aleph_0}$) is also attained, while 04's and 10's monomial counts give only $\aleph_0$. With a countable coefficient field all three source bounds fail:

Example 10.5 (None of the monomial or weighted counts is attained [merge]). Let $k = D = \mathbb{Q}$, $\Gamma = \bigoplus_{n < \omega} \mathbb{Q}e_n$ with the least nonzero index dominant, and $A = \mathcal{A}_{\mathbb{Q},\mathbb{Q}}(\Gamma)$ with full Hahn supports. For $x = X^{e_0}$ and $c = e_0/2$, $H_c(\Gamma) = \bigoplus_{n \geq 1} \mathbb{Q}e_n$, and L_c contains the $2^{\aleph_0}$ distinct series $\sum_{n \in B} X^{e_n}$, $B \subseteq \omega \setminus \{0\}$ (the exponents e_n decrease with n , so each support is reverse well ordered). Since Γ is countable, $|A| \leq 2^{\aleph_0} \cdot \aleph_0^{\aleph_0} = 2^{\aleph_0}$. So the exact detection threshold of X^{e_0} is $2^{\aleph_0}$, while 04's bound $|H_{e_0}(\Gamma)_{>0}|$, 07's bound $|\mathbb{Q}^\times| \cdot |H_{e_0}(\Gamma)_{>0}|$ and 10's bound $|U_\Gamma(\{e_0\})|$ are all $\aleph_0$.

These examples settle the two example-level questions of the sources and leave the general ones open: 04's question on optimal small targets in a fixed exponent group is answered for its own example $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$ (the answer is $2^{\aleph_0}$), and for 10's question (when is the bound of Proposition 10.1(ii) attained?) Example 10.5 shows, in ZFC, that the bound need not be attained. The same phenomenon appears in 03's ring $A_{\aleph_1}^{03}$ of Section 11.1: there $|U_\Gamma(S)| \leq \aleph_1$, while the exact threshold is $2^{\aleph_0}$, so if the continuum hypothesis fails, 10's bound is not attained there either. Remark 10.6 records the corresponding correction to a sentence of 04.

Remark 10.6 (A correction to 04’s commentary). 04 remarks that in $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$ “this is only a countable family” which “cannot contradict the existence of a sufficiently large infinite set target, such as the ring itself”. The statement is true, but the ring itself has $2^{\aleph_0}$ elements and, by [Example 10.4](#), no smaller target detects $X^{(1,0)}$; the countable family is not the obstruction that decides the answer.

10.3 Criteria that kill the whole ideal below a cardinal

To pass from monomials to the whole ideal one also needs a gap: every element must be divisible by a positive monomial inside the ring. 03’s example $\sum_{n \geq 1} X^{1/n}$ over $\Gamma = \mathbb{R}$ shows that this can fail ([Example 3.2](#)), and 09’s uncountable-support series f_* ([Remark 11.21](#)) shows the same for full supports over a long group.

Theorem 10.7 (Quantitative criterion [08] [merge]). *Let Γ be divisible, κ an infinite cardinal, and suppose*

- (a) *every nonzero $f \in \Pi(F)$ has a positive exponent $\epsilon \in \Gamma$ strictly below its support;*
- (b) *for every $c > 0$ in Γ , the set $\{a \in \Gamma : a \ll c\}$ has at least κ elements.*

Then every ring homomorphism from A to a ring of cardinality $< \kappa$ factors uniquely through $\text{ct} : A \rightarrow D$. [merge] The same conclusion holds, with the same proof, if (b) is replaced by (b’): $|L_c(F)| \geq \kappa$ for every $c > 0$.

Proof. Fix $c > 0$. Under (b), for two distinct $a > b \ll c$ the series $q_{c;a,b}$ equals the rational expression $X^c/(X^a - X^b)$ in $k((X^\Gamma))$, whose numerator and denominator lie in the field F ; so $q_{c;a,b} \in \Pi(F) \subseteq A$, and a nonzero image of X^c would make the images of κ monomials distinct. Under (b’) use [Proposition 10.3\(ii\)](#) with $c/2$ in place of c . So every positive monomial is killed. For $f \in \Pi(F)$, (a) gives $f = X^\epsilon g$ with $g = X^{-\epsilon} f \in F$ of positive support, so f is killed. $\square$

Condition (a) is a gap property of the chosen field and (b) a supply of lower Archimedean scales; neither says that homomorphisms preserve strongly summable families (08). 08 states the theorem for a subfield $K \subseteq \mathbb{R}$ and constants $\mathbb{Z}$; the proof uses neither restriction, and it is stated here in the setting of this section.

Theorem 10.8 (Cointiality criterion [11]). *Let Γ be divisible, κ uncountable, and $\text{coin}(\Gamma_{>0}) \geq \kappa$ (every set of fewer than κ positive elements has a positive strict lower bound). Put $R_{\Gamma, < \kappa} = D + \{s \in \Pi_k(\Gamma) : |\text{supp } s| < \kappa\}$, a ring for every infinite κ . Then every unital map $R_{\Gamma, < \kappa} \rightarrow S$ with $|S| < \kappa$ factors uniquely through ct .*

Proof. Let $s \neq 0$ have $|\text{supp } s| < \kappa$. Choose $\delta > 0$ strictly below $\text{supp } s$, and then a positive strict lower bound η of the countable set $\{\delta/n : n \geq 1\}$ (possible since $\kappa > \aleph_0$); every $a \in E = (0, \eta)$ satisfies $na < \delta$. The set E has at least κ elements: otherwise it would have a positive strict lower bound b , but then $b \in E$ (as $b < \eta/2 \in E$), which is absurd. For distinct $a, b \in E$, [Lemma 4.7](#) gives $s \in (X^a - X^b)R_{\Gamma, < \kappa}$ with a quotient of support size at most $\max(|\text{supp } s|, \aleph_0) < \kappa$. Since $|S| < |E|$, two of the monomials collide and s is killed. $\square$

This theorem concerns an ordinary set-sized source and cannot exclude its identity representation; the target bound is indispensable. Regularity of κ is not needed for the theorem itself, only for the examples of [Section 11.5](#) (11).

Lemma 10.9 (Cyclic modules [08]). *Let A be a commutative ring, I an ideal and κ a cardinal such that every ring homomorphism from A onto a ring of cardinality $< \kappa$ kills I . Then $IM = 0$ for every A -module M with $|M| < \kappa$.*

Proof. For $m \in M$, $A/\text{Ann}(m) \rightarrow Am$, $a \mapsto am$, is a bijection, so the quotient ring $A/\text{Ann}(m)$ has fewer than κ elements, and $I \subseteq \text{Ann}(m)$. $\square$

Applying the ring statement to $\text{End}_{\mathbb{Z}}(M)$ would need an estimate of $|\text{End}_{\mathbb{Z}}(M)|$, which may exceed κ even when $|M| < \kappa$; the lemma avoids that hidden cardinal-arithmetic assumption (08). [Proposition 10.3\(ii\)](#) gives the module bound directly (06, 03, 09).

10.4 The organizing theorem: the threshold is the size of the ring

The five constructions of [Section 11](#) are summarized here; their notation is tagged by source because they are different rings.

ring	exponent group	supports	coefficients	hypotheses	size	gen. of Π_A
A_{κ}^{03}	$\bigoplus_{\kappa^{\text{op}+\kappa}} \mathbb{Q}$, 2 arms	countable	$\mathbb{R}$	cf $\kappa > \aleph_0$	$\kappa^{\aleph_0}$	cf κ
A_{κ}^{06}	$\text{span}_{\mathbb{Q}}\{\omega^{\pm(\alpha+1)}\}$, 2 arms	$< \kappa$	$\mathbb{R}$	κ reg. unctbl.	$\kappa^{<\kappa}$	κ
A_{κ}^{08}	$\bigoplus_{\xi < \kappa} \mathbb{Q}$, 1 arm	countable, controlled	$\mathbb{R} \cap \overline{\mathbb{Q}}$	κ infinite	κ	cf κ
$A_{\lambda, D}^{09}$	$\bigoplus_{\omega_1 \times \lambda} \mathbb{Q}$, 1 arm	countable	$k \in \{\mathbb{R}, \mathbb{C}\}$	$\lambda^{\aleph_0} = \lambda$	λ	$\aleph_1$
$R_{\Gamma, < \kappa}^{11}$	$\bigoplus_{\alpha < \kappa} \mathbb{Q}$, 1 arm	$< \kappa$	$ K \leq \max(\kappa, 2^{\aleph_0})$	κ reg. unctbl.	$\kappa^{<\kappa}$	κ

All are ZFC constructions. The sizes are proved in [Section 11](#); for A_{κ}^{06} and $R_{\Gamma, < \kappa}^{11}$ the size $\kappa^{<\kappa}$ is a merge result (06 and 11 compute the size only under $\kappa^{<\kappa} = \kappa$).

Theorem 10.10 (Detection threshold equals size [03, 08, 09] [merge]). *Let A be any of the five rings of the table, with purely infinite ideal Π_A , and let $0 \neq x \in \Pi_A$. Then*

$$\min\{|S| : \varphi : A \rightarrow S, \varphi(x) \neq 0\} = \min\{|M| : M \text{ an } A\text{-module}, xM \neq 0\} = |A|.$$

Every homomorphism from A to a ring of cardinality $< |A|$ factors uniquely through ct , every A -module of cardinality $< |A|$ is annihilated by Π_A , and every quotient A/J either is $D/\text{ct}(J)$ with $\Pi_A \subseteq J$ or has cardinality exactly $|A|$. The same holds for the Gaussian versions of A_{κ}^{03} , A_{κ}^{06} and $A_{\lambda, \mathbb{Z}[i]}^{09}$.

Proof. The identity map and the regular module detect x and have cardinality $|A|$. For the lower bound, each construction has a divisible exponent group and the gap property: x has a positive strict lower bound g of its support ([Lemmas 11.3, 11.7, 11.15 and 11.17](#) and, for $R_{\Gamma, < \kappa}^{11}$, $\text{coin}(\Gamma_{>0}) = \kappa$ with [Proposition 11.22](#)). With $c = g/2$, [Proposition 10.3\(iii\)](#) shows that every ring or module detecting x has at least $|L_c(F)|$ elements, and in each construction $|L_c(F)| = |A|$ for every $c > 0$ ([Lemmas 11.4, 11.10 and 11.17](#) and [Proposition 11.23](#); for A_{κ}^{08} , $L_c(F) \subseteq F$ has at most κ elements and contains the κ monomials $X^{e_{\xi}}$ with ξ beyond the least index of c , by [Lemma 11.13](#)). A homomorphism into a smaller ring therefore kills Π_A and factors through ct , and a smaller module is annihilated. If $\Pi_A \not\subseteq J$ the quotient map detects an element of Π_A , so $|A/J| \geq |A|$; the reverse inequality is trivial. $\square$

For A_κ^{03} , A_κ^{08} and $A_{\lambda,D}^{09}$ the theorem is proved in the sources (03 with the field bound; 08 and 09 by counting monomials, which suffices because there the number of small monomials already equals $|A|$). For A_κ^{06} 06 proves the lower bound κ and exactness under $\kappa^{<\kappa} = \kappa$, and for $R_{\Gamma,<\kappa}^{11}$ 11 proves sharpness under $\kappa^{<\kappa} = \kappa$ and $|K| \leq \kappa$; the unconditional exact value $\kappa^{<\kappa}$ is new here (Theorem 11.11 and Proposition 11.23). The quotient alternative is 08's and 09's for their rings and a one-line consequence for the others.

Remark 10.11 (Reconciling the thresholds [merge]).

- *Realized cardinals.* Since $(\kappa^{\aleph_0})^{\aleph_0} = \kappa^{\aleph_0}$, and conversely a cardinal θ with $\theta^{\aleph_0} = \theta$ has $\text{cf } \theta > \aleph_0$ by König's theorem (so $\kappa = \theta$ is allowed in 03), the thresholds realized by 03 are exactly the cardinals θ with $\theta^{\aleph_0} = \theta$; these are 09's cardinals λ . 06 and 11 realize $\kappa^{<\kappa}$ for regular uncountable κ . 08 realizes every infinite cardinal, and is the only construction reaching cardinals below $2^{\aleph_0}$ or with $\kappa^{\aleph_0} > \kappa$: if rX^c lies in the ring for every real r , every target detecting X^{2^c} has at least $2^{\aleph_0}$ elements by Lemma 4.1 with $L = \mathbb{R}$, and 08 escapes this with real algebraic coefficients and a countable-stage closure.
- *The same ring, different bounds.* $A_{\aleph_1}^{06}$ and $A_{\aleph_1}^{03}$ are isomorphic: both are the countable-support real Hahn integer parts over the lexicographic sum $\bigoplus_J \mathbb{Q}$ with J of order type $\omega_1^{\text{op}} + \omega_1$ (06's generators satisfy $\epsilon_\beta \ll \epsilon_\alpha \ll h_\alpha \ll h_\beta$ for $\alpha < \beta$). 06 proves the threshold is at least $\aleph_1$ and exactly $\aleph_1$ under CH; 03 proves it is exactly $2^{\aleph_0}$ in ZFC; 06's own example already notes that without CH its ring need not have size $\aleph_1$. These agree.
- *Generators.* The generator counts are $\text{cf } \kappa$ (03, 08), κ (06, and for 11's ring by the argument of (H3) in Section 12) and $\aleph_1$ (09). In each case the proof exhibits a coinital chain of monomials in the positive cone of the exponent group of that length; the counts differ because the constructions differ, and no general formula is asserted for other exponent groups or support restrictions.
- *Fraction fields.* $\text{Frac } A = F$ is proved for 03 and 06, which use a second (reversed) arm to clear negative exponents; 08 explicitly does not claim it for its one-armed controlled field, and 09 and 11 do not address it.
- *Hypotheses.* Regularity (06) and $\text{cf } \kappa > \aleph_0$ (03) are used for simultaneous gaps below fewer than κ (respectively countably many) full supports; 08 avoids both through its finite-coordinate invariant, which confines each element to finitely many coordinates.

11 Set-sized integer parts with exact thresholds

Five constructions replace the proper-class supply of scales by a cardinal. They are different rings and are tagged by source. Each is an explicit subring of $\mathbf{Oz}$ (or of $\mathbf{Oz}[i]$), and each is a ZFC construction of sets. For every one of them Theorem 10.10 gives the exact detection threshold; this section supplies the constructions and the size computations it uses. Manuscript 05 (sibling report [46]) asked which set-sized exponent groups admit analogues of the common-divisor and set-target theorems (its “exact size boundary” question); 03, 06, 08, 09 and 11 each answer it with a sharp family, and none claims a classification of all exponent groups or support restrictions.

11.1 Countable supports over a two-armed group [03]

For an ordered abelian group G and $k \in \{\mathbb{R}, \mathbb{C}\}$ let $H_\omega(k, G)$ be the Hahn series over k with *at most countable* reverse well-ordered support (any countable reverse order type is allowed; the subscript

means countable support, not order type ω). It is the $\aleph_1$ -bounded Hahn field in the terminology of [23].

Lemma 11.1 (Countable-support Hahn fields [03]). *$H_\omega(k, G)$ is a field. If G is divisible, $H_\omega(\mathbb{R}, G)$ is real closed, $H_\omega(\mathbb{C}, G)$ is algebraically closed, and $H_\omega(\mathbb{C}, G) = H_\omega(\mathbb{R}, G)[i]$.*

Proof. Finitely many series lie in the full Hahn field over the countable subgroup H generated by their supports, all of whose supports are countable; so sums, products and inverses stay in $H_\omega(k, G)$. For closedness, place the coefficient supports of a polynomial in their divisible hull, still countable; the full Hahn field over a divisible group with real closed (algebraically closed) coefficients is real closed (algebraically closed) [23, Fact 2.1.1], and it lies inside $H_\omega(k, G)$. Real and imaginary parts split a countable support into two subsets. $\square$

With $A(D, k, G) = D + \{f \in H_\omega(k, G) : \text{supp } f \subseteq G_{>0}\}$, the constant term is a split surjection, the units are $D^\times$ (Lemma 2.4), and for $(D, k) = (\mathbb{Z}, \mathbb{R})$ the ring is a discretely ordered integer part of $H_\omega(\mathbb{R}, G)$: write $x = P + r + \epsilon$ by positive, zero and negative exponents and take $P + \lfloor r \rfloor$, except $P + r - 1$ when $r \in \mathbb{Z}$ and $\epsilon < 0$ (discarding all negative exponents would send $-X^{-g}$ to 0, which is not its floor). The same floor rule gives the omnific integer part of **No** (03, 06, 08; compare [44]).

The construction. Let κ be an infinite cardinal with $\mu = \text{cf } \kappa > \aleph_0$, $\theta = \kappa^{\aleph_0}$, $J_\kappa = \kappa^{\text{op}} + \kappa$ with indices p_α (reversed arm) and q_α (forward arm), and

$$G_\kappa = \bigoplus_{j \in J_\kappa} \mathbb{Q}e_j,$$

ordered lexicographically with the *least* index of a nonzero vector deciding its sign; earlier indices are larger scales ($\iota(e_{j'}) < e_j$ for $j < j'$). $\iota(e_{p_\alpha}) = \omega^\alpha$, $\iota(e_{q_\alpha}) = \omega^{-(\alpha+1)}$ extends to an ordered additive embedding $G_\kappa \hookrightarrow \mathbf{No}$, and $\sum a_g X^g \mapsto \sum a_g \omega^{\iota(g)}$ is an ordered field embedding. Put

$$F_\kappa^{03} = H_\omega(\mathbb{R}, G_\kappa), \quad A_\kappa^{03} = A(\mathbb{Z}, \mathbb{R}, G_\kappa) = F_\kappa^{03} \cap \mathbf{Oz}, \quad \Pi_\kappa^{03} = \ker(\text{ct} \upharpoonright_{A_\kappa^{03}}),$$

and use the superscript $\mathbb{C}$ for the parallel construction over $\mathbb{C}$ and $\mathbb{Z}[i]$: $F_\kappa^{03, \mathbb{C}} = F_\kappa^{03}[i]$, $A_\kappa^{03, \mathbb{C}} = A_\kappa^{03}[i]$, $\Pi_\kappa^{03, \mathbb{C}} = \Pi_\kappa^{03} + i\Pi_\kappa^{03}$.

Lemma 11.2 (Room on both sides [03]).

- (i) Every family of fewer than μ elements of G_κ is bounded above and below in G_κ .
- (ii) Every family of fewer than μ positive elements has a positive strict lower bound.
- (iii) For every $h > 0$, a final segment of the forward arm, of cardinality κ , consists of basis vectors $e_{q_\alpha} \ll h$.

Proof. Fewer than μ vectors have fewer than μ support indices, bounded below κ in each arm. If β exceeds all reversed-arm ordinals occurring, p_β precedes all occurring indices, so e_{p_β} exceeds the absolute value of each vector; this gives (i). If γ exceeds all forward-arm ordinals occurring, q_γ follows every occurring index, and each positive vector, having a positive leading coefficient at an earlier index, exceeds $e_{q_\gamma} > 0$; this gives (ii), and (iii) is the case of one vector. $\square$

The reversed arm supplies a denominator large enough to shift a countable support upward; the forward arm supplies a positive divisor below a countable positive support and many indistinguishable smaller scales (03).

Lemma 11.3 (Small families have common divisors [03]). *Every family of fewer than μ elements of Π_κ^{03} (or $\Pi_\kappa^{03,\mathbb{C}}$) has a common divisor X^g , $g > 0$, with all quotients in the ideal.*

Proof. The union of fewer than μ countable supports has fewer than μ elements, μ being regular and uncountable; apply Lemma 11.2(ii) and translate by $-g$. $\square$

Lemma 11.4 (Every smaller-scale field has size θ [03]). *For either coefficient pair, the field, the ring, its purely infinite ideal and every $L_h = H_\omega(k, H_h(G_\kappa))$ ($h > 0$) have cardinality $\theta = \kappa^{\aleph_0}$.*

Proof. There are at most $\kappa^{\aleph_0}$ countable supports and at most $(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0} \leq \theta$ coefficient choices for each. Conversely, for a forward tail $T \subseteq \kappa$ with $e_{q_\alpha} \ll h$ ($\alpha \in T$), each countable $S \subseteq T$ gives $f_S = \sum_{\alpha \in S} X^{e_{q_\alpha}} \in L_h$ (increasing ordinals give decreasing exponents), and $L_h \supseteq k$; since $\kappa^{\aleph_0} = \max(2^{\aleph_0}, |\kappa|^{\leq \aleph_0})$ (Lemma 11.6), $|L_h| \geq \theta$. Multiplication by $X^{h/2}$ embeds L_h in the ideal. $\square$

Theorem 11.5 (Cardinal realization [03]). *For $A = A_\kappa^{03}$ (or $A_\kappa^{03,\mathbb{C}}$) with ideal Π_A :*

- (i) $|F_\kappa^{03}| = |A| = \theta$ and $\text{Frac}(A) = F_\kappa^{03}$;
- (ii) every unital map $A \rightarrow R$ with $|R| < \theta$ factors uniquely through ct (for the Gaussian ring the factorization through the Gaussian constant term is unique for a given map; different square roots of -1 give different maps);
- (iii) every A -module of cardinality $< \theta$ is annihilated by Π_A , and for every $0 \neq f \in \Pi_A$ the least cardinality of a ring target detecting f , and of a module on which f acts nontrivially, is exactly θ ;
- (iv) Π_A needs exactly μ generators; it is flat with $\Pi_A^2 = \Pi_A$, the module $D = A/\Pi_A$ is not flat, and $\text{Tor}_n^A(D, N) = 0$ for every D -module N and $n > 0$.

No inaccessible cardinal, CH or other additional axiom is used.

Proof. (i) Sizes by Lemma 11.4. For $x \in F_\kappa^{03}$, Lemma 11.2(i) gives $h \in G_\kappa$ larger than $-g$ for every exponent g of x , so $x = X^h x / X^h$ with numerator and denominator in A . (ii) and (iii) are Theorem 10.10: by Lemma 11.3 each $f \in \Pi_A$ is $X^g b$ with $b \in \Pi_A$, and Proposition 10.3 with Lemma 11.4 gives the lower bound θ (03's own proof is exactly this). (iv) Generators: choose a cofinal sequence $(\alpha_\xi)_{\xi < \mu}$ in κ ; the countably many index supports of the exponents of $f \in \Pi_A$ are bounded on the forward arm, so $X^{e_{q_{\alpha_\xi}}}$ divides f in A for large ξ , and these μ monomials generate; a family of fewer than μ elements lies in some $X^h A$ by Lemma 11.3, which misses $X^{h/2}$. Flatness, idempotence and Tor are Theorem 12.2. $\square$

For $\kappa = \aleph_1$: $2^{\aleph_0} \leq \aleph_1^{\aleph_0} \leq (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$, so $A_{\aleph_1}^{03}$ has the size of the continuum, every target of size below the continuum sees only the constant term, whether or not CH holds, and the ideal needs $\aleph_1$ generators. For every infinite ρ , $\kappa = 2^\rho$ gives $\theta = 2^\rho$ with $\text{cf}(2^\rho) > \rho$; so arbitrarily large thresholds are realized in ZFC. This is a realization theorem for this family, not a classification of ordered exponent groups. The case $\text{cf } \kappa = \aleph_0$ is excluded because countable supports can then run cofinally through an arm (03).

Lemma 11.6 (Cardinal facts [03]). *For every infinite κ , $\kappa^{\aleph_0} = \max(2^{\aleph_0}, |\kappa|^{\leq \aleph_0})$; for every infinite ρ , $(2^\rho)^{\aleph_0} = 2^\rho$ and $\text{cf}(2^\rho) > \rho$.*

Proof. A countable sequence is determined by its countable range and a sequence in that range; conversely binary sequences embed in $\kappa^{\mathbb{N}}$ and each countable subset is the range of an enumeration (choice). $(2^\rho)^{\aleph_0} = 2^{\rho \aleph_0} = 2^\rho$. If 2^ρ were a union of $\tau \leq \rho$ families E_ξ of binary functions on ρ , each of size $< 2^\rho$, split ρ into τ blocks of size ρ and on block ξ choose a pattern that is not the restriction of any member of E_ξ ; the assembled function lies in no E_ξ (König). $\square$

11.2 Supports of size $< \kappa$ over a two-sided group [06]

Fix an uncountable regular cardinal κ . In the additive group of **No** put

$$h_\alpha = \omega^{\alpha+1}, \quad \epsilon_\alpha = \omega^{-(\alpha+1)} \quad (\alpha < \kappa), \quad \Gamma_\kappa^{06} = \text{span}_{\mathbb{Q}}\{h_\alpha, \epsilon_\alpha : \alpha < \kappa\}$$

(finite rational combinations of distinct Conway monomials, hence linearly independent). This is a divisible ordered group of cardinality κ , and $\epsilon_\beta \ll \epsilon_\alpha \ll h_\alpha \ll h_\beta$ for $\alpha < \beta$; here $\ll$ refers to the outer exponents.

Lemma 11.7 (Simultaneous upper and lower scales [06]). *For every $T \subseteq \Gamma_\kappa^{06}$ with $|T| < \kappa$ there is $\beta < \kappa$ with $\epsilon_\beta \ll |t| \ll h_\beta$ for all $t \in T \setminus \{0\}$.*

Proof. The generators occurring in the finite expressions of the members of T have fewer than κ indices, bounded below some β by regularity. The leading generator of a nonzero expression determines its Archimedean class, which lies strictly between those of ϵ_β and h_β . $\square$

For $k \in \{\mathbb{R}, \mathbb{C}\}$ put $K_{\kappa,k}^{06} = \{f \in k((X^{\Gamma_\kappa^{06}})) : |\text{supp } f| < \kappa\}$, $K_\kappa^{06} = K_{\kappa,\mathbb{R}}^{06}$, $\Pi_\kappa^{06} = \{f \in K_\kappa^{06} : \text{supp } f > 0\}$ and $A_\kappa^{06} = \mathbb{Z} + \Pi_\kappa^{06}$ (06's R_κ); $X^g \mapsto \omega^g$ makes them subsets of **No** and **No**[i].

Proposition 11.8 (Field closure and integer part [06]). *$K_{\kappa,\mathbb{R}}^{06}$ is real closed, $K_{\kappa,\mathbb{C}}^{06} = K_{\kappa,\mathbb{R}}^{06}[i]$ is algebraically closed, $A_\kappa^{06} = K_\kappa^{06} \cap \mathbf{Oz}$, $\text{Frac}(A_\kappa^{06}) = K_\kappa^{06}$, and A_κ^{06} is a discretely ordered integer part of K_κ^{06} .*

Proof. Inverting a series of support size $\mu < \kappa$ (factor the leading term and use the geometric series) gives support in a translate of the monoid generated by at most $\max(\mu, \aleph_0) < \kappa$ elements; sums and products also keep the bound. For closedness, the coefficients of a polynomial use fewer than κ exponents; the rational span Δ of these has $|\Delta| < \kappa$, the full field $\mathbb{C}((X^\Delta))$ is algebraically closed [27, Corollary 4] and lies in $K_{\kappa,\mathbb{C}}^{06}$; the real field with its leading-term order has this algebraically closed complexification, so it is real closed. For $f \in K_\kappa^{06}$, Lemma 11.7 gives h_β above $|g|$ for every exponent g , so $X^{h_\beta} f \in \Pi_\kappa^{06}$ and f is a quotient. The floor rule of Section 11.1 gives the integer part, and a nonzero element with positive leading exponent exceeds every real, so there is nothing strictly between 0 and 1. $\square$

Field closure uses only that κ is uncountable; regularity is used for simultaneous gaps and generation (06). The first-kappa report proves the closedness of $\{f \in k((t^\Gamma)) : |\text{supp } f| < \kappa\}$ for every uncountable κ , possibly singular (`fkc:thm:main` in [45]); 06's closure argument is not a novelty claim, and singular support bounds may preserve field closure while failing the simultaneous gap property used here. The two families h_α and ϵ_α are deliberate: the ϵ_α provide smaller positive gaps and the h_α shift supports upward, and a one-sided family would not automatically give the fraction-field assertion (06).

Proposition 11.9 (A chain of principal ideals [06]). *$\Pi_\kappa^{06} = \bigcup_{\alpha < \kappa} X^{\epsilon_\alpha} A_\kappa^{06}$ is a strictly increasing union; the ideal is idempotent and its least number of generators is κ .*

Proof. For $\alpha < \beta$, $\epsilon_\alpha - \epsilon_\beta > 0$ gives the inclusion, strict because the reverse quotient has a negative exponent. A gap $\epsilon_\alpha < \text{supp } f$ (Lemma 11.7) gives $f \in X^{\epsilon_\alpha} \Pi_\kappa^{06}$, hence the union and idempotence. Fewer than κ elements have fewer than κ support exponents in total (regularity), so they lie in some $X^a A_\kappa^{06}$, which misses $X^{a/2}$. $\square$

Lemma 11.10 (Size of the ring and of every smaller-scale field [merge]). *For every $c > 0$ in Γ_κ^{06} , the field $L_c = \{f \in K_\kappa^{06} : \text{supp } f \subseteq H_c(\Gamma_\kappa^{06})\}$ has cardinality at least $\kappa^{<\kappa}$, and*

$$|K_{\kappa, \mathbb{R}}^{06}| = |K_{\kappa, \mathbb{C}}^{06}| = |A_\kappa^{06}| = |\Pi_\kappa^{06}| = \kappa^{<\kappa}.$$

Proof. The finite expression of c uses indices below some $\alpha_0 < \kappa$, and its Archimedean class is that of its leading generator; so $\epsilon_\alpha \ll c$ for every $\alpha > \alpha_0$. For $B \subseteq (\alpha_0, \kappa)$ with $|B| < \kappa$, the series $\sum_{\alpha \in B} X^{\epsilon_\alpha}$ has support $\{\epsilon_\alpha : \alpha \in B\}$, reverse well ordered because ϵ_α decreases as α increases, of size $|B| < \kappa$, and inside H_c ; so it lies in L_c , which is a field by Proposition 10.3(i). Distinct B give distinct series, and the number of subsets of size $< \kappa$ of a set of size κ is $\sum_{\nu < \kappa} \kappa^\nu = \kappa^{<\kappa}$. Hence $|L_c| \geq \kappa^{<\kappa}$, and $X^c L_c \subseteq \Pi_\kappa^{06}$ gives the same lower bound for the ideal, ring and fields. Conversely a series is given by a support of size $\nu < \kappa$ and a coefficient function: at most $\kappa^\nu \cdot (2^{\aleph_0})^\nu$ choices, which is $\leq \kappa^\nu$ for infinite ν and $\leq \kappa \cdot 2^{\aleph_0} \leq \kappa^{\aleph_0} \leq \kappa^{<\kappa}$ for finite ν (as $\kappa > \aleph_0$). Summing over the $\leq \kappa$ values of ν gives at most $\kappa^{<\kappa}$. $\square$

Theorem 11.11 (The exact threshold of A_κ^{06} [06] [merge]). *Let κ be uncountable and regular. Every homomorphism $A_\kappa^{06} \rightarrow S$ with $|S| < \kappa^{<\kappa}$ factors through ct , every A_κ^{06} -module M with $|M| < \kappa^{<\kappa}$ satisfies $\Pi_\kappa^{06} M = 0$, and for each $0 \neq x \in \Pi_\kappa^{06}$ the least cardinality of a ring target detecting x , and of a module on which x acts nontrivially, is exactly $\kappa^{<\kappa}$. The same holds for $A_\kappa^{06}[i]$ with $\mathbb{Z}[i]$.*

Proof. Theorem 10.10 with Lemmas 11.7 and 11.10: each $0 \neq x \in \Pi_\kappa^{06}$ has a gap $\epsilon_\beta < \text{supp } x$, and with $c = \epsilon_\beta/2$ the field L_c has cardinality at least $\kappa^{<\kappa} = |A_\kappa^{06}|$. $\square$

06 proves the lower bound κ (with the explicit telescope family of the κ monomials $X^{a+\epsilon_\alpha}$, each witness having countable support) and, under the extra hypothesis $\kappa^{<\kappa} = \kappa$, that $|A_\kappa^{06}| = \kappa$ and the threshold is exactly κ ; that is the special case of the theorem where $\kappa^{<\kappa} = \kappa$. The lower bounds of 06 need no cardinal arithmetic, and neither does the exact value $\kappa^{<\kappa}$, which comes from 06's own collision lemma (compare 03's field bound). This answers the first half of 06's question "Cardinal arithmetic without the matching hypothesis": the least size of a target detecting a fixed nonzero $x \in \Pi_\kappa^{06}$ is $\kappa^{<\kappa}$ in ZFC. The second half ("a different integer-part construction of size exactly κ with the same lower-bound property under weaker hypotheses") is answered by 08 in ZFC for every infinite κ (Theorem 11.16), for a ring of size exactly κ whose fraction field, unlike $\text{Frac } A_\kappa^{06} = K_\kappa^{06}$, is not claimed to be its ambient field.

Example 11.12 (The first uncountable cardinal [06]). Under CH, $\aleph_1^{<\aleph_1} = \aleph_1^{\aleph_0} = 2^{\aleph_0} = \aleph_1$, so $A_{\aleph_1}^{06}$ has size $\aleph_1$: every action on a countable abelian group is an ordinary integer action, while the regular module of size $\aleph_1$ sees all purely infinite elements. Without CH the lower bound at $\aleph_1$ holds, the ring has size $\aleph_1^{\aleph_0} = 2^{\aleph_0}$, and by Theorem 11.11 the exact threshold is $2^{\aleph_0}$; this ring is isomorphic to $A_{\aleph_1}^{03}$ (Remark 10.11).

11.3 A controlled field of size exactly κ [08]

Let κ be any infinite cardinal and $\Gamma_\kappa^{08} = \bigoplus_{\xi < \kappa} \mathbb{Q}e_\xi$, ordered lexicographically with the least nonzero index deciding the sign, so $0 < ne_\eta < e_\xi$ for $\xi < \eta$; it is divisible of cardinality κ , and each nonzero element has a finite coordinate support.

Lemma 11.13 (Scale and support bounds [08]). *For every $c > 0$ there are κ positive $a \ll c$ (namely e_ξ for ξ beyond the least nonzero index of c). If $S \subseteq (\Gamma_\kappa^{08})_{>0}$ has fewer than cf κ elements, some $\beta < \kappa$ has $e_\beta \ll s$ for all $s \in S$.*

Proof. A union of fewer than cf κ finite subsets of κ is bounded; take β above it. $\square$

Let $K = \mathbb{R} \cap \overline{\mathbb{Q}}$, $H = K((X^{\Gamma_\kappa^{08}}))$ and $H_{\text{ctbl}} \subseteq H$ the series with at most countable support. H_{ctbl} is a real closed subfield: finitely many series lie in $K((X^\Delta))$ for the countable divisible subgroup Δ generated by their supports, which is real closed [23, Fact 2.1.1] and has only countable supports. A *truncation* of a series is its restriction to an initial segment of its support in decreasing order; a countable support has countably many. Let F_0 be the real closure in H_{ctbl} of the field generated by K and all X^g ; given F_n , adjoin all truncations of its elements and let F_{n+1} be the real closure in H_{ctbl} of the resulting field; put $F = \bigcup_n F_n$.

Lemma 11.14 (The controlled field [08]). *F is real closed and truncation-closed, has cardinality κ , consists of series with countable support, and contains K and every monomial.*

Proof. An algebraic extension of an infinite field of size κ has size $\leq \kappa$ (κ polynomials, finitely many roots each), and adjoining countably many truncations of each of κ elements adds at most κ elements; so every F_n , and F , has size κ , and at least κ because of the monomials. Truncations of an element of F_n lie in F_{n+1} , and finitely many coefficients lie in a common real closed stage. $\square$

Lemma 11.15 (Finite-coordinate invariant [08]). *Every $f \in F$ has $\text{supp } f \subseteq \Gamma_E = \bigoplus_{\xi \in E} \mathbb{Q}e_\xi$ for some finite $E \subseteq \kappa$ (its support need not be finite).*

Proof. For finite E , $H_E = K((X^{\Gamma_E}))$ is real closed and lies in H_{ctbl} (Γ_E is countable); the H_E form a directed family. An element of the field generated by the monomials is a rational expression in finitely many of them, so lies in some H_E . An element of F_0 is algebraic over that field, hence over some H_E (finitely many coefficients); as H_E is real closed and H_{ctbl} is an ordered field containing it, it has no proper ordered algebraic extension there, so the element lies in H_E . Truncations of elements of H_E stay in H_E , finite rational expressions stay in $H_{E \cup E'}$, and real closure is handled by the same algebraicity argument; induct over n . $\square$

This invariant removes regularity: even at a singular κ of countable cofinality, the coordinates in one element are confined to a finite set. The proof uses countable supports to control the number of truncations and finite coordinate sets to find smaller positive scales (08).

The map $\iota(\sum q_\xi e_\xi) = \sum q_\xi \omega^{-\xi}$ is an increasing additive embedding of Γ_κ^{08} into $\mathbf{No}$ (the least nonzero coordinate decides the sign on both sides), and $\Phi(\sum a_g X^g) = \sum a_g \omega^{\iota(g)}$ is an ordered field embedding $H \hookrightarrow \mathbf{No}$ preserving truncations and constant terms. Put $F_\kappa^{08} = \Phi(F)$, $A_\kappa^{08} = F_\kappa^{08} \cap \mathbf{Oz} = \{f \in F : \text{supp } f \geq 0, \text{ct}(f) \in \mathbb{Z}\}$ and $\Pi_\kappa^{08} = \ker(\text{ct}|_{A_\kappa^{08}})$. Since $\iota(e_0) = 1$, the monomial X^{e_0} is the actual surreal ω . A_κ^{08} is a discretely ordered integer part of F of cardinality κ : truncate x to its positive part $P \in F$ and apply the floor rule of Section 11.1 to $x = P + r + \epsilon$ with $r \in K$.

Theorem 11.16 (Sharp models at every infinite cardinal [08]). *For every infinite cardinal κ :*

- (i) F_κ^{08} is a truncation-closed real closed subfield of $\mathbf{No}$ with countable supports, $|F_\kappa^{08}| = |A_\kappa^{08}| = \kappa$, and A_κ^{08} is an integer part of it;
- (ii) every homomorphism from A_κ^{08} to a ring of cardinality $< \kappa$ factors through ct ;
- (iii) every quotient ring of A_κ^{08} is a quotient $\mathbb{Z}/n\mathbb{Z}$ ($n \geq 0$) under ct or has cardinality exactly κ ;

- (iv) Π_κ^{08} needs exactly cf κ generators;
- (v) every unital A_κ^{08} -module of cardinality $< \kappa$ has action $f \cdot m = \text{ct}(f)m$;
- (vi) the positive-characteristic maximal residue fields are the $\mathbb{F}_p$, with maximal ideals pA_κ^{08} ; characteristic-zero maximal residue fields exist and all have cardinality exactly κ .

No hypothesis such as CH, GCH, $\kappa^{\aleph_0} = \kappa$ or an inaccessible cardinal is used.

Proof. (i) is Lemma 11.14 and the paragraph above. Each $f \in \Pi_\kappa^{08}$ has support in some Γ_E (Lemma 11.15), so e_β , $\beta > \max E$, is a positive strict lower bound; and Lemma 11.13 gives κ scales below every c . These are the hypotheses of Theorem 10.7, which gives (ii). (iii) If $|A/J| < \kappa$, (ii) applied to $A \rightarrow A/J$ gives $\Pi \subseteq J$; otherwise $|A/J| \leq |A| = \kappa$. (iv) With a cofinal $C \subseteq \kappa$ of size cf κ , for $f \in \Pi_\kappa^{08}$ choose $\beta \in C$ above its coordinate set; then $X^{-e_\beta} f \in \Pi_\kappa^{08}$, so $\{X^{e_\beta} : \beta \in C\}$ generates. Fewer than cf κ elements have coordinate sets bounded by some β , so they lie in (X^{e_β}) , which misses $X^{e_\beta/2}$. (v) is Lemma 10.9 with (ii). (vi) Coefficient division gives $n\Pi_\kappa^{08} = \Pi_\kappa^{08}$, so $nA = n\mathbb{Z} + \Pi_\kappa^{08}$ and $A/pA \cong \mathbb{F}_p$ is already a field; a maximal ideal of characteristic p contains pA and equals it. The element $u = 1 + X^{e_0}$ is a nonunit (Lemma 2.4), so the set-sized ring A has a maximal ideal $M \ni u$ (ordinary maximal-ideal theorem); if a nonzero integer n lay in M , then $X^{e_0} \in n\Pi_\kappa^{08} \subseteq M$ and $1 \in M$. A characteristic-zero residue field of size $< \kappa$ would receive a map through $\mathbb{Z}$ by (ii), with image $\mathbb{Z}$, not a field; and it has size $\leq \kappa$ as a quotient. $\square$

At $\kappa = \aleph_0$ the theorem restricts only finite targets, and the characteristic-zero residue fields are countable and detect infinite monomials. For regular κ the generator count is κ ; at a singular κ it is smaller, so a countably generated ideal can be invisible to all ring targets below a much larger singular κ of countable cofinality. Two resources are separated: monomial generators coinitial in the support scales (cf κ) and pairwise collision witnesses below a fixed scale (κ) (08).

In the residue field of a maximal ideal containing $1 + \omega$ the image of ω is -1 , not 0, and that field has size κ ; so for every infinite κ the subring $A_\kappa^{08} \subseteq \mathbf{Oz}$ has a characteristic-zero field quotient of cardinality κ in which $\omega \mapsto -1$. This quotient map cannot extend to $\mathbf{Oz}$, where every set-sized image sends ω to 0: an explicit extension obstruction between a set-sized integer part and the class ring (08). 08 does not identify $\text{Frac}(A_\kappa^{08})$ with F_κ^{08} , imposes no initial-segment condition or birthday cutoff on the embedding, and designs the construction for truncation, support control and cardinality, not for closure under a derivation or an exponential; the isomorphism types of the residue fields are not classified.

11.4 Countable supports over a long one-armed group [09]

Fix a cardinal λ with $\lambda^{\aleph_0} = \lambda$; then automatically $\lambda \geq 2^{\aleph_0}$, since $\lambda^{\aleph_0} \geq 2^{\aleph_0}$ (09 states $\lambda \geq 2^{\aleph_0}$ as a separate hypothesis; it is redundant). For example $\lambda = 2^\nu$ for any infinite ν . Neither regularity nor CH is assumed. Let $L_\lambda = \omega_1 \times \lambda$ with the lexicographic order and $\Gamma_\lambda^{09} = \bigoplus_{\ell \in L_\lambda} \mathbb{Q}e_\ell$, the least nonzero index deciding the sign.

Lemma 11.17 (A large small-exponent tail [09]). *For every countable $S \subseteq (\Gamma_\lambda^{09})_{>0}$ there is $\alpha_0 < \omega_1$ with $e_{(\alpha,\beta)} \in U_\Gamma(S)$ for all $\alpha_0 \leq \alpha < \omega_1$ and $\beta < \lambda$; in particular $U_\Gamma(S)$ contains λ elements, and every countable positive set has a positive strict lower bound.*

Proof. The countable union of the finite index supports has first coordinates bounded below some $\alpha_0 < \omega_1$; each $s \in S$ has its least index before every (α, β) with $\alpha \geq \alpha_0$, with positive coefficient there. $\square$

For $k \in \{\mathbb{R}, \mathbb{C}\}$ and unital $D \subseteq k$ let $H_{\lambda,k}$ be the countable-support Hahn field over $\Gamma_{\lambda}^{09}, \Pi_{\lambda,k}^{09}$, its positive-support part and $A_{\lambda,D}^{09} = D + \Pi_{\lambda,k}^{09}$. It is a domain with units $D^{\times}$, the quotients of [Lemma 4.7](#) of its elements stay in it (countable supports), and $|A_{\lambda,D}^{09}| = \lambda$: there are at most $\lambda^{\aleph_0} = \lambda$ countable supports and $|k|^{\aleph_0} = 2^{\aleph_0} \leq \lambda$ coefficient choices, and λ positive basis monomials. With $j(\alpha, \beta) = \lambda \cdot \alpha + \beta$ (ordinal operations), $\iota(\sum q_{\ell} e_{\ell}) = \sum q_{\ell} \omega^{-j(\ell)}$ is an ordered embedding and $\sum c_{\gamma} X^{\gamma} \mapsto \sum c_{\gamma} \omega^{\iota(\gamma)}$ embeds $H_{\lambda,\mathbb{R}}$ in $\mathbf{No}$ and $H_{\lambda,\mathbb{C}}$ in $\mathbf{No}[i]$, sending $A_{\lambda,\mathbb{Z}}^{09}$ into $\mathbf{Oz}$ and $A_{\lambda,\mathbb{Z}[i]}^{09}$ into $\mathbf{Oz}[i]$; here $X^{e_{\ell}} \mapsto \omega^{\omega^{-j(\ell)}}$, the group coordinates describing exponents.

Theorem 11.18 (Cardinal barrier [09]). *Let $A = A_{\lambda,D}^{09}$ and $\Pi_A = \Pi_{\lambda,k}^{09}$.*

- (i) *If $\varphi : A \rightarrow B$ is a unital ring map with $\varphi(\Pi_A) \neq 0$, then $|\varphi(A)| \geq \lambda$.*
- (ii) *If M is an A -module with $\Pi_A M \neq 0$, then $|M| \geq \lambda$.*
- (iii) *All maps to rings of size $< \lambda$, and all modules of size $< \lambda$, factor through ct .*
- (iv) *For every ideal J , either $\Pi_A \subseteq J$ and $A/J \cong D/\text{ct}(J)$, or $|A/J| = \lambda$.*

In particular, in ZFC there is an explicit $A \subseteq \mathbf{Oz}$ of cardinality $2^{\aleph_0}$ (take $\lambda = 2^{\aleph_0}$, $D = \mathbb{Z}$) whose maps to countable rings factor through ct , whose quotients are finite, countably infinite, or of cardinality $2^{\aleph_0}$ (the last for every quotient not induced from $\mathbb{Z}$), and whose countable modules are abelian groups with constant-term action.

Proof. For $f \in \Pi_A$ detected by φ or by a vector, $\text{supp } f$ is countable, [Lemma 11.17](#) supplies λ small exponents, and [Lemma 4.8](#) with [Lemma 4.7](#) injects them into $\varphi(A)$ (with the module structure given by φ and the vector 1) or into M . (iii) follows, and (iv) as in [Theorem 10.10](#). $\square$

When the continuum exceeds $\aleph_1$ the last statement also excludes all intermediate quotient cardinalities; this is conditional only on that cardinal inequality (09).

Lemma 11.19 (Recovering the coefficient field [09]). *Let $\varphi : A \rightarrow F$ be a unital map to a field with $\varphi(\Pi_A) \neq 0$. There is a unique embedding $\sigma : k \hookrightarrow F$ with $\varphi(rg) = \sigma(r)\varphi(g)$ for $r \in k$, $g \in \Pi_A$, and $\sigma|_D = \varphi|_D$; in particular F has characteristic 0.*

Proof. Take $f \in \Pi_A$ with $\varphi(f) \neq 0$ and $\sigma(r) = \varphi(rf)/\varphi(f)$ (note $rf \in \Pi_A$ even if $r \notin A$). It is additive and unital, $(rf)(sf) = (rsf)f$ gives multiplicativity, and it is injective as a map of fields. The identity $(rg)f = (rf)g$ gives the displayed formula and independence of f . $\square$

Theorem 11.20 (Residue fields at the threshold [09]). *Let $A = A_{\lambda,D}^{09}$.*

- (i) *For every $0 \neq f \in \Pi_A$ there is a maximal ideal $\mathfrak{m}$ with $f \equiv 1 \pmod{\mathfrak{m}}$ and $|A/\mathfrak{m}| = \lambda$.*
- (ii) *Every maximal ideal not containing Π_A has a residue field of cardinality λ with a canonical copy of k .*
- (iii) *For $D = \mathbb{Z}$, the maximal ideals containing Π_A are the pA , with residue fields $\mathbb{F}_p$, and every other maximal ideal has a characteristic-zero residue field of cardinality λ .*
- (iv) *There are at least λ distinct maximal ideals not containing Π_A .*

Consequently the least cardinality of a field (and of a module) detecting a given $0 \neq f \in \Pi_A$ is exactly λ ; and for $D = \mathbb{Z}$ the intersection of all finite-index ideals is $\Pi_A \neq 0$ while $\text{Jac}(A) = 0$.

Proof. (i) $1 - f$ is a nonconstant nonunit; a maximal ideal of the set-sized ring A containing it sends f to 1, so the residue field has size $\geq \lambda$ by [Theorem 11.18](#) and $\leq \lambda$ as a quotient. (ii) is [Theorem 11.18](#) with [Lemma 11.19](#); (iii) uses $pA = p\mathbb{Z} + \Pi_A$. (iv) For $\ell < \ell'$ in L_λ put $\gamma = e_\ell$, $a = e_{\ell'}$; then $na < \gamma$ for all n , and $q = -\sum_{n \geq 1} X^{\gamma-na} \in \Pi_A$ satisfies $(1 - X^a)q = X^\gamma$, so $1 = (1 - X^\gamma) + (1 - X^a)q$; the λ proper ideals $(1 - X^{e_\ell})$ are pairwise comaximal, and maximal ideals above them (set-sized choice) are distinct and do not contain Π_A . For the consequences, the residue field of (i) is a field target and a module on which f acts as the identity. Every finite target has size $< \lambda$, so the finite residual is Π_A ; if $0 \neq x = n + f$, a prime not dividing $n \neq 0$, or (i) when $n = 0$, gives a maximal ideal avoiding x . $\square$

All of this concerns set-sized rings and is not an application of the maximal-ideal theorem to **Oz**. The distinction between finite ring probes and field probes is sharp: the extra information is carried by residue fields of the full workspace cardinality (09). The number of maximal ideals and the isomorphism types of the residue fields are not determined; distinct maximal ideals need not give nonisomorphic fields. $\Pi_{\lambda,k}^{09} = \bigcup_{\alpha < \omega_1} X^{e_{(\alpha,0)}} A$ is an increasing chain, and the least number of generators of the ideal is exactly $\aleph_1$ (the chain gives $\aleph_1$ generators; a countable family lies in $X^{2h} A$ for a small h by [Lemma 11.17](#), which misses X^h); flatness is in [Theorem 12.2](#).

Remark 11.21 (Why the support restriction is real [09]). In the unrestricted Hahn field over Γ_λ^{09} the series $f_* = \sum_{\alpha < \omega_1} X^{e_{(\alpha,0)}}$ has reverse well-ordered but uncountable support with no positive lower bound in Γ_λ^{09} (a late enough $e_{(\alpha,0)}$ lies below any given $h > 0$); it is not divisible by any positive monomial within the nonnegative-support ring and is not in $A_{\lambda,D}^{09}$. No assertion is made about the quotient classification of that larger unrestricted ring.

11.5 Supports of size $< \kappa$ over a one-armed group [11]

[Theorem 10.8](#) (11) applies to any divisible set group Γ with $\text{coin}(\Gamma_{>0}) \geq \kappa$. The coinitality condition says that every set of fewer than κ positive elements has a positive strict lower bound.

Proposition 11.22 (Long lexicographic groups [11]). *For κ uncountable and regular and $\Gamma = \bigoplus_{\alpha < \kappa} \mathbb{Q}e_\alpha$ with the least nonzero index dominant, $|\Gamma| = \kappa$ and $\text{coin}(\Gamma_{>0}) = \kappa$. The group Γ_κ^{06} also has coinitality κ , so $A_\kappa^{06} = R_{\Gamma_\kappa^{06}, < \kappa}^{11}$ with $(D, K) = (\mathbb{Z}, \mathbb{R})$ is an instance of [Theorem 10.8](#).*

Proof. The e_β with β beyond the least index of a positive vector lie below it, so $(e_\beta)_{\beta < \kappa}$ is coinital; fewer than κ positive vectors have boundedly many leading indices (regularity), and a later basis vector lies below all of them. For Γ_κ^{06} use [Lemma 11.7](#). $\square$

11 proves sharpness under the displayed extra hypotheses $|K| \leq \kappa$ and $\kappa^{<\kappa} = \kappa$: then $|R_{\Gamma, < \kappa}^{11}| = \kappa$ and its identity map is a target of size κ not factoring through ct. 11 states explicitly that these hypotheses are assumptions for the sharpness example, not assertions about all uncountable regular cardinals, and that regularity enters only the lexicographic example.

Proposition 11.23 (Exact size and threshold [merge]). *In [Proposition 11.22](#), for every set field K and every $c > 0$ the field $L_c = \{f \in K((X^\Gamma)) : |\text{supp } f| < \kappa, \text{ supp } f \subseteq H_c(\Gamma)\}$ has at least $\kappa^{<\kappa}$ elements. If $|K| \leq \max(\kappa, 2^{\aleph_0})$ (which includes 11's hypothesis $|K| \leq \kappa$), then $|R_{\Gamma, < \kappa}^{11}| = \kappa^{<\kappa}$, and the least cardinality of a ring target or module detecting any nonzero element of the positive-support ideal is exactly $\kappa^{<\kappa}$.*

Proof. Exactly as in [Lemma 11.10](#): the series $\sum_{\alpha \in B} X^{e_\alpha}$, B a subset of size $< \kappa$ of the indices beyond the leading index of c , lie in L_c (e_α decreases with α), and the $< \kappa$ -support series form a field

because inverses have support in a monoid generated by fewer than κ elements. The upper bound counts supports of size $\nu < \kappa$ and coefficient functions: for infinite ν , $|K|^\nu \leq \max(\kappa^\nu, 2^\nu) = \kappa^\nu$, and for finite ν , $|K|^\nu \leq \max(\kappa, 2^{\aleph_0}) \leq \kappa^{\aleph_0} \leq \kappa^{<\kappa}$; also $|D| \leq |K|$, and the support choices number κ^ν . The threshold follows from [Proposition 10.3](#), the coinitality gap, and the identity map. $\square$

So the sharpness of [Theorem 10.8](#) in the lexicographic example holds without the hypothesis $\kappa^{<\kappa} = \kappa$ once the bound is stated as $\kappa^{<\kappa}$, which equals κ exactly when $\kappa^{<\kappa} = \kappa$; the coefficient field is still assumed to satisfy $|K| \leq \max(\kappa, 2^{\aleph_0})$. With $(D, K) = (\mathbb{Z}, \mathbb{R})$ this ring is a Hahn integer part of the kind treated in this section; 11 does not embed it in **No**, and no embedding is used in this section. Manuscript 17 supplies one: $e_\alpha \mapsto \omega^{-\alpha}$ embeds Γ in $(\mathbf{No}, +)$ as an ordered group, and hence $R_{\Gamma, <\kappa}^{11}$ in **Oz** for $(D, K) = (\mathbb{Z}, \mathbb{R})$ and in **Oz** $[i]$ for $(\mathbb{Z}[i], \mathbb{C})$ ([Lemma 13.30](#)).

12 Flat ideals and homological reflection

All rings, modules, resolutions, tensor products and Tor, Ext groups in this section are ordinary set-sized objects over the set-sized rings of [Section 11](#). No class-sized resolution is formed, and no directed-colimit theorem is applied to the proper-class union $\Pi = \bigcup_{a>0} \omega^a \mathbf{Oz}$ (03, 06, 09). The package is 06's, proved there for A_κ^{06} ; 03 and 09 prove its flatness and Tor parts for their rings. The proofs use only the following properties, which all five constructions have:

- (H1) A is a commutative domain, $A = D \oplus I$ with $I = \ker \text{ct}$ an ideal and $D = A/I$;
- (H2) A is built over a divisible exponent group Γ , and every $f \in I$ has the form $f = X^g b$ with $g \in \Gamma_{>0}$ and $b \in I$; hence $I = \bigcup_{h \in \Gamma_{>0}} X^h A$, a directed set-indexed union of principal ideals;
- (H3) I is not finitely generated.

(H2) is the gap property used in [Theorem 10.10](#) (a positive strict lower bound g of $\text{supp } f$ gives $b = X^{-g} f \in I$). (H3) holds because the least number of generators is $\text{cf } \kappa$, κ , $\text{cf } \kappa$, $\aleph_1$, respectively κ ([Theorem 11.5\(iv\)](#), [Proposition 11.9](#), [Theorem 11.16\(iv\)](#), the chain after [Theorem 11.20](#), and for $R_{\Gamma, <\kappa}^{11}$ the coinital chain $(X^{e_\alpha})_{\alpha < \kappa}$ of [Proposition 11.22](#): fewer than κ elements have a common monomial divisor X^h , and $X^h A$ misses $X^{h/2}$); these counts are infinite.

Lemma 12.1 (Projective ideals of a domain [06]). *A nonzero projective ideal of a commutative domain is finitely generated.*

Proof. Fix $0 \neq a \in J$. For A -linear $f : J \rightarrow A$ and $x \in J$, $af(x) = f(ax) = xf(a)$, so $f(a) = 0$ forces $f = 0$. A dual basis (u_s, f_s) has $x = \sum_s f_s(x)u_s$ with finitely many nonzero terms for each x ; only finitely many $f_s(a)$ are nonzero, all other f_s vanish identically, and the finitely many corresponding u_s generate J . $\square$

Theorem 12.2 (Homological package [03, 06, 09] [merge]). *Let (A, I, D) satisfy (H1)–(H3). Then:*

- (i) I is flat, $I^2 = I$ (each element is a product of two elements of I), and I is not projective;
- (ii) for every D -module N (inflated along ct), $I \otimes_A N = 0$ and $\text{Tor}_n^A(D, N) = 0$ for $n > 0$; in particular $\text{Tor}_n^A(D, D) = 0$;
- (iii) D is not flat: $\text{fd}_A D = 1$, and $\text{Tor}_1^A(D, A/xA) \cong D$ for every $0 \neq x \in I$;
- (iv) for all D -modules N, M and all $n \geq 0$, inflation induces natural isomorphisms $\text{Ext}_A^n(N, M) \cong \text{Ext}_D^n(N, M)$;

(v) $\text{pd}_A D \geq 2$;

(vi) the sequence $0 \rightarrow I \rightarrow A \xrightarrow{\text{ct}} D \rightarrow 0$ does not split, so $\text{Ext}_A^1(D, I) \neq 0$;

(vii) if $(x_\alpha)_{\alpha \in \Lambda}$ generates I , $P = \bigoplus_\alpha Ae_\alpha \rightarrow I$, $e_\alpha \mapsto x_\alpha$, and J is its kernel, then $\text{Ext}_A^2(D, J) \neq 0$.

Proof. (i) Each $X^h A$ is free of rank one because A is a domain, the union in (H2) is directed ($X^h A \subseteq X^{h'} A$ for $h' < h$), and a filtered colimit of flat modules is flat [29, Lemma 10.39.3]. The factorization $f = X^g b$ writes f as a product of two elements of I , so $I^2 = I$. Projectivity would contradict (H3) by Lemma 12.1. (ii) Since $IN = 0$, $I \otimes_A N \cong (I/I^2) \otimes_D N = 0$ (the map $x \otimes n \mapsto (x + I^2) \otimes n$ is well defined in reverse because $(ab) \otimes n = b \otimes an = 0$ for $a, b \in I$); tensoring the flat resolution $0 \rightarrow I \rightarrow A \rightarrow D \rightarrow 0$ with N computes all positive Tor groups as 0. (iii) The same resolution gives $\text{fd}_A D \leq 1$. For $0 \neq x \in I$, $0 \rightarrow A \xrightarrow{x} A \rightarrow A/xA \rightarrow 0$ is a free resolution; after $\otimes_A D$ the first map is 0 because $x \in I$, so $\text{Tor}_1^A(D, A/xA) \cong D \neq 0$ and D is not flat. Equivalently, the injection $xA \hookrightarrow A$ becomes the zero map $D \rightarrow D$ after tensoring (03, 09). (iv) Take a projective A -resolution $P_\bullet \rightarrow N$. By (ii), $D \otimes_A P_\bullet$ has homology N in degree 0 only, and its terms are projective D -modules (direct summands of free ones), so it is a projective D -resolution; the adjunction $\text{Hom}_A(P_\bullet, M) \cong \text{Hom}_D(D \otimes_A P_\bullet, M)$ is an isomorphism of complexes, and comparison of resolutions makes the result natural. (v) If $\text{pd}_A D \leq 1$, the kernel I of the free surjection $A \rightarrow D$ would be projective (Schanuel), against (i). (vi) A section s gives $r = s(1)$ with $\text{ct}(r) = 1$ and $xr = s(x \cdot 1) = 0$ for $0 \neq x \in I$, so $r = 0$ in the domain A . (vii) $P \rightarrow I$ is surjective; $0 \rightarrow J \rightarrow P \rightarrow I \rightarrow 0$ does not split (else I would be a summand of a free module, hence projective), giving $0 \neq \text{Ext}_A^1(I, J)$; applying $\text{Hom}_A(-, J)$ to $0 \rightarrow I \rightarrow A \rightarrow D \rightarrow 0$ with A projective gives $\text{Ext}_A^1(I, J) \cong \text{Ext}_A^2(D, J)$, represented by the spliced sequence $0 \rightarrow J \rightarrow P \rightarrow A \rightarrow D \rightarrow 0$. $\square$

In standard terms, $A \rightarrow D$ is a homological ring epimorphism ($D \otimes_A D = D$, all higher self-Tor vanish) but not a flat one (03, 06); this terminology is optional. The exact value of $\text{pd}_A D$ is not determined in general: only $\text{pd}_A D \geq 2$ and $\text{fd}_A D = 1$ are proved (06); when I is countably generated, $\text{pd}_A D = 2$ (Proposition 13.12(v), from 12's telescope). Hypothesis (H2) is where the totally ordered exponent cone enters: for 12's coordinatewise cones the flat dimension of D is any $d \geq 1$, and the two kinds of statement are compared in Remark 13.16. The nonzero $\text{Tor}_1^A(D, A/xA)$ does not contradict (ii), because A/xA is not a D -module: $I \not\subseteq xA$, since a gap below $\text{supp } x$ exhibits a smaller positive monomial of I outside xA (06). $I/I^2 = 0$ is not a finite-presentation or smoothness statement (09).

Theorem 12.3 (Exact size of cohomological witnesses [06] [merge]). *Let A be one of the five rings of Section 10.4, with $D = \mathbb{Z}$ (or any admissible D), and $\tau = |A|$. For every A -module M with $|M| < \tau$ and every $n > 0$, $\text{Ext}_A^n(D, M) = 0$, and for any two A -modules of size $< \tau$ the Ext groups over A are those of the underlying D -modules. For $n \in \{1, 2\}$,*

$$\min\{|M| : \text{Ext}_A^n(D, M) \neq 0\} = \tau.$$

For A_κ^{06} this minimum is $\kappa^{<\kappa}$ in ZFC; under $\kappa^{<\kappa} = \kappa$ it is κ , which is 06's statement. For A_κ^{03} , A_κ^{08} and $A_{\lambda, D}^{09}$ it is $\kappa^{\aleph_0}$, κ and λ .

Proof. Modules of size $< \tau$ are annihilated by I (Theorem 10.10), so they are inflated D -modules and Theorem 12.2(iv) applies; D is free over itself. For the upper bounds, $|I| = \tau$ and $\text{Ext}_A^1(D, I) \neq 0$. For $n = 2$ choose a generating family of cardinality at most τ (the generator counts of (H3) are all $\leq \tau$); then $|P| \leq \tau$, so $|J| \leq \tau$, and $\text{Ext}_A^2(D, J) \neq 0$ with the lower bound forces $|J| = \tau$. $\square$

The Ext groups are computed in the full category of A -modules, not only among small modules; this is why [Theorem 12.2\(iv\)](#) is an ambient derived-functor statement and not merely an equivalence of small module categories (06). The two witnesses show that a quotient can look homologically projective against every small coefficient module while failing projectivity in the ambient category. The whole argument works for the Gaussian versions with $D = \mathbb{Z}[i]$, where Ext_D means extensions of $\mathbb{Z}[i]$ -modules (06). Finally, for $0 \neq x \in \Pi_\kappa^{06}$ the module $A_\kappa^{06}/xA_\kappa^{06}$ is detected by I (it is not a D -module), so its carrier has at least $\kappa^{<\kappa}$ elements; 06 stated this with the qualifier “under the cardinal-arithmetic hypothesis”, which is not needed.

13 Homological and tensor algebra of omnific cores

This section is the part contributed by manuscripts 12 and 14 ([Section 1.3](#)). Both study ordinary commutative algebra of rings of the form “constants plus a tail” inside $\mathbf{Oz}$ and $\mathbf{Oz}[i]$, and they share one set-sized model: 14’s finite-support ring A_0 is 12’s rank-one coordinate-cone core A_1 , and 12’s two-generator relation module is the case $r = 2$ of 14’s lattice syzygy. Each is printed once ([Lemma 13.1](#) and [Theorem 13.9](#)). Manuscript 12 computes flat and projective dimensions of set-sized finite-support cores ([Sections 13.1 to 13.3](#) and [13.5](#)); 14 computes tensor products, symmetric and exterior powers, Rees algebras, homomorphisms and duals of coefficient-lattice modules, over the full class ring and over A_0 ([Sections 13.4](#) and [13.6 to 13.8](#)). Manuscript 17 of the third batch re-derives, independently, 14’s matrix, syzygy, Hom, tensor-torsion and power-growth theorems and 12’s intersection; these are printed once, with 17 added to their sources, and 17’s new material (presentations and relation counts, the Tor identification, moduli, orders, the lexicographic models inside $\mathbf{Oz}$) is [Section 13.9](#). Manuscript 21 of a fifth batch determines the projective dimensions that 12 bounds and computes the Tor algebra of lattice quotients ([Section 13.10](#)). Labels carry the sub-prefixes `osq:hd:` (12), `osq:tn:` (14), `osq:rel:` (17) and `osq:da:` (21).

Scope. The rings of [Sections 13.1 to 13.3](#) and [13.5](#) and 14’s model A_0 have *finite* supports. They are therefore outside the universal theorem, like the finite-support rings of [Section 9](#): the augmentation that sends every monomial to 1 is a unital map to the coefficient field that detects the tail ([Proposition 13.2](#)). What 12 calls arithmetic invisibility is invisibility to *finite* rings and to modules on which the tail acts by zero; these rings are not invisible to set-sized rings in the sense of [Theorem 5.1](#).

Notation. 12’s augmentation ε is `ct`; its tail M is Π_Λ (the letter M is 14’s coefficient lattice); its ordered-cone ring $T = D \oplus P$ with monomials t^γ is $T_\Gamma = D \oplus \Pi_\Gamma^{\text{fin}}$ with monomials X^γ (in this report t^g means X^{-g} , [Section 2.2](#)); its test modules N_F are V_F ; its cyclic module $Q = A/(X_1, cX_1)$ is written A/H . 14’s ideal $\mathcal{I}$, its rings $B = k + \mathcal{I}$, $A = D + \mathcal{I}$ and its field F are Π_k , $\mathcal{A}_{k,k}$, $\mathcal{A}_{D,k}$ and $k((X^{\mathbf{No}}))$; its set-sized P , A_0 , B_0 are $\Pi_\mathbb{Q}^{\text{fin}}$, $T_\mathbb{Q}$ and $k + \Pi_\mathbb{Q}^{\text{fin}}$; its multiplication map $\mu_{M,N}$, kernel $K(M, N)$ and carrier $E(M, N)$ are $\mathfrak{m}_{M,N}$, $\mathcal{T}(M, N)$ and $\mathbf{E}(M, N)$; its multiplier ring $\mathcal{O}(M)$ and orders $\mathcal{O}$ in number fields K are $\mathfrak{o}(M)$ and $\mathfrak{o} \subset K_0$ (the letter $\mathcal{O}$ denotes the finite surreal rings); the saturated lattice L of its matrix theorem is W_D . 17’s positive-support ideal $\mathfrak{m}$ (which it warns is not the infinitesimal ideal; here $\mathfrak{m}$ is the negative-support ideal) is Π , its $F = \text{Frac}(D)$ is written $\text{Frac}(D)$ (14’s F was the fraction field $k((X^{\mathbf{No}}))$), its lattices L, M are M, N , its ideals $J_L^\gamma = \omega^\gamma(L + \mathfrak{m})$ are $J_\gamma(M) = X^\gamma L(M)$, its multiplier ring $E(L)$ and conductor $(M : L)$ are $\mathfrak{o}(M)$ and $(N : M)$, its multiplication map $\mu_{L,M}$ and kernel $T(L, M)$ are $\mathfrak{m}_{M,N}$ and $\mathcal{T}(M, N)$ (the letter μ was 14’s generator count), its kernel dimensions d, r are q, ℓ , and its set-sized models A_κ , $\mathfrak{m}_\kappa$ are $R_{\Gamma_\kappa, <\kappa}^{11}$ and

Π_κ^{11} .

13.1 Coordinate-cone and ordered-cone cores

Throughout Sections 13.1 to 13.3, K is a field and $D \subseteq K$ a unital subring, so D is a domain. Let Λ be $[d] = \{1, \dots, d\}$ ($d \geq 1$) or $\mathbb{N}_{>0}$, and let $P_\Lambda = \mathbb{Q}_{\geq 0}^{(\Lambda)}$ be the monoid of finitely supported vectors $\alpha = (\alpha_i)_{i \in \Lambda}$ with nonnegative rational coordinates. In the monoid algebra $K[P_\Lambda]$ (finite sums, $X^\alpha X^\beta = X^{\alpha+\beta}$) put

$$\Pi_\Lambda = \bigoplus_{0 \neq \alpha \in P_\Lambda} KX^\alpha, \quad A_\Lambda = D \oplus \Pi_\Lambda, \quad (13.1)$$

and write A_d, Π_d for $\Lambda = [d]$ and $X_i = X^{\mathbf{e}_i}$. The coefficient at X^0 is a split surjective ring homomorphism $\text{ct} : A_\Lambda \rightarrow D$ with kernel Π_Λ , because a sum of two nonnegative vectors is zero only when both are. For a nonzero divisible subgroup $\Gamma \subseteq \mathbb{R}$ put

$$\Pi_\Gamma^{\text{fin}} = \bigoplus_{\gamma \in \Gamma_{>0}} KX^\gamma, \quad T_\Gamma = D \oplus \Pi_\Gamma^{\text{fin}}, \quad (13.2)$$

the finite-support ring of Section 9 over the fixed ordered group Γ , again with $\text{ct} : T_\Gamma \rightarrow D$. The difference between the two families is the exponent cone: coordinatewise nonnegative vectors in A_Λ , the positive cone of a totally ordered group in T_Γ .

Lemma 13.1 (Domains and embeddings [12; 14]).

- (i) $K[P_\Lambda]$, A_Λ and T_Γ are domains, and Π_Λ is a nonzero prime ideal.
- (ii) Let $(\gamma_i)_{i \in \Lambda}$ be positive surreals linearly independent over $\mathbb{Q}$. Then $X^\alpha \mapsto \omega^{\sum_i \alpha_i \gamma_i}$ defines an injective ring homomorphism $\mathbb{R}[P_\Lambda] \rightarrow \mathbf{No}$, and, applied to real and imaginary parts, $\mathbb{C}[P_\Lambda] \rightarrow \mathbf{No}[i]$; it maps $\mathbb{Z} + \Pi_\Lambda$ into $\mathbf{Oz}$ and $\mathbb{Z}[i] + \Pi_\Lambda$ into $\mathbf{Oz}[i]$. The γ_i may be taken real: $\gamma_i = \theta^{i-1}$ for a transcendental real $\theta > 1$. Likewise $X^\gamma \mapsto \omega^\gamma$ embeds T_Γ in $\mathbf{Oz}$ or $\mathbf{Oz}[i]$ for $(D, K) = (\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$.
- (iii) If the γ_i are positive reals linearly independent over $\mathbb{Q}$ and $\Gamma = \sum_i \mathbb{Q}\gamma_i$, then $X_i \mapsto X^{\gamma_i}$ is an injective ring homomorphism $A_\Lambda \hookrightarrow T_\Gamma$ compatible with ct . For $d = 1$ and $\gamma_1 = 1$ it is an isomorphism $A_1 \cong T_\mathbb{Q}$; this ring, with $(D, K) = (\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$, is 14's set-sized model A_0 .

Proof. (i) Finitely many elements of $K[P_\Lambda]$ involve finitely many coordinates and a common denominator n of their exponents, so they lie in the polynomial ring $K[X_{i_1}^{1/n}, \dots, X_{i_s}^{1/n}]$, a domain. In T_Γ the largest exponents of two nonzero finite sums add and their leading coefficients multiply. $A_\Lambda/\Pi_\Lambda = D$ is a domain and $X_1 \in \Pi_\Lambda$. (ii) The map $\ell(\alpha) = \sum_i \alpha_i \gamma_i$ is additive, injective by linear independence, and positive on nonzero $\alpha \in P_\Lambda$; so the monomials go to distinct Conway monomials, $\omega^u \omega^v = \omega^{u+v}$ and finite distributivity give a ring homomorphism, and uniqueness of finite normal forms gives injectivity. Nonconstant terms go to positive exponents and the constant coefficient is an integer (Gaussian integer), which is membership in $\mathbf{Oz}$ ($\mathbf{Oz}[i]$). A rational relation among the θ^{i-1} would give a nonzero rational polynomial vanishing at θ . The case of T_Γ is the same with $\ell = \text{id}$. (iii) ℓ is an injective monoid map $P_\Lambda \rightarrow \Gamma_{\geq 0}$ with $\ell^{-1}(0) = 0$; for $d = 1$ it is $\mathbb{Q}_{\geq 0} \rightarrow \mathbb{Q}_{\geq 0}$, the identity. $\square$

Thus d counts independent monomial directions, not Archimedean scales: with $\gamma_i = \theta^{i-1}$ all exponents lie in the single Archimedean class of 1 (12). Every element of every ring in this subsection is a finite sum; no topology, convergence or infinite summation is involved.

Proposition 13.2 (Finite targets only [12]). *Let A be A_Λ or T_Γ with tail Π , and $(D, K) = (\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$.*

- (i) *For $0 \neq a \in D$, $a\Pi = \Pi$ and $A/aA \cong D/aD$. Every unital map from A to a finite ring, and every finite A -module, factors through ct ; $\hat{A} \cong \hat{D}$ for the profinite completions, $\varprojlim_k A/p^k A \cong \varprojlim_k D/p^k D$ for every ordinary prime p , $\bigcap_k p^k A = \Pi$, and $\text{Tor}_{>0}^A(D, E) = 0$ for every finite A -module E .*
- (ii) *The augmentation $\sum_\alpha c_\alpha X^\alpha \mapsto \sum_\alpha c_\alpha$ is a unital ring homomorphism $A \rightarrow K$ sending every monomial to 1; it does not factor through ct .*

Proof. (i) Divide the coefficients of a tail by a ; so $aA = aD \oplus \Pi$. A finite ring has a positive characteristic n , so a map to it kills $n\Pi = \Pi$; a finite module is killed by some $n \geq 1$, and $m = n(m/n)$ for $m \in \Pi$. The ideals $nA = nD \oplus \Pi$ are cofinal among ideals of finite index, and $\bigcap_n nD = 0 = \bigcap_k p^k D$ (coordinatewise for $\mathbb{Z}[i]$). A finite module is therefore a D -module, and the Tor assertion is Corollary 13.5 below for A_Λ and Theorem 12.2(ii) with Theorem 13.13 for T_Γ . (ii) is the computation of Theorem 9.1(ii): for finite supports, the coefficient sum of a product is the product of the coefficient sums. $\square$

Hence all the cores, and 14's A_0 , have the same finite quotients and profinite completion as D , while Corollary 13.8 below shows that their homological dimensions differ; this limits any reconstruction of an omnific subring from its finite arithmetic shadows (12). Item (ii) is the reason the word “invisible” must be read with (i)’s restriction.

13.2 Flat face ideals and a Boolean resolution

For a finite nonempty $S \subseteq \Lambda$ put $p_S = \prod_{i \in S} X_i$ and let

$$I_S = \bigoplus_{\alpha \in P_\Lambda, \alpha_i > 0 \ (i \in S)} KX^\alpha, \quad (13.3)$$

an ideal of A_Λ with $I_S = \bigcap_{i \in S} I_{\{i\}}$ and $\Pi_\Lambda = \sum_i I_{\{i\}}$.

Lemma 13.3 (Face ideals [12]). *Let $A = A_\Lambda$ and $\Pi = \Pi_\Lambda$.*

- (i) *$I_S = \bigcup_{n \geq 0} p_S^{2^{-n}} A$, an increasing chain of free rank-one A -modules; hence I_S is flat.*
- (ii) *$\Pi^2 = \Pi$, $I_S^2 = I_S$ and $\Pi I_S = I_S$; for every D -module E , regarded as an A -module through ct , $I_S \otimes_A E = 0$.*
- (iii) *Neither Π nor any I_S is finitely generated.*

Proof. (i) Multiplication by $p_S^{2^{-n}}$ is injective (Lemma 13.1), and $p_S^{2^{-n}} = p_S^{2^{-n-1}} p_S^{2^{-n-1}}$ gives the chain. Given $f \in I_S$, choose n with 2^{-n} below every coordinate indexed by S of the finitely many exponents of f ; then $f/p_S^{2^{-n}}$ has nonnegative exponents, positive on S , so it lies in $\Pi \subseteq A$ although its coefficients are arbitrary in K . A filtered colimit of flat modules is flat [29, Lemma 10.39.3]. (ii) $kX^\alpha = (kX^{\alpha/2})X^{\alpha/2}$, both factors in I_S when α is positive on S . Since $\Pi E = 0$, $I_S \otimes_A E \cong (I_S/\Pi I_S) \otimes_D E = 0$. (iii) Let $|\alpha| = \sum_i \alpha_i$. If finitely many elements generated Π (or I_S), let $\delta > 0$ be the least total degree of their monomials; every element of the generated ideal has only monomials of total degree at least δ , since multiplication by A adds nonnegative degrees and cancellation creates no new monomial. But $X_1^q \in \Pi$ (respectively $p_S^q \in I_S$) with q small has smaller total degree. $\square$

Flatness of every I_S does not imply flatness of their sum Π ; the failure begins at $d = 2$ (Corollary 13.8). For $p \geq 1$ put $C_p = \bigoplus_{S \subseteq \Lambda, |S|=p} I_S$ and $C_0 = A$; for $S = \{s_1 < \dots < s_p\}$ let d_p send f in the summand I_S to $\sum_{r=1}^p (-1)^{r-1} f$ in the summands $I_{S \setminus \{s_r\}}$ (for $p = 1$ the target is A and the map is the inclusion). Deleting two indices in either order gives opposite signs, so $d_{p-1}d_p = 0$.

Theorem 13.4 (Coordinate-intersection resolution [12]). *The augmented complex*

$$\cdots \longrightarrow C_2 \xrightarrow{d_2} C_1 \xrightarrow{d_1} A_\Lambda \xrightarrow{\text{ct}} D \longrightarrow 0$$

is exact and every C_p is flat. For $\Lambda = [d]$, $C_p = 0$ for $p > d$, and this is a flat resolution of D of length d ; also for $\Lambda = \mathbb{N}_{>0}$ the complex is exact.

Proof. All maps preserve the exponent grading. In degree $\mathbf{0}$ only A and D contribute, and ct is the identity of D . Fix $\alpha \neq \mathbf{0}$ with support $T = \{i : \alpha_i > 0\}$. The degree- α part of I_S is K exactly when $\emptyset \neq S \subseteq T$, so the degree- α part of the complex is the augmented chain complex of the full simplex on T with coefficients K , a p -subset in degree p . With $t = \min T$, the homotopy that inserts t (with the sign placing it in the ordered list) into subsets not containing t , is zero on subsets containing t , and sends the augmentation vector to the vertex $\{t\}$, satisfies $dh + hd = 1$. Each element involves finitely many degrees and summands, so degreewise exactness is exactness. Flatness is Lemma 13.3(i) and closure under direct sums. $\square$

Corollary 13.5 (Arithmetic modules are Tor-acyclic [12]). *For every D -module E , $D \otimes_{A_\Lambda} E \cong E$ and $\text{Tor}_j^{A_\Lambda}(D, E) = 0$ for $j > 0$; in particular $A_\Lambda \rightarrow D$ is a homological ring epimorphism.*

Proof. Tensor the resolution with E : every positive term vanishes by Lemma 13.3(ii), and flat resolutions compute Tor . $\square$

This concerns every module on which the tail acts by zero; it is why such modules cannot see the flat dimension computed next (12).

Proposition 13.6 (Formal lifting and differentials [12; 13]). *Let A be A_Λ or T_Γ with tail Π . Then $A \rightarrow D$ is formally étale for nilpotent lifting problems, but neither flat nor finitely presented as a ring map, and $\Omega_{A/D} \otimes_A D = 0$: every D -derivation of A into a module killed by Π vanishes.*

Proof. If $f : A \rightarrow R$ is an A -algebra structure map and $D \rightarrow R/J$ an A -algebra map with $J^n = 0$, then $f(\Pi) \subseteq J$, so $f(\Pi) = f(\Pi^n) \subseteq J^n = 0$ ($\Pi^2 = \Pi$ by halving exponents) and f factors uniquely through D ; this is the idempotent-ideal mechanism of [30, Tag 060H]. Multiplication by a nonzero monomial of Π is injective on A but becomes 0 on $D = D \otimes_A A$, so D is not flat; A/Π is finitely presented only if Π is finitely generated, which it is not (Lemma 13.3(iii)); for T_Γ , a finite family of tails has a least exponent among its monomials). A derivation ∂ as stated satisfies $\partial(uv) = u\partial v + v\partial u = 0$ for $u, v \in \Pi$, and $\Pi = \Pi^2$. $\square$

The mechanism is classical and not a novelty claim (12). Manuscript 13 proves the same statements for the full omnific ring in a two-universe reading (Remark 13.16), and Theorem 7.4 is the class-ring analogue for derivations. “Nilpotent” is a test condition on R : the cores are domains.

13.3 The multigraded Tor and exact flat dimensions

For a finite nonempty $F \subseteq \Lambda$ let $J_F = (X_i : i \in F)A_\Lambda$ and $V_F = A_\Lambda/J_F$. For finite S put $\mathbf{e}_S = \sum_{i \in S} \mathbf{e}_i$; the notation $K\langle \mathbf{e}_S \rangle$ means a copy of the D -module K placed in multidegree $\mathbf{e}_S$ (not a subring of A_Λ).

Theorem 13.7 (Boolean-face Tor formula [12]). *For Λ finite or countably infinite and F as above,*

$$\begin{aligned} \mathrm{Tor}_0^{A_\Lambda}(D, V_F) &\cong D\langle \mathbf{0} \rangle, & \mathrm{Tor}_1^{A_\Lambda}(D, V_F) &\cong \bigoplus_{i \in F} D\langle \mathbf{e}_i \rangle, \\ \mathrm{Tor}_p^{A_\Lambda}(D, V_F) &\cong \bigoplus_{S \subseteq F, |S|=p} K\langle \mathbf{e}_S \rangle & (p \geq 2). \end{aligned}$$

In particular $\mathrm{Tor}_p = 0$ for $p > |F|$.

Proof. Tensor Theorem 13.4 with V_F ; the positive terms become $I_S/J_F I_S$ and the degree-zero term is V_F . Fix $\alpha \neq \mathbf{0}$ with support T . If $\alpha_j > 1$ for some $j \in F$, then $\alpha - \mathbf{e}_j$ is nonnegative, nonzero and positive wherever α is, so the whole coefficient space at α lies in $X_j I_S$ and in $X_j A$; every term vanishes in this degree. So assume $\alpha_j \leq 1$ for $j \in F$ and put $U = \{j \in F : \alpha_j = 1\}$. For $\emptyset \neq S \subseteq T$ the degree- α part of $I_S/J_F I_S$ is a full copy of K exactly when $U \subseteq S$ (multiplication by X_j reaches this degree from I_S only when $\alpha_j = 1$ and $j \notin S$), and zero otherwise.

Case $U = \emptyset$. $(V_F)_\alpha = K$ and all faces survive; the augmented simplex is exact. *Case $U \neq \emptyset \neq T \setminus U$.* $(V_F)_\alpha = 0$, since subtracting $\mathbf{e}_j$ ($j \in U$) leaves a nonzero vector; inserting some $t \in T \setminus U$ contracts the complex of faces $U \subseteq S \subseteq T$. *Case $T = U$, $|U| \geq 2$.* Only K in degree $|U|$ survives, with no incoming or outgoing maps ($(V_F)_{\mathbf{e}_U} = 0$); this gives $K\langle \mathbf{e}_U \rangle$. *Case $T = U = \{j\}$.* In degree $\mathbf{e}_j$ the complex is $K \rightarrow (V_F)_{\mathbf{e}_j} = K/D$, because J_F has coefficient space D (not K) at $\mathbf{e}_j$: the cofactor of X_j is a constant. The map is the quotient map, with kernel D and zero cokernel; this gives $D\langle \mathbf{e}_j \rangle$ in degree one. In degree $\mathbf{0}$ only D in degree zero remains. $\square$

For $|S| = p \geq 2$ the class of $k \in K$ is represented by $kX^{\mathbf{e}_S}$ in $I_S/J_F I_S$; in degree one only coefficients $d \in D$ give cycles dX_i . Treating the constant coefficient as if it lay in K gives a wrong answer (12). Scaling coordinates by positive rationals b_i ($i \in F$) is an automorphism of A_Λ fixing the coefficients, so the same formula holds for $A_\Lambda/(X_i^{b_i} : i \in F)$ with $\mathbf{e}_S$ replaced by $\sum_{i \in S} b_i \mathbf{e}_i$ (12).

Corollary 13.8 (Exact flat dimensions [12]). *For $\Lambda = [d]$, $\mathrm{fd}_{A_d} D = d$ and $\mathrm{fd}_{A_d} \Pi_d = d - 1$; for $\Lambda = \mathbb{N}_{>0}$ both are infinite. All positive self-Tor groups $\mathrm{Tor}_{>0}^{A_d}(D, D)$ vanish.*

Proof. The resolution gives $\mathrm{fd} D \leq d$, and $\mathrm{Tor}_d(D, V_{[d]})$ is D ($d = 1$) or K ($d \geq 2$), nonzero. Truncation gives $\mathrm{fd} \Pi_d \leq d - 1$; for $d \geq 2$, $0 \rightarrow \Pi_d \rightarrow A_d \rightarrow D \rightarrow 0$ gives $\mathrm{Tor}_{d-1}(\Pi_d, V_{[d]}) \cong \mathrm{Tor}_d(D, V_{[d]}) \neq 0$, and for $d = 1$, $\Pi_1 = I_{\{1\}}$ is flat. In infinite rank take $|F| = n$ for every n . The last assertion is Corollary 13.5. $\square$

For $d = 2$ the groups are D^2 and K (the latter represented by kX_1X_2), and for $d = 3$ they are D^3, K^3, K : vertices, edges and the full face (12). The finite audit of 12 checks this formula for $D = \mathbb{Q} \subset K = \mathbb{Q}(\sqrt{2})$ (Section 19.1).

13.4 Coefficient lattices and their syzygies

The next results need only a ring of the following shape, which covers 12's cores, 12's ordered-cone rings and 14's rings.

Setting (L). K is a field, $D \subseteq K$ a unital subring, and $A = D \oplus \Pi$ is one of: (a) A_Λ with $\Pi = \Pi_\Lambda$; (b) T_Γ with $\Pi = \Pi_\Gamma^{\text{fin}}$; (c) the class ring $\mathcal{A}_{D,K}$ with $\Pi = \Pi_K$ (full surreal exponents, K a set field). In each case $B = K \oplus \Pi$ is a ring containing A , Π is an ideal of B and a K -vector space, $K \cap \Pi = 0$, all three lie in a field, and Π is not finitely generated in (a), (b) and not generated by a set in (c) (Lemma 13.3(iii), Proposition 13.6, Proposition 3.3(iii)). A *coefficient lattice* is $M = Dm_1 + \cdots + Dm_r \subseteq K$ with $m_1, \dots, m_r$ linearly independent over D (equivalently over $\text{Frac } D$); $r = \text{rank } M$. Put

$$L(M) = M + \Pi \subseteq B, \quad J_g(M) = X^g L(M) \subseteq A$$

for a nonzero monomial $X^g \in \Pi$ (a vector $g \neq \mathbf{0}$ in (a), $g \in \Gamma_{>0}$ in (b), $g > 0$ in (c)). For $(D, K) = (\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$, D is a principal ideal domain and every nonzero finitely generated D -submodule of K is free, so these are exactly 14's coefficient lattices.

Theorem 13.9 (Generators and the first syzygy [14; 12; 17] [merge]). *In Setting (L), let M have basis $m_1, \dots, m_r$.*

(i) $L(M) = \sum_j A m_j$, $\Pi L(M) = \Pi$ and $L(M)/\Pi L(M) \cong M$; the least number of generators of $L(M)$ is r , although $L(M)$ has rank one over the fraction field.

(ii) The sequence

$$0 \longrightarrow \Pi^{r-1} \xrightarrow{\sigma} A^r \xrightarrow{\pi} L(M) \longrightarrow 0, \quad \pi(a) = \sum_j a_j m_j,$$

$$\sigma(u_2, \dots, u_r) = \left(-\sum_{j \geq 2} \frac{m_j}{m_1} u_j, u_2, \dots, u_r \right),$$

is exact. For $r = 1$, $L(M) \cong A$; for $r \geq 2$, $L(M)$ is not finitely presented in (a), (b), and in (c) the kernel is not generated by any set.

(iii) (12's certificate.) Let $c \in K \setminus \text{Frac}(D)$ and $H = (X^g, cX^g)A$. Then $X^g A \cap cX^g A = X^g \Pi$ and $0 \rightarrow \Pi \xrightarrow{m_1 \mapsto (-cm_1, m_1)} A^2 \xrightarrow{(a,b) \mapsto X^g a + cX^g b} H \rightarrow 0$ is exact; in (a) and (b), A is not coherent.

Proof. (i) Since $0 \neq m_1 \in K$ and Π is a K -space, $m_1 \Pi = \Pi$; so $\sum_j A m_j \supseteq M + m_1 \Pi = L(M)$, and the reverse inclusion is clear. $\Pi L(M) \subseteq \Pi$ because Π is an ideal of B , and $\Pi L(M) \supseteq \Pi m_1 = \Pi$. As $M \cap \Pi \subseteq K \cap \Pi = 0$, $L(M)/\Pi \cong M \cong D^r$. Generators of $L(M)$ give generators of the D -module $L(M)/\Pi L(M)$, and a free module of rank r over a nonzero commutative ring needs at least r generators (reduce modulo a maximal ideal). (ii) If $\sum_j a_j m_j = 0$ with $a_j = d_j + u_j$, the components in K and Π give $\sum_j d_j m_j = 0$, so every $d_j = 0$ by independence, and $u_1 = -\sum_{j \geq 2} (m_j/m_1) u_j$; conversely these tuples lie in $\ker \pi$, and $(m_j/m_1) u_j \in \Pi$. σ is injective. For $r \geq 2$ the kernel projects onto Π , which is not finitely (set-) generated, and a finitely presented module has finitely generated kernels for every surjection from a finite free module. (iii) Apply (ii) to $M = D + Dc$ (the elements $1, c$ are independent over D because $c \notin \text{Frac } D$) and multiply by X^g , which is injective: $H = X^g L(M)$. If $X^g(d + u) = cX^g(e + v)$ then $d = ce$ forces $d = e = 0$, so $X^g A \cap cX^g A = X^g(\Pi \cap c\Pi) = X^g \Pi$, not finitely generated; in a coherent ring the intersection of two finitely generated ideals is finitely generated [31, Tag 05CU]. $\square$

Theorem 13.9(iii) is 12's noncoherence certificate for A_Λ and T_Γ with any unital $D \subseteq K$; (i) and (ii) are 14's theorem for $\mathbf{Oz}$, $\mathbf{Oz}[i]$ and A_0 with D a principal ideal domain. Stating both at once in Setting (L) is the merge's; the proofs are the sources'. All compatible relations between two generators with the same monomial and incompatible constant lines are forced into the infinitely divisible tail (12). No novelty is attached to noncoherence as such: rings of the form $D + M$ and their

coherence have a long history, including Dobbs–Papick [14]; 14 uses no valuation-ring hypothesis on B , and the useful information is the exact syzygy (14). Manuscript 17 proves (i), (ii) and the intersection formula of (iii) again, independently, for $\mathbf{Oz}$, $\mathbf{Oz}[i]$ and the models $R_{\Gamma_\kappa, < \kappa}^1$, and adds that in case (c) the module $L(M)$ with $r \geq 2$ has no presentation by sets (Theorem 13.29); it credits the kernel shape Π^{r-1} and the flat dimension one of the $D + M$ setting to Dobbs [13] and the intersection mechanism to Dobbs–Papick.

Proposition 13.10 (Antisymmetric torsion and nonflatness [14] [merge]). *In Setting (L), let M, N be coefficient lattices and $\mathbf{m}_{M,N} : M \otimes_D N \rightarrow K$ the multiplication map. On $\mathbf{E}(M, N) = \Pi \oplus (M \otimes_D N)$ the rule*

$$(d + h) \cdot (u, t) = (du + hu + h \mathbf{m}_{M,N}(t), dt) \quad (d \in D, h \in \Pi) \quad (13.4)$$

is an A -module structure, and $(m + u) \otimes (n + v) \mapsto (mv + un + uv, m \otimes n)$ is a well-defined A -linear map $\beta : L(M) \otimes_A L(N) \rightarrow \mathbf{E}(M, N)$. If $\text{rank } M \geq 2$ and m_1, m_2 are basis vectors, then $\tau = m_1 \otimes m_2 - m_2 \otimes m_1$ is a nonzero element of $L(M) \otimes_A L(M)$ with product 0. Consequently, in (a) and (b), $L(M)$ and every ideal $J_g(M)$ are not flat, hence neither projective nor invertible.

Proof. For $a = d + h$, $b = e + g'$ in A one computes that both $a \cdot (b \cdot (u, t))$ and $(ab) \cdot (u, t)$ have first coordinate $abu + (dg' + he + hg')\mathbf{m}_{M,N}(t)$ and second coordinate det ; additivity and $1 \cdot (u, t) = (u, t)$ are clear, and $h \mathbf{m}_{M,N}(t) \in \Pi$ because Π is a K -space. Write $x = m + u \in L(M)$, $y = n + v \in L(N)$ (uniquely, since $M \cap \Pi = 0$); the formula is $(xy - mn, m \otimes n)$, which is biadditive. For $a = d + h \in A$, the constant part of ax is dm , so the value at (ax, y) is $(axy - dm n, dm \otimes n)$, which equals $a \cdot (xy - mn, m \otimes n)$ by (13.4), and the value at (x, ay) is the same. Hence β exists. $\beta(\tau) = (0, \tau)$, and $\tau \neq 0$ in the free D -module $M \otimes_D M$ with basis $m_i \otimes m_j$. For flatness, $J = J_g(M) \cong L(M)$ is an ideal of A ($X^g M \subseteq \Pi$ and $X^g \Pi \subseteq \Pi$); tensoring the injection $J \hookrightarrow A$ with J gives the multiplication $J \otimes_A J \rightarrow J$, which kills the nonzero element corresponding to τ . Projective modules and invertible ideals are flat. $\square$

In (c) the tensor product is 14's explicit carrier (Section 13.6); there the same element shows that tensoring $J_g(M) \hookrightarrow A$ with $J_g(M)$ is not injective, which is 14's statement that $L(M)$ is not flat over $\mathbf{Oz}$. Over A_0 this is 14's corollary; allowing any unital $D \subseteq K$ and 12's rings is the merge's (the proof is 14's, and no principal-ideal hypothesis is used in it).

13.5 Flat and projective dimensions; the ordered cone

Theorem 13.11 (Lattice modules and cyclic quotients [12, 14] [merge]). *Let A be A_d ($d \geq 1$) or T_Γ , and put $\delta = d$, respectively $\delta = 1$. Let M be a coefficient lattice of rank $r \geq 2$ and $J = J_g(M)$. Then*

$$\text{fd}_A L(M) = \delta, \quad \text{fd}_A(A/J) = \delta + 1,$$

and for every A -module N and $j \geq 2$, $\text{Tor}_j^A(L(M), N) \cong \text{Tor}_j^A(D, N)^{r-1} \cong \text{Tor}_{j+1}^A(A/J, N)$. Over $A_{\mathbb{N}_{>0}}$ both dimensions are infinite. In particular, for $c \in K \setminus \text{Frac}(D)$ and $H = (X_1, cX_1)$, $\text{fd}_{A_d}(A_d/H) = d + 1$ and $\text{fd}_{T_\Gamma}(T_\Gamma/(X^g, cX^g)) = 2$, and over $A_0 = T_\mathbb{Q}$, $\text{fd } L(M) = 1$ and $\text{fd}(A_0/J_e(M)) = 2$. If moreover $c^2 \in D$, then $\text{Tor}_2^A(A/H, A/H) \cong D + cD \cong D \oplus D$.

Proof. By Theorem 13.9(ii) there is an exact sequence $0 \rightarrow \Pi^{r-1} \rightarrow A^r \rightarrow L(M) \rightarrow 0$, and $\text{fd } \Pi = \delta - 1$ (Corollary 13.8; for T_Γ , Π is flat by Theorem 13.13 below). Hence $\text{fd } L(M) \leq \delta$ and, from $0 \rightarrow J \rightarrow A \rightarrow A/J \rightarrow 0$ with $J \cong L(M)$, $\text{fd}(A/J) \leq \delta + 1$. The long exact sequences give, for $j \geq 2$, $\text{Tor}_j(L(M), N) \cong \text{Tor}_{j-1}(\Pi, N)^{r-1} \cong \text{Tor}_j(D, N)^{r-1}$ and $\text{Tor}_{j+1}(A/J, N) \cong \text{Tor}_j(J, N)$.

If $\delta = d \geq 2$, $\text{Tor}_d(D, V_{[d]}) \cong K \neq 0$ ([Theorem 13.7](#)) gives both lower bounds. If $\delta = 1$, $L(M)$ is not flat ([Proposition 13.10](#)), so $\text{fd } L(M) = 1$ and $\text{Tor}_2(A/J, N) \cong \text{Tor}_1(J, N) \neq 0$ for some N . In infinite rank, $\text{Tor}_n(D, V_F) \neq 0$ for $|F| = n$. The special cases are $M = D + Dc$ with $J = H$, and M of rank at least two over $T_{\mathbb{Q}}$. For the last assertion (12), with $U = D + cD$ one has $H = UX^g \oplus X^g\Pi \subseteq \Pi$, $H\Pi = X^g\Pi$ (as $\Pi^2 = \Pi$ and $c\Pi = \Pi$) and $cH \subseteq H$ (as $c^2 \in D$). The two sequences give $\text{Tor}_2(A/H, A/H) \cong \text{Tor}_1(H, A/H) \cong \ker(\Pi/X^g\Pi \rightarrow (A/H)^2, m \mapsto (-cm, m))$, and the kernel consists of the classes of $m \in H$, that is $H/X^g\Pi \cong U$. $\square$

Correction to 12 and the relation to 14. 12 states $\text{fd}(A_d/H) = d+1$ and $\text{fd}(T_{\Gamma}/H) = 2$ under the extra hypothesis $c^2 \in D$, which it uses only to compute $\text{Tor}_2(A/H, A/H)$; for $d \geq 2$ its own shift argument never uses it, and for $d = 1$ and T_{Γ} the lower bound 2 follows from 14's antisymmetric torsion tensor. The hypothesis is needed only for the formula $\text{Tor}_2(A/H, A/H) \cong D \oplus D$. 14's flat-dimension proposition for A_0 is the case $A = T_{\mathbb{Q}} = A_1$ of the theorem. Since a nonzero Tor in degree n forces weak global dimension at least n : if $K \neq \text{Frac } D$ (for instance $(D, K) = (\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$ with $c = \sqrt{2}$), then $\text{w.gl.dim } A_d \geq d+1$, $\text{w.gl.dim } T_{\Gamma} \geq 2$, and the countable-rank core has infinite weak global dimension (12, 14). Neither source claims the exact weak global dimension of any of these rings.

Proposition 13.12 (Projective dimensions [12] [merge]).

- (i) (Telescope, 12.) Let A be a domain and $I = \bigcup_{n \geq 0} x_n A$ for nonzero x_n with $x_n = x_{n+1}u_n$, $u_n \in A$. With $F = \bigoplus_{n \geq 0} A e_n$, the sequence $0 \rightarrow F \xrightarrow{\Delta} F \xrightarrow{\pi} I \rightarrow 0$, $\Delta(e_n a) = e_n a - e_{n+1}u_n a$, $\pi(e_n a) = x_n a$, is exact; so $\text{pd}_A I \leq 1$. This applies to every I_S ($x_n = p_S^{2^{-n}}$) and to $\Pi_{\Gamma}^{\text{fin}}$ ($x_n = X^{2^{-n}g}$, $g \in \Gamma_{>0}$).
- (ii) [merge] $\text{pd}_{A_{\Lambda}} I_S = 1$ and $\text{pd}_{T_{\Gamma}} \Pi_{\Gamma}^{\text{fin}} = 1$.
- (iii) $d \leq \text{pd}_{A_d} D \leq d+1$ (12), and [merge] $\text{pd}_{A_1} D = \text{pd}_{T_{\Gamma}} D = 2$. (The upper bound is attained for every d : [Corollary 13.47](#), 21.)
- (iv) [merge] Over T_{Γ} (in particular over A_0), for a lattice of rank at least two, $\text{pd } L(M) = 2$ and $\text{pd}(T_{\Gamma}/J_g(M)) = 3$. Over A_d , $d \leq \text{pd } L(M) \leq d+1$ and $d+1 \leq \text{pd}(A_d/J_g(M)) \leq d+2$ (12, for $M = D + Dc$). (The upper bounds are attained: [Theorem 13.48](#), 21.)
- (v) [merge] Let (A, I, D) satisfy (H1)–(H3) of [Section 12](#), with A consisting of series of nonnegative support (as do all rings of [Section 11](#) and T_{Γ}). If I is countably generated, equivalently if $\Gamma_{>0}$ has a countable coinital subset, then $\text{pd}_A I = 1$ and $\text{pd}_A D = 2$.

Proof. (i) $\pi\Delta = 0$ since $x_{n+1}u_n = x_n$; π is onto. Δ is injective: at the least index n_0 of a nonzero vector the coefficient is unchanged. If $\pi(v) = 0$, use $e_n a \equiv e_{n+1}u_n a$ modulo $\text{im } \Delta$ to move v to a single index N : $v \equiv e_N a$ with $x_N a = \pi(v) = 0$, so $a = 0$. For T_{Γ} , every tail has a positive exponent below its finitely many exponents, hence below some $2^{-n}g$. (ii) These ideals are nonzero and not finitely generated ([Lemma 13.3](#)(iii) and its analogue for T_{Γ}), so they are not projective by [Lemma 12.1](#). (iii) Every C_p of [Theorem 13.4](#) has projective dimension at most 1 by (i); splicing (or dimension shifting along the resolution) gives $\text{pd } D \leq d+1$, and $\text{pd} \geq \text{fd} = d$. For T_{Γ} (and $A_1 \cong T_{\mathbb{Q}}$), $0 \rightarrow \Pi \rightarrow T_{\Gamma} \rightarrow D \rightarrow 0$ gives $\text{pd } D \leq \text{pd } \Pi + 1 = 2$; if $\text{pd } D \leq 1$, Π would be projective (Schanuel), against (ii). (iv) $0 \rightarrow \Pi^{r-1} \rightarrow T_{\Gamma}^r \rightarrow L(M) \rightarrow 0$ gives $\text{pd } L(M) \leq 2$; if $\text{pd } L(M) \leq 1$, the kernel Π^{r-1} , hence its summand Π , would be projective. So $\text{pd } L(M) = 2$, and since $J = J_g(M) \cong L(M)$ is not projective, $\text{pd}(T_{\Gamma}/J) = \text{pd } J + 1 = 3$. Over A_d the upper bounds come from $\text{pd } \Pi_d \leq d$ (the truncated resolution, by (i)), and the lower bounds from [Theorem 13.11](#), since $\text{pd} \geq \text{fd}$. (v)

If I is generated by countably many $f_n = X^{g_n} b_n$ ((H2)), then for $h \in \Gamma_{>0}$ the element X^h lies in $\sum_{n \leq N} f_n A \subseteq X^g A$ with $g = \min_{n \leq N} g_n$, and $X^{h-g} \in A$ forces $g \leq h$ because A has no negative exponents: the g_n are coinital. Conversely, for a coinital decreasing sequence h_n , (H2) gives $I = \bigcup_n X^{h_n} A$ with $X^{h_n} = X^{h_{n+1}} X^{h_n - h_{n+1}}$, a countable chain as in (i). So $\text{pd } I \leq 1$; I is not projective by (H3) and Lemma 12.1; hence $\text{pd } I = 1$ and, as in (iii), $\text{pd } D = 2$. $\square$

Items (ii)–(v) sharpen 12’s bounds $1 \leq \text{pd}_{A_1} D \leq 2$ and are consistent with 06’s $\text{pd} \geq 2$ (Theorem 12.2(v)). For $d \geq 2$ these items left the value $\text{pd}_{A_d} D \in \{d, d+1\}$ undetermined; 21 shows that it is $d+1$ for every d , and that the lattice bounds of (iv) are attained (Corollary 13.47 and Theorem 13.48). Hypothesis (H2) in (v) cannot be dropped: A_d has a countably generated tail and satisfies (H1) and (H3), but not (H2), and $\text{pd}_{A_d} D = d+1$ (Remark 13.49, a merge remark). Among the five constructions of Section 11, (v) applies exactly to A_κ^{08} with $\text{cf } \kappa = \aleph_0$: the tail of each construction needs as many generators as the length of its coinital chain (Remark 10.11), namely $\text{cf } \kappa$ for 03 and 08, κ for 06 and 11, $\aleph_1$ for 09, and only 08 allows countable cofinality. Question 18.3 is re-scoped accordingly.

Theorem 13.13 (Ordered-cone cores [12]). *($T_\Gamma, \Pi_\Gamma^{\text{fin}}, D$) satisfies (H1)–(H3) of Section 12. Consequently $\Pi_\Gamma^{\text{fin}} = \bigcup_{\delta \in \Gamma_{>0}} X^\delta T_\Gamma$ is flat and idempotent but not projective, $\text{Tor}_{>0}^{T_\Gamma}(D, N) = 0$ for every D -module N , $\text{fd}_{T_\Gamma} D = 1$ with $\text{Tor}_1(D, T_\Gamma/X^\gamma T_\Gamma) \cong D$, and the other items of Theorem 12.2 hold; $T_\Gamma \rightarrow D$ is formally étale and not flat, $\text{pd}_{T_\Gamma} D = 2$, and T_Γ is not coherent.*

Proof. (H1) is Lemma 13.1(i). For (H2), a nonzero tail f has finitely many positive exponents; with g half the least of them, $f = X^g(X^{-g}f)$ and $X^{-g}f$ is again a tail, its coefficients being arbitrary in K . (H3) holds because a finite family of tails has a least exponent among its monomials and cannot generate a monomial of smaller exponent. Apply Theorem 12.2, Proposition 13.6, Proposition 13.12(iii) and Theorem 13.9(iii). $\square$

Theorem 13.14 (Same exponent group, different dimension [12]). *Let $(\gamma_i)_{i \in \Lambda}$ be positive reals linearly independent over $\mathbb{Q}$, $\Gamma = \sum_i \mathbb{Q}\gamma_i$, and $A_\Lambda \hookrightarrow T_\Gamma$ the inclusion of Lemma 13.1(iii). Then $\Pi_\Lambda T_\Gamma = \Pi_\Gamma^{\text{fin}}$ and $T_\Gamma \otimes_{A_\Lambda} D \cong D$. If $|\Lambda| \geq 2$ the extension is not flat: $(X_1 A_\Lambda \cap X_2 A_\Lambda) T_\Gamma \neq X_1 T_\Gamma \cap X_2 T_\Gamma$. The flat dimension of D drops from d to 1 in rank d , and from ∞ to 1 in countable rank.*

Proof. For $\gamma \in \Gamma_{>0}$ choose a positive rational q with $q\gamma_1 < \gamma$; then $kX^\gamma = X_1^q \cdot kX^{\gamma - q\gamma_1} \in \Pi_\Lambda T_\Gamma$, so $\Pi_\Lambda T_\Gamma = \Pi_\Gamma^{\text{fin}}$ and $T_\Gamma \otimes_{A_\Lambda} A_\Lambda / \Pi_\Lambda \cong T_\Gamma / \Pi_\Gamma^{\text{fin}} = D$. Relabel so that $\gamma_1 < \gamma_2$. Every monomial of an element of $X_1 A_\Lambda \cap X_2 A_\Lambda$ has first and second coordinate at least 1, hence real exponent at least $\gamma_1 + \gamma_2$, and multiplying by T_Γ only adds nonnegative exponents; but $X^{\gamma_2} = X_1 X^{\gamma_2 - \gamma_1} \in X_1 T_\Gamma \cap X_2 T_\Gamma$. A flat extension preserves finite intersections of ideals: tensor the exact sequence $0 \rightarrow I \cap I' \rightarrow A \rightarrow A/I \oplus A/I'$ with it. The dimensions are Corollary 13.8 and Theorem 13.13. $\square$

Example 13.15 (Two directions inside **Oz** [12]). With $\gamma_1 = 1$, $\gamma_2 = \sqrt{2}$ and $(D, K) = (\mathbb{Z}, \mathbb{R})$, the core $A_2 = \mathbb{Z} + \bigoplus_{a,b \in \mathbb{Q}_{\geq 0}, a+b>0} \mathbb{R} \omega^{a+b\sqrt{2}}$ lies in the ring $T_{\mathbb{Q}+\mathbb{Q}\sqrt{2}} \subseteq \mathbf{Oz}$, which also contains $\omega^{\sqrt{2}-1} \notin A_2$; the identity $\omega^{\sqrt{2}} = \omega \cdot \omega^{\sqrt{2}-1}$ is the extra intersection element, and the two arithmetic quotients have flat dimensions 2 and 1. Replacing $(\mathbb{Z}, \mathbb{R})$ by $(\mathbb{Z}[i], \mathbb{C})$ gives the same example in $\mathbf{Oz}[i]$.

Remark 13.16 (Flat dimension one is a property of the cone [merge]). The report now contains two kinds of statements about the flat dimension of the arithmetic quotient, and they concern different rings.

- $\text{fd}_A D = 1$ holds whenever (H1)–(H3) hold: the tail is a directed union of principal monomial ideals, because the positive exponents form the positive cone of a totally ordered group and every family of tails of the relevant size (finite for finite supports, of fewer than κ or countably many elements in the constructions, a set for the class ring) has a common monomial divisor in the ring. This covers the five constructions of Section 11 (Theorem 12.2), including 06’s ring A_κ^{06} ; the full omnific ring in the two-universe reading, where 13 proves that the tail is flat, idempotent and not generated by a family from the smaller universe, that $\text{Tor}_{>0}(D, D) = 0$ and that D is not flat, so $\text{fd } D = 1$; 14’s A_0 ; and 12’s T_Γ (Theorem 13.13).
- For the coordinatewise cone $\mathbb{Q}_{\geq 0}^d$ the flat dimension is d (Corollary 13.8): under the embedding, with $\gamma_1 < \gamma_2$, the quotient $\omega^{\epsilon(\gamma_2 - \gamma_1)}$ of X_2^ϵ by X_1^ϵ lies in $\mathbf{Oz}$ but not in A_d , so the tail of A_d is not a directed union of principal ideals; it is a sum of d flat ideals that is not flat for $d \geq 2$.

So “flat dimension one” must never be asserted for an arbitrary subring of $\mathbf{Oz}$: $A_d \subseteq T_\Gamma \subseteq \mathbf{Oz}$, with all exponents in one Archimedean class, have flat dimensions d and 1, and $A_d \rightarrow T_\Gamma$ is not flat. Being a subring of a ring with small quotient flat dimension imposes no bound on the subring, and conversely the cores say nothing about the homological dimensions of the full class ring $\mathbf{Oz}$, which none of the sources computes (12, 14). The projective dimension of D behaves differently again: under (H1)–(H3) it is 2 when the tail is countably generated, in particular for T_Γ , and is otherwise known only to be at least 2; for A_d it is $d + 1$, one more than the flat dimension (Proposition 13.12, with 21’s Corollary 13.47; before 21 it was known only to be d or $d + 1$).

13.6 The tensor normal form

For the rest of this section (14) fix $(D, k) = (\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$, and let (A, Π, B) be either the class rings $(\mathcal{A}_{D,k}, \Pi_k, \mathcal{A}_{k,k})$, that is $A = \mathbf{Oz}$ or $\mathbf{Oz}[i]$, or the set-sized model $(T_\mathbb{Q}, \Pi_\mathbb{Q}^{\text{fin}}, k + \Pi_\mathbb{Q}^{\text{fin}})$ with $T_\mathbb{Q} = A_0$. Coefficient lattices are those of Setting (L). Every result below holds in both cases (14); its proofs use only that D is a principal ideal domain in k , that $A = D \oplus \Pi$ with Π a k -linear ideal of B , and that every $u \in \Pi$ is a product $u = ca$ with $c, a \in \Pi$ (Proposition 3.3(ii); in $T_\mathbb{Q}$ split off a smaller monomial). This interface is 14’s proposal for a formalization (Section 19.2). For the class rings a tensor product means an explicit carrier with the balanced universal property proved by finite identities (for two lattices, $\Pi \times (M \otimes_D N)$); no quotient of a proper class by a class relation and no class colimit is used (14).

Lemma 13.17 (Absorption [14]). *For every lattice N and $0 \neq n_0 \in N$, multiplication is an isomorphism $\Pi \otimes_A L(N) \cong \Pi$ with inverse $u \mapsto (u/n_0) \otimes n_0$; indeed $u \otimes v = (uv/n_0) \otimes n_0$ for $u \in \Pi$, $v \in L(N)$.*

Proof. Write $u = ca$ with $c, a \in \Pi$. Since $v/n_0 \in B$, $b = a(v/n_0) \in \Pi \subseteq A$, and $u \otimes v = c \otimes av = c \otimes bn_0 = cb \otimes n_0 = (uv/n_0) \otimes n_0$. So every finite tensor sum is the image of its product, multiplication is onto ($u = (u/n_0)n_0$), and the two maps are mutually inverse. For a balanced bilinear map b' into a class module, $u \mapsto b'(u/n_0, n_0)$ is the required factorization, unique by the same identity. $\square$

Theorem 13.18 (Exact tensor normal form [14; 17]). *The map β of Proposition 13.10 is an A -module isomorphism $L(M) \otimes_A L(N) \cong E(M, N)$. Multiplication corresponds to $(u, t) \mapsto u + \mathfrak{m}_{M,N}(t)$, and there is an exact sequence*

$$0 \longrightarrow \mathcal{T}(M, N) \longrightarrow L(M) \otimes_A L(N) \longrightarrow L(MN) \longrightarrow 0, \quad \mathcal{T}(M, N) = \ker(M \otimes_D N \rightarrow MN),$$

with $\mathcal{T}(M, N)$ embedded as $(0, \mathcal{T}(M, N))$. This is exactly the A -torsion submodule; if it is nonzero its annihilator is exactly Π , and $\text{rank}_D \mathcal{T}(M, N) = \text{rank } M \cdot \text{rank } N - \text{rank}(MN)$.

Proof. Fix $0 \neq n_0 \in N$, let $j(w) = (w/n_0) \otimes n_0$ on Π and let s be induced by the constant inclusions on $M \otimes_D N$. Then $(u, t) \mapsto j(u) + s(t)$ is inverse to β : in an expanded pure tensor the terms with an infinite first factor reduce by [Lemma 13.17](#), a term $m \otimes v$ with $v = ca \in \Pi$ equals $cm \otimes a$, and all mixed terms become $j(mv + un + uv)$, while the constant term is $s(m \otimes n)$. The same finite identities give the universal property for balanced maps into class modules, so this carrier is the tensor product in the class case. The image of multiplication is $MN + \Pi = L(MN)$, and since $\Pi \cap k = 0$ its kernel is $(0, \mathcal{T}(M, N))$, killed by Π by [\(13.4\)](#); the quotient lies in a field and is torsion free, so the kernel is all the torsion. It is a submodule of a finite free D -module, hence free, and a nonzero $d \in D$ kills none of its nonzero elements. Ranks: $M \otimes_D N \rightarrow MN$ is onto. $\square$

For $M = D + D\alpha$ with $\alpha \notin \text{Frac } D$, $\tau = 1 \otimes \alpha - \alpha \otimes 1$ is a nonzero tensor with $h\tau = 0$ for all $h \in \Pi$; if $\alpha^2 = c \in D$, then τ and $\alpha \otimes \alpha - c(1 \otimes 1)$ form a D -basis of the multiplication kernel (14). Manuscript 17 proves the exact sequence and the torsion statement independently over its set-sized models, as a consequence of its Tor identification ([Theorem 13.34](#)). Here “torsion” means A -torsion: $\mathcal{T}(M, N)$ is a free abelian group (14).

Proposition 13.19 (Several factors [14]). *For lattices $M_1, \dots, M_n$, $\bigotimes_{j,A} L(M_j) \cong \Pi \oplus \bigotimes_{j,D} M_j$ with the action (13.4) for the n -fold product map; the torsion is its kernel, killed by Π . For equal factors, permutations act trivially on the first coordinate and by permuting tensor factors on the second.*

Proof. Expand a pure tensor: the all-constant term keeps its D -tensor coordinate, and every other term contains a factor from Π and reduces, by repeated absorption (moving a tail scalar to an adjacent factor), to its product in one copy of Π . So the multilinear map has coordinates $(\prod_j (m_j + u_j) - \prod_j m_j, \bigotimes_j m_j)$, with inverse as in [Theorem 13.18](#). $\square$

Theorem 13.20 (Exterior and symmetric powers [14]). *Let $r = \text{rank } M$.*

- (i) *For $j \geq 2$, $\bigwedge_A^j L(M) \cong \bigwedge_D^j M$ with A acting through ct : a free D -module of rank $\binom{r}{j}$ with annihilator Π when nonzero.*
- (ii) *For $n \geq 1$, $\text{Sym}_A^n L(M) \cong \Pi \oplus \text{Sym}_D^n M$ with the action (13.4) for the product map $\mathbf{m}_n : \text{Sym}_D^n M \rightarrow k$; the product map has image $L(M^n)$ (M^n the D -span of n -fold products) and torsion kernel $\ker \mathbf{m}_n$, of rank $\binom{r+n-1}{n} - \text{rank}(M^n)$, annihilated by Π .*

Proof. (i) For $u \in \Pi$ and $0 \neq m \in M$, $u/m \in A$ and $u \wedge m = (u/m)m \wedge m = 0$; for $u, v \in \Pi$ both are multiples of one m_0 , so $u \wedge v = 0$. Hence every wedge with a tail factor vanishes, $h(m_1 \wedge \dots \wedge m_j) = 0$ for $h \in \Pi$, the constant inclusion induces a surjection $\bigwedge_D^j M \rightarrow \bigwedge_A^j L(M)$, and taking constants in all factors is an alternating A -multilinear left inverse. (ii) By [Proposition 13.19](#) a permutation difference has the form $(0, t - \sigma t)$, with product zero, so multiplying it by an element of Π gives zero and imposes no relation on the first coordinate; the quotient is $\Pi \oplus \text{Sym}_D^n M$, and the kernel argument of [Theorem 13.18](#) applies. $\square$

A rank-one ideal of $\mathbf{Oz}$ can thus have a nonzero r th exterior power for every $r \geq 2$; these powers vanish after passing to the fraction field, in agreement with the field-valued rank (14).

13.7 Rees equations

For $e > 0$ the Rees algebra of $L(M)$ is $\mathcal{R}(L(M)) = A \oplus \bigoplus_{n \geq 1} L(M)^n T^n \subseteq F[T]$ (F the fraction field); degreewise multiplication by X^{ne} identifies it, and the symmetric algebra, with those of the

ideal $J_e(M)$ [34, Tag 052P]. An ideal is *of linear type* if $\text{Sym} \rightarrow \mathcal{R}$ is an isomorphism, and the *first nonlinear degree* is the least degree of a nonzero element of the kernel after the linear relations are imposed (14).

Theorem 13.21 (All nonlinear Rees equations are arithmetic [14]). *Let $m_1, \dots, m_r$ be a basis of M and $\mathfrak{q}_M = \ker(D[X_1, \dots, X_r] \rightarrow k[T], X_i \mapsto m_i T)$. The constant-coefficient map $D[X_1, \dots, X_r] \rightarrow \text{Sym}_A L(M)$ is injective, and under it $\ker(\text{Sym}_A L(M) \rightarrow \mathcal{R}(L(M))) = \mathfrak{q}_M$; this is all the A -torsion of the symmetric algebra, it is killed by Π , and it is generated by any set of homogeneous generators of $\mathfrak{q}_M$, finitely many of which suffice. With $\mathfrak{l}_M = (hX_j - \frac{m_j}{m_1} hX_1 : h \in \Pi, 2 \leq j \leq r) \subseteq A[X_1, \dots, X_r]$,*

$$\text{Sym}_A L(M) \cong A[\mathbf{X}]/\mathfrak{l}_M, \quad \mathcal{R}(L(M)) \cong A[\mathbf{X}]/(\mathfrak{l}_M + \mathfrak{q}_M A[\mathbf{X}]).$$

Proof. The symmetric algebra is presented by the linear relations of the generator map [33, Tag 00DM], which are those of Theorem 13.9(ii). In degree $n \geq 1$, Theorem 13.20(ii) identifies the constant polynomials with $\text{Sym}_D^n M$ and the kernel with $\ker \mathfrak{m}_n$, the degree- n part of $\mathfrak{q}_M$; in degree 0 the map is the identity of A . The quotient embeds in the domain $F[T]$, so the kernel is all the torsion. A polynomial in $A[\mathbf{X}]$ acts on the kernel through its coefficientwise constant term, so generators over $D[\mathbf{X}]$ generate the kernel, and $D[\mathbf{X}]$ is Noetherian. $\square$

The nonlinear kernel is finitely generated while, for $r \geq 2$, the linear relation module is not finitely (set-) generated (Theorem 13.9); the two statements are compatible (14).

Theorem 13.22 (The algebraic degree is the first nonlinear degree [14]). *Let $\alpha \in k \setminus \text{Frac}(D)$, $M = D + D\alpha$ and $J = (X^e, \alpha X^e) \subseteq A$.*

- (i) *If α is transcendental over $\text{Frac}(D)$, then J is of linear type, although it is not flat and not finitely presented.*
- (ii) *If α is algebraic of degree d (necessarily $d \geq 2$), let $f \in D[Z]$ be primitive irreducible with $f(\alpha) = 0$ and $F_f(X, Y) = X^d f(Y/X)$. Then $\mathcal{R}(J) \cong A[X, Y]/(hY - \alpha hX \ (h \in \Pi), F_f(X, Y))$; the kernel of $\text{Sym}_A J \rightarrow \mathcal{R}(J)$ is generated by F_f , and the first nonlinear degree is exactly d .*

Proof. A homogeneous G of degree n lies in $\mathfrak{q}_M$ iff $G(1, \alpha) = 0$. In the transcendental case $G = 0$. In the algebraic case the minimal polynomial of α divides $g(Z) = G(1, Z)$ over $\text{Frac } D$, and Gauss's lemma for the unique factorization domain D and the primitivity of f give $g = fh$ with $h \in D[Z]$, $\deg h \leq n - d$; homogenizing, $G = F_f \cdot X^{n-d} h(Y/X)$. Conversely $F_f(1, \alpha) = 0$. Apply Theorem 13.21; nonflatness and failure of finite presentation are Proposition 13.10 and Theorem 13.9. $\square$

Corollary 13.23 (A family of every degree [14; 17]). *For $d \geq 2$ let $J_d = (\omega, 2^{1/d}\omega) \subseteq \mathbf{O}_Z$. Then $\mathcal{R}_{\mathbf{O}_Z}(J_d) \cong \mathbf{O}_Z[X, Y]/(hY - 2^{1/d}hX \ (h \in \Pi), Y^d - 2X^d)$, there is no nonlinear equation in degrees $2, \dots, d-1$, and for $n \geq 1$*

$$\begin{aligned} \mu_{\mathbf{O}_Z}(J_d^n) &= \min(n+1, d), & \text{rank}_{\mathbb{Z}} \ker(\text{Sym}_{\mathbf{O}_Z}^n J_d \rightarrow J_d^n) &= \max(0, n-d+1), \\ \text{rank}_{\mathbb{Z}} \ker(J_d^{\otimes n} \rightarrow J_d^n) &= 2^n - \min(n+1, d), \end{aligned}$$

where μ is the least number of generators. All these kernels are killed by Π .

Proof. $Z^d - 2$ is irreducible by Eisenstein at 2, so $1, \alpha, \dots, \alpha^{d-1}$ are independent and higher powers reduce by $\alpha^d = 2$; the lattice spanned by $1, \alpha, \dots, \alpha^n$ has rank $\min(n+1, d)$, and J_d^n is the monomial scaling of its lattice module. Apply Theorem 13.9(i), Theorem 13.20(ii) and Proposition 13.19. $\square$

For $d = 5$ the two-generated ideal $(\omega, 2^{1/5}\omega)$ has powers needing $2, 3, 4, 5, 5, \dots$ generators, a symmetric kernel of ranks $0, 0, 0, 0, 1, 2, \dots$, and a nonzero antisymmetric tensor already in its tensor square: symmetric and tensor obstructions occur at different stages (14). Manuscript 17 proves the generator count $\min(n+1, d)$ independently, from the product law $J_g(M)J_h(N) = J_{g+h}(MN)$, for every α of degree d over $\text{Frac}(D)$, and adds that it is $n+1$ for every n when α is transcendental; the lattices M^n need not stabilize when their ranks do, and the count is not a reconstruction theorem (17).

13.8 Homomorphisms, ideal classes and duals

For lattices put $(N : M) = \{c \in k : cM \subseteq N\}$ and $\mathfrak{o}(M) = (M : M)$.

Theorem 13.24 (Scalar homomorphisms and isomorphism types [14; 17]). *Identifying a scalar with multiplication by it, $\text{Hom}_A(L(M), L(N)) = \Pi + (N : M)$, composition is multiplication, and $L(M) \cong L(N)$ if and only if $N = cM$ for some $c \in k^\times$. Hence $\text{End}_A L(M) = \Pi + \mathfrak{o}(M)$ and $\text{Aut}_A L(M) = \mathfrak{o}(M)^\times$. The ring $\mathfrak{o}(M)$ is finite free over D and integral over D ; in the real case $\mathbb{Q}\mathfrak{o}(M) \subseteq \mathbb{R}$ is a number field of degree at most $\text{rank } M$ (not necessarily totally real).*

Proof. An A -linear map φ between nonzero A -submodules of the fraction field is multiplication by a scalar: for nonzero a, b in a common ideal (after a monomial denominator), $a\varphi(b) = b\varphi(a)$. If $qL(M) \subseteq L(N)$ and $0 \neq m_0 \in M$, then $q = (qm_0)/m_0 \in B$; write $q = c + h$ with $c \in k$, $h \in \Pi$; the inclusion is equivalent to $cM \subseteq N$. An isomorphism has q and q^{-1} in B , so $q \in B^\times = k^\times$ (Lemma 2.4), and nonzero constants preserve Π . Multiplication by m_0 embeds $\mathfrak{o}(M)$ in M ; Cayley–Hamilton for the matrix of multiplication by c on a basis of M gives integrality; in the real case the rational span is a finite-dimensional domain, hence a field, of dimension at most $\text{rank } M$. $\square$

Manuscript 17 proves the Hom, isomorphism and automorphism statements independently for $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ (for the ideals $J_\gamma(M)$, with $\text{Hom}_A(J_\gamma(M), J_\eta(N)) = \omega^{\eta-\gamma}((N : M) + \Pi)$), and sharpens the degree bound: the field spanned by $\mathfrak{o}(M)$ has degree dividing $\text{rank } M$ (Theorem 13.39).

Let $K_0 \subset \mathbb{R}$ be a number field and $\mathfrak{o} \subset K_0$ an order. An invertible fractional $\mathfrak{o}$ -ideal is a lattice, and for these modules put $U \odot V = (U \otimes_A V)/(\text{elements killed by a nonzero element of } A)$; by Theorem 13.18, $L(M) \odot L(N) \cong L(MN)$.

Theorem 13.25 (Ideal classes survive torsion-free multiplication [14]). *$P \mapsto L(P)$ distinguishes exactly the ideal classes of invertible fractional $\mathfrak{o}$ -ideals, and under $\odot$ these isomorphism types form a group isomorphic to $\text{Pic}(\mathfrak{o})$ with identity $[L(\mathfrak{o})]$. If $[K_0 : \mathbb{Q}] \geq 2$ this is not a subgroup of $\text{Pic}(A)$ and its identity is not $[A]$.*

Proof. If $L(P) \cong L(Q)$ then $Q = cP$ with $c \in \mathbb{R}^\times$ (Theorem 13.24), and $c = (cp)/p \in K_0^\times$ for $0 \neq p \in P$; the converse is clear. $\odot$ corresponds to ideal multiplication, $\mathfrak{o}P = P$ and $PP^{-1} = \mathfrak{o}$. For $[K_0 : \mathbb{Q}] \geq 2$, $L(P)$ needs $[K_0 : \mathbb{Q}]$ generators and is not flat (Proposition 13.10), so it is not invertible, and $L(\mathfrak{o})$ needs two or more generators while $A = L(\mathbb{Z})$ needs one. $\square$

The group has its own idempotent identity inside a larger multiplicative semigroup; it is not a unital homomorphism into the Picard group of A . In the class setting the isomorphism is read as an explicit set-indexed family of representatives with its equivalence criterion (14).

Example 13.26 (A class of order two [14]). In $K_0 = \mathbb{Q}(\sqrt{10})$ with $\mathfrak{o} = \mathbb{Z} + \mathbb{Z}\sqrt{10}$ let $P = 2\mathbb{Z} + \mathbb{Z}\sqrt{10}$. It is an $\mathfrak{o}$ -ideal, $P^2 = 4\mathbb{Z} + 2\sqrt{10}\mathbb{Z} + 10\mathbb{Z} = 2\mathfrak{o}$, so P is invertible. It is not principal: $P = c\mathfrak{o}$ with

$c = a + b\sqrt{10} \in P$ would give $|a^2 - 10b^2| = [\mathfrak{o} : P] = 2$, impossible modulo 5. Hence $L(P) \not\cong L(\mathfrak{o})$ and $L(P) \odot L(P) \cong L(\mathfrak{o})$; both have rank one and need two generators, and the tensor square of $L(P)$ has torsion of rank $4 - 2 = 2$. As ideals, $\omega L(P) = (2\omega, \sqrt{10}\omega)$ and $\omega L(\mathfrak{o}) = (\omega, \sqrt{10}\omega)$.

Theorem 13.27 (Uniform duals [14]). *($A : \Pi) = B$ and $(A : B) = \Pi$ (in both cases of this subsection). For a lattice M of rank at least two, $L(M)^* = \text{Hom}_A(L(M), A) = \Pi$, $L(M)^{**} = B$, the canonical bidual map is the inclusion $L(M) \hookrightarrow B$, and $B/L(M) \cong k/M$ with A acting through ct . Moreover $\Pi^* = B$, $B^* = \Pi$, $\Pi^{**} = \Pi$, $B^{**} = B$, and $J_e(M)^* = X^{-e}\Pi$, $J_e(M)^{**} = X^e B$.*

Proof. If $q\Pi \subseteq A$ and q had a negative exponent γ , then for $0 < \delta < -\gamma$ the coefficient of $X^{\gamma+\delta}$ in qX^δ would be the same nonzero coefficient; so $q \in B$ (in $T_{\mathbb{Q}}$, first $qX^1 \in T_{\mathbb{Q}}$ shows that q has finite rational support). If $qB \subseteq A$ then $q = d + u \in A$ and $dk \subseteq D$ forces $d = 0$. By Theorem 13.24, $L(M)^* = \Pi + (D : M)$, and $(D : M) = 0$ because a nonzero c would embed M in D . Evaluation at x sends $u \in \Pi$ to ux , so the bidual scalar of x is x . Scaling by X^e reverses in the first dual and restores in the second. $\square$

So the two nonisomorphic modules of Example 13.26 have the same dual and reflexive hull: their difference lies below the hull and is seen by coefficient arithmetic, not by rank or double dualization (14).

Theorem 13.28 (Common-scale constant matrices [14; 17]). *Let $C \in k^{m \times n}$, $e > 0$, $W = \ker_k C$, $q = \dim_k W$, $W_D = W \cap D^n$ and $\ell = \text{rank}_D W_D$. Then $\ker_A(X^e C) \cong A^\ell \oplus \Pi^{q-\ell}$. The image of $X^e C : A^n \rightarrow A^m$ is finitely presented if and only if $\ell = q$, that is, if and only if W is defined over $\text{Frac}(D)$, and it is then free of rank $n - q$.*

Proof. For $x = d + u$, $Cx = 0$ splits into $Cd = 0$ and $Cu = 0$ (constants against tails). W_D is saturated in D^n , hence a direct summand; a D -basis of it is k -independent and extends to a k -basis of W ; in these coordinates the kernel is $A^\ell \oplus \Pi^{q-\ell}$ (Gaussian elimination with constants preserves Π). If $q > \ell$ the kernel projects onto Π and is not finitely (set-) generated. If $q = \ell$, a constant change of basis over D makes the first q columns zero and the others independent over the fraction field. $\square$

The matrix theorem concerns constant matrices at one common monomial scale; several support levels need more analysis (14, 17). Manuscript 17 proves it independently for $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ and adds that $\ell = \dim_{\text{Frac } D}(W \cap \text{Frac}(D)^n)$ (a nonzero vector of $W \cap \text{Frac}(D)^n$ has a common denominator in D), so that the kernel is set-generated exactly when W is spanned by vectors over $\text{Frac}(D)$; its computation of q and ℓ and the relation counts over the models are in Section 13.9. Its companion, 14's criterion that a point $[r_1 : \dots : r_n] \in \mathbb{P}^{n-1}(k)$ has a unimodular representative in A^n if and only if it is $\text{Frac}(D)$ -rational, is in the real case the rationality theorem for primitive constant directions of the Diophantine report [46], which 14 does not credit (it credits only that report's example $[1 : \sqrt{2}]$, to a companion manuscript); the Gaussian and principal-ideal form is printed there with that theorem, not here.

The Gaussian case and the reach of the results (14). All proofs above use only a principal ideal domain $D \subset k$ and the common-factor property, so the Gaussian versions over $\mathbf{Oz}[i] = \mathbb{Z}[i] + \Pi_{\mathbb{C}}$ are proved, not merely analogous: ranks are $\mathbb{Z}[i]$ -ranks, multiplication kernels are killed by $\Pi_{\mathbb{C}}$, and in Theorem 13.22 the degree is taken over $\mathbb{Q}(i)$. For real algebraic α it equals the degree over $\mathbb{Q}$ ($[K_0(i) : \mathbb{Q}(i)] = [K_0 : \mathbb{Q}]$ for $K_0 \subset \mathbb{R}$), so Corollary 13.23 holds over $\mathbf{Oz}[i]$ with the same ranks over $\mathbb{Z}[i]$; for $\sqrt{2} \notin \mathbb{Q}(i)$ the ideal $(\omega, \sqrt{2}\omega) \subseteq \mathbf{Oz}[i]$ has the single quadratic equation $Y^2 - 2X^2$.

Complex conjugation transports the construction for M to that for $\overline{M}$. These results concern the omnific *rings*: over the surcomplex field itself all these module calculations are vector-space calculations and the arithmetic torsion vanishes. They do not classify arbitrary finitely generated omnific ideals, whose generators may lie at several support levels.

13.9 Relations, Tor and moduli of lattice ideals

This subsection is manuscript 17's new material; its results that repeat 12 and 14 are printed above, with 17 added to their sources. As in [Section 13.6](#), (D, k) is $(\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$, so that $\text{Frac } D$ is $\mathbb{Q}$ or $\mathbb{Q}(i)$; a lattice is a coefficient lattice $M \subset k$ of Setting (L), of rank r , and $J_g(M) = X^g L(M)$. The class ring is $A = \mathcal{A}_{D,k}$, that is **Oz** or **Oz** $[i]$. 17 works in class theory with global choice; over the class ring, Hom means the class of scalar codes of [Theorem 13.24](#), and ordinary Tor groups, flat resolutions and weak dimensions are taken over set-sized rings only (17). A module P has a *set-presentation* if there is an exact sequence $A^{(T)} \rightarrow A^{(S)} \rightarrow P \rightarrow 0$ of finite-support free modules on sets S, T ; equivalently, some surjection from a set-indexed free module onto P has a set-generated kernel.

Theorem 13.29 (No presentation by sets [17]). *Let $A = \mathcal{A}_{D,k}$, let M be a lattice of rank $r \geq 2$ and $g > 0$.*

- (i) *$J_g(M)$ and $L(M)$ have no set-presentation. In particular, for $\alpha \in k \setminus \text{Frac } D$ and $\gamma > 0$ the two-generated ideal $(\omega^\gamma, \alpha\omega^\gamma)$ has none, and the intersection $\omega^\gamma A \cap \alpha\omega^\gamma A = \omega^\gamma \Pi_k$ of two principal ideals has no set of generators. Thus **Oz** and **Oz** $[i]$ are neither coherent nor finite-conductor domains, in the strong sense that an intersection of two principal ideals need not be generated by a set.*
- (ii) *$J_g(M)$ fails the equational criterion of flatness: $a_j = X^g m_j$ and $x = (X^g m_2, -X^g m_1, 0, \dots, 0)$ satisfy $\sum_j a_j x_j = 0$, but there is no factorization $x_j = \sum_t b_{jt} y_t$ with $y_t \in J_g(M)$, $b_{jt} \in A$ and $\sum_j a_j b_{jt} = 0$ for every t .*

Proof. (i) Multiplying [Theorem 13.9\(ii\)](#) by the nonzerodivisor X^g gives an exact sequence $0 \rightarrow \Pi_k^{r-1} \rightarrow A^r \xrightarrow{\pi} J \rightarrow 0$ for $J = J_g(M)$. Suppose that $\pi' : A^{(S)} \rightarrow J$ is onto and that its kernel H is generated by a set. In the pullback $P = \{(x, y) \in A^r \oplus A^{(S)} : \pi(x) = \pi'(y)\}$ the second projection is onto $A^{(S)}$ with kernel $\ker \pi \oplus 0$, and it splits without any choice over a proper class: an element $X^g(\sum_j d_j m_j + u)$ of J , with $d_j \in D$ and $u \in \Pi_k$ unique, has the preimage $\sigma(\cdot) = (d_1 + u/m_1, d_2, \dots, d_r)$ under π , and $e_s \mapsto (\sigma(\pi'(e_s)), e_s)$ is a section. The first projection is onto A^r with kernel $0 \oplus H$, and it splits because A^r is free of finite rank. Hence $\Pi_k^{r-1} \oplus A^{(S)} \cong P \cong H \oplus A^r$ (Schanuel's lemma). The right side is generated by a set, hence so is the image of the left side under the projection onto its first summand Π_k , contrary to [Proposition 3.3\(iii\)](#). The same holds for $L(M) \cong J$. The two-generated ideal is $J_\gamma(D + D\alpha)$, the intersection formula is [Theorem 13.9\(iii\)](#), and $\omega^\gamma \Pi_k \cong \Pi_k$. Coherence asks that finitely generated ideals be finitely presented, and the finite-conductor property that intersections of two principal ideals be finitely generated.

(ii) A factorization would make each column $(b_{1t}, \dots, b_{rt})$ an element of $\ker \pi$, whose coordinates lie in Π_k ([Theorem 13.9\(ii\)](#)). Then every x_j would lie in $\Pi_k J = X^g \Pi_k L(M) = X^g \Pi_k$ ([Theorem 13.9\(i\)](#)), whereas $x_1/X^g = m_2$ is a nonzero constant. $\square$

The kernel shape Π^{r-1} , flat dimension one and the intersection mechanism are classical for $D + M$ rings (Dobbs [13], Dobbs–Papick [14]); 17 presents neither as new and does not infer any

factorization property from noncoherence. Over a set-sized ring, (ii) is the equational criterion [29, Lemma 10.39.11]; over the class ring it is stated without a category of class tensor products (17).

Lemma 13.30 (The lexicographic models inside **Oz** [17]; [merge] for $\kappa = \aleph_0$). *Let κ be an infinite regular cardinal and $\Gamma_\kappa = \bigoplus_{\alpha < \kappa} \mathbb{Q}e_\alpha$, ordered by the sign of the coefficient at the least index where it is nonzero (the group of Proposition 11.22).*

- (i) $e_\alpha \mapsto \omega^{-\alpha}$ extends to an injective order-preserving homomorphism $\iota : \Gamma_\kappa \rightarrow (\mathbf{No}, +)$.
- (ii) Every subset of $\Gamma_\kappa^{>0}$ with fewer than κ elements has a positive strict lower bound; the e_α are coinital in $\Gamma_\kappa^{>0}$, whose coinitality is exactly κ .
- (iii) $X^g \mapsto \omega^{\iota(g)}$ embeds $R_\kappa^{11} := R_{\Gamma_\kappa, < \kappa}^{11}$, with $(D, K) = (\mathbb{Z}, \mathbb{R})$, respectively $(\mathbb{Z}[i], \mathbb{C})$, in **Oz**, respectively **Oz**[i], as the ring of normal forms with constant term in D and fewer than κ further exponents, all in $\iota(\Gamma_\kappa)_{>0}$. We write Π_κ^{11} for its positive-support ideal.

Proof. (i) Here $\omega^{-\alpha}$ is the Conway monomial of the negative of the ordinal α , so $\iota(e_0) = 1$. For $\alpha < \beta$ the ratio $\omega^{-\alpha}/\omega^{-\beta} = \omega^{\beta-\alpha}$ is infinite, so a nonzero finite rational combination $\sum_\alpha q_\alpha \omega^{-\alpha}$ has the sign of its term of least index, which gives injectivity and monotonicity. (ii) Fewer than κ elements involve fewer than κ indices, finitely many each; by regularity all lie below some $\beta < \kappa$, and then $0 < e_\beta$ lies below each of the elements. A single positive element lies above e_β for β beyond its indices, so the e_α are coinital, and by the first assertion no family of fewer than κ positive elements is. (iii) A Hahn series over the ordered subgroup $\iota(\Gamma_\kappa) \subseteq \mathbf{No}$ is a Conway normal form, with the same sums and products (Sections 2.2 and 2.3); sums and products of series with fewer than κ exponents have fewer than κ exponents because κ is infinite and regular. $\square$

11 does not embed its ring in **No** (Section 11.5); (iii) places the ring of Propositions 11.22 and 11.23 inside **Oz**. For $\kappa = \aleph_0$ the ring $R_{\aleph_0}^{11}$ consists of finite-support series and, like the finite-support cores of 12 and 14, lies outside Theorem 5.1: the coefficient sum is a unital map to K detecting its tail, by the proof of Proposition 13.2(ii). In both cases R_κ^{11} has the properties listed in Setting (L) of Section 13.4, with $B = K + \Pi_\kappa^{11}$ and Π_κ^{11} not finitely generated, so Theorem 13.9(i),(ii) and Proposition 13.10 hold for it with the same proofs (17 proves them for its models).

Theorem 13.31 (Exact relation counts [17]; [merge] for $\kappa = \aleph_0$). *Let κ be an infinite regular cardinal, $A = R_\kappa^{11}$ and $\Pi = \Pi_\kappa^{11}$.*

- (i) $\Pi^2 = \Pi = \bigcup_{\delta \in \Gamma_\kappa^{>0}} X^\delta A$, a directed union of free modules of rank one; Π is flat, and $\text{gen}_A(\Pi) = \kappa$.
- (ii) For $C \in k^{m \times n}$ and $e \in \Gamma_\kappa^{>0}$, with q and ℓ as in Theorem 13.28, $\ker_A(X^e C) \cong A^\ell \oplus \Pi^{q-\ell}$; it needs exactly ℓ generators if $q = \ell$ and exactly κ if $q > \ell$.
- (iii) For a lattice M of rank $r \geq 2$ and $g \in \Gamma_\kappa^{>0}$, the kernel of every surjection from a finite free A -module onto $J_g(M)$ needs exactly κ generators; for the natural surjection it is Π^{r-1} .

Proof. (i) A support of fewer than κ positive exponents has a positive strict lower bound δ (Lemma 13.30(ii)), and dividing by X^δ leaves fewer than κ positive exponents; so $f = X^\delta(X^{-\delta}f)$ with both factors in Π , and choosing $e_\alpha < \delta$ puts f in $X^{e_\alpha}A$. Thus the κ monomials X^{e_α} generate Π , the union is directed ($X^\delta A \subseteq X^{\delta'} A$ for $\delta' < \delta$), each $X^\delta A$ is free because A is a domain, and a filtered colimit of flat modules is flat [29, Lemma 10.39.3]. If fewer than κ elements generated Π , the union of their supports would have fewer than κ elements, hence a positive strict lower bound δ , and the ideal they generate would lie in $X^\delta A$, which does not contain $X^{\delta/2}$. (ii) The proof of Theorem 13.28 uses only the splitting of A into constants and a k -linear ideal, and applies verbatim.

If $q > \ell$, projection onto one summand Π and (i) give at least κ generators, and κ suffice for a finite direct sum; if $q = \ell$ the kernel is free of rank ℓ , which needs exactly ℓ generators (tensor with the fraction field). (iii) The natural kernel is Π^{r-1} by [Theorem 13.9\(ii\)](#). For a surjection $A^m \rightarrow J_g(M)$ with kernel H , Schanuel's lemma gives $H \oplus A^r \cong \Pi^{r-1} \oplus A^m$. If H had fewer than κ generators, so would the right side, and projection onto one copy of Π would contradict (i); conversely H is a direct summand, hence a quotient, of a module generated by κ elements. $\square$

For uncountable κ , (i) is the case $R_{\Gamma, < \kappa}^{11}$ of [Theorem 12.2\(i\)](#) together with the generator count recorded in (H3). Statement (iii) concerns numbers of generators; the cardinality thresholds $\kappa^{< \kappa}$ of [Proposition 11.23](#) are a different invariant of the same ring. In the class ring the relation module needs more than a set of generators ([Theorem 13.29](#)); a set-sized model measures the same obstruction by the regular cardinal κ (17).

Remark 13.32 (The countable case [merge]). 17 states [Lemma 13.30](#) and [Theorem 13.31](#) for uncountable regular κ , but the proofs use only that κ is infinite and regular, and $\aleph_0$ is regular. For $\kappa = \aleph_0$ every $L(M)$ with $r \geq 2$ is countably presented but not finitely presented, and every finite presentation of it has a relation module needing exactly $\aleph_0$ generators. The tail $\Pi_{\aleph_0}^{11}$ is countably generated, so [Proposition 13.12\(v\)](#) gives $\text{pd } \Pi_{\aleph_0}^{11} = 1$ and $\text{pd } D = 2$, and the argument of [Proposition 13.12\(iv\)](#) gives $\text{pd } L(M) = 2$ and $\text{pd}(A/J_g(M)) = 3$ for $r \geq 2$. For uncountable κ the tail needs κ generators and these projective dimensions stay open ([Questions 18.3](#) and [18.15](#)).

Proposition 13.33 (Flat dimensions over the models [17]). *Let κ be infinite and regular, $A = R_\kappa^{11}$ and M a lattice of rank r . Then $\text{Tor}_j^A(J_g(M), -) = 0$ for $j \geq 2$. If $r \geq 2$, then $\text{fd}_A J_g(M) = 1$ and $\text{fd}_A(A/J_g(M)) = 2$; hence the weak global dimension of A is at least 2.*

Proof. The natural presentation $0 \rightarrow \Pi^{r-1} \rightarrow A^r \rightarrow J_g(M) \rightarrow 0$ is a flat resolution by [Theorem 13.31\(i\)](#), so $\text{fd } J_g(M) \leq 1$ and $\text{fd}(A/J_g(M)) \leq 2$. For $r \geq 2$, [Theorem 13.34](#) and [Corollary 13.36](#) below give $\text{Tor}_2(A/J_g(M), A/J_g(M)) \cong \text{Tor}_1(J_g(M), A/J_g(M)) \neq 0$. $\square$

This is the analogue over R_κ^{11} of [Theorem 13.11](#) for T_Γ (14's A_0 , where 14 proves it); both rings have a totally ordered exponent cone, as [Remark 13.16](#) requires. No upper bound on the weak global dimension of R_κ^{11} and no global dimension is claimed; Dobbs's global-dimension theorem for valuation pullbacks does not apply, because $K + \Pi_\kappa^{11}$ is not a valuation ring (17).

Theorem 13.34 (Tor as coefficient multiplication [17; 21]; [merge] over the cores). *Let A be R_κ^{11} for an infinite regular κ (with $K = k$), or one of the set-sized rings A_Λ , T_Γ of Setting (L)(a), (b) with any unital $D \subseteq K$. Let M, N be lattices of ranks r, s , let $X^g, X^h \in \Pi$ be nonzero monomials, $I = J_g(M)$ and $J = J_h(N)$. Then, with A acting on the right through ct ,*

$$\text{Tor}_1^A(I, A/J) \cong \text{Tor}_2^A(A/I, A/J) \cong \mathcal{T}(M, N) = \ker(\mathfrak{m}_{M,N} : M \otimes_D N \rightarrow K),$$

a D -module of rank $rs - \text{rank}(MN)$ (ranks are dimensions after tensoring with $\text{Frac } D$). There is an exact sequence

$$0 \longrightarrow \mathcal{T}(M, N) \longrightarrow I \otimes_A J \xrightarrow{\text{mult}} X^{g+h}(MN + \Pi) \longrightarrow 0,$$

whose first term is the torsion submodule of $I \otimes_A J$ and is killed by Π .

Proof. Let $m_1, \dots, m_r$ be a basis of M . By [Theorem 13.9\(ii\)](#) the natural presentation of I is $0 \rightarrow K_M \rightarrow A^r \rightarrow I \rightarrow 0$ with $K_M = \{u \in \Pi^r : \sum_j m_j u_j = 0\}$. Since Π is an ideal of B and a

K -space, K_M is a B -module and $\mu K_M = K_M$ for $0 \neq \mu \in K$; as $J = X^h(N + \Pi)$ with $N \neq 0$, this gives $JK_M = X^h K_M$. Tensoring the presentation with A/J gives

$$\mathrm{Tor}_1^A(I, A/J) \cong \ker(K_M/JK_M \rightarrow (A/J)^r) = (K_M \cap J^r)/X^h K_M.$$

Let $x \in K_M \cap J^r$ and write $x = X^h y$ with $y \in (N + \Pi)^r$, uniquely since X^h is a nonzerodivisor, and $c = \mathrm{ct}(y) \in N^r$. From $\sum_j m_j y_j = 0$ and $K \cap \Pi = 0$ one gets $\sum_j m_j c_j = 0$. The map $x \mapsto c$ is additive and onto the solutions of this equation in N^r : for such c , $x = X^h c$ lies in $\Pi^r \cap J^r$ and in K_M . Its kernel is $X^h K_M$: if $c = 0$ then $y \in \Pi^r$ and $\sum_j m_j y_j = 0$, so $y \in K_M$; conversely elements of $X^h K_M$ have $c = 0$. The basis identifies N^r with $M \otimes_D N$ by $(c_j) \mapsto \sum_j m_j \otimes c_j$, and the equation with $\ker \mathfrak{m}_{M,N}$; a change of basis of M changes both sides by the same invertible matrix. For $a \in A$ the constant term of ay is $\mathrm{ct}(a)c$, which is the action. The sequence $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$ and $\mathrm{Tor}_{>0}(A, -) = 0$ give $\mathrm{Tor}_2(A/I, A/J) \cong \mathrm{Tor}_1(I, A/J)$, and $0 \rightarrow \mathcal{T}(M, N) \rightarrow M \otimes_D N \rightarrow MN \rightarrow 0$ gives the rank.

Tensoring $0 \rightarrow J \rightarrow A \rightarrow A/J \rightarrow 0$ with I identifies $\mathrm{Tor}_1(I, A/J)$ with the kernel of the multiplication $I \otimes_A J \rightarrow I$. Its image is $IJ = X^{g+h}(M + \Pi)(N + \Pi) = X^{g+h}(MN + \Pi)$: products with a factor in Π lie in Π , and $u = m(u/m)$ for $u \in \Pi$ and $0 \neq m \in M$. The kernel is killed by Π because A acts through ct , and the image, a submodule of a field, is torsion free; so the kernel is exactly the torsion. $\square$

For R_κ^{11} with uncountable κ this is 17's theorem. Its proof uses only the properties of Setting (L) and the presentation of [Theorem 13.9\(ii\)](#), which is why it applies to 12's cores and to 14's $A_0 = T_{\mathbb{Q}}$; that extension is the merge's. The exact sequence is 14's [Theorem 13.18](#) for the class ring and for A_0 , where the class tensor product is an explicit carrier; the Tor identifications are 17's. The proof shows which coefficient relations survive: a relation $\sum_j m_j \otimes c_j$ is represented in $I \otimes_A J$ by $\sum_j (X^g m_j) \otimes (X^h c_j)$, whose product is zero although the tensor is not (17). Manuscript 21, whose pin precedes 17's placement, proves the identification of Tor_2 with $\mathcal{T}(M, N)$, the action through ct and the rank again, independently and by the same kernel computation, over $T_{\mathbb{Q}}$ and over the class ring in a two-universe reading; it adds that $\mathrm{Tor}_n = 0$ for $n \geq 3$ there, and the first Tor ([Theorem 13.51](#)).

Corollary 13.35 (12's cyclic modules without $c^2 \in D$ [merge]). *In the setting of [Theorem 13.11](#), let $c \in K \setminus \mathrm{Frac}(D)$, $M = D + Dc$ and $H = X^g L(M) = (X^g, cX^g)$. Then $\mathrm{Tor}_2^A(A/H, A/H) \cong \mathcal{T}(M, M)$ has rank $4 - \mathrm{rank}(M^2)$, which is 2 if c is quadratic over $\mathrm{Frac}(D)$ and 1 otherwise. If $c^2 \in D$, it is free with basis $1 \otimes c - c \otimes 1$ and $c \otimes c - c^2(1 \otimes 1)$, which is 12's $\mathrm{Tor}_2(A/H, A/H) \cong D \oplus D$.*

Proof. Apply [Theorem 13.34](#) with $N = M$. The span $M^2 = D + Dc + Dc^2$ has rank 2 exactly when $c^2 \in \mathrm{Frac}(D) + \mathrm{Frac}(D)c$. If $c^2 \in D$, the multiplication map sends $a(1 \otimes 1) + b(1 \otimes c) + b'(c \otimes 1) + d(c \otimes c)$ to $(a + dc^2) + (b + b')c$, and since $1, c$ are independent over D its kernel is as stated. $\square$

Thus 12's hypothesis $c^2 \in D$ is needed for the particular isomorphism $\mathrm{Tor}_2 \cong D + cD$ of [Theorem 13.11](#), not for the nonvanishing or the rank, which [Corollary 13.35](#) gives for every $c \notin \mathrm{Frac}(D)$.

Corollary 13.36 (The self-Tor range [17; 21]). *In [Theorem 13.34](#) let $N = M$, $J = I$ and $\ell = \mathrm{rank}(M^2)$. Then*

$$\mathrm{rank} \mathrm{Tor}_2^A(A/I, A/I) = r^2 - \ell, \quad \frac{r(r-1)}{2} \leq r^2 - \ell \leq r(r-1).$$

For $(D, k) = (\mathbb{Z}, \mathbb{R})$ and $(\mathbb{Z}[i], \mathbb{C})$ both ends occur for every $r \geq 1$.

Proof. Multiplication by a fixed nonzero element embeds M in M^2 , and M^2 is spanned by the $r(r+1)/2$ products $m_i m_j$, $i \leq j$; so $r \leq \ell \leq r(r+1)/2$. For the lower end take $M = D + D\alpha_2 + \cdots + D\alpha_r$ with $\alpha_2, \dots, \alpha_r \in k$ algebraically independent over $\text{Frac } D$: the products are the $r(r+1)/2$ distinct monomials of degree at most two, independent over $\text{Frac } D$. For the upper end take for M an order of rank r , so that $M^2 = M$: over $\mathbb{Q}$ the order $\mathbb{Z}[2^{1/r}]$ ($X^r - 2$ is Eisenstein at 2), over $\mathbb{Q}(i)$ the $\mathbb{Z}[i]$ -order generated by a root of $X^r - (1+i)$ (Eisenstein at the Gaussian prime $1+i$). $\square$

21 proves the rank $r^2 - \ell$ and the value $r(r-1)$ for orders again, independently, and splits Tor_2 into alternating and symmetric relations ([Corollary 13.55](#)); it does not state the range.

Proposition 13.37 (The maximal self-obstruction detects a field [17]). *In [Corollary 13.36](#) suppose $1 \in M$ and put $V = \text{Frac}(D)M$. Then $\text{rank } M^2 = r$ if and only if V is a field (for the multiplication of k), if and only if $\text{rank } \text{Tor}_2^A(A/I, A/I) = r(r-1)$. This does not imply that M itself is a ring.*

Proof. As $1 \in V$, $V \subseteq V^2$, and equality of dimensions means that V is closed under multiplication; then each $0 \neq v \in V$ acts injectively, hence surjectively, on V , so $vw = 1$ for some $w \in V$. Conversely a field has $V^2 = V$. The last equivalence is [Corollary 13.36](#). For $M = \mathbb{Z} + \mathbb{Z}/\sqrt{2}$ one has $1 \in M$ and $\text{rank } M^2 = 2$, but $(1/\sqrt{2})^2 = 1/2 \notin M$. $\square$

Theorem 13.38 (Rank-two moduli [17]). *For $\alpha \in k \setminus \text{Frac}(D)$ put $M_\alpha = D + D\alpha$. For $\alpha, \beta \in k \setminus \text{Frac}(D)$ and $\gamma, \eta > 0$, the ideals $J_\gamma(M_\alpha) = (\omega^\gamma, \alpha\omega^\gamma)$ and $J_\eta(M_\beta)$ of $\mathcal{A}_{D,k}$ are isomorphic if and only if $\beta = (a\alpha + b)/(c\alpha + d)$ for some $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}_2(D)$. There are exactly $2^{\aleph_0}$ isomorphism classes of these two-generated ideals; all have rank one over the fraction field and relation module isomorphic to Π_k .*

Proof. By [Theorem 13.24](#) the ideals are isomorphic exactly when $M_\beta = sM_\alpha$ for some $s \in k^\times$, that is when $(1, \beta)$ and $(s, s\alpha)$ are bases of one free D -module; this means $1 = s(c\alpha + d)$ and $\beta = s(a\alpha + b)$ for a matrix with $ad - bc \in D^\times$, and $c\alpha + d \neq 0$ because $\alpha \notin \text{Frac}(D)$. Conversely such a matrix and $s = (c\alpha + d)^{-1}$ give the equality of lattices. The group $\text{GL}_2(D)$ is countable and $k \setminus \text{Frac}(D)$ has $2^{\aleph_0}$ elements, so the orbits are countable and there are $2^{\aleph_0}$ of them. The last assertions are [Theorem 13.9](#). $\square$

The count $2^{\aleph_0}$ and the $\text{GL}_2(D)$ form are not in 14, whose [Theorem 13.24](#) they refine.

Theorem 13.39 (Orders from endomorphism rings [17]). *Let M be a lattice of rank r , $V = \text{Frac}(D)M$ and $K_V = \{c \in k : cV \subseteq V\}$. Then K_V is a field, finite over $\text{Frac } D$, and $[K_V : \text{Frac } D]$ divides r . The ring $\mathfrak{o}(M)$ is an order in K_V : it is finite free and integral over D , with $\text{Frac } \mathfrak{o}(M) = K_V$ and $\text{rank}_D \mathfrak{o}(M) = [K_V : \text{Frac } D]$. In the class ring $\text{End}_A(J_\gamma(M)) = \mathfrak{o}(M) + \Pi_k = \mathcal{A}_{\mathfrak{o}(M),k}$, so $\text{End}_A(J_\gamma(M))/\Pi_k \cong \mathfrak{o}(M)$ is its universal set-sized quotient.*

Proof. K_V contains $\text{Frac } D$ and is closed under the ring operations; for $0 \neq c \in K_V$, multiplication by c is injective on the finite-dimensional V , hence surjective, so $c^{-1} \in K_V$. For $0 \neq v \in V$, $c \mapsto cv$ embeds K_V in V , and V is a K_V -vector space, so the degree divides r . $\mathfrak{o}(M)$ embeds in $\text{End}_D(M) \cong D^{r \times r}$, so it is finite free, and Cayley–Hamilton gives integrality ([Theorem 13.24](#)). For $c \in K_V$ the matrix of c in a basis of M has entries in $\text{Frac } D$; a common denominator $0 \neq a \in D$ gives $ac \in \mathfrak{o}(M)$, so $K_V \subseteq \text{Frac } \mathfrak{o}(M) \subseteq K_V$. The endomorphism ring is [Theorem 13.24](#), and its universal set-sized quotient is given by [Theorem 5.1](#) with the coefficient ring $\mathfrak{o}(M)$, which need not be a principal ideal domain. $\square$

Corollary 13.40 (Every order occurs [17]). *If $\mathfrak{o} \subset k$ is an order over D in a finite extension of $\text{Frac } D$, then $\text{End}_A(J_\gamma(\mathfrak{o})) = \mathfrak{o} + \Pi_k$ and $\text{Aut}_A(J_\gamma(\mathfrak{o})) = \mathfrak{o}^\times$.*

Proof. $c\mathfrak{o} \subseteq \mathfrak{o}$ gives $c = c \cdot 1 \in \mathfrak{o}$, and conversely; apply [Theorem 13.24](#). $\square$

14's [Theorem 13.25](#) realizes the ideal classes of such an order as well; the corollary realizes the order itself as the constant quotient of an endomorphism ring.

Example 13.41 (A unit that is not an omnific integer [17]). For $J = (\omega, \sqrt{2}\omega) \subseteq \mathbf{Oz}$, $\text{End}_{\mathbf{Oz}}(J) = \mathbb{Z}[\sqrt{2}] + \Pi$, and multiplication by $1 + \sqrt{2}$ is an automorphism of J , with inverse $\sqrt{2} - 1$ and matrix $\begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$ (determinant -1) on the generators $\omega, \sqrt{2}\omega$; its powers give infinitely many automorphisms, although $1 + \sqrt{2} \notin \mathbf{Oz}$. Over $\mathbf{Oz}[i]$ the same ideal has endomorphism order $\mathbb{Z}[i, \sqrt{2}]$. If $[\text{Frac}(D)(\alpha) : \text{Frac } D] \geq 3$ or α is transcendental, then $\mathfrak{o}(D + D\alpha) = D$: an element $a + b\alpha$ of it must send α into $D + D\alpha$, which forces $b = 0$; so $(\omega, \alpha\omega)$ has only the automorphisms ± 1 (four in the Gaussian case).

Theorem 13.42 (Modules killed by the tail [17; 21]). *Let $A = R_\kappa^{11}$, M a lattice of rank r , and P a D -module on which A acts through ct . Then $J_g(M) \otimes_A P \cong M \otimes_D P \cong P^r$ and $\text{Tor}_j^A(J_g(M), P) = 0$ for $j \geq 1$. Over the class ring $\mathcal{A}_{D,k}$, every set-sized unital module P is of this kind ([Theorem 5.1\(iv\)](#)), and $M \otimes_D P$ is a set-sized model of $J_g(M) \otimes P$: every balanced map on $J_g(M) \times P$ factors through $M \times P$.*

Proof. Since $\Pi P = 0$, $J \otimes_A P \cong (J/\Pi J) \otimes_D P$ with $J/\Pi J \cong M$ ([Theorem 13.9\(i\)](#)). The kernel $K_M \cong \Pi^{r-1}$ of the natural presentation satisfies $\Pi K_M = K_M$ (as $\Pi^2 = \Pi$), so $K_M \otimes_A P = 0$, which gives $\text{Tor}_1(J, P) = 0$; for $j \geq 2$, $\text{Tor}_j(J, P) \cong \text{Tor}_{j-1}(K_M, P) = 0$ because K_M is flat. Over the class ring a balanced map kills $\Pi J \times P$ for the same reason. $\square$

This extends [Theorem 12.2\(ii\)](#) and [Corollary 13.5](#) from D to the lattice ideals. For $r \geq 2$ it does not make $J_g(M)$ flat: the test module $A/J_g(M)$, which is not killed by the tail, detects the obstruction ([Corollary 13.36](#)). On every set-sized test the ideal looks free of rank r , although its rank over the fraction field is one; no full-class Tor functor is inferred (17). Manuscript 21 obtains the same Tor groups, in the equivalent form $\text{Tor}_1(A/J_g(M), P) \cong M \otimes_D P$ and $\text{Tor}_{\geq 2}(A/J_g(M), P) = 0$, independently over $T_{\mathbb{Q}}$ and over the class ring in a two-universe reading, together with a square-zero model ([Theorem 13.57](#)).

Computing q and ℓ (17). If the entries of $C \in k^{m \times n}$ lie in a known $\text{Frac}(D)$ -span with a basis $\theta_1, \dots, \theta_t$ that is independent over $\text{Frac } D$, write $C = \sum_j \theta_j C_j$ with $C_j \in \text{Frac}(D)^{m \times n}$; then $Cx = 0$ for $x \in \text{Frac}(D)^n$ exactly when every $C_j x = 0$, so

$$\ell = n - \text{rank}_{\text{Frac } D} \begin{pmatrix} C_1 \\ \vdots \\ C_t \end{pmatrix}, \quad q = n - \text{rank}_k C,$$

and [Theorems 13.28](#) and [13.31](#) decide the kernel type and its generator count. This assumes an exact description of the coefficient span, as number-field data provide; it is not an algorithm for arbitrary real input or floating-point data (17).

Examples (17). All Tor statements here are over R_κ^{11} (or the cores). For $M = \mathbb{Z} + \mathbb{Z}\sqrt{2}$ the multiplication matrix in the bases $(1 \otimes 1, 1 \otimes \alpha, \alpha \otimes 1, \alpha \otimes \alpha)$, $\alpha = \sqrt{2}$, has rows $(1, 0, 0, 2)$ and $(0, 1, 1, 0)$, and $\mathcal{T}(M, M)$ has basis $\alpha \otimes 1 - 1 \otimes \alpha$ (commutativity) and $\alpha \otimes \alpha - 2(1 \otimes 1)$ (the quadratic equation), so $\text{Tor}_2(A/J, A/J) \cong \mathbb{Z}^2$ for $J = (\omega, \sqrt{2}\omega)$; the ideal has two generators, its relation module needs κ generators, and all its powers have two generators; over $\mathbb{Z}[i]$ the same matrix gives $\mathbb{Z}[i]^2$. For $\alpha = \sqrt[3]{2}$ and for a transcendental τ the matrices coincide (rank three, kernel spanned by the antisymmetric tensor), and so do the orders $(\mathbb{Z})$, but the power sequences $\min(n+1, 3)$ and $n+1$ differ: generator counts, first relation modules, orders and self-Tor ranks do not determine one another. For $M = \mathbb{Z} + \mathbb{Z}\sqrt{2}$ and $N = \mathbb{Z} + \mathbb{Z}\sqrt{3}$, $\text{rank } MN = 4$, so $\mathcal{T}(M, N) = 0$ and $\text{Tor}_2(A/J_g(M), A/J_h(N)) = 0$, although each self-Tor has rank two; the same holds over $\mathbb{Z}[i]$, since $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$ stay independent over $\mathbb{Q}(i)$. 21 gives the same pair independently, as the first case of its linear-disjointness test ([Theorem 13.53](#)).

13.10 Exact projective dimensions and derived arithmetic

This subsection is manuscript 21's material ([Section 1.3](#)). 21 pins 9693b28, at which this section held 12's and 14's results but not yet 17's [Section 13.9](#). Its first part answers the first clause of [Question 18.12](#): for every $d \geq 1$, $\text{pd}_{A_d} D = d+1$, with an explicit countably generated free resolution and an explicit copy of $D^{\mathbb{N}}/D^{(\mathbb{N})}$ in the top Ext group; it also determines the projective dimensions of the coordinate face quotients and of the lattice modules over A_d , which [Proposition 13.12](#)(iii), (iv) bound. Its second part computes the Tor groups of two lattice quotients over $T_{\mathbb{Q}}$ and over the class ring read in two universes. There its second-Tor formula with the rank count, the pair $\mathbb{Z}[\sqrt{2}]$, $\mathbb{Z}[\sqrt{3}]$ and the Tor groups against modules killed by the tail repeat 17's [Theorems 13.34](#) and [13.42](#) and [Corollary 13.36](#), which 21 could not see; they are printed there, with 21 credited as a further source, and are only cited here. New in the second part are the first-Tor window formula, a flat DGA resolution with the complete self-Tor product, the linear-disjointness test and the framed reconstruction of orders. Labels carry the sub-prefix **osq:da:**.

Notation. 21's field k and ring A_d are K and A_d of [Section 13.1](#); its tail $\mathcal{I}_d$ is Π_d , and its embedding $\gamma_i = \omega^{i-1}$ is an instance of [Lemma 13.1](#)(ii) (the report uses θ^{i-1} ; both are valid). Its lattices L, M and its module $M_L = L + \mathcal{I}_d$ are M, N and $L(M)$: the letters are swapped. Its monomial $t_a = \omega^a$ ($a > 0$) is X^a here, so its ideal $J_{a,L} = t_a(L + \mathcal{I})$ is $J_a(M)$; the letter t is not used, because in this report t^g means X^{-g} ([Section 2.2](#)). Its face quotient B_U is F_U (the letter B is the ring $K \oplus \Pi$), its $c = |U|$ is written $|U|$ (the letter c is an element of $K \setminus \text{Frac } D$), its ideals Q_n are $J_U^{(n)}$, its test module $A/(X_i : i \in U)$ is V_U of [Section 13.3](#), and its complexes K_n, F , shift σ , telescope $\mathcal{T}$ and dual differential δ_N are Kos_n , Kos , sh , Tel and $\Delta_P^\vee$ (the letter K is the field, $\mathcal{T}$ the multiplication kernel). In the second part its rings R_0 and R are $T_{\mathbb{Q}}$ and $\mathcal{A}_{D,k}$, its ideal $\mathcal{I}$ is Π , its kernel $\mathcal{K}(L, M) = \ker(L \otimes_D M \rightarrow LM)$ is $\mathcal{T}(M, N)$ (the same kernel, since $MN \subseteq k$), its syzygy module W_L is Syz_M , its quotients S, T are $A/I, A/J$ (T is 12's ordered-cone ring), its $H_j = \text{Tor}_j(S, S)$ are T_j (the letter H is 12's cyclic module), its boundary map λ is $\mathfrak{b}$ (λ is a cardinal in [Section 11.4](#)), its quotient Q_2 is Ind_2 , its boundary lattice $C_{a,b}$ and window $\mathcal{H}_k(0, c)$ are $\mathbb{L}_{g,h}$ and $\Pi_{(0,e)}$, its fields E, F over F_0 are K_1, K_2 over $\text{Frac } D$, and its orders O and kernels $\mathcal{K}_O$ are $\mathfrak{o}$ and $\mathcal{T}(\mathfrak{o}, \mathfrak{o})$.

Face quotients and a free telescope. Fix $d \geq 1$, $A = A_d$, $\Pi = \Pi_d$ and a nonempty $U \subseteq [d]$. With the face ideals I_S of (13.3) put

$$\mathfrak{f}_U = \sum_{i \in U} I_{\{i\}}, \quad \mathbf{F}_U = A/\mathfrak{f}_U.$$

The ideal $\mathfrak{f}_U$ is spanned by the terms KX^α with $\alpha_i > 0$ for some $i \in U$, so $\mathbf{F}_U$ keeps exactly the terms whose coordinates in U vanish: $\mathbf{F}_U \cong A_{d-|U|}$, with the same (D, K) , if $|U| < d$, and $\mathbf{F}_{[d]} = A/\Pi = D$. For $1 \leq p \leq |U|$ let $C_p^U = \bigoplus_{S \subseteq U, |S|=p} I_S$, with the differentials of Theorem 13.4, and $C_0^U = A$.

Proposition 13.43 (Face quotients: flat resolution and flat dimension [21]). *The complex $0 \rightarrow C_{|U|}^U \rightarrow \cdots \rightarrow C_1^U \rightarrow A \rightarrow \mathbf{F}_U \rightarrow 0$ is exact, every C_p^U is flat, and $\text{fd}_A \mathbf{F}_U = |U|$.*

Proof. For $U = [d]$ this is Theorem 13.4 and Corollary 13.8 (12). In general fix a multidegree α and put $T = \{i \in U : \alpha_i > 0\}$. If $T = \emptyset$, no positive term has a nonzero component in degree α , and $A \rightarrow \mathbf{F}_U$ is bijective there. Otherwise $(\mathbf{F}_U)_\alpha = 0$, and the degree- α part of I_S ($S \subseteq U$) is K exactly when $S \subseteq T$; so this part of the complex is the augmented chain complex of the full simplex on T with coefficients K , which the homotopy of Theorem 13.4 contracts. Flatness is Lemma 13.3(i), so $\text{fd} \mathbf{F}_U \leq |U|$. For the lower bound tensor with $V_U = A/J_U$, $J_U = (X_i : i \in U)A$, and look at the multidegree $\mathbf{e}_U$. The top term $I_U \otimes_A V_U = I_U/J_U I_U$ has coefficient space K there, because subtracting $\mathbf{e}_i$ ($i \in U$) from $\mathbf{e}_U$ gives a vector that vanishes at i , which is not an exponent of I_U . If $|U| \geq 2$, the face $U \setminus \{i\}$ is nonempty and $kX^{\mathbf{e}_U} = X_i \cdot kX^{\mathbf{e}_{U \setminus \{i\}}} \in X_i I_{U \setminus \{i\}}$, so the differential out of the top term vanishes in this degree; there is no term above it, and $\text{Tor}_{|U|}^A(\mathbf{F}_U, V_U) \neq 0$. If $U = \{i\}$, the map in degree $\mathbf{e}_i$ is $K \rightarrow (V_U)_{\mathbf{e}_i} = K/D$, because the elements of $X_i A$ of that degree are DX_i ; its kernel D is a nonzero $\text{Tor}_1^A(\mathbf{F}_U, V_U)$. $\square$

For $n \geq 0$ put $J_U^{(n)} = (X_i^{2^{-n}} : i \in U)A$ and $u_n = p_U^{2^{-n-1}}$; since $u_n X_i^{2^{-n-1}} = X_i^{2^{-n}} p_{U \setminus \{i\}}^{2^{-n-1}}$, one has $u_n J_U^{(n+1)} \subseteq J_U^{(n)}$. Let Kos_n be the Koszul complex over A of the $|U|$ elements $X_i^{2^{-n}}$ ($i \in U$): it is free on basis vectors e_S ($S \subseteq U$) in degree $|S|$, with $\partial e_S = \sum_{r=1}^p (-1)^{r-1} X_{s_r}^{2^{-n}} e_{S \setminus \{s_r\}}$ for $S = \{s_1 < \cdots < s_p\}$. The maps $f_n(e_S) = p_S^{2^{-n-1}} e_S$ (with $f_n(e_\emptyset) = e_\emptyset$) are chain maps $\text{Kos}_n \rightarrow \text{Kos}_{n+1}$, because $X_i^{2^{-n-1}} p_S^{2^{-n-1}} = X_i^{2^{-n}} p_{S \setminus \{i\}}^{2^{-n-1}}$ for $i \in S$.

Theorem 13.44 (A free telescope resolution [21]).

- (i) *In each degree $p \geq 1$ the direct system (Kos_n, f_n) has colimit C_p^U , through the maps sending e_S at stage n to $p_S^{2^{-n}}$ in the summand I_S , and the colimit of the Koszul differentials is the differential of Proposition 13.43; in degree 0 the system is constant with value A .*
- (ii) *Let $\text{Kos} = \bigoplus_{n \geq 0} \text{Kos}_n$, let $\text{sh} : \text{Kos} \rightarrow \text{Kos}$ send the n th summand to the $(n+1)$ st by f_n , and put $\Delta = 1 - \text{sh}$. The mapping cone $\text{Tel}_j = \text{Kos}_j \oplus \text{Kos}_{j-1}$, $d(x, y) = (\partial x + \Delta y, -\partial y)$, is a complex of countably generated free A -modules in degrees $0, \dots, |U| + 1$, and the augmentations of the Kos_n make it a free resolution of $\mathbf{F}_U$ of length $|U| + 1$.*

No regularity of $(X_i^{2^{-n}})_{i \in U}$ is used, and the individual Kos_n need not be acyclic: for $D \neq K$ the sequence need not be regular (for $c \in K \setminus D$, X_2 kills the nonzero class of cX_1 in $A_2/X_1 A_2$).

Proof. (i) Multiplication by $p_S^{2^{-n}}$ embeds the copy Ae_S at stage n in I_S (A is a domain); the embeddings are compatible with f_n because $p_S^{2^{-n-1}} p_S^{2^{-n-1}} = p_S^{2^{-n}}$, and their union is I_S by Lemma 13.3(i). Under these maps ∂e_S goes to $\sum_r (-1)^{r-1} X_{s_r}^{2^{-n}} p_{S \setminus \{s_r\}}^{2^{-n-1}} = \sum_r (-1)^{r-1} p_S^{2^{-n}}$, the alternating sum of

the inclusions. (ii) $d^2 = 0$ because $\partial\Delta = \Delta\partial$. Δ is injective in every degree: at the least index of a nonzero vector its component is unchanged. Its cokernel is the colimit C^U of (i), so $(x, y) \mapsto [x]$ is a surjective chain map $\text{Tel} \rightarrow C^U$. Its kernel consists of the pairs $(\Delta z, y)$; in the coordinates (z, y) the differential is $(z, y) \mapsto (\partial z + y, -\partial y)$, and $h(z, y) = (0, z)$ satisfies $dh + hd = 1$. So $\text{Tel} \rightarrow C^U$ is a quasi-isomorphism, and $C^U \rightarrow F_U$ is one by Proposition 13.43. Each Kos_j is a countable direct sum of finite free modules and vanishes for $j > |U|$. For the example, $X_2 c X_1 = X_1 \cdot c X_2$ with $c X_2 \in \Pi$, while the elements of $X_1 A_2$ of degree $\mathbf{e}_1$ are $D X_1$. $\square$

For $|U| = 1$ this is 12's telescope $0 \rightarrow F \rightarrow F \rightarrow I_S \rightarrow 0$ of Proposition 13.12(i), placed in the cone; 21 assembles the telescopes of all faces into one resolution (21 credits countable telescopes as a standard tool, citing Positselski [28]).

Theorem 13.45 (The top Ext group [21]). *For an A -module P put $G_n(P) = P/J_U^{(n)}P$; multiplication by u_n maps $G_{n+1}(P)$ to $G_n(P)$. Then, naturally in P ,*

$$\text{Ext}_A^{|U|+1}(F_U, P) \cong \text{coker}\left(\Delta_P^\vee : \prod_{n \geq 0} G_n(P) \longrightarrow \prod_{n \geq 0} G_n(P), (y_n)_n \longmapsto (y_n - u_n y_{n+1})_n\right),$$

which is $\varprojlim^1$ of the inverse system $(G_n(P))_n$.

Proof. The last nonzero term of Tel is $\text{Tel}_{|U|+1} = \text{Kos}_{|U|} = \bigoplus_n A e_{U,n}$, where $e_{U,n}$ is e_U at stage n , so $\text{Hom}_A(\text{Tel}_{|U|+1}, P) = \prod_n P$. The dual of $d(0, y) = (\Delta y, -\partial y)$ sends a functional (φ, ψ) on $\text{Tel}_{|U|} = \text{Kos}_{|U|} \oplus \text{Kos}_{|U|-1}$ to $y \mapsto \varphi(\Delta y) - \psi(\partial y)$. Since $\Delta e_{U,n} = e_{U,n} - u_n e_{U,n+1}$, the first part gives $(z_n - u_n z_{n+1})_n$ with $z_n = \varphi(e_{U,n})$ arbitrary; the second gives in component n a signed sum $\sum_{i \in U} \pm X_i^{2^{-n}} \psi(e_{U \setminus \{i\}, n})$, and these range over all of $\prod_n J_U^{(n)}P$. The top cohomology of $\text{Hom}_A(\text{Tel}, P)$ is therefore $\prod_n P$ modulo these elements, which is the stated cokernel. $\square$

Theorem 13.46 (A copy of $D^\mathbb{N}/D^{(\mathbb{N})}$ in the top Ext group [21]). *Send $(c_n) \in D^\mathbb{N}$ to the class of the constant representatives $(c_n) \in \prod_n G_n(A)$ in $\text{Ext}_A^{|U|+1}(F_U, A)$. This D -linear map has kernel exactly $D^{(\mathbb{N})}$, the finitely supported sequences, so $D^\mathbb{N}/D^{(\mathbb{N})} \hookrightarrow \text{Ext}_A^{|U|+1}(F_U, A)$.*

Proof. Suppose (c_n) is in the image of $\Delta_A^\vee$, with representatives $y_n \in A$ of a preimage: $y_n - u_n y_{n+1} - c_n \in J_U^{(n)}$. Since $u_0 \cdots u_{j-1} = p_U^{1-2^{-j}}$ and $p_U^{1-2^{-j}} J_U^{(j)} \subseteq J_U^{(0)} = J_U$, multiplying the j th relation by $u_0 \cdots u_{j-1}$ and adding for $j = 0, \dots, N$ gives

$$y_0 - \sum_{j=0}^N c_j p_U^{1-2^{-j}} - p_U^{1-2^{-N-1}} y_{N+1} \in J_U.$$

Every monomial of an element of J_U has some coordinate in U at least 1, and every monomial of $p_U^{1-2^{-N-1}} y_{N+1}$ has all its coordinates in U at least $1 - 2^{-N-1}$. The monomials $m_j = p_U^{1-2^{-j}}$ ($j \leq N$) are pairwise distinct and have all coordinates in U below $1 - 2^{-N-1}$; so the coefficient of m_j in the single finite sum y_0 is c_j , for every j (take $N \geq j$). Hence $c_j \neq 0$ for only finitely many j . Conversely, for finitely supported c the finite sums $y_n = \sum_{j \geq n} c_j p_U^{2^{-n}-2^{-j}}$ lie in A (their constant term is $c_n \in D$ and their other monomials are positive on U), and $y_n - u_n y_{n+1} = c_n$. $\square$

Corollary 13.47 (Exact projective dimensions of the coordinate cores [21]). *For every $d \geq 1$ and every nonempty $U \subseteq [d]$, $\text{pd}_{A_d} F_U = |U| + 1$ and $\text{fd}_{A_d} F_U = |U|$. In particular*

$$\text{pd}_{A_d} D = d + 1, \quad \text{pd}_{A_d} \Pi_d = d,$$

and $\text{Ext}_{A_d}^{d+1}(D, A_d) \cong \text{Ext}_{A_d}^d(\Pi_d, A_d)$ contains a copy of $D^\mathbb{N}/D^{(\mathbb{N})}$.

Proof. The constant sequence $(1, 1, \dots)$ is not finitely supported, so [Theorem 13.46](#) gives $\text{Ext}_A^{|U|+1}(\mathbf{F}_U, A) \neq 0$ and $\text{pd } \mathbf{F}_U \geq |U| + 1$; [Theorem 13.44](#) gives the reverse inequality, and [Proposition 13.43](#) the flat dimension. For $U = [d]$, the sequence $0 \rightarrow \Pi_d \rightarrow A_d \rightarrow D \rightarrow 0$ gives $\text{Ext}^j(\Pi_d, -) \cong \text{Ext}^{j+1}(D, -)$ for $j \geq 1$, hence $\text{pd } \Pi_d = d$. $\square$

So for every d the projective dimension of the arithmetic quotient exceeds its flat dimension by exactly one, as in rank one, and the upper bound of [Proposition 13.12\(iii\)](#) is attained. For $d = 2$ and $U = \{1, 2\}$, a preimage of the constant sequence 1 would need its first representative to contain all the monomials $1, (X_1 X_2)^{1/2}, (X_1 X_2)^{3/4}, (X_1 X_2)^{7/8}, \dots$, whose coordinates are below 1; so $\text{Ext}_{A_2}^3(D, A_2) \neq 0$, although the flat resolution of [Theorem 13.4](#) has length two and the finite Koszul complexes are not resolutions (21, its worked appendix). Every finite initial segment of the equations $y_n - u_n y_{n+1} = 1$ has a solution, obtained by choosing a last value and solving backwards; the obstruction is that a solution of all of them would force infinitely many monomials into y_0 , so no finite computation proves it (21).

Theorem 13.48 (Lattice modules over the coordinate cores [21] [merge]). *Let $d \geq 1$, let $D \subseteq K$ be any unital subring, M a coefficient lattice of Setting (L) of rank $r \geq 2$, $\mathbf{0} \neq g \in P_{[d]}$ and $J = J_g(M) \subseteq A_d$. Then*

$$\text{pd}_{A_d} L(M) = d + 1, \quad \text{pd}_{A_d}(A_d/J) = d + 2,$$

and for every A_d -module P there are natural isomorphisms $\text{Ext}_{A_d}^{d+1}(L(M), P) \cong \text{Ext}_{A_d}^{d+2}(A_d/J, P) \cong \text{Ext}_{A_d}^{d+1}(D, P)^{r-1}$; for $P = A_d$ these groups contain $(D^{\mathbb{N}}/D^{(\mathbb{N})})^{r-1}$. If $K \neq \text{Frac } D$, the global dimension of A_d is at least $d + 2$. For $r = 1$, $L(M) \cong A_d$ and A_d/J has projective and flat dimension 1.

Proof. [Theorem 13.9\(ii\)](#) gives $0 \rightarrow \Pi_d^{r-1} \rightarrow A_d^r \rightarrow L(M) \rightarrow 0$, so $\text{Ext}^{j+1}(L(M), P) \cong \text{Ext}^j(\Pi_d, P)^{r-1}$ for $j \geq 1$; with $j = d$ and [Corollary 13.47](#) this is $\text{Ext}^{d+1}(D, P)^{r-1}$, it vanishes for $j > d$, and it is nonzero for $P = A_d$ by [Theorem 13.46](#). Multiplication by X^g is injective, so $J \cong L(M)$, and $0 \rightarrow J \rightarrow A_d \rightarrow A_d/J \rightarrow 0$ shifts once more. If $c \in K \setminus \text{Frac } D$, the lattice $D + Dc$ has rank two, so some module has projective dimension $d + 2$. For $r = 1$, $L(M) = m_1 A_d$ and J is principal and nonzero. $\square$

21 states the theorem for a principal ideal domain D , which it uses only to give M a basis; in Setting (L) a lattice has a basis by definition, and extending the statement to every unital $D \subseteq K$ is the merge's (the proof is 21's). The flat dimensions d and $d + 1$ of $L(M)$ and A_d/J are [Theorem 13.11](#) (12, 14), which 21 proves again over A_d ; the weak global dimension of A_d is at least $d + 1$ ([Section 13.5](#)). Neither global dimension is determined (21).

Remark 13.49 (Consistency with the earlier bounds; (H2) cannot be dropped from [Proposition 13.12\(v\)](#) [merge]).

- (i) For $d = 1$, $\text{pd}_{A_1} D = 2$ is the merge value of [Proposition 13.12\(iii\)](#), and $\text{pd } L(M) = 2$, $\text{pd}(A_1/J_g(M)) = 3$ are its item (iv) for $T_{\mathbb{Q}} = A_1$; 21 adds an explicit free resolution and an explicit nonzero class. For every d the values of [Corollary 13.47](#) and [Theorem 13.48](#) are the upper ends of the bounds $d \leq \text{pd } D \leq d + 1$, $d \leq \text{pd } L(M) \leq d + 1$ and $d + 1 \leq \text{pd}(A_d/J_g(M)) \leq d + 2$ of [Proposition 13.12\(iii\)](#), (iv).
- (ii) The tail Π_d is countably generated, by the monomials $X_i^{2^{-n}}$ ($i \in [d]$, $n \geq 0$): a term kX^α with $\alpha_i > 0$ equals $X_i^{2^{-n}} \cdot kX^{\alpha - 2^{-n}\mathbf{e}_i}$ for $2^{-n} < \alpha_i$, and the second factor lies in Π_d . Moreover

- A_d satisfies (H1) (Lemma 13.1(i)) and (H3) (Lemma 13.3(iii)), and under the embeddings of Lemma 13.1(ii) its elements have nonnegative support. For $d \geq 2$ it fails (H2): if $X_1 + X_2 = X^g b$ with $g > 0$ and $b \in \Pi_d$, then under an embedding $X^\beta \mapsto \omega^{\ell(\beta)}$ of Lemma 13.1(ii) the exponents $\gamma_1 - g, \gamma_2 - g$ of b are $\ell(\beta_1), \ell(\beta_2)$ with $\beta_1, \beta_2 \in P_{[d]}$; linear independence gives $\beta_1 - \beta_2 = \mathbf{e}_1 - \mathbf{e}_2$, so the second coordinate of β_2 is at least 1, and $g = \ell(\mathbf{e}_2 - \beta_2) \leq 0$ because $\mathbf{e}_2 - \beta_2$ has no positive coordinate. (Under (H2) the tail would also be flat, by Theorem 12.2(i), whereas $\text{fd } \Pi_d = d - 1 \geq 1$ by Corollary 13.8; compare Remark 13.16.) Since $\text{pd}_{A_d} D = d + 1 > 2$, the conclusion $\text{pd } D = 2$ of Proposition 13.12(v) fails for A_d : a countably generated tail does not suffice without (H2), and the rings A_d realize every value $\text{pd } D \geq 2$.
- (iii) T_Γ and 17's countable model $R_{\aleph_0}^{11}$ satisfy (H1)–(H3) and have $\text{pd } D = 2$ (Theorem 13.13, Remark 13.32). For the same exponent group, the contrast of Theorem 13.14 between the coordinate cone and the totally ordered cone is now $d + 1$ against 2 for the projective dimension, as it is d against 1 for the flat dimension.
 - (iv) Modules killed by the tail do not see the projective dimension: $\text{Tor}_{>0}^{A_d}(D, P) = 0$ for every D -module P (Corollary 13.5, 12), although $\text{pd}_{A_d} D = d + 1$; the witness of Theorem 13.46 uses the target A_d itself (21).

Nothing here concerns the uncountable constructions of Section 11 or R_κ^{11} with κ uncountable, which satisfy (H2) and whose projective dimensions stay open (Question 18.3).

Lattice quotients over the ordered core and the class ring. For the rest of this subsection (D, k) is $(\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$, and A is one of two rings: $T_\mathbb{Q}$ (14's A_0 , 21's R_0), with positive rational scales; or the class ring $\mathcal{A}_{D,k}$, that is $\mathbf{Oz}$ or $\mathbf{Oz}[i]$, with positive surreal scales, read in two universes as in the universe-relative reading of Section 2.1. In that reading 21 fixes Grothendieck universes $\mathcal{U} \in \mathcal{V}$ containing k , uses the surreals with $\mathcal{U}$ -small sign sequences and the normal forms with $\mathcal{U}$ -small supports, and forms all modules, resolutions, tensor products and Tor and Ext groups in $\mathcal{V}$, where this ring is a set; this is 13's reading of Remark 13.16. It is explicit size bookkeeping, not an assertion that a proper class is an ordinary ring, and it differs from 17's convention in Section 13.9, under which ordinary Tor is formed over set-sized rings only. In both cases every $\mathcal{U}$ -small (for $T_\mathbb{Q}$: finite) family of tails has a common monomial divisor X^ϵ , $\epsilon > 0$, so $\Pi = \bigcup_{\epsilon > 0} X^\epsilon A$ is a directed union of free ideals, hence flat, and $\Pi^2 = \Pi$ (Theorem 13.13 for $T_\mathbb{Q}$; the proof of Proposition 3.3, with $\mathcal{U}$ -small in place of set-sized, for the class ring). These are the only rings of this part; for A_d with $d \geq 2$ the tail is not flat and nothing below is asserted there. Let $M, N \subset k$ be lattices of ranks r, s with bases $m_1, \dots, m_r$ and $n_1, \dots, n_s$, let $g, h > 0$ be scales, $I = J_g(M)$ and $J = J_h(N)$. For $u \in A^r$ put $m(u) = \sum_j m_j u_j \in B$, and let $\text{Syz}_M = \{u \in \Pi^r : m(u) = 0\} \cong \Pi^{r-1}$ (Theorem 13.9(ii)).

Theorem 13.50 (A flat DGA resolution [21]). *The complex $\mathbf{P}_\bullet$ with $\mathbf{P}_0 = A$, $\mathbf{P}_1 = A^r$, $\mathbf{P}_2 = \text{Syz}_M$, $\mathbf{P}_j = 0$ for $j \geq 3$, differential $u \mapsto X^g m(u)$ on $\mathbf{P}_1$ and the inclusion on $\mathbf{P}_2$, is a flat resolution of A/I . With the A -module structure, the product*

$$u \cdot v = X^g(m(u)v - m(v)u) \in \mathbf{P}_2 \quad (u, v \in \mathbf{P}_1),$$

and all products of positive degrees summing to at least three equal to zero, it is a unital, associative, graded-commutative differential graded algebra, and the augmentation $\mathbf{P}_\bullet \rightarrow A/I$ is a quasi-isomorphism of differential graded algebras.

Proof. Exactness is Theorem 13.9(ii) together with $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$; $\mathbf{P}_0, \mathbf{P}_1$ are free and $\mathbf{P}_2 \cong \Pi^{r-1}$ is flat. Each coordinate of $u \cdot v$ lies in $X^g(M + \Pi)A \subseteq \Pi$, and $m(u \cdot v) =$

$X^g(m(u)m(v) - m(v)m(u)) = 0$, so the product lies in Syz_M ; it is bilinear and alternating, so degree-one squares are zero. For $u, v \in P_1$, $d(u \cdot v) = X^g m(u)v - X^g m(v)u = d(u)v - u d(v)$. For $u \in P_1$ and $x \in P_2$ the product is zero, and $d(u)x - u \cdot d(x) = X^g m(u)x - X^g(m(u)x - m(x)u) = 0$ because $m(x) = 0$; the case $x \cdot u$ is the same with the sign of degree two. All other cases are A -linearity or vanish for degree reasons, and so do associativity and graded commutativity. The augmentation is the quotient map in degree 0 and zero above, so it is multiplicative. $\square$

A bounded flat resolution computes derived tensor products, so $P_\bullet \otimes_A A/J$ computes $\text{Tor}^A(A/I, A/J)$, and for $J = I$ its multiplication induces the product of the Tor algebra (21, citing [32]). 21 does not claim that this model is a cofibrant simplicial commutative algebra, that it determines higher operations, or any E_∞ -formality statement.

Theorem 13.51 (All Tor groups of two lattice quotients [17; 21]).

- (i) $\text{Tor}_2^A(A/I, A/J) \cong \mathcal{T}(M, N)$, a free D -module of rank $rs - \text{rank}(MN)$ on which A acts through ct , independently of the scales g, h ; and $\text{Tor}_n^A(A/I, A/J) = 0$ for $n \geq 3$.
- (ii) $\text{Tor}_0^A(A/I, A/J) = A/(I + J)$ and $\text{Tor}_1^A(A/I, A/J) \cong (I \cap J)/IJ$. Put $e = \min(g, h)$ and let $L_{g,h}$ be N if $g < h$, M if $h < g$, and $M \cap N$ if $g = h$. Then, as A -modules and as D -modules respectively,

$$\text{Tor}_1^A(A/I, A/J) \cong \frac{L_{g,h} + \Pi}{X^e(MN + \Pi)}, \quad \text{Tor}_1^A(A/I, A/J) \cong L_{g,h} \oplus \Pi_{(0,e)} \oplus (k/MN)X^e,$$

where $\Pi_{(0,e)}$ is the group of tails supported in the open interval $(0, e)$. The second decomposition is coefficientwise; it is not an A -module splitting.

Proof. (i) For $T_{\mathbb{Q}}$ the isomorphism and the rank are [Theorem 13.34](#) (17, with the merge over T_{Γ}). For the class ring in the reading above the proof of [Theorem 13.34](#) applies verbatim, since it uses only Setting (L) and the natural presentation; this is 21's computation of the homology $(\text{Syz}_M \cap J^r)/X^h \text{Syz}_M$ of $P_\bullet \otimes_A A/J$ in degree two. The kernel is free, being a submodule of the finite free D -module $M \otimes_D N$ over a principal ideal domain. The resolution $P_\bullet$ is flat of length two, which gives $\text{Tor}_{\geq 3} = 0$ (for $T_{\mathbb{Q}}$ this is also $\text{fd}(A/I) = 2$, [Theorem 13.11](#)). (ii) Tensoring $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$ with A/J gives $\text{Tor}_1(A/I, A/J) = \ker(I/IJ \rightarrow A/J) = (I \cap J)/IJ$; no flatness is used. If $g < h$ then $J \subseteq I$, since $X^{h-g}(N + \Pi) \subseteq \Pi$; so $I \cap J = J$ and $I + J = I$, and symmetrically if $h < g$. If $g = h$, separating the coefficient at X^g from the higher terms gives $I \cap J = X^g((M \cap N) + \Pi)$ and $I + J = X^g((M + N) + \Pi)$. In all cases $I \cap J = X^{\max(g,h)}(L_{g,h} + \Pi)$, and $IJ = X^{g+h}(MN + \Pi)$ (proof of [Theorem 13.34](#)); dividing by $X^{\max(g,h)}$ gives the first formula. The denominator consists of the elements with coefficient in MN at X^e and arbitrary terms above e , and no terms below; splitting a representative into its constant term, its terms in $(0, e)$, its term at e and its terms above e (truncation keeps supports admissible) gives the second. A tail multiplier can move a term of the window to the boundary or beyond it, so the decomposition is not A -linear. $\square$

So the first Tor keeps a whole interval of scales and a boundary quotient k/MN , which can have torsion, while the second is a finite-rank coefficient module that does not see the scales at all; neither should be replaced by a leading-term approximation (21). In particular $\text{Tor}_1 \neq 0$ always, so the vanishing of Tor_2 is never Tor-independence.

Lemma 13.52 (The compositum [21]). *Let $\mathfrak{o}_1, \mathfrak{o}_2 \subset k$ be orders over D : unital subrings, finite free over D , whose $\text{Frac}(D)$ -spans K_1, K_2 are fields. Then the $\text{Frac}(D)$ -span of the product lattice $\mathfrak{o}_1 \mathfrak{o}_2$ is the compositum $K_1 K_2$, so $\text{rank}_D(\mathfrak{o}_1 \mathfrak{o}_2) = [K_1 K_2 : \text{Frac } D]$.*

Proof. The span V of all products xy ($x \in K_1$, $y \in K_2$) contains K_1 and K_2 (both contain 1), is closed under multiplication and is finite-dimensional over $\text{Frac } D$. As a subring of the field k it is a domain, so a nonzero element acts injectively, hence surjectively, on V , and is invertible in V . So V is a field containing K_1, K_2 and contained in their compositum, hence equal to it; it is also the span of the products of elements of $\mathfrak{o}_1$ and $\mathfrak{o}_2$. $\square$

Theorem 13.53 (Second Tor detects linear disjointness [21]). *For orders $\mathfrak{o}_1, \mathfrak{o}_2$ as in Lemma 13.52 and all scales g, h ,*

$$\text{rank}_D \text{Tor}_2^A(A/J_g(\mathfrak{o}_1), A/J_h(\mathfrak{o}_2)) = [K_1 : \text{Frac } D][K_2 : \text{Frac } D] - [K_1 K_2 : \text{Frac } D],$$

and this Tor_2 vanishes if and only if K_1 and K_2 (with their given embeddings in k) are linearly disjoint over $\text{Frac } D$.

Proof. Combine Theorem 13.51(i) with Lemma 13.52. After tensoring with $\text{Frac } D$ the kernel becomes that of the surjection $K_1 \otimes_{\text{Frac } D} K_2 \rightarrow K_1 K_2$, whose injectivity is linear disjointness; a torsion-free module with zero localization is zero. $\square$

For $\mathfrak{o}_1 = \mathbb{Z}[\sqrt{2}]$ and $\mathfrak{o}_2 = \mathbb{Z}[\sqrt{3}]$ this is 17's example after Theorem 13.42 (rank zero), and for $\mathfrak{o}_1 = \mathfrak{o}_2 = \mathbb{Z}[\sqrt{2}]$ the rank is two; 21 adds that the scales may be $(1, 2)$ over $T_{\mathbb{Q}}$ or $(1, \omega)$ over the class ring, and that $\text{Tor}_1 \neq 0$ in both cases. For $D = \mathbb{Z}[i]$ the test is over $\mathbb{Q}(i)$. The criterion is a translation between two structures: it creates no extensions and does not decide linear disjointness without access to the arithmetic of the fields (21). Its diagonal counterpart, the maximal self-Tor rank detecting a field, is 17's Proposition 13.37.

The self-Tor algebra. Take $N = M$, $h = g$, so $J = I$, and write $\mathbb{T}_j = \text{Tor}_j^A(A/I, A/I)$: $\mathbb{T}_0 = A/I$, $\mathbb{T}_1 = I/I^2$ and $\mathbb{T}_2 \cong \mathcal{T}(M, M)$. The boundary map

$$\mathbf{b} : \mathbb{T}_1 \longrightarrow M, \quad [X^g(m + w)] \longmapsto m \quad (m \in M, w \in \Pi),$$

is well defined because $I^2 = X^{2g}(M^2 + \Pi) \subseteq X^g\Pi$, and it induces $\mathbb{T}_1/\Pi\mathbb{T}_1 \cong M$ because $\Pi I = X^g\Pi$. For the product, $\mathbb{T}_2$ is identified with $\mathcal{T}(M, M)$ by sending the surviving coefficient vector (c_j) of Theorem 13.34 to $\sum_j c_j \otimes m_j$, the flip of the convention of that proof; this fixes the sign below.

Theorem 13.54 (The self-Tor product [21]). *For $x, y \in \mathbb{T}_1$,*

$$xy = \mathbf{b}(x) \otimes \mathbf{b}(y) - \mathbf{b}(y) \otimes \mathbf{b}(x) \in \mathcal{T}(M, M).$$

All products of positive degrees summing to at least three vanish, and A acts on $\mathbb{T}_2$ through ct . With the A/I -module $\mathbb{T}_1$, these rules determine the graded algebra $\text{Tor}^A(A/I, A/I)$.

Proof. Compute in $\mathbf{P}_{\bullet} \otimes_A A/I$ (Theorem 13.50). Degree-one classes are represented by $u, v \in A^r$ with constant vectors $\text{ct}(u), \text{ct}(v) \in D^r$; the class of u corresponds to $[X^g m(u)] \in I/I^2$, so $\mathbf{b}([u]) = m(\text{ct } u)$, and likewise for v . The product vector $X^g(m(u)v - m(v)u)$ lies in $\text{Syz}_M \cap I^r$; after division by X^g its constant coefficients are $c_j = m(\text{ct } u)\text{ct}(v)_j - m(\text{ct } v)\text{ct}(u)_j \in M$, and $\sum_j c_j \otimes m_j = m(\text{ct } u) \otimes m(\text{ct } v) - m(\text{ct } v) \otimes m(\text{ct } u)$. The algebra is zero above degree two, and the action is that of Theorem 13.34. $\square$

Corollary 13.55 (Alternating and symmetric relations [21]). *Let Ind_2 be $\mathbb{T}_2$ modulo the D -span of the products $\mathbb{T}_1\mathbb{T}_1$, and $\ell = \text{rank}_D(M^2)$. There is an exact sequence of D -modules*

$$0 \longrightarrow \bigwedge_D^2 M \longrightarrow \mathbb{T}_2 \longrightarrow \ker(\text{Sym}_D^2 M \rightarrow M^2) \longrightarrow 0,$$

so $\text{Ind}_2 \cong \ker(\text{Sym}_D^2 M \rightarrow M^2)$, $\text{rank } \mathbb{T}_2 = r^2 - \ell$ and $\text{rank } \text{Ind}_2 = r(r+1)/2 - \ell$. For an order $(M^2 = M)$ the ranks are $r(r-1)$ and $r(r-1)/2$: the products account for half of $\mathbb{T}_2$, and the other half consists of symmetric multiplication relations.

Proof. $\mathbf{b}$ is onto, so by [Theorem 13.54](#) the products span the image of the injective map $x \wedge y \mapsto x \otimes y - y \otimes x$ of $\bigwedge_D^2 M$ into $M \otimes_D M$, which lies in the kernel because multiplication in k is commutative. The quotient of $M \otimes_D M$ by this image is $\text{Sym}_D^2 M$, multiplication factors through it, and every symmetric relation lifts to $\mathcal{T}(M, M)$. The ranks follow. $\square$

The rank $r^2 - \ell$ and its range are 17's [Corollary 13.36](#); new here are the product and the split into alternating and symmetric relations (14's [Theorem 13.20](#) computes exterior and symmetric powers of $L(M)$, not a Tor product). For a lattice that is not a ring the symmetric half can vanish: if $M = D + D\alpha$ with $1, \alpha, \alpha^2$ independent over $\text{Frac } D$, then $\ell = 3$, $\mathbb{T}_2 \cong D$ is spanned by the alternating tensor, and $\text{Ind}_2 = 0$ (compare 17's examples $\sqrt[3]{2}$ and τ). So the minimal polynomial of an element of degree at least three cannot be read from a two-generator $\mathbb{T}_2$ alone (21).

Framed reconstruction of orders. For an order $\mathfrak{o} \subset k$ over D and $I = J_g(\mathfrak{o})$, the *framed boundary data* are the scale X^g , the identification $\mathbb{T}_1/\text{IIT}_1 \cong \mathfrak{o}$ given by $\mathbf{b}$, under which the class of the generator X^g is $1 \in \mathfrak{o}$, and the resulting embedding $\mathbb{T}_2 \cong \mathcal{T}(\mathfrak{o}, \mathfrak{o}) \subseteq \mathfrak{o} \otimes_D \mathfrak{o}$. Equivalently, the two ideal sequences identify $\mathbb{T}_2$ with the kernel of the multiplication $I \otimes_A I \rightarrow I$, and the coefficient boundaries give the embedding ([Theorem 13.34](#)); it is an explicit map, not a chosen isomorphism of free groups.

Theorem 13.56 (Framed reconstruction [21]). *The D -module $\mathfrak{o}$, its element 1 and the submodule $\mathcal{T}(\mathfrak{o}, \mathfrak{o}) \subseteq \mathfrak{o} \otimes_D \mathfrak{o}$ determine the multiplication of $\mathfrak{o}$:*

$$xy = z \iff x \otimes y - 1 \otimes z \in \mathcal{T}(\mathfrak{o}, \mathfrak{o}).$$

A D -module isomorphism $f : \mathfrak{o} \rightarrow \mathfrak{o}'$ between orders is a unital ring isomorphism if and only if $f(1) = 1$ and $(f \otimes f)(\mathcal{T}(\mathfrak{o}, \mathfrak{o})) = \mathcal{T}(\mathfrak{o}', \mathfrak{o}')$.

Proof. The multiplication $\mathfrak{o} \otimes_D \mathfrak{o} \rightarrow \mathfrak{o}$ is split by $z \mapsto 1 \otimes z$, so for given x, y exactly one z puts $x \otimes y - 1 \otimes z$ in its kernel, namely $z = xy$. A unital ring isomorphism preserves the kernel. Conversely, if $z = xy$, applying $f \otimes f$ gives $f(x) \otimes f(y) - 1 \otimes f(z) \in \mathcal{T}(\mathfrak{o}', \mathfrak{o}')$, and uniqueness in $\mathfrak{o}'$ gives $f(z) = f(x)f(y)$. $\square$

The reconstruction is elementary once the kernel is located; the substance is that this embedded relation module is the second Tor over an omnific ring (21). It is a different mechanism from 17's [Theorem 13.39](#), which recovers the order as the constant quotient of $\text{End}_A(J_\gamma(\mathfrak{o}))$. Because D is integrally closed and $\mathfrak{o}$ is integral over D , $\mathfrak{o} \cap \text{Frac } D = D$, so 1 extends to a D -basis $e_0 = 1, e_1, \dots, e_{r-1}$ of $\mathfrak{o}$, and

$$e_i \otimes e_j - e_0 \otimes (e_i e_j) \quad (1 \leq i \leq r-1, 0 \leq j \leq r-1)$$

is a D -basis of $\mathcal{T}(\mathfrak{o}, \mathfrak{o})$: each has its own pivot $e_i \otimes e_j$ with coefficient 1, subtracting them reduces any tensor to some $e_0 \otimes z$, and such a tensor lies in the kernel only if $z = 0$. The structure constants,

the traces, the trace form and the discriminant are then finite matrix computations. For $\mathbb{Z}[\sqrt{m}]$ ($m > 0$ not a square) the basis is $e_1 \otimes e_0 - e_0 \otimes e_1$ (up to sign a product of degree-one classes) and $e_1 \otimes e_1 - m e_0 \otimes e_0$ (the square relation, which survives in Ind_2); the trace form is $\text{diag}(2, 2m)$ with discriminant $4m$. So $\mathbb{Z}[\sqrt{2}]$ and $\mathbb{Z}[\sqrt{3}]$ have abstract second Tor groups both isomorphic to $\mathbb{Z}^2$, while their framed relations recover the discriminants 8 and 12. For $\mathbb{Z}[\varphi]$, $\varphi^2 = \varphi + 1$, the second relation is $e_1 \otimes e_1 - e_0 \otimes e_1 - e_0 \otimes e_0$ and the discriminant is 5. For $\mathbb{Z}[\sqrt[3]{2}]$ the rank is six, the three alternating tensors and $e_1 \otimes e_1 - e_0 \otimes e_2$, $e_1 \otimes e_2 - 2e_0 \otimes e_0$, $e_2 \otimes e_2 - 2e_0 \otimes e_1$ form an integral basis, the trace form has rows $(3, 0, 0)$, $(0, 0, 6)$, $(0, 6, 0)$, and the discriminant is -108 . Over $\mathbb{Z}[i]$ the order $\mathbb{Z}[i][\sqrt{2}]$ has the same two relations as $\mathbb{Z}[\sqrt{2}]$ and discriminant 8; complex conjugation carries $J_g(M)$ to $J_g(\overline{M})$ and transports all the maps coefficientwise (21).

Theorem 13.57 (The derived constant-term shadow [17; 21]). *Let E be a D -module on which A acts through ct . Then in the derived category of D -modules*

$$A/I \otimes_A^{\mathbf{L}} E \cong E \oplus (M \otimes_D E)[1],$$

so $\text{Tor}_0^A(A/I, E) = E$, $\text{Tor}_1^A(A/I, E) \cong M \otimes_D E$ and $\text{Tor}_n^A(A/I, E) = 0$ for $n \geq 2$. For $E = D$ the algebra $\mathbf{P}_\bullet \otimes_A D$ is the square-zero differential graded algebra $D \oplus M[1]$.

Proof. $\Pi \otimes_A E \cong (\Pi/\Pi^2) \otimes_D E = 0$, so $\text{Syz}_M \otimes_A E = 0$. Hence $\mathbf{P}_\bullet \otimes_A E$ is E in degree 0 and E^r in degree 1, with zero differential because its coefficients $X^g m_j$ lie in Π ; the basis identifies E^r with $M \otimes_D E$. For $E = D$ the degree-two term is zero, so all products of degree-one elements vanish. $\square$

The Tor groups are 17's Theorem 13.42 (over R_κ^{11}), shifted by one degree; 21 proves them over $T_{\mathbb{Q}}$ and over the class ring in the reading above, and the square-zero model is new. Consequently all orders of one rank, for instance those of discriminants 8, 12 and 5, have isomorphic derived constant-term shadows, although their framed self-Tor data recover them: derived constant-term reduction keeps the additive lattice and erases its multiplication (21). Over the class ring every set-sized module is of this kind (Theorem 5.1(iv)); 21 does not need that theorem. On the finite-support core $T_{\mathbb{Q}}$ it would be false that every set-sized target factors through ct , since the augmentation sending every monomial to 1 is a map to k (Proposition 13.2(ii)); the theorem concerns the specified constant-term modules only (21).

14 Large internal fields and the binomial quotient

The universal theorem (Theorem 5.1) describes what a set-sized target can see of $\mathbf{Oz}$: every purely infinite element dies, and only the constant term survives. This section, taken from manuscript 07 together with one proposition of manuscript 06, shows how much survives in a large target once a single purely infinite element becomes a *unit*. The answer is a great deal: the target then contains a real closed Hahn field into which all of $\mathbf{No}$ embeds as an ordered field. In the nonzero binomial quotient $\mathcal{Q}_a = \mathbf{Oz}/(1 + \omega^a)$, which has no nonzero set-sized ring image at all, the same mechanism produces an algebraically closed class field containing a copy of $\mathbf{No}[i]$, exact finite cyclotomic algebras over it, and an atomless Boolean algebra of idempotents.

The dependence of the results is as follows (07). An ordered additive compression of $\mathbf{No}$ and the closedness of set-sized Hahn fields give a copy of $\mathbf{No}$ inside the compressed field $\mathcal{F}_a$ (Proposition 14.1 and Theorem 14.4); localization transfers $\mathcal{F}_a$ into every quotient in which a suitable purely infinite element is invertible (Theorem 14.7); a support argument gives exact binomial kernels (Theorem 14.15); and finite Chinese remainder decompositions give the internal

field $\mathcal{E}_a$ and the Cantor algebra (Corollary 14.17 and Theorem 14.19). The absence of small images of $\mathcal{Q}_a$ comes from the other branch, Theorems 5.1 and 6.1. No factorization conjecture enters either chain. What manuscript 07 calls its Main Theorem B is Theorem 14.7 together with Theorem 14.4, and its Main Theorem C is the combination of Lemma 14.14, Corollary 14.17, and Theorem 14.19 with the absence of small images of $\mathcal{Q}_a$ recorded before Lemma 14.14.

14.1 Compressed exponent fields

Fix $a > 0$. Recall from (2.4) and Lemma 4.2 that $H_a = \{h \in \mathbf{No} : n|h| < a \text{ for all ordinary } n \geq 1\}$ is a convex divisible subgroup of $(\mathbf{No}, +)$ containing a proper class of positive elements, and that $\mathcal{F}_a(k) = k((X^{H_a}))$ is a field for every set field k . For $k = \mathbb{R}$ and $k = \mathbb{C}$ we realize these fields inside the surreals and surcomplex numbers,

$$\begin{aligned}\mathcal{F}_a &= \mathcal{F}_a(\mathbb{R}) = \{f \in \mathbf{No} : \text{supp } f \subseteq H_a\}, \\ \mathcal{F}_a(i) &= \mathcal{F}_a(\mathbb{C}) = \{f \in \mathbf{No}[i] : \text{supp } f \subseteq H_a\}.\end{aligned}\tag{14.1}$$

Taking real and imaginary parts of coefficients shows $\mathcal{F}_a(i) = \mathcal{F}_a + i\mathcal{F}_a$. Manuscript 07 calls them *compressed exponent fields*; the phrase refers to the restriction on exponents, not to finiteness of supports. Both are proper classes. Manuscript 07 checks directly that H_a is a subgroup by the inequality $n|h + h'| \leq n|h| + n|h'| < a/2 + a/2$, using the defining inequalities at $2n$; this is part of Lemma 4.2.

Proposition 14.1 (Closedness of the compressed fields [07]). *The class $\mathcal{F}_a$ is a real closed subfield of $\mathbf{No}$, and $\mathcal{F}_a(i)$ is an algebraically closed subfield of $\mathbf{No}[i]$.*

Proof. Since H_a is a subgroup, sums and products of series supported in H_a are supported in H_a ; the inverse of a nonzero Hahn series has support in the subgroup generated by its own support (factor out the leading monomial and use the Hahn geometric series), so it stays in H_a as well. Hence $\mathcal{F}_a$ and $\mathcal{F}_a(i)$ are subfields (Lemma 4.2).

Let P be a nonconstant polynomial with coefficients in $\mathcal{F}_a(i)$. The union of the supports of its finitely many coefficients is a set; its rational span is a set-sized divisible ordered subgroup $G \subseteq H_a$, because H_a is a divisible subgroup. The series with support in G form the set-sized Hahn fields $\mathbb{R}((\omega^G)) \subseteq \mathcal{F}_a$ and $\mathbb{C}((\omega^G)) \subseteq \mathcal{F}_a(i)$, whose operations are those of $\mathbf{No}$ and $\mathbf{No}[i]$, and P has coefficients in $\mathbb{C}((\omega^G))$. For divisible G this field is algebraically closed by the Hahn-field theorem [27, Corollary 4] (stated there for well-ordered supports; the substitution $t^g = \omega^{-g}$ converts). So P has a root in $\mathbb{C}((\omega^G)) \subseteq \mathcal{F}_a(i)$, and $\mathcal{F}_a(i)$ is algebraically closed.

For the real field, note first that $\mathbb{R}((\omega^G))$ is real closed. If $0 < f \in \mathbb{R}((\omega^G))$, write a square root of f in $\mathbb{C}((\omega^G))$ as $u + iv$ with $u, v \in \mathbb{R}((\omega^G))$; then $u^2 - v^2 = f$ and $2uv = 0$. If $u = 0$ then $f = -v^2 \leq 0$, which is impossible; so $v = 0$ and $u^2 = f$. If Q has odd degree and coefficients in $\mathbb{R}((\omega^G))$, coefficientwise conjugation is an automorphism of $\mathbb{C}((\omega^G))$ fixing Q , so the nonreal roots of Q occur in conjugate pairs with equal multiplicities, and an odd number of roots counted with multiplicity must be real. An ordered field in which positive elements are squares and odd-degree polynomials have roots is real closed. Now let $0 < f \in \mathcal{F}_a$, or let Q be of odd degree over $\mathcal{F}_a$; choosing G as above for f or for the coefficients of Q , the required square root or root lies in $\mathbb{R}((\omega^G)) \subseteq \mathcal{F}_a$. Hence $\mathcal{F}_a$ is real closed. Only set-sized instances of the classical theorem have been used. $\square$

Remark 14.2 (Bounded in the ambient field, yet a proper class [07]). Every positive $f \in \mathcal{F}_a$ satisfies $\omega^{-a} < f < \omega^a$: its leading exponent h satisfies $|h| < a/2$ and its leading coefficient is a positive

real. A field contained in such an ambient interval need not be small, since the interval is itself a proper class; $\mathcal{F}_a$ is a proper class, and by [Theorem 14.4](#) it even contains an ordered-field copy of $\mathbf{No}$. Bounds in the surrounding field do not give bounds inside an isomorphic copy of $\mathbf{No}$.

14.2 An ordered additive compression of the surreals

Knowing that $\mathcal{F}_a$ contains Hahn fields of every set size ([Lemma 4.2](#)) would already show considerable largeness. Manuscript 07 does more: it embeds all of $\mathbf{No}$ into $\mathcal{F}_a$ as an ordered field. Define

$$\vartheta(x) = \frac{1}{2} \left(\frac{x}{1+|x|} - 3 \right), \quad \vartheta : \mathbf{No} \longrightarrow (-2, -1). \quad (14.2)$$

For $x \geq 0$ the inner fraction is $x/(1+x) = 1 - 1/(1+x) \in [0, 1)$, and for $x < 0$ it is $x/(1-x) = -1 + 1/(1-x) \in (-1, 0)$. Each branch is strictly increasing and they agree at 0, so ϑ is strictly increasing with values in $(-2, -1)$. For a real-coefficient normal form put

$$\Psi \left(\sum_{\gamma} r_{\gamma} \omega^{\gamma} \right) = \sum_{\gamma} r_{\gamma} \omega^{\vartheta(\gamma)}, \quad \Psi_a(x) = a\Psi(x). \quad (14.3)$$

(Manuscript 07 writes E_a for Ψ_a ; the letter $\mathcal{E}_a$ is reserved here for the internal field of [Corollary 14.17](#).)

Lemma 14.3 (Ordered additive compression [07]). *The map $\Psi_a : (\mathbf{No}, +) \rightarrow (H_a, +)$ is an injective, order-preserving, $\mathbb{R}$ -linear homomorphism of additive groups.*

Proof. A strictly increasing map sends a reverse well-ordered set to a reverse well-ordered set, so the support $\vartheta(\text{supp } x)$ is a reverse well-ordered set and $\Psi(x)$ exists. Since ϑ is injective, distinct exponents stay distinct, and the coefficient of $\Psi(x)$ at $\vartheta(\gamma)$ is x_{γ} ; hence coefficientwise addition, with all cancellations, and multiplication by real scalars are preserved. If $x \neq 0$, the greatest exponent γ_0 of $\text{supp } x$ is sent to the greatest exponent $\vartheta(\gamma_0)$ of $\text{supp } \Psi(x)$ with the same coefficient; since the sign of a nonzero normal form is the sign of its leading coefficient, Ψ maps positive elements to positive elements, and being additive it is injective and order preserving. The leading exponent of a nonzero $\Psi(x)$ lies in $(-2, -1)$, so $\Psi(x)$ is infinitesimal and $n|\Psi(x)| < 1$ for every ordinary $n \geq 1$. Multiplying by $a > 0$ gives $n|\Psi_a(x)| < a$, that is, $\Psi_a(x) \in H_a$, and multiplication by the positive surreal a preserves additivity, $\mathbb{R}$ -linearity and order. $\square$

The map ϑ is not additive, and Ψ is not asserted to be multiplicative. The construction deliberately has two levels: first an additive exponent group is compressed, and only then is the additive map used as a substitution on monomials.

Theorem 14.4 (A full surreal field in every compressed field [07]). *The formula*

$$\Theta_a \left(\sum_{\gamma} r_{\gamma} \omega^{\gamma} \right) = \sum_{\gamma} r_{\gamma} \omega^{\Psi_a(\gamma)} \quad (14.4)$$

defines an injective ordered-field homomorphism $\Theta_a : \mathbf{No} \hookrightarrow \mathcal{F}_a$ fixing $\mathbb{R}$. It preserves every set-indexed strongly summable family. Extending coefficients, $x + iy \mapsto \Theta_a(x) + i\Theta_a(y)$, gives an injective field homomorphism $\mathbf{No}[i] \hookrightarrow \mathcal{F}_a(i)$ fixing $\mathbb{C}$.

Proof. The support of the right side of (14.4) is $\Psi_a(\text{supp } x)$, the image of a reverse well-ordered set under a strictly increasing map into H_a ([Lemma 14.3](#)); so the output exists and lies in $\mathcal{F}_a$. If (x_{λ}) is a strongly summable set-indexed family, the union of the supports is reverse well ordered and

each exponent receives finitely many contributions; since Ψ_a is injective and strictly increasing, the same holds for the family $(\Theta_a(x_\lambda))$ with the same contributions at $\Psi_a(\gamma)$ as at γ , so $\Theta_a(\sum_\lambda x_\lambda) = \sum_\lambda \Theta_a(x_\lambda)$.

For multiplication, write $x = \sum r_\gamma \omega^\gamma$ and $y = \sum s_\delta \omega^\delta$. Because Ψ_a is additive and injective, $\Psi_a(\gamma) + \Psi_a(\delta) = \Psi_a(\varepsilon)$ holds exactly when $\gamma + \delta = \varepsilon$, and every sum $\Psi_a(\gamma) + \Psi_a(\delta)$ lies in the image of Ψ_a . Hence the coefficient of $\Theta_a(x)\Theta_a(y)$ at $\Psi_a(\varepsilon)$ is $\sum_{\gamma+\delta=\varepsilon} r_\gamma s_\delta = (xy)_\varepsilon$, and its coefficients at exponents outside the image vanish; so $\Theta_a(xy) = \Theta_a(x)\Theta_a(y)$. Additivity is clear, and constants are fixed because $\Psi_a(0) = 0$. A nonzero x has a nonzero image whose leading coefficient is that of x , so Θ_a is injective and preserves signs, hence order. It is an injective unital ring map between fields, hence a field embedding. The complex extension is the same coefficient-and-support argument with complex coefficients. $\square$

Example 14.5 ([07]). Since $\vartheta(0) = -3/2$, one has $\Psi_a(1) = a\omega^{-3/2}$ and

$$\Theta_a(\omega) = \omega^{a\omega^{-3/2}}.$$

No preservation of the ordinal embedding, of birthdays or simplicity, or of the surreal exponential is asserted or needed: Θ_a is an explicit ordered-field embedding with a controlled normal-form support image. Constructions of self-similar surreal subclasses and strongly linear maps have a substantial prior literature, notably the surreal substructures of Bagayoko and van der Hoeven [3]. Manuscript 07 uses the elementary construction above to make the transfer theorem independent of a general classification of surreal substructures, and does not claim that surreal self-similarity itself is new.

14.3 Transfer by inverting a purely infinite element

Lemma 14.6 (Multiplication by a purely infinite denominator [07]). *Let $0 \neq g \in \Pi$ and choose $a > 0$ with $a < \gamma$ for every $\gamma \in \text{supp } g$ (possible by Lemma 3.1). If $f \in \mathcal{F}_a$, then $gf \in \Pi$. The same holds for $0 \neq g \in \Pi_{\mathbb{C}}$ and $f \in \mathcal{F}_a(i)$.*

Proof. Every $h \in \text{supp } f$ satisfies $|h| < a/2$. Every exponent occurring in the product before cancellation has the form $\gamma + h$ with $\gamma \in \text{supp } g$, and $\gamma + h > a - a/2 > 0$. Hahn multiplication is defined and cancellation cannot create an exponent outside the sum of the supports, so gf has strictly positive support and real (respectively complex) coefficients. $\square$

Theorem 14.7 (Localization and quotient transfer [07]). *Under the hypotheses of Lemma 14.6, the subfield $\mathcal{F}_a$ of $\mathbf{No}$ lies in the localization $\mathbf{Oz}[1/g] \subseteq \mathbf{No}$. If $q : \mathbf{Oz} \rightarrow B$ is a unital homomorphism to a nonzero class ring and $q(g)$ is invertible in B , then*

$$\iota_g(f) = q(gf)q(g)^{-1} \quad (f \in \mathcal{F}_a) \quad (14.5)$$

is an injective field homomorphism $\mathcal{F}_a \hookrightarrow B$, and $\iota_g \circ \Theta_a$ is an injective field homomorphism $\mathbf{No} \hookrightarrow B$. For $q : \mathbf{Oz}[i] \rightarrow B$ and $0 \neq g \in \Pi_{\mathbb{C}}$ with $q(g)$ invertible, the same formula gives an injective field homomorphism $\mathcal{F}_a(i) \hookrightarrow B$, and hence a copy of $\mathbf{No}[i]$ in B . In either case B is a proper class.

Proof. Inside $\mathbf{No}$, $f = (gf)/g$ with $gf \in \Pi \subseteq \mathbf{Oz}$ by Lemma 14.6, so $f \in \mathbf{Oz}[1/g]$. The elements of $q(\mathbf{Oz})$ commute pairwise, and $q(g)^{-1}$ commutes with every element y commuting with $q(g)$ (multiply $q(g)y = yq(g)$ on both sides by $q(g)^{-1}$). Hence $q(\mathbf{Oz})$ and $q(g)^{-1}$ generate a commutative

subring of B , and all computations below take place there; no commutativity of B is needed. The map (14.5) is additive and $\iota_g(1) = 1$. For $f, f' \in \mathcal{F}_a$ the elements gf, gf', gff' lie in $\mathbf{Oz}$ and $(gf)(gf') = g \cdot (gff')$, so

$$\iota_g(f)\iota_g(f') = q(gf)q(gf')q(g)^{-2} = q(g)q(gff')q(g)^{-2} = \iota_g(ff').$$

(Equivalently, ι_g is the restriction to $\mathcal{F}_a$ of the extension of q to $\mathbf{Oz}[1/g]$ given by the universal property of localization.) If $f \neq 0$ and $\iota_g(f) = 0$, then $1_B = \iota_g(f)\iota_g(f^{-1}) = 0$, contradicting $B \neq 0$; so ι_g is injective. Composing with Theorem 14.4 gives the copy of $\mathbf{No}$. The complex case is identical, with Lemma 14.6 for $\Pi_{\mathbb{C}}$ and the complex extension of Θ_a . Finally, if B were set-sized, Theorem 5.1 would give $q(g) = 0$, which is not invertible in a nonzero ring. $\square$

Remark 14.8 ([07]). If the target B is noncommutative, the image of the localization still lies in a commutative subring, as the proof shows; no commutativity condition on all of B is needed. The hypothesis on a is always available in the full surreal exponent class, even when the exponents of g approach zero inside a smaller ordered group (Lemma 3.1). No inference of this kind is valid for an arbitrary fixed set-sized exponent group without an additional lower-bound hypothesis.

Corollary 14.9 (Quotients by a unit-constant perturbation [07]). *If $0 \neq g \in \Pi$, then $\mathbf{Oz}/(1+g)$ is nonzero and contains a class copy of $\mathbf{No}$. If $0 \neq g \in \Pi_{\mathbb{C}}$, then $\mathbf{Oz}[i]/(1+g)$ is nonzero and contains a class copy of $\mathbf{No}[i]$. Neither quotient has a unital homomorphism to a nonzero set-sized ring.*

Proof. The element $1+g$ is nonconstant, hence a nonunit by Lemma 2.4, so the quotients are nonzero. In the quotient $g = -1$ is a unit, so Theorem 14.7 applies. Since $\text{ct}(1+g) = 1$, the ideal $(1+g)$ has constant-term image all of $\mathbb{Z}$ (respectively $\mathbb{Z}[i]$), and Theorem 6.1 makes the universal set-sized image the zero ring; a unital map from the zero ring exists only to the zero ring. $\square$

Corollary 14.10 (A conditional statement about nonarithmetic primes [07]). *Let P be a prime ideal of $\mathbf{Oz}$ such that $P \cap \mathbb{Z} = 0$ and $P \not\subseteq \Pi$. Then $\mathbf{Oz}/P$ contains a class copy of $\mathbf{No}$ and is a $\mathbb{Q}$ -algebra. Likewise, if P is a prime ideal of $\mathbf{Oz}[i]$ with $P \cap \mathbb{Z}[i] = 0$ and $P \not\subseteq \Pi_{\mathbb{C}}$, then $\mathbf{Oz}[i]/P$ contains a class copy of $\mathbf{No}[i]$ and is a $\mathbb{Q}(i)$ -algebra.*

Proof. Choose $f \in P$ with nonzero constant term c . The series f/c still lies in the omnific ring: its constant term is 1 and its other coefficients are real (respectively complex). Since $f = c \cdot (f/c)$, P is prime and $c \notin P$, we get $f/c \in P$. Write $f/c = 1+g$ with g purely infinite; $g \neq 0$ because $c \notin P$. In the quotient $g = -1$ is a unit, and Theorem 14.7 applies. For every nonzero ordinary coefficient d ($d \in \mathbb{Z}$, respectively $d \in \mathbb{Z}[i]$), g/d lies in the ring, and in the quotient $d \cdot (-g/d) = -g = 1$; so d is invertible, which gives the $\mathbb{Q}$ -, respectively $\mathbb{Q}(i)$ -, algebra structure. $\square$

The corollary is deliberately conditional. Manuscript 07 does not assert that prime ideals with these properties exist; see Section 14.8.

Example 14.11 (Invertibility cannot be replaced by nonvanishing [07]). The quotient $\mathbf{Oz}/(\omega^a)$ is nonzero, and the class of $\omega^{a/2}$ is a nonzero element of square zero: it is nonzero because $\omega^{a/2}/\omega^a = \omega^{-a/2} \notin \mathbf{Oz}$. Since $(\omega^a) \subseteq \Pi$, this quotient still maps onto $\mathbb{Z}$ by constant terms, and so it cannot contain a unital copy of $\mathbb{Q}$: composing such a copy with that map would give a unital map $\mathbb{Q} \rightarrow \mathbb{Z}$. A surviving nilpotent and a surviving unit are therefore substantively different.

The same mechanism, with a fraction field in place of the quotient, applies to every prime ideal not containing the purely infinite ideal. Here X^g is the monomial of $k((X^{\mathbf{No}}))$, which is ω^g when $k = \mathbb{R}$ or $k = \mathbb{C}$.

Proposition 14.12 (A scaled field in every non-arithmetic residue field [06]). *Let k be a set field, $D \subseteq k$ a unital subring, $A = \mathcal{A}_{D,k}$, and let P be a prime class ideal of A with $\Pi_k \not\subseteq P$. There exists $a > 0$ for which the class field $\mathcal{F}_a(k) = k((X^{H_a}))$ embeds in $\text{Frac}(A/P)$.*

Proof. Choose $f \in \Pi_k \setminus P$ and write $f = X^g h$ with $g > 0$ and $h \in \Pi_k$ (Lemma 3.1 applied to $\text{supp } f$). Then $X^g \notin P$, since otherwise $f \in P$. Put $a = g/2$ and $m = X^a$. As $m^2 = X^g \notin P$ and P is prime, $\bar{m} \neq 0$ in the domain A/P . Since $m\mathcal{F}_a(k) \subseteq \Pi_k \subseteq A$ (Lemma 4.2), we may define

$$\iota(x) = \frac{\overline{mx}}{\bar{m}} \in \text{Frac}(A/P) \quad (x \in \mathcal{F}_a(k)).$$

Additivity and $\iota(1) = 1$ are immediate. Multiplicativity follows by reducing the identity $(mx)(my) = m(mxy)$ in A and dividing by $\bar{m}^2$. For $x \neq 0$, the identity $(mx)(m/x) = m^2$ in A shows $\overline{mx} \neq 0$, because $\bar{m}^2 \neq 0$. Thus the unital field homomorphism ι is injective. $\square$

This is the scaled-field collision mechanism of Lemma 4.1 in a fraction field. The conclusion says more than that the residue field is large: it contains a specific Hahn field built on all exponent scales infinitesimal relative to one positive a . It is a necessary structural condition on such residue fields; manuscript 06 states that it does not classify those prime ideals, assert that they exist, or assert that every field of this form is a residue field.

Remark 14.13 (The composed embedding [merge]). Let P be a prime class ideal of $\mathbf{Oz}$ with $\Pi \not\subseteq P$. Then there is an injective field homomorphism $\mathbf{No} \hookrightarrow \text{Frac}(\mathbf{Oz}/P)$ whose first factor is an ordered-field embedding,

$$\mathbf{No} \xrightarrow{\Theta_a} \mathcal{F}_a \xrightarrow{\iota} \text{Frac}(\mathbf{Oz}/P).$$

Indeed, Proposition 14.12 with $(D, k) = (\mathbb{Z}, \mathbb{R})$ gives $a > 0$ and an injective field homomorphism $\iota : \mathcal{F}_a \rightarrow \text{Frac}(\mathbf{Oz}/P)$, and Theorem 14.4 gives Θ_a ; a composite of injective field homomorphisms is one. With $(D, k) = (\mathbb{Z}[i], \mathbb{C})$ and the complex extension of Θ_a , every prime P of $\mathbf{Oz}[i]$ with $\Pi_{\mathbb{C}} \not\subseteq P$ gives $\mathbf{No}[i] \hookrightarrow \text{Frac}(\mathbf{Oz}[i]/P)$. Neither source states this composition.

The hypotheses of Proposition 14.12 and Corollary 14.10 are different. If $P \cap \mathbb{Z} = 0$ and $P \not\subseteq \Pi$, then $\Pi \not\subseteq P$: otherwise, for $x \in P$ we would have $x - \text{ct}(x) \in \Pi \subseteq P$, hence $\text{ct}(x) \in P \cap \mathbb{Z} = 0$, so $P \subseteq \Pi$. Conversely, $\Pi \not\subseteq P$ forces $P \cap \mathbb{Z} = 0$, since a nonzero $n \in P$ would give $\Pi = n\Pi \subseteq P$ (Proposition 3.3), but it allows $P \subseteq \Pi$, for instance $P = 0$, where the statement is the trivial $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ (Proposition 3.4). Thus 07's hypothesis is the stronger one and yields the stronger conclusion (a copy of $\mathbf{No}$ inside $\mathbf{Oz}/P$ itself, and a $\mathbb{Q}$ -algebra structure), while 06's hypothesis also covers primes contained in Π , with the conclusion only in the fraction field.

14.4 The binomial quotient and its localization

Fix $a > 0$ and write

$$\mathcal{Q}_a = \mathbf{Oz}/(1 + \omega^a), \quad q_a : \mathbf{Oz} \longrightarrow \mathcal{Q}_a$$

for the quotient map (07's B_a). The element $1 + \omega^a$ is a nonunit (Lemma 2.4), so $\mathcal{Q}_a \neq 0$; and since $\text{ct}(1 + \omega^a) = 1$, Theorem 6.1 shows that $\mathcal{Q}_a$ has no unital homomorphism to a nonzero set-sized ring. A purely arithmetic calculation shows why: in a set-sized target ω^a becomes 0 (Theorem 5.1), so $1 + \omega^a$ becomes 1, and a target in which it vanishes has $1 = 0$. Its failure to have small ring images is therefore not caused by its being zero, a field, or a domain; it is none of these by Corollary 14.22. By Corollary 14.9 it contains a copy of $\mathbf{No}$; the rest of this section identifies much more of its internal structure.

The support description of the localization

$$\mathbf{Oz}[\omega^{-a}] = \{f \in \mathbf{No} : \text{supp } f \subseteq [-Na, \infty) \text{ for some ordinary } N \geq 0\}$$

is [Proposition 6.11](#); manuscripts 04 and 07 both prove it. (07 calls this ring L_a .)

Lemma 14.14 (Localization of the binomial quotient [07]).

(i) $\omega^a \mathcal{F}_a \subseteq \Pi$; in particular $\mathcal{F}_a \subseteq \mathbf{Oz}[\omega^{-a}]$.

(ii) There is a unique ring homomorphism $\tilde{q}_a : \mathbf{Oz}[\omega^{-a}] \rightarrow \mathcal{Q}_a$ extending q_a . It is surjective with kernel $(1 + \omega^a)\mathbf{Oz}[\omega^{-a}]$, so

$$\mathcal{Q}_a \cong \mathbf{Oz}[\omega^{-a}]/(1 + \omega^a)\mathbf{Oz}[\omega^{-a}]. \quad (14.6)$$

(iii) The restriction of $\tilde{q}_a$ to $\mathcal{F}_a$ is the injective field homomorphism

$$\iota_a(f) = -q_a(\omega^a f) \quad (f \in \mathcal{F}_a). \quad (14.7)$$

Proof. (i) Every $h \in H_a$ satisfies $a + h > 0$, so translating a support in H_a by a gives a strictly positive support ([Lemma 4.2](#)); then $f = \omega^{-a}(\omega^a f)$.

(ii) The subring of $\mathbf{No}$ generated by $\mathbf{Oz}$ and ω^{-a} consists of the finite sums $\sum_{j \leq N} h_j \omega^{-ja} = \omega^{-Na} \sum_{j \leq N} h_j \omega^{(N-j)a}$, so every element has the form $\omega^{-Na} h$ with $N \geq 0$ and $h \in \mathbf{Oz}$. Any extension of q_a sends ω^{-a} to $q_a(\omega^a)^{-1} = -1$, which proves uniqueness; define $\tilde{q}_a(\omega^{-Na} h) = (-1)^N q_a(h)$. This is well defined: if $\omega^{-Na} h = \omega^{-Ma} h'$ with $N \leq M$, then $h' = \omega^{(M-N)a} h$ and $(-1)^M q_a(h') = (-1)^M (-1)^{M-N} q_a(h) = (-1)^N q_a(h)$. Writing two elements over a common power ω^{-Na} shows that $\tilde{q}_a$ is a ring homomorphism; it extends q_a and is surjective. If $\tilde{q}_a(\omega^{-Na} h) = 0$, then $q_a(h) = 0$, so $h = (1 + \omega^a)y$ with $y \in \mathbf{Oz}$ and $\omega^{-Na} h = (1 + \omega^a)\omega^{-Na} y \in (1 + \omega^a)\mathbf{Oz}[\omega^{-a}]$. Conversely $\tilde{q}_a((1 + \omega^a)\omega^{-Na} y) = (-1)^N q_a((1 + \omega^a)y) = 0$ for $y \in \mathbf{Oz}$.

(iii) By (i) and (ii), $\tilde{q}_a(f) = \tilde{q}_a(\omega^{-a} \cdot \omega^a f) = -q_a(\omega^a f)$. A unital ring homomorphism from a field to a nonzero ring is injective. $\square$

Henceforth $\mathcal{F}_a$ is identified with its image $\iota_a(\mathcal{F}_a) \subseteq \mathcal{Q}_a$ whenever we work inside $\mathcal{Q}_a$; all coefficient fields below are embedded with the identity of $\mathcal{Q}_a$. This identification is not constant-term extraction: a nonintegral real scalar r is represented in $\mathcal{Q}_a$ by the omnific integer $-r\omega^a$.

14.5 Exact binomial kernels

For an ordinary integer $n \geq 1$ put $x_n = q_a(\omega^{a/n}) \in \mathcal{Q}_a$. Certainly $x_n^n = q_a(\omega^a) = -1$. The substantive question is whether the full ambient omnific ring, with its arbitrary surreal exponents, creates additional polynomial relations over $\mathcal{F}_a$.

Theorem 14.15 (Exact binomial kernel [07]). *For every $n \geq 1$, evaluation at x_n induces an injective ring homomorphism*

$$\mathcal{F}_a[T]/(T^n + 1) \hookrightarrow \mathcal{Q}_a. \quad (14.8)$$

Equivalently, the kernel of $\mathcal{F}_a[T] \rightarrow \mathcal{Q}_a$, $T \mapsto x_n$, is exactly $(T^n + 1)$.

Proof. The polynomial $T^n + 1$ lies in the kernel. Dividing by this monic polynomial over the field $\mathcal{F}_a$, it suffices to show that no nonzero remainder

$$R(T) = \sum_{j=0}^{n-1} r_j T^j \quad (r_j \in \mathcal{F}_a)$$

vanishes at x_n . Put $\xi = \omega^{a/n} \in \mathbf{Oz}$; then $R(\xi) \in \mathbf{Oz}[\omega^{-a}]$ by Lemma 14.14(i), and $\tilde{q}_a(R(\xi)) = R(x_n)$.

The quotient in $\mathbf{No}$. The series $U = \sum_{k \geq 0} (-1)^k \omega^{-(k+1)a}$ has the decreasing support $\{-(k+1)a\}$, and $(1 + \omega^a)U = \sum_{k \geq 0} (-1)^k \omega^{-(k+1)a} + \sum_{k \geq 0} (-1)^k \omega^{-ka} = 1$, each exponent receiving at most two contributions that cancel except at exponent 0. Hence $U = (1 + \omega^a)^{-1}$ in $\mathbf{No}$, and multiplying the finite sum $R(\xi) = \sum_j r_j \omega^{ja/n}$ by U gives

$$\frac{R(\xi)}{1 + \omega^a} = \sum_{k \geq 0} \sum_{j=0}^{n-1} (-1)^k r_j \omega^{(j/n-k-1)a}, \quad (14.9)$$

provided the right side is a strongly summable family, which we now check together with the absence of cancellation between blocks.

Block cosets. The support of the block indexed by (j, k) lies in the coset $c_{j,k} + H_a$ with centre $c_{j,k} = (j/n - k - 1)a$, because $\text{supp } r_j \subseteq H_a$. Distinct pairs give distinct cosets. Indeed, $c_{j,k} - c_{j',k'} = (m/n)a$ with $m = j - j' - n(k - k') \in \mathbb{Z}$, and $(m/n)a \in H_a$ only for $m = 0$, since for $m \neq 0$ one has $n \cdot |(m/n)a| = |m|a \geq a$. From $j - j' = n(k - k')$ and $|j - j'| < n$ we get $j = j'$ and $k = k'$. Cosets of H_a are disjoint, so distinct blocks have disjoint supports, and each exponent receives a contribution from at most one block.

Reverse well-ordering. As (j, k) runs over $\{0, \dots, n-1\} \times \mathbb{N}$, the integer $j - n(k+1)$ runs bijectively over the integers $m \leq -1$, so the centres are the points $(m/n)a$, $m \leq -1$, a strictly descending discrete progression with step a/n . Every offset $h \in H_a$ satisfies $4n|h| < a$, that is, $|h| < a/(4n)$, so the block with centre $(m/n)a$ lies in the open interval of radius $a/(4n)$ about its centre; these intervals are pairwise disjoint and ordered as their centres. Given a nonempty subset S of the union of the supports, the set of m whose block meets S is a nonempty set of integers bounded above by -1 , so it has a largest element m^* ; the elements of S in block m^* exceed all other elements of S , and since that block has reverse well-ordered support, S has a greatest element. All supports and the index family are sets. Hence the right side of (14.9) is a legitimate strongly summable Hahn expression, and (14.9) holds because Hahn multiplication distributes over strongly summable families. (Equivalently, multiplying the right side by $1 + \omega^a$ cancels consecutive k -blocks and leaves exactly $R(\xi)$.)

Unbounded-below support. Choose j with $r_j \neq 0$ and $h \in \text{supp } r_j$. For every $k \geq 0$ the exponent $h + (j/n - k - 1)a$ lies only in block (j, k) , so it occurs in (14.9) with the nonzero coefficient $(-1)^k (r_j)_h$. Since $h < a$ and $j/n < 1$, this exponent is less than $a + a - (k+1)a = (1-k)a$, which is below $-Na$ as soon as $k \geq N+1$. Thus the support of (14.9) is not contained in $[-Na, \infty)$ for any ordinary N , and by Proposition 6.11 the quotient $R(\xi)/(1 + \omega^a)$ does not belong to $\mathbf{Oz}[\omega^{-a}]$.

Conclusion. If $R(x_n) = 0$, then $R(\xi) \in \ker \tilde{q}_a = (1 + \omega^a)\mathbf{Oz}[\omega^{-a}]$ by Lemma 14.14(ii), so $R(\xi) = (1 + \omega^a)u$ with $u \in \mathbf{Oz}[\omega^{-a}]$. Division in the field $\mathbf{No}$ identifies u with (14.9), a contradiction. Hence $R(x_n) \neq 0$. $\square$

Remark 14.16 ([07]). The proof does not restrict the exponents of an alleged ambient factor u to rational multiples of a ; the full support characterization of $\mathbf{Oz}[\omega^{-a}]$ is exactly what rules out such hidden factors. A calculation solely in a finite group algebra (the span of the monomials $\omega^{ja/n}$) would not establish the theorem.

Corollary 14.17 (Internal algebraically closed field [07]). *Let $u = x_2 = q_a(\omega^{a/2})$. The subring $\mathcal{E}_a = \mathcal{F}_a[u] \subseteq \mathcal{Q}_a$ is a field, isomorphic over $\mathcal{F}_a$ to $\mathcal{F}_a(i)$ by $i \mapsto u$. It is algebraically closed and contains a class copy of $\mathbf{No}[i]$.*

Proof. By [Theorem 14.15](#) with $n = 2$, $\mathcal{E}_a \cong \mathcal{F}_a[T]/(T^2+1)$. Since $\mathcal{F}_a$ is real closed ([Proposition 14.1](#)), -1 is not a square in it, $T^2 + 1$ is irreducible, and the quotient is the field $\mathcal{F}_a + i\mathcal{F}_a = \mathcal{F}_a(i)$, which is algebraically closed by [Proposition 14.1](#). The complex extension of Θ_a ([Theorem 14.4](#)) embeds $\mathbf{No}[i]$ in it. $\square$

There is no order contradiction. Although $\omega^{a/2}$ is positive in $\mathbf{Oz}$, its square becomes -1 in $\mathcal{Q}_a$; the order of $\mathbf{Oz}$ does not descend to $\mathcal{Q}_a$. The corollary is a statement about rings and fields, not about ordered quotients.

More generally, finite real factorization and the Chinese remainder theorem give

$$\mathcal{F}_a[T]/(T^n + 1) \cong \begin{cases} \mathcal{E}_a^m, & n = 2m, \\ \mathcal{F}_a \times \mathcal{E}_a^m, & n = 2m + 1. \end{cases} \quad (14.10)$$

The roots of $T^n + 1$ are the n distinct ordinary complex numbers $\exp(i\pi(2k+1)/n)$, which lie in $\mathbb{C} \subseteq \mathcal{F}_a(i)$. For even n none is real, and the m conjugate pairs give m distinct monic quadratic factors with real coefficients, irreducible over $\mathcal{F}_a$ because their roots are not in $\mathcal{F}_a$. For odd n the only real root is -1 , and the remaining roots give m such quadratics. All roots are simple, so the factors are pairwise coprime, and each quadratic quotient is isomorphic to $\mathcal{F}_a(i) \cong \mathcal{E}_a$. The isomorphisms in (14.10) need not identify a chosen imaginary unit in the same way at different stages; the dyadic tower below makes coherent choices.

14.6 The dyadic tower and its Cantor algebra

For $r \geq 1$ define

$$\xi_r = q_a(\omega^{a/2^r}) = x_{2^r}, \quad \mathcal{S}_r = \mathcal{F}_a[\xi_r] \subseteq \mathcal{Q}_a$$

(07's $\mathcal{S}_r$). Then $\xi_r^2 = \xi_{r-1}$ for $r \geq 2$, $\mathcal{S}_r \subseteq \mathcal{S}_{r+1}$, and $\mathcal{S}_1 = \mathcal{E}_a$, so each $\mathcal{S}_r$ is an $\mathcal{E}_a$ -algebra. In this subsection i denotes the element $u = \xi_1$ of $\mathcal{E}_a$; under $\mathcal{E}_a \cong \mathcal{F}_a(i)$ the ordinary complex field $\mathbb{C} \subseteq \mathcal{F}_a(i)$ corresponds to $\mathbb{R} + \mathbb{R}u \subseteq \mathcal{E}_a$, and ordinary complex numbers are read in $\mathcal{E}_a$ through this identification.

Lemma 14.18 (A single stage over the fixed complex field [07]). *Let $m = 2^{r-1}$. As $\mathcal{E}_a$ -algebras,*

$$\mathcal{S}_r \cong \mathcal{E}_a[T]/(T^m - i), \quad T \mapsto \xi_r. \quad (14.11)$$

Proof. The $\mathcal{E}_a$ -algebra map $\mathcal{E}_a[T] \rightarrow \mathcal{S}_r$, $T \mapsto \xi_r$, is surjective, since $\mathcal{E}_a[\xi_r] = \mathcal{F}_a[\xi_r^m, \xi_r] = \mathcal{S}_r$, and it kills $T^m - i$ because $\xi_r^m = \xi_1 = i$. For injectivity on the quotient, suppose $\sum_{j < m} (a_j + b_j i) \xi_r^j = 0$ with $a_j, b_j \in \mathcal{F}_a$. Substituting $i = \xi_r^m$ gives $\sum_{j < m} a_j \xi_r^j + \sum_{j < m} b_j \xi_r^{j+m} = 0$, a polynomial relation over $\mathcal{F}_a$ of degree less than $2m = 2^r$ satisfied by $\xi_r = x_{2^r}$. By [Theorem 14.15](#) with $n = 2^r$ all its coefficients vanish, so all a_j and b_j are zero. $\square$

The roots of $T^m - i$ can be chosen in the ordinary complex subfield of $\mathcal{E}_a$:

$$\zeta_{r,k} = \exp\left(i \frac{\pi/2 + 2\pi k}{2^{r-1}}\right) \quad (k \in \mathbb{Z}/2^{r-1}\mathbb{Z}). \quad (14.12)$$

The expression is evaluated in $\mathbb{C}$ and then embedded in $\mathcal{E}_a$; no exponential function on surcomplex arguments is assumed. These m roots are distinct, and

$$\zeta_{r+1,k}^2 = \zeta_{r,k \bmod 2^{r-1}} \quad (k \in \mathbb{Z}/2^r\mathbb{Z}).$$

Since the roots are distinct, the Chinese remainder theorem and (14.11) give an isomorphism of $\mathcal{E}_a$ -algebras

$$\Phi_r : \mathcal{S}_r \longrightarrow \mathcal{E}_a^{\mathbb{Z}/2^{r-1}\mathbb{Z}}, \quad P(\xi_r) \longmapsto (P(\zeta_{r,k}))_k \quad (P \in \mathcal{E}_a[T]). \quad (14.13)$$

The inclusion $\mathcal{S}_r \subseteq \mathcal{S}_{r+1}$ replaces ξ_r by ξ_{r+1}^2 , so $P(\xi_r) = P(\xi_{r+1}^2)$ has stage- $(r+1)$ coordinates $P(\zeta_{r+1,k}^2) = P(\zeta_{r,k \bmod m})$. In coordinates the transition map is therefore exactly

$$(c_k)_{k \bmod m} \longmapsto (c_{k \bmod m})_{k \bmod 2m},$$

which repeats each coordinate on its two lifts. This compatibility, and not merely the existence of separate finite product decompositions, is what identifies the direct limit.

Theorem 14.19 (Cantor algebra theorem [07]). *Let $\text{LC}(\mathbb{Z}_2, \mathcal{E}_a)$ be the class of functions from the 2-adic integers $\mathbb{Z}_2$ to $\mathcal{E}_a$ that factor through some finite quotient $\mathbb{Z}/2^N\mathbb{Z}$, with pointwise operations; thus $\text{LC}(\mathbb{Z}_2, \mathcal{E}_a) = \bigcup_{N \geq 0} \mathcal{E}_a^{\mathbb{Z}/2^N\mathbb{Z}}$ with the transition maps repeating each coordinate on its two lifts. There is an explicit $\mathcal{E}_a$ -algebra isomorphism*

$$\bigcup_{r \geq 1} \mathcal{S}_r \cong \text{LC}(\mathbb{Z}_2, \mathcal{E}_a), \quad (14.14)$$

under which ξ_r corresponds to the function $z \mapsto \zeta_{r, z \bmod 2^{r-1}}$. The left side is the subalgebra of $\mathcal{Q}_a$ generated by $\mathcal{F}_a$ and the classes of $\omega^{a/2^r}$, $r \geq 1$.

Proof. Identify $\mathcal{E}_a^{\mathbb{Z}/2^{r-1}\mathbb{Z}}$ with the functions $\mathbb{Z}_2 \rightarrow \mathcal{E}_a$ factoring through $\mathbb{Z}/2^{r-1}\mathbb{Z}$, via $c \mapsto (z \mapsto c_{z \bmod 2^{r-1}})$. By the computation of the transition maps, the function attached to an element of $\mathcal{S}_r$ by Φ_r is the same as the one attached by Φ_{r+1} , so the Φ_r combine into a map Φ on the increasing union; it is an $\mathcal{E}_a$ -algebra homomorphism, and it is injective because each Φ_r is. Conversely, every function factoring through $\mathbb{Z}/2^N\mathbb{Z}$ lies in the image of the bijection Φ_{N+1} , so Φ is surjective. Evaluating T at the roots gives the function attached to ξ_r . Finally, $\xi_{r'} = \xi_r^{2^{r-r'}}$ $\in \mathcal{S}_r$ for $r' \leq r$, so the union of the $\mathcal{S}_r$ is the subalgebra generated by $\mathcal{F}_a$ and all ξ_r . $\square$

The isomorphism is an algebraic statement about a specified subalgebra; it is not a classification of all of $\mathcal{Q}_a$.

Remark 14.20 (Locally constant functions [07]). The topological terminology is justified: a locally constant function on the ordinary compact space $\mathbb{Z}_2$ has a finite subcover of neighbourhoods of constancy and hence factors through one finite quotient, and conversely such a factorization is locally constant. This uses only the ordinary Cantor space; no topology is imposed on the proper-class field $\mathcal{E}_a$, and one can equally take the finite-factorization condition as the definition.

Remark 14.21 (Evaluation at a point [07]). Evaluation at a point $z \in \mathbb{Z}_2$ is a homomorphism from the subalgebra (14.14) onto $\mathcal{E}_a$. Its target is a proper-class field, not a set-sized ring, and no assertion is made that it extends to all of $\mathcal{Q}_a$. The construction is therefore fully compatible with the absence of small ring images of $\mathcal{Q}_a$.

For $m = 2^{r-1}$ and $k \in \mathbb{Z}/m\mathbb{Z}$ define, in $\mathcal{S}_r$,

$$e_{r,k} = \frac{1}{m} \sum_{\ell=0}^{m-1} \left(\frac{\xi_r}{\zeta_{r,k}} \right)^\ell. \quad (14.15)$$

At the root $\zeta_{r,j}$ the sum has value 1 for $j = k$ and 0 otherwise, by the finite geometric-sum identity for the m th root of unity $\zeta_{r,j}/\zeta_{r,k}$. Thus $\Phi_r(e_{r,k})$ is the indicator of the coordinate k , and

$$e_{r,k}^2 = e_{r,k}, \quad e_{r,k}e_{r,j} = 0 \ (k \neq j), \quad \sum_{k \bmod m} e_{r,k} = 1, \quad (14.16)$$

$$e_{r,k} = e_{r+1,k} + e_{r+1,k+m}. \quad (14.17)$$

Each $e_{r,k}$ is nonzero because Φ_r is injective, which rests on [Theorem 14.15](#). The refinement formula (14.17) follows from the coordinate transition, or by checking values at all roots of stage $r + 1$.

The first nontrivial instance can be written without trigonometric notation. Put

$$y = \xi_2, \quad u = \xi_1 = y^2, \quad \zeta = \frac{1+u}{\sqrt{2}} \in \mathcal{E}_a.$$

Then $u^2 = -1$ and $\zeta^2 = (1 + 2u + u^2)/2 = u = y^2$, so $(y/\zeta)^2 = 1$, and

$$e_+ = \frac{1}{2} \left(1 + \frac{y}{\zeta} \right), \quad e_- = \frac{1}{2} \left(1 - \frac{y}{\zeta} \right) \quad (14.18)$$

satisfy $e_{\pm}^2 = e_{\pm}$, $e_+e_- = \frac{1}{4}(1 - (y/\zeta)^2) = 0$ and $e_+ + e_- = 1$. Nonvanishing is not inferred from the suggestive formula: the roots at stage 2 are $\zeta_{2,0} = \zeta$ and $\zeta_{2,1} = -\zeta$, so $e_+ = e_{2,0}$, $e_- = e_{2,1}$, and Φ_2 identifies $\mathcal{S}_2 \cong \mathcal{E}_a \times \mathcal{E}_a$ and sends e_+, e_- to $(1, 0)$ and $(0, 1)$. This is why the exact-kernel theorem is an essential part of the argument.

Corollary 14.22 (Boolean algebra and failure of finiteness conditions [07]). *The idempotents of $\bigcup_r \mathcal{S}_r$ form precisely the Boolean algebra of clopen subsets of $\mathbb{Z}_2$, and this Boolean algebra is atomless. The ring $\mathcal{Q}_a$ has an infinite sequence of pairwise orthogonal nonzero idempotents, is neither a domain nor a local ring, and fails the ascending chain condition on ideals.*

Proof. A field has only the idempotents 0 and 1, so an idempotent of $\text{LC}(\mathbb{Z}_2, \mathcal{E}_a)$ is the indicator of a set that is a union of residue classes modulo some 2^N , that is, of a clopen subset of $\mathbb{Z}_2$; conversely, by compactness every clopen subset is a finite union of such residue classes, and its indicator is idempotent. Every nonempty clopen set contains a residue class modulo some 2^N , which splits into two nonempty residue classes modulo 2^{N+1} ; so there are no atoms.

For $j \geq 0$ the sets $C_j = \{z \in \mathbb{Z}_2 : z \equiv 2^j \pmod{2^{j+1}}\}$ (the z of 2-adic valuation exactly j) are nonempty, clopen and pairwise disjoint. Their indicators give pairwise orthogonal nonzero idempotents $f_j \in \mathcal{Q}_a$. The ideals

$$(f_0) \subsetneq (f_0, f_1) \subsetneq (f_0, f_1, f_2) \subsetneq \cdots$$

of $\mathcal{Q}_a$ strictly increase: if $f_{n+1} = \sum_{j \leq n} c_j f_j$ with $c_j \in \mathcal{Q}_a$, multiplication by f_{n+1} would give $f_{n+1} = 0$. This proves the failure of the ascending chain condition without forming any class-sized generating family. A nontrivial idempotent e satisfies $e(1 - e) = 0$, so $\mathcal{Q}_a$ is not a domain; and in a local ring (for every x , one of x and $1 - x$ is a unit) either e or $1 - e$ is a unit, which forces $e = 1$ or $e = 0$. $\square$

Grouping a sufficiently fine dyadic partition of $\mathbb{Z}_2$ into any prescribed number $d \geq 1$ of nonempty blocks yields a unital subalgebra of $\mathcal{Q}_a$ isomorphic to $\mathcal{E}_a^d$. Manuscript 07 claims neither that all idempotents of $\mathcal{Q}_a$ lie in the Cantor algebra (14.14) nor that $\mathcal{Q}_a$ itself is a ring of locally constant functions. No infinite sum of the orthogonal sequence (f_j) is used.

14.7 Gaussian binomial quotients

Let

$$\mathcal{Q}_a^{\mathbb{C}} = \mathbf{Oz}[i]/(1 + \omega^a), \quad q_a^{\mathbb{C}} : \mathbf{Oz}[i] \rightarrow \mathcal{Q}_a^{\mathbb{C}}.$$

Since $\mathbf{Oz}[i] = \mathbf{Oz} + i\mathbf{Oz}$ and ω^{-a} is real, $\mathbf{Oz}[i][\omega^{-a}] = \mathbf{Oz}[\omega^{-a}] + i\mathbf{Oz}[\omega^{-a}]$, so by [Proposition 6.11](#) it consists of the surcomplex normal forms whose supports are bounded below by some $-Na$. It contains $\mathcal{F}_a(i)$, and the proof of [Lemma 14.14](#) applies verbatim with $\mathbf{Oz}[i]$ in place of $\mathbf{Oz}$: the extension $\tilde{q}_a^{\mathbb{C}}$ of $q_a^{\mathbb{C}}$ has kernel $(1 + \omega^a)\mathbf{Oz}[i][\omega^{-a}]$, and it embeds the coefficient field by

$$f \mapsto -q_a^{\mathbb{C}}(\omega^a f) \quad (f \in \mathcal{F}_a(i)).$$

For $f = i$ this is the Gaussian imaginary unit of the quotient, since $-q_a^{\mathbb{C}}(\omega^a i) = -q_a^{\mathbb{C}}(\omega^a)q_a^{\mathbb{C}}(i) = q_a^{\mathbb{C}}(i)$.

Theorem 14.23 (Complex exact kernels and dyadic algebra [07]). *For every $n \geq 1$ there is an injective ring homomorphism*

$$\mathcal{F}_a(i)[T]/(T^n + 1) \hookrightarrow \mathcal{Q}_a^{\mathbb{C}}, \quad T \mapsto q_a^{\mathbb{C}}(\omega^{a/n}),$$

and its source is isomorphic to $\mathcal{F}_a(i)^n$. The increasing union of the dyadic stages $n = 2^r$, $r \geq 0$, is isomorphic to $\text{LC}(\mathbb{Z}_2, \mathcal{F}_a(i))$. The ring $\mathcal{Q}_a^{\mathbb{C}}$ is nonzero, has no unital homomorphism to a nonzero set-sized ring, and contains a class copy of $\mathbf{No}[i]$.

Proof. The proof of [Theorem 14.15](#) uses only supports and the nonvanishing of coefficients, not their reality, so it applies verbatim over $\mathcal{F}_a(i)$ and $\mathbf{Oz}[i][\omega^{-a}]$: a nonzero remainder R with coefficients in $\mathcal{F}_a(i)$ gives the expansion (14.9), whose blocks lie in the distinct cosets $(j/n - k - 1)a + H_a$, whose union of supports is reverse well ordered, and whose support is not bounded below by any $-Na$; so $R(\omega^{a/n}) \notin (1 + \omega^a)\mathbf{Oz}[i][\omega^{-a}]$. The polynomial $T^n + 1$ has n distinct roots $\exp(i\pi(2k+1)/n)$ in $\mathbb{C} \subseteq \mathcal{F}_a(i)$, so the Chinese remainder theorem gives $\mathcal{F}_a(i)[T]/(T^n + 1) \cong \mathcal{F}_a(i)^n$ by evaluation at the roots.

For $n = 2^r$ label the roots $\eta_{r,k} = \exp(i(\pi + 2\pi k)/2^r)$, $k \in \mathbb{Z}/2^r\mathbb{Z}$. Then $\eta_{r,k}^2 = \eta_{r-1, k \bmod 2^{r-1}}$, and the stage- $(r-1)$ generator $q_a^{\mathbb{C}}(\omega^{a/2^{r-1}})$ is the square of the stage- r generator; so, exactly as in [Theorem 14.19](#), the coordinate transitions repeat each entry on its two lifts, and the union is $\text{LC}(\mathbb{Z}_2, \mathcal{F}_a(i))$. The ring is nonzero because $1 + \omega^a$ is not among the units $\pm 1, \pm i$ of $\mathbf{Oz}[i]$ ([Lemma 2.4](#)); it has no nonzero set-sized image by [Theorem 6.1](#), since $\text{ct}(1 + \omega^a) = 1$ generates $\mathbb{Z}[i]$; and it contains a copy of $\mathbf{No}[i]$ by the complex extension of Θ_a ([Theorem 14.4](#)). $\square$

Coefficientwise complex conjugation preserves $\mathbf{Oz}[i]$ and, because $1 + \omega^a$ is real, the ideal $(1 + \omega^a)$; hence it descends to an involution of $\mathcal{Q}_a^{\mathbb{C}}$. It commutes with the coefficient-field embedding $f \mapsto -q_a^{\mathbb{C}}(\omega^a f)$ because the multiplier $-\omega^a$ is real. This supplies a compatible involution without requiring an order or an analytic interpretation of the quotient.

14.8 Boundaries of the large-field results

The small-image theorem and the large-field results concern different hypotheses and reinforce one another (07). A homomorphism to a set-sized ring kills every purely infinite element ([Theorem 5.1](#)); a homomorphism to a nonzero ring that makes even one purely infinite element invertible has a huge field in its target ([Theorem 14.7](#)). That target is necessarily a proper class, but the explicit field embedding gives considerably more than this size conclusion, and the Cantor algebra describes zero-divisor structure that no cardinality argument captures. Manuscript 07 insists on the following distinctions.

A nonzero element versus a unit. A nonzero nilpotent, as in [Example 14.11](#), does not produce a field. Invertibility is what allows the formula $f = (gf)/g$ to be transported to a quotient. No cancellation in a ring with zero divisors has been used.

A specified subalgebra versus the entire quotient. The exact kernel theorem identifies the subalgebras $\mathcal{F}_a[x_n]$, and their dyadic union is a Cantor algebra. Surreal normal forms with other supports can still create further elements of $\mathcal{Q}_a$. Neither a classification of all idempotents of $\mathcal{Q}_a$ nor a description of its prime ideals has been proved.

Class ideals versus set-theoretic Zorn arguments. Zorn’s lemma applies to sets of candidates. The ring $\mathcal{Q}_a$ is a proper class, so the usual existence argument for maximal ideals cannot simply be repeated over its class of ideals. Statements such as [Corollary 14.10](#) are deliberately conditional: it is not concluded that a proper-class nonarithmetic maximal ideal exists merely because $(1 + \omega^a)$ is proper. Likewise (06) [Proposition 14.12](#) is a necessary condition on residue fields, not an existence theorem.

Strong sums versus limits. Every infinite sum used above, namely the geometric series for $(1 + \omega^a)^{-1}$ and the block expansion [\(14.9\)](#), has a set-sized reverse well-ordered support and finitely many contributions at each exponent, and is verified coefficientwise. Finite geometric remainders of the kind in [Lemma 4.3](#) do not tend to zero, and no limit argument is used.

Finite checks. The program accompanying manuscript 07 checks exact finite geometric identities, finite binomial remainders, the first nontrivial idempotents [\(14.18\)](#) and finite coordinate refinement maps (07 records that all 621 named checks passed). It does not represent arbitrary surreal normal forms, and it does not verify the ambient support argument, namely that the infinite expansion [\(14.9\)](#) lies outside $\mathbf{Oz}[\omega^{-a}]$; that is a mathematical proof in the text. For a formalization, 07 recommends keeping the support characterization of $\mathbf{Oz}[\omega^{-a}]$ and the exact binomial kernel separate from the routine finite Chinese remainder calculations; no corresponding Lean declarations are claimed.

15 Integral closure, complete integral closure, and invisible equations

The universal theorem uses exponent scales *below* the supports of purely infinite elements. This section uses a scale *above* all negative exponents, which yields one denominator for a whole set-generated algebra (03). The consequences separate three notions that coincide for Noetherian domains: membership in $\mathbf{Oz}$, integrality over $\mathbf{Oz}$, and almost integrality over $\mathbf{Oz}$. The complete integral closure of $\mathbf{Oz}$ is all of $\mathbf{No}$, while its integral closure $\mathcal{N}$ is a proper, fine-dense subring that no set of generators produces (03). A monic equation of 08 has no root in $\mathbf{Oz}[i]$ although its image in every set-sized ring has the root 1; its positive root is an explicit element of $\mathcal{N} \setminus \mathbf{Oz}$. The section ends with 19’s description of $\mathcal{N}$ and $\mathcal{N}_{\mathbb{C}}$ modulo the ideals generated by Π , which classifies their set-sized domain images ([Section 15.8](#)), and with 20’s proof that the Boolean algebra of that description is atomless and a proper class ([Section 15.9](#)), and with 23’s fixed cyclic workspace, whose extended ideal is not radical although the Boolean algebra of its reduced fibre is still atomless ([Section 15.10](#)).

Standing notation. As in [Section 11.1](#), κ is an infinite cardinal with $\mu = \text{cf}(\kappa) > \aleph_0$, $\theta = \kappa^{\aleph_0}$, G_κ is the two-armed lexicographic group of 03, $F_\kappa^{03} = H_\omega(\mathbb{R}, G_\kappa)$ is the field of countably supported Hahn series over G_κ , identified with its image in $\mathbf{No}$, and

$$A_\kappa^{03} = F_\kappa^{03} \cap \mathbf{Oz} = \mathbb{Z} + \Pi_\kappa^{03},$$

where Π_κ^{03} consists of the elements of F_κ^{03} with strictly positive support. We write X^g ($g \in G_\kappa$) for the monomials of F_κ^{03} ; under the identification they are surreal monomials. The Gaussian versions are $F_\kappa^{03}[i] = H_\omega(\mathbb{C}, G_\kappa)$, $A_\kappa^{03}[i] = \mathbb{Z}[i] + \Pi_\kappa^{03} + i\Pi_\kappa^{03}$ and the ideal $\Pi_\kappa^{03} + i\Pi_\kappa^{03}$. The field F_κ^{03} is real closed and $F_\kappa^{03}[i]$ is algebraically closed ([Section 11.1](#)). We write $\overline{\mathbb{Z}}$ for the ring of complex algebraic integers and $\overline{\mathbb{Z}}_\mathbb{R} = \overline{\mathbb{Z}} \cap \mathbb{R}$ for the real ones. For an inclusion of rings $A \subseteq R$ the *conductor* is $(A : R) = \{a \in A : aR \subseteq A\}$.

15.1 Uniform denominators

The clearing statement [Proposition 3.4\(i\)](#) clears one set of surreals. Applied to a well-chosen set it clears a whole algebra.

Theorem 15.1 (One denominator for a set-generated algebra [03]). *For every set $S \subseteq \mathbf{No}$ there is a positive monomial $d = \omega^h \in \Pi$ with*

$$d \mathbf{Oz}[S] \subseteq \Pi. \quad (15.1)$$

For every set $S \subseteq \mathbf{No}[i]$ there is such a d with

$$d \mathbf{Oz}[i][S] \subseteq \Pi_\mathbb{C}. \quad (15.2)$$

In particular d is a nonzero element of the conductor $(\mathbf{Oz} : \mathbf{Oz}[S])$, respectively $(\mathbf{Oz}[i] : \mathbf{Oz}[i][S])$.

Proof. Although $\mathbf{Oz}[S]$ is in general a proper class, the ring $\mathbb{Z}[S]$ is a set: it consists of the values of ordinary integer polynomials in finitely many members of S . By [Proposition 3.4\(i\)](#) there is $h > 0$ with $\omega^h \mathbb{Z}[S] \subseteq \Pi$. Every element of $\mathbf{Oz}[S]$ is a finite sum $\sum_j a_j b_j$ with $a_j \in \mathbf{Oz}$ and $b_j \in \mathbb{Z}[S]$, and

$$\omega^h \sum_j a_j b_j = \sum_j a_j (\omega^h b_j) \in \Pi$$

because Π is an ideal of $\mathbf{Oz}$ ([Lemma 2.2](#)). The complex statement is the same argument with the set $\mathbb{Z}[i][S]$, the complex clause of [Proposition 3.4\(i\)](#) and the ideal $\Pi_\mathbb{C}$ of $\mathbf{Oz}[i]$. Finally $\Pi \subseteq \mathbf{Oz}$ and $\Pi_\mathbb{C} \subseteq \mathbf{Oz}[i]$. $\square$

Remark 15.2 (Conductors [03]). The theorem gives more than a nonzero element of the conductor $(\mathbf{Oz} : \mathbf{Oz}[S])$: it gives a monomial whose products with the whole algebra land in the strictly positive-support ideal Π . The real clearing statement and its proof are due to manuscript 05 (now in the sibling report [46]); the algebra formulation and the closure consequences below are 03's.

Theorem 15.3 (The bounded-size version [03]). *Let A be A_κ^{03} or $A_\kappa^{03}[i]$, with ambient field $F = F_\kappa^{03}$, respectively $F_\kappa^{03}[i]$, constant ring $D = \mathbb{Z}$, respectively $\mathbb{Z}[i]$, and ideal $P = \Pi_\kappa^{03}$, respectively $\Pi_\kappa^{03} + i\Pi_\kappa^{03}$. For every set $S \subseteq F$ with $|S| < \mu = \text{cf}(\kappa)$ there is $h > 0$ in G_κ with*

$$X^h A[S] \subseteq P. \quad (15.3)$$

In particular $\text{Frac}(A_\kappa^{03}) = F_\kappa^{03}$ and $\text{Frac}(A_\kappa^{03}[i]) = F_\kappa^{03}[i]$.

Proof. The ring $D[S]$ has cardinality at most $\max(|S|, \aleph_0) < \mu$, since μ is uncountable. Each of its elements has countable support (in the complex case, take the union of the supports of the real and imaginary coordinates). The union U of these supports is a union of fewer than μ countable sets, so $|U| < \mu$ because μ is regular and uncountable. By Lemma 11.2(i) the family $\{-g : g \in U\} \cup \{0\}$, which has fewer than μ members, has an upper bound $b \in G_\kappa$; put $h = b + e$ for any $e > 0$ in G_κ . Then $h > 0$ and $h > -g$ for every $g \in U$. Multiplication by X^h translates each support by h , so it maps every element of $D[S]$ to a countably supported series with strictly positive support, that is, into P . Every element of $A[S]$ is a finite sum $\sum_j a_j b_j$ with $a_j \in A$ and $b_j \in D[S]$, and $X^h \sum_j a_j b_j = \sum_j a_j (X^h b_j) \in P$ because P is an ideal of A .

For $x \in F$ apply this to $S = \{x\}$: then $x = (X^h x)/X^h$ is a quotient of elements of A , and $X^h \in P \subseteq A$ is nonzero. $\square$

This supplies the fraction-field clause $\text{Frac}(A_\kappa^{03}) = F_\kappa^{03}$ of 03's cardinal realization (compare Theorem 11.5). Clearing one element is weaker than clearing *all powers* of that element by the same denominator; both theorems supply the latter as well, since $x^n \in A[x]$ for every n .

15.2 Almost integrality and complete integral closure

Definition 15.4 ([03]). Let A be a domain contained in a field F . An element $x \in F$ is *almost integral* over A if there is $0 \neq d \in A$ with $dx^n \in A$ for every ordinary $n \geq 0$. When $F = \text{Frac}(A)$, the set A^* of these elements is the *complete integral closure* of A . The *integral closure* $\text{IC}_F(A)$ consists instead of the roots in F of monic polynomials in $A[T]$.

These are the standard notions of commutative algebra; they differ in non-Noetherian settings, and we take the terminology from [4] without importing any result of that paper as a proof. The quantifier over powers ranges over ordinary finite powers, not over ordinal iterations or surreal exponentiation. Each individual witness d and each monic polynomial is set-sized data, so the definitions apply verbatim to the class rings $\mathbf{Oz}$ and $\mathbf{Oz}[i]$.

Theorem 15.5 (Every ambient element is almost integral [03]). *Relative to their fraction fields,*

$$\mathbf{Oz}^* = \mathbf{No}, \quad \mathbf{Oz}[i]^* = \mathbf{No}[i], \quad (15.4)$$

$$(A_\kappa^{03})^* = F_\kappa^{03}, \quad (A_\kappa^{03}[i])^* = F_\kappa^{03}[i]. \quad (15.5)$$

More generally, every ring R with $A \subseteq R \subseteq F$, where (A, F) is one of the four pairs $(\mathbf{Oz}, \mathbf{No})$, $(\mathbf{Oz}[i], \mathbf{No}[i])$, $(A_\kappa^{03}, F_\kappa^{03})$, $(A_\kappa^{03}[i], F_\kappa^{03}[i])$, has $\text{Frac}(R) = F$ and $R^* = F$.

Proof. The fraction fields are the indicated ambient fields by Proposition 3.4(ii) and Theorem 15.3, so there are no further elements to consider. For x in the ambient field apply Theorem 15.1 or Theorem 15.3 to $S = \{x\}$. The single nonzero denominator d supplied there satisfies $dx^n \in dA[x] \subseteq A$ for every ordinary $n \geq 0$, so x is almost integral. If $A \subseteq R \subseteq F$, then $F = \text{Frac}(A) \subseteq \text{Frac}(R) \subseteq F$, and the same $d \in A \subseteq R$ gives $dx^n \in A \subseteq R$. $\square$

Example 15.6 (Two explicit denominators [03]). The rational $1/2$ is almost integral over $\mathbf{Oz}$, witnessed by $d = \omega$: indeed $\omega/2^n \in \Pi$ for all $n \geq 0$. The monomial ω^{-1} is almost integral, witnessed by $d = \omega^\omega$: indeed $\omega^{\omega-n} \in \Pi$ for every ordinary n , since $\omega - n > 0$. Neither claim uses a limit as n tends to an infinite ordinal. Neither element is integral (Lemma 15.7 below).

15.3 Integral closure: reciprocals, small units and density

The geometric statements are proved first for an arbitrary integer part of a real closed field; this isolates what is a consequence of integer-part geometry rather than of the proper-class size of **No** (03).

Lemma 15.7 (An integral reciprocal was already present [03]). *Let A be a domain in a field F and $0 \neq a \in A$. If a^{-1} is integral over A , then $a^{-1} \in A$. Hence the integral closure of A in F contains the inverse of no nonunit of A .*

Proof. Suppose $(a^{-1})^n + c_{n-1}(a^{-1})^{n-1} + \dots + c_0 = 0$ with $c_j \in A$ and $n \geq 1$. Multiplying by a^{n-1} gives

$$a^{-1} = -c_{n-1} - c_{n-2}a - \dots - c_0a^{n-1} \in A. \quad \square$$

Consequently $1/2$ is not integral over **Oz**, A_κ^{03} , **Oz** $[i]$ or $A_\kappa^{03}[i]$, despite [Example 15.6](#): 2 is a nonunit of each of these rings by [Lemma 2.4](#) (the units of $A_\kappa^{03} \subseteq \mathbf{Oz}$ are among $\mathbf{Oz}^\times = \{\pm 1\}$, and similarly in the Gaussian case). More generally, for $h > 0$ and $0 \neq c \in \mathbb{R}$ the negative monomial $c\omega^{-h}$ is not integral over **Oz**, and for $0 \neq c \in \mathbb{C}$ it is not integral over **Oz** $[i]$: its reciprocal $c^{-1}\omega^h$ lies in the purely infinite ideal Π , respectively $\Pi_\mathbb{C}$, and is a nonunit. The same holds for cX^{-h} over A_κ^{03} and $A_\kappa^{03}[i]$ when $h > 0$ in G_κ .

Lemma 15.8 (Integer parts [03]). *Recall that a subring A of an ordered field F is an integer part if for every $x \in F$ there is a unique $\lfloor x \rfloor \in A$ with $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$. Then **Oz** is an integer part of **No** and A_κ^{03} is an integer part of F_κ^{03} . Every nonzero element of **Oz** has absolute value at least 1, and every nonzero element of **Oz** $[i]$ has surreal modulus $|a + ib| = \sqrt{a^2 + b^2}$ at least 1.*

Proof. A nonzero element of **Oz** is either a nonzero ordinary integer or has positive degree ([Lemma 2.4](#)); in the second case its absolute value exceeds every real number. Hence $|a| \geq 1$ for $0 \neq a \in \mathbf{Oz}$, and for $a + ib \in \mathbf{Oz}[i]$ with $a, b \in \mathbf{Oz}$ not both zero, $\sqrt{a^2 + b^2} \geq \max(|a|, |b|) \geq 1$. This lower bound also gives uniqueness of floors: two elements of **Oz** in a half-open interval of length 1 differ by an element of absolute value less than 1. For existence, write the normal form of $x \in \mathbf{No}$ uniquely as $x = P + r + \epsilon$ with $P \in \Pi$, $r \in \mathbb{R}$ and $\text{supp } \epsilon$ strictly negative, so that ϵ is zero or infinitesimal. Take $\lfloor x \rfloor = P + \lfloor r \rfloor$, except when $r \in \mathbb{Z}$ and $\epsilon < 0$, where $\lfloor x \rfloor = P + r - 1$. In every case $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$. For $x \in F_\kappa^{03}$ the same decomposition has $P \in \Pi_\kappa^{03}$, so the floor lies in A_κ^{03} . $\square$

The exceptional case in the floor formula matters: discarding all negative exponents sends $-\omega^{-1}$ to 0, which is not its floor.

Why integral elements form a ring. That the integral elements over a commutative ring A form a ring, and that integrality is transitive, is finite algebra, which we recall because it is applied to class rings. If x satisfies a monic equation of degree n over A , then $A[x] = A + Ax + \dots + Ax^{n-1}$ is a finitely generated A -module, by reducing powers with that equation. If x and y are integral, $A[x, y]$ is generated by finitely many monomials $w_1 = 1, w_2, \dots, w_N$ in x and y . For $z \in A[x, y]$ write $zw_k = \sum_l a_{kl}w_l$ with $a_{kl} \in A$; the determinant trick shows that $\det(zI - (a_{kl}))$ annihilates every w_l , in particular $w_1 = 1$, so it is zero, and this is a monic equation for z . The same argument, applied to the finitely many coefficients of a monic equation and to the element, proves transitivity. Only finitely many elements are involved at each step, so the arguments apply unchanged to **Oz**, **Oz** $[i]$ and their overrings in **No** and **No** $[i]$ (03).

Theorem 15.9 (Small integral units [03]). *Let F be a real closed ordered field, $A \subseteq F$ an integer part, and N the integral closure of A in F . For every $H \in A_{>0}$ the element $u_H = \sqrt{H^2 + 1} - H$ is a positive unit of N satisfying*

$$u_H^2 + 2Hu_H - 1 = 0, \quad u_H^{-1} = u_H + 2H, \quad 0 < u_H < \frac{1}{2H}. \quad (15.6)$$

For every $\epsilon \in F_{>0}$ there is such a unit with $0 < u_H < \epsilon$.

Proof. Since $\sqrt{H^2 + 1} > H > 0$, the element u_H is positive. Squaring $u_H + H = \sqrt{H^2 + 1}$ gives the first equation, which is monic over A ; so $u_H \in N$. The same equation says $u_H(u_H + 2H) = 1$, and $u_H + 2H \in N$ because N is a ring containing A ; so u_H is a unit of N . Rationalizing,

$$u_H = \frac{1}{\sqrt{H^2 + 1} + H} < \frac{1}{2H}.$$

An integer part is cofinal in F : $\lfloor 1/(2\epsilon) \rfloor + 1 \in A$ is positive and exceeds $1/(2\epsilon)$. Taking H to be this element gives $u_H < 1/(2H) < \epsilon$. $\square$

Theorem 15.10 (Dense normalization of an integer part [03]). *Under the hypotheses of Theorem 15.9, N is dense in F for the order topology. More precisely, for every $x \in F$ and every $\epsilon \in F_{>0}$ there is $y \in N$ with*

$$0 \leq x - y < \epsilon. \quad (15.7)$$

Proof. Choose a positive unit $u < \epsilon$ of N by Theorem 15.9 and let $n = \lfloor x/u \rfloor \in A$. Multiplying $n \leq x/u < n + 1$ by $u > 0$ gives $0 \leq x - un < u < \epsilon$; put $y = un \in N$. To meet a prescribed open interval (a, b) , apply this with $x = (a + b)/2$ and $\epsilon < (b - a)/2$; then $y \in (x - \epsilon, x] \subseteq (a, b)$. $\square$

No completion and no sequential argument is used. The same proof is valid for the order topology of **No** (the fine topology), with all positive surreal radii. Results on dense subfields and integer parts in other settings provide context [22]; the object here is the integral closure of a specified integer part, not an arbitrary dense subfield, and the normalization theorem is not attributed to that work. The small-unit argument is a general elementary lemma about integer parts, and 03 makes no priority claim for it.

The normalizations. We write

$$\mathcal{N} = \text{IC}_{\mathbf{No}}(\mathbf{Oz}), \quad \mathcal{N}_\kappa = \text{IC}_{F_\kappa^{03}}(A_\kappa^{03}). \quad (15.8)$$

These are 03's B and B_κ . Manuscript 03 wrote $\text{Int}_F(A)$ for the integral closure of A in F ; we write $\text{IC}_F(A)$ because Int denotes rings of integer-valued polynomials, such as $\text{Int}(\mathbb{Z}^r)$, elsewhere in this report. Note that $\mathcal{N}_\kappa$ is defined relative to F_κ^{03} ; only $\mathcal{N}_\kappa \subseteq \mathcal{N} \cap F_\kappa^{03}$ is automatic, and 03 claims no descent equality. The equality $\mathcal{N} \cap F_\kappa^{03} = \mathcal{N}_\kappa$ does hold: it follows from 20's support projection (Corollary 15.59, a merge result).

Theorem 15.11 (Set-generated intermediate algebras are uniformly discrete [03]). *For every set $S \subseteq \mathbf{No}$ the algebra $C = \mathbf{Oz}[S]$ has a positive surreal separation bound r :*

$$|x - y| \geq r \quad \text{for all distinct } x, y \in C.$$

*It is a closed, uniformly discrete subclass of **No** in the fine order topology. The same holds for $A_\kappa^{03}[S] \subseteq F_\kappa^{03}$, in the order topology of F_κ^{03} , whenever $|S| < \mu$.*

Proof. Choose the positive monomial d of [Theorem 15.1](#), respectively $d = X^h$ of [Theorem 15.3](#). For distinct $x, y \in C$ the product $d(x - y)$ is a nonzero element of Π (respectively Π_κ^{03}), hence of the integer part, so $|d(x - y)| \geq 1$ by [Lemma 15.8](#). Thus $|x - y| \geq 1/d =: r$.

Uniform discreteness gives an isolating interval of radius $r/3$ around each point. For closedness, let $z \notin C$. If no point of C lies within $r/3$ of z , that neighbourhood misses C . Otherwise let $c \in C$ be such a point; every other point of C is at distance greater than $2r/3$ from z by the triangle inequality, so a radius smaller than both $r/3$ and $|z - c|$ gives a neighbourhood of z disjoint from C . $\square$

This is a class-size statement about a proper-class base ring; it is not the familiar assertion that a finite extension of a discrete ring is discrete in an Archimedean topology (03).

Theorem 15.12 (Proper dense normalization, zero conductor, no set of generators [03]). *For the full omnific ring,*

$$\mathbf{Oz} \subsetneq \mathcal{N} \subsetneq \mathbf{No}, \quad \overline{\mathcal{N}}^{\text{fine}} = \mathbf{No}, \quad (\mathbf{Oz} : \mathcal{N}) = 0, \quad (15.9)$$

and $\mathcal{N}$ is not generated as an $\mathbf{Oz}$ -algebra by any set of its elements. For the set-sized ring A_κ^{03} , the normalization $\mathcal{N}_\kappa$ satisfies $A_\kappa^{03} \subsetneq \mathcal{N}_\kappa \subsetneq F_\kappa^{03}$, is dense in F_κ^{03} , has conductor $(A_\kappa^{03} : \mathcal{N}_\kappa) = 0$, and requires at least μ algebra generators over A_κ^{03} .

Proof. The element $\sqrt{2} \in \mathbb{R} \subseteq F_\kappa^{03}$ is integral, being a root of $T^2 - 2$, and it is in neither integer part, whose real elements are the ordinary integers; so the first inclusions are strict. The element $1/2$ is not integral by [Lemma 15.7](#), so the second inclusions are strict. The fields $\mathbf{No}$ and F_κ^{03} are real closed and $\mathbf{Oz}$, A_κ^{03} are integer parts ([Lemma 15.8](#)); density is [Theorem 15.10](#).

For the conductor, fix $a \neq 0$ in the integer part. Choose a positive unit u of the normalization with $u < 1/|a|$ ([Theorem 15.9](#)). Then au lies in the normalization and $0 < |au| < 1$, so au is not in the integer part by [Lemma 15.8](#). Hence a is not in the conductor, and this holds for every $a \neq 0$.

If $\mathcal{N} = \mathbf{Oz}[S]$ for a set S , then [Theorem 15.1](#) would give a nonzero element d of the conductor, a contradiction. (Equivalently, $\mathcal{N}$ would be uniformly discrete by [Theorem 15.11](#), contradicting density.) The same argument with [Theorem 15.3](#) shows that $\mathcal{N}_\kappa \neq A_\kappa^{03}[S]$ whenever $|S| < \mu$. $\square$

Integral closure therefore changes the fine geometry of the ring, although $\mathcal{N}$ still contains neither $1/2$ nor any nonzero negative monomial $c\omega^{-h}$. For $\kappa = \aleph_1$ the theorem says, without the continuum hypothesis, that the normalization cannot be obtained by adjoining countably many elements to the continuum-sized ring $A_{\aleph_1}^{03}$ (here $\theta = \aleph_1^{\aleph_0} = 2^{\aleph_0}$). For general κ the theorem gives only the *lower bound* μ on the number of algebra generators of $\mathcal{N}_\kappa$, not an exact generation number; the exact count μ of 03's cardinal realization ([Theorem 11.5](#)) concerns the ideal Π_κ^{03} , not the normalization.

Remark 15.13 (Dense does not mean sequentially approximable [03]). The fine topology of $\mathbf{No}$ has the size obstructions discussed in the foundations report [42]. The density statement chooses a point separately for each positive surreal radius; it does not produce a nonconstant convergent set-indexed net. Indeed, every fixed set of elements of $\mathcal{N}$ lies in a single uniformly discrete algebra $\mathbf{Oz}[S]$ by [Theorem 15.11](#). The dense normalization requires access to more than any fixed set of its additional generators.

15.4 Constant slices, truncation, and a unit equation

The results above do not give a complete normal-form test for integrality. They do give exact tests on the nonnegative-support slice, and they show why leading-term tests fail (03). Recall

$\mathcal{A}_{\mathbb{R},\mathbb{R}} = \mathbb{R} + \Pi$ (03's $P_{\mathbb{R}}$), on which ct is a ring homomorphism onto $\mathbb{R}$ by Lemma 2.2, although $\mathcal{A}_{\mathbb{R},\mathbb{R}}$ is not an integer part; its countable version is $\mathbb{R} + \Pi_{\kappa}^{03}$.

Theorem 15.14 (The real constant and support slices [03]).

$$\mathcal{N} \cap \mathcal{A}_{\mathbb{R},\mathbb{R}} = \overline{\mathbb{Z}}_{\mathbb{R}} + \Pi = \mathcal{A}_{\overline{\mathbb{Z}}_{\mathbb{R}},\mathbb{R}}, \quad \mathcal{N} \cap \mathbb{R} = \overline{\mathbb{Z}}_{\mathbb{R}}, \quad \mathcal{N} \cap \mathbb{Q} = \mathbb{Z}. \quad (15.10)$$

The same formulas hold for $\mathcal{N}_{\kappa}$, with $\mathcal{A}_{\mathbb{R},\mathbb{R}}$ and Π replaced by $\mathbb{R} + \Pi_{\kappa}^{03}$ and Π_{κ}^{03} in the first formula.

Proof. Let $x = r + P$ with $r \in \mathbb{R}$ and $P \in \Pi$, and suppose $x^n + a_{n-1}x^{n-1} + \dots + a_0 = 0$ with $a_j \in \mathbf{Oz}$. All terms lie in $\mathcal{A}_{\mathbb{R},\mathbb{R}}$, and applying the ring homomorphism ct gives $r^n + \text{ct}(a_{n-1})r^{n-1} + \dots + \text{ct}(a_0) = 0$, a monic equation over $\mathbb{Z}$. So $r \in \overline{\mathbb{Z}}_{\mathbb{R}}$. Conversely, a real algebraic integer r is integral over $\mathbb{Z} \subseteq \mathbf{Oz}$, and $P \in \Pi \subseteq \mathbf{Oz}$, so $r + P$ is integral over $\mathbf{Oz}$ because integral elements form a ring. This is the first equality; intersecting with $\mathbb{R}$ gives the second, and the third holds because a rational root of a monic integer polynomial is an integer. For $\mathcal{N}_{\kappa}$ the coefficients lie in $A_{\kappa}^{03} \subseteq \mathbf{Oz}$, the element lies in $\mathbb{R} + \Pi_{\kappa}^{03}$, and the same argument applies; conversely $\overline{\mathbb{Z}}_{\mathbb{R}} \subseteq \mathbb{R} \subseteq F_{\kappa}^{03}$ and $\Pi_{\kappa}^{03} \subseteq A_{\kappa}^{03}$. $\square$

Example 15.15 ([03]). The element $\pi\omega + \sqrt{2}$ is integral over $\mathbf{Oz}$, whereas $\pi\omega + \pi$ is not. The nature of a coefficient at a *positive* exponent is irrelevant to the test; only the ordinary constant coefficient is constrained. The element $\omega + 1/2$ is almost integral (Theorem 15.5) but not integral, since $1/2 \notin \overline{\mathbb{Z}}_{\mathbb{R}}$.

Proposition 15.16 (Normalization is not truncation closed [03]). Let $h > 0$ and $H = \omega^h$ (or $H = X^h$ with $h > 0$ in G_{κ}). The integral unit u_H of Theorem 15.9 has the normal form

$$u_H = \sum_{n \geq 1} \binom{1/2}{n} \omega^{-(2n-1)h} = \frac{1}{2}\omega^{-h} - \frac{1}{8}\omega^{-3h} + \frac{1}{16}\omega^{-5h} - \frac{5}{128}\omega^{-7h} + \frac{7}{256}\omega^{-9h} - \dots \quad (15.11)$$

Its leading monomial $\frac{1}{2}\omega^{-h}$ is not integral over the integer part. Therefore neither $\mathcal{N}$ nor $\mathcal{N}_{\kappa}$ is closed under leading normal-form truncation.

Proof. By Lemma 15.24 below with $m = 2$ and $\gamma = 2h$, the series $s = \omega^h \mathbf{b}_2(\omega^{-2h})$ satisfies $s^2 = \omega^{2h}(1 + \omega^{-2h}) = H^2 + 1$. Its leading term is $\omega^h > 0$, so s is the positive square root $\sqrt{H^2 + 1}$ (an ordered field has at most one positive square root of a given element). Hence

$$u_H = s - \omega^h = \omega^h (\mathbf{b}_2(\omega^{-2h}) - 1) = \sum_{n \geq 1} \binom{1/2}{n} \omega^{h-2nh},$$

a series with countable, reverse well-ordered support $\{-(2n-1)h : n \geq 1\}$. The first five binomial coefficients are $\binom{1/2}{1} = \frac{1}{2}$, $\binom{1/2}{2} = -\frac{1}{8}$, $\binom{1/2}{3} = \frac{1}{16}$, $\binom{1/2}{4} = -\frac{5}{128}$ and $\binom{1/2}{5} = \frac{7}{256}$. In the countable model the same computation takes place in F_{κ}^{03} . The leading monomial $\frac{1}{2}\omega^{-h}$ is the reciprocal of the nonunit $2\omega^h$ of the integer part, so it is not integral by Lemma 15.7. $\square$

This is stronger than the statement that some integral elements have negative exponents: their negative tails can be essential, since deleting all but the leading term destroys integrality. In the opposite direction, Theorem 15.14 gives a complete constant-term criterion when negative exponents are absent (03).

Proposition 15.17 (A unit equation acquires solutions only after normalization [03]). *Let A be $\mathbf{Oz}$ or A_κ^{03} and $0 \neq H \in A$. The equation*

$$U^2 + 2HU - 1 = 0 \quad (15.12)$$

has no solution in A . For $H > 0$ its two roots in the real closed ambient field are u_H and $-2H - u_H$; both are integral units and belong to the normalization.

Proof. A solution in A would satisfy $U(U + 2H) = 1$, so U would be a unit of A , hence $U = \pm 1$ (Lemma 2.4; $A_\kappa^{03} \subseteq \mathbf{Oz}$). Substituting $U = 1$ gives $2H = 0$ and substituting $U = -1$ gives $-2H = 0$, both contrary to $H \neq 0$. The two roots are u_H and $-2H - u_H$ because their sum is $-2H$. Each is integral by the monic equation, and their product is -1 , so each is a unit of the normalization. $\square$

Thus membership, integrality and almost integrality give three different arithmetic environments even for a single quadratic equation with a parameter. This does not contradict constant-term transfer for systems with ordinary integer coefficients: here a possibly infinite coefficient H is fixed, and changing the ambient ring changes the allowed witnesses (03).

15.5 The surcomplex and Gaussian versions

The Gaussian omnific ring $\mathbf{Oz}[i] = \mathbf{Oz} + i\mathbf{Oz} = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}$ is the natural Gaussian analogue of $\mathbf{Oz}$; it is not discretely ordered. It has the split Gaussian constant term $\text{ct} : \mathbf{Oz}[i] \rightarrow \mathbb{Z}[i]$ (Lemma 2.2), its units are $\pm 1, \pm i$ (Lemma 2.4), and each nonzero element has surreal modulus at least 1 (Lemma 15.8). Put

$$\mathcal{N}_{\mathbb{C}} = \text{IC}_{\mathbf{No}[i]}(\mathbf{Oz}[i]), \quad \mathcal{N}_{\kappa, \mathbb{C}} = \text{IC}_{F_\kappa^{03}[i]}(A_\kappa^{03}[i]), \quad (15.13)$$

03's C and C_κ . Since i is integral over $\mathbf{Oz}$, transitivity shows that these are also the integral closures of $\mathbf{Oz}$ in $\mathbf{No}[i]$ and of A_κ^{03} in $F_\kappa^{03}[i]$. Complex conjugation is a ring automorphism of $\mathbf{No}[i]$ preserving $\mathbf{Oz}[i]$ (and of $F_\kappa^{03}[i]$ preserving $A_\kappa^{03}[i]$), so it preserves $\mathcal{N}_{\mathbb{C}}$ and $\mathcal{N}_{\kappa, \mathbb{C}}$. Consequently

$$\mathcal{N}_{\mathbb{C}} \cap \mathbf{No} = \mathcal{N}, \quad \mathcal{N}_{\kappa, \mathbb{C}} \cap F_\kappa^{03} = \mathcal{N}_\kappa. \quad (15.14)$$

Complexification nevertheless does not commute with normalization, and the failure is located at the prime 2.

Proposition 15.18 (The complexification defect is killed by 2 [03]). *There are strict inclusions of additive subclasses*

$$\mathcal{N}[i] \subsetneq \mathcal{N}_{\mathbb{C}} \subsetneq \tfrac{1}{2}\mathcal{N}[i], \quad \mathcal{N}_{\mathbb{C}}[1/2] = \mathcal{N}[1/2][i]. \quad (15.15)$$

The last object of the chain is a fractional module, not asserted to be a ring. In particular $2\mathcal{N}_{\mathbb{C}} \subseteq \mathcal{N}[i]$ although $\mathcal{N}_{\mathbb{C}} \neq \mathcal{N}[i]$. The same inclusions hold for $\mathcal{N}_\kappa$ and $\mathcal{N}_{\kappa, \mathbb{C}}$, and the set-sized quotient $\mathcal{N}_{\kappa, \mathbb{C}}/\mathcal{N}_\kappa[i]$ is a nonzero $\mathcal{N}_\kappa$ -module annihilated by 2.

Proof. Every element of $\mathcal{N}[i]$ is integral over $\mathbf{Oz}[i]$, so $\mathcal{N}[i] \subseteq \mathcal{N}_{\mathbb{C}}$. If $z = x + iy \in \mathcal{N}_{\mathbb{C}}$ with $x, y \in \mathbf{No}$, then $\bar{z} \in \mathcal{N}_{\mathbb{C}}$, so

$$2x = z + \bar{z} \in \mathcal{N}_{\mathbb{C}} \cap \mathbf{No} = \mathcal{N}, \quad 2y = -i(z - \bar{z}) \in \mathcal{N}_{\mathbb{C}} \cap \mathbf{No} = \mathcal{N}$$

by (15.14). Hence $2\mathcal{N}_{\mathbb{C}} \subseteq \mathcal{N}[i]$, that is, $\mathcal{N}_{\mathbb{C}} \subseteq \tfrac{1}{2}\mathcal{N}[i]$. Inverting 2 in $2\mathcal{N}_{\mathbb{C}} \subseteq \mathcal{N}[i] \subseteq \mathcal{N}_{\mathbb{C}}$ gives $\mathcal{N}_{\mathbb{C}}[1/2] = \mathcal{N}[i][1/2] = \mathcal{N}[1/2][i]$.

The root of unity $\zeta = (-1 + i\sqrt{3})/2$ satisfies $\zeta^2 + \zeta + 1 = 0$, so $\zeta \in \mathcal{N}_{\mathbb{C}}$; its real coordinate $-1/2$ is not in $\mathcal{N}$ by Lemma 15.7, so $\zeta \notin \mathcal{N}[i]$. Finally $1/2 \in \frac{1}{2}\mathcal{N}[i]$ is not in $\mathcal{N}_{\mathbb{C}}$, again by the reciprocal lemma, since 2 is a nonunit of $\mathbf{Oz}[i]$. The proof is unchanged in $F_{\kappa}^{03}[i]$, which contains ζ ; there ζ represents a nonzero class of $\mathcal{N}_{\kappa, \mathbb{C}}/\mathcal{N}_{\kappa}[i]$, and this quotient of $\mathcal{N}_{\kappa}$ -modules is annihilated by 2 because $2\mathcal{N}_{\kappa, \mathbb{C}} \subseteq \mathcal{N}_{\kappa}[i]$. $\square$

Theorem 15.19 (Dense Gaussian normalization and its constant slice [03]). *The ring $\mathcal{N}_{\mathbb{C}}$ is a proper dense subring of $\mathbf{No}[i]$ for the surreal-modulus topology, its conductor $(\mathbf{Oz}[i] : \mathcal{N}_{\mathbb{C}})$ is zero, and it cannot be generated over $\mathbf{Oz}[i]$ by a set. Moreover, with $\mathcal{A}_{\mathbb{C}, \mathbb{C}} = \mathbb{C} + \Pi_{\mathbb{C}}$,*

$$\mathcal{N}_{\mathbb{C}} \cap \mathcal{A}_{\mathbb{C}, \mathbb{C}} = \overline{\mathbb{Z}} + \Pi_{\mathbb{C}} = \mathcal{A}_{\overline{\mathbb{Z}}, \mathbb{C}}, \quad \mathcal{N}_{\mathbb{C}} \cap \mathbb{C} = \overline{\mathbb{Z}}. \quad (15.16)$$

For $\mathcal{N}_{\kappa, \mathbb{C}}$ the analogous properness, density, slice (with $\Pi_{\kappa}^{03} + i\Pi_{\kappa}^{03}$ in place of $\Pi_{\mathbb{C}}$) and zero-conductor statements hold, and fewer than μ algebra generators over $A_{\kappa}^{03}[i]$ do not suffice.

Proof. Let $z = x + iy \in \mathbf{No}[i]$ and $\epsilon > 0$. Choose a positive real integral unit $u < \epsilon/2$ of $\mathcal{N}$ (Theorem 15.9) and put

$$n = \lfloor x/u \rfloor, \quad m = \lfloor y/u \rfloor, \quad w = u(n + im).$$

The floors lie in $\mathbf{Oz}$ and $u \in \mathcal{N} \subseteq \mathcal{N}_{\mathbb{C}}$, so $w \in \mathcal{N}_{\mathbb{C}}$. Each coordinate of $z - w$ lies in $[0, u)$, hence $|z - w| < \sqrt{2}u < \epsilon$. This proves density. The element $1/2$ is excluded by Lemma 15.7, which proves properness.

For $0 \neq a \in \mathbf{Oz}[i]$ choose such a u with $0 < u < 1/|a|$. Then $au \in \mathcal{N}_{\mathbb{C}}$ and $0 < |au| < 1$, so $au \notin \mathbf{Oz}[i]$ by Lemma 15.8; thus the conductor is zero. If $\mathcal{N}_{\mathbb{C}} = \mathbf{Oz}[i][S]$ for a set S , Theorem 15.1 would give a nonzero element of the conductor; for $\mathcal{N}_{\kappa, \mathbb{C}}$ and fewer than μ generators, Theorem 15.3 does the same. The two absorption theorems also show directly that every such small-generated algebra has a uniform modulus separation bound, as in Theorem 15.11.

Finally, let $z = c + P$ with $c \in \mathbb{C}$ and $P \in \Pi_{\mathbb{C}}$, integral over $\mathbf{Oz}[i]$. All terms of its monic equation lie in $\mathcal{A}_{\mathbb{C}, \mathbb{C}}$, and applying the ring homomorphism $\text{ct} : \mathcal{A}_{\mathbb{C}, \mathbb{C}} \rightarrow \mathbb{C}$ shows that c is integral over $\mathbb{Z}[i]$; since $\mathbb{Z}[i]$ is integral over $\mathbb{Z}$, $c \in \overline{\mathbb{Z}}$. Conversely such a c is integral over the base, and P already belongs to it. Restricting to constants gives the second equality. The same arguments work in $F_{\kappa}^{03}[i]$, using the real integer part A_{κ}^{03} of F_{κ}^{03} for the floors. $\square$

The original integer-part rings have many finite quotients: $\mathbf{Oz}/n\mathbf{Oz} \cong \mathbb{Z}/n\mathbb{Z}$ for ordinary $n \geq 1$ by (3.2), and likewise $A_{\kappa}^{03}/nA_{\kappa}^{03} \cong \mathbb{Z}/n\mathbb{Z}$, because the purely infinite ideal is divisible by every nonzero ordinary integer (compare Theorem 6.2). Their normalizations behave oppositely.

Theorem 15.20 (No nonzero finite ring targets [03]). *None of $\mathcal{N}$, $\mathcal{N}_{\kappa}$, $\mathcal{N}_{\mathbb{C}}$, $\mathcal{N}_{\kappa, \mathbb{C}}$ admits a unital homomorphism to a nonzero finite ring. In particular none of them has a proper ideal of finite index.*

Proof. Suppose such a homomorphism exists. Its image is a nonzero finite commutative ring, even if the target is not commutative, and a maximal ideal of the image gives a unital homomorphism onto a finite field $\mathbb{F}_q$. The ordinary monic polynomial $T^q - T - 1$ is negative at 0 and positive at 2, so it has a real root $\alpha \in (0, 2)$. This α is a real algebraic integer and belongs to all four normalizations by Theorems 15.14 and 15.19. Its image $\bar{\alpha}$ would satisfy $\bar{\alpha}^q - \bar{\alpha} - 1 = 0$ in $\mathbb{F}_q$, whereas every element of $\mathbb{F}_q$ satisfies $\bar{\alpha}^q - \bar{\alpha} = 0$; so $1 = 0$ in $\mathbb{F}_q$, which is impossible. A proper ideal of finite index would give a unital homomorphism onto a nonzero finite ring. $\square$

The proof uses only the real algebraic integer constants. Accordingly this finite-quotient obstruction is a general consequence of containing those constants, not a phenomenon specific to surreal supports. It says nothing about infinite set-sized homomorphic images of the normalizations, and 03 classifies none (03). Manuscript 19 classifies the domain images and factors every set-sized image through $\mathcal{N}/\Pi\mathcal{N}$ (Theorem 15.46 and Corollary 15.45).

15.6 Set-valued valuations cannot contain the omnific ring

A *valuation* of a field F (possibly a proper class, such as $\mathbf{No}$ or $\mathbf{No}[i]$) is a map $v : F^\times \rightarrow \Gamma$ to an ordered abelian group with $v(xy) = v(x) + v(y)$ and $v(x + y) \geq \min(v(x), v(y))$ whenever $x + y \neq 0$; its valuation ring consists of 0 and the elements of nonnegative value. No compatibility with the ordering of F is assumed. The value group is *set-sized* if Γ is a set.

Lemma 15.21 (No nonzero set-valued ordered quotient of $(\mathbf{No}, +)$ [03]). *Every order-preserving additive class map $w : (\mathbf{No}, +) \rightarrow \Gamma$ into a set-sized ordered abelian group is zero.*

Proof. Suppose $w(x) \neq 0$ for some x . Replacing x by $-x$ if necessary, $w(x) > 0$; then $x \neq 0$, and $x < 0$ would give $w(x) \leq w(0) = 0$, so $h := x > 0$ and $w(h) > 0$. Choose a cardinal $\lambda > |\Gamma|$ and consider the set of surreals αh (surreal products) for ordinals $\alpha < \lambda$. If $\alpha < \beta$, then $\beta \geq \alpha + 1$ in the surreal order, the ordinal successor being the surreal sum with 1. Hence $\beta h - \alpha h \geq h$ and, by additivity and monotonicity,

$$w(\beta h) - w(\alpha h) = w(\beta h - \alpha h) \geq w(h) > 0.$$

So the λ elements αh have pairwise distinct images in Γ , contradicting $\lambda > |\Gamma|$. Only a set of ordinals was used. $\square$

Theorem 15.22 (Full-class valuation obstruction [03]). *Let $v : \mathbf{No}^\times \rightarrow \Gamma$ be a valuation with set-sized value group. If $v(a) \geq 0$ for every $0 \neq a \in \mathbf{Oz}$, then v is trivial. Likewise, a valuation of $\mathbf{No}[i]$ with set-sized value group whose valuation ring contains $\mathbf{Oz}[i]$ is trivial.*

Proof. Define $w(g) = v(\omega^g)$ for $g \in \mathbf{No}$. The monomial law $\omega^{g+g'} = \omega^g \omega^{g'}$ makes w additive. If $g > 0$ then $\omega^g \in \Pi \subseteq \mathbf{Oz}$, so $w(g) \geq 0$; hence $g \leq g'$ implies $w(g') - w(g) = w(g' - g) \geq 0$, and w is order preserving. By Lemma 15.21, $w = 0$.

For $x \neq 0$, clear the set $\{x, x^{-1}\}$ by Proposition 3.4(i): there is a monomial $d = \omega^h$ with $dx, dx^{-1} \in \Pi \subseteq \mathbf{Oz}$. Since $v(d) = w(h) = 0$,

$$v(x) = v(dx) \geq 0, \quad -v(x) = v(x^{-1}) = v(dx^{-1}) \geq 0,$$

so $v(x) = 0$. The complex proof uses the same real monomials and the complex clause of Proposition 3.4(i), with $dx, dx^{-1} \in \Pi_{\mathbb{C}} \subseteq \mathbf{Oz}[i]$. $\square$

Remark 15.23 (Why the size restriction is essential [03]). The theorem does not say that $\mathbf{No}$ has no interesting valuations. Its normal-form valuation $x \mapsto -\max \text{supp } x$ takes values in the proper-class group $(\mathbf{No}, +)$ and is negative on positive-exponent monomials, so it does not satisfy the containment hypothesis anyway. Nor is the theorem a consequence of complete integral closure alone: in the lexicographically ordered group $\mathbb{Z} \oplus \mathbb{Z}$, the values $v(d) = (1, 0)$ and $v(x) = (0, -1)$ would give $v(dx^n) = (1, -n) > 0$ for every ordinary n although $v(x) < 0$. Almost integrality by itself cannot eliminate valuations of higher rank.

For ordinary set-sized domains the integral closure is characterized by valuation overrings with set-sized value groups [35, Lemmas 10.50.3 and 10.50.11]. That valuative criterion of integrality cannot be imported to a class ring while retaining only set-sized value groups: $1/2$ is not integral over $\mathbf{Oz}$, yet Theorem 15.22 rules out every nontrivial valuation of that restricted kind whose ring contains $\mathbf{Oz}$. No assertion is made about the existence or classification of *proper-class-valued* valuation overrings separating nonintegral surreal elements.

15.7 An equation invisible to every set-sized image

Manuscript 08 writes $B_m(t)$ for the binomial series; we write $\mathbf{b}_m(t)$.

Lemma 15.24 (The binomial identity [08]). *For an ordinary integer $m \geq 2$ the formal power series*

$$\mathbf{b}_m(t) = \sum_{n \geq 0} \binom{1/m}{n} t^n \in \mathbb{Q}[[t]]$$

satisfies $\mathbf{b}_m(t)^m = 1 + t$. For every $\gamma \in \mathbf{No}_{>0}$ the substitution $t = \omega^{-\gamma}$ is a valid Hahn substitution, and $\mathbf{b}_m(\omega^{-\gamma})^m = 1 + \omega^{-\gamma}$ in $\mathbf{No}$. The same holds for $t = X^{-\gamma}$ with $\gamma > 0$ in any ordered abelian group, in particular in F_κ^{03} (this last sentence is 03's use of the binomial expansion in its Hahn fields, not 08's).

Proof. Write $b_n = \binom{1/m}{n}$; then $b_0 = 1$ and $(n+1)b_{n+1} = (1/m - n)b_n$. The coefficient of t^n in $(1+t)\mathbf{b}'_m$ is $(n+1)b_{n+1} + nb_n = b_n/m$, so $(1+t)\mathbf{b}'_m = \frac{1}{m}\mathbf{b}_m$. Put $c = \mathbf{b}_m^m = \sum_n c_n t^n$. Then $(1+t)c' = m\mathbf{b}_m^{m-1}(1+t)\mathbf{b}'_m = c$ and $c_0 = 1$. Comparing coefficients gives $(n+1)c_{n+1} = (1-n)c_n$ for $n \geq 0$, so $c_1 = 1$, $c_2 = 0$ and hence $c_n = 0$ for all $n \geq 2$. Thus $c = 1 + t$.

After substitution the support lies in $\{-n\gamma : n \geq 0\}$, which is reverse well ordered. In any fixed finite product each exponent $-n\gamma$ receives finitely many contributions, because an ordinary integer $n \geq 0$ has only finitely many ordered decompositions into a fixed number of nonnegative integers. Hence $\sum_n c_n t^n \mapsto \sum_n c_n \omega^{-n\gamma}$ is a ring homomorphism $\mathbb{Q}[[t]] \rightarrow \mathbf{No}$, and the formal identity remains a Hahn identity. $\square$

Theorem 15.25 (Universal small-target failure of root detection [08]). *Fix $\gamma \in \mathbf{No}_{>0}$ and an ordinary integer $m \geq 2$. The equation*

$$T^m = \omega^\gamma + 1 \tag{15.17}$$

has no root in $\mathbf{Oz}[i]$, and therefore none in $\mathbf{Oz}$. Nevertheless, for every unital ring homomorphism $\varphi : \mathbf{Oz} \rightarrow R$ with R set-sized, the image equation $T^m = \varphi(\omega^\gamma) + 1_R$ has the solution $T = 1_R$. The same holds for unital homomorphisms with domain $\mathbf{Oz}[i]$.

Proof. Put $r = \omega^{\gamma/m} \mathbf{b}_m(\omega^{-\gamma}) \in \mathbf{No}$. By Lemma 15.24, $r^m = \omega^\gamma(1 + \omega^{-\gamma}) = \omega^\gamma + 1$. For every ordinary complex m th root of unity ζ , the element $\zeta r \in \mathbf{No}[i]$ is a root. These are m distinct roots (as $r \neq 0$), so they exhaust the roots of the degree- m polynomial $T^m - \omega^\gamma - 1$ in the field $\mathbf{No}[i]$; no algebraic-closedness argument is needed. The complex normal form of each root begins

$$\zeta r = \zeta \omega^{\gamma/m} + \frac{\zeta}{m} \omega^{\gamma/m - \gamma} + \sum_{n \geq 2} \zeta \binom{1/m}{n} \omega^{\gamma/m - n\gamma}. \tag{15.18}$$

The exponent $\gamma/m - \gamma$ is strictly negative because $m \geq 2$, its coefficient ζ/m is nonzero, and it differs from every other exponent $\gamma/m - n\gamma$ ($n \neq 1$) of the expansion. So this negative-exponent term cannot cancel, and no root lies in $\mathbf{Oz}[i] = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}$, whose elements have nonnegative support.

On the other hand, by [Theorem 5.1](#) every unital homomorphism φ from $\mathbf{Oz}$ to a set-sized ring kills Π , so $\varphi(\omega^\gamma) = 0$ and the image equation is $T^m = 1_R$, solved by 1_R . For a unital homomorphism with domain $\mathbf{Oz}[i]$, the universal theorem with $D = \mathbb{Z}[i]$ and $k = \mathbb{C}$ (see [Corollary 5.3](#) for the description of these maps) again gives $\varphi(\omega^\gamma) = 0$. $\square$

The nonexistence of a root in $\mathbf{Oz}[i]$ is also a case of 16's image-gap theorem ([Remark 16.37\(c\)](#)). This is not merely a failure of a local-global test by congruences modulo ordinary primes. Even allowing *every set-sized ring* as a target does not reveal the missing negative tail in [\(15.18\)](#). The obstruction belongs to the full normal-form arithmetic and survives in the proper-class ring itself [\(08\)](#).

Corollary 15.26 ([08; 03]). *The ring $\mathbf{Oz}$ is not integrally closed in its fraction field $\mathbf{No}$.*

Proof. The positive square root of $\omega + 1$ lies in $\mathbf{No} = \text{Frac}(\mathbf{Oz})$ ([Proposition 3.4\(ii\)](#)). It is integral over $\mathbf{Oz}$, being a root of the monic polynomial $T^2 - (\omega + 1)$, but it is not in $\mathbf{Oz}$ by [Theorem 15.25](#) with $m = 2$ and $\gamma = 1$. $\square$

The same corollary is contained in 03's [Theorem 15.12](#), with the witness $\sqrt{2}$, and manuscript 05 of the sibling report [\[46\]](#) proves it with the same witness $\sqrt{2}$. This elementary corollary is included to clarify the distinction between an integer part and an integral closure; it is not offered as a new factorization theorem [\(08\)](#).

Remark 15.27 (Explicit elements of $\mathcal{N} \setminus \mathbf{Oz}$ [merge]). Let r be the positive root of $T^m - (\omega^\gamma + 1)$ in $\mathbf{No}$, for $\gamma > 0$ and $m \geq 2$; it is the element $r = \omega^{\gamma/m} \mathbf{b}_m(\omega^{-\gamma})$ of the proof of [Theorem 15.25](#). Then

$$r \in \mathcal{N} \setminus \mathbf{Oz}, \quad \zeta r \in \mathcal{N}_{\mathbb{C}} \setminus \mathbf{Oz}[i] \quad (\zeta^m = 1).$$

Indeed r is a root of a monic polynomial with coefficients in $\mathbf{Oz}$, so it lies in $\mathcal{N}$ by definition, and ζr lies in $\mathcal{N}_{\mathbb{C}}$ for the same reason; neither lies in the respective integer ring by [Theorem 15.25](#). Thus 08's equation exhibits explicit elements of the normalization whose absence from $\mathbf{Oz}$ is invisible to every set-sized image: under every unital homomorphism from $\mathbf{Oz}$ to a set-sized ring the defining equation of r becomes $T^m = 1$, which already has the root 1. Equivalently, if ψ is a unital homomorphism from $\mathcal{N}$ (or from $\mathbf{Oz}[r]$) to a set-sized ring, then its restriction to $\mathbf{Oz}$ kills ω^γ by [Theorem 5.1](#), so $\psi(r)^m = 1$. Unlike $\sqrt{2}$, the elements r are not constants: by [\(15.18\)](#) they have infinite leading term $\omega^{\gamma/m}$ and a nonzero negative tail. Nothing is claimed in this remark about the set-sized images of $\mathcal{N}$ themselves; they are the subject of [Section 15.8](#), where [Remark 15.47](#) sharpens $\psi(r)^m = 1$ for domain targets.

15.8 The arithmetic fibre of the normalization

This subsection is manuscript 19's. It studies the normalizations $\mathcal{N}$ and $\mathcal{N}_{\mathbb{C}}$ (19's $N_{\mathbb{R}}$ and $N_{\mathbb{C}}$) modulo the ideals that Π generates in them, and it answers part of [Question 18.11](#). We write $\mathcal{N}_{\bullet}$ for $\mathcal{N}_{\mathbb{R}} = \mathcal{N}$ or $\mathcal{N}_{\mathbb{C}}$, $\Pi \mathcal{N}_{\bullet}$ for the extended ideal of finite sums of products (19's $K_{\mathbb{R}}$, $K_{\mathbb{C}}$), and

$$\mathbf{B}_{\bullet} = \mathcal{N}_{\bullet} / \Pi \mathcal{N}_{\bullet}, \quad \mathbf{B}_{\bullet}^{\mathbb{Q}} = \mathbf{B}_{\bullet} \otimes_{\mathbb{Z}} \mathbb{Q}$$

(19's B_{ϵ} , E_{ϵ} ; 03 had used B for the normalization itself, here $\mathcal{N}$, and the letters $\mathcal{E}$ and $\mathbf{E}$ are taken). $\text{Idem}(R)$ is the Boolean algebra of idempotents of a commutative ring R (19's $\mathcal{E}$), $\overline{\mathbb{Z}}$ is the ring of all complex algebraic integers, $\overline{\mathbb{Q}}$ its fraction field and $\overline{\mathbb{F}}_p$ an algebraic closure of $\mathbb{F}_p$. Quotients by class ideals are read as in [Section 2.1](#). 19 did not see 03's text ([Section 1.3](#)), and four of its results are 03's,

printed above and credited here: the constant slices $\mathcal{N} \cap \mathbb{R} = \overline{\mathbb{Z}}_{\mathbb{R}}$ and $\mathcal{N}_{\mathbb{C}} \cap \mathbb{C} = \overline{\mathbb{Z}}$ (Theorems 15.14 and 15.19), the absence of nonzero finite images (Theorem 15.20; a different proof is Theorem 15.46 below), the strict inclusion $\mathcal{N}[i] \subsetneq \mathcal{N}_{\mathbb{C}}$ (Proposition 15.18; 19's witness is $w_{\pm}$ of Proposition 15.43 in place of ζ_3), and infinitesimal integral units (Theorem 15.9; 19's unit of Proposition 15.34 has a different residue). Its reproof of the universal theorem for $\mathbf{Oz}$ is credited in Section 5.1.

Lemma 15.28 (Contraction [19, 20]). $\Pi\mathcal{N}_{\bullet} \cap \mathbf{Oz} = \Pi$, and $\Pi\mathcal{N}_{\bullet}$ is divisible by every nonzero ordinary integer. Hence $\mathbf{B}_{\bullet}$ is a nonzero ring containing $\mathbb{Z}$, torsion free over $\mathbb{Z}$, and integral over $\mathbb{Z}$.

Proof. Suppose $0 \neq n \in \mathbb{Z}$ lies in $\Pi\mathcal{N}_{\bullet}$, say $n = \sum_{j \leq r} a_j b_j$ with $a_j \in \Pi$ and $b_j \in \mathcal{N}_{\bullet}$. The ring $C = \mathbf{Oz}[b_1, \dots, b_r]$ is generated as an $\mathbf{Oz}$ -module by finitely many elements $c_1 = 1, c_2, \dots, c_s$, and $nc_l = \sum_m h_{lm} c_m$ with all $h_{lm} \in \Pi$ because $n \in \Pi C$. The adjugate of $nI - (h_{lm})$ shows that $\det(nI - (h_{lm}))$ kills every c_l , in particular $c_1 = 1$; expanding, $n^s + j_{s-1}n^{s-1} + \dots + j_0 = 0$ with $j_{\nu} \in \Pi$, and ct gives $n^s = 0$, which is absurd. If $a \in \Pi\mathcal{N}_{\bullet} \cap \mathbf{Oz}$, then $\text{ct}(a) = a - (a - \text{ct}(a)) \in \Pi\mathcal{N}_{\bullet}$, so $\text{ct}(a) = 0$. Divisibility: $\sum_j a_j b_j / n = \sum_j (a_j / n) b_j$. So $nx \in \Pi\mathcal{N}_{\bullet}$ with $x \in \mathcal{N}_{\bullet}$ forces $x \in \Pi\mathcal{N}_{\bullet}$, which is torsion freeness. Reducing a monic equation of $x \in \mathcal{N}_{\bullet}$ over $\mathbf{Oz}$ modulo $\Pi\mathcal{N}_{\bullet}$ replaces each coefficient a by $\text{ct}(a) \in \mathbb{Z}$. $\square$

A large normal overring need not preserve the contraction of an ideal; the determinant argument prevents $\Pi\mathcal{N}_{\bullet}$ from containing 1, although it contains elements that differ from 1 by an infinitesimal (19).

Theorem 15.29 (Radicality [19, 20]). If $x \in \mathcal{N}_{\bullet}$ and $x^m \in \Pi\mathcal{N}_{\bullet}$ for some ordinary $m \geq 1$, then $x \in \Pi\mathcal{N}_{\bullet}$. Hence $\mathbf{B}_{\bullet}$ and $\mathbf{B}_{\bullet}^{\mathbb{Q}}$ are reduced, $\mathbf{B}_{\bullet} \rightarrow \mathbf{B}_{\bullet}^{\mathbb{Q}}$ is injective, and $\mathbf{B}_{\bullet}^{\mathbb{Q}}$ is algebraic over $\mathbb{Q}$.

Proof. Write $x^m = \sum_j a_j b_j$ with $a_j \in \Pi$, $b_j \in \mathcal{N}_{\bullet}$. By Proposition 3.3(i) there is $a > 0$ with every $a_j \in \omega^a \Pi$; put $d = \omega^{a/m}$, so that $a_j / d^m \in \Pi$ for all j . Then $(x/d)^m = \sum_j (a_j / d^m) b_j \in \mathcal{N}_{\bullet}$, so x/d is a root of a monic polynomial over $\mathcal{N}_{\bullet}$ and lies in $\mathcal{N}_{\bullet}$; hence $x = d(x/d) \in \Pi\mathcal{N}_{\bullet}$. Reducedness passes to the localization, injectivity is torsion freeness, and a rational multiple of an element integral over $\mathbb{Z}$ is algebraic. $\square$

Theorem 15.30 (Rational saturation [19, 20]). $\mathbf{B}_{\bullet}$ is integrally closed in $\mathbf{B}_{\bullet}^{\mathbb{Q}}$, and $\mathbf{B}_{\bullet} = \{y \in \mathbf{B}_{\bullet}^{\mathbb{Q}} : y \text{ is integral over } \mathbb{Z}\}$.

Proof. Represent y by x/n with $x \in \mathcal{N}_{\bullet}$ and $0 \neq n \in \mathbb{Z}$, and let y satisfy a monic equation over $\mathbf{B}_{\bullet}$; lift it to a monic $f \in \mathcal{N}_{\bullet}[X]$. The kernel of $\mathcal{N}_{\bullet} \otimes \mathbb{Q} \rightarrow \mathbf{B}_{\bullet}^{\mathbb{Q}}$ is the localization of $\Pi\mathcal{N}_{\bullet}$, which is $\Pi\mathcal{N}_{\bullet}$ itself by divisibility (Lemma 15.28); so $c = f(x/n) \in \Pi\mathcal{N}_{\bullet}$, and x/n is a root of the monic $f(X) - c \in \mathcal{N}_{\bullet}[X]$, hence lies in $\mathcal{N}_{\bullet}$. Elements of $\mathbf{B}_{\bullet}$ are integral over $\mathbb{Z}$ by Lemma 15.28, and an element of $\mathbf{B}_{\bullet}^{\mathbb{Q}}$ integral over $\mathbb{Z}$ is integral over $\mathbf{B}_{\bullet}$. $\square$

Proposition 15.31 (The extended ideal [19]). $\Pi\mathcal{N}_{\bullet} = (\Pi\mathcal{N}_{\bullet})^2 = \bigcup_{\delta > 0} \omega^{\delta} \mathcal{N}_{\bullet}$, and $\Pi\mathcal{N}_{\bullet}$ is not generated by any set of its elements.

Proof. For $x = \sum_j a_j b_j$ choose a positive monomial d with $a_j / d \in \Pi$ (Proposition 3.3(i)); then $x/d \in \Pi\mathcal{N}_{\bullet}$ and $x = d \cdot (x/d)$. If a set G generated $\Pi\mathcal{N}_{\bullet}$, then for each $g \in G$ the class $D_g = \{\delta > 0 : g \in \omega^{\delta} \mathcal{N}_{\bullet}\}$ is nonempty and downward closed, and its elements of least birthday form a set; a positive δ below the union of these sets lies in every D_g , so $G \subseteq \omega^{\delta} \mathcal{N}_{\bullet}$. Then $\omega^{\delta/2} \in \Pi$ would lie in $\omega^{\delta} \mathcal{N}_{\bullet}$, so $\omega^{-\delta/2}$ and hence $\omega^{-\delta}$ would lie in $\mathcal{N}_{\bullet}$, and $1 \in \omega^{\delta} \mathcal{N}_{\bullet} \subseteq \Pi\mathcal{N}_{\bullet}$, contrary to Lemma 15.28. $\square$

19 chooses one expression per generator with global choice; the least-birthday selection makes the argument choice free. This parallels 03's statement that $\mathcal{N}$ is not set-generated as an $\mathbf{Oz}$ -algebra (Theorem 15.12), which concerns a different object.

Theorem 15.32 (Real-rooted lifting [19]). *Let $P(X) = X^d + \sum_{k < d} a_k X^k \in \mathcal{N}[X]$ with $d \geq 1$, let $M \in \mathbf{No}$ with $M \geq 1$ and $M \geq |a_k|$ for all k , put $\eta_d = \frac{1}{4}(\frac{3}{4})^{d-1}$ and $C_d = d(d+1)^{d-1}$, and let $T \in \Pi$ be a positive monomial with $T > C_d M / \eta_d$. Then*

$$Q(X) = P(X) + \prod_{j=1}^d (X - jT) - X^d$$

satisfies $Q - P \in \Pi[X]$ and has exactly one root in each interval $((j - \frac{1}{4})T, (j + \frac{1}{4})T)$, $1 \leq j \leq d$; these are d distinct, positive, infinite roots, and they lie in $\mathcal{N}$.

Proof. Put $R_T = \prod_j (X - jT)$. At an endpoint $x = (j \pm \frac{1}{4})T$ the factor $x - jT$ has modulus $T/4$ and the others at least $3T/4$, so $|R_T(x)| \geq \eta_d T^d$, and R_T has opposite signs at the two endpoints of the j th interval. Since $|x| \leq (d+1)T$ and $T > 1$, $|P(x) - x^d| \leq \sum_{k < d} |a_k| |x|^k \leq C_d M T^{d-1} < \eta_d T^d$, so Q has the sign of R_T at every endpoint, and the intermediate value property of polynomials over the real closed field $\mathbf{No}$ gives a root in each of the d disjoint intervals; there are no others. The roots exceed $\frac{3}{4}T$, which is infinite; they are integral over $\mathcal{N}$, hence in $\mathcal{N}$. The coefficients of $R_T - X^d$ are integer multiples of positive powers of T . $\square$

Corollary 15.33 (Absolute integral closedness [19]). *$\mathbf{B}_{\mathbb{R}}$, $\mathcal{N}_{\mathbb{C}}$ and $\mathbf{B}_{\mathbb{C}}$ are absolutely integrally closed: every monic polynomial over them splits into linear factors. $\mathcal{N}$ itself is not, since $X^2 + 1$ has no root in $\mathcal{N} \subseteq \mathbf{No}$.*

Proof. Lift a monic polynomial over $\mathbf{B}_{\mathbb{R}}$ to $\mathcal{N}$, apply Theorem 15.32 and reduce the factorization; the change of coefficients lies in $\Pi[X]$. Over $\mathcal{N}_{\mathbb{C}}$ the roots in the algebraically closed field $\mathbf{No}[i]$ are integral, hence in $\mathcal{N}_{\mathbb{C}}$; reduce for $\mathbf{B}_{\mathbb{C}}$. $\square$

The lifted roots need not stay distinct modulo $\Pi\mathcal{N}$: the lifting preserves the factorization, not the separation of residues (19).

Proposition 15.34 (An infinitesimal unit whose residue squares to -1 [19]). *Let $T \in \Pi$ be a positive monomial and $v_T = (T - \sqrt{T^2 - 4})/2 = 2/(T + \sqrt{T^2 - 4})$. Then $v_T^2 - Tv_T + 1 = 0$, v_T is a positive infinitesimal unit of $\mathcal{N}$ with inverse $T - v_T$, $v_T \notin \mathbf{Oz}$, and its image $\mathbf{v} \in \mathbf{B}_{\mathbb{R}}$ satisfies $\mathbf{v}^2 = -1$. For $T = \omega$,*

$$v_{\omega} = \omega^{-1} + \omega^{-3} + 2\omega^{-5} + 5\omega^{-7} + 14\omega^{-9} + \cdots = \sum_{n \geq 0} \frac{1}{n+1} \binom{2n}{n} \omega^{-(2n+1)}.$$

Consequently $\mathcal{N} \rightarrow \mathbf{B}_{\mathbb{R}}$ is not compatible with any ordered-ring quotient, and $\Pi\mathcal{N}$ is not convex in $\mathbf{No}$.

Proof. The equation is the quadratic formula; it is monic over $\mathbf{Oz}$ and says $v_T(T - v_T) = 1$. The second expression gives $0 < v_T < 2/T$, and $\mathbf{Oz}$ has no positive element below 1. Modulo $\Pi\mathcal{N}$, $v_T^2 + 1 = Tv_T \in \Pi\mathcal{N}$. Writing $v_{\omega} = \omega^{-1}C(\omega^{-2})$ turns the equation into $C = 1 + zC^2$, whose solution in $\mathbb{Q}[[z]]$ has the Catalan coefficients; the substitution $z = \omega^{-2}$ is a Hahn substitution as in Lemma 15.24, and the sum is a strong sum, not a limit of partial sums. Finally $0 < 1 < Tv_T = 1 + v_T^2$ with $Tv_T \in \Pi\mathcal{N}$ and $1 \notin \Pi\mathcal{N}$. $\square$

03's unit $u_H = \sqrt{H^2 + 1} - H$ with $H \in \Pi$ (Theorem 15.9) has a residue $\bar{u}_H$ with $\bar{u}_H^2 = 1$; 19's residue squares to -1 . Normalization adds a real infinitesimal whose arithmetic residue behaves like an imaginary unit; no imaginary surreal is inserted into the real field, and it is the quotient that loses the order (19).

Lemma 15.35 (A coefficient copy of $\overline{\mathbb{Q}}$ [19, 20]). *There is an embedding $\overline{\mathbb{Q}} \hookrightarrow \mathbf{B}_\bullet^\mathbb{Q}$ whose restriction to $\overline{\mathbb{Z}}$ lands in $\mathbf{B}_\bullet$.*

Proof. Real algebraic integers lie in $\mathcal{N}$. A nonzero one, a , satisfies $ab = m$ for a real algebraic integer b and some $0 \neq m \in \mathbb{Z}$; as $\mathbb{Z} \subseteq \mathbf{B}_\bullet$ is torsion free, the image of a is a unit of $\mathbf{B}_\bullet^\mathbb{Q}$. So the field $k_0 = \overline{\mathbb{Q}} \cap \mathbb{R}$ of real algebraic numbers embeds. In the real case $X \mapsto v$ defines $k_0[X]/(X^2 + 1) \rightarrow \mathbf{B}_\mathbb{R}^\mathbb{Q}$, injective because its source is the field $k_0(i) \cong \overline{\mathbb{Q}}$; in the complex case use $i \in \mathcal{N}_\mathbb{C}$. Algebraic integers map into $\mathbf{B}_\bullet$ by Theorem 15.30. $\square$

In the real case the embedding depends on the choice of v ; it is an embedding of algebraic numbers into the quotient rationalization, not an embedding of $\mathbb{C}$ into $\mathbf{No}$, whose existence is not asserted (19).

Lemma 15.36 (Spectral idempotents [19, 20]). *Let E be a reduced algebraic $\mathbb{Q}$ -algebra containing a unital copy of $\overline{\mathbb{Q}}$. Every $x \in E$ is $\sum_{j \leq r} \lambda_j e_j$ with distinct $\lambda_j \in \overline{\mathbb{Q}}$ and orthogonal idempotents e_j with $\sum_j e_j = 1$, and finitely many elements have such expressions on one common partition.*

Proof. Let p be the squarefree part of a nonzero rational polynomial killing x ; $p(x)$ is nilpotent, hence 0. Factor $p = c \prod_j (X - \lambda_j)$ over $\overline{\mathbb{Q}}$ and put $e_j = L_j(x)$ with the Lagrange polynomials $L_j = \prod_{k \neq j} (X - \lambda_k) / (\lambda_j - \lambda_k)$; modulo p , $L_j^2 \equiv L_j$, $L_j L_k \equiv 0$, $\sum_j L_j \equiv 1$ and $X \equiv \sum_j \lambda_j L_j$. For several elements multiply their partitions. $\square$

Definition 15.37 (Finite-partition Boolean powers [19, 20]). For a Boolean algebra $\mathfrak{E}$ (possibly a proper class) and a commutative ring R , an element of $R[\mathfrak{E}]$ is a finite partition $1 = e_1 \vee \dots \vee e_r$ into disjoint elements of $\mathfrak{E}$ with values $d_1, \dots, d_r \in R$; values on zero parts are ignored, two such objects are equal when their values agree on every nonzero intersection of parts, and the ring operations are computed on a common refinement. For idempotents, $e \wedge f = ef$, $\neg e = 1 - e$ and $e \vee f = e + f - ef$. No infinite sums occur.

Theorem 15.38 (The fibre is a Boolean power of the algebraic integers [19, 20]). *With the coefficient embedding of Lemma 15.35, $\text{Idem}(\mathbf{B}_\bullet^\mathbb{Q}) = \text{Idem}(\mathbf{B}_\bullet)$, and evaluation $(\lambda_j, e_j)_j \mapsto \sum_j \lambda_j e_j$ gives isomorphisms*

$$\mathbf{B}_\bullet^\mathbb{Q} \cong \overline{\mathbb{Q}}[\text{Idem}(\mathbf{B}_\bullet)], \quad \mathbf{B}_\bullet \cong \overline{\mathbb{Z}}[\text{Idem}(\mathbf{B}_\bullet)].$$

Every finite family of elements of $\mathbf{B}_\bullet$ lies in a subring isomorphic to $\overline{\mathbb{Z}}^r$, embedded by an orthogonal partition of 1.

Proof. Idempotents are roots of $X^2 - X$, so Theorem 15.30 puts those of $\mathbf{B}_\bullet^\mathbb{Q}$ in $\mathbf{B}_\bullet$. Evaluation is onto by Lemma 15.36 and Theorem 15.29. On a nonzero common part e of two representations, equality gives $(\lambda - \mu)e = 0$, and a nonzero $\lambda - \mu \in \overline{\mathbb{Q}}$ is a unit; so evaluation is injective. If $x = \sum_j \lambda_j e_j \in \mathbf{B}_\bullet$ with every $e_j \neq 0$ and $f \in \mathbb{Z}[X]$ is monic with $f(x) = 0$, then $f(\lambda_j)e_j = 0$, so $f(\lambda_j) = 0$ and $\lambda_j \in \overline{\mathbb{Z}}$; conversely a finite sum of commuting integral elements is integral, and lies in $\mathbf{B}_\bullet$ by Theorem 15.30. On a common partition with nonzero parts, the map from $\overline{\mathbb{Z}}^r$ is injective. $\square$

The Boolean description gives [Corollary 15.33](#) again: refine the coefficients of a monic polynomial to one partition and split it over $\overline{\mathbb{Z}}$ on each part. $\mathbf{B}_\bullet$ is neither one finite product of copies of $\overline{\mathbb{Z}}$ nor an unrestricted product; the finite-partition restriction is part of the theorem, and 19 does not determine $\text{Idem}(\mathbf{B}_\mathbb{R})$ (19). 20 shows that it is nontrivial, atomless and a proper class with explicit independent families ([Theorems 15.72](#) and [15.73](#)); its isomorphism type is not determined.

Theorem 15.39 (Finitely generated ideals of the fibre [19]). *Every finitely generated ideal of $\overline{\mathbb{Z}}$ is principal (a classical fact [11]). Every finitely generated ideal I of $\mathbf{B}_\bullet$ is principal, $I = g\mathbf{B}_\bullet$, and there is an idempotent e with $\text{Ann}(g) = (1 - e)\mathbf{B}_\bullet$ and $I \cong e\mathbf{B}_\bullet$; so I is projective and $\mathbf{B}_\bullet$ is semihereditary.*

Proof. For $I = (a_1, \dots, a_r)\overline{\mathbb{Z}} \neq 0$ choose a number field L containing the a_j , put $I_0 = (a_1, \dots, a_r)O_L$, and use finiteness of the class group [25, Theorem 4.4] to get $h \geq 1$ and $c \in O_L$ with $I_0^h = cO_L$. With $t \in \overline{\mathbb{Z}}$, $t^h = c$, every $x \in I_0$ has $(x/t)^h = x^h/c \in O_L$, so $I \subseteq t\overline{\mathbb{Z}}$; and $(I/t)^h = \overline{\mathbb{Z}}$ forces $I/t = \overline{\mathbb{Z}}$. For $\mathbf{B}_\bullet$ put the generators on a common partition $e_1, \dots, e_q$, write $x_i = \sum_j a_{ij}e_j$, choose generators g_j of $(a_{1j}, \dots, a_{rj})\overline{\mathbb{Z}}$ and put $g = \sum_j g_j e_j$ and $e = \sum_{g_j \neq 0} e_j$. The coefficient identities on each part show $I = g\mathbf{B}_\bullet$; since $\overline{\mathbb{Z}}$ is a domain, $\text{Ann}(g) = (1 - e)\mathbf{B}_\bullet$, and multiplication by g identifies $e\mathbf{B}_\bullet$ with I . $\square$

No Bézout property of $\mathbf{Oz}$ or of $\mathcal{N}_\bullet$ is claimed; the theorem concerns their arithmetic fibre (19).

Proposition 15.40 (Nilpotents modulo a prime [19]). *For every ordinary prime p and $n \geq 2$, $\mathbf{B}_\bullet/p\mathbf{B}_\bullet$ has an element of nilpotence index exactly n ; it is nonzero and not reduced.*

Proof. Take $\alpha = p^{1/n} \in \overline{\mathbb{Z}} \subseteq \mathbf{B}_\bullet$. If $\alpha^k \in p\mathbf{B}_\bullet$ with $1 \leq k < n$, then $\alpha^k/p \in \mathbf{B}_\bullet$ is integral over $\mathbb{Z}$, and so is its n th power $p^{k-n} \in \mathbb{Q} \setminus \mathbb{Z}$, which is absurd; the case $k = 0$ is $1 \notin p\mathbf{B}_\bullet$. $\square$

Lemma 15.41 (Rational comparison [19, 20]). $\mathcal{N}_\mathbb{C} \otimes \mathbb{Q} = (\mathcal{N} \otimes \mathbb{Q})[i]$ and $\mathbf{B}_\mathbb{C}^\mathbb{Q} \cong \mathbf{B}_\mathbb{R}^\mathbb{Q}[X]/(X^2 + 1)$; the map $\mathbf{B}_\mathbb{R} \rightarrow \mathbf{B}_\mathbb{C}$ is injective, that is, $\Pi\mathcal{N}_\mathbb{C} \cap \mathcal{N} = \Pi\mathcal{N}$.

Proof. For $z \in \mathcal{N}_\mathbb{C}$, $z + \bar{z}$ and $(z - \bar{z})/i$ are real and integral, so lie in $\mathcal{N}$, and $2z$ is in $\mathcal{N}[i]$ (as in [Proposition 15.18](#)); with $\mathcal{N}[i] \subseteq \mathcal{N}_\mathbb{C}$ this gives the first equality, and $1, i$ are independent over $\mathcal{N} \otimes \mathbb{Q}$. Since $\Pi\mathcal{N}$ is rationally divisible, $\Pi\mathcal{N}_\mathbb{C} \otimes \mathbb{Q} = \Pi\mathcal{N} \otimes \mathbb{Q} + i(\Pi\mathcal{N} \otimes \mathbb{Q})$, and the quotient is $\mathbf{B}_\mathbb{R}^\mathbb{Q}[i]$, into which $\mathbf{B}_\mathbb{R}^\mathbb{Q}$ injects. $\square$

With $\mathbf{v}$ from [Proposition 15.34](#), $\mathbf{B}_\mathbb{R}^\mathbb{Q}$ contains $\frac{1}{2}$ and the unit $\mathbf{v}$, and the Chinese remainder map

$$\Psi : \mathbf{B}_\mathbb{R}^\mathbb{Q}[i] \rightarrow \mathbf{B}_\mathbb{R}^\mathbb{Q} \times \mathbf{B}_\mathbb{R}^\mathbb{Q}, \quad a + ib \mapsto (a + \mathbf{v}b, a - \mathbf{v}b), \quad (15.19)$$

is an isomorphism with inverse $(r, s) \mapsto \frac{1}{2}(r + s) + \frac{i}{2\mathbf{v}}(r - s)$.

Theorem 15.42 (Integral doubling [19, 20]). Ψ restricts to an isomorphism $\mathbf{B}_\mathbb{C} \cong \mathbf{B}_\mathbb{R} \times \mathbf{B}_\mathbb{R}$ of $\mathbf{B}_\mathbb{R}$ -algebras, with $\mathbf{B}_\mathbb{R}$ acting diagonally and complex conjugation exchanging the factors. Hence $\mathbf{B}_\mathbb{C}/m\mathbf{B}_\mathbb{C} \cong (\mathbf{B}_\mathbb{R}/m\mathbf{B}_\mathbb{R})^2$ for every ordinary $m \geq 2$.

Proof. By [Theorem 15.30](#), $\mathbf{B}_\mathbb{C}$ is the set of elements of $\mathbf{B}_\mathbb{C}^\mathbb{Q}$ integral over $\mathbb{Z}$. A pair is integral over the diagonal $\mathbb{Z}$ exactly when both coordinates are (multiply two monic annihilators), and the integral elements of $\mathbf{B}_\mathbb{R}^\mathbb{Q}$ are $\mathbf{B}_\mathbb{R}$. The rest is read off from (15.19). $\square$

Factoring $X^2 + 1$ over $\mathbf{B}_{\mathbb{R}}$ and applying the Chinese remainder theorem there would not prove this, because 2 need not be a unit; rationalization followed by the characterization of integral elements is what gives the integral statement (19).

Proposition 15.43 (The integral idempotents [19]). *With $v = v_T$ as in Proposition 15.34, $w_{\pm} = (1 \mp iv)/2$ lie in $\mathcal{N}_{\mathbb{C}}$; their residues are complementary nonzero idempotents of $\mathbf{B}_{\mathbb{C}}$, sent by Ψ to $(1, 0)$ and $(0, 1)$. They are not in $\mathcal{N}[i]$.*

Proof. From $v^2 + 1 = Tv$: $w_{\pm}^2 - w_{\pm} = -Tv/4$, $w_+ + w_- = 1$ and $w_+w_- = Tv/4$, where $Tv/4 = (T/4)v \in \Pi\mathcal{N}$. So each $w_{\pm}$ is integral over $\mathcal{N}$, hence over $\mathbf{Oz}$, and the residues are complementary idempotents; Ψ gives the coordinate vectors, using $v^2 = -1$. The real coordinate of $w_{\pm}$ is $\frac{1}{2} \notin \mathcal{N}$ (Lemma 15.7). $\square$

So $w_{\pm}$ is a second witness for $\mathcal{N}[i] \subsetneq \mathcal{N}_{\mathbb{C}}$ (03's is ζ_3): normalization adds integral elements written with a denominator 2, certified by $Tv/4 \in \Pi\mathcal{N}$ (19).

Corollary 15.44 (Characteristic two, trace and norm [19]). *Modulo 2 the residues of $w_{\pm}$ still split $\mathbf{B}_{\mathbb{C}}/2\mathbf{B}_{\mathbb{C}} \cong (\mathbf{B}_{\mathbb{R}}/2\mathbf{B}_{\mathbb{R}})^2$ into two nonzero factors, although $i \mapsto (v, -v)$ has equal coordinates modulo 2. Field trace and norm induce $\text{Tr}, \text{Nm} : \mathbf{B}_{\mathbb{C}} \rightarrow \mathbf{B}_{\mathbb{R}}$, equal to $(r, s) \mapsto r + s$ and $(r, s) \mapsto rs$; the fixed ring of conjugation is the diagonal $\mathbf{B}_{\mathbb{R}}$, and the trace is onto, also modulo 2.*

Proof. The factors are nonzero by Proposition 15.40. For $z \in \mathcal{N}_{\mathbb{C}}$, $z + \bar{z}$ and $z\bar{z}$ lie in $\mathcal{N}$, and changing z modulo $\Pi\mathcal{N}_{\mathbb{C}}$ changes them by elements of $\Pi\mathcal{N}_{\mathbb{C}} \cap \mathcal{N} = \Pi\mathcal{N}$ (Lemma 15.41). Conjugation swaps the coordinates, and $(r, 0)$ has trace r . $\square$

The mod-2 splitting is a product of two nonreduced rings, not of fields. The norm-one units of $\mathbf{B}_{\mathbb{C}}$ form a group isomorphic to $\mathbf{B}_{\mathbb{R}}^{\times}$ via $r \mapsto (r, r^{-1})$, on which conjugation acts by inversion; it is not asserted that such units lift to norm-one units of $\mathcal{N}_{\mathbb{C}}$ (19).

Corollary 15.45 (Set-sized images factor through the fibre [19, 20]). *Every unital homomorphism from $\mathcal{N}_{\bullet}$ to a set-sized ring kills $\Pi\mathcal{N}_{\bullet}$ and factors uniquely through $\mathbf{B}_{\bullet}$.*

Proof. Its restriction to $\mathbf{Oz}$ kills Π (Theorem 5.1), hence it kills every finite sum of products in $\Pi\mathcal{N}_{\bullet}$. $\square$

$\mathbf{B}_{\bullet}$ is not asserted to be a set: this is a factorization property for each set-sized target, not a universal object in a category of set-sized rings (19). In fact $\mathbf{B}_{\bullet}$ is a proper class, and no set-indexed family of set-sized targets is jointly faithful on it (20; Theorems 15.73 and 15.79 and Corollaries 15.77 and 15.81).

Theorem 15.46 (Domain images [19]). *Let $\varphi : \mathcal{N}_{\bullet} \rightarrow S$ be a unital homomorphism to a set-sized ring whose image R is a nonzero domain. Then $R \cong \overline{\mathbb{Z}}$ if $\text{char } R = 0$ and $R \cong \overline{\mathbb{F}}_p$ if $\text{char } R = p > 0$. In particular every such image is countable, no nonzero finite ring is a unital image of $\mathcal{N}_{\bullet}$ (Theorem 15.20), and no set-sized formally real field is a unital target.*

Proof. By Corollary 15.45 and Theorem 15.38, φ factors through $\overline{\mathbb{Z}}[\text{Idem}(\mathbf{B}_{\bullet})]$. In a domain each idempotent goes to 0 or 1, so exactly one part of each finite partition goes to 1, and R is the image of $\overline{\mathbb{Z}}$. In characteristic 0 this map is injective: a nonzero $\alpha \in \overline{\mathbb{Z}}$ divides a nonzero ordinary integer. In characteristic p , R is a domain algebraic over $\mathbb{F}_p$, hence a field, and it is algebraically closed because monic polynomials over $\overline{\mathbb{Z}}$ split; so $R \cong \overline{\mathbb{F}}_p$. A nonzero finite image, which is commutative,

has a finite field as a quotient, and that would be a finite domain image; in a formally real field the image of v (or of i) would square to -1 . $\square$

This answers, for domain targets, the clause of [Question 18.11](#) asking for the infinite set-sized images of $\mathcal{N}$, and it contains 03's absence of finite images with a different proof (03's uses a root of $T^q - T - 1$). Images with zero divisors are not classified beyond the factorization through $\mathbf{B}_\bullet$, and it is not claimed that all set-sized images are countable (19).

Remark 15.47 (Images of 08's invisible roots [merge]). Let $r = \omega^{\gamma/m} \mathbf{b}_m(\omega^{-\gamma}) \in \mathcal{N}$ be the root of $T^m = \omega^\gamma + 1$ of [Remark 15.27](#). Its residue in $\mathbf{B}_\mathbb{R}$ satisfies $\bar{r}^m = 1$, so under [Theorem 15.38](#) it is $\sum_j \zeta_j e_j$ with $\zeta_j^m = 1$ on every nonzero part; under every unital map from $\mathcal{N}$ to a set-sized domain, r goes to an m th root of unity of $\overline{\mathbb{Z}}$ or $\overline{\mathbb{F}}_p$. This sharpens " $\psi(r)^m = 1$ " of [Remark 15.27](#) for domain targets.

Lemma 15.48 (Class ultrafilters [19, 20]). *In Gödel–Bernays class theory with global choice, every Boolean algebra whose elements are sets, even a proper class, has an ultrafilter (a class) containing any prescribed nonzero element.*

Proof. Enumerate the algebra along a global well-order, by an ordinal or by the class of ordinals, starting from the prescribed element e . At stage α add b_α if the decisions so far together with b_α have the finite intersection property, and otherwise $\neg b_\alpha$. If both failed, finite meets c, d of earlier decisions would have $c \wedge b_\alpha = 0 = d \wedge \neg b_\alpha$, hence $c \wedge d = 0$, which is impossible; limit stages preserve the finite intersection property. The history at each stage is a set, so ordinary transfinite recursion with the global well-order as a class parameter defines the class of decisions, which generates an ultrafilter containing e . $\square$

No class form of Zorn's lemma and no measurable cardinal is used; one part of each finite partition is chosen, without any countable completeness (19).

Theorem 15.49 (Detection by maps to $\overline{\mathbb{Z}}$ [19, 20]). *Assume global choice. For an ultrafilter U on $\text{Idem}(\mathbf{B}_\bullet)$, $\sum_j \lambda_j e_j \mapsto \lambda_{j_0}$ with $e_{j_0} \in U$ is a surjective unital homomorphism $\overline{\mathbb{Z}}[\text{Idem}(\mathbf{B}_\bullet)] \rightarrow \overline{\mathbb{Z}}$. Every $x \in \mathcal{N}_\bullet \setminus \Pi \mathcal{N}_\bullet$ is sent to a nonzero element by a surjective class homomorphism $\mathcal{N}_\bullet \rightarrow \overline{\mathbb{Z}}$. So $\Pi \mathcal{N}_\bullet$ is the common kernel of the unital maps from $\mathcal{N}_\bullet$ to countable domains, and already of those to $\overline{\mathbb{Z}}$, in the pointwise sense (no set or class of all such maps is formed).*

Proof. Exactly one part of a finite partition lies in U , and refinement preserves the selected value; so evaluation is a well-defined surjective homomorphism. If $x \notin \Pi \mathcal{N}_\bullet$, its image has a nonzero part e with a nonzero coefficient α ; take $U \ni e$ by [Lemma 15.48](#), and $x \mapsto \alpha$. Conversely maps to countable domains kill $\Pi \mathcal{N}_\bullet$ by [Corollary 15.45](#). $\square$

Corollary 15.50 (Field targets [19]). *Assume global choice. A set-sized field L receives a unital homomorphism from $\mathcal{N}_\bullet$ if and only if L contains an algebraic closure of its prime field.*

Proof. A map forces a copy of $\overline{\mathbb{Z}}$, hence of $\overline{\mathbb{Q}}$, or a copy of $\overline{\mathbb{F}}_p$ ([Theorem 15.46](#)). Conversely compose an evaluation $\mathcal{N}_\bullet \rightarrow \overline{\mathbb{Z}}$ with $\overline{\mathbb{Z}} \subseteq \overline{\mathbb{Q}} \subseteq L$, or with $\overline{\mathbb{Z}} \rightarrow \overline{\mathbb{Z}}/\mathfrak{p} \cong \overline{\mathbb{F}}_p$ for a prime $\mathfrak{p}$ of $\overline{\mathbb{Z}}$ over p (lying over), followed by an inclusion. $\square$

Theorem 15.51 (A finite-axiom form [19]). *Let F be a real closed field (a set or a class) containing the real algebraic numbers and $A \subseteq F$ a unital subring with a ring retraction $c : A \rightarrow \mathbb{Z}$ and kernel J such that (i) J is divisible by every nonzero integer; (ii) for all $a_1, \dots, a_r \in J$ and $m \geq 1$*

there is $0 \neq d \in J$ with $a_j/d^m \in A$ for all j ; (iii) the positive elements of J are cofinal in $F_{>0}$. Let $N_{\mathbb{R}}, N_{\mathbb{C}}$ be the integral closures of A in F and $F(i)$ and $\mathbf{B}_{\bullet} = N_{\bullet}/JN_{\bullet}$. Then [Lemma 15.28](#) and [Theorems 15.29, 15.30, 15.38 and 15.42](#) and [Corollary 15.33](#) hold with $(\mathbf{Oz}, \Pi, \text{ct})$ replaced by (A, J, c) , and [Theorem 15.32](#) holds with T any sufficiently large positive element of J , the roots being positive and distinct but not necessarily infinite.

Proof. The contraction uses only the retraction and finite integrality, torsion freeness uses (i), radicality uses exactly (ii), and saturation uses (i) and normality. By (iii) there is $T \in J$ with $T > 2$ for [Proposition 15.34](#) and as large as needed for the lifting; v_T is a unit with residue square -1 whether or not it is infinitesimal. Real algebraic integers are integral over $\mathbb{Z} \subseteq A$, so the coefficient embedding, the spectral idempotents, the Boolean description and the doubling apply unchanged. $\square$

The theorem does not give that v_T is infinitesimal, nor that $JN_{\bullet}$ has no set of generators (that needs the set-family divisibility of the full class), nor [Corollary 15.45](#), which needs arbitrarily large sets of exponent scales below a fixed exponent, a cardinal condition absent from (i)–(iii). In a fixed real-exponent Hahn integer part the support $\{1/n : n \geq 1\}$ blocks the monomial proof of (ii); this is a failure of the argument, not a counterexample to any conclusion (19). (Status (batch 32): the cyclic workspace $\mathbb{Z} + \omega^a \mathbb{R}[\omega^a]$ lies outside the theorem, since its Laurent field is not real closed and (ii) fails there, and in it the conclusion of [Theorem 15.29](#) does fail: the extended ideal is not radical, [Theorem 15.86](#) and [Remark 15.119](#); for real closed fixed workspaces such as the Hahn field over $\mathbb{Q}a$ the question stays open, [Question 18.21](#), while the bounded-denominator Puiseux ring satisfies (i)–(iii), [Remark 15.102](#).) For the reduced fibre $C(G)$ of the full Hahn field over a nonzero set group (defined in (15.20) below), 20 obtains the Boolean-power form without (ii) and without deciding radicality ([Proposition 15.76](#)).

Remark 15.52 (Candidates for idempotents [merge]). $\text{Idem}(\mathbf{B}_{\mathbb{R}})$ is nontrivial exactly when some $x \in \mathcal{N}$ has $x^2 - x \in \Pi\mathcal{N}$ while $x, 1 - x \notin \Pi\mathcal{N}$. For $s = \sqrt{1 + \omega}$ and $x = (1 + s)/2$ one has $x^2 - x = \omega/4$, so $x \in \mathcal{N}$ (a root of $X^2 - X - \omega/4$) and its residue is an idempotent. By [Proposition 15.31](#), $x \in \Pi\mathcal{N}$ exactly when $x/\omega^{\delta} \in \mathcal{N}$ for some $\delta > 0$; so this residue is nontrivial exactly when neither x/ω^{δ} nor $(1 - x)/\omega^{\delta}$ lies in $\mathcal{N}$ for any $\delta > 0$, an instance of the open membership question. Likewise 03's unit gives the idempotent $(1 + \bar{u}_H)/2$ of $\mathbf{B}_{\mathbb{R}}$ ([Theorem 15.30](#)), nontrivial exactly when $u_H \mp 1 \notin \Pi\mathcal{N}$. Neither case is decided by this remark ([Question 18.17](#)).

This reduction to the membership question is superseded: 20 proves nontriviality, and much more, by support projection and two specializations, without a membership test ([Section 15.9](#)), and both candidates are nontrivial, the first being the second for $H = \omega^{1/2}$ ([Proposition 15.68](#), a merge result).

15.9 Proper-class Boolean branching of the fibre

This subsection is manuscript 20's ([Section 1.3](#)). It answers [Question 18.17\(a\)](#): $\text{Idem}(\mathbf{B}_{\mathbb{R}})$ is nontrivial and atomless, it is a proper class with no set-sized order-dense subset, it contains an explicit independent family indexed by all ordinals, and one idempotent splits every nonzero element of any prescribed set-sized subring of $\mathbf{B}_{\mathbb{R}}$. The tool is a support projection that makes integrality and radical membership local to the exponent group of an element, followed by two specializations of a quadratic branch algebra at one fresh scale. The splitting theorems use neither the Boolean-power theorem nor the universal theorem; only the statements about maps at the end of the subsection do (20).

20 read 19 as a preceding draft (it was not in the repository at 20's pin) and reproves what it needs of it, with the same constructions. Those results are printed once, above, and carry the tag [19, 20] where 20 proves the whole statement: the contraction, radicality and saturation (Lemma 15.28 and Theorems 15.29 and 15.30), the coefficient copy, spectral idempotents and Boolean power (Lemmas 15.35 and 15.36, Definition 15.37, and Theorem 15.38), the rational comparison and integral doubling (Lemma 15.41 and Theorem 15.42), the factorization of set-sized images, class ultrafilters and detection (Corollary 15.45, Lemma 15.48, and Theorem 15.49). 20 also proves most of Proposition 15.34 (all but the Catalan expansion), of Proposition 15.43 (all but $w_{\pm} \notin \mathcal{N}[i]$) and of Theorem 15.46 (the classification, not the consequences for finite and formally real targets), and the part of Corollary 15.44 saying that $w_{\pm}$ separate the two nonzero factors modulo 2. 20's set bounds and common monomial division are Lemma 3.1 and Proposition 3.3(i), its constant term and $\Pi^2 = \Pi$ are Lemma 2.2 and Proposition 3.3(ii), and its proof of the universal theorem for **Oz** is credited at Theorem 5.1. Its dominant-scale lemma is 16's Laurent-block theorem, which was not in the repository at 20's pin; it is credited in Lemma 15.62.

Notation. For a set-sized additive subgroup $G \subseteq \mathbf{No}$ let $K_G = \mathbb{R}((X^G)) \subseteq \mathbf{No}$ be the real Hahn field with supports in G , as in Section 16.6 with $k = \mathbb{R}$ (20's F_G), and put

$$\begin{aligned} \mathcal{A}(G) &= \mathbf{Oz} \cap K_G, & \Pi(G) &= \Pi \cap K_G, \\ \mathcal{N}(G) &= \text{IC}_{K_G}(\mathcal{A}(G)), & \mathcal{C}(G) &= \mathcal{N}(G) / \sqrt{\Pi(G)\mathcal{N}(G)} \end{aligned} \quad (15.20)$$

(20's A_G, J_G, N_G, C_G). Thus $\mathcal{A}(G) = \mathcal{A}_{\mathbb{Z}, \mathbb{R}}(G)$ and $\Pi(G) = \Pi_{\mathbb{R}}(G)$ in the notation of Section 2.3; the parentheses keep $\mathcal{N}(G)$ apart from 03's $\mathcal{N}_{\kappa}$ and from $\mathcal{N}_{\bullet}$, and the letter C is 03's (for **Oz**[S] and for its complex normalization). 20's $N_{\mathbb{R}}, N_{\mathbb{C}}, \mathcal{J}, K_{\mathbb{R}}, K_{\mathbb{C}}, B_{\mathbb{R}}, B_{\mathbb{C}}, E, \mathcal{E}$ are $\mathcal{N}, \mathcal{N}_{\mathbb{C}}, \Pi, \Pi\mathcal{N}, \Pi\mathcal{N}_{\mathbb{C}}, \mathbf{B}_{\mathbb{R}}, \mathbf{B}_{\mathbb{C}}, \mathbf{B}_{\mathbb{R}}^{\mathbb{Q}}, \text{Idem}(\mathbf{B}_{\mathbb{R}})$; its unit u and residue v are v_T and $\mathbf{v}$. Its coefficient projection π_G is written pr_G (π_b is a detecting map in Theorem 5.12); its dominant exponent a and group $H = G \oplus \mathbb{Z}a$ are 16's fresh exponent b and $\tilde{G} = G \oplus \mathbb{Z}b$ (H_a is the scale subgroup (2.4), and H is 03's parameter in u_H); its monomial $T = \omega^a$ is $t = \omega^b$, with T the formal Laurent variable as in Theorem 16.31; its branches s_0, s_1, s_r are $\varrho_0, \varrho_1, \varrho_r$ (s is 16's centre and the root of Remark 15.52); its specializations $\theta_{\pm}$ are $\sigma_{\pm}$ ($\theta = \kappa^{\aleph_0}$ in 03's construction); its splitting element $w(T)$ is w_t ($w_{\pm}$ are 19's idempotents). Square roots are the positive ones in **No**, and extended ideals consist of finite sums of products.

Lemma 15.53 (Monic radical certificates [20]). *Let $D \subseteq C$ be an integral extension of commutative rings (sets or classes) and $I \subseteq D$ an ideal. An element $x \in C$ lies in $\sqrt{IC}$ if and only if $x^m + c_{m-1}x^{m-1} + \dots + c_0 = 0$ for some ordinary $m \geq 1$ and $c_0, \dots, c_{m-1} \in I$.*

Proof. Let $x^q = \sum_{l \leq r} a_l b_l$ with $a_l \in I$ and $b_l \in C$. The ring $C_0 = D[x, b_1, \dots, b_r]$ is generated as a D -module by finitely many elements $c_1 = 1, c_2, \dots, c_s$, and $x^q c_{\nu} = \sum_l a_l (b_l c_{\nu}) \in IC_0 = \sum_{\mu} I c_{\mu}$, so $x^q c_{\nu} = \sum_{\mu} h_{\nu\mu} c_{\mu}$ with all $h_{\nu\mu} \in I$. The adjugate of $x^q \mathbf{1}_s - (h_{\nu\mu})$ shows that $P(x^q) = 0$ for the characteristic polynomial P of $(h_{\nu\mu})$, using $c_1 = 1$; the coefficient of X^{s-j} in P ($j \geq 1$) is a sum of products of j entries, hence in I . Then $P(X^q)$ is a monic polynomial of degree qs killing x whose nonleading coefficients lie in I (or are 0). Conversely the displayed relation gives $x^m = -\sum_j c_j x^j \in IC$. Only finitely many elements occur, so the argument applies to class rings. $\square$

Corollary 15.54 (Exact membership certificate [20]). *For $x \in \mathcal{N}_{\bullet}$: $x \in \Pi\mathcal{N}_{\bullet}$ if and only if $x^m + \sum_{j < m} c_j x^j = 0$ for some $m \geq 1$ and $c_j \in \Pi$.*

Proof. $\mathcal{N}_\bullet$ is integral over $\mathbf{Oz}$; apply [Lemma 15.53](#) with $(D, C, I) = (\mathbf{Oz}, \mathcal{N}_\bullet, \Pi)$ and the radicality of $\Pi\mathcal{N}_\bullet$ ([Theorem 15.29](#)). $\square$

The certificate lets one project the coefficients of a vanishing relation without projecting a homomorphism or an ideal expression (20).

Lemma 15.55 (Support projection [20]). *For a set-sized subgroup $G \subseteq \mathbf{No}$ put $\text{pr}_G(\sum_g r_g \omega^g) = \sum_{g \in G} r_g \omega^g$ on $\mathbf{No}[i]$ (coefficients in $\mathbb{C}$). Then pr_G is additive, fixes $K_G[i]$ pointwise, and*

$$\text{pr}_G(yx) = \text{pr}_G(y) x \quad (y \in \mathbf{No}[i], x \in K_G[i]). \quad (15.21)$$

Moreover $\text{pr}_G(\mathbf{Oz}) = \mathcal{A}(G)$ and $\text{pr}_G(\Pi) = \Pi(G)$.

Proof. A subset of a reverse well-ordered set is reverse well ordered, and additivity and the fixing are coefficientwise. A contribution to an exponent $h \in G$ of yx comes from exponents g of y and s of x with $g + s = h$; since $s \in G$ and G is a group, $g = h - s \in G$. So exactly the terms of y kept by pr_G contribute to the exponents in G , and each coefficient of a Hahn product is a finite sum, so no exchange of infinite sums is needed. Selection keeps positive exponents and the constant coefficient, and it fixes $\mathcal{A}(G)$ and $\Pi(G)$. $\square$

Remark 15.56 (Not a ring retraction [20]). $\text{pr}_{\{0\}}(\omega) = \text{pr}_{\{0\}}(\omega^{-1}) = 0$ although $\text{pr}_{\{0\}}(\omega \cdot \omega^{-1}) = 1$; only (15.21), with one factor supported in G , is used. This is the warning of [Remark 2.3](#) for $G = \{0\}$.

Theorem 15.57 (Exact support descent [20]). *For set-sized subgroups $G \subseteq G' \subseteq \mathbf{No}$,*

$$\mathcal{N} \cap K_G = \mathcal{N}(G), \quad \Pi\mathcal{N} \cap K_G = \sqrt{\Pi(G)\mathcal{N}(G)}, \quad (15.22)$$

$$\mathcal{N}(G') \cap K_G = \mathcal{N}(G), \quad \sqrt{\Pi(G')\mathcal{N}(G')} \cap K_G = \sqrt{\Pi(G)\mathcal{N}(G)}. \quad (15.23)$$

Proof. Let $x \in K_G$ satisfy $x^m + \sum_{j < m} a_j x^j = 0$ with $a_j \in \mathbf{Oz}$. Every x^j lies in K_G , so (15.21) gives $x^m + \sum_{j < m} \text{pr}_G(a_j) x^j = 0$ with coefficients in $\mathcal{A}(G)$; hence $x \in \mathcal{N}(G)$. The reverse inclusion is clear, and the same argument with $a_j \in \mathcal{A}(G')$ gives $\mathcal{N}(G') \cap K_G = \mathcal{N}(G)$. If $x \in \Pi\mathcal{N} \cap K_G$, project the certificate of [Corollary 15.54](#): its coefficients go to $\Pi(G)$, $x \in \mathcal{N}(G)$ by the first part, and [Lemma 15.53](#) with $(\mathcal{A}(G), \mathcal{N}(G), \Pi(G))$ gives $x \in \sqrt{\Pi(G)\mathcal{N}(G)}$. Conversely $\Pi(G)\mathcal{N}(G) \subseteq \Pi\mathcal{N}$, which is radical. The local radical equality is the same argument with [Lemma 15.53](#) applied to $(\mathcal{A}(G'), \mathcal{N}(G'), \Pi(G'))$ and then to $(\mathcal{A}(G), \mathcal{N}(G), \Pi(G))$. $\square$

Corollary 15.58 (Reduced fibres embed [20]). *For $G \subseteq G'$ inclusion induces injections $\mathbb{C}(G) \hookrightarrow \mathbb{C}(G') \hookrightarrow \mathbb{B}_{\mathbb{R}}$. Larger surreal supports cannot annihilate a nonzero reduced-fibre class from a smaller support group.*

Proof. These are the contracted-ideal equalities of [Theorem 15.57](#). $\square$

So an integrality witness using arbitrary surreal scales can be replaced by one using only the exponent group of the element: in 19's membership question, which allows witnesses at larger scales, larger scales never help. [Theorem 15.57](#) is a descent statement, not a decision procedure for $\mathcal{N}(G)$ (20). The second equality in (15.22) deliberately keeps the radical: it is not asserted that $\Pi(G)\mathcal{N}(G)$ is radical in a fixed workspace (20). (Status (batch 32): for $G = \mathbb{Z}a$ it is not radical, so there the radical cannot be removed, [Theorem 15.86](#) and [Remark 15.119](#).) The same projection decides the descent asked in [Question 18.11](#).

Corollary 15.59 (Descent for 03's fields [merge]). *With the notation of Section 15,*

$$\mathcal{N} \cap F_\kappa^{03} = \mathcal{N}_\kappa, \quad \Pi\mathcal{N} \cap F_\kappa^{03} = \sqrt{\Pi_\kappa^{03}\mathcal{N}_\kappa}, \quad \mathcal{N}_\mathbb{C} \cap F_\kappa^{03}[i] = \mathcal{N}_{\kappa,\mathbb{C}}, \quad \Pi\mathcal{N}_\mathbb{C} \cap F_\kappa^{03}[i] = \sqrt{\Pi_\kappa^{03}\mathcal{N}_{\kappa,\mathbb{C}}},$$

the radicals being taken in $\mathcal{N}_\kappa$ and $\mathcal{N}_{\kappa,\mathbb{C}}$.

Proof. Identify F_κ^{03} with its image in **No** (Section 11.1): the series with countable support in the subgroup $\iota(G_\kappa)$. Let $x \in F_\kappa^{03}[i]$ be integral over **Oz**, and let G' be the subgroup generated by the supports of its real and imaginary parts. It is countable and contained in $\iota(G_\kappa)$, so every element of $K_{G'}$ has countable support in $\iota(G_\kappa)$: $K_{G'} \subseteq F_\kappa^{03}$ and $\mathcal{A}(G') \subseteq \mathbf{Oz} \cap F_\kappa^{03} = A_\kappa^{03}$. Projecting a monic equation of x over **Oz** by $\text{pr}_{G'}$ (Lemma 15.55) gives one over $\mathcal{A}(G') \subseteq A_\kappa^{03}$, so x is integral over A_κ^{03} . By (15.8) and (15.13) (and the remark after it that $\mathcal{N}_{\kappa,\mathbb{C}}$ is the integral closure of A_κ^{03} in $F_\kappa^{03}[i]$) this proves the first and third equalities; the reverse inclusions hold because $A_\kappa^{03} \subseteq \mathbf{Oz}$. If moreover $x \in \Pi\mathcal{N}_\bullet$, the certificate of Corollary 15.54 has coefficients in Π , and its projection has coefficients in $\Pi(G') \subseteq \Pi_\kappa^{03}$; Lemma 15.53, applied to A_κ^{03} and the ring $\mathcal{N}_\kappa$ or $\mathcal{N}_{\kappa,\mathbb{C}}$, which is integral over it, puts x in the stated radical. Conversely $\Pi_\kappa^{03}\mathcal{N}_\kappa \subseteq \Pi\mathcal{N}$ and $\Pi_\kappa^{03}\mathcal{N}_{\kappa,\mathbb{C}} \subseteq \Pi\mathcal{N}_\mathbb{C}$, and these are radical. $\square$

The proof uses only that F_κ^{03} contains the full Hahn field over the subgroup generated by the support of each of its elements; it is a descent, not a membership test for $\mathcal{N}_\kappa$.

Theorem 15.60 (One-scale rational functions [20]). *For $a > 0$ and $T = \omega^a$,*

$$\mathcal{N} \cap \mathbb{R}(T) = \overline{\mathbb{Z}}_\mathbb{R} + T\mathbb{R}[T], \quad \Pi\mathcal{N} \cap \mathbb{R}(T) = T\mathbb{R}[T].$$

Proof. A rational function of T expands at infinity with support in $\mathbb{Z}a$, so $\mathbb{R}(T) \subseteq K_{\mathbb{Z}a}$. A reverse well-ordered subset of $\{na : n \geq 0\}$ is finite, so $\mathcal{A}(\mathbb{Z}a) = \mathbb{Z} + T\mathbb{R}[T]$ and $\Pi(\mathbb{Z}a) = T\mathbb{R}[T]$. If $f \in \mathbb{R}(T)$ is integral over **Oz**, it is integral over $\mathbb{Z} + T\mathbb{R}[T] \subseteq \mathbb{R}[T]$ by Theorem 15.57, hence lies in $\mathbb{R}[T]$, which is integrally closed (a monic equation for p/q in lowest terms, multiplied by q^m , forces $q \mid p^m$). Evaluating the monic equation at $T = 0$ makes $f(0)$ a real algebraic integer. Conversely $f(0) \in \overline{\mathbb{Z}}_\mathbb{R} \subseteq \mathcal{N}$ (Theorem 15.14) and $f - f(0) \in T\mathbb{R}[T] \subseteq \Pi$. If also $f \in \Pi\mathcal{N}$, project its certificate onto $\mathbb{Z}a$: the coefficients lie in $T\mathbb{R}[T]$, and evaluation at 0 gives $f(0)^m = 0$. Finally $T\mathbb{R}[T] \subseteq \Pi \subseteq \Pi\mathcal{N}$. $\square$

Example 15.61 ([20]). $(\omega^2 - 1)^{-1}$ is not integral over **Oz**, although a monic witness over **Oz** could use coefficients at arbitrary surreal scales; 19's $v_\omega = (\omega - \sqrt{\omega^2 - 4})/2$ (Proposition 15.34) is integral, and its infinite Laurent tail is algebraic but not rational in ω , as Theorem 15.60 requires.

Lemma 15.62 (Laurent blocks exhaust the enlarged field [16, 20]). *Let $G \subseteq \mathbf{No}$ be a set-sized subgroup, $b > G$ fresh (Lemma 16.30), $\tilde{G} = G \oplus \mathbb{Z}b$ and $t = \omega^b$. The map ι_b of Theorem 16.31 is a field isomorphism $K_G((T^{-1})) \rightarrow K_{\tilde{G}}$ with $\iota_b(T) = t$, and*

$$\iota_b^{-1}(\mathcal{A}(\tilde{G})) = \mathcal{A}(G) + TK_G[T], \quad \iota_b^{-1}(\Pi(\tilde{G})) = \Pi(G) + TK_G[T].$$

In particular a coefficient in $\Pi(\tilde{G})$ is a polynomial in t over K_G whose value at $T = 0$ lies in $\Pi(G)$.

Proof. Theorem 16.31 (16) gives an injective field homomorphism into $K_{\tilde{G}}$. It is onto (20): an exponent of $\tilde{G}$ is uniquely $g + jb$ with $g \in G$, $j \in \mathbb{Z}$; in a reverse well-ordered support the occurring j are bounded above, since increasing j gives larger exponents by freshness, and the part with a fixed j is ω^{jb} times an element of K_G . The ring statements are Corollary 16.32 with $D = \mathbb{Z}$: $\iota_b(\sum_j c_j T^j) \in \mathbf{Oz}$ exactly when $c_j = 0$ for $j < 0$ and $c_0 \in \mathbf{Oz}$, and the positive-index blocks are purely infinite, so the constant term is $\text{ct}(c_0)$. $\square$

Lemma 15.63 (Independent quadratic branches [20]). *Let k be an ordered field and, in $k((T^{-1}))$, let $\varrho_0 = T(1 - T^{-2})^{1/2}$ and $\varrho_1 = \frac{T}{2}(1 - 4T^{-2})^{1/2}$ be the binomial branches. Then*

$$k[T, \varrho_0, \varrho_1] \cong k[T, U, V]/(U^2 - T^2 + 1, V^2 - T^2/4 + 1),$$

a free $k[T]$ -module with basis $1, \varrho_0, \varrho_1, \varrho_0\varrho_1$, and there are k -algebra homomorphisms

$$\sigma_{\pm} : k[T, \varrho_0, \varrho_1] \rightarrow k[i] = k[X]/(X^2 + 1), \quad T \mapsto 0, \quad \varrho_0 \mapsto i, \quad \varrho_1 \mapsto \pm i.$$

Proof. The binomial series exist in characteristic 0, and $\varrho_0^2 = d_0 = T^2 - 1$, $\varrho_1^2 = d_1 = T^2/4 - 1$. At $T = 1$, d_0 has a simple zero and $d_1 \neq 0$; at $T = 2$ the roles are reversed. So d_0, d_1 and d_0d_1 are not squares in $k(T)$, since a square has even order at every linear factor. If d_1 were a square $(x + y\varrho_0)^2$ in $k(T)(\varrho_0)$, then $2xy = 0$, and $y = 0$ or $x = 0$ would make d_1 or d_0d_1 a square in $k(T)$. Hence $[k(T)(\varrho_0, \varrho_1) : k(T)] = 4$. The presented algebra is free over $k[T]$ on the four monomials, by reduction with the two monic relations, and their images are linearly independent over $k(T)$; so the map is injective. The substitutions respect both relations, which defines $\sigma_{\pm}$. $\square$

Remark 15.64 (Specializations, not evaluations [20]). $\sigma_{\pm}$ are specializations of a finite branch algebra whose presentation has been proved. They are not evaluations of Laurent series at $T = 0$, where these have no meaning, nor maps $K_{\tilde{G}} \rightarrow k[i]$, and no evaluation map on all surreal numbers is used.

For a fresh b and $t = \omega^b$ put, with ϱ_0, ϱ_1 read through ι_b (they are the positive roots in $\mathbf{No}$, having positive leading coefficients),

$$\varrho_0 = \sqrt{t^2 - 1}, \quad \varrho_1 = \sqrt{t^2/4 - 1}, \quad \mathbf{w}_t = \frac{1 - \varrho_0\varrho_1}{2}, \quad \mathbf{w}_t^2 - \mathbf{w}_t = \frac{t^4 - 5t^2}{16} \in \Pi. \quad (15.24)$$

So $\mathbf{w}_t \in \mathcal{N}$, and $\mathbf{e}_t = [\mathbf{w}_t]$ is an idempotent of $\mathbf{B}_{\mathbb{R}}$. The element $\mathbf{w}_t = -t^2/4 + 9/8 + 9t^{-2}/32 + \dots$ has an infinite Laurent tail, but its certificate is a polynomial in t with zero constant coefficient (20).

Theorem 15.65 (One-scale uniform splitting [20]). *Let $G \subseteq \mathbf{No}$ be a set-sized subgroup, $b > G$ fresh and $t = \omega^b$. For every $x \in \mathcal{N} \cap K_G$,*

$$x\mathbf{w}_t \in \Pi\mathcal{N} \iff x \in \Pi\mathcal{N} \iff x(1 - \mathbf{w}_t) \in \Pi\mathcal{N}.$$

Equivalently, multiplication by $\mathbf{e}_t$ and by $1 - \mathbf{e}_t$ kills no nonzero element of $\mathbf{C}(G) \subseteq \mathbf{B}_{\mathbb{R}}$.

Proof. The implications from the middle hold because $\Pi\mathcal{N}$ is an ideal. Suppose $z = x\mathbf{w}_t \in \Pi\mathcal{N}$ and take a certificate $z^m + \sum_{j < m} c_j z^j = 0$, $c_j \in \Pi$ (Corollary 15.54). Since $z \in K_{\tilde{G}}$, Lemma 15.55 gives $z^m + \sum_{j < m} p_j z^j = 0$ with $p_j = \text{pr}_{\tilde{G}}(c_j) \in \Pi(\tilde{G})$, and by Lemma 15.62 each p_j is ι_b of a polynomial in $\Pi(G) + TK_G[T]$. So the relation holds in $\iota_b(K_G[T, \varrho_0, \varrho_1])$, and Lemma 15.63 with $k = K_G$ lets us apply σ_+ . It fixes x and sends $\mathbf{w}_t$ to $(1 - i^2)/2 = 1$, so $x^m + \sum_{j < m} p_j(0)x^j = 0$ in K_G , with $p_j(0) \in \Pi(G) \subseteq \Pi$; hence $x \in \Pi\mathcal{N}$ by Corollary 15.54. If instead $x(1 - \mathbf{w}_t) \in \Pi\mathcal{N}$, use σ_- , which sends $\mathbf{w}_t$ to $(1 - i(-i))/2 = 0$ and $1 - \mathbf{w}_t$ to 1. $\square$

Theorem 15.66 (Uniform splitting [20]). *For every set-sized unital subring $R \subseteq \mathbf{B}_{\mathbb{R}}$ there is an idempotent $e \in \mathbf{B}_{\mathbb{R}}$ with $re \neq 0$ and $r(1 - e) \neq 0$ for every $0 \neq r \in R$. Consequently $R \times R \rightarrow \mathbf{B}_{\mathbb{R}}$, $(r, s) \mapsto re + s(1 - e)$, is an injective unital homomorphism extending the diagonal copy of R .*

Proof. Represent the elements of R by elements of $\mathcal{N}$ (a set of representatives exists, for instance the least-rank ones of Section 2.1), let G be the group generated by the union of their supports, choose $b > G$ and put $e = \mathbf{e}_{\omega^b}$. Theorem 15.65 gives $re \neq 0 \neq r(1 - e)$. The map is a unital homomorphism because $e(1 - e) = 0$, and if $re + s(1 - e) = 0$, multiplying by e and by $1 - e$ gives $re = 0 = s(1 - e)$, so $r = s = 0$. $\square$

This splits all nonzero elements of a prescribed set at once, with no exception caused by zero divisors of R (20).

Example 15.67 (A nontrivial idempotent at $t = \omega$ [20]). With $G = \{0\}$ and $t = \omega$, the residue $[(1 - \sqrt{\omega^2 - 1}\sqrt{\omega^2/4 - 1})/2]$ is neither 0 nor 1: apply [Theorem 15.65](#) to $x = 1 \notin \Pi\mathcal{N}$. The identity (15.24) proves idempotence; the two specializations prove the separately needed nontriviality.

One radical suffices, and it can be 03's. This decides the two candidates of [Remark 15.52](#).

Proposition 15.68 (03's unit splits, and the candidates are nontrivial [merge]). *Let G , b , t be as in [Theorem 15.65](#), let $h \in TK_G[T]$ be nonzero with positive leading coefficient, and put $H = \iota_b(h)$, a positive element of Π . With 03's unit $u_H = \sqrt{H^2 + 1} - H$ ([Theorem 15.9](#)), the element $y_H = (1 + u_H)/2$ lies in $\mathcal{N}$, its residue e_H is an idempotent of $\mathbb{B}_{\mathbb{R}}$, and for every $x \in \mathcal{N} \cap K_G$*

$$xy_H \in \Pi\mathcal{N} \iff x \in \Pi\mathcal{N} \iff x(1 - y_H) \in \Pi\mathcal{N}.$$

In particular, for every $a > 0$ and every positive $H \in \Pi$ with $\text{supp } H \subseteq \mathbb{Z}a$, the idempotent $(1 + \bar{u}_H)/2$ of [Remark 15.52](#) is neither 0 nor 1, that is, $u_H \mp 1 \notin \Pi\mathcal{N}$; and $x = (1 + \sqrt{1 + \omega})/2$ has a residue that is neither 0 nor 1.

Proof. From $u_H^2 + 2Hu_H - 1 = 0$, $y_H^2 - y_H = (u_H^2 - 1)/4 = -Hu_H/2$, and $H/2 \in \Pi$, $u_H \in \mathcal{N}$; so y_H is a root of $X^2 - X + Hu_H/2 \in \mathcal{N}[X]$, hence in $\mathcal{N}$, and e_H is idempotent. Put $\varrho = \iota_b(h(1 + h^{-2})^{1/2}) = \sqrt{H^2 + 1}$ (the binomial series in $h^{-2} \in T^{-2}K_G[[T^{-1}]]$). The polynomial $h^2 + 1$ is not a square in $K_G(T)$: a square root would lie in $K_G[T]$, and $(g - h)(g + h) = 1$ would make $g \pm h$ constant and h constant. So $1, \varrho$ are linearly independent over $K_G(T)$, $K_G[T, \varrho] \cong K_G[T, U]/(U^2 - h^2 - 1)$ is free over $K_G[T]$ with basis $1, U$, and since $h(0) = 0$ there are K_G -algebra maps $\sigma'_{\pm} : K_G[T, \varrho] \rightarrow K_G$ with $T \mapsto 0$, $\varrho \mapsto \pm 1$; they send $u_H = \varrho - h$ to ± 1 and y_H to 1, respectively 0. The proof of [Theorem 15.65](#) now applies verbatim with w_t replaced by y_H and $\sigma_{\pm}$ by $\sigma'_{\pm}$. For the candidates take $G = \{0\}$ and $b = a$, so that $H = h(\omega^a)$ with $h \in T\mathbb{R}[T]$, and $x = 1$: then $e_H \neq 0, 1$, and $e_H = (1 + \bar{u}_H)/2$ differs from 0 and 1 exactly when $u_H \mp 1 \notin \Pi\mathcal{N}$ (both $\Pi\mathcal{N}$ and $\mathbb{B}_{\mathbb{R}}$ are torsion free, [Lemma 15.28](#)). Finally $(1 + \sqrt{1 + \omega})/2 = y_H + H/2$ for $H = \omega^{1/2}$, and $H/2 \in \Pi$, so its residue is e_H . $\square$

The two candidates are therefore one idempotent (for $H = \omega^{1/2}$), and [Remark 15.52](#)'s reduction to the membership question is superseded: nontriviality follows from support projection and two specializations, without a membership test. For $H \in \Pi$ whose support is not contained in one cyclic group the method needs H to be a polynomial in a fresh monomial over a group containing the rest of its support, and general H is not decided here. 20's two-radical splitter (15.24) is not minimal in this sense, which is not an error.

Lemma 15.69 (Multiquadratic extensions [20, 23]). *Over a field L of characteristic different from 2, classes $d_1, \dots, d_n$ independent in $L^{\times}/L^{\times 2}$ give an extension $L(\sqrt{d_1}, \dots, \sqrt{d_n})$ of degree 2^n , with the square-root monomials as a basis and all 2^n independent sign automorphisms.*

Proof. Induction on n . If $y^2 = d_n$ in the extension by the first $n - 1$ roots, every sign automorphism sends y to $\pm y$, so y is a common eigenvector; the monomial basis has distinct sign characters, so y is a scalar multiple of one monomial, and squaring expresses the class of d_n as a product of earlier classes, a contradiction. So the last step has degree 2 and supplies the new basis elements and sign change. $\square$

Theorem 15.70 (Rank extension of set-sized Hahn fibres [20]). *Let G be a set-sized ordered abelian group, $K_G = \mathbb{R}((X^G))$, and define $\mathcal{A}(G) = \mathbb{Z} + \Pi_{\mathbb{R}}(G)$, $\Pi(G) = \Pi_{\mathbb{R}}(G)$, $\mathcal{N}(G)$ and $\mathcal{C}(G)$ as in (15.20). Let $\tilde{G} = G \oplus \mathbb{Z}$ with the new coordinate dominant, and make the same definitions for $\tilde{G}$. Then $\mathcal{C}(G) \rightarrow \mathcal{C}(\tilde{G})$ is injective, there is a diagonal-extending injection $\mathcal{C}(G) \times \mathcal{C}(G) \hookrightarrow \mathcal{C}(\tilde{G})$, and for every set $P \subseteq K_G^{>0} \setminus \{1\}$ there is an injective $\mathcal{C}(G)$ -algebra map*

$$\mathcal{C}(G)[X_r : r \in P]/(X_r^2 - X_r : r \in P) \hookrightarrow \mathcal{C}(\tilde{G}).$$

So one dominant coordinate gives a free Boolean family of cardinality $|K_G|$ over the smaller reduced fibre.

Proof. A monic $\Pi(G)$ -certificate for 1 (Lemma 15.53) would give $1 = 0$ after ct; so $\mathcal{C}(G) \neq 0$. The proofs of Theorem 16.31 and Lemmas 15.55 and 15.62 apply verbatim to $G \subseteq \tilde{G}$, with T the monomial of the new coordinate and $k = K_G$; the local equality (15.23) then gives the first injection, using radical ideals only and no global radicality. The second is proved exactly as Theorem 15.65. For the third put $\varrho_0 = \sqrt{T^2 - 1}$, $\varrho_r = \frac{T}{r}(1 - r^2 T^{-2})^{1/2}$ and $w_r = (1 - \varrho_0 \varrho_r)/2$ ($r \in P$), binomial branches with positive leading coefficients, so that

$$w_r^2 - w_r = \frac{T^4/r^2 - (1 + r^{-2})T^2}{4} \in Tk[T] \subseteq \Pi(\tilde{G}),$$

and $w_r \in \mathcal{N}(\tilde{G})$ has an idempotent residue e_r . For finite $P_0 \subseteq P$ the radicands $T^2 - 1$ and $T^2/r^2 - 1$ ($r \in P_0$) have the simple zeros ± 1 and $\pm r$, pairwise disjoint because the r are positive, distinct and $\neq 1$; so every nonempty product of them has a simple zero and is not a square, and Lemma 15.69 gives degree $2^{1+|P_0|}$. As in Lemma 15.63, the branch algebra has only its monic quadratic relations, and $T \mapsto 0$, $\varrho_0 \mapsto i$, $\varrho_r \mapsto \pm i$ can be chosen independently in $k[i]$, realizing every assignment of 0 and 1 to the e_r . If $0 \neq \bar{x} \in \mathcal{C}(G)$ and p is a finite word in the e_r and their complements, a certificate for $xp \in \sqrt{\Pi(\tilde{G})\mathcal{N}(\tilde{G})}$ specialized with $p \mapsto 1$ would be a $\Pi(G)$ -certificate for x ; so $\bar{x}p \neq 0$. An element of the source uses finitely many variables and has a unique expansion on their Boolean atoms, and multiplying a vanishing image by each atom kills its coefficient; so the map is injective. The positive part of an infinite ordered field minus one point has the cardinality of the field. $\square$

Neither K_G nor $K_{\tilde{G}}$ need be real closed, and $\Pi(G)\mathcal{N}(G)$ is not assumed radical; the theorem concerns reduced fibres. For $G = \{0\}$, $K_G = \mathbb{R}$ and $\mathcal{C}(G) = \overline{\mathbb{Z}}_{\mathbb{R}}$, so one Laurent scale already carries continuum many independent idempotents over $\overline{\mathbb{Z}}_{\mathbb{R}}$ (20). 23 shows that $\text{Idem}(\mathcal{C}(\mathbb{Z}a))$ is atomless, of cardinality and order density exactly $2^{\aleph_0}$, with another explicit independent family (Corollary 15.110 and Theorem 15.111).

Theorem 15.71 (Relative Boolean freeness [20]). *For every set-sized unital subring $R \subseteq \mathbb{B}_{\mathbb{R}}$ and every set I there is an injective R -algebra homomorphism*

$$R[X_i : i \in I]/(X_i^2 - X_i : i \in I) \hookrightarrow \mathbb{B}_{\mathbb{R}}.$$

Proof. Well-order I as an ordinal κ . At stage $\alpha < \kappa$ let R_α be the ring generated by R and the e_β , $\beta < \alpha$; it is a set, and Theorem 15.66 gives a splitter e_α for it. For $0 \neq r \in R$ the product of r with any finite choice of e_α or $1 - e_\alpha$, in increasing order of indices, stays nonzero, since each partial product lies in the ring present before the next index. In finitely many variables the polynomials $p_\varepsilon = \prod_j X_j^{\varepsilon_j} (1 - X_j)^{1-\varepsilon_j}$ ($\varepsilon \in \{0, 1\}^n$) are orthogonal idempotents summing to 1, and every element is uniquely $\sum_\varepsilon r_\varepsilon p_\varepsilon$ (evaluate at zero-one vectors); a vanishing image, multiplied by the image of each p_ε , forces every $r_\varepsilon = 0$. A polynomial in infinitely many variables uses finitely many. $\square$

For $|I| = n$ the source is R^{2^n} ; for infinite I it is the directed union of these finite products, not the unrestricted product $R^{\{0,1\}^I}$ (20).

Theorem 15.72 (An explicit independent class [20]). *For every ordinal α put $t_\alpha = \omega^{\omega^\alpha}$ and $e_\alpha = \mathbf{e}_{t_\alpha} = \left\lfloor (1 - \sqrt{t_\alpha^2 - 1} \sqrt{t_\alpha^2/4 - 1})/2 \right\rfloor$. Every finite Boolean word in distinct e_α and their complements is nonzero in $\mathbf{B}_\mathbb{R}$.*

Proof. w_{t_α} has support in $\mathbb{Z}\omega^\alpha$. The group generated by the supports for $\beta < \alpha$ lies in the rational span of $\{\omega^\beta : \beta < \alpha\}$ in the exponent field, whose members are finite combinations with leading exponent below α ; so ω^α exceeds them all and is fresh over that group. Theorem 15.65, applied inductively along increasing indices, proves every finite word nonzero. Only individual earlier supports are used, never an infinite Boolean product, and the family is defined by a formula, without class-valued recursion. $\square$

Theorem 15.73 (Atomless, a proper class, no set-sized dense core [20]). *$\text{Idem}(\mathbf{B}_\mathbb{R})$ is atomless and a proper class, and no set $\mathcal{D} \subseteq \text{Idem}(\mathbf{B}_\mathbb{R}) \setminus \{0\}$ has the property that every nonzero idempotent lies above a member of $\mathcal{D}$.*

Proof. For $f \neq 0$ split the set-sized ring $\mathbb{Z}[f]$: fe and $f(1 - e)$ are nonzero orthogonal idempotents with sum f . If $\text{Idem}(\mathbf{B}_\mathbb{R})$ were a set, the ring it generates would be a set; a splitter e of that ring lies in it and is nonzero, and splitting $r = e$ would give $e(1 - e) \neq 0$. (Alternatively Theorem 15.72 injects the ordinals.) If $\mathcal{D}$ were such a set, a splitter e of the ring it generates is nonzero, and no $d \in \mathcal{D}$ has $d \leq e$, since $d(1 - e) \neq 0$. $\square$

With Theorem 15.38 this sharpens 19's description: $\mathbf{B}_\mathbb{R} \cong \overline{\mathbb{Z}}[\text{Idem}(\mathbf{B}_\mathbb{R})]$ with the ring of all complex algebraic integers as coefficients and an atomless proper-class Boolean parameter with no set-sized dense core; the complexity lies in how arithmetic branches are organized, not in new algebraic numbers on one branch (20). Free subalgebras of every set size do not make the whole class algebra free or complete, nor show that it realizes every set-sized Boolean type (20).

Corollary 15.74 (Idempotent matrices split [20]). *Every nonzero idempotent matrix $P \in M_n(\mathbf{B}_\mathbb{R})$ is $eP + (1 - e)P$ with eP , $(1 - e)P$ nonzero orthogonal idempotents, for a scalar idempotent e .*

Proof. Split the set-sized ring generated by the entries of P ; a nonzero entry survives multiplication by e and by $1 - e$, and centrality of e with $P^2 = P$ gives idempotence and orthogonality. $\square$

In the idempotent-matrix description of finitely generated projective modules, every nonzero one therefore decomposes nontrivially. This is a finite matrix statement over a class ring; no category of class-sized modules is formed, and no projective or global dimension is computed (20).

Remark 15.75 (No global choice in the splitting results [merge]). 20 works in Gödel–Bernays class theory with global choice, but Theorems 15.66 and 15.71 to 15.73 and Corollary 15.74 need only choice for sets (for well-ordering I in Theorem 15.71). The splitter can be chosen canonically: take as representatives of the elements of R all their members of least rank (a set), let G be the group generated by all their supports and b the simplest surreal above $G \cup \{0\}$; Theorem 15.65 applies to every representative. The recursion in Theorem 15.71 then makes no choices, and Theorem 15.72 is explicit. Theorem 15.79 below uses no global choice either; Theorem 15.80, like Theorem 15.49, uses class ultrafilters.

Proposition 15.76 (Reduced fibres of fixed Hahn fields [20]). *For a nonzero set-sized ordered abelian group G , with the notation of Theorem 15.70, $C(G) \cong \overline{\mathbb{Z}}[\text{Idem}(C(G))]$. For $G = \{0\}$, $C(G) = \overline{\mathbb{Z}}_{\mathbb{R}}$.*

Proof. Let $I = \Pi(G)\mathcal{N}(G)$ and $L = \sqrt{I}$. I is divisible by nonzero integers, and so is L : if $x^m \in I$, then $(x/n)^m \in I \subseteq \mathcal{N}(G)$, so $x/n \in \mathcal{N}(G)$ by normality and then $x/n \in L$. A monic $\Pi(G)$ -certificate for a nonzero integer, followed by ct , would make a nonzero power of it vanish; so $L \cap \mathbb{Z} = 0$, and $C(G)$ is nonzero, reduced, torsion free and integral over $\mathbb{Z}$. The proof of Theorem 15.30 applies unchanged. A positive monomial $T = \omega^g$, $0 < g \in G$, gives $v_T \in K_G$ by its binomial Laurent expansion, whether or not K_G is real closed, with $v_T^2 + 1 = Tv_T \in I$; so the coefficient embedding and the spectral argument of Lemmas 15.35 and 15.36 and Theorem 15.38 apply. For $G = \{0\}$: $K_G = \mathbb{R}$, $\mathcal{A}(G) = \mathbb{Z}$ and $\Pi(G) = 0$. $\square$

This complements Theorem 15.51, which describes the unreduced $N_{\bullet}/JN_{\bullet}$ under the divisibility axiom (ii) that a fixed Hahn field may fail. No atomlessness of $\text{Idem}(C(G))$ for a fixed G follows: the proof for **No** gains a fresh scale beyond a prescribed set of supports, which a fixed workspace need not supply, and reduction by $\sqrt{\Pi(G)\mathcal{N}(G)}$ cannot silently be replaced by reduction by $\Pi(G)\mathcal{N}(G)$ (20). (Status (batch 32): for cyclic G both points are decided by 23 without a fresh scale: $\text{Idem}(C(\mathbb{Z}a))$ is atomless, by ramification at unused finite places, and the replacement is impossible because $\Pi(\mathbb{Z}a)\mathcal{N}(\mathbb{Z}a)$ is not radical, Corollary 15.110 and Theorem 15.86; see also Remark 15.114.)

Corollary 15.77 (The surcomplex fibre [20]). *Every set-sized subring of $B_{\mathbb{C}}$ has a uniform splitter, and Theorems 15.71 and 15.73 and Corollary 15.74 hold for $B_{\mathbb{C}}$.*

Proof. Identify $B_{\mathbb{C}} = B_{\mathbb{R}} \times B_{\mathbb{R}}$ (Theorem 15.42). For a set-sized subring S let R be the ring generated by both coordinate projections of S and e a splitter of R ; the diagonal idempotent (e, e) splits every nonzero pair, since some coordinate is nonzero. The other proofs use only uniform splitting. $\square$

Atomlessness does not give $B_{\mathbb{R}} \cong B_{\mathbb{R}} \times B_{\mathbb{R}}$, which would need more information about the global Boolean algebra; the integral doubling is $B_{\mathbb{C}} \cong B_{\mathbb{R}} \times B_{\mathbb{R}}$ (20).

Maps to set-sized rings. The remaining results use the universal theorem through Corollary 15.45, and the last one uses class ultrafilters. For a class ultrafilter U on $\text{Idem}(B_{\mathbb{R}})$, the evaluation ev_U of Theorem 15.49 is a surjective unital map $B_{\mathbb{R}} \rightarrow \overline{\mathbb{Z}}$ fixing the coefficient copy; 20 calls it a *branch map*.

Lemma 15.78 (Class ultrafilters through a prescribed set [19, 20]). *With global choice, a set of elements of a Boolean algebra (possibly a proper class) with the finite intersection property is contained in an ultrafilter (a class).*

Proof. As for Lemma 15.48, adding the prescribed set to the decisions: at stage α add b_{α} if the prescribed set and the earlier decisions together with b_{α} keep the finite intersection property, and $\neg b_{\alpha}$ otherwise; if both failed, finite meets c, d from the prescribed set and the earlier decisions would satisfy $c \wedge b_{\alpha} = 0 = d \wedge \neg b_{\alpha}$, hence $c \wedge d = 0$. $\square$

Theorem 15.79 (An idempotent invisible to every set of small targets [20]). *Let $(\varphi_i)_{i \in I}$ be a set-indexed family, given by one class relation, of unital homomorphisms $\varphi_i : \mathcal{N} \rightarrow S_i$ to set-sized rings. There is $x \in \mathcal{N}$ whose residue is a nonzero idempotent of $B_{\mathbb{R}}$ with $\varphi_i(x) = 0$ for all i . Hence $\bigcap_i \ker \varphi_i \supsetneq \Pi \mathcal{N}$, and $B_{\mathbb{R}}$ does not embed in a set-indexed product of set-sized rings.*

Proof. The maps factor through $\mathbf{B}_{\mathbb{R}}$ (Corollary 15.45), and together give $\psi : \mathbf{B}_{\mathbb{R}} \rightarrow S = \prod_i S_i$, a set. Take a cardinal $\kappa > |S|$ and the independent family $(e_\alpha)_{\alpha < \kappa}$ of Theorem 15.72; two members $e_\alpha \neq e_\beta$ have the same image p , an idempotent. Their symmetric difference $d = e_\alpha + e_\beta - 2e_\alpha e_\beta = e_\alpha(1 - e_\beta) + (1 - e_\alpha)e_\beta$ is an idempotent, nonzero by independence, and $\psi(d) = 2p - 2p^2 = 0$ in every characteristic. Any lift $x \in \mathcal{N}$ of d works. $\square$

Each nonzero residue is detected by a map to the countable domain $\overline{\mathbb{Z}}$ (Theorem 15.49), so in the elementwise sense of Theorem 5.12 $\mathbf{B}_{\mathbb{R}}$ is residually set-sized [merge]; but no set-indexed family is jointly faithful. The obstruction concerns the size of the index set, not of any one target, and it exhibits a nonzero idempotent in every such common kernel (20).

Theorem 15.80 (Many branches through prescribed data [20]). *Assume global choice. Fix a coefficient copy $\overline{\mathbb{Z}} \subseteq \mathbf{B}_{\mathbb{R}}$ and a branch map $\varphi : \mathbf{B}_{\mathbb{R}} \rightarrow \overline{\mathbb{Z}}$ fixing it. For every set $S \subseteq \mathbf{B}_{\mathbb{R}}$ and every cardinal κ there are at least 2^κ branch maps agreeing with φ on S .*

Proof. Choose finite-partition representations of the elements of S , let $\mathfrak{E}_0$ be the set-sized Boolean algebra generated by the parts used and $B_0 = \overline{\mathbb{Z}}[\mathfrak{E}_0]$, a set-sized subring containing S and the coefficient copy; φ restricted to $\mathfrak{E}_0$ is an ultrafilter U_0 . Take $(e_\alpha)_{\alpha < \kappa}$ free over B_0 (Theorem 15.71). For $\eta : \kappa \rightarrow \{0, 1\}$, the set $U_0 \cup \{e_\alpha : \eta(\alpha) = 1\} \cup \{1 - e_\alpha : \eta(\alpha) = 0\}$ has the finite intersection property: a finite meet from U_0 is a nonzero idempotent of B_0 , and its product with any finite assignment word stays nonzero by relative freeness. Extend it to a class ultrafilter (Lemma 15.78) and evaluate. The map selects the same part as φ in every partition from $\mathfrak{E}_0$, so it agrees with φ on B_0 ; different η give maps differing on some e_α . The recursion is uniform in the set parameter η , so this is a set-indexed class family of maps. $\square$

Fixing the coefficient copy only removes an irrelevant automorphism of $\overline{\mathbb{Z}}$: an arbitrary characteristic-zero domain map can be normalized by composing with the inverse of the automorphism it induces on the copy (20).

Corollary 15.81 (Surcomplex maps [20]). *Theorems 15.79 and 15.80 hold for $\mathcal{N}_{\mathbb{C}}$, $\Pi\mathcal{N}_{\mathbb{C}}$ and $\mathbf{B}_{\mathbb{C}}$; in the extension statement the factor of $\mathbf{B}_{\mathbb{C}} \cong \mathbf{B}_{\mathbb{R}} \times \mathbf{B}_{\mathbb{R}}$ selected by the given branch can be kept fixed.*

Proof. Corollary 15.77 gives uniform splitting and relative freeness in $\mathbf{B}_{\mathbb{C}}$, and the proof of Theorem 15.79 is unchanged. A domain map out of a product selects one factor; keep it, project the prescribed set into it, and apply Theorem 15.80. $\square$

$\mathcal{N} \subseteq \mathbf{No}$ has no idempotents other than 0, 1, while $\mathbf{B}_{\mathbb{R}}$ has a proper class of them and contains the residue $\mathbf{v}$ with $\mathbf{v}^2 = -1$; this is possible because $\Pi\mathcal{N}$ is not convex (Proposition 15.34). The quotient is not standard part, and an idempotent residue does not say that its chosen representative is idempotent (20). What 20 leaves open is recorded in Questions 18.17 and 18.18 and its limitations in Section 18.2.

15.10 Ramification in a cyclic workspace

This subsection is manuscript 23's (Section 1.3). It settles the fixed-group clause of Question 18.17(b) for every nonzero cyclic exponent group $G = \mathbb{Z}a$, with opposite answers for its two parts: the extended ideal $\Pi(\mathbb{Z}a)\mathcal{N}(\mathbb{Z}a)$ is *not* radical, since the branch $\mathbf{z}_n = T(1 + T^{1-n})^{1/n}$ has nilpotency index exactly n modulo it (Theorem 15.86), whereas $\text{Idem}(\mathbf{C}(\mathbb{Z}a))$ is atomless, of cardinality and order density $\mathfrak{c} = 2^{\aleph_0}$ (Corollary 15.110 and Theorem 15.111). It describes $\mathcal{N}(\mathbb{Z}a)$ completely by a

residue test at the finitely many branches over $T = 0$ of each finite subextension (Theorems 15.91 and 15.93); this answers the fixed-group clause of Question 18.11 for cyclic groups, as a criterion and not as an algorithm. The mechanism is a distinction between two places. The elements are Laurent series at $T = \infty$, where $T = \omega^a$ is infinitely large, while integrality and nilpotence modulo the extended ideal are governed by the algebraic places over $T = 0$; a series at infinity is never evaluated term by term at zero (23). Atomlessness needs no fresh dominant scale, which 20's argument uses and a fixed workspace lacks (the paragraph after Proposition 15.76): the new idempotents come from ramification at unused finite places of $k(T)$. 23 adds exact thresholds for the refinement $T = U^m$, a reduced bounded-denominator Puiseux limit with no reduced finite stage, the local structure of the coefficient normalization, the associated graded ring and the completion, a periodic free resolution over the coefficient fibre that fails over the arithmetic fibre, and a countable fragment whose Boolean algebra is the Cantor algebra. Every result holds for the real pair and for its Gaussian counterpart.

23 pins 343dc2c (23 September 2026), which contains the writing of 20's subsection but not that of 22's; it read this report's normalization, fibre and branching material (its audit names Questions 18.11 and 18.17 and Theorems 15.57 and 15.60). Printed once, under the existing label and with 23 added as a proving source, is 20's multiquadratic lemma (Lemma 15.69; 23's Lemma 7.4, with the same proof). 23's Corollary 4.5 (Corollary 15.94) contains, for the real pair, the first equality of 20's Theorem 15.60, which 23 credits and recovers as a check; its Gaussian form and its statement for the fixed ideal are new. At its pin this report also contained, after Theorem 15.70, the remark that one Laurent scale carries continuum many independent idempotents over $\overline{\mathbb{Z}}_{\mathbb{R}}$: with $G = \{0\}$, Theorem 15.70 gives an independent family of size $\mathfrak{c}$ in $\text{Idem}(\mathbb{C}(\mathbb{Z}a))$ (20). 23 does not cite it. The lower-bound half of 23's cardinality statement for the reduced fibre is therefore 20's; 23's family, its independence in the unreduced and coefficient fibres, the exact value $\mathfrak{c}$, atomlessness, order density, uniform splitting and the count $2^{\mathfrak{c}}$ are new. The radicality of the Puiseux limit also follows from 19's finite-axiom theorem Theorem 15.51, which was in the report at 23's pin and which 23 does not cite (Remark 15.102, a merge observation); 23's refinement proof is kept as the first route, and the stage-by-stage statements of Theorem 15.100 are new. The numbering of 23 cited here is that of its own manuscript.

Notation and renamings. Throughout, (k, D) is $(\mathbb{R}, \mathbb{Z})$ or $(\mathbb{C}, \mathbb{Z}[i])$, $a > 0$ is a fixed surreal exponent, and $T = \omega^a$ is the actual monomial, as in Theorem 15.60. $K_{\mathbb{Z}a} = k((X^{\mathbb{Z}a}))$ is 16's Hahn field with coefficients k and supports in $\mathbb{Z}a$ (for $k = \mathbb{R}$ it is 20's $K_{\mathbb{Z}a}$); by Lemma 15.62 with $G = \{0\}$ and $b = a$ it is the Laurent field $k((T^{-1}))$, the Laurent variable being identified with ω^a , so its elements are actual surreal, respectively surcomplex, numbers, and $\mathfrak{c} = |K_{\mathbb{Z}a}|$. For the real pair (15.20) gives $\mathcal{A}(\mathbb{Z}a) = \mathbb{Z} + T\mathbb{R}[T]$, $\Pi(\mathbb{Z}a) = T\mathbb{R}[T]$, $\mathcal{N}(\mathbb{Z}a) = \text{IC}_{K_{\mathbb{Z}a}}(\mathcal{A}(\mathbb{Z}a))$ and $\mathbb{C}(\mathbb{Z}a)$. For the Gaussian pair the same symbols denote $\mathcal{A}(\mathbb{Z}a) = \mathbf{Oz}[i] \cap K_{\mathbb{Z}a}$, $\Pi(\mathbb{Z}a) = \Pi_{\mathbb{C}} \cap K_{\mathbb{Z}a}$, the integral closure $\mathcal{N}(\mathbb{Z}a)$ of $\mathcal{A}(\mathbb{Z}a)$ in $K_{\mathbb{Z}a}$, and $\mathbb{C}(\mathbb{Z}a)$ defined by (15.20) verbatim; the pair is fixed and not shown in the notation. Put

$$\begin{aligned} \mathcal{B} &= \text{IC}_{K_{\mathbb{Z}a}}(k[T]), & \mathfrak{l} &= \Pi(\mathbb{Z}a)\mathcal{N}(\mathbb{Z}a), & \mathfrak{J} &= \sqrt{T\mathcal{B}} \subseteq \mathcal{B}, \\ \mathbf{B}(\mathbb{Z}a) &= \mathcal{N}(\mathbb{Z}a)/\mathfrak{l}, & \mathbf{B}^{\text{co}}(\mathbb{Z}a) &= \mathcal{B}/T\mathcal{B}, & \mathbf{C}^{\text{co}}(\mathbb{Z}a) &= \mathcal{B}/\mathfrak{J}, \end{aligned} \quad (15.25)$$

the radical $\mathfrak{J}$ being taken in $\mathcal{B}$. Theorems 15.91 and 15.93 show $\mathfrak{J} \subseteq \mathcal{N}(\mathbb{Z}a)$, $\mathfrak{l} = T\mathcal{B}$ and $\mathbb{C}(\mathbb{Z}a) = \mathcal{N}(\mathbb{Z}a)/\mathfrak{J}$. $\mathbf{B}(\mathbb{Z}a)$ is the *arithmetic fibre*, the fixed-workspace analogue of 19's $\mathbf{B}_{\bullet}$, and $\mathbf{C}(\mathbb{Z}a)$ is its reduction; $\mathbf{B}^{\text{co}}(\mathbb{Z}a)$ is the *coefficient fibre*, with reduction $\mathbf{C}^{\text{co}}(\mathbb{Z}a)$. The coefficient fibre is the arithmetic fibre of the pair (k, k) , since $k[T] = \mathcal{A}_{k,k}(\mathbb{Z}a)$ and $T\mathcal{B} = \Pi_k(\mathbb{Z}a)\mathcal{B}$. K^{alg} is the subfield of $K_{\mathbb{Z}a}$ of elements algebraic over $k(T)$. For a finite extension $k(T) \subseteq L \subseteq K^{\text{alg}}$ put $\mathcal{B}_L = \text{IC}_L(k[T])$; for a maximal ideal $\mathfrak{q}$ of $\mathcal{B}_L$ write $\mathfrak{q} \mid T$ if $\mathfrak{q} \cap k[T] = (T)$, $\ell_{\mathfrak{q}} = \mathcal{B}_L/\mathfrak{q}$ for its residue field, $f \bmod \mathfrak{q}$

for the residue of f , and

$$v_q(f) = \frac{\text{ord}_q(f)}{\text{ord}_q(T)}, \quad v_q(T) = 1. \quad (15.26)$$

For an integer $m \geq 1$, $U = \omega^{a/m}$, so $T = U^m$, and $\mathbb{Z}a/m$ is the group generated by a/m ; the objects defined with U in place of T are $K_{\mathbb{Z}a/m}$, $\mathcal{A}(\mathbb{Z}a/m)$, $\Pi(\mathbb{Z}a/m)$, $\mathcal{N}(\mathbb{Z}a/m)$, $\mathcal{I}(\mathbb{Z}a/m)$ and $\mathcal{B}(\mathbb{Z}a/m)$, so that $\mathcal{I} = \mathcal{I}(\mathbb{Z}a)$. 23's symbols are renamed as follows.

- Its F , A , P , N , B , I , E are $K_{\mathbb{Z}a}$, $\mathcal{A}(\mathbb{Z}a)$, $\Pi(\mathbb{Z}a)$, $\mathcal{N}(\mathbb{Z}a)$, $\mathcal{B}$, $\mathcal{I}$, K^{alg} ($\mathcal{B}$ is the ring $D + \mathfrak{m}_k$ and $\mathcal{B}_\bullet$ the full-class fibre; I and J are local ideals in Proposition 15.76 and Theorem 15.51; $\mathcal{E}$ is 07's field and E an ambient field). Its arithmetic fibre $\mathcal{R} = N/I$, its reduction $\mathcal{R}_{\text{red}}$, its coefficient fibre $\mathcal{C} = B/TB$ and $\mathcal{C}_{\text{red}}$ are $\mathcal{B}(\mathbb{Z}a)$, $\mathcal{C}(\mathbb{Z}a)$, $\mathcal{B}^{\text{co}}(\mathbb{Z}a)$ and $\mathcal{C}^{\text{co}}(\mathbb{Z}a)$ ($\mathcal{R}$ is 14's Rees algebra). Its J and its nilradical $\mathfrak{n} = J/TB$ are $\mathcal{J}$ and $\mathcal{J}/T\mathcal{B}$.
- Its B_L and residue field $\kappa(\mathfrak{q})$ are $\mathcal{B}_L$ and ℓ_q (κ is a cardinal); its minimum order μ_T and truncated order h_T are ν_T and $\hat{\nu}_T$ (μ is a cardinal letter, h an exponent); its filtration $\mathcal{F}^r \mathcal{R}$ is Fil^r ($\mathcal{F}$ is a family in Proposition 3.3); its $V = B_q$ with maximal ideal $\mathfrak{m}$ is $\mathcal{B}_q$ with maximal ideal $\mathfrak{q}\mathcal{B}_q$ ($\mathfrak{m}$ is the negative-support ideal).
- Its branches z_n , h_c , e_c and w are z_n , ς_c , y_c and 20's ϱ_0 of Lemma 15.63 (a sequence z_n occurs in the proof of Theorem 13.45; e_r , e_t , e_α , e_H , $w_\pm$ and w_t are 19's and 20's idempotents and splitters); the residue of y_c is $\bar{y}_c$.
- Its refined rings $F_m, A_m, P_m, N_m, I_m, \mathcal{R}_m$ are the objects over $\mathbb{Z}a/m$ above; its Puiseux rings $F_{\text{Pui}}, A_{\text{Pui}}, P_{\text{Pui}}, N_{\text{Pui}}, I_{\text{Pui}}$ are $K_{\text{Pui}}, \mathcal{A}_{\text{Pui}}, \Pi_{\text{Pui}}, \mathcal{N}_{\text{Pui}}, \mathcal{I}_{\text{Pui}}$; its $F_{\mathbb{Q}}, A_{\mathbb{Q}}, P_{\mathbb{Q}}$ are $K_{\mathbb{Q}a}$, $\mathcal{A}(\mathbb{Q}a)$, $\Pi(\mathbb{Q}a)$.
- Its D_k^{int} is $\overline{\mathbb{Z}}_{\mathbb{R}}$ or $\overline{\mathbb{Z}}$; its coefficient fragments k_r , k_c are 19's $k_0 = \overline{\mathbb{Q}} \cap \mathbb{R}$ and $\overline{\mathbb{Q}}$; its module $M = \mathcal{C}/(x)$ is $\mathcal{M}_n$ (M is a lattice); its graded variable X is W (X^g are formal monomials).

23's finite extensions L keep their letter (as the set-sized field of Corollary 15.50; not 14's module $L(M)$, 16's lower set or 03's $L_c(F)$), and so do its primes $\mathfrak{q}$ (not 14's kernels $\mathfrak{q}_M$). The letter c is a positive real parameter, $\mathfrak{c}$ the continuum. Four tempting readings are false. (1) The Laurent constant coefficient is not a residue at a place over zero: $z_2 = T + \frac{1}{2} - \dots$ lies in $\mathcal{N}(\mathbb{Z}a)$ although $\frac{1}{2}$ is not an algebraic integer (Example 15.87). (2) The valuations v_q have $v_q(T) = 1$; the Laurent valuation at infinity has value -1 at T , and neither extends the other (Remark 15.85). (3) $k\mathcal{N}(\mathbb{Z}a) = \mathcal{B}$ is a finite k -linear span: it does not say $k \subseteq \mathcal{N}(\mathbb{Z}a)$, and $\mathcal{I} = T\mathcal{B}$ is larger than $T\mathcal{N}(\mathbb{Z}a)$. (4) The injection $\mathcal{C}(\mathbb{Z}a) \hookrightarrow \mathcal{B}_{\mathbb{R}}$ of Corollary 15.58 concerns the reduced fibre; on $\mathcal{B}(\mathbb{Z}a)$ the map to $\mathcal{B}_{\mathbb{R}}$ kills exactly the nilradical (Remark 15.119).

The workspace and its finite normalizations.

Proposition 15.82 (The cyclic integer part [23]). **Oz** $\cap K_{\mathbb{Z}a} = \mathbb{Z} + T\mathbb{R}[T]$ for the real pair and **Oz** $[i] \cap K_{\mathbb{Z}a} = \mathbb{Z}[i] + T\mathbb{C}[T]$ for the Gaussian pair. In both cases $\Pi(\mathbb{Z}a) = Tk[T]$ and $\text{Frac}(\mathcal{A}(\mathbb{Z}a)) = k(T)$.

Proof. An omnific normal form has no negative exponents and an integer constant coefficient, and a reverse well-ordered subset of $\{na : n \geq 1\}$ is finite; so the positive part is a polynomial in T without constant term (as in the proof of Theorem 15.60). The Gaussian case applies this to both coordinates. Finally T and cT lie in $\mathcal{A}(\mathbb{Z}a)$ for $c \in k$, so $c = (cT)/T \in \text{Frac}(\mathcal{A}(\mathbb{Z}a))$ and $k(T) \subseteq \text{Frac}(\mathcal{A}(\mathbb{Z}a))$; the reverse inclusion is clear. $\square$

For the real pair $\mathcal{A}(\mathbb{Z}a)$ is an integer part of $K_{\mathbb{Z}a}$, although $K_{\mathbb{Z}a}$ is not real closed: keep the positive powers of f , round its real constant down, and subtract one more if the constant was an integer and the negative-power tail is negative; this is the unique $u \in \mathcal{A}(\mathbb{Z}a)$ with $u \leq f < u + 1$. It does not extend ct from $\mathcal{A}(\mathbb{Z}a)$ to $\mathcal{N}(\mathbb{Z}a)$ (23).

For every finite extension $k(T) \subseteq L \subseteq K^{\text{alg}}$ (separable, in characteristic 0), $\mathcal{B}_L$ is a finite $k[T]$ -module and a Dedekind domain whose nonzero prime localizations are discrete valuation rings, its residue fields are finite over those of $k[T]$, and every prime of $k[T]$ has a prime above it [36, 37, 39]. There are finitely many $\mathfrak{q} \mid T$, and $\ell_{\mathfrak{q}}$ is $\mathbb{C}$ if $k = \mathbb{C}$ and is $\mathbb{R}$ or $\mathbb{C}$ if $k = \mathbb{R}$.

Lemma 15.83 (Finite detection and descent [23]). *The rings $\mathcal{B}_L$ form a directed system with $\mathcal{B} = \bigcup_L \mathcal{B}_L$, $\mathcal{B} \cap L = \mathcal{B}_L$ and $\text{Frac}(\mathcal{B}) = K^{\text{alg}}$. For $f \in \mathcal{B}_L$ and an integer $r \geq 1$,*

$$f \in T^r \mathcal{B} \iff f \in T^r \mathcal{B}_L \iff v_{\mathfrak{q}}(f) \geq r \text{ for every } \mathfrak{q} \mid T. \quad (15.27)$$

Proof. An integral element is algebraic and lies in a finite extension, and integrality over $k[T]$ is a property of the element, so $\mathcal{B} \cap L = \mathcal{B}_L$. Clearing the denominators of a monic equation, a nonzero polynomial multiple of any element of K^{alg} is integral over $k[T]$. Next, $f \in T^r \mathcal{B}$ means $f/T^r \in \mathcal{B} \cap L = \mathcal{B}_L$. A Dedekind domain is the intersection of its localizations at nonzero primes; at primes not above T the element T is a unit, and at the primes above T regularity of f/T^r is the stated inequality. $\square$

Lemma 15.84 (Finite branch locus [23]). *A finite extension $L/k(T)$ is unramified at all but finitely many nonzero primes of $k[T]$.*

Proof. Multiply a primitive element by a nonzero polynomial to make it integral over $k[T]$. Its minimal polynomial is monic and separable, with nonzero discriminant Δ . After inverting Δ the finite free algebra it generates is étale over $k[T][\Delta^{-1}]$, hence normal, hence the normalization there; only the finitely many prime divisors of Δ can ramify [38]. $\square$

Remark 15.85 (Two unrelated valuations [23]). The Laurent valuation at infinity has value -1 at T , the valuations (15.26) have value 1. The latter are valuations of finite algebraic function fields, not continuous extensions of the Laurent topology to an evaluation at $T = 0$. An expansion with infinitely many negative powers can be regular at a place over zero, and an element finite in the surreal order, such as z_2/T below, can have a pole there. Confusing the two would invalidate every argument below (23).

Explicit nilpotents of every finite order. For an ordinary integer $n \geq 2$ put

$$z_n = T \sum_{j \geq 0} \binom{1/n}{j} T^{j(1-n)}, \quad z_n^n = T^n + T = T(1 + T^{n-1}). \quad (15.28)$$

The exponents are integers decreasing to $-\infty$, the formal binomial identity holds in characteristic 0, and the coefficients are rational; in $K_{\mathbb{Z}a}$ for $k = \mathbb{R}$ this is the positive root with leading term T , and the same branch is used for $k = \mathbb{C}$.

Theorem 15.86 (Exact cyclic nilpotency [23]). *$z_n \in \mathcal{N}(\mathbb{Z}a)$, $z_n^n \in \mathfrak{l}$, and $z_n^j \notin \mathfrak{l}$ for $1 \leq j < n$. Hence $\mathfrak{l} = \Pi(\mathbb{Z}a)\mathcal{N}(\mathbb{Z}a)$ is not radical, and the residue of z_n in $\mathcal{B}(\mathbb{Z}a)$ has nilpotency index exactly n .*

Proof. (15.28) is a monic equation over $\mathcal{A}(\mathbb{Z}a)$ with right side in $\Pi(\mathbb{Z}a)$, so $z_n \in \mathcal{N}(\mathbb{Z}a)$ and $z_n^n \in \mathcal{I}$. Suppose $z_n^j = \sum_{\ell \leq r} p_\ell b_\ell$ with $1 \leq j < n$, $p_\ell \in Tk[T]$ and $b_\ell \in \mathcal{N}(\mathbb{Z}a)$. Choose a finite L containing z_n and all b_ℓ ; each b_ℓ is integral over $k[T]$, hence in $\mathcal{B}_L$. At any $\mathfrak{q} \mid T$ every summand has value at least 1, while $1 + T^{n-1}$ is a unit and (15.28) gives $v_{\mathfrak{q}}(z_n^j) = j/n < 1$, a contradiction. $\square$

This proof uses only a finite ideal expression, the normalization of one finite function field and one valuation; it does not use the identity $\mathcal{I} = T\mathcal{B}$ of Theorem 15.93, and it is a self-contained negative answer to fixed-cyclic radicality (23).

Example 15.87 (The first obstruction [23]). For $n = 2$,

$$z_2 = \sqrt{T^2 + T} = T + \frac{1}{2} - \frac{1}{8T} + \frac{1}{16T^2} - \frac{5}{128T^3} + \cdots,$$

and its residue in $\mathcal{B}(\mathbb{Z}a)$ is nonzero with square zero. Although z_2 is integral over $\mathcal{A}(\mathbb{Z}a)$, its Laurent constant coefficient $\frac{1}{2}$ is not an algebraic integer, so integrality over $\mathcal{A}(\mathbb{Z}a)$ cannot be tested on the constant coefficient at infinity. And z_2/T is infinitesimally close to 1 but not integral over $\mathcal{A}(\mathbb{Z}a)$: it has value $-\frac{1}{2}$ at every place over zero. In the full class $z_2 = \omega^{a/2}\sqrt{1 + \omega^a}$ with $\sqrt{1 + \omega^a} \in \mathcal{N}$, so $z_2 \in \Pi\mathcal{N}$, in accordance with Theorem 15.29 [merge].

Corollary 15.88 (Unbounded nilpotence [23]). *The nilradical of $\mathcal{B}(\mathbb{Z}a)$ is not nilpotent and not finitely generated; $\mathcal{B}(\mathbb{Z}a)$ and $\mathcal{N}(\mathbb{Z}a)$ are not Noetherian.*

Proof. The nilpotency indices of Theorem 15.86 are unbounded. A nil ideal generated by $x_1, \dots, x_s$ with $x_i^{d_i} = 0$ has vanishing $(1 + \sum_i (d_i - 1))$ th power, since every monomial of that degree contains some $x_i^{d_i}$. A quotient of a Noetherian ring is Noetherian. $\square$

Theorem 15.89 (Exact order after refinement [23]). *For $n \geq 2$, $m \geq 1$ and $j \geq 1$, $z_n^j \in \mathcal{I}(\mathbb{Z}a/m)$ if and only if $mj \geq n$. So under $\mathcal{B}(\mathbb{Z}a) \rightarrow \mathcal{B}(\mathbb{Z}a/m)$, induced by $\mathcal{A}(\mathbb{Z}a) \subseteq \mathcal{A}(\mathbb{Z}a/m)$, the residue of z_n has index $\lceil n/m \rceil$.*

Proof. At a place above $U = 0$, normalized by $v(U) = 1$, the relation $z_n^n = U^m(1 + U^{m(n-1)})$ gives $v(z_n^j) = mj/n$, while every element of $\mathcal{I}(\mathbb{Z}a/m)$ has value at least 1, as in Theorem 15.86. Conversely, if $mj \geq n$, then $y = z_n^j/U$ satisfies $y^n = U^{mj-n}(1 + U^{m(n-1)})^j$, a polynomial with constant coefficient 0 (if $mj > n$) or 1 (if $mj = n$), which lies in $\mathcal{A}(\mathbb{Z}a/m)$; so y is integral over $\mathcal{A}(\mathbb{Z}a/m)$ and $z_n^j = Uy \in \mathcal{I}(\mathbb{Z}a/m)$. $\square$

For z_{12} and $m = 1, 2, 3, 4, 6, 12$ the indices are 12, 6, 4, 3, 2, 1; index 1 means that the residue has become 0. Every refined workspace still has nilpotents of unbounded index (z_n with n large), so no single refinement removes all nilpotents; that each one is removed by some refinement (Theorem 15.99) has the other order of quantifiers (23).

The residue criterion. $\mathcal{I}$ is in general larger than $T\mathcal{N}(\mathbb{Z}a)$: $\Pi(\mathbb{Z}a)$ contains arbitrary coefficients in k , which need not lie in $\mathcal{N}(\mathbb{Z}a)$. The following lemma accounts for them.

Lemma 15.90 (Scaling [23]). $T\mathcal{B} \subseteq \mathcal{N}(\mathbb{Z}a)$.

Proof. If $b^d + p_{d-1}(T)b^{d-1} + \cdots + p_0(T) = 0$ with $p_j \in k[T]$, multiplying by T^d gives $(Tb)^d + Tp_{d-1}(T)(Tb)^{d-1} + \cdots + T^d p_0(T) = 0$, a monic equation with nonleading coefficients in $Tk[T] \subseteq \mathcal{A}(\mathbb{Z}a)$. $\square$

Theorem 15.91 (Residue criterion [23]). *Let $k(T) \subseteq L \subseteq K^{\text{alg}}$ be finite and $f \in \mathcal{B}_L$. Then $f \in \mathcal{N}(\mathbb{Z}a)$ if and only if $f \bmod \mathfrak{q} \in \ell_{\mathfrak{q}}$ is integral over D for every $\mathfrak{q} \mid T$. In particular $\sqrt{T\mathcal{B}_L} \subseteq \mathcal{N}(\mathbb{Z}a)$ and $\mathcal{J} \subseteq \mathcal{N}(\mathbb{Z}a)$.*

Proof. If f satisfies a monic equation over $\mathcal{A}(\mathbb{Z}a)$, reduce it modulo $\mathfrak{q} \mid T$: a coefficient $d + Tp(T)$ becomes $d \in D$. Conversely, for each of the finitely many $\mathfrak{q} \mid T$ choose a monic $Q_{\mathfrak{q}} \in D[Y]$ killing $f \bmod \mathfrak{q}$, and let Q be their product; then $Q(f) \in \mathfrak{q}$ for all $\mathfrak{q} \mid T$. With $T\mathcal{B}_L = \prod_{\mathfrak{q} \mid T} \mathfrak{q}^{e_{\mathfrak{q}}}$ and $r \geq \max e_{\mathfrak{q}}$, $Q(f)^r = Tb$ with $b \in \mathcal{B}_L$. By Lemma 15.90 Tb is integral over $\mathcal{A}(\mathbb{Z}a)$, $Y^r - Tb$ shows that $Q(f)$ is integral over $\mathcal{A}(\mathbb{Z}a)[Tb]$, and $Q(Y) - Q(f)$ is monic over $\mathcal{A}(\mathbb{Z}a)[Q(f)]$ and kills f ; transitivity gives $f \in \mathcal{N}(\mathbb{Z}a)$. Elements of $\sqrt{T\mathcal{B}_L}$ have zero residues. An element of $\mathcal{J}$ lies in some $\mathcal{B}_L$ with a power in $T\mathcal{B}$, hence in $T\mathcal{B}_L$ by Lemma 15.83. $\square$

The proof holds for every coefficient field of characteristic 0 and every unital D in it; the real and complex hypotheses enter only in Theorem 15.93, through the small dimensions of the residue fields (23).

Remark 15.92 (A finite certificate, not an oracle [23]). The sufficiency proof yields an explicit monic certificate once the normalization and the residues are known: if $b = Q(f)^r/T$ satisfies a monic equation of degree d over $k[T]$, Lemma 15.90 gives a monic H of degree d over $\mathcal{A}(\mathbb{Z}a)$ killing $Q(f)^r$, and $H(Q(Y)^r)$ kills f and has degree $dr \deg Q$. This is a finite algebraic criterion for a represented algebraic Laurent element, not a decision procedure for arbitrary real coefficients or arbitrary normal forms: an implementation must represent L , identify the chosen branch at infinity, normalize at the finite places and decide integrality of the residue constants, and substituting $T = 0$ in a possibly singular defining equation need not separate the branches of its normalization (23).

Theorem 15.93 (Coefficient span and the extended ideal [23]). *$k\mathcal{N}(\mathbb{Z}a) = \mathcal{B}$ (finite k -linear span) and $\mathfrak{l} = T\mathcal{B}$. Hence $\mathcal{B}(\mathbb{Z}a) = \mathcal{N}(\mathbb{Z}a)/T\mathcal{B} \hookrightarrow \mathcal{B}^{\text{co}}(\mathbb{Z}a)$, the two fibres have the same nilradical $\mathcal{J}/T\mathcal{B}$, and $\mathcal{C}(\mathbb{Z}a) = \mathcal{N}(\mathbb{Z}a)/\mathcal{J} \hookrightarrow \mathcal{C}^{\text{co}}(\mathbb{Z}a)$.*

Proof. Clearly $k\mathcal{N}(\mathbb{Z}a) \subseteq \mathcal{B}$. Let $b \in \mathcal{B}_L$. By the Chinese remainder theorem $\mathcal{B}_L/\sqrt{T\mathcal{B}_L} \cong \prod_{\mathfrak{q} \mid T} \ell_{\mathfrak{q}}$, a finite product of copies of $\mathbb{C}$ (if $k = \mathbb{C}$) or of $\mathbb{R}$ and $\mathbb{C}$ (if $k = \mathbb{R}$). The coordinate vectors, together for $k = \mathbb{R}$ with the vector equal to i in one complex factor and 0 elsewhere, form a k -basis whose entries are integral over D . Lift them to $b_1, \dots, b_s \in \mathcal{B}_L$; they lie in $\mathcal{N}(\mathbb{Z}a)$ by Theorem 15.91. Choose $c_j \in k$ with $b - \sum_j c_j b_j \in \sqrt{T\mathcal{B}_L} \subseteq \mathcal{N}(\mathbb{Z}a)$; so $b \in k\mathcal{N}(\mathbb{Z}a)$. As $T \in \mathcal{N}(\mathbb{Z}a)$, $k[T]\mathcal{N}(\mathbb{Z}a) = k\mathcal{N}(\mathbb{Z}a) = \mathcal{B}$ and $\mathfrak{l} = Tk[T]\mathcal{N}(\mathbb{Z}a) = T\mathcal{B}$. The injection is immediate, $\mathcal{J} \subseteq \mathcal{N}(\mathbb{Z}a)$ by Theorem 15.91, and nilpotence modulo $T\mathcal{B}$ means the same in both rings; the radical of $\mathfrak{l}$ in $\mathcal{N}(\mathbb{Z}a)$ is $\mathcal{J} \cap \mathcal{N}(\mathbb{Z}a) = \mathcal{J}$. $\square$

Corollary 15.94 (Rational functions [23]). *$\mathcal{N}(\mathbb{Z}a) \cap k(T) = \overline{\mathbb{Z}}_k + Tk[T]$ and $\mathfrak{l} \cap k(T) = Tk[T]$, where $\overline{\mathbb{Z}}_k$ is $\overline{\mathbb{Z}}_{\mathbb{R}}$ for the real pair and $\overline{\mathbb{Z}}$ for the Gaussian pair.*

Proof. $k[T]$ is integrally closed in $k(T)$, and its only place over T is $T = 0$, so Theorem 15.91 asks exactly that $f(0)$ be integral over D . For the ideal use $\mathfrak{l} = T\mathcal{B}$ and $\mathcal{B} \cap k(T) = k[T]$. $\square$

For the real pair the first equality is 20's Theorem 15.60, credited by 23 and recovered as a check on the residue criterion. 20's second equality concerns the full-class ideal, $\Pi\mathcal{N} \cap \mathbb{R}(T) = T\mathbb{R}[T]$; by Theorem 15.57 this is $\mathcal{J} \cap \mathbb{R}(T)$, and 23's $\mathfrak{l} \cap \mathbb{R}(T) = T\mathbb{R}[T]$ is the same set because $T\mathbb{R}[T] \subseteq \mathfrak{l} \subseteq \mathcal{J}$. The Gaussian pair is new.

The ramification filtration and exact refinement.

Definition 15.95 (Minimum finite-place order [23]). For $0 \neq f \in \mathcal{B}$ choose a finite $L \ni f$ and put $\nu_T(f) = \min_{\mathfrak{q}|T} v_{\mathfrak{q}}(f) \in \mathbb{Q}_{\geq 0}$; put $\nu_T(0) = +\infty$.

Lemma 15.96 (Invariance and inequalities [23]). $\nu_T(f)$ does not depend on L . For $f, g \in \mathcal{B}$ and $r \geq 1$, $\nu_T(f + g) \geq \min\{\nu_T(f), \nu_T(g)\}$, $\nu_T(fg) \geq \nu_T(f) + \nu_T(g)$ and $\nu_T(f^r) = r\nu_T(f)$; and

$$f \in T^r \mathcal{B} \iff \nu_T(f) \geq r, \quad f \in \mathcal{J} \iff \nu_T(f) > 0. \quad (15.29)$$

Proof. If $L \subseteq L'$, every $\mathfrak{q} | T$ of $\mathcal{B}_L$ has a prime of $\mathcal{B}_{L'}$ above it, and with both valuations normalized at T the larger restricts to the smaller; so the set of values gains repetitions but no new values. Compare two choices in a compositum. The inequalities hold at each prime of one finite field containing both elements. The ideal criterion is (15.27); a positive minimum becomes at least 1 after multiplication by a large integer, a zero minimum never does. $\square$

Definition 15.97 (The truncated order and its filtration [23]). For $\bar{f} \in \mathcal{B}(\mathbb{Z}a)$ put $\hat{\nu}_T(\bar{f}) = \min\{1, \nu_T(f)\} \in [0, 1] \cap \mathbb{Q}$, and for rational $0 \leq r \leq 1$ put $\text{Fil}^r = \{x \in \mathcal{B}(\mathbb{Z}a) : \hat{\nu}_T(x) \geq r\}$.

Representatives differing by an element of $T\mathcal{B}$ have the same values below 1, so $\hat{\nu}_T$ is well defined, and $\hat{\nu}_T(\bar{f}) = 1$ exactly when $\bar{f} = 0$. Moreover $\hat{\nu}_T(x + y) \geq \min\{\hat{\nu}_T(x), \hat{\nu}_T(y)\}$ and $\hat{\nu}_T(xy) \geq \min\{1, \hat{\nu}_T(x) + \hat{\nu}_T(y)\}$; the Fil^r are ideals with $\text{Fil}^0 = \mathcal{B}(\mathbb{Z}a)$, $\text{Fil}^1 = 0$, $\text{Fil}^r \text{Fil}^s \subseteq \text{Fil}^{\min(1, r+s)}$ and $\text{Nil}(\mathcal{B}(\mathbb{Z}a)) = \bigcup_{0 < r \leq 1} \text{Fil}^r$. This is a rationally indexed filtration, not a valuation: the product inequality can be strict because different branches realize the two minima, and nontrivial complementary idempotents (constructed below) have order 0 while their product has order 1 (23).

Corollary 15.98 (The index of any nilpotent [23]). For $0 \neq f \in \mathcal{N}(\mathbb{Z}a)$ with $\nu_T(f) > 0$ the residue of f has nilpotency index $\lceil 1/\nu_T(f) \rceil$; if $\nu_T(f) = 0$ it is not nilpotent.

Proof. Its r th power vanishes exactly when $r\nu_T(f) \geq 1$, by Theorem 15.93 and (15.29). $\square$

Theorem 15.99 (General refinement law [23]). For $m \geq 1$ and $f \in \mathcal{N}(\mathbb{Z}a)$, with ν_U normalized by $v(U) = 1$,

$$\nu_U(f) = m\nu_T(f), \quad f \in \mathfrak{l}(\mathbb{Z}a/m) \iff \nu_T(f) \geq \frac{1}{m}; \quad (15.30)$$

equivalently $\hat{\nu}_U(\bar{f}) = \min\{1, m\hat{\nu}_T(\bar{f})\}$ under $\mathcal{B}(\mathbb{Z}a) \rightarrow \mathcal{B}(\mathbb{Z}a/m)$. Every nilpotent residue is killed by some refinement, and a non-nilpotent residue by none.

Proof. Let $L \ni f$ be finite and $L' = Lk(U) \subseteq K_{\mathbb{Z}a/m}$. The integral closures of $k[T]$ and of $k[U]$ in L' coincide ($k[U]$ is integral over $k[T]$), every prime of $\mathcal{B}_L$ above T extends, and the extensions are exactly the primes above $U = 0$. A valuation normalized by $v(U) = 1$ sends T to m , so it is m times the old normalization on L ; every old prime occurs, so the minima scale by m . The second formula is Theorem 15.93 over $\mathbb{Z}a/m$ with (15.29). If $\nu_T(f) > 0$ take $m \geq 1/\nu_T(f)$. $\square$

This is an exact finite certificate for the disappearance of a represented nilpotent: compute its positive minimum order at the finitely many branches over zero and take a root refinement at least its reciprocal. The bound is a threshold for ideal membership, not a ramification index or a common multiple of ramification indices; for $\mathbf{z}_n$ every branch has order $1/n$ (23).

The bounded-denominator Puiseux limit. For $m \mid m'$ the inclusions $K_{\mathbb{Z}a/m} \subseteq K_{\mathbb{Z}a/m'}$ form a directed system. Put

$$\begin{aligned} K_{\text{Pui}} &= \bigcup_m K_{\mathbb{Z}a/m}, & \mathcal{A}_{\text{Pui}} &= \bigcup_m \mathcal{A}(\mathbb{Z}a/m), & \Pi_{\text{Pui}} &= \bigcup_m \Pi(\mathbb{Z}a/m), \\ \mathcal{N}_{\text{Pui}} &= \text{IC}_{K_{\text{Pui}}}(\mathcal{A}_{\text{Pui}}), & \mathfrak{l}_{\text{Pui}} &= \Pi_{\text{Pui}} \mathcal{N}_{\text{Pui}}. \end{aligned} \quad (15.31)$$

Each element of K_{Pui} has exponents with a bounded common denominator, so K_{Pui} is strictly smaller than the Hahn field $K_{\mathbb{Q}a}$: $\sum_{n \geq 1} T^{1/n}$ has reverse well-ordered support in $\mathbb{Q}a$ and no common denominator.

Theorem 15.100 (Reduction by rational scales [23]). $\mathcal{N}_{\text{Pui}} = \bigcup_m \mathcal{N}(\mathbb{Z}a/m)$, $\mathfrak{l}_{\text{Pui}} = \bigcup_m \mathfrak{l}(\mathbb{Z}a/m)$ and $\mathcal{N}_{\text{Pui}}/\mathfrak{l}_{\text{Pui}} \cong \text{colim}_m \mathcal{B}(\mathbb{Z}a/m)$. The ideal $\mathfrak{l}_{\text{Pui}}$ is radical, and for each m the kernel of $\mathcal{B}(\mathbb{Z}a/m) \rightarrow \mathcal{N}_{\text{Pui}}/\mathfrak{l}_{\text{Pui}}$ is exactly $\text{Nil}(\mathcal{B}(\mathbb{Z}a/m))$. Finitely many nilpotent residues are killed at one common later stage, and no finite stage is reduced.

Proof. An element of K_{Pui} and the finitely many coefficients of an integrality equation over $\mathcal{A}_{\text{Pui}}$ lie in a common stage; so do the finitely many factors of an extended-ideal expression. This gives the two unions and the colimit. If $f^r \in \mathfrak{l}_{\text{Pui}}$, move f and the expression to a common stage m ; the residue of f in $\mathcal{B}(\mathbb{Z}a/m)$ is nilpotent, so Theorem 15.99 (over $\mathbb{Z}a/m$) kills it at a later stage and $f \in \mathfrak{l}_{\text{Pui}}$. A residue that dies in the colimit dies at a finite later stage, so its minimum order was positive by (15.30) and it was nilpotent. For finitely many residues take a common refinement above all thresholds. Each finite stage is a cyclic workspace and contains the family of Theorem 15.86. $\square$

Remark 15.101 (A set-sized positive case [23]). For $k = \mathbb{R}$, K_{Pui} is real closed and $K_{\text{Pui}}[i]$ is the complex Puiseux field, algebraically closed by the Newton–Puiseux theorem [26]; the real case also follows from the characterization of real closed fields by algebraic closedness after adjoining i . $\mathcal{A}_{\text{Pui}}$ is its truncation integer part. So radicality is restored in a set-sized real closed surreal subfield, without a proper class of scales. This concerns the bounded-denominator support class only: it decides nothing for the Hahn field $K_{\mathbb{Q}a}$ (Question 18.21), and it does not say that the colimit fibre is isomorphic to a reduced finite-stage fibre; the kernel on each stage is known, but the image may grow (23).

Remark 15.102 (A second route through the finite-axiom theorem [merge]). For the real pair, the field K_{Pui} , the ring $\mathcal{A}_{\text{Pui}} = \mathbb{Z} + \Pi_{\text{Pui}}$, the retraction ct and its kernel Π_{Pui} (the F , A , c and J of Theorem 15.51) satisfy its hypotheses: K_{Pui} is real closed and contains the real algebraic numbers; (i) Π_{Pui} is divisible by nonzero integers; (ii) if $x_1, \dots, x_r \in \Pi(\mathbb{Z}a/n)$ and $m \geq 1$, then $d = T^{1/(2mn)} \in \Pi_{\text{Pui}}$ and every x_j/d^m has T -exponents at least $\frac{1}{n} - \frac{1}{2n} > 0$ with denominator dividing $2n$, so lies in $\mathcal{A}_{\text{Pui}}$; (iii) the powers T^j are cofinal. So 19's theorem gives the radicality of $\mathfrak{l}_{\text{Pui}}$ for the real pair again, and more: $\mathcal{N}_{\text{Pui}}/\mathfrak{l}_{\text{Pui}}$ is integrally closed in its rationalization and is the Boolean power $\overline{\mathbb{Z}}[\text{Idem}(\mathcal{N}_{\text{Pui}}/\mathfrak{l}_{\text{Pui}})]$, and the fibre of the integral closure $\mathcal{N}'$ of $\mathcal{A}_{\text{Pui}}$ in $K_{\text{Pui}}[i]$ is absolutely integrally closed and splits as the square of the real one (Theorems 15.30, 15.38 and 15.42 and Corollary 15.33). The Gaussian pair is covered too: $\mathbb{Z}[i] + \Pi_{\text{Pui}}[i] = \mathcal{A}_{\text{Pui}}[i]$ is integral over $\mathcal{A}_{\text{Pui}}$, so its integral closure in $K_{\text{Pui}}[i]$ is $\mathcal{N}'$, and its extended ideal $\Pi_{\text{Pui}}[i]\mathcal{N}' = \Pi_{\text{Pui}}\mathcal{N}'$ is the ideal of Theorem 15.51. Its Boolean algebra is atomless of cardinality $\mathfrak{c}$: an idempotent of the colimit is the residue of an idempotent of some $\mathcal{B}(\mathbb{Z}a/m)$, which Theorem 15.109 (over $\mathbb{Z}a/m$) splits into two nonzero idempotents; these are not nilpotent and so survive by Theorem 15.100, and likewise the independent family of Theorem 15.111 survives, while $|K_{\text{Pui}}| = \mathfrak{c}$. This is an instance of the second clause of Question 18.17(b): real-rooted lifting gives a Boolean-power fibre here, with the monomial divisibility (ii), although set-family divisibility fails ($T^{1/n} \in \Pi_{\text{Pui}}$ for all n and no T^δ divides all of

them in Π_{Pui}). The cyclic workspace itself is outside [Theorem 15.51](#): $K_{\mathbb{Z}a}$ is not real closed and (ii) fails there.

Local structure and the nilradical. The valuation facts used are standard [\[35\]](#).

Theorem 15.103 (Local structure of the coefficient normalization [\[23\]](#)). *$\mathcal{B}$ is a Prüfer domain. For every maximal ideal $\mathfrak{q} \ni T$ of $\mathcal{B}$, the valuation domain $\mathcal{B}_{\mathfrak{q}}$ has value group $\mathbb{Q}$, normalized by $v(T) = 1$, and residue field isomorphic to $\mathbb{C}$, for both pairs; it has rank one and is not a discrete valuation ring.*

Proof. A maximal ideal $\mathfrak{q}$ of $\mathcal{B}$ contracts to a maximal ideal of $k[T]$, as $\mathcal{B}$ is integral over $k[T]$, and with $\mathfrak{q}_L = \mathfrak{q} \cap \mathcal{B}_L$ one has $\mathcal{B}_{\mathfrak{q}} = \bigcup_L (\mathcal{B}_L)_{\mathfrak{q}_L}$, a directed union of discrete valuation rings with common fraction field K^{alg} : for a nonzero element put numerator and denominator in one stage, where it or its inverse is regular. So $\mathcal{B}_{\mathfrak{q}}$ is a valuation domain. If $T \in \mathfrak{q}$, its normalized value group is a union of groups $\frac{1}{e_L}\mathbb{Z}$, so lies in $\mathbb{Q}$, and $v(z_n) = 1/n$ for all $n \geq 2$ by [\(15.28\)](#); so it is $\mathbb{Q}$, Archimedean and not discrete. The residue field is algebraic over k , the union of the finite residue fields; for $k = \mathbb{R}$, 20 's $\varrho_0 = T(1 - T^{-2})^{1/2} \in \mathcal{B}$ has $\varrho_0^2 = T^2 - 1$, so its residue squares to -1 and the residue field is $\mathbb{C}$. $\square$

Remark 15.104 (A real domain with a nonreal arithmetic fibre [\[23\]](#)). ϱ_0 is even integral over $\mathcal{A}(\mathbb{Z}a)$, since the constant coefficient of $T^2 - 1$ is $-1 \in D$, and its residue in $\mathcal{B}(\mathbb{Z}a)$ squares to -1 . So the real arithmetic fibre is not formally real although $\mathcal{N}(\mathbb{Z}a)$ lies in an ordered field; $\mathbb{I}$ is not order-convex, and the quotient is not a standard-part construction. [23](#) notes that the full-class normalization shows the same phenomenon; its witness there is 19 's v_T ([Proposition 15.34](#)), which for $T = \omega^a$ also lies in $K_{\mathbb{Z}a}$ [\[merge\]](#), while ϱ_0 is a second witness inside the cyclic workspace.

Theorem 15.105 (The nilradical is idempotent [\[23\]](#)). *$J^2 = J$, as an ideal of $\mathcal{B}$ and as an ideal of $\mathcal{N}(\mathbb{Z}a)$. So the common nilradical $J/T\mathcal{B}$ of $\mathcal{B}(\mathbb{Z}a)$ and $\mathcal{B}^{\text{co}}(\mathbb{Z}a)$ is nonzero and equal to its square, and it is neither nilpotent nor finitely generated.*

Proof. At a maximal ideal not containing T both J and J^2 localize to the unit ideal. At $\mathfrak{q} \ni T$ the value group is $\mathbb{Q}$, so $\sqrt{T\mathcal{B}_{\mathfrak{q}}} = \mathfrak{q}\mathcal{B}_{\mathfrak{q}}$; for $0 \neq b$ in it choose rational $0 < \delta < v(b)$ and u with $v(u) = \delta$, so that $b = u \cdot (b/u)$ is a product of two elements of the maximal ideal, which is therefore idempotent. Equality of ideals is local, so $J^2 = J$ in $\mathcal{B}$. As $J \subseteq \mathcal{N}(\mathbb{Z}a)$, a coefficient from $\mathcal{B}$ in a finite sum of products can be absorbed into one factor, which stays in J ; so the same holds over $\mathcal{N}(\mathbb{Z}a)$. The rest is [Theorem 15.86](#) and [Corollary 15.88](#). $\square$

This does not conflict with Nakayama's lemma, since the ideal is not finitely generated, and an idempotent ideal need not consist of idempotents: here all its elements are nilpotent [\(23\)](#).

Theorem 15.106 (The associated graded ring [\[23\]](#)). *For every $r \geq 1$, $\mathbb{I}^r = T^r\mathcal{B}$ and $\mathbb{I}^r/\mathbb{I}^{r+1} \cong \mathcal{B}^{\text{co}}(\mathbb{Z}a)$ as $\mathcal{B}(\mathbb{Z}a)$ -modules, the action being through the injection of [Theorem 15.93](#). Multiplication gives a graded isomorphism*

$$\text{gr}_{\mathbb{I}}(\mathcal{N}(\mathbb{Z}a)) \cong \mathcal{B}(\mathbb{Z}a) \oplus W \mathcal{B}^{\text{co}}(\mathbb{Z}a)[W] = \{g \in \mathcal{B}^{\text{co}}(\mathbb{Z}a)[W] : g(0) \in \mathcal{B}(\mathbb{Z}a)\}, \quad (15.32)$$

and in particular $\mathbb{I}/\mathbb{I}^2 \cong \mathcal{B}^{\text{co}}(\mathbb{Z}a) \neq 0$ and $\text{Tor}_1^{\mathcal{N}(\mathbb{Z}a)}(\mathcal{B}(\mathbb{Z}a), \mathcal{B}(\mathbb{Z}a)) \cong \mathcal{B}^{\text{co}}(\mathbb{Z}a)$.

Proof. $\mathbf{l} = T\mathcal{B}$ gives $\mathbf{l}^r \subseteq T^r\mathcal{B}$, and $T^rb = (Tb)T \cdots T$ is a product of r elements of $\mathbf{l}$. Multiplication by T^r identifies $\mathcal{B}/T\mathcal{B}$ with $T^r\mathcal{B}/T^{r+1}\mathcal{B}$ ($\mathcal{B}$ is a domain), $\mathcal{N}(\mathbb{Z}a)$ acts through its image in $\mathcal{B}/T\mathcal{B}$, and products of these identifications multiply classes in $\mathbf{B}^{\text{co}}(\mathbb{Z}a)$. Tensoring $0 \rightarrow \mathbf{l} \rightarrow \mathcal{N}(\mathbb{Z}a) \rightarrow \mathbf{B}(\mathbb{Z}a) \rightarrow 0$ with $\mathbf{B}(\mathbb{Z}a)$, the map $\mathbf{l} \otimes \mathbf{B}(\mathbb{Z}a) \cong \mathbf{l}/\mathbf{l}^2 \rightarrow \mathbf{B}(\mathbb{Z}a)$ is zero, which gives the Tor_1 . $\square$

The conormal module is a further reason not to transfer identities from the full-class ideal, which satisfies $\Pi\mathcal{N} = (\Pi\mathcal{N})^2$ (Proposition 15.31), to a cyclic workspace: $\mathbf{l}$ has a genuine descending sequence of powers, although the nilradical of its quotient is idempotent (23).

Corollary 15.107 (Adic completion as a pullback [23]). *With $\widehat{\mathcal{B}} = \varprojlim_r \mathcal{B}/T^r\mathcal{B}$ and $\widehat{\mathcal{N}} = \varprojlim_r \mathcal{N}(\mathbb{Z}a)/\mathbf{l}^r$, $\widehat{\mathcal{N}} \cong \widehat{\mathcal{B}} \times_{\mathbf{B}^{\text{co}}(\mathbb{Z}a)} \mathbf{B}(\mathbb{Z}a)$, and $\mathcal{B} \rightarrow \widehat{\mathcal{B}}$ and $\mathcal{N}(\mathbb{Z}a) \rightarrow \widehat{\mathcal{N}}$ are injective.*

Proof. Since $T\mathcal{B} \subseteq \mathcal{N}(\mathbb{Z}a)$, the image of $\mathcal{N}(\mathbb{Z}a)/\mathbf{l}^r$ in $\mathcal{B}/T^r\mathcal{B}$ is the inverse image of $\mathbf{B}(\mathbb{Z}a) \subseteq \mathbf{B}^{\text{co}}(\mathbb{Z}a)$; pass to the limit. A nonzero element of $\mathcal{B}$ has finite ν_T , so lies in only finitely many $T^r\mathcal{B}$. $\square$

Boolean splitting without a new scale. For a real $c > 0$ put

$$\varsigma_c = T(1 + cT^{-2})^{1/2}, \quad y_c = \frac{1 + \varsigma_c/\sqrt{c}}{2}, \quad \varsigma_c^2 = T^2 + c, \quad y_c^2 - y_c = \frac{T^2}{4c}. \quad (15.33)$$

The last equation is monic over $\mathcal{A}(\mathbb{Z}a)$, so $y_c \in \mathcal{N}(\mathbb{Z}a)$ and $\bar{y}_c$ is an idempotent of $\mathbf{B}(\mathbb{Z}a)$, for both pairs. Its nontriviality and independence from given data are proved by choosing c outside the finite branch locus of the data.

Lemma 15.108 (A quadratic branch splitting every zero-place branch [23]). *Let $k(T) \subseteq L \subseteq K^{\text{alg}}$ be finite. For all but finitely many real $c > 0$: (i) $T^2 + c$ is not a square in L , and $[L(\varsigma_c) : L] = 2$; (ii) every $\mathfrak{q} \mid T$ of $\mathcal{B}_L$ has extensions to $\mathcal{B}$ at which the residue of ς_c is $\sqrt{c}$, and extensions at which it is $-\sqrt{c}$; (iii) at these, with $v(T) = 1$, $(v(y_c), v(1 - y_c))$ is $(0, 2)$, respectively $(2, 0)$.*

Proof. Only finitely many primes of $k[T]$ ramify in L (Lemma 15.84); choose c so that the prime factors of $T^2 + c$ avoid them, which excludes finitely many c (for $k = \mathbb{R}$ the $T^2 + c$ are distinct irreducible quadratics, for $k = \mathbb{C}$ their pairs of roots are disjoint). Then $T^2 + c$ has order 1 at every prime of $\mathcal{B}_L$ above one of its factors, so it is not a square in L . Hence $\mathcal{B}_L[\varsigma_c]$ is free over $\mathcal{B}_L$ on $1, \varsigma_c$, and modulo $\mathfrak{q} \mid T$ it is $\ell_{\mathfrak{q}}[Y]/(Y^2 - c) \cong \ell_{\mathfrak{q}} \times \ell_{\mathfrak{q}}$, since $\sqrt{c} \in k$ and $2\sqrt{c} \neq 0$. Both primes of this algebra extend to $\mathcal{B}$ by lying over, in set-sized integral extensions. There y_c has residue 1, respectively 0, and $y_c(1 - y_c) = -T^2/(4c)$ fixes the other value as 2. $\square$

Irreducibility is forced at a previously unused finite place, while the two branches occur over zero, and the Laurent expansion at infinity is kept throughout: a new Boolean coordinate with the exponent group still exactly $\mathbb{Z}a$ (23).

Theorem 15.109 (Uniform relative splitting [23]). *Let S be a unital subring of $\mathbf{B}(\mathbb{Z}a)$ with $|S| < \mathfrak{c}$. There is a real $c > 0$ such that, for $e = \bar{y}_c$, the ring homomorphism*

$$S \times S \rightarrow \mathbf{B}(\mathbb{Z}a), \quad (s_0, s_1) \mapsto s_0(1 - e) + s_1e, \quad (15.34)$$

is injective. The same holds with $\mathbf{B}(\mathbb{Z}a)$ replaced by $\mathbf{C}(\mathbb{Z}a)$, $\mathbf{B}^{\text{co}}(\mathbb{Z}a)$ or $\mathbf{C}^{\text{co}}(\mathbb{Z}a)$.

Proof. Choose a representative $f_s \in \mathcal{N}(\mathbb{Z}a)$ of each $s \in S$. The finite subsets of representatives generate at most $\max(|S|, \aleph_0) < \mathfrak{c}$ finite extensions of $k(T)$, each excluding finitely many c in [Lemma 15.108](#); choose $c > 0$ outside their union. (No regularity of $\mathfrak{c}$ is used.) The map is a ring homomorphism because e is idempotent. Let $(s_0, s_1) \neq (0, 0)$, say $s_j \neq 0$, and put both representatives in one of the chosen fields L . As $f_{s_j} \notin T\mathcal{B}$, some $\mathfrak{q} \mid T$ has $v_{\mathfrak{q}}(f_{s_j}) < 1$. At an extension of [Lemma 15.108](#) where the idempotent multiplying s_j has value 0 and the other value 2, the summand of s_j has value below 1 and the other at least 2, so the sum is not in $T\mathcal{B}$. For $\mathbb{C}(\mathbb{Z}a)$ a nonzero class has a branch of value exactly 0, and the sum has value 0, so it is not in J . For the coefficient fibres take representatives in $\mathcal{B}$, whose values are still nonnegative. $\square$

If all representatives of a subring lie in one finite L , one finite exceptional set suffices, whatever the size of the subring; this is used for countable coefficient fields in [Theorem 15.118](#) (23).

Corollary 15.110 (Atomlessness [23]). *The Boolean algebras $\text{Idem}(\mathbb{B}(\mathbb{Z}a))$, $\text{Idem}(\mathbb{C}(\mathbb{Z}a))$, $\text{Idem}(\mathbb{B}^{\text{co}}(\mathbb{Z}a))$ and $\text{Idem}(\mathbb{C}^{\text{co}}(\mathbb{Z}a))$ are atomless.*

Proof. A nonzero idempotent b generates a countable unital ring; [Theorem 15.109](#) gives e with be and $b(1 - e)$ nonzero, disjoint and with sum b . $\square$

Theorem 15.111 (Independent idempotents and cardinal invariants [23]). *The family $(\bar{y}_c)_{c \in \mathbb{R}_{>0}}$ is Boolean independent in each of the four fibres of [Corollary 15.110](#). Their Boolean algebras have cardinality $\mathfrak{c}$ and order density $\mathfrak{c}$: no set of fewer than $\mathfrak{c}$ nonzero idempotents has a member below every nonzero idempotent.*

Proof. For distinct $c_1, \dots, c_r > 0$ the square classes of the $T^2 + c_j$ in $k(T)$ are independent, since a nonempty product has odd order at a prime factor of only one of them; by [Lemma 15.69](#) adjoining the ς_{c_j} gives degree 2^r . So $k[T, \varsigma_{c_1}, \dots, \varsigma_{c_r}]$ is free over $k[T]$ on the square-root monomials (the monic relations span, the degree gives independence), and modulo T it is the product of 2^r copies of k , one for each assignment $\varsigma_{c_j} \mapsto \pm\sqrt{c_j}$. Every such prime extends to $\mathcal{B}$, and the corresponding Boolean word in the y_{c_j} and $1 - y_{c_j}$ has residue 1 there, so it is not in J and is nonzero in all four fibres. As K^{alg} is algebraic over $k(T)$, $|K^{\text{alg}}| \leq |k(T)|\aleph_0 = \mathfrak{c}$, which bounds the fibres. If $\mathcal{D}$ were order dense of size $< \mathfrak{c}$, the ring S it generates has $|S| < \mathfrak{c}$, and a splitter e of S is nonzero while $d(1 - e) \neq 0$ for every $0 \neq d \in \mathcal{D}$, so no $d \leq e$. $\square$

Corollary 15.112 (The number of zero-place branches [23]). *$\mathcal{B}$ has exactly $2^{\mathfrak{c}}$ maximal ideals containing T , hence at least $2^{\mathfrak{c}}$ distinct valuation domains as in [Theorem 15.103](#).*

Proof. For $\eta : \mathbb{R}_{>0} \rightarrow \{0, 1\}$, the ideal generated by T and the elements $y_c - \eta(c)$ is proper, since every finite subset of these relations holds at a prime of the proof of [Theorem 15.111](#); a maximal ideal containing it determines η . There are at most $2^{|\mathcal{B}|} = 2^{\mathfrak{c}}$ subsets of $\mathcal{B}$, and distinct maximal ideals give distinct localizations. $\square$

Remark 15.113 (What these invariants do not determine [23]). An independent family of size $\mathfrak{c}$ does not make the algebra free on it; uniform splitting adds one independent coordinate over a prescribed small subring and does not realize arbitrary infinite Boolean types; neither completeness nor infinite joins follow, and the isomorphism type for $k = \mathbb{R}$ or $\mathbb{C}$ is open ([Question 18.22](#)). The nilradical does not obstruct comparison with the reduced fibre: in a commutative ring idempotents lift uniquely across a nil ideal (if $b^2 - b = t$ is nilpotent, $2b - 1$ is a unit since $(2b - 1)^2 = 1 + 4t$, and $b - t/(2b - 1)$ squares the error up to a unit; two idempotents differing by a nilpotent have nilpotent idempotent cross terms $e(1 - f)$, $f(1 - e)$, hence zero ones). This standard fact is not claimed (23).

Remark 15.114 (The Boolean-power form of the cyclic reduced fibre [merge]). For the real pair [Proposition 15.76](#) gives $C(\mathbb{Z}a) \cong \overline{\mathbb{Z}}[\text{Idem}(C(\mathbb{Z}a))]$, and by [Corollary 15.110](#) and [Theorem 15.111](#) the Boolean parameter is atomless of cardinality and order density $\mathfrak{c}$, with $2^{\mathfrak{c}}$ branches over zero. The same proof of [Proposition 15.76](#) applies verbatim over the coefficient field $k_0 = \overline{\mathbb{Q}} \cap \mathbb{R}$ of [Theorem 15.118](#): it uses normality, divisibility of the ideal by nonzero integers, v_T (whose Laurent coefficients are rational) and the real algebraic integers, all present there. So the reduced fibre of the k_0 -fragment is $\overline{\mathbb{Z}}[\mathfrak{B}]$ with $\mathfrak{B}$ the Cantor algebra, the ring of locally constant functions from Cantor space $\{0, 1\}^{\mathbb{N}}$ to the discrete ring $\overline{\mathbb{Z}}$. By [Remark 15.113](#) the idempotents of the unreduced fibres are the same.

Exact zero divisors and periodic homology.

Theorem 15.115 (Exact annihilators in the coefficient fibre [23]). *Fix $n \geq 2$ and let $\bar{z}_n$ be the residue of z_n in $B^{\text{co}}(\mathbb{Z}a)$. For $1 \leq j < n$, $\text{Ann}_{B^{\text{co}}(\mathbb{Z}a)}(\bar{z}_n^j) = \bar{z}_n^{n-j} B^{\text{co}}(\mathbb{Z}a)$.*

Proof. The right side annihilates since $\bar{z}_n^n = 0$. If $y \in \mathcal{B}$ represents an annihilator, $yz_n^j = Tb$ with $b \in \mathcal{B}$; multiplying by z_n^{n-j} and cancelling T in the domain $\mathcal{B}$ gives $y(1 + T^{n-1}) = bz_n^{n-j}$, and $1 + T^{n-1} \equiv 1$ modulo $T\mathcal{B}$. $\square$

Locally, in $\mathcal{B}_q/T\mathcal{B}_q$, annihilation by $\bar{z}_n^j$ means value at least $(n-j)/n$, which is the principal ideal of $\bar{z}_n^{n-j}$; the cancellation proof needs no local-to-global theorem (23).

Theorem 15.116 (A periodic resolution [23]). *With $B^{\text{co}} = B^{\text{co}}(\mathbb{Z}a)$ and $\mathcal{M}_n = B^{\text{co}}/(\bar{z}_n)$, the complex*

$$\cdots \xrightarrow{\bar{z}_n} B^{\text{co}} \xrightarrow{\bar{z}_n^{n-1}} B^{\text{co}} \xrightarrow{\bar{z}_n} B^{\text{co}} \longrightarrow \mathcal{M}_n \longrightarrow 0, \quad (15.35)$$

with the two maps alternating, is a free resolution, $\text{Tor}_q^{B^{\text{co}}}(\mathcal{M}_n, \mathcal{M}_n) \cong \mathcal{M}_n \neq 0$ for all $q \geq 0$, $\text{pd } \mathcal{M}_n = \text{fd } \mathcal{M}_n = \infty$, and $B^{\text{co}}(\mathbb{Z}a)$ has infinite global and weak global dimension.

Proof. Consecutive maps compose to multiplication by $\bar{z}_n^n = 0$, and [Theorem 15.115](#) (for $j = 1$ and $j = n - 1$) identifies every kernel with the preceding image. Tensoring with $\mathcal{M}_n$ makes every differential zero, so the homology is $\mathcal{M}_n$ in every degree, and $\mathcal{M}_n \neq 0$ because the nilpotent $\bar{z}_n$ is not a unit of the nonzero ring B^{co} . Nonvanishing Tor in all degrees excludes every finite flat, hence every finite projective, resolution. $\square$

Periodicity from a pair of exact zero divisors is classical and related to matrix factorizations [15]; the content here is the explicit pair in these non-Noetherian fibres, proved without a theorem that needs a regular Noetherian local ring (23).

Proposition 15.117 (Arithmetic annihilators [23]). *Under $B(\mathbb{Z}a) \hookrightarrow B^{\text{co}}(\mathbb{Z}a)$, for $1 \leq j < n$,*

$$\text{Ann}_{B(\mathbb{Z}a)}(\bar{z}_n^j) = \bar{z}_n^{n-j} B^{\text{co}}(\mathbb{Z}a) \subseteq B(\mathbb{Z}a), \quad \bar{z}_n^{n-j} B(\mathbb{Z}a) \subsetneq \bar{z}_n^{n-j} B^{\text{co}}(\mathbb{Z}a).$$

So (15.35) with $B(\mathbb{Z}a)$ in place of $B^{\text{co}}(\mathbb{Z}a)$ is not exact.

Proof. For $b \in \mathcal{B}$ and $1 \leq r < n$ every value of $z_n^r b$ over zero is at least $r/n > 0$, so $z_n^r b \in J \subseteq \mathcal{N}(\mathbb{Z}a)$ and $\bar{z}_n^r B^{\text{co}}(\mathbb{Z}a) \subseteq B(\mathbb{Z}a)$; intersect [Theorem 15.115](#) with $B(\mathbb{Z}a)$. If $\bar{z}_n^{n-j}/2 = \bar{z}_n^{n-j} u$ with $u \in B(\mathbb{Z}a)$, then $u - \frac{1}{2}$ kills $\bar{z}_n^{n-j}$ in $B^{\text{co}}(\mathbb{Z}a)$, so lies in $\bar{z}_n^j B^{\text{co}}(\mathbb{Z}a)$, and a representative of u has residue $\frac{1}{2}$ at every place over zero; but $\frac{1}{2}$ is integral over neither $\mathbb{Z}$ nor $\mathbb{Z}[i]$, contrary to [Theorem 15.91](#). $\square$

For $n = 2$: the residue of $z_2/2$ is killed by that of z_2 but is not in the principal arithmetic ideal it generates; the missing coefficient $\frac{1}{2}$ is the obstruction. 23 computes the weak global dimension of $B^{\text{co}}(\mathbb{Z}a)$ and deliberately leaves that of $B(\mathbb{Z}a)$ open (Question 18.24).

A countable fragment with a determined Boolean algebra. Let $k_0 = \overline{\mathbb{Q}} \cap \mathbb{R}$ (as in Lemma 15.35), a real closed field, and $\overline{\mathbb{Q}}$. Repeat the definitions of $K_{\mathbb{Z}a}$, $\mathcal{A}(\mathbb{Z}a)$, $\mathcal{N}(\mathbb{Z}a)$, $\mathcal{B}$, $\mathfrak{l}$ and the four fibres with k replaced by k_0 (and $D = \mathbb{Z}$), respectively by $\overline{\mathbb{Q}}$ (and $D = \mathbb{Z}[i]$). These are smaller coefficient fragments, not the workspaces above; their Laurent fields are still uncountable, but the relative algebraic subfields are countable.

Theorem 15.118 (Algebraic-coefficient fragments [23]). *In either fragment: (i) $\mathcal{N}(\mathbb{Z}a)$ and $\mathcal{B}$ are countable, $\mathfrak{l} = T\mathcal{B}$, and Theorems 15.86 and 15.99 hold; (ii) every localization of $\mathcal{B}$ at a maximal ideal containing T has value group $\mathbb{Q}$ and residue field $\overline{\mathbb{Q}}$; (iii) the Boolean algebras of idempotents of the four fibres are each isomorphic to the algebra of clopen subsets of Cantor space $\{0, 1\}^{\mathbb{N}}$, the countable atomless Boolean algebra.*

Proof. There are countably many polynomials over the countable field of rational functions, each with finitely many roots, so the relative algebraic closure is countable, and it contains $\mathcal{N}(\mathbb{Z}a)$ and $\mathcal{B}$. The normalization and residue arguments use characteristic 0 and, for the coefficient span, that finite residue extensions are the field itself or its extension by i (real case) and the field itself (algebraically closed case); the basis vectors with entries $0, 1, i$ are integral over D . So $\mathfrak{l} = T\mathcal{B}$ and the criterion hold. z_n and all refinement certificates have rational coefficients; z_n supplies the values $1/n$ and ϱ_0 a square root of -1 , giving the residue field $k_0(i) = \overline{\mathbb{Q}}$, respectively $\overline{\mathbb{Q}}$. To split a nonzero idempotent put a representative in a finite L and restrict the parameters of Lemma 15.108 to positive rational squares, of which only finitely many are excluded. So all four algebras are countable and atomless. Two countable atomless Boolean algebras are isomorphic: an isomorphism of finite subalgebras extends to one more prescribed element, since atomlessness allows exactly the needed splits of atoms on the other side, and alternating in both directions along enumerations gives an isomorphism. The finite unions of cylinder sets of $\{0, 1\}^{\mathbb{N}}$ form a countable atomless algebra, and by compactness they are all the clopen sets. $\square$

The uniqueness of the countable atomless Boolean algebra is classical; the application is to these fibres, and the uncountable algebras of the full-coefficient workspaces are not classified (Question 18.22). The fragment is a concrete countable model in which nilpotence, refinement and Boolean splitting coexist and can be approached computationally (23).

Comparison with the full class.

Remark 15.119 (What larger scales kill [merge]). By Theorem 15.57 and Theorem 15.93, $\Pi\mathcal{N} \cap K_{\mathbb{Z}a} = \sqrt{\mathfrak{l}} = \mathfrak{J}$ for the real pair. So the natural map $B(\mathbb{Z}a) \rightarrow B_{\mathbb{R}}$ has kernel exactly the nilradical $\mathfrak{J}/T\mathcal{B}$, nonzero by Theorem 15.86, and the radical in the second equality of (15.22) cannot be removed for $G = \mathbb{Z}a$; the reduced fibre $C(\mathbb{Z}a)$ embeds (Corollary 15.58). The comparison of ideals is: $\Pi\mathcal{N}$ is radical with $\Pi\mathcal{N} = (\Pi\mathcal{N})^2$ (Theorem 15.29 and Proposition 15.31); $\mathfrak{l}$ is not radical and $\mathfrak{l}/\mathfrak{l}^2 \neq 0$ (Theorems 15.86 and 15.106), while its radical $\mathfrak{J}$ satisfies $\mathfrak{J}^2 = \mathfrak{J}$ (Theorem 15.105); and $\mathfrak{l}_{\text{Pui}}$ is radical (Theorem 15.100). So the conclusion that Theorem 15.29 proves for the full class fails in a fixed workspace; it does not contradict Theorem 15.51, whose hypotheses the cyclic workspace fails (Remark 15.102), and it says nothing about real closed fixed workspaces such as $K_{\mathbb{Q}a}$ (Question 18.21).

23 records the resolution as follows: for every $G = \mathbb{Z}a$ the extended ideal is not radical and its fibre has nilpotents of every order, which is not a classification for arbitrary ordered groups or support classes; the cyclic fibre is atomless with exact cardinality and order density $\mathfrak{c}$, without its isomorphism type; the residue criterion is a finite-place test for algebraic Laurent elements, not an oracle for arbitrary coefficients or Hahn input; in every cyclic workspace $\mathfrak{l} \subsetneq \sqrt{\mathfrak{l}}$, so removing the radical from [Theorem 15.57](#) would give a stronger, false statement there, and that theorem is compared with, not used by, the cyclic proofs; the bounded-denominator Puiseux normalization has radical extended ideal, and the full rational-exponent Hahn field is left open. Its dependencies are: finite normalization gives $\mathbf{z}_n$ and the exact refinement; scaling and the residue criterion give $k\mathcal{N}(\mathbb{Z}a) = \mathcal{B}$, hence $\mathfrak{l} = T\mathcal{B}$, hence the filtration and the general refinement law; the finite ramification locus gives the fresh quadratic splitting and the Boolean results; and $\mathfrak{l} = T\mathcal{B}$ gives the exact annihilators and the periodic homology over $\mathbf{B}^{\text{co}}(\mathbb{Z}a)$, with deliberately no arrow to $\mathbf{B}(\mathbb{Z}a)$, which [Proposition 15.117](#) disproves. All rings and modules in these proofs are sets; the surreal a only names the exponent group; composita, primes, localizations and branches are taken in set-sized rings and fields, with ordinary choice (the maximal-ideal theorem) and no class Zorn lemma, class module category or measurable cardinal. Neither the continuum hypothesis nor regularity of $\mathfrak{c}$ is used. The Newton–Puiseux theorem is used only to identify the limit field as real closed or algebraically closed; the reducedness of its fibre does not need it (23).

16 Arithmetic of polynomial maps on the omnific lattices

This section is manuscript 10’s arithmetic of finite-degree polynomial and rational maps on the real lattice $\mathbf{Oz} \subset \mathbf{No}$ and the Gaussian lattice $\mathbf{Oz}[i] \subset \mathbf{No}[i]$, followed by manuscript 16’s fresh-scale results for $D + \Pi_k$ with any unital constant ring D ([Sections 16.6](#) and [16.7](#)) and by manuscript 18’s ordinary-output rigidity ([Section 16.8](#)), with manuscript 22’s single-scale certificates at the end ([Section 16.10](#)). It uses from the core only the universal theorem [Theorem 5.1](#), the quotient list [Theorem 6.2](#) with the identity (3.2), the constant-term lemma [Lemma 2.2](#), the clearing of supports [Proposition 3.4\(i\)](#) and the unit lists of [Lemma 2.4](#).

Renamings. Manuscript 10’s symbols are renamed as follows. Its ideals $\mathfrak{P}$ and $\mathfrak{P}_{\mathbb{C}}$ are Π and $\Pi_{\mathbb{C}}$; its constant-term map π is ct ; its projection Π on polynomial rings (the *shadow*) is ct_* , the coefficientwise constant term; its Gaussian lattice is $\mathbf{Oz}[i]$ inside $\mathbf{No}[i]$; its ring $R(D, K)$ is $\mathcal{A}_{D,K}$; its binomial polynomials $B_k(X)$ are written $\binom{X}{k}$; its $L_k = \text{lcm}(1, \dots, k)$ is written lcm_k ; and its rings S_r and I_r are written $\mathcal{N}_r(\mathbf{Oz})$ and $\text{Int}(\mathbb{Z}^r)$ (the letter I_r would collide with ideals). The numbered results keep 10’s content.

Notation. Throughout, $r \geq 1$ is an ordinary integer, $\mathbf{X} = (X_1, \dots, X_r)$, and bold letters denote r -tuples. For an ordinary $k \geq 0$,

$$\binom{X}{k} = \frac{X(X-1) \cdots (X-k+1)}{k!}, \quad \binom{X}{0} = 1,$$

and for a multi-index $\mathbf{k} = (k_1, \dots, k_r)$ we put $\binom{\mathbf{X}}{\mathbf{k}} = \prod_{j=1}^r \binom{X_j}{k_j}$. We write $\Delta_j F(\mathbf{X}) = F(\mathbf{X} + \mathbf{e}_j) - F(\mathbf{X})$, and $\Delta = \Delta_1$ when $r = 1$. Put $\text{lcm}_0 = 1$ and $\text{lcm}_k = \text{lcm}(1, \dots, k)$ for $k \geq 1$. Recall $\mathcal{A}_{\mathbb{Q}, \mathbb{R}} = \mathbb{Q} + \Pi$ and $\mathcal{A}_{\mathbb{Q}(i), \mathbb{C}} = \mathbb{Q}(i) + \Pi_{\mathbb{C}}$; by [Lemma 2.2](#) the constant term ct is a ring homomorphism on both, with kernels Π and $\Pi_{\mathbb{C}}$, so Π is an ideal of $\mathbb{Q} + \Pi$ and $\Pi_{\mathbb{C}}$ is an ideal of $\mathbb{Q}(i) + \Pi_{\mathbb{C}}$. Its

coefficientwise extension

$$\text{ct}_* : (\mathbb{Q} + \Pi)[\mathbf{X}] \rightarrow \mathbb{Q}[\mathbf{X}], \quad \text{ct}_* : (\mathbb{Q}(i) + \Pi_{\mathbb{C}})[\mathbf{X}] \rightarrow \mathbb{Q}(i)[\mathbf{X}]$$

is therefore a ring homomorphism with kernel $\Pi[\mathbf{X}]$, respectively $\Pi_{\mathbb{C}}[\mathbf{X}]$. Define

$$\begin{aligned} \mathcal{N}_r(\mathbf{Oz}) &= \{F \in \mathbf{No}[\mathbf{X}] : F(\mathbf{Oz}^r) \subseteq \mathbf{Oz}\}, & \text{Int}(\mathbb{Z}^r) &= \{f \in \mathbb{Q}[\mathbf{X}] : f(\mathbb{Z}^r) \subseteq \mathbb{Z}\}, \\ \mathcal{N}_r(\mathbf{Oz}[i]) &= \{F \in \mathbf{No}[i][\mathbf{X}] : F(\mathbf{Oz}[i]^r) \subseteq \mathbf{Oz}[i]\}, & \text{Int}(\mathbb{Z}[i]^r) &= \{f \in \mathbb{Q}(i)[\mathbf{X}] : f(\mathbb{Z}[i]^r) \subseteq \mathbb{Z}[i]\}. \end{aligned}$$

These are rings under pointwise operations. All degrees, finite sums and factorials are ordinary.

16.1 Lattice separation and an omitted congruence type

Lemma 16.1 (Lattice separation and cofinality [10]). *For $\Lambda = \mathbf{Oz}$ and for $\Lambda = \mathbf{Oz}[i]$, every nonzero $x \in \Lambda$ satisfies $|x| \geq 1$. The finite elements of $\mathbf{Oz}$ are exactly the ordinary integers. The positive elements of $\mathbf{Oz}$ are cofinal in $\mathbf{No}$. The units are $\mathbf{Oz}^\times = \{\pm 1\}$ and $\mathbf{Oz}[i]^\times = \{\pm 1, \pm i\}$ (Lemma 2.4).*

Proof. A nonzero real purely infinite series has a positive leading exponent, so its absolute value exceeds every positive real. Hence $n + u$ with $n \in \mathbb{Z}$ and $0 \neq u \in \Pi$ is infinite, the finite elements of $\mathbf{Oz}$ are the ordinary integers, and every nonzero element of $\mathbf{Oz}$ has absolute value at least 1. In the Gaussian case write $x = a + ib$ with $a, b \in \mathbf{Oz}$, not both zero; the surcomplex modulus dominates either coordinate, $|x| = \sqrt{a^2 + b^2} \geq \max(|a|, |b|) \geq 1$.

Given $y \in \mathbf{No}$, choose an exponent $g > 0$ larger than the leading exponent of y (any $g > 0$ if $y = 0$). Then $\omega^g \in \mathbf{Oz}$ is positive and exceeds $n|y|$ for every ordinary n , in particular $\omega^g > |y|$. This proves cofinality. The unit lists are Lemma 2.4. $\square$

Remark 16.2 (Modulus and standard part [10]). The surcomplex modulus needs only real closedness of $\mathbf{No}$: $|a + ib| = \sqrt{a^2 + b^2}$ for $a, b \in \mathbf{No}$, and it is multiplicative and satisfies the triangle inequality. The standard part of a finite surcomplex number is taken coordinatewise in $\mathbb{C}$; on the finite rings $\mathcal{O}_{\mathbb{R}}$ and $\mathcal{O}_{\mathbb{C}}$ it is the map st of Lemma 2.2.

Proposition 16.3 (A finitely satisfiable countable congruence type [10]). *Fix an ordinary prime p and $\alpha \in \mathbb{Z}_p \setminus \mathbb{Z}$. For $e \geq 1$ let a_e be the representative of $\alpha \bmod p^e$ in $\{0, \dots, p^e - 1\}$. The countable family of ring formulas*

$$\exists y (x - a_e = p^e y), \quad e = 1, 2, \dots, \quad (16.1)$$

is finitely satisfiable in $\mathbb{Z} \subset \mathbf{Oz}$ but has no realization in $\mathbf{Oz}$. An explicit choice is

$$\alpha = \sum_{j \geq 0} p^{2^j} \in \mathbb{Z}_p. \quad (16.2)$$

Proof. For a finite subfamily let E be its largest index and take $x = a_E$. Since the residues are compatible, $a_E \equiv a_e \pmod{p^e}$ for $e \leq E$, which gives witnesses $y \in \mathbb{Z}$ for every formula of the subfamily. If $x \in \mathbf{Oz}$ realized all formulas, applying the ring homomorphism ct would give an ordinary integer $\text{ct}(x)$ with $\text{ct}(x) \equiv a_e \equiv \alpha \pmod{p^e}$ for every e (by (3.2), $x - a_e \in p^e \mathbf{Oz}$ means $\text{ct}(x) - a_e \in p^e \mathbb{Z}$). In $\mathbb{Z}_p$ this forces $\text{ct}(x) = \alpha$, contradicting $\alpha \notin \mathbb{Z}$.

For (16.2): the powers p^{2^j} are distinct, so the base- p digits of α are 1 at the positions 2^j and 0 elsewhere. They are not eventually zero, so α is not a nonnegative integer; and they are not eventually $p - 1$ (for $p = 2$ the positions $2^j + 1 < 2^{j+1}$ with $j \geq 1$ carry the digit 0), so α is not a negative integer. $\square$

Remark 16.4 (Ring types, not saturation [10]). This elementary omitted type concerns the ring structure. It shows that the order-theoretic richness of the full surreal class does not give a corresponding realization property for countable ring types. No use of the term “saturated model” for a proper-class structure is needed, and none is made.

16.2 Polynomials preserving the omnific lattices

Lemma 16.5 (Omnific binomial closure [10; 22]). *For every $x \in \mathbf{Oz}$ and every ordinary $k \geq 0$, $\binom{x}{k} \in \mathbf{Oz}$. More generally, if $f \in \mathbb{Q}[\mathbf{X}]$ and $f(\mathbb{Z}^r) \subseteq \mathbb{Z}$, then $f(\mathbf{Oz}^r) \subseteq \mathbf{Oz}$.*

Proof. Write $x_j = n_j + u_j$ with $n_j \in \mathbb{Z}$ and $u_j \in \Pi$. Expanding each monomial of f shows that $f(\mathbf{n} + \mathbf{u}) - f(\mathbf{n})$ is a finite sum of terms, each a rational multiple of a product of ordinary integers and of powers of the u_j containing at least one factor u_j . Since Π is an ideal of $\mathbb{Q} + \Pi$, every such term lies in Π ; thus

$$f(\mathbf{n} + \mathbf{u}) - f(\mathbf{n}) \in \Pi.$$

As $f(\mathbf{n}) \in \mathbb{Z}$, we get $f(\mathbf{x}) \in \mathbb{Z} + \Pi = \mathbf{Oz}$. The binomial case is the case $f = \binom{X}{k}$, which takes integer values at all ordinary integers, including negative ones: $\binom{-n}{k} = (-1)^k \binom{n+k-1}{k}$. $\square$

Thus $\mathbf{Oz}$ is a binomial ring in the usual torsion-free sense: it is $\mathbb{Z}$ -torsion free and closed under $x \mapsto \binom{x}{k}$ [16]. The terminology is classical; the support argument above explains its application here without importing an abstract free-ring theorem.

Theorem 16.6 (Exact finite-grid test [10; 16; 22]). *Let $F \in \mathbf{No}[\mathbf{X}]$ with $\deg_{X_j} F \leq d_j$ for $j = 1, \dots, r$. The following are equivalent:*

- (i) $F(\mathbf{Oz}^r) \subseteq \mathbf{Oz}$;
- (ii) $F(n_1, \dots, n_r) \in \mathbf{Oz}$ for all ordinary integers $0 \leq n_j \leq d_j$;
- (iii) in the unique expansion

$$F(\mathbf{X}) = \sum_{\mathbf{k} \leq \mathbf{d}} c_{\mathbf{k}} \binom{\mathbf{X}}{\mathbf{k}} \quad (16.3)$$

all coefficients $c_{\mathbf{k}}$ belong to $\mathbf{Oz}$.

Explicitly,

$$c_{\mathbf{k}} = (\Delta_1^{k_1} \cdots \Delta_r^{k_r} F)(\mathbf{0}) = \sum_{\ell \leq \mathbf{k}} (-1)^{\sum_j (k_j - \ell_j)} \left(\prod_{j=1}^r \binom{k_j}{\ell_j} \right) F(\ell). \quad (16.4)$$

Proof. The polynomials $\binom{X}{0}, \dots, \binom{X}{d}$ form a basis of the univariate polynomials of degree at most d over the field $\mathbf{No}$: their degrees are $0, \dots, d$ and their leading coefficients are nonzero. Taking products, the $\binom{\mathbf{X}}{\mathbf{k}}$ with $\mathbf{k} \leq \mathbf{d}$ form a basis of the polynomials with $\deg_{X_j} \leq d_j$, which gives existence and uniqueness of (16.3). Since $\Delta_m^{(X)} = \binom{X}{m-1}$ for $m \geq 1$ and $\Delta_0^{(X)} = 0$, one has $\Delta_j^{k_j} \binom{X_j}{m_j} = \binom{X_j}{m_j - k_j}$ for $m_j \geq k_j$ and 0 otherwise; as $\binom{0}{m} = 0$ for $m > 0$, applying $\Delta_1^{k_1} \cdots \Delta_r^{k_r}$ to (16.3) and evaluating at $\mathbf{0}$ leaves exactly $c_{\mathbf{k}}$. Writing $\Delta_j = \tau_j - 1$ with the shift τ_j and expanding by the binomial theorem gives the last expression in (16.4).

(i) $\Rightarrow$ (ii) is restriction to ordinary grid points. Under (ii), the last expression in (16.4) is an integral linear combination of grid values in $\mathbf{Oz}$, which proves (iii). Under (iii), each factor $\binom{x_j}{k_j}$ with $x_j \in \mathbf{Oz}$ lies in $\mathbf{Oz}$ by Lemma 16.5, and evaluating (16.3) proves (i). $\square$

The test is finite for a given finite-degree polynomial, although the preservation property quantifies over the entire class $\mathbf{Oz}^r$. It is a mathematical test with exact surreal input; no claim is made that arbitrary surreal coefficients have an effective finite encoding.

Theorem 16.7 (The real numerical-polynomial ring [10; 16; 18; 22]). *Every $F \in \mathcal{N}_r(\mathbf{Oz})$ has a unique decomposition*

$$F = U + f, \quad U \in \Pi[\mathbf{X}], \quad f \in \text{Int}(\mathbb{Z}^r), \quad (16.5)$$

so that $\mathcal{N}_r(\mathbf{Oz}) = \Pi[\mathbf{X}] \oplus \text{Int}(\mathbb{Z}^r)$ additively. The coefficientwise constant term $\text{ct}_*(F) = f$ is a surjective ring homomorphism $\mathcal{N}_r(\mathbf{Oz}) \rightarrow \text{Int}(\mathbb{Z}^r)$ with kernel $\Pi[\mathbf{X}]$, split by the inclusion $\text{Int}(\mathbb{Z}^r) \subseteq \mathcal{N}_r(\mathbf{Oz})$. For $\mathbf{x} \in \mathbf{Oz}^r$,

$$\text{ct}(F(\mathbf{x})) = \text{ct}_*(F)(\text{ct}(x_1), \dots, \text{ct}(x_r)). \quad (16.6)$$

Proof. Let $F \in \mathcal{N}_r(\mathbf{Oz})$. By Theorem 16.6 its Newton coefficients $c_{\mathbf{k}}$ lie in $\mathbf{Oz}$; write $c_{\mathbf{k}} = m_{\mathbf{k}} + u_{\mathbf{k}}$ with $m_{\mathbf{k}} \in \mathbb{Z}$ and $u_{\mathbf{k}} \in \Pi$, and put

$$U = \sum_{\mathbf{k}} u_{\mathbf{k}} \binom{\mathbf{X}}{\mathbf{k}}, \quad f = \sum_{\mathbf{k}} m_{\mathbf{k}} \binom{\mathbf{X}}{\mathbf{k}}.$$

The binomials have rational coefficients and Π is a $\mathbb{Q}$ -vector space, so $U \in \Pi[\mathbf{X}]$; products of binomials are integer valued on $\mathbb{Z}^r$, so $f \in \text{Int}(\mathbb{Z}^r)$. Conversely, a polynomial in $\Pi[\mathbf{X}]$ takes values in Π on $\mathbf{Oz}^r$, because Π is an ideal of $\mathbb{Q} + \Pi \supseteq \mathbf{Oz}$, and $\text{Int}(\mathbb{Z}^r) \subseteq \mathcal{N}_r(\mathbf{Oz})$ by Lemma 16.5; hence $\Pi[\mathbf{X}] + \text{Int}(\mathbb{Z}^r) \subseteq \mathcal{N}_r(\mathbf{Oz})$. The sum is direct because a coefficient that is both rational and purely infinite is zero.

By (16.5) all coefficients of F lie in $\mathbb{Q} + \Pi$, so $\mathcal{N}_r(\mathbf{Oz})$ is a subring of $(\mathbb{Q} + \Pi)[\mathbf{X}]$ and ct_* restricts to a ring homomorphism on it. It sends U to 0 and fixes f , so its restriction is the projection onto the second summand; its kernel is $\Pi[\mathbf{X}]$ and the inclusion of $\text{Int}(\mathbb{Z}^r)$ is a ring section. Finally write $x_j = n_j + u_j$ with $n_j = \text{ct}(x_j)$. Then $U(\mathbf{x}) \in \Pi$ and $f(\mathbf{x}) - f(\mathbf{n}) \in \Pi$ by the proof of Lemma 16.5, while $f(\mathbf{n}) \in \mathbb{Z}$. Hence $\text{ct}(F(\mathbf{x})) = f(\mathbf{n})$, which is (16.6). $\square$

The decomposition is an additive direct sum whose first summand is an ideal; it is *not* a direct product of rings.

Example 16.8 ([10]). The polynomial

$$F(X) = 3 \binom{X}{2} + \omega^{\sqrt{2}} X^3 + \left(\sum_{m \geq 1} \omega^{1/m} \right) (X - 1)$$

is omnific-integer valued and has shadow $\text{ct}_*(F) = 3 \binom{X}{2}$. By contrast $\omega^{-1}X$ fails already at $X = 1$. No negative-exponent coefficient can be hidden in a finite-degree lattice-preserving polynomial: by Theorem 16.7 every coefficient of an element of $\mathcal{N}_r(\mathbf{Oz})$ lies in $\mathbb{Q} + \Pi$.

Theorem 16.9 (Gaussian numerical polynomials [10; 16; 18; 22]). *For every ordinary $r \geq 1$,*

$$\mathcal{N}_r(\mathbf{Oz}[i]) = \Pi_{\mathbb{C}}[\mathbf{X}] \oplus \text{Int}(\mathbb{Z}[i]^r) \quad (16.7)$$

additively. The coefficientwise constant term ct_* is a surjective ring homomorphism $\mathcal{N}_r(\mathbf{Oz}[i]) \rightarrow \text{Int}(\mathbb{Z}[i]^r)$ with kernel $\Pi_{\mathbb{C}}[\mathbf{X}]$, split by inclusion, and for $\mathbf{x} \in \mathbf{Oz}[i]^r$, with $\text{ct} : \mathbf{Oz}[i] \rightarrow \mathbb{Z}[i]$ the Gaussian constant term,

$$\text{ct}(F(\mathbf{x})) = \text{ct}_*(F)(\text{ct}(x_1), \dots, \text{ct}(x_r)). \quad (16.8)$$

Proof. Let $F \in \mathcal{N}_r(\mathbf{Oz}[i])$ with $\deg_{X_j} F \leq d_j$. The Newton expansion (16.3) and formula (16.4) hold over the field $\mathbf{No}[i]$ by the same basis argument. The ordinary grid points lie in $\mathbf{Oz}[i]^r$, so (16.4) shows $c_{\mathbf{k}} \in \mathbf{Oz}[i] = \mathbb{Z}[i] + \Pi_{\mathbb{C}}$. Expanding the binomials, each monomial coefficient of F is a $\mathbb{Q}$ -linear combination of the $c_{\mathbf{k}}$, hence lies in $\mathbb{Q}(i) + \Pi_{\mathbb{C}}$. Put $f = \text{ct}_*(F) \in \mathbb{Q}(i)[\mathbf{X}]$ and $U = F - f \in \Pi_{\mathbb{C}}[\mathbf{X}]$. For an ordinary Gaussian tuple $\mathbf{z} \in \mathbb{Z}[i]^r$ we have $U(\mathbf{z}) \in \Pi_{\mathbb{C}}$, because $\Pi_{\mathbb{C}}$ is an ideal of $\mathbb{Q}(i) + \Pi_{\mathbb{C}}$, and $F(\mathbf{z}) = w + v$ with $w \in \mathbb{Z}[i]$, $v \in \Pi_{\mathbb{C}}$. Then $f(\mathbf{z}) - w = v - U(\mathbf{z})$ lies in $\mathbb{Q}(i) \cap \Pi_{\mathbb{C}} = 0$, so $f(\mathbf{z}) = w \in \mathbb{Z}[i]$. Thus $f \in \text{Int}(\mathbb{Z}[i]^r)$.

Conversely write a Gaussian omnific input as $\mathbf{z} + \mathbf{u}$ with $\mathbf{z} \in \mathbb{Z}[i]^r$ and $u_j \in \Pi_{\mathbb{C}}$. For $f \in \text{Int}(\mathbb{Z}[i]^r)$ the expansion argument of Lemma 16.5 gives $f(\mathbf{z} + \mathbf{u}) - f(\mathbf{z}) \in \Pi_{\mathbb{C}}$, so $f(\mathbf{z} + \mathbf{u}) \in \mathbb{Z}[i] + \Pi_{\mathbb{C}} = \mathbf{Oz}[i]$; and a polynomial in $\Pi_{\mathbb{C}}[\mathbf{X}]$ takes values in $\Pi_{\mathbb{C}}$ on $\mathbf{Oz}[i]^r$. This proves (16.7); the sum is direct since $\mathbb{Q}(i) \cap \Pi_{\mathbb{C}} = 0$. As in Theorem 16.7, $\mathcal{N}_r(\mathbf{Oz}[i])$ is a subring of $(\mathbb{Q}(i) + \Pi_{\mathbb{C}})[\mathbf{X}]$, on which ct_* is a ring homomorphism; this gives multiplicativity, the kernel and the section. The same computation, $\text{ct}(F(\mathbf{z} + \mathbf{u})) = f(\mathbf{z})$, proves (16.8). $\square$

Remark 16.10 (Why the real Newton test cannot simply be copied [10]). The ordinary binomial polynomials need not be Gaussian-integer valued:

$$\binom{i}{2} = \frac{i(i-1)}{2} = \frac{-1-i}{2} \notin \mathbb{Z}[i].$$

Consequently values on an ordinary integer grid do not by themselves certify preservation of $\mathbf{Oz}[i]$ (the grid values are necessary, not sufficient). The Gaussian theorem retains the ordinary ring $\text{Int}(\mathbb{Z}[i]^r)$ rather than imposing a false binomial basis on it; the arithmetic of $\text{Int}(\mathbb{Z}[i]^r)$ is not replaced by the simpler real Newton basis.

Theorem 16.11 (Small-target reflection for numerical polynomials [10]). *Every class ring homomorphism φ from $\mathcal{N}_r(\mathbf{Oz})$ to a set-sized ring A factors uniquely through*

$$\text{ct}_* : \mathcal{N}_r(\mathbf{Oz}) \longrightarrow \text{Int}(\mathbb{Z}^r).$$

The analogous assertion holds for $\mathcal{N}_r(\mathbf{Oz}[i])$ with universal image $\text{Int}(\mathbb{Z}[i]^r)$. Therefore the set-sized quotient presentations of either numerical-polynomial ring are precisely the quotient presentations of its ordinary shadow pulled back along ct_ : their kernels are the class ideals $\text{ct}_*^{-1}(\mathfrak{a})$ for ideals $\mathfrak{a}$ of $\text{Int}(\mathbb{Z}^r)$, respectively $\text{Int}(\mathbb{Z}[i]^r)$, with quotients $\text{Int}(\mathbb{Z}^r)/\mathfrak{a}$, respectively $\text{Int}(\mathbb{Z}[i]^r)/\mathfrak{a}$.*

Proof. The restriction of φ to the ideal Π of constant polynomials is a (possibly nonunital) class ring homomorphism $\Pi \rightarrow A$, hence zero by Theorem 5.1. Every element of $\Pi[\mathbf{X}]$ is a finite sum of products $u \mathbf{X}^{\mathbf{m}}$ with $u \in \Pi$ and $\mathbf{X}^{\mathbf{m}} \in \mathcal{N}_r(\mathbf{Oz})$, so $\varphi(u \mathbf{X}^{\mathbf{m}}) = \varphi(u) \varphi(\mathbf{X}^{\mathbf{m}}) = 0$ and φ kills $\Pi[\mathbf{X}] = \ker \text{ct}_*$ (Theorem 16.7). Hence $\varphi = \psi \circ \text{ct}_*$ with ψ the restriction of φ to the section $\text{Int}(\mathbb{Z}^r)$; ψ is unique because ct_* is surjective. The Gaussian case is identical, with $\Pi_{\mathbb{C}}$ and Theorem 16.9.

If φ is surjective onto a set-sized ring with kernel J , then ψ is surjective and $J = \text{ct}_*^{-1}(\ker \psi)$. Conversely, for an ideal $\mathfrak{a}$ of the shadow ring, the composite of ct_* with the quotient map onto the set $\text{Int}(\mathbb{Z}^r)/\mathfrak{a}$ is a set-sized quotient presentation with kernel $\text{ct}_*^{-1}(\mathfrak{a})$. This is the kernel argument of Theorem 6.2. $\square$

16.3 Congruences modulo every omnific integer

Definition 16.12 ([10]). A function $F : \mathbf{Oz} \rightarrow \mathbf{Oz}$ is *congruence preserving* if $a - b$ divides $F(a) - F(b)$ in $\mathbf{Oz}$ for all $a, b \in \mathbf{Oz}$; equivalently, for every $m \in \mathbf{Oz}$, $a \equiv b \pmod{m\mathbf{Oz}}$ implies

$F(a) \equiv F(b) \pmod{m\mathbf{Oz}}$. We write $\text{CP}(\mathbf{Oz})$ for the congruence-preserving elements of $\mathcal{N}_1(\mathbf{Oz})$ and $\text{CP}(\mathbb{Z})$ for the elements $f \in \text{Int}(\mathbb{Z})$ such that $a - b$ divides $f(a) - f(b)$ in $\mathbb{Z}$ for all $a, b \in \mathbb{Z}$.

The two formulations agree: the forward implication is transitivity of divisibility, and the reverse one takes $m = a - b$. When $a = b$ the condition reads $0 \mid 0$, which holds. Differences and finite sums of congruence-preserving functions are congruence preserving.

Lemma 16.13 (Integral divided differences of weighted binomials [10]). *For every ordinary $k \geq 1$, the identity*

$$\frac{\binom{X+T}{k} - \binom{X}{k}}{T} = \sum_{j=1}^k \binom{X}{k-j} \frac{1}{j} \binom{T-1}{j-1} \quad (16.9)$$

holds in $\mathbb{Q}[X, T]$. In particular $\text{lcm}_k \binom{X}{k}$ is congruence preserving on $\mathbf{Oz}$.

Proof. Vandermonde's identity gives $\binom{X+T}{k} = \sum_{j=0}^k \binom{X}{k-j} \binom{T}{j}$. For $j \geq 1$ the falling-factorial formula gives $\binom{T}{j} = \frac{T}{j} \binom{T-1}{j-1}$. Subtracting the term $j = 0$ and cancelling the polynomial factor T gives (16.9).

After multiplication by lcm_k , every scalar lcm_k / j with $1 \leq j \leq k$ is an ordinary integer, so by Lemma 16.5 the right side evaluates into $\mathbf{Oz}$ for all $X, T \in \mathbf{Oz}$. Taking $X = b$ and $T = a - b$ gives $\text{lcm}_k \binom{a}{k} - \text{lcm}_k \binom{b}{k} = (a - b)w$ with $w \in \mathbf{Oz}$, including for nonzero infinite $a - b$; the case $a = b$ is trivial. $\square$

Theorem 16.14 (All-modulus Newton criterion [10]). *Let $F \in \mathcal{N}_1(\mathbf{Oz})$ have Newton expansion $F = \sum_{k=0}^d c_k \binom{X}{k}$, $c_k \in \mathbf{Oz}$. The following are equivalent:*

- (i) F is congruence preserving on $\mathbf{Oz}$;
- (ii) $c_k \in \text{lcm}_k \mathbf{Oz}$ for every $0 \leq k \leq d$;
- (iii) $\text{ct}(c_k) \in \text{lcm}_k \mathbb{Z}$ for every $0 \leq k \leq d$;
- (iv) the ordinary numerical polynomial $\text{ct}_*(F)$ is congruence preserving on $\mathbb{Z}$.

Consequently, within the respective numerical-polynomial rings,

$$\text{CP}(\mathbf{Oz}) = \Pi[X] \oplus \text{CP}(\mathbb{Z}). \quad (16.10)$$

Proof. (ii) $\Rightarrow$ (i). Write $c_k = \text{lcm}_k d_k$ with $d_k \in \mathbf{Oz}$. Each summand $d_k \text{lcm}_k \binom{X}{k}$ is congruence preserving by Lemma 16.13 (multiply the divisibility by d_k), and a finite sum of congruence-preserving functions is congruence preserving.

(i) $\Rightarrow$ (ii). We show $c_k \in \text{lcm}_k \mathbf{Oz}$ by induction on k . For $k = 0$ it is automatic since $\text{lcm}_0 = 1$. Suppose it holds for all $j < k$. Then $\sum_{j < k} c_j \binom{X}{j}$ is congruence preserving by the previous paragraph, so

$$G(X) = F(X) - \sum_{j < k} c_j \binom{X}{j} = \sum_{j \geq k} c_j \binom{X}{j}$$

is congruence preserving. Since $\binom{n}{j} = 0$ for ordinary $0 \leq n < j$ and $\binom{k}{k} = 1$, we get $G(0) = \dots = G(k-1) = 0$ and $G(k) = c_k$. For each $1 \leq m \leq k$, comparing the inputs k and $k - m$ gives $m \mid G(k) - G(k - m) = c_k$ in $\mathbf{Oz}$. By (3.2), the ordinary integer $\text{ct}(c_k)$ is divisible by every $m \leq k$, hence by lcm_k , and again by (3.2), $c_k \in \text{lcm}_k \mathbf{Oz}$.

(ii) $\Leftrightarrow$ (iii) is (3.2). For (iii) $\Leftrightarrow$ (iv), note that $\text{ct}_*(F) = \sum_k \text{ct}(c_k) \binom{X}{k}$ is the Newton expansion of $\text{ct}_*(F)$, since the binomials have rational coefficients. The same two arguments over $\mathbb{Z}$ (the identity

(16.9) with $X, T \in \mathbb{Z}$, and the same induction with divisibility in $\mathbb{Z}$ show that an $f \in \text{Int}(\mathbb{Z})$ with Newton coefficients $m_k \in \mathbb{Z}$ is congruence preserving on $\mathbb{Z}$ if and only if $m_k \in \text{lcm}_k \mathbb{Z}$ for all k . Applied to $m_k = \text{ct}(c_k)$, this is (iii) $\Leftrightarrow$ (iv).

For (16.10), decompose $F = U + f$ by Theorem 16.7. The Newton coefficients of $U \in \Pi[X]$ are integral combinations of values $U(\ell) \in \Pi$, and $\Pi \subseteq \text{lcm}_k \mathbf{Oz}$ by (3.2); so $U \in \text{CP}(\mathbf{Oz})$ by (ii). An $f \in \text{Int}(\mathbb{Z})$ lies in $\text{CP}(\mathbf{Oz})$ exactly when it lies in $\text{CP}(\mathbb{Z})$, by (i) $\Leftrightarrow$ (iv) applied to $f = \text{ct}_*(f)$. Since $\text{ct}_*(F) = f$, the equivalence (i) $\Leftrightarrow$ (iv) for F now gives $F \in \text{CP}(\mathbf{Oz})$ if and only if $f \in \text{CP}(\mathbb{Z})$. $\square$

The divisibility condition in (i) includes infinite differences and therefore every infinite omnific modulus; no extra coefficient condition appears at those moduli. The ordinary least-common-multiple pattern is classical, and related finite-ring formulations are proved in [9]; Newton bases and integer-valued polynomials are treated in [8]. The content specific to the present domain is that the pattern suffices for differences at every surreal scale, not only for congruences modulo ordinary integers.

Example 16.15 (Two kinds of denominators [10]). The polynomial $12\binom{X}{4}$ is congruence preserving ($\text{lcm}_4 = 12$), although its leading coefficient $1/2$ is not an ordinary integer. On the other hand

$$F(X) = \binom{X}{2} + \omega^{\sqrt{2}} X^3$$

is integer valued on all of $\mathbf{Oz}$ but not congruence preserving: $F(2) - F(0) = 1 + 8\omega^{\sqrt{2}}$ has odd constant term and so is not divisible by 2 in $\mathbf{Oz}$, by (3.2). The purely infinite term does not repair the odd constant term.

Remark 16.16 (What Theorem 16.14 does not say [10]). The theorem does not say that reduction modulo an infinite omnific integer is an ordinary finite congruence; such a quotient need not have any set-sized presentation (by Theorem 6.2 it has one only when it contains Π). It says that the polynomial operation respects those congruences, with an explicit quotient supplied by (16.9). It also does not classify all congruence-preserving class functions $\mathbf{Oz} \rightarrow \mathbf{Oz}$: the finite Newton expansion is essential to the statement and the proof, and no infinite interpolation formula or full-class summation is implicit.

16.4 p -adic characters and completions

In this subsection p is an ordinary prime, $e \geq 1$ and $r \geq 1$ are ordinary integers, and $C(\mathbb{Z}_p^r, M)$ denotes the continuous functions from $\mathbb{Z}_p^r$ with its p -adic topology to M ($M = \mathbb{Z}/p^e\mathbb{Z}$ or $\mathbb{F}_p$ discrete, or $M = \mathbb{Z}_p$). All p -adic fields, rings and function spaces here are ordinary set-sized structures.

Lemma 16.17 (Continuous extension [10]). *Every $f \in \text{Int}(\mathbb{Z}^r)$ extends, by its rational-coefficient formula, to a continuous function $\mathbb{Z}_p^r \rightarrow \mathbb{Z}_p$, and this continuous extension is unique.*

Proof. A rational polynomial defines a continuous function $\mathbb{Q}_p^r \rightarrow \mathbb{Q}_p$. Ordinary integer tuples are dense in $\mathbb{Z}_p^r$ and take values in $\mathbb{Z} \subseteq \mathbb{Z}_p$; since $\mathbb{Z}_p$ is closed in $\mathbb{Q}_p$, all values on $\mathbb{Z}_p^r$ lie in $\mathbb{Z}_p$. Density also gives uniqueness. $\square$

Reduction of values modulo p^e is therefore a ring homomorphism $\text{Int}(\mathbb{Z}^r) \rightarrow C(\mathbb{Z}_p^r, \mathbb{Z}/p^e\mathbb{Z})$. Its surjectivity needs an approximation argument; we prove it explicitly by indicator polynomials

rather than assume it, so no unproved approximation assertion is used. For $0 \leq a < p^k$ write $a = a_0 + a_1p + \cdots + a_{k-1}p^{k-1}$ with digits $0 \leq a_j < p$, and put

$$E_{a,k}(X) = \prod_{j=0}^{k-1} \left[1 - \left(\binom{X}{p^j} - a_j \right)^{p-1} \right] \in \text{Int}(\mathbb{Z}), \quad I_{a,k,e}(X) = E_{a,k}(X)^{p^{e-1}}. \quad (16.11)$$

Lemma 16.18 (Digit and indicator congruences [10]). *For every $x \in \mathbb{Z}_p$, with $\mathbf{1}_{a,k}(x) = 1$ if $x \equiv a \pmod{p^k}$ and $\mathbf{1}_{a,k}(x) = 0$ otherwise,*

$$E_{a,k}(x) \equiv \mathbf{1}_{a,k}(x) \pmod{p}, \quad I_{a,k,e}(x) \equiv \mathbf{1}_{a,k}(x) \pmod{p^e}.$$

Proof. First let x be a nonnegative ordinary integer with base- p digits x_j . In $\mathbb{F}_p[T]$ the Frobenius identity gives

$$(1+T)^x = \prod_{j \geq 0} (1+T)^{x_j p^j} = \prod_{j \geq 0} (1+T^{p^j})^{x_j}.$$

The coefficient of T^{p^j} on the right is x_j : the factors of index below j have combined degree at most $\sum_{i < j} (p-1)p^i = p^j - 1$, and factors of index above j contribute only exponents 0 or at least p^{j+1} . Hence $\binom{x}{p^j} \equiv x_j \pmod{p}$. Both sides are continuous functions $\mathbb{Z}_p \rightarrow \mathbb{F}_p$ of x (Lemma 16.17; the j th digit depends only on $x \pmod{p^{j+1}}$), and the nonnegative integers are dense in $\mathbb{Z}_p$, so the congruence holds for all $x \in \mathbb{Z}_p$ with x_j the j th p -adic digit. By Fermat, $t^{p-1} = 1$ for $0 \neq t \in \mathbb{F}_p$, so the j th factor of $E_{a,k}(x)$ is $\equiv 1$ if $x_j = a_j$ and $\equiv 0$ otherwise. The product is the indicator of $x_j = a_j$ for all $j < k$, that is, of $x \equiv a \pmod{p^k}$.

For the lift, let $z \in \mathbb{Z}_p$. If $z \equiv 0 \pmod{p}$ then $z^{p^{e-1}} \equiv 0 \pmod{p^e}$, because $p^{e-1} \geq e$. If $z = 1 + pw$, then $(1 + pw)^{p^{e-1}} \equiv 1 \pmod{p^e}$ by induction from $(1 + p^s v)^p \equiv 1 \pmod{p^{s+1}}$ for $s \geq 1$: in $(1 + p^s v)^p = 1 + p^{s+1}v + \sum_{i=2}^p \binom{p}{i} p^{si} v^i$ the middle binomial coefficients are divisible by p and $sp \geq s+1$; this is also valid for $p = 2$. Applying these two cases to $z = E_{a,k}(x)$ gives the second congruence. $\square$

Theorem 16.19 (Exact finite-level function rings [10]). *The coefficientwise constant term and p -adic evaluation induce natural ring isomorphisms*

$$\mathcal{N}_r(\mathbf{OZ})/p^e \mathcal{N}_r(\mathbf{OZ}) \simeq \text{Int}(\mathbb{Z}^r)/p^e \text{Int}(\mathbb{Z}^r) \simeq C(\mathbb{Z}_p^r, \mathbb{Z}/p^e \mathbb{Z}). \quad (16.12)$$

Proof. By Theorem 16.7, $\mathcal{N}_r(\mathbf{OZ}) = \Pi[\mathbf{X}] \oplus \text{Int}(\mathbb{Z}^r)$ additively, and $p^e \Pi[\mathbf{X}] = \Pi[\mathbf{X}]$ because Π is a $\mathbb{Q}$ -vector space. Hence $p^e \mathcal{N}_r(\mathbf{OZ}) = \Pi[\mathbf{X}] \oplus p^e \text{Int}(\mathbb{Z}^r)$, and the ring homomorphism ct_* induces the first isomorphism.

For the second, the map sends f to $x \mapsto f(x) \pmod{p^e}$ (Lemma 16.17). If $f = p^e g$ with $g \in \text{Int}(\mathbb{Z}^r)$, its reduction is zero. Conversely, if f vanishes modulo p^e at every p -adic tuple, its values at ordinary integer tuples are divisible by p^e , so $f/p^e \in \text{Int}(\mathbb{Z}^r)$. The kernel is exactly $p^e \text{Int}(\mathbb{Z}^r)$.

Let $g \in C(\mathbb{Z}_p^r, \mathbb{Z}/p^e \mathbb{Z})$. The target is discrete, so every point has a residue box $\mathbf{y} + p^k \mathbb{Z}_p^r$ on which g is constant; by compactness of $\mathbb{Z}_p^r$ finitely many of these boxes cover it. Let k be the largest of their levels. Every level- k box $\mathbf{y} + p^k \mathbb{Z}_p^r$ lies in the covering box that contains $\mathbf{y}$, so g is constant on every residue box modulo p^k . For $\mathbf{a} \in \{0, \dots, p^k - 1\}^r$ choose an ordinary integer $c_{\mathbf{a}}$ lifting the value of g on the box of $\mathbf{a}$, and put

$$f(\mathbf{X}) = \sum_{\mathbf{a}} c_{\mathbf{a}} \prod_{j=1}^r I_{a_j, k, e}(X_j) \in \text{Int}(\mathbb{Z}^r).$$

By Lemma 16.18, at $\mathbf{x}$ in the box of $\mathbf{a}$ the product indexed by $\mathbf{a}$ is $\equiv 1$ and all others are $\equiv 0$ modulo p^e , so $f(\mathbf{x}) \equiv g(\mathbf{x}) \pmod{p^e}$. This proves surjectivity. $\square$

Corollary 16.20 (The p -adic completion [10]). *There is a canonical isomorphism of topological rings*

$$\varprojlim_{e \geq 1} \mathcal{N}_r(\mathbf{Oz})/p^e \mathcal{N}_r(\mathbf{Oz}) \simeq C(\mathbb{Z}_p^r, \mathbb{Z}_p), \quad (16.13)$$

where the left side carries the inverse-limit topology of the discrete quotients and the right side the uniform p -adic topology. The canonical map $\mathcal{N}_r(\mathbf{Oz}) \rightarrow C(\mathbb{Z}_p^r, \mathbb{Z}_p)$ sends F to the continuous extension of $\text{ct}_*(F)$, has dense image, and has kernel $\Pi[\mathbf{X}]$.

Proof. The isomorphisms of [Theorem 16.19](#) commute with reduction of values from level $e + 1$ to level e . A coherent sequence of functions $g_e : \mathbb{Z}_p^r \rightarrow \mathbb{Z}/p^e\mathbb{Z}$ determines a unique function $g : \mathbb{Z}_p^r \rightarrow \mathbb{Z}_p$ pointwise, and g is continuous exactly when every $g_e = g \bmod p^e$ is continuous, since the sets $y + p^e\mathbb{Z}_p$ form a base of the topology of $\mathbb{Z}_p$. This identifies the inverse limit with $C(\mathbb{Z}_p^r, \mathbb{Z}_p)$ as rings. The kernel of the projection to level e corresponds to the functions that vanish modulo p^e everywhere, which are the basic neighbourhoods of 0 in the uniform p -adic topology; so the topologies agree. Surjectivity at every finite level gives density of the image.

If the continuous extension of $f \in \text{Int}(\mathbb{Z}^r)$ is zero, then f vanishes on $\mathbb{Z}^r$, and a rational polynomial vanishing on $\mathbb{Z}^r$ is zero (write it as a polynomial in X_r with coefficients in $\mathbb{Q}[X_1, \dots, X_{r-1}]$; for each ordinary tuple $(n_1, \dots, n_{r-1})$ the resulting univariate polynomial has infinitely many roots, so all coefficients vanish on $\mathbb{Z}^{r-1}$; induct on r). With [Theorem 16.7](#) the kernel on $\mathcal{N}_r(\mathbf{Oz})$ is exactly $\Pi[\mathbf{X}]$. Equivalently $\bigcap_e p^e \text{Int}(\mathbb{Z}^r) = 0$, because an ordinary integer divisible by every p^e is zero. $\square$

Remark 16.21 (Not a profinite ring completion [10]). For $r \geq 1$ the characteristic functions of residue boxes at increasing depths are infinitely many distinct elements of $C(\mathbb{Z}_p^r, \mathbb{F}_p)$. Therefore $\mathcal{N}_r(\mathbf{Oz})/p\mathcal{N}_r(\mathbf{Oz})$ is infinite (and discrete in its quotient topology), as are all the rings in (16.12). The completion (16.13) is a p -adic completion, not an inverse limit of finite rings. This differs from the genuine profinite completion of $\mathbf{Oz}$: its finite quotient presentations are the rings $\mathbb{Z}/n\mathbb{Z}$ with kernels $n\mathbf{Oz}$ ([Theorem 6.2](#)), so $\varprojlim_n \mathbf{Oz}/n\mathbf{Oz} \simeq \widehat{\mathbb{Z}}$.

The ordinary polynomial-approximation content is part of the classical theory of integer-valued polynomials [8]. The explicit proof above also identifies exactly which omnific coefficients disappear, rather than only invoking density for the ordinary subring. The compactness of $\mathbb{Z}_p^r$ used here is ordinary p -adic compactness, not compactness of an omnific interval.

Theorem 16.22 (p -adic classification of characters [10]). *Every unital class ring homomorphism $\chi : \mathcal{N}_r(\mathbf{Oz}) \rightarrow \mathbb{F}_p$ has the form*

$$\chi_\alpha(F) = \text{ct}_*(F)(\alpha) \bmod p \quad (16.14)$$

for a unique $\alpha \in \mathbb{Z}_p^r$, and every $\alpha \in \mathbb{Z}_p^r$ defines such a character. The prime ideals of $\mathcal{N}_r(\mathbf{Oz})$ containing $p\mathcal{N}_r(\mathbf{Oz})$ are exactly the kernels $\ker \chi_\alpha$, with distinct α giving distinct kernels; every such prime ideal is maximal with quotient $\mathbb{F}_p$.

Proof. The map χ_α is a composite of ring homomorphisms ([Theorem 16.7](#) and [Lemma 16.17](#)), so it is a character. Conversely, a unital χ kills $p\mathcal{N}_r(\mathbf{Oz})$, so by [Theorem 16.19](#) with $e = 1$ it is $\psi \circ \rho$, where $\rho(F) = \text{ct}_*(F) \bmod p$ and $\psi : C(\mathbb{Z}_p^r, \mathbb{F}_p) \rightarrow \mathbb{F}_p$ is a unital ring homomorphism. At level k , the indicators of the p^{kr} residue boxes modulo p^k are pairwise orthogonal idempotents with sum 1. Their images are idempotents of $\mathbb{F}_p$, hence 0 or 1, pairwise orthogonal with sum 1, so exactly one box of level k is selected. A box indicator is the sum of the indicators of its children at level $k + 1$, and a child indicator c satisfies $c = c \cdot b$ for its parent b ; so the selected boxes are nested. They are nonempty compact sets with diameters tending to 0, so by completeness of $\mathbb{Z}_p^r$ they have exactly

one common point α . Every $h \in C(\mathbb{Z}_p^r, \mathbb{F}_p)$ is constant on the residue boxes of some level k (as in the proof of [Theorem 16.19](#)), so h is an $\mathbb{F}_p$ -linear combination of box indicators and $\psi(h)$ is the value of h on the selected box, that is, $h(\alpha)$. Hence $\chi = \chi_\alpha$. Distinct points are separated by a residue-box indicator, which is $\rho(F)$ for some F by surjectivity; so α is unique.

The prime ideals of $\mathcal{N}_r(\mathbf{Oz})$ containing $p\mathcal{N}_r(\mathbf{Oz})$ correspond, through the explicit set-sized presentation ρ , to the prime ideals of $C(\mathbb{Z}_p^r, \mathbb{F}_p)$. Every element z of this ring satisfies $z^p = z$ pointwise. The quotient by a prime ideal is a commutative domain in which every element is a root of $X^p - X$; such a domain has at most p elements, and it contains the p distinct images of the constants, so it is $\mathbb{F}_p$. Thus every such prime is maximal with quotient $\mathbb{F}_p$, and the quotient map is a character, hence some χ_α . A unital surjection onto $\mathbb{F}_p$ is determined by its kernel, because $\mathbb{F}_p$ has only the identity automorphism; with the uniqueness of α this shows that distinct parameters give distinct kernels. $\square$

Corollary 16.23 (Most characters are not omnific evaluations [10]). *A character $\mathcal{N}_r(\mathbf{Oz}) \rightarrow \mathbb{F}_p$ obtained by evaluating at a tuple $\mathbf{x} \in \mathbf{Oz}^r$ and then applying a unital ring homomorphism $\mathbf{Oz} \rightarrow \mathbb{F}_p$ has parameter $\alpha = (\text{ct}(x_1), \dots, \text{ct}(x_r)) \in \mathbb{Z}^r \subset \mathbb{Z}_p^r$. Conversely every parameter in $\mathbb{Z}^r$ arises in this way. Parameters in $\mathbb{Z}_p^r \setminus \mathbb{Z}^r$ give characters that are not evaluation at any omnific tuple.*

Proof. By [Theorem 5.1](#) the unital homomorphism $\mathbf{Oz} \rightarrow \mathbb{F}_p$ is uniquely ct followed by reduction modulo p . By (16.6), its composite with evaluation at $\mathbf{x}$ is $F \mapsto \text{ct}_*(F)(\text{ct}(\mathbf{x})) \bmod p$, the character with parameter $\text{ct}(\mathbf{x})$. An ordinary integer tuple is itself an allowed input, which gives the converse. The last assertion follows from the uniqueness of the parameter in [Theorem 16.22](#). $\square$

For $r = 1$ the lacunary p -adic integer (16.2) of the omitted type in [Proposition 16.3](#) gives a concrete nonevaluation character. There are uncountably many parameters and only countably many integer ones. The enormous class of omnific inputs does not enlarge the evaluation parameters beyond the ordinary integers, because the constant term of every omnific input is an ordinary integer.

16.5 Rational self-maps and gaps at arbitrary surreal scales

Throughout this subsection (E, Λ) is one of the pairs $(\mathbf{No}, \mathbf{Oz})$ or $(\mathbf{No}[i], \mathbf{Oz}[i])$; in the second the absolute value is the surreal-valued modulus. The input and output lattices are the full classes, not a fixed set-sized truncation. We write $\mathcal{N}_1(\Lambda)$ for $\mathcal{N}_1(\mathbf{Oz})$, respectively $\mathcal{N}_1(\mathbf{Oz}[i])$, and $E(X)$ for the rational functions in one variable over the class field E . We use repeatedly the following estimate, valid in E by the properties of the modulus: if $Q(X) = \sum_{j \leq n} q_j X^j \in E[X]$ has $q_n \neq 0$ and $|y| \geq R_Q := 1 + 2 \sum_{j < n} |q_j/q_n|$, then

$$\frac{1}{2}|q_n||y|^n \leq |Q(y)| \leq \frac{3}{2}|q_n||y|^n, \quad (16.15)$$

because $|\sum_{j < n} q_j y^j| \leq \sum_{j < n} |q_j||y|^{n-1} \leq \frac{1}{2}|q_n||y|^n$.

Lemma 16.24 (Finite differences of a proper rational function [10]). *Let $r \in E(X)$ be nonzero and proper, and let $\Delta r(X) = r(X+1) - r(X)$. For every ordinary $k \geq 0$, $\Delta^k r$ is nonzero and proper. If the leading Laurent term of r at infinity is cX^{-m} with $m \geq 1$, then the leading term of $\Delta^k r$ is*

$$(-1)^k c m(m+1) \cdots (m+k-1) X^{-m-k}, \quad (16.16)$$

the empty product for $k = 0$ being 1.

Proof. Compute in the field of formal Laurent series in X^{-1} over a set-sized subfield of E containing the finitely many coefficients involved; the rational functions over that subfield embed in it, a rational function is proper exactly when its expansion has only negative exponents, and the shift $X \mapsto X + 1$ becomes the substitution $X^{-1} \mapsto X^{-1}(1 + X^{-1})^{-1}$. The binomial series gives

$$(X + 1)^{-n} - X^{-n} = X^{-n}((1 + X^{-1})^{-n} - 1) = -nX^{-n-1} + \text{terms of lower degree in } X.$$

Since $r = cX^{-m} + (\text{terms } X^{-n}, n > m)$, and Δ sends each such lower term to terms of degree at most $-n - 1 < -m - 1$, the leading term of Δr is $-cmX^{-m-1}$. Induction on k gives (16.16). The coefficient is nonzero in characteristic zero and the exponent is negative, which proves both assertions. $\square$

Theorem 16.25 (Polynomiality of rational lattice self-maps [10; 16; 18]). *Let $f \in E(X)$ satisfy $f(x) \in \Lambda$ for every $x \in \Lambda$ at which f is defined. Then $f \in E[X]$. In particular its reduced rational presentation has no finite poles anywhere in E .*

Proof. Divide the numerator by the denominator to write $f = P + r$ with $P \in E[X]$ and r proper, and suppose $r \neq 0$. Choose an ordinary $k \geq 1$ larger than $\deg P$ (any $k \geq 1$ if $P = 0$). Then $\Delta^k P = 0$, so $g := \Delta^k f = \Delta^k r$ is nonzero and proper by Lemma 16.24. Write f and g in reduced form, $f = N_f/D_f$ and $g = N_g/D_g$, with $\deg N_g < \deg D_g$.

We find $B \in \mathbf{No}$ such that for every $x \in \mathbf{No}$ with $x > B$ all of $f(x), \dots, f(x + k)$ are defined, $g(x)$ is defined, and

$$0 < |g(x)| < 1. \quad (16.17)$$

By (16.15), $D_f(y) \neq 0$ for $|y| \geq R_{D_f}$, $N_g(y) \neq 0$ for $|y| \geq R_{N_g}$, and for $|y| \geq \max(1, R_{N_g}, R_{D_g})$,

$$|g(y)| \leq \frac{\frac{3}{2}|\ell(N_g)||y|^{\deg N_g}}{\frac{1}{2}|\ell(D_g)||y|^{\deg D_g}} \leq 3 \frac{|\ell(N_g)|}{|\ell(D_g)|} |y|^{-1},$$

where $\ell(\cdot)$ is the leading coefficient. Take B larger than 1, than $R_{D_f}, R_{N_g}, R_{D_g}$ and than $3|\ell(N_g)/\ell(D_g)|$; for real $x > B$ the points $x, \dots, x + k$ are real and larger than B , so every requirement holds. These are finitely many inequalities in the ordered field $\mathbf{No}$.

By Lemma 16.1 choose $x \in \mathbf{Oz}$ with $x > B$. Then $x + j \in \mathbf{Oz} \subseteq \Lambda$ for $0 \leq j \leq k$, f is defined at these points, and

$$g(x) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} f(x + j) \in \Lambda.$$

This contradicts (16.17) and the lattice separation of Lemma 16.1. Therefore $r = 0$ and $f = P$. $\square$

The finite-difference argument is a general discrete-lattice method. It is included as a supporting result, not asserted to be a new ordinary theorem about rational functions on $\mathbb{Z}$. 16 proves the theorem for every unital constant ring with one fresh input (Theorem 16.42); the two routes are kept.

Recall that $y \in E$ is finite if $|y|$ is bounded by an ordinary integer and infinitesimal if $|y| < 1/n$ for every ordinary $n \geq 1$; the standard part of a finite element is the unique real (respectively complex) number at infinitesimal distance, taken coordinatewise in $\mathbf{No}[i]$ (Remark 16.2).

Lemma 16.26 (Finite roots and reduction [10]). *A root in E of a monic polynomial with finite coefficients is finite. The standard part is a ring homomorphism on the finite elements of E and carries such a root to a root of the coefficientwise standard-part polynomial. For finite $z \in \mathbf{No}[i]$,*

$$\text{st}(|z|) = |\text{st}(z)|.$$

Proof. Let $z^d + \sum_{j<d} c_j z^j = 0$ with $|c_j| \leq C$ for an ordinary integer C . If $|z| > 1 + dC$, then

$$\left| \sum_{j<d} c_j z^j \right| \leq dC|z|^{d-1} < |z|^d,$$

which excludes a root; so every root is finite. Sums and products of finite elements with infinitesimal differences have infinitesimal differences, so st is a ring homomorphism on the finite elements (Lemma 2.2 for $\mathcal{O}_{\mathbb{R}}$ and $\mathcal{O}_{\mathbb{C}}$) and commutes with evaluation of polynomials; applying it to the equation gives the second assertion. For finite $z \in \mathbf{No}[i]$, $|z| \leq |\Re z| + |\Im z|$ is finite; taking standard parts in $|z|^2 = (\Re z)^2 + (\Im z)^2$ gives $\text{st}(|z|)^2 = |\text{st}(z)|^2$, and $\text{st}(|z|) \geq 0$, so uniqueness of nonnegative real square roots gives the identity. $\square$

For an ordinary $d \geq 2$ define the ordinary integer

$$q_{E,d} = \begin{cases} 1, & E = \mathbf{No} \text{ and } d \text{ odd,} \\ 2, & E = \mathbf{No} \text{ and } d \text{ even,} \\ d, & E = \mathbf{No}[i]. \end{cases} \quad (16.18)$$

Theorem 16.27 (A bounded number of lattice preimages at every scale [10]). *Let $F \in \mathcal{N}_1(\Lambda)$ have degree $d \geq 2$, and let $H \in \mathbf{No}_{>0}$. There is a positive surreal T_0 such that, for every $T \in \mathbf{Oz}$ with $T > T_0$, the class*

$$\{u \in \Lambda : |F(u) - F(T)| \leq H\} \quad (16.19)$$

has at most $q_{E,d}$ elements. In particular $F(\Lambda)$ has at most $q_{E,d}$ points in the closed ball of radius H about $F(T)$.

Proof. Write $F(X) = a_d X^d + \sum_{j<d} a_j X^j$ with $a_d \neq 0$, and put $T_0 = \omega \cdot \max(1, H/|a_d|, |a_j/a_d| \ (j < d))$. For $T > T_0$ we have $T > 1$ and, for every ordinary $n \geq 1$, $T > n|a_j/a_d|$ and $T > nH/|a_d|$. Since $d - j \geq 1$ and $d - 1 \geq 1$, it follows that

$$b_j := \frac{a_j}{a_d T^{d-j}} \ (j < d) \quad \text{and} \quad \frac{H}{|a_d| T^{d-1}} \quad \text{are infinitesimal.} \quad (16.20)$$

Fix such a T and an element u of (16.19), and put $z = u/T$. Dividing $F(u) - F(T)$ by $a_d T^d$ gives

$$z^d + \sum_{j<d} b_j z^j = 1 + \sum_{j<d} b_j + \delta, \quad |\delta| \leq \frac{H}{|a_d| T^d} \leq \frac{H}{|a_d| T^{d-1}}, \quad (16.21)$$

so δ is infinitesimal. Thus z is a root of a monic polynomial with finite coefficients, hence finite by Lemma 16.26, and taking standard parts gives $\text{st}(z)^d = 1$. There are at most d such residues in $\mathbb{C}$; among real residues only 1 is possible for odd d , and only ± 1 for even d . So there are at most $q_{E,d}$ possible residues $\text{st}(u/T)$.

It remains to show that each residue belongs to at most one lattice input. Let u, v be elements of (16.19) with $\text{st}(u/T) = \text{st}(v/T) = \zeta$, and put $z = u/T$, $w = v/T$. Write $F(u) - F(v) = (u - v)Q_F(u, v)$ with $Q_F(u, v) = \sum_{j=1}^d a_j \sum_{\ell=0}^{j-1} u^{j-1-\ell} v^\ell$. Direct expansion gives

$$\frac{Q_F(u, v)}{a_d T^{d-1}} = \sum_{\ell=0}^{d-1} z^{d-1-\ell} w^\ell + \sum_{j=1}^{d-1} b_j \sum_{\ell=0}^{j-1} z^{j-1-\ell} w^\ell. \quad (16.22)$$

Its standard part is $d\zeta^{d-1}$, of modulus d because $\zeta^d = 1$; by Lemma 16.26 the standard part of its modulus is d , so

$$|Q_F(u, v)| > \frac{d}{2}|a_d|T^{d-1}.$$

If $u \neq v$, the triangle inequality gives $|F(u) - F(v)| \leq 2H$, hence

$$|u - v| = \frac{|F(u) - F(v)|}{|Q_F(u, v)|} < \frac{4H}{d|a_d|T^{d-1}} < 1,$$

the middle quantity being infinitesimal by (16.20). This contradicts lattice separation for $0 \neq u - v \in \Lambda$ (Lemma 16.1). So $u = v$, and (16.19) has at most $q_{E,d}$ elements. Every image point in the ball is the image of an element of (16.19), which gives the last assertion. $\square$

The proof uses the finite residue polynomial $Z^d - 1$ only as a label for possible clusters. It assumes no convergence of an iterative root-finding scheme, no compactness of an omnific interval, and no existence theorem for new roots in the full field.

Theorem 16.28 (Arbitrary-scale holes in nonlinear polynomial images [10]). *Let $F \in \mathcal{N}_1(\Lambda)$ have degree $d \geq 2$. For every $H > 0$ and every $B \geq 0$ in $\mathbf{No}$ there is $c \in \Lambda$ with $|c| > B$ such that*

$$\{y \in E : |y - c| \leq H\} \cap F(\Lambda) = \emptyset. \quad (16.23)$$

For the real lattice this is an empty closed interval; for the Gaussian lattice it is an empty closed surcomplex disk.

Proof. Put $q = q_{E,d}$ and choose, by Lemma 16.1, a positive $M \in \mathbf{Oz}$ with $M > H$. Let $L = M + 1$ and $R = 3qL + M$. Apply Theorem 16.27 with radius R to get T_0 , and choose $T \in \mathbf{Oz}$ with $T > T_0$ and also

$$|F(T)| > B + 3qL.$$

This is possible: by (16.15), for $T \geq \max(1, R_F)$ one has $|F(T)| \geq \frac{1}{2}|a_d|T^d \geq \frac{1}{2}|a_d|T$, so any $T \in \mathbf{Oz}$ exceeding T_0 , 1 , R_F and $2(B + 3qL)/|a_d|$ works (Lemma 16.1).

Set $y = F(T) \in \Lambda$ and consider the $q + 1$ lattice points $c_j = y + 3jL$, $j = 0, \dots, q$. The closed balls of radius M about them are pairwise disjoint, since distinct centres are at distance at least $3L > 2M$. Each lies in the closed ball of radius R about y , since $|x - y| \leq |x - c_j| + 3jL \leq M + 3qL$. That ball contains at most q points of $F(\Lambda)$ by Theorem 16.27, so one of the $q + 1$ smaller balls contains none. Its centre $c = c_j$ satisfies $|c| \geq |y| - 3qL > B$, and since $H < M$ the ball of radius H about c is disjoint from $F(\Lambda)$ as well. $\square$

Failure of surjectivity is therefore not a single small defect near zero: the gaps exceed any prescribed infinite surreal radius and occur beyond any prescribed bound in modulus, that is, at arbitrarily large moduli.

Corollary 16.29 (All rational bijections of the two lattices [10; 16; 18]). *A rational function $F \in E(X)$ defining a bijection $\Lambda \rightarrow \Lambda$ has the form*

$$F(X) = uX + b, \quad b \in \Lambda, \quad u \in \Lambda^\times,$$

and every such map is a bijection. Thus the rational bijections of $\mathbf{Oz}$ are $X \mapsto \pm X + b$ with $b \in \mathbf{Oz}$, and those of $\mathbf{Oz}[i]$ are $X \mapsto uX + b$ with $b \in \mathbf{Oz}[i]$ and $u \in \{\pm 1, \pm i\}$.

Proof. By [Theorem 16.25](#), F is a polynomial, so $F \in \mathcal{A}_1(\Lambda)$. A constant is not a bijection of Λ . Degree at least two is excluded by [Theorem 16.28](#), because the centre c there is itself a lattice point missing from the image. Hence $F(X) = aX + b$ with $b = F(0) \in \Lambda$ and $a = F(1) - F(0) \in \Lambda$. Surjectivity gives $x \in \Lambda$ with $F(x) = b + 1$, so $ax = 1$ and a is a unit of Λ . Conversely, an affine map with slope $u \in \Lambda^\times$ has the lattice-preserving inverse $X \mapsto u^{-1}(X - b)$. The unit lists are those of [Lemma 2.4](#). $\square$

Here “rational” means a rational function in one variable over the ambient field E . Arbitrary functions $\Lambda \rightarrow \Lambda$, and maps such as coordinatewise conjugation on $\mathbf{Oz}[i]$, are not being classified. 16 shows that surjectivity alone already forces the affine form, for every unital constant ring ([Theorem 16.43](#)).

16.6 Fresh scales: image gaps, fibers and surjective self-maps

This subsection and the next are manuscript 16’s. They work over an arbitrary constant ring and replace the lattice-separation mechanism of [Section 16.5](#) by a *fresh scale*: an exponent above an entire set-sized group of old exponents. Throughout, $k \in \{\mathbb{R}, \mathbb{C}\}$, $E_k = k((X^{\mathbf{No}}))$ (so $E_{\mathbb{R}} = \mathbf{No}$, $E_{\mathbb{C}} = \mathbf{No}[i]$), $D \subseteq k$ is a unital subring, and $\mathcal{A}_{D,k} = D + \Pi_k$ as in [Section 2.3](#); $\mathcal{A}_{\mathbb{Z},\mathbb{R}} = \mathbf{Oz}$ and $\mathcal{A}_{\mathbb{Z}[i],\mathbb{C}} = \mathbf{Oz}[i]$. The ring $\mathcal{A}_{k,k} = k + \Pi_k$ of normal forms without negative exponents (16’s $\mathcal{I}_k$) is *not* the ring $\mathcal{O}_k$ of finite surreals: $\omega \in \mathcal{A}_{k,k}$ and $\omega^{-1} \notin \mathcal{A}_{k,k}$. 16’s $\mathcal{P}_k$ and $\mathcal{A}_D$ are Π_k and $\mathcal{A}_{D,k}$; its defect polynomial $D_{\zeta,c}$ is $d_{\zeta,c}$ (the letter D is the constant ring), its centre μ is s , its affine part L_ζ is a_ζ and its forbidden coefficient A_ζ is c_ζ . For a set-sized additive subgroup $G \subseteq \mathbf{No}$, $K_G = k((X^G))$ is the Hahn field with supports in G , identified with its normal-form image in E_k ($X^g \mapsto \omega^g$).

Lemma 16.30 (Fresh exponents [16; 22]). *Every set-sized family of elements of E_k lies in K_G for some set-sized subgroup $G \subseteq \mathbf{No}$. There is $b > 0$ with $b > g$ for all $g \in G$ (a fresh exponent), and then $jb + g > 0$ for $j > 0$ and $jb + g < 0$ for $j < 0$ ($j \in \mathbb{Z}$, $g \in G$); in particular $G \cap \mathbb{Z}b = \{0\}$.*

Proof. Let G be the subgroup generated by the union of the supports; it is a set, and K_G is a set-sized Hahn field. Choose b strictly above the set $G \cup \{0\}$ ([Lemma 3.1](#)); since $G = -G$, $b + g > 0 > -b + g$, and the same follows for all nonzero integral multiples of b . $\square$

Freshness dominates the whole group, not only finitely many leading exponents: to dominate $G = \mathbb{Q}$ one may take $b = \omega$, while no ordinary real suffices (16).

Theorem 16.31 (Laurent blocks [16; 22]). *Let $b > G$ be fresh and $t = \omega^b$. The map $\iota_b : K_G((T^{-1})) \rightarrow E_k$, $\sum_{j \leq m} c_j T^j \mapsto \sum_{j \leq m} c_j \omega^{jb}$, is an injective field homomorphism, the identity on K_G , with $\iota_b(T) = t$. Each nonzero c_j occupies the support block $jb + \text{supp } c_j$; blocks with different j are disjoint and ordered by j . In particular t is transcendental over K_G .*

Proof. For $j < \ell$ and $g, h \in G$, $(\ell b + h) - (jb + g) = (\ell - j)b + (h - g) > 0$. A nonempty subset of the union of the blocks has a greatest element: take the greatest outer index of a block meeting it (the indices are bounded above), then the greatest element in that block. So the image is a legitimate Hahn sum even when the c_j have infinite supports. Addition is coefficientwise; in a product a fixed outer index receives contributions from finitely many pairs of outer indices, each an ordinary Hahn product. A nonzero coefficient gives a nonempty block that cannot cancel, so ι_b is injective; a unital injective ring map from a field preserves inverses, and injectivity on $K_G[T]$ is transcendence. $\square$

Corollary 16.32 (Exact support test [16]). *For $L = \sum_{j \leq m} c_j T^j$, $\iota_b(L) \in \mathcal{A}_{D,k}$ if and only if $c_j = 0$ for all $j < 0$ and $c_0 \in \mathcal{A}_{D,k}$; the same holds with $\mathcal{A}_{k,k}$ in place of $\mathcal{A}_{D,k}$. Positive-index blocks are purely infinite.*

Proof. Negative-index blocks consist of negative exponents and cannot cancel; the block of index 0 is c_0 ; positive-index blocks do not affect the negative terms or the constant term. $\square$

Theorem 16.33 (One fresh value forces polynomiality [16]). *Let $F \in K_G(T)$. Its denominator does not vanish at t , and $F(t) \in \mathcal{A}_{D,k}$ if and only if $F \in K_G[T]$ and the constant coefficient of F lies in $\mathcal{A}_{D,k}$. The same holds with t replaced by ut for any $u \in K_G^\times$; these inputs lie in Π_k . If the proper part of F begins $aT^{-n} + O(T^{-n-1})$ with $a \neq 0$, $n \geq 1$, then $F(t)$ has the nonzero block $-nb + \text{supp } a$, and if the constant coefficient of the polynomial part has no negative terms, the leading negative exponent of $F(t)$ is $\max \text{supp } a - nb$, with the leading coefficient of a .*

Proof. Transcendence gives the first assertion. Expand F at infinity by polynomial division and a geometric inverse; its image under ι_b is $F(t)$, and a rational function has no negative Laurent coefficients exactly when it is a polynomial. Apply Corollary 16.32. Substituting uT multiplies c_j by u^j and fixes c_0 ; ut occupies the positive block $b + G$. The last assertion is the block order of Theorem 16.31. $\square$

One fresh value detects negative Laurent tails, not integer-valuedness: $F(X) = X/2$ passes the test but fails at $X = 1$ (16).

Lemma 16.34 (Branches of a shifted value [16]). *Let $d \geq 2$, let K be a field of characteristic 0 containing the d th roots of unity, and $f = a_d X^d + \dots + a_0 \in K[X]$ with $a_d \neq 0$. For every $c \in K$ the equation $f(Y) = f(T) + c$ has exactly d distinct roots in $K((T^{-1}))$, indexed by $\zeta^d = 1$, of the form*

$$Y_{\zeta,c} = \zeta T + (\zeta - 1)s + O(T^{-1}), \quad s = \frac{a_{d-1}}{da_d}.$$

Proof. With $u = T^{-1}$ and $W = uY$, multiplying by u^d gives $\sum_j a_j u^{d-j}(W^j - 1) - cu^d = 0$, which at $u = 0$ is $a_d(W^d - 1) = 0$ with simple roots ζ (the W -derivative there is $da_d \zeta^{d-1} \neq 0$). The coefficient of u^n is then solved uniquely step by step, giving $W_\zeta \in K[[u]]$ with $W_\zeta(0) = \zeta$, and $Y = TW_\zeta(T^{-1})$; the d distinct leading coefficients give d distinct roots, which exhaust the roots of a polynomial of degree d . Comparing coefficients of T^{d-1} in $f(\zeta T + w + \dots) = f(T) + c$ gives $da_d \zeta^{d-1} w + a_{d-1} \zeta^{d-1} = a_{d-1}$, that is $w = (\zeta - 1)s$. $\square$

Put $\mathbf{a}_\zeta(T) = \zeta T + (\zeta - 1)s$ and $\mathbf{d}_{\zeta,c}(T) = f(\mathbf{a}_\zeta(T)) - f(T) - c$. The choices of ζ and s cancel the terms of degrees d and $d - 1$, so $\deg \mathbf{d}_{\zeta,c} \leq d - 2$ unless $\mathbf{d}_{\zeta,c} = 0$. Since $\mathbf{a}_\zeta$ fixes $-s$,

$$\mathbf{d}_{\zeta,c}(-s) = -c : \tag{16.24}$$

a nonzero output shift cannot be produced by an affine change of input, even if the coefficients of f are infinite or infinitesimal (16). Degrees and leading coefficients lc below refer to the formal variable T .

Theorem 16.35 (Forbidden layers [16]). *In Lemma 16.34 let $c \neq 0$. For each $\zeta^d = 1$ put $q_\zeta = \deg \mathbf{d}_{\zeta,c}$, $r_\zeta = d - 1 - q_\zeta$ and $\mathbf{c}_\zeta = -\text{lc}(\mathbf{d}_{\zeta,c})/(da_d \zeta^{d-1})$. Then $\mathbf{d}_{\zeta,c} \neq 0$, $1 \leq r_\zeta \leq d - 1$, $\mathbf{c}_\zeta \neq 0$ and*

$$Y_{\zeta,c} = \mathbf{a}_\zeta(T) + \mathbf{c}_\zeta T^{-r_\zeta} + O(T^{-r_\zeta-1}); \quad \text{in particular} \quad Y_{1,c} = T + \frac{c}{da_d} T^{1-d} + O(T^{-d}),$$

so the bound $d - 1$ is attained for every $c \neq 0$.

Proof. By (16.24) the defect is nonzero, of degree between 0 and $d-2$. The remainder $R = Y_{\zeta,c} - \mathbf{a}_\zeta$ lies in $T^{-1}K[[T^{-1}]]$ and is nonzero; let χT^m ($m \leq -1$) be its first term. Taylor expansion gives $0 = \mathbf{d}_{\zeta,c} + f'(\mathbf{a}_\zeta)R + \sum_{j \geq 2} f^{(j)}(\mathbf{a}_\zeta)R^j/j!$. The linear term has degree $d-1+m$ and leading coefficient $da_d\zeta^{d-1}\chi$, while the j th term has degree at most $d-j+jm < d-1+m$ because $(j-1)(1-m) > 0$. Hence $q_\zeta = d-1+m$ and $\text{lc}(\mathbf{d}_{\zeta,c}) + da_d\zeta^{d-1}\chi = 0$. For $\zeta = 1$, $\mathbf{a}_1 = T$ and $\mathbf{d}_{1,c} = -c$. $\square$

Theorem 16.36 (Uniform image gap [16]). *Let $f \in K_G[X]$ have degree $d \geq 2$, let $b > G$ be fresh and $t = \omega^b$. Then*

$$f(\mathcal{A}_{k,k}) \cap (f(t) + K_G) = \{f(t)\}, \quad \text{hence} \quad f(\mathcal{A}_{D,k}) \cap (f(t) + K_G) = \{f(t)\}.$$

For every $0 \neq c \in K_G$, every root in $\mathbf{No}[i]$ of $f(Y) = f(t) + c$ has a nonzero block $-r_\zeta b + \text{supp } c_\zeta$ of negative exponents, with $1 \leq r_\zeta \leq d-1$ as in Theorem 16.35.

Proof. Work over $\mathbb{C}((X^G))$ if $k = \mathbb{R}$ (the roots of unity are ordinary constants, so no new exponents arise). Lemma 16.34 and Theorem 16.31 give d distinct roots of $f(Y) - f(t) - c$ in $\mathbf{No}[i]$, hence all of them. For $c \neq 0$ each has, by Theorem 16.35, a nonzero coefficient in a block of negative index, which lies below exponent 0 and cannot cancel against the affine constant or another block; so no root lies in $\mathcal{A}_{\mathbb{C},\mathbb{C}}$, and none in $\mathcal{A}_{k,k} \supseteq \mathcal{A}_{D,k}$. For $c = 0$, $t \in \Pi_k$ gives the value $f(t)$. $\square$

The same b works for every $c \in K_G$ at once, and G need not be Archimedean, countable or divisible (16).

Remark 16.37 (Special cases printed elsewhere [16; 08]). Three nonsolvability statements proved separately are instances of Theorem 16.36 with $G = \{0\}$, where every $b > 0$ is fresh and $K_G = k$. (a) $x^2 = \omega^2 + 1$ has no solution in $\mathbf{Oz}$: take $f = X^2$, $t = \omega$, $c = 1$; this is an example of the Diophantine report [46]. (b) For $a \in \mathbb{R}_{>0}$, $\gamma > 0$, $0 \neq \beta \in \mathbb{Z}$ and $n \geq 2$, $X^n = a\omega^\gamma + \beta$ has no solution in $\mathbf{Oz}$: put $X = a^{1/n}Y$ with $Y \in \mathcal{A}_{\mathbb{R},\mathbb{R}}$, $f = aY^n$, $t = \omega^{\gamma/n}$, $c = \beta$; this is the two-term root obstruction of the Diophantine report [46], whose short proof by the binomial expansion is kept there. (c) For $\gamma > 0$ and $m \geq 2$, $X^m = \omega^\gamma + 1$ has no root in $\mathbf{Oz}[i]$: take $k = \mathbb{C}$, $f = X^m$, $t = \omega^{\gamma/m}$, $c = 1$; this is 08's invisible equation (Theorem 15.25), whose explicit expansion (15.18) exhibits the forbidden block of layer $r_1 = m-1$. Theorem 16.36 is also consistent with 10's local count (Theorem 16.27): for fresh t the fiber of $f(t)$ has at most $e(f) \leq d$ points in the complex case and at most one (d odd) or two (d even) in the real case (Theorem 16.40 below), within 10's bound $q_{E,d}$.

Corollary 16.38 (Omnific non-solvability [16]). *If f maps $\mathcal{A}_{D,k}$ into itself and has degree at least two, then with K_G and t as above, $f(Y) = f(t) + c$ has no solution $Y \in \mathcal{A}_{D,k}$ for any $0 \neq c \in K_G \cap \mathcal{A}_{D,k}$, although its right side lies in $\mathcal{A}_{D,k}$. For $f \in k[X]$ one may take $G = \{0\}$ and any $b > 0$.*

Definition 16.39 (Centered polynomial [16]). For f of degree $d \geq 2$ with centre $s = a_{d-1}/(da_d)$ write $h(U) = f(U - s) = \sum_{j=0}^d h_j U^j$ (so $h_{d-1} = 0$) and $e(f) = \gcd\{j \geq 1 : h_j \neq 0\}$.

Theorem 16.40 (Exact fresh fibers [16]). *For $\zeta^d = 1$, $f(\mathbf{a}_\zeta(T)) = f(T)$ if and only if $h(\zeta U) = h(U)$ if and only if $\zeta^{e(f)} = 1$; if this fails, the branch $Y_{\zeta,0}$ has a nonzero negative Laurent term, and if it holds, $Y_{\zeta,0} = \mathbf{a}_\zeta$. Consequently, for $f \in K_G[X]$ of degree at least two and fresh t ,*

$$\{y \in \mathcal{A}_{D,k} : f(y) = f(t)\} = \{\zeta t + (\zeta - 1)s : \zeta \in k, \zeta^{e(f)} = 1, (\zeta - 1)s \in \mathcal{A}_{D,k}\},$$

distinct ζ giving distinct points. If s has a nonzero negative-exponent part the fiber is $\{t\}$; if $s \in \Pi_k$, all $e(f)$ points occur when $k = \mathbb{C}$, and one or two when $k = \mathbb{R}$, according as $e(f)$ is odd or even.

Proof. $a_\zeta(T) + s = \zeta(T + s)$ gives the first equivalence; the second is $\zeta^j = 1$ for all supported j , that is $\zeta^{e(f)} = 1$ by Bézout. If the defect vanishes, a_ζ is a root with leading coefficient ζ , hence the branch by uniqueness of the simple-root lift; otherwise the remainder is nonzero and the argument of [Theorem 16.35](#) applies. Every root is a branch; nonsymmetric branches have a forbidden block; a symmetric one is $\zeta t + (\zeta - 1)s$, whose fresh term cannot affect the old negative terms or the constant, so it lies in $\mathcal{A}_{D,k}$ exactly when $(\zeta - 1)s$ does (and $\zeta \in k$: a nonreal ζ gives a nonreal leading coefficient). Distinct ζ have distinct fresh leading coefficients; multiplying a nonzero negative part of s by $\zeta - 1 \neq 0$ keeps it; the count is that of roots of unity in k . $\square$

Example 16.41 (Worked fibers and defects [16]). (a) For $f = X^d$, $s = 0$ and $e(f) = d$: the fresh fiber of t^d in $\mathbf{Oz}[i]$ is $\{\zeta t : \zeta^d = 1\}$ (the constant term of ζt is 0, so ζ need not be a Gaussian integer), and for $0 \neq c \in k$ the shifted roots are $\zeta t(1 + ct^{-d})^{1/d} = \zeta t + \frac{\zeta c}{d} t^{1-d} + \frac{\zeta(1-d)c^2}{2d^2} t^{1-2d} + \dots$, with first negative layer exactly $d - 1$ (a formal binomial expansion followed by the block embedding). (b) $f = X^3 + X^2$ has $s = \frac{1}{3}$, $h = U^3 - \frac{1}{3}U + \frac{2}{27}$ and $e(f) = 1$, so $f(t)$ has the single preimage t in $\mathbf{Oz}$ and in $\mathbf{Oz}[i]$; this does not make f injective ($f(0) = f(-1)$). (c) $f = \omega X^2 + X$ has the infinitesimal centre $1/(2\omega)$; at $t = \omega^\omega$ the other algebraic preimage of $f(t)$ is $-t - \omega^{-1}$, and f is in fact injective on $\mathbf{Oz}$ and $\mathbf{Oz}[i]$, since $f(x) = f(y)$, $x \neq y$, forces $x + y = -\omega^{-1}$. (d) For $f = X^3 + \lambda X + \nu$ ($s = 0$), $d_{\zeta,c} = \lambda(\zeta - 1)T - c$: the identity branch first detects the shift at layer two ($\frac{c}{3}T^{-2}$), a nontrivial branch with $\lambda \neq 0$ at layer one ($-\lambda(\zeta - 1)T^{-1}/(3\zeta^2)$). (e) For $f = X^4 + pX^2 + qX + r$, $d_{\zeta,c} = p(\zeta^2 - 1)T^2 + q(\zeta - 1)T - c$; the first negative terms are $cT^{-3}/4$ at $\zeta = 1$, $-qT^{-2}/2$ at $\zeta = -1$ ($q \neq 0$) and $ipT^{-1}/2$ at $\zeta = i$ ($p \neq 0$); when a displayed coefficient vanishes the next term of the finite defect decides. (f) Freshness cannot be weakened casually: with $G = \mathbb{Q}$, $F(T) = \omega/T$ takes the value 1 at the non-fresh $T = \omega$; and $f = X^2$, $t = \omega$ with the shift $2\omega + 1$, which is not below a fresh exponent, gives $(t + 1)^2$, an omnific square.

Theorem 16.42 (Rational self-maps over any constant ring [16]; second route to [Theorem 16.25](#)). *Let $F \in E_k(X)$ be a rational function, in lowest terms, whose values lie in $\mathcal{A}_{D,k}$ at every point of $\mathcal{A}_{D,k}$ where it is defined. Then F is a polynomial. Hence the rational self-maps of $\mathcal{A}_{D,k}$ are exactly its integer-valued polynomials.*

Proof. Put the finitely many coefficients in some K_G ([Lemma 16.30](#)) and take a fresh $t = \omega^b \in \Pi_k \subseteq \mathcal{A}_{D,k}$. The denominator does not vanish at t , and $F(t) \in \mathcal{A}_{D,k}$, so F is a polynomial by [Theorem 16.33](#). $\square$

For $D = \mathbb{Z}, \mathbb{Z}[i]$ this is 10's [Theorem 16.25](#); the two proofs are different routes. 10's uses finite differences and lattice separation ($|x| \geq 1$ for $0 \neq x \in \mathbf{Oz}$), which fails for $D = \mathbb{Q}$ or $D = k$; 16's uses one fresh input and works for every D . The bare statement that integer-valued rational functions are polynomials is not new in principle: it is the D-ring phenomenon of Gunji–McQuillan [21], as recalled by Liu [24, Proposition 1.12]; for $\mathbf{Oz}$ and $\mathbf{Oz}[i]$, whose unit groups are finite, a nonconstant polynomial cannot take all its values in the unit group, which is the link with that criterion. 16 avoids importing a theorem about set-sized domains into a proper-class ring and supplies an explicit fresh witness (16). 22 credits the same antecedent.

Theorem 16.43 (Surjective rational self-maps [16]; second route to [Corollary 16.29](#)). *A rational function over E_k that is defined at every point of $\mathcal{A}_{D,k}$ and maps $\mathcal{A}_{D,k}$ into itself is surjective onto $\mathcal{A}_{D,k}$ if and only if it is $X \mapsto uX + a$ with $a \in \mathcal{A}_{D,k}$ and $u \in D^\times$; every such map is bijective, and under composition they form the affine group $\mathcal{A}_{D,k} \rtimes D^\times$, $(a, u)(a', u') = (a + ua', uu')$. For $\mathbf{Oz}$ these are $X \mapsto \pm X + a$, and for $\mathbf{Oz}[i]$ they are $X \mapsto uX + a$ with $u \in \{\pm 1, \pm i\}$.*

Proof. By [Theorem 16.42](#) the map is a polynomial; constants are not surjective. If its degree is at least two, then for a fresh t the element $F(t) + 1 \in \mathcal{A}_{D,k}$ is not a value, by [Theorem 16.36](#). So $F = uX + a$ with $a = F(0)$ and $u = F(1) - F(0)$ in $\mathcal{A}_{D,k}$, and surjectivity at $a + 1$ gives $ux = 1$ with $x \in \mathcal{A}_{D,k}$, so $u \in D^\times$ ([Lemma 2.4](#)). The inverse is $X \mapsto u^{-1}(X - a)$. $\square$

For $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ this contains 10's classification of the rational bijections; 16 assumes only surjectivity and allows every unital $D \subseteq k$, and its proof uses the image gap where 10's uses holes of every radius ([Theorem 16.28](#)). Both are kept.

Corollary 16.44 (Omitted values form a proper class [16]). *If F is an integer-valued polynomial of $\mathcal{A}_{D,k}$ of degree at least two, then $\mathcal{A}_{D,k} \setminus F(\mathcal{A}_{D,k})$ is a proper class; changing F on a set of inputs cannot make it surjective.*

Proof. Given a set $S \subseteq \mathcal{A}_{D,k}$, enlarge G to contain the supports of S and of the coefficients of F and take $b > G$. The omitted value $F(\omega^b) + 1$ has leading exponent $db + g_d$ (g_d the leading exponent of the leading coefficient), which is not in G , so it is not in S . Changing F on a set adds at most a set of values. $\square$

Complex conjugation bijects $\mathbf{Oz}[i]$ but is not a rational function of one surcomplex variable (a rational function agreeing with X on the ordinary integers is X); admitting $F(\bar{X})$ adds exactly the maps $u\bar{X} + a$. The theorem is one-variable: $(x, y) \mapsto (x + y^2, y)$ is a nonlinear polynomial bijection of $\mathcal{A}_{D,k}^2$ (16).

16.7 Finite tests, fresh probes and sampling

For a unital $D \subseteq k$ and $r \geq 1$ put $\text{Int}(D^r) = \{g \in \text{Frac}(D)[\mathbf{X}] : g(D^r) \subseteq D\}$ and $\mathcal{N}_r(\mathcal{A}_{D,k}) = \{F \in E_k[\mathbf{X}] : F(\mathcal{A}_{D,k}^r) \subseteq \mathcal{A}_{D,k}\}$, so that $\mathcal{N}_r(\mathbf{Oz})$ and $\mathcal{N}_r(\mathbf{Oz}[i])$ are those of [Section 16](#).

Proposition 16.45 (Constant-ring transfer [16]; 10 for $D = \mathbb{Z}, \mathbb{Z}[i]$). *For every unital $D \subseteq k$ and $r \geq 1$, $\mathcal{N}_r(\mathcal{A}_{D,k}) = \text{Int}(D^r) \oplus \Pi_k[\mathbf{X}]$ additively, the second summand an ideal, and the coefficientwise constant term is a ring homomorphism onto $\text{Int}(D^r)$, split by inclusion.*

Proof. Let $F \in \mathcal{N}_r(\mathcal{A}_{D,k})$ have degree at most d in each variable. The Newton coefficients (16.4) are integral combinations of values at $\{0, \dots, d\}^r$, hence lie in $\mathcal{A}_{D,k}$, and $(d!)^r \binom{\mathbf{X}}{\mathbf{k}} \in \mathbb{Z}[\mathbf{X}]$, so every monomial coefficient of F lies in $\text{Frac}(D) + \Pi_k$. Split $F = g + p$ with $g \in \text{Frac}(D)[\mathbf{X}]$, $p \in \Pi_k[\mathbf{X}]$. The constant term is a ring homomorphism on $\text{Frac}(D) + \Pi_k$ ([Lemma 2.2](#)), so for $\mathbf{a} \in D^r$, $g(\mathbf{a}) = \text{ct}(F(\mathbf{a})) \in D$; thus $g \in \text{Int}(D^r)$. Conversely, for $\mathbf{x} = \mathbf{a} + \mathbf{z}$ with $\mathbf{a} \in D^r$, $\mathbf{z} \in \Pi_k^r$, every term of $g(\mathbf{a} + \mathbf{z}) - g(\mathbf{a})$ contains a factor z_j , so it lies in the ideal Π_k of $\mathcal{A}_{k,k}$, and $p(\mathbf{x}) \in \Pi_k$; hence $F(\mathbf{x}) \in g(\mathbf{a}) + \Pi_k$. Uniqueness is $k \cap \Pi_k = 0$, and the section is the inclusion. $\square$

For $D = \mathbb{Z}$ and $\mathbb{Z}[i]$ this is [Theorems 16.7](#) and [16.9](#) (10, and independently 18 in n variables); 16 proves it for $r = 1$ and for several variables with any D . The finite-difference identity needs no convergence argument: both sides agree at the grid points (16).

Theorem 16.46 (Optimal point tests [16]; 10, 18, 22). *Let $L \subseteq \mathbb{N}^n$ be a finite nonempty lower set ($\alpha \in L$ and $\beta \leq \alpha$ coordinatewise imply $\beta \in L$) and $V_L = \text{span}_{\mathbf{No}}\{\mathbf{X}^\alpha : \alpha \in L\}$. For $F \in V_L$ the following are equivalent: $F(\mathbf{Oz}^n) \subseteq \mathbf{Oz}$; $F(\alpha) \in \mathbf{Oz}$ for all $\alpha \in L$; all coefficients of the expansion $F = \sum_{\alpha \in L} c_\alpha \binom{\mathbf{X}}{\alpha}$ lie in $\mathbf{Oz}$. Moreover $|L|$ is the least size of a set of points of $\mathbf{Oz}^n$, arbitrary omnific points allowed, whose values test membership for all of V_L . In particular, for one variable and degree at most d , the $d + 1$ points $0, \dots, d$ suffice and no d points do.*

Proof. The $\binom{\mathbf{X}}{\alpha}$, $\alpha \in L$, form a basis of V_L : each has leading monomial $\mathbf{X}^\alpha / \prod_j \alpha_j!$ and its other monomials have smaller indices, which stay in L . In a linear extension of the coordinatewise order the evaluation matrix $((\binom{\beta}{\alpha}))$ on L is integral and unitriangular, so $c_\alpha = \sum_{\beta \leq \alpha} (-1)^{|\alpha| - |\beta|} \prod_j \binom{\alpha_j}{\beta_j} F(\beta)$. Each $\binom{X_j}{\alpha_j}$ preserves $\mathbf{Oz}$ (Lemma 16.5). If fewer than $|L|$ points are proposed, their evaluation maps have a common nonzero kernel element $h \in V_L$; a polynomial vanishing on $\mathbb{Z}^n$ is zero, so $h(\mathbf{n}) \neq 0$ for some $\mathbf{n} \in \mathbb{Z}^n$, and $h/(2h(\mathbf{n}))$ passes all proposed tests but takes the value $\frac{1}{2}$ at $\mathbf{n}$. $\square$

For rectangular L the equivalences are 10's Theorem 16.6; general lower sets (for instance total degree at most d , with $\binom{n+d}{n}$ points) and the optimality statement are 16's; 18 proves the total-degree case and the binomial basis independently. So the binomial polynomials form a basis of the $\mathbf{Oz}$ -module $\mathcal{N}_1(\mathbf{Oz})$, each element having finitely many nonzero coordinates, and the countable set $\mathbb{Z}$ tests all of $\mathcal{N}_1(\mathbf{Oz})$; no topological density is asserted (16).

Proposition 16.47 (A finite Gaussian test [16]). *Let $F \in \mathbf{No}[i][\mathbf{X}]$ have total degree at most d , $M = d!$ and $C_M = \{a + bi : a, b \in \mathbb{Z}, 0 \leq a, b < M\}$. Then F preserves $\mathbf{Oz}[i]$ if and only if its values lie in $\mathbf{Oz}[i]$ on $L_d \cup C_M^n$, where $L_d = \{\alpha \in \mathbb{N}^n : |\alpha| \leq d\}$. This is an explicit finite certificate, not a claim of minimal size.*

Proof. The values on L_d give $F = g + p$ with $g \in M^{-1}\mathbb{Z}[i][\mathbf{X}]$ and $p \in \Pi_{\mathbb{C}}[\mathbf{X}]$ (the denominators $\prod_j \alpha_j!$ divide $d!$). With $h = Mg \in \mathbb{Z}[i][\mathbf{X}]$, $g(\mathbf{z}) \in \mathbb{Z}[i]$ means $h(\mathbf{z}) \equiv 0 \pmod{M\mathbb{Z}[i]}$, which depends only on the residues of the coordinates, and C_M is a complete residue system. So $g \in \text{Int}(\mathbb{Z}[i]^n)$; apply Proposition 16.45. $\square$

The ordinary grid alone is not enough over $\mathbf{Oz}[i]$: $\binom{i}{2} = (-1 - i)/2$ (Remark 16.10).

Lemma 16.48 (Independent fresh coordinates [16]). *For a set-sized subgroup G there are positive $b_1, \dots, b_n$ whose classes are linearly independent over $\mathbb{Q}$ modulo G , and the $t_j = \omega^{b_j}$ are algebraically independent over K_G .*

Proof. Choose b_j above the set-sized rational span of $G, b_1, \dots, b_{j-1}$; a nontrivial relation would contradict the freshness of the last participating b_j . Distinct monomials $\prod_j t_j^{\alpha_j}$ have exponents in distinct cosets of G , so their K_G -multiples have disjoint supports. $\square$

With such b_j , let $H = G + \sum_j \mathbb{Q}b_j$, choose $B > H$, and let $\xi^{(i)} \in \mathcal{A}_{D,k}^n$ have i th coordinate ω^B and j th coordinate ω^{b_j} for $j \neq i$.

Theorem 16.49 (Coordinate-dominant probes [16]). *If $F \in K_G(X_1, \dots, X_n)$ and $F(\xi^{(i)}) \in \mathcal{A}_{D,k}$ for $1 \leq i \leq n$, then $F \in K_G[X_1, \dots, X_n]$. Consequently every rational function mapping $\mathcal{A}_{D,k}^n$ into $\mathcal{A}_{D,k}$ at its points of definition is a polynomial.*

Proof. Fix i and view F as a rational function of X_i over $E_i = K_G(X_j : j \neq i)$. Substituting $X_j = t_j$ ($j \neq i$) embeds E_i in K_H (Lemma 16.48); a reduced numerator and denominator stay coprime (their Bézout identity survives) and keep their degrees. As ω^B is fresh over K_H , Theorem 16.33 makes the denominator constant in X_i , so $F \in E_i[X_i]$ for each i . If $F = P/Q$ in lowest terms in the unique factorization domain $K_G[\mathbf{X}]$ had nonconstant Q , an irreducible factor of positive degree in some X_i would remain a denominator over E_i (Gauss's lemma). $\square$

Corollary 16.50 (Workspace-uniform rational certificates [16]). *Fix K_G and $d \geq 0$. For rational functions over K_G with numerator of total degree at most d in lowest terms, being a self-map of*

$\mathbf{Oz}^n \rightarrow \mathbf{Oz}$ is tested by L_d together with the n probes $\xi^{(i)}$ (at most $\binom{n+d}{n} + n$ values; $d+2$ for $n=1$), and of $\mathbf{Oz}[i]^n \rightarrow \mathbf{Oz}[i]$ by $L_d \cup C_d^n$ together with the probes.

Proof. The probes force a polynomial, of total degree at most d ; then apply [Theorem 16.46](#) or [Proposition 16.47](#). Necessity is clear since all test points lie in the domain. $\square$

These certificates are uniform for a whole coefficient workspace K_G ; they are relative to exact coefficient data, zero tests and a certificate of freshness, and are not an algorithm for arbitrary presentations of surreal coefficients (16). For $n=1$ the certificate is also 22's; 22's certificate in several variables, with one fresh exponent, is compared with this one in [Remark 16.71](#).

Theorem 16.51 (No set-sized sample tests all rational maps [16; 22]). *Let Λ be $\mathbf{Oz}$ or $\mathbf{Oz}[i]$ and $S \subseteq \Lambda$ a set. There is $\beta > 0$ such that $R_S(X) = \omega^\beta / (X - \frac{1}{2})$ is defined on all of Λ , satisfies $R_S(S) \subseteq \Pi_k$, and $R_S(\omega^{2\beta}) = \sum_{m \geq 0} 2^{-m} \omega^{-(2m+1)\beta} \notin \Lambda$.*

Proof. $\frac{1}{2} \notin \Lambda$. Clear the set of supports of the $(s - \frac{1}{2})^{-1}$, $s \in S$ ([Proposition 3.4\(i\)](#)) to get β . At $x = \omega^{2\beta}$, $R_S(x) = \omega^{-\beta} (1 - \frac{1}{2} \omega^{-2\beta})^{-1}$ is a nonzero infinitesimal. $\square$

Thus no set of points, with all values known, detects all rational self-maps, even among functions of numerator degree 0 and denominator degree 1 with no pole on Λ : the single function $\omega / (X - \frac{1}{2})$ takes purely infinite values at every ordinary and Gaussian integer, yet $\omega / (\omega^2 - \frac{1}{2}) = \omega^{-1} + \frac{1}{2} \omega^{-3} + \dots \notin \mathbf{Oz}$. For polynomials, by contrast, the ordinary integers suffice in every degree ([Theorem 16.46](#)) (16).

Theorem 16.52 (Co-small integrality [16]). *If $F \in E_k(X)$ is defined and $\mathcal{A}_{D,k}$ -valued outside a set-sized subset of $\mathcal{A}_{D,k}$, then F is an integer-valued polynomial of $\mathcal{A}_{D,k}$: a rational function that is not a self-map fails on a proper class of inputs (poles counted as failures).*

Proof. Let S contain all failures and put its supports and the coefficients of F into K_G . If F is not a polynomial, a fresh $t \notin S$ gives $F(t) \notin \mathcal{A}_{D,k}$ ([Theorem 16.33](#)). If F is a polynomial with $F(a) \notin \mathcal{A}_{D,k}$, enlarge G by the supports of a ; the constant coefficient of $F(a + T)$ is $F(a)$, so $F(a + t) \notin \mathcal{A}_{D,k}$ by [Corollary 16.32](#), and $a + t \notin S$. $\square$

[Theorems 16.51](#) and [16.52](#) do not conflict: the first builds a bad function adapted to a set of successful tests, the second says its failures form a proper class; neither provides one set meeting the failures of every function (16). 22 proves [Theorem 16.51](#) again with another witness ([Remark 16.73](#)), and [Theorem 16.52](#) for $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ in every number of variables ([Theorem 16.75](#)).

Proposition 16.53 (Hulls [16]). *For $S \subseteq \mathcal{A}_{D,k}$ let $\text{Hull}_{\text{rat}}(S)$ be the set of $a \in \mathcal{A}_{D,k}$ such that every rational function over E_k defined and $\mathcal{A}_{D,k}$ -valued on S is defined and $\mathcal{A}_{D,k}$ -valued at a (poles off S allowed), and define Hull_{pol} with polynomials. Then $\text{Hull}_{\text{rat}}(S) = S$ for every set S , while $\text{Hull}_{\text{pol}}(\mathbb{Z}) = \mathbf{Oz}$ and $\text{Hull}_{\text{pol}}(\mathbb{Z}[i]) = \mathbf{Oz}[i]$.*

Proof. For $a \notin S$, clearing the supports of the $(s - a)^{-1}$ gives β with $\omega^\beta / (X - a)$ purely infinite on S and a pole at a . The polynomial statements follow from [Proposition 16.45](#): values on D force the ordinary part into $\text{Int}(D)$, and the purely infinite part is harmless. $\square$

The pole allowed in the hull is not what drives [Theorem 16.51](#), whose functions have no pole on the ring (16). In a fixed smaller Hahn field whose value group has no exponent above the coefficient group, the fresh-scale argument cannot be reused; this is one of 16's questions ([Question 18.14](#)), and 22 states exactly when a fixed Hahn field has such an exponent ([Proposition 16.86](#)).

16.8 Ordinary outputs, canonical translations and composition

This subsection is manuscript 18's new material. Its description of $\mathcal{N}_r(\mathbf{Oz})$ and $\mathcal{N}_r(\mathbf{Oz}[i])$, its simplex test for $\mathbf{Oz}^n$ (the lower set L_d of Theorem 16.46) with the binomial basis, its rational collapse and its classification of surjective rational self-maps are 10's and 16's results proved again independently; they are credited at Theorems 16.7, 16.9, 16.25 and 16.46 and Corollary 16.29. Throughout, (E, Λ, D) is $(\mathbf{No}, \mathbf{Oz}, \mathbb{Z})$ or $(\mathbf{No}[i], \mathbf{Oz}[i], \mathbb{Z}[i])$, Π_Λ is Π or $\Pi_{\mathbb{C}}$, and $F_0 = \text{Frac } D$. 18's ideal $\mathcal{J}$, ring A , field K , field k_0 and class $\text{Int}(A)$ are Π_Λ , Λ , E , F_0 and $\mathcal{N}_1(\Lambda)$; its reduction $\bar{f}$ is $\text{ct}_*(f)$; its $\text{pi}(a) = a - \text{ct}(a)$ is written $\mathring{a}$ (for $a \in F_0 + \Pi_\Lambda$); and its *ordinary-output locus* $E_f = f^{-1}(D)$ and target loci $E_b(f) = f^{-1}(b + D)$ are $\mathcal{L}_f$ and $\mathcal{L}_{f,b}$ (the letter E is the field). The coefficients of every $f \in \mathcal{N}_1(\Lambda)$ lie in $F_0 \oplus \Pi_\Lambda$ (Theorems 16.7 and 16.9), so $\text{ct}_*(f) \in \text{Int}(D)$ and $\mathring{a}$ are defined for them.

The first results hold for an abstract *discretely ordered ring* B , an ordered domain in which 1 is the least positive element. Put $(R_*, D_*) = (B, \mathbb{Z})$ or $(B[i], \mathbb{Z}[i])$; an element of an ordered field extension of $\text{Frac } B$, or of its quadratic extension with the modulus $|x + iy| = \sqrt{x^2 + y^2}$ computed in a real closure, is *finite* if its modulus is bounded by an ordinary integer and *infinitesimal* if it is below every $1/n$.

Lemma 16.54 (Discrete lattices [18]). *The finite elements of B are the ordinary integers, and distinct elements of B are at distance at least 1; the same holds for $B[i]$ with $\mathbb{Z}[i]$ and the modulus. $B_{\mathbb{Q}} = \{b/n : b \in B, n \geq 1\}$ and $B[i]_{\mathbb{Q}}$ contain no nonzero infinitesimal. For $B = \mathbf{Oz}$ this is Lemma 16.1.*

Proof. A finite b lies between two consecutive ordinary integers or equals one, and discreteness excludes the first case; apply this to differences. Both coordinates of a finite Gaussian element are finite, and a nonzero difference has a coordinate of modulus at least 1. A nonzero b/n has modulus at least $1/n$. $\square$

Lemma 16.55 (Coefficient gap and one cluster [18]).

- (i) *If $\deg f \leq d$ and $f(0), \dots, f(d) \in R_*$, then all coefficients of f lie in $(R_*)_{\mathbb{Q}}$; a nonzero leading coefficient is not infinitesimal.*
- (ii) *If f has degree $d \geq 1$ and a leading coefficient that is not infinitesimal, and $x_0, \dots, x_d \in R_*$ are distinct with all $f(x_j)$ finite, then all differences $x_i - x_j$ are finite.*

Proof. (i) Lagrange interpolation at $0, \dots, d$ has rational basis coefficients; apply Lemma 16.54. (ii) If the nodes fell into at least two classes of finite distance, then for each i some $|x_i - x_j|$ would be infinite and the others at least 1, so every summand of

$$\text{lc}(f) = \sum_{i=0}^d \frac{f(x_i)}{\prod_{j \neq i} (x_i - x_j)},$$

the coefficient of X^d in Lagrange interpolation, would be infinitesimal, and so would $\text{lc}(f)$. $\square$

Theorem 16.56 (Sharp ordinary-output dichotomy [18]). *Let f be a polynomial of degree $d \geq 1$ with $f(R_*) \subseteq R_*$, and $\mathcal{L}_f = \{x \in R_* : f(x) \in D_*\}$. Exactly one of the following holds:*

- (i) $\#\mathcal{L}_f \leq d$;
- (ii) $f(X) = g(X - a)$ for some $a \in R_*$ and $g \in \text{Int}(D_*)$ of degree d , and then $\mathcal{L}_f = a + D_*$, a coset determined by f .

Proof. Let $\mathcal{L}_f$ contain $d + 1$ points. By Lemma 16.55(i) the leading coefficient is not infinitesimal, and by (ii) the points lie in one class of finite distance: they are $a + n_j$ with $n_j \in D_*$ (Lemma 16.54). Then $g(X) = f(X + a)$ takes values in D_* at the $d + 1$ points n_j , so $g \in \text{Frac}(D_*)[X]$; for every $n \in D_*$, $g(n) = f(a + n) \in R_*$ is an ordinary rational (or Gaussian rational) element of R_* , hence finite, hence in D_* . So $g \in \text{Int}(D_*)$ and $a + D_* \subseteq \mathcal{L}_f$. If $x \in \mathcal{L}_f$ and $y = x - a$ were infinite, the ordinary coefficients of g would give $|g(y)| \geq \frac{1}{2}|\text{lc}(g)||y|^d$, which is infinite, although $g(y) = f(x) \in D_*$; so y is finite and lies in D_* . The alternatives exclude each other because D_* is infinite. $\square$

The clustering and the descent to $\text{Frac}(D_*)[X]$ use only $f(0), \dots, f(d) \in R_*$ and $d + 1$ ordinary outputs; the step $g \in \text{Int}(D_*)$ uses preservation of all of R_* . For **Oz** that global hypothesis is itself finitely testable (Theorem 16.46); for a general discretely ordered ring this must not be assumed (18).

Corollary 16.57 ([18]). *$f(R_*) \cap D_*$ has at most d elements or equals $g(D_*)$ with g as in Theorem 16.56. If $R_* \neq D_*$, no nonconstant polynomial preserving R_* maps R_* into D_* .*

Proof. Apply f to $\mathcal{L}_f$. If $\mathcal{L}_f = R_*$, then R_* is infinite and equals a coset $a + D_*$ containing 0, so $R_* = D_*$. $\square$

Proposition 16.58 (The bound is sharp [18]). *If B has a positive infinite element H , then for $d \geq 2$ the polynomial $\prod_{j=1}^d (X - jH)$, and for $d = 1$ the polynomial HX , preserves B and $B[i]$, and its ordinary-output locus consists of exactly d points ($H, \dots, dH$, respectively 0).*

Proof. For $d \geq 2$ the d roots lie in different classes of finite distance, so by Theorem 16.56 there is no further point. If Hx is finite and $x \neq 0$, then $|x| \geq 1$ and $|Hx|$ is infinite. $\square$

The degree-one replacement is necessary: $X - H$ has the locus $H + D_*$. The bound concerns inputs, not distinct outputs: all d roots above go to 0 (18).

Theorem 16.59 (Canonical translation [18]). *Let $f \in \mathcal{N}_1(\Lambda)$ have degree $d \geq 1$. Either $\#f^{-1}(D) \leq d$, or there is a unique $\eta \in \Pi_\Lambda$ with*

$$f(X) = \text{ct}_*(f)(X - \eta),$$

and then $\deg \text{ct}_(f) = d$ and $f^{-1}(D) = \eta + D$.*

Proof. **Oz** is discretely ordered (Lemma 16.1); apply Theorem 16.56. In its second case write $a = c + \eta$ with $c \in D$ and $\eta \in \Pi_\Lambda$, so $f(X) = h(X - \eta)$ with $h(Y) = g(Y - c) \in \text{Int}(D)$. The coefficients of $h(X - \eta)$ are polynomials in η with coefficients in F_0 , and ct is a ring homomorphism on $F_0 + \Pi_\Lambda$ with $\text{ct}(\eta) = 0$, so $\text{ct}_*(f) = h$. If $\eta + D = \eta' + D$ then $\eta - \eta' \in D \cap \Pi_\Lambda = 0$. $\square$

Theorem 16.60 (The two highest coefficients [18]). *Let $f = \sum_{j \leq d} a_j X^j \in \mathcal{N}_1(\Lambda)$ with $a_d \neq 0$.*

(i) *If $a_d \notin F_0$, then $\#f^{-1}(D) \leq d$.*

(ii) *If $a_d \in F_0$, put $\eta_f = -\hat{a}_{d-1}/(da_d) \in \Pi_\Lambda$. Then $f^{-1}(D)$ is infinite if and only if $f(X + \eta_f) = \text{ct}_*(f)(X)$; otherwise it has at most d elements.*

Proof. In the canonical form the leading coefficient is that of $\text{ct}_*(f)$, which is in F_0 ; this proves (i). Comparing coefficients of X^{d-1} in $f(X) = \text{ct}_*(f)(X - \eta)$ gives $a_{d-1} = \text{ct}(a_{d-1}) - da_d\eta$, so $\eta = \eta_f$; this holds for $d = 1$ too. The converse is the identity itself and Theorem 16.59. Π_Λ is closed under division by the nonzero ordinary scalar da_d . $\square$

The test is a finite algebraic certificate relative to exact coefficient operations (two purely infinite parts, one translation, finitely many comparisons), not an algorithm for arbitrary presentations of surreal numbers (18). For example, $f = \binom{X-\eta}{3}$ with $\eta = \omega + \omega^{1/2}/3$ has $\text{ct}_*(f) = \binom{X}{3}$ and $f^{-1}(\mathbb{Z}) = \eta + \mathbb{Z}$, and η_f recovers η ; whereas $\omega X^d + X$ ($d \geq 2$) has $\text{ct}_*(f) = X$ but, its leading coefficient being infinite, at most d ordinary outputs: coefficientwise reduction controls congruences, the canonical form controls exact values (18). In the notation of Definition 16.39, η_f is minus the purely infinite part of 16's centre $s = a_{d-1}/(da_d)$: indeed $s = \text{ct}(a_{d-1})/(da_d) + \mathring{a}_{d-1}/(da_d)$ when $a_d \in F_0$.

Theorem 16.61 (At most one rich target [18]). *Let $f \in \mathcal{N}_1(\Lambda)$ have degree $d \geq 2$ and put $\mathcal{L}_{f,b} = f^{-1}(b + D)$ for $b \in \Lambda$. The following are equivalent: (i) some $\mathcal{L}_{f,b}$ has more than d elements; (ii) $f(X) = \text{ct}_*(f)(X - \eta) + \beta$ for some $\eta, \beta \in \Pi_\Lambda$. Then η and β are unique, $\beta = f(\eta) - \text{ct}(f(\eta))$, and $\mathcal{L}_{f,b} = \eta + D$ if $\mathring{b} = \beta$, while $\#\mathcal{L}_{f,b} \leq d$ otherwise.*

Proof. If $\mathcal{L}_{f,b}$ has more than d points, Theorem 16.59 applied to $f - b$ gives $f(X) - b = (\text{ct}_*(f) - \text{ct}(b))(X - \eta)$, which is (ii) with $\beta = \mathring{b}$. Conversely (ii) and the growth of $\text{ct}_*(f)$ give $f^{-1}(\beta + D) = \eta + D$. Since $d \geq 2$, subtracting a constant does not change a_{d-1} , so Theorem 16.60 gives the same η for every rich target, and then $\beta = f(\eta) - \text{ct}(f(\eta))$ because $\text{ct}_*(f)(0) \in D$. $\square$

Corollary 16.62 (One test for all targets [18]). *For $d \geq 2$ and $a_d \in F_0$, let $Q(X) = f(X + \eta_f)$. A rich target exists if and only if every nonconstant coefficient of Q lies in F_0 ; then $Q = \text{ct}_*(f) + \beta$ with $\beta = Q(0) - \text{ct}(Q(0))$.*

Proof. This is (ii) of Theorem 16.61 translated back, using $\text{ct}(\eta_f) = 0$, so that $\text{ct}_*(Q) = \text{ct}_*(f)$. $\square$

Degree one is different: for $f = aX + c$ with $0 \neq a \in D$ every target $\beta + D$ has the infinite preimage $(\beta - \mathring{c})/a + D$, and an infinite linear coefficient gives at most one preimage in each target. For $(X - \omega)^2 + \omega^2$ the unique rich target is $\omega^2 + \mathbb{Z}$, with preimage $\omega + \mathbb{Z}$; for X^2 it is $\mathbb{Z}$, and the target $\omega + \mathbb{Z}$ has exactly the two preimages $\pm\omega^{1/2}$ (18). This agrees with 16's fresh fibre of $f(t) = \omega$ at $t = \omega^{1/2}$ (Theorem 16.40) and with 10's bound $q_{\text{No},2} = 2$ (Theorem 16.27); 18 counts preimages of whole D -cosets, 10 of balls at large scale, 16 of fresh values.

Theorem 16.63 (Composition alignment [18]). *Let $f = p(X - \eta) + \beta$ and $g = q(X - \xi) + \gamma$ with $p, q \in \text{Int}(D)$ of degrees $m, n \geq 2$ and $\eta, \beta, \xi, \gamma \in \Pi_\Lambda$. Then $h = f \circ g$ has a target with more than mn preimages if and only if $\gamma = \eta$. In that case $h(X) = (p \circ q)(X - \xi) + \beta$ and $h^{-1}(\beta + D) = \xi + D$; otherwise $\#h^{-1}(b + D) \leq mn$ for every $b \in \Lambda$.*

Proof. Put $\delta = \gamma - \eta$ and $Y = X - \xi$, so $h(Y + \xi) = p(q(Y) + \delta) + \beta$. If $\delta = 0$ the assertions are immediate. Let $\delta \neq 0$, $a = \text{lc}(p)$ and $c = \text{lc}(q)$. Expanding $p(q + \delta) - p(q) = \sum_{j \geq 1} p^{(j)}(q)\delta^j/j!$, every δ -dependent term has degree at most $(m-1)n \leq mn - 2$; so the leading coefficient ac^m and the coefficient of Y^{mn-1} are ordinary, and Theorem 16.60 gives the input translation ξ for h . In degree $(m-1)n \geq 1$ the only δ -dependent contribution is the leading term of $p'(q)\delta$, so that coefficient has purely infinite part

$$mac^{m-1}\delta \neq 0.$$

Since this coefficient is not constant, Corollary 16.62 fails, and Theorem 16.61 applies to the degree- mn polynomial h . $\square$

In particular, if $f = p(X - \eta)$ and $g = q(X - \xi)$ both have infinite ordinary-output loci, $f \circ g$ has one exactly when $\eta = 0$. For example $(X - \omega)^2$ and X^2 both have infinite loci, but $(X^2 - \omega)^2$ has

the ordinary-output locus $\{\omega^{1/2}, -\omega^{1/2}\}$. The hypotheses $m, n \geq 2$ are needed: a linear inner map absorbs an input translation and a linear outer map moves an output translation without creating a nonconstant obstruction (18).

Proposition 16.64 (A Gaussian simplex test [18]; comparison [merge]). *A polynomial $F \in \mathbf{No}[i][Z_1, \dots, Z_n]$ of total degree at most d preserves $\mathbf{Oz}[i]^n$ if and only if $F(\mathbf{a} + i\mathbf{b}) \in \mathbf{Oz}[i]$ for all $\mathbf{a}, \mathbf{b} \in \mathbb{N}^n$ with $|\mathbf{a}| + |\mathbf{b}| \leq d$, a set of $\binom{2n+d}{2n}$ points. For $d \geq 3$ it has fewer points than the certificate $L_d \cup C_{d!}^n$ of Proposition 16.47, which contains the $(d!)^{2n}$ points of $C_{d!}^n$; for $d = 1$, and for $(n, d) = (1, 2)$, 16's certificate is the smaller one. More generally, every set $S \subseteq \mathbb{Z}[i]$ that tests membership in $\text{Int}(\mathbb{Z}[i])$ for polynomials over $\mathbb{Q}(i)$ of degree at most d also tests preservation of $\mathbf{Oz}[i]$ for polynomials over $\mathbf{No}[i]$ of degree at most d .*

Proof. $G(\mathbf{U}, \mathbf{V}) = F(\mathbf{U} + i\mathbf{V})$ is a polynomial of total degree at most d in $2n$ variables, so by the basis argument of Theorem 16.46 for the lower set $L_d \subseteq \mathbb{N}^{2n}$ its coefficients in the binomial basis $\binom{\mathbf{U}, \mathbf{V}}{\alpha}$ are integral combinations of its values on L_d , that is of F at the stated points; they lie in $\mathbf{Oz}[i]$, and $\binom{u}{k} \in \mathbf{Oz}$ for $u \in \mathbf{Oz}$, so $F(\mathbf{u} + i\mathbf{v}) \in \mathbf{Oz}[i]$ for $\mathbf{u}, \mathbf{v} \in \mathbf{Oz}^n$. Necessity is clear. For the comparison, $\binom{2n+d}{2n} < 2^{2n+d}$, while $(d!)^{2n} = 36^n > 8 \cdot 4^n = 2^{2n+3}$ for $d = 3$, and for $d \geq 4$, $d! \geq 2^d$ gives $(d!)^{2n} \geq 2^{2nd} > 2^{2n+d}$ since $2nd - 2n - d = (2n-1)(d-1) - 1 \geq 2$. For $n = 1$ the certificates of 16 and 18 have 2 and 3 points when $d = 1$, and 5 and 6 when $d = 2$. For the last assertion, interpolate F at $d+1$ points of S (a testing set has at least $d+1$ points): its coefficients lie in $\mathbb{Q}(i) + \Pi_{\mathbb{C}}$, so $F = g + p$ with $g \in \mathbb{Q}(i)[X]$ and $p \in \Pi_{\mathbb{C}}[X]$, and $g(s) = \text{ct}(F(s)) \in \mathbb{Z}[i]$ for $s \in S$; hence $g \in \text{Int}(\mathbb{Z}[i])$ and F preserves $\mathbf{Oz}[i]$ by Proposition 16.45. $\square$

The comparison of the two certificates is the merge's; neither is claimed minimal. The ordinary test $\{0, \dots, d\}$ fails over $\mathbf{Oz}[i]$ ($\binom{i}{2} \notin \mathbb{Z}[i]$). By Volkov and Petrov [40], as cited by 18, one-variable testing sets of size $O(d)$ exist for $\text{Int}(\mathbb{Z}[i])$, while $d+1$ points do not suffice for large d ; by the last assertion they test $\mathbf{Oz}[i]$ as well. These are classical inputs about $\mathbb{Z}[i]$, not claims of 18. 22 proves the converse of the last assertion and extends it to $\mathbb{Z}$ and to every number of variables (Theorem 16.82).

Theorem 16.65 (Matrices and jets [18]). *For $s \geq 1$ let $\mathcal{M}_s(\Lambda) = \{f \in E[X] : f(M) \in \Lambda^{s \times s} \text{ for all } M \in \Lambda^{s \times s}\}$ (scalar coefficients) and $\mathcal{M}_s(D) = \{g \in F_0[X] : g(D^{s \times s}) \subseteq D^{s \times s}\}$; let $\Lambda[\varepsilon]_s = \Lambda[\varepsilon]/(\varepsilon^{s+1})$, $\text{Jet}_s(\Lambda) = \{f \in E[X] : f(\Lambda[\varepsilon]_s) \subseteq \Lambda[\varepsilon]_s\}$ and $\text{Jet}_s(D)$ likewise with F_0 and D . Then*

$$\mathcal{M}_s(\Lambda) = \Pi_{\Lambda}[X] \oplus \mathcal{M}_s(D), \quad \text{Jet}_s(\Lambda) = \Pi_{\Lambda}[X] \oplus \text{Jet}_s(D), \quad \bigcap_s \mathcal{M}_s(\Lambda) = \bigcap_s \text{Jet}_s(\Lambda) = \Lambda[X].$$

Moreover $f \in \text{Jet}_s(\Lambda)$ if and only if the Hasse derivatives $f^{[r]} = \sum_{j \geq r} \binom{j}{r} a_j X^{j-r}$, $r \leq s$, lie in $\mathcal{N}_1(\Lambda)$, and for $\deg f \leq d$ the single nilpotent Jordan block N of size $d+1$ forces all coefficients into Λ , since $f(N)_{1,j+1} = a_j$.

Proof. Scalar matrices give $f \in \mathcal{N}_1(\Lambda)$, so $f = h + g$ with $h \in \Pi_{\Lambda}[X]$ and $g = \text{ct}_*(f) \in \text{Int}(D)$. Entrywise ct is a ring homomorphism on matrices over $F_0 + \Pi_{\Lambda}$ and gives $\text{ct}(f(M)) = g(\text{ct}(M))$; so $g \in \mathcal{M}_s(D)$. Conversely $h(M)$ has entries in Π_{Λ} and $g(M)$ has entries with constant terms those of $g(\text{ct}(M)) \in D^{s \times s}$. For jets, $f(a + u) = \sum_{r \leq s} f^{[r]}(a)u^r$ for $a \in \Lambda$ and nilpotent u with coefficients in Λ , and ct commutes with Hasse derivatives, which preserve $\Pi_{\Lambda}[X]$; apply Theorems 16.7 and 16.9. The Jordan block and evaluation at ε in $\Lambda[\varepsilon]_d$ read off the coefficients. $\square$

For example $\binom{X}{2}$ preserves $\mathbf{Oz}$ but $\binom{N}{2} = -\frac{1}{2}N$ for $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, and $(X^2 - X)/(1 + i)$ preserves $\mathbf{Oz}[i]$ (as $x^2 \equiv x$ modulo $1 + i$ on $\mathbb{Z}[i]$) but has the entry $-1/(1 + i)$ at N . These are applications of the constant-term structure and of nilpotent evaluation; the all-size coefficient test is a general algebraic observation, not a new matrix theorem, and the scalar-coefficient convention is one of those surveyed by Frisch [19] (18).

Theorem 16.66 (Rational functions on discretely ordered rings [18]). *Let B be a discretely ordered ring and $R_* = B$ or $B[i]$. A rational function in one variable, over an ordered field extension of $\text{Frac } B$ or its quadratic extension, that takes values in R_* at all but finitely many points of R_* where it is defined, is a polynomial and maps all of R_* into R_* . A rational function in finitely many variables that is defined and R_* -valued on all of R_*^n is a polynomial.*

Proof. Descent. If $r = P/Q \in L(X)$, $\deg P \leq m$, $\deg Q \leq n$, takes values in a subfield F at infinitely many points of F , choose $m + n + 1$ such nodes t_j with values v_j ; the $m + n + 1$ linear equations $A(t_j) = v_j B(t_j)$ in the $m + n + 2$ coefficients of A, B have a nonzero solution over F , $B \neq 0$ (else A would have too many roots), and $PB - AQ$ vanishes at the $m + n + 1$ nodes, so $r = A/B \in F(X)$. *Collapse.* Write $r = q + u$ with q a polynomial and u proper over $\text{Frac}(R_*)$, and suppose $u \neq 0$. For an ordinary $s > \deg q$, $\Delta^s r = \Delta^s u$ is nonzero and proper (Lemma 16.24); B is cofinal in $\text{Frac } B$ ($|a/b| \leq |a|$), so $|\Delta^s u(x)| < 1$ for all sufficiently large $x \in B$. Avoiding the finitely many exceptions, their shifts by $0, \dots, s$ and the poles, $\Delta^s r(x) = \sum_j (-1)^{s-j} \binom{s}{j} r(x+j) \in R_*$ has modulus below 1, hence is 0, for infinitely many x , which is impossible for the nonzero $\Delta^s u$. *Filling.* For $a \in R_*$ choose N with $a + N, \dots, a + N + \deg r$ outside the exceptions; Newton interpolation of $r(a + N + X)$ at $0, \dots, \deg r$, evaluated at $X = -N$, writes $r(a)$ as an integral combination of allowed values. *Several variables.* If $r = P/Q$ in lowest terms had a denominator of positive degree in X_1 , a specialization of the other variables to ordinary integers avoiding a nonzero resultant condition keeps the degrees and the coprimality, and gives a nonpolynomial one-variable function that is R_* -valued on R_* , contrary to the first part. $\square$

For Λ this is Theorem 16.25 (10), Theorems 16.42 and 16.49 (16), and 16's Theorem 16.52, which allows a set-sized exceptional set, is stronger there; 18 adds the abstract discretely ordered ring and the finitely many exceptions. The collapse is not new in principle: it is the D-ring phenomenon of Gunji–McQuillan [21], as recalled by Liu [24, Proposition 1.12], since a discretely ordered ring has unit group $\{\pm 1\}$ and a nonconstant polynomial cannot be unit-valued; 18's direct proof avoids importing maximal-ideal arguments into a proper-class ring. 18 then reproves the classification of surjective rational self-maps (Corollary 16.29 and Theorem 16.43) by a third route: a surjective f has an infinite ordinary-output locus, so $f = \text{ct}_*(f)(X - \eta)$ by Theorem 16.59, $\text{ct}_*(f)$ is surjective on D , and counting lattice points forces $\text{ct}_*(f) = uX + c$ with $u \in D^\times$. Injectivity alone is weaker: X^3 is injective on $\mathbf{Oz}$ but not surjective.

Corollary 16.67 (Fixed Hahn workspaces [18]). *Let Γ be a set-sized ordered abelian group, not necessarily divisible, J_Γ the series in $\mathbb{R}((T^\Gamma))$ with support in $\Gamma_{>0}$ (large monomials) and $A_\Gamma = \mathbb{Z} \oplus J_\Gamma$, with the Gaussian version $\mathbb{Z}[i] \oplus (J_\Gamma + iJ_\Gamma)$. Then Theorems 16.56, 16.59 to 16.61, 16.63, 16.65 and 16.66, the description of the integer-valued polynomials, the finite tests and the classification of surjective rational self-maps hold for these rings.*

Proof. A_Γ is discretely ordered (a nonzero element of J_Γ is infinite), ct is a ring homomorphism on $\mathbb{Q} \oplus J_\Gamma$, and J_Γ is closed under ordinary scalars; the proofs use nothing else, the descent step supplying coefficients in the full Hahn field, and complex moduli being computed in a real closure. $\square$

So 18's arguments, which use only the discrete gap, survive in a fixed Hahn field, where 16's fresh-scale arguments cannot be reused (Section 16.9). For $\Gamma = \{0\}$ the dichotomy is its ordinary case, and the sharp examples need an infinite element. Several variables behave differently: XY preserves Λ^2 , but its ordinary-output locus contains both axes, proper classes not contained in one translate of D^2 ; what survives is the linewise statement that $t \mapsto f(\mathbf{a} + t\mathbf{v})$ obeys the one-variable dichotomy. The discrete gap fails in **No** and in the finite surreals, and finite testing concerns polynomials of known degree only: no finite sample certifies rational functions (compare Theorem 16.51), and entire Hahn series and expressions of nonstandard degree are outside the theorems (18).

16.9 Scope of the polynomial arithmetic

Two mechanisms act in this section and have different hypotheses. The reflection theorem Theorem 16.11 transports the size-sensitive divisibility mechanism of Theorem 5.1 through finite polynomial algebra; that mechanism applies to arbitrary set coefficient fields in the class Hahn construction. The results of Section 16.5 use instead *finite reduction plus lattice separation*: at a large scale T , polynomial preimages of a prescribed output ball reduce to roots of $Z^d - 1$, and a nonzero reduced divided difference would force two inputs of one cluster to lie less than 1 apart, which the lattice forbids. This gives Theorems 16.27 and 16.28 without a count of points in a surreal interval or a real-valued asymptotic density; it uses the real or surcomplex modulus, the standard part and the unit-spaced lattices. Neither mechanism should be used as a substitute for the other. 16 adds a third: *fresh scales* (Sections 16.6 and 16.7). A monomial above an entire set-sized exponent group separates formal Laurent powers into disjoint support blocks, so negative Laurent tails cannot be hidden by cancellation; this needs no lattice separation, works for every unital $D \subseteq k$, and is a support argument, not a convergence argument in the fine topology. It needs the full surreal exponent class: in a fixed Hahn field whose value group has no exponent above the coefficient group it cannot be reused (16). 18 adds a fourth, *finite interpolation in a discretely ordered ring* (Section 16.8): the Lagrange formula for the leading coefficient, whose summands become infinitesimal when ordinary outputs occur in two infinitely separated clusters, while a coefficient that is a rational combination of lattice values cannot be infinitesimal. It uses neither the modulus estimates at a large scale nor fresh exponents, and it survives in every fixed Hahn workspace (Corollary 16.67). 22 (Section 16.10) combines 16's fresh scale with a resultant specialization at ordinary integer coordinates, so that one fresh exponent suffices in every number of variables, and states exactly when a fixed Hahn workspace supplies that exponent (Proposition 16.86).

Manuscript 10 proposes Theorem 16.11 and the arbitrary-scale bound Theorem 16.27 as new; the other results of this section are explicit consequences or transfers of classical arithmetic facts (Newton interpolation, binomial-ring facts [16], the least-common-multiple pattern [9], p -adic approximation by integer-valued polynomials [8]), proved here to make the scope complete. The results do not classify all class ideals of $\mathbf{Oz}$ or of $\mathcal{N}_r(\mathbf{Oz})$, all automorphisms of $\mathbf{Oz}$, or the factorization of arbitrary omnific integers, and they neither construct nor use a global surcomplex exponential. They do not assert a general transfer principle from $\mathbb{Z}$ to $\mathbf{Oz}$; the omitted type of Proposition 16.3 is a concrete warning against such a principle. The Gaussian theorem Theorem 16.9 does not replace the arithmetic of $\text{Int}(\mathbb{Z}[i]^r)$ by a binomial basis, and no Gaussian analogue of the congruence criterion Theorem 16.14 is given. For $r \geq 1$ the completion of Corollary 16.20 is a p -adic, not a profinite, completion. The classification of Corollary 16.29 concerns rational functions only. Manuscript 10 records a question suggested by this material: for the full omnific integers, determine how much an automorphism can do inside the ideal Π while fixing the ordinary shadow ct (every ring endomorphism of $\mathbf{Oz}$ preserves ct , by the uniqueness in Theorem 5.1). Manuscript 16 presents

the image-gap, forbidden-layer and fresh-fiber package and the two sampling statements as its proposed contributions, credits rational collapse to the D-ring theory and Newton interpolation to the classical literature, and proves only one-variable surjectivity rigidity. Manuscript 18 presents the ordinary-output dichotomy, the canonical translation, the uniqueness of a rich target and composition alignment as its proposed contributions, and treats the description of $\mathcal{N}_r$, the matrix and jet reductions and the rational collapse as applications or classical material. Manuscript 22 presents the single-scale detector, the multivariate certificate and its common-scale uniformity, the pole-free construction defeating every set of tests, the removal of set-sized exceptions and their joint real and Gaussian formulation as its proposed contributions, and credits the D-ring collapse and the binomial basis; several of these were already 16's (Section 16.10).

16.10 Finite tests at a single new scale

This subsection is manuscript 22's (Section 1.3). 22 certifies that a rational function of m variables preserves $\mathbf{Oz}$ or $\mathbf{Oz}[i]$ by its values on an ordinary box and on finitely many points that have one coordinate at a single fresh monomial and ordinary integers elsewhere. It adds the removal of set-sized exceptions in every number of variables, a tensor description of the numerical polynomials, an exact transfer of ordinary constant test sets, finite generators of image ideals, a birational consequence and a criterion for fresh scales in a fixed Hahn workspace. 22 did not read this report, which at its pin already contained 16's and 18's subsections (Section 1.4), and about half of it re-derives them. Printed once, under the existing labels and with 22 added as a proving source, are: the fresh exponent and the Laurent blocks (Lemma 16.30 and Theorem 16.31; 22's Lemma 3.1, with the fresh exponent built as ω times a surreal bound of the supports, and Theorem 3.2); the one-point detector (Corollary 16.32 and Theorem 16.33, for $\mathcal{A}_{k,k}$, $\mathbf{Oz}$ and $\mathbf{Oz}[i]$; its Corollary 3.3); omnific binomial closure, with the constant term of $\binom{a}{n}$ equal to $\binom{\text{ct}(a)}{n}$, the grid test, and its corollary that $F(\mathbf{Oz}^m) \subseteq \mathbf{Oz}$ if and only if $F(\mathbb{Z}^m) \subseteq \mathbf{Oz}$ (Lemma 16.5 and Theorem 16.6; its Lemma 4.1, Theorem 4.2, Corollary 4.3); the optimal number $\prod_j (d_j + 1)$ of fixed polynomial tests (Theorem 16.46 for a rectangular lower set; its Theorem 4.4); the decomposition of the numerical polynomials with its split specialization and evaluation formula (Theorems 16.7 and 16.9 and Proposition 16.45; its Proposition 4.5 and the decomposition in its Theorem 7.3), and the binomial basis of $\mathcal{N}_m(\mathbf{Oz})$ over $\mathbf{Oz}$ (Theorem 16.6(iii)); multivariate rational collapse (the last assertion of Theorem 16.49, and Theorem 16.66; its Corollary 5.3); the case $m = 1$ of its certificate, with the $d + 2$ values $0, \dots, d, t$ (Corollary 16.50); the absence of a set-sized test for rational maps (Theorem 16.51; its Theorem B, with a second witness, Remark 16.73); the case $m = 1$ of its removal of exceptions (Theorem 16.52, there for every D); the Gaussian denominators $\binom{i}{2} \notin \mathbb{Z}[i]$ and $(Z^2 - Z)/(1 + i) \in \text{Int}(\mathbb{Z}[i])$ (Remark 16.10 and the paragraph after Theorem 16.65; its Example 7.6); the warnings on ct (Remark 2.3; its Remark 2.1); fractions and units (Proposition 3.4 and Lemma 2.4; its Lemma 2.2); and the D-ring antecedent (after Theorem 16.42). 22's Theorem 7.1 is the coordinate-degree twin of 18's simplex test (Theorem 16.78). 22 reads class statements as Section 2.1 does, in a class theory carrying $\mathbf{No}$ or with a lower and an upper universe; every support, coefficient family and group it generates is a set, and a fresh exponent is chosen only after such a set has been assembled (its Remark 2.3). The numbering of 22 cited here is that of its own manuscript.

Notation and renamings. Throughout, (D, k, Λ, E) is $(\mathbb{Z}, \mathbb{R}, \mathbf{Oz}, \mathbf{No})$ or $(\mathbb{Z}[i], \mathbb{C}, \mathbf{Oz}[i], \mathbf{No}[i])$, $F_0 = \text{Frac } D$, and $m \geq 1$ is the number of variables (10 writes r , 16 and 18 write n). A rational function is written in lowest terms over E ; it is *defined* at a point if its reduced denominator does not vanish there, so a removable factor of a displayed expression is never a pole, and it *preserves* Λ

if it is defined and Λ -valued on all of Λ^m . 22's symbols are renamed as follows.

- Its positive-support classes $\mathcal{J}_{\mathbb{R}}, \mathcal{J}_{\mathbb{C}}$ are $\Pi, \Pi_{\mathbb{C}}$ (together Π_k); its $\mathcal{A}_k$ is $\mathcal{A}_{k,k}$; its Gaussian ring $\mathbf{Og}$ is $\mathbf{Oz}[i]$; and its $B \in \{\mathbf{Oz}, \mathbf{Oz}[i]\}$ is Λ (B is 18's discretely ordered ring and $\mathcal{B}$ the ring $D + \mathfrak{m}_k$).
- Its $\text{Int}(B^m)$, the ambient-coefficient polynomials preserving B , is $\mathcal{N}_m(\Lambda)$, and its $\text{Int}^R(B^m)$, the rational functions preserving B , is $\mathcal{N}_m^{\text{rat}}(\Lambda)$; the letter Int is kept for the ordinary rings $\text{Int}(D^m) = \{f \in F_0[\mathbf{X}] : f(D^m) \subseteq D\}$. Its $E = \text{Frac } D$ is F_0 (E is the ambient field, as in [Section 16.8](#)), and its binomial products B_{ν} are $\binom{\mathbf{X}}{\nu}$.
- Its coefficient group Γ and Hahn field $K_k(\Gamma)$ are 16's G and K_G ; its exponent β with $\Gamma \ll \beta$, its monomial $T = \omega^{\beta}$ and its embedding $\mathcal{E}_T$ are 16's fresh $b > G$, $t = \omega^b$ and ι_b ([Lemma 16.30](#) and [Theorem 16.31](#)).
- Its field L of characteristic 0 is K (L is a lower set in [Theorem 16.46](#)); its bad-specialization polynomial H_j is $\mathfrak{h}_j$ (H is a group in [Theorem 16.49](#)); and its degree bound $D = 2d^2 + 2d$ is N_d (D is the constant ring, as in 22's own Sections 7–9).
- Its box $\mathcal{B}_{m,d}$, remote grid $\mathcal{H}_{m,D}(T)$ and test set $\mathcal{T}_{m,d}(T)$ are $\square_{m,d}$, $R_{m,d}(t)$ and $\text{Test}_{m,d}(t)$ ($\mathcal{B}$ is $D + \mathfrak{m}_k$, $\mathcal{T}(M, N)$ a multiplication kernel and $\mathcal{T}_j$ 21's self-Tor groups); its Gaussian grid G_d is C_{d+1} in the notation of [Proposition 16.47](#).
- Its exceptional set E is S ; its deceiver's τ , $\eta = \omega\beta$, $A = \omega^{\eta}$, λ and F_S are t , ωb , $\omega^{\omega b}$, a and $\tilde{R}_S$ (η is 18's translation and R_S 16's witness); the ambient field K of its Corollary 8.2 is E .
- In its Section 9, its value group Λ and coefficient subgroup $\Gamma \subseteq \Lambda$ are Γ and G , as in [Corollary 16.67](#); its ring $B_{\Lambda} = D \oplus \mathcal{J}_k(\Lambda)$ is $\mathcal{A}_{D,k}(\Gamma)$ of [\(2.2\)](#); and its convex subgroup $C(\Gamma)$ is $\text{cvx}_{\Gamma}(G)$ (C_M is 16's residue grid and C is 03's letter).

Three tempting readings are false. (1) 22's $\Gamma \ll \beta$, which says $|\gamma| < \beta$ for every $\gamma \in \Gamma$, is freshness $b > G$; it is not the relation $h \ll a$ of [\(2.4\)](#), which compares two positive exponents up to ordinary multiples. (2) Passing at t does not certify integer-valuedness: $X/2$ is $\mathbf{Oz}$ -valued at t but not at 1; the fresh point detects denominators, and the ordinary box does the arithmetic. (3) C_{d+1} is a grid of interpolation nodes for coordinate degree d , whereas 16's $C_{d!}$ is a complete residue system; the first works because Newton interpolation needs $d + 1$ nodes in each real coordinate, the second because integrality of $h/d!$ depends only on residues.

Lemma 16.68 (Rectangular grids [22]). *Let K be a field of characteristic 0 and $0 \neq h \in K[Y_1, \dots, Y_r]$ with $\deg_{Y_i} h \leq e_i$. Then h does not vanish at some point of $\prod_{i=1}^r \{0, 1, \dots, e_i\}$.*

Proof. For $r = 0$ a nonzero constant is nonzero, and for $r = 1$ a nonzero polynomial of degree at most e_1 has at most e_1 roots. For $r \geq 2$ expand in Y_r , choose a nonzero coefficient, find by induction a point of the first $r - 1$ grids at which it does not vanish, and apply the case $r = 1$ to the specialized polynomial in Y_r . $\square$

Let $P, Q \in K[X_1, \dots, X_m]$ be nonzero and coprime, $p_i = \deg_{X_i} P$, $q_i = \deg_{X_i} Q$, and choose j with $q_j > 0$. By Gauss's lemma P and Q stay coprime in $K(X_i : i \neq j)[X_j]$, so the resultant $\text{Res}_{X_j}(P, Q)$ is a nonzero polynomial in the other variables. Put

$$\mathfrak{h}_j = \text{lc}_{X_j}(P) \text{lc}_{X_j}(Q) \text{Res}_{X_j}(P, Q). \quad (16.25)$$

The leading coefficients make the degrees survive specialization, and the resultant prevents a common factor from appearing afterwards.

Lemma 16.69 (Good specializations [22]). *For $i \neq j$,*

$$\deg_{X_i} h_j \leq (q_j + 1)p_i + (p_j + 1)q_i. \quad (16.26)$$

If $\mathbf{a}$ assigns values in K to the variables other than X_j and $h_j(\mathbf{a}) \neq 0$, then the specializations $P_{\mathbf{a}}, Q_{\mathbf{a}} \in K[X_j]$ have degrees p_j, q_j and are coprime, so $P_{\mathbf{a}}/Q_{\mathbf{a}}$ is not a polynomial.

Proof. The Sylvester determinant is homogeneous of degree q_j in the coefficients of P and of degree p_j in those of Q , which have degree at most p_i , respectively q_i , in X_i ; so $\deg_{X_i} \text{Res}_{X_j}(P, Q) \leq q_j p_i + p_j q_i$, and the two leading coefficients add at most $p_i + q_i$. This covers $p_j = 0$, where the resultant is P^{q_j} . Where $h_j \neq 0$ both leading coefficients survive, so the specialized Sylvester matrix is that of $P_{\mathbf{a}}, Q_{\mathbf{a}}$, and its nonzero determinant is their resultant; they are coprime, and the denominator has degree $q_j > 0$. $\square$

For $m, d \geq 1$ put $N_d = 2d^2 + 2d$ and, for a fresh monomial $t = \omega^b$,

$$\begin{aligned} \square_{m,d} &= \{0, 1, \dots, d\}^m, & R_{m,d}(t) &= \bigcup_{j=1}^m \{0, \dots, N_d\}^{j-1} \times \{t\} \times \{0, \dots, N_d\}^{m-j}, \\ \text{Test}_{m,d}(t) &= \square_{m,d} \cup R_{m,d}(t). \end{aligned} \quad (16.27)$$

For $m = 1$ this is $\{0, 1, \dots, d, t\}$.

Theorem 16.70 (Single-scale finite certificate [22]). *Let $F = P/Q \in \mathbf{No}(X_1, \dots, X_m)$ be in lowest terms with $\deg_{X_i} P \leq d$ and $\deg_{X_i} Q \leq d$ for all i , where $d \geq 1$. Let $G \subseteq \mathbf{No}$ be a set-sized subgroup containing the supports of all coefficients of P and Q , $b > G$ fresh and $t = \omega^b$. Then F preserves $\mathbf{Oz}$ if and only if it is defined and $\mathbf{Oz}$ -valued at every point of $\text{Test}_{m,d}(t)$, a set of at most*

$$(d+1)^m + m(2d^2 + 2d + 1)^{m-1}$$

points. The same t and the same test set serve every rational function over K_G with these degree bounds.

Proof. Necessity is clear; assume that the tests pass and that $P \neq 0$. The coefficients lie in K_G , and P, Q are coprime over K_G , since a common factor there would be one over $\mathbf{No}$. Suppose Q is not constant and choose j with $q_j > 0$. By (16.26), $\deg_{X_i} h_j \leq (d+1)d + (d+1)d = N_d$ for $i \neq j$, so Lemma 16.68 gives ordinary integers $\mathbf{a} \in \{0, \dots, N_d\}^{m-1}$ with $h_j(\mathbf{a}) \neq 0$, and $F_{\mathbf{a}} = P_{\mathbf{a}}/Q_{\mathbf{a}} \in K_G(X_j)$ is not a polynomial (Lemma 16.69). Its denominator does not vanish at t , and $F_{\mathbf{a}}(t) \notin \mathcal{A}_{\mathbb{R}, \mathbb{R}} \supseteq \mathbf{Oz}$ (Theorem 16.33). This is the value of F at the point of $R_{m,d}(t)$ whose j th coordinate is t and whose other coordinates are $\mathbf{a}$, a failed test. So Q is constant, F is a polynomial with $\deg_{X_i} F \leq d$, and the tests on $\square_{m,d}$ give $F(\mathbf{Oz}^m) \subseteq \mathbf{Oz}$ by Theorem 16.6. The box has $(d+1)^m$ points and each of the m remote grids has $(N_d + 1)^{m-1}$. Only K_G , the degree bounds and $b > G$ were used. $\square$

The certificate is relative to exact field operations and exact membership in $\mathbf{Oz}$; it is not an algorithm for arbitrary descriptions of surreal numbers, and choosing b above an arbitrary support is a construction, not an effective operation (22). When the degree data are known, the grid in direction j may use $\{0, \dots, (q_j + 1)p_i + (p_j + 1)q_i\}$ in coordinate $i \neq j$, only directions with $q_j > 0$ are needed, and the box may use $\max(p_i, q_i)$ in coordinate i , or the actual degrees once the denominator is known to be constant; this refinement is available only after the presentation has been inspected and does not change the coefficient-uniform statement. The bound is $d + 2$ for $m = 1$ and $(d+1)^2 + 2(2d^2 + 2d + 1) = 5d^2 + 6d + 3$ for $m = 2$, that is 3 and 14 for $d = 1$. These count exact ring-valued evaluations at particular surreal points, not floating-point samples or bit operations, and the bound is not claimed optimal (22).

Remark 16.71 (Comparison with 16's probes [merge]). For $m = 1$, $\text{Test}_{1,d}(t) = \{0, \dots, d, t\}$ is 16's certificate (Corollary 16.50). For $m \geq 2$, 16's coordinate-dominant probes (Theorem 16.49) force a polynomial without any bound on the denominator, and together with the rectangular lower set $\square_{m,d}$ of Theorem 16.46 they test the functions of Theorem 16.70 with $(d+1)^m + m$ values, fewer than $(d+1)^m + m(2d^2 + 2d + 1)^{m-1}$. 16's probes use $m+1$ new exponents ($b_1, \dots, b_m$ independent modulo G , and one exponent above all of them) and put a fresh monomial in every coordinate; 22's points use one fresh exponent, in one coordinate, with ordinary integers elsewhere, and a failed test comes with an explicit specialization $\mathbf{a}$ at which the one-variable restriction has a negative block. Both are sizes of explicit certificates; neither is claimed minimal.

Corollary 16.72 (Common tests for a set of functions [22]). *Every set-sized family of rational functions in m variables over $\mathbf{No}$ has one countable set of points testing preservation of $\mathbf{Oz}$ for all its members, and one finite set of the form $\text{Test}_{m,d}(t)$ if the coordinate degrees of the reduced presentations are bounded by d .*

Proof. Put all coefficient supports of the family into one set-sized G (Lemma 16.30) and take $t = \omega^b$ with $b > G$. In the bounded case apply Theorem 16.70; in general the countable set $\mathbb{Z}^m \cup \bigcup_{j=1}^m \mathbb{Z}^{j-1} \times \{t\} \times \mathbb{Z}^{m-j}$ contains every grid needed for every member. $\square$

So Theorem 16.51 is sharp with respect to the quantification over coefficients: every set of functions has common tests, and the class of all functions has no set of common tests (22). For bounded degree over one workspace K_G this is also the uniformity of Corollary 16.50.

Remark 16.73 (A second witness for Theorem 16.51 [22]). Let $S \subseteq \Lambda$ be a set, G a set-sized group containing the supports of its elements, $b > G$ and $t = \omega^b$, and put

$$\tilde{R}_S(X) = \frac{\omega^{\omega b}}{X + t + \frac{1}{2}} = \frac{2\omega^{\omega b}}{2X + 2t + 1}, \quad (16.28)$$

where ωb is a product in $\mathbf{No}$ and $\omega^{\omega b}$ a monomial. Numerator and denominator have coefficients in $\mathbf{Oz}$ and are coprime, and the only pole $-t - \frac{1}{2}$ has constant coefficient $-\frac{1}{2} \notin D$, so it is not in Λ . For $s \in S$, $u = (s + \frac{1}{2})/t$ is infinitesimal (its exponents are $g - b < 0$), and $\tilde{R}_S(s) = \omega^{\omega b} t^{-1} \sum_{n \geq 0} (-u)^n$ has its n th summand supported in $\omega b - (n+1)b + G$, whose elements exceed $(\omega - n - 2)b > 0$; the blocks are ordered as in Theorem 16.31, so $\tilde{R}_S(s) \in \Pi_k$. For $a > 0$ above the group generated by G , b and ωb , and $x = \omega^a \in \mathbf{Oz}$, the value $\tilde{R}_S(x)$ is nonzero with all exponents at most $\omega b - a < 0$, so it is not in Λ . For $S \subseteq \Lambda^m$ apply this to the first coordinates and let the function depend on X_1 only; so Theorem 16.51 holds in every number of variables and for $\mathbf{Oz}[i]$, still with numerator degree 0, denominator degree 1, coefficients in $\mathbf{Oz}$ and no pole on Λ^m . 16's $R_S = \omega^\beta / (X - \frac{1}{2}) = 2\omega^\beta / (2X - 1)$ also has coefficients in $\mathbf{Oz}$; its pole is at $\frac{1}{2}$, 22's at the fresh scale. The failure of either witness is not caused by a high degree or by a pole at an untested input, but by coefficients at scales above the prescribed inputs (22).

Example 16.74 (Every ordinary integer passes [22]). $F(X) = \omega^\omega / (X + \omega + \frac{1}{2}) = 2\omega^\omega / (2X + 2\omega + 1)$ is defined on $\mathbf{Oz}$, and for $n \in \mathbb{Z}$, $F(n) = \sum_{r \geq 0} (-n - \frac{1}{2})^r \omega^{\omega - r - 1} \in \Pi$; but $F(\omega^{\omega^2})$ has leading exponent $\omega - \omega^2 < 0$. The ordinary integers, an exact test set for polynomials, miss this obstruction, and ω^{ω^2} is the remote test for the group generated by 1 and ω . 16's $\omega / (X - \frac{1}{2})$ (after Theorem 16.51) is an example of the same kind.

Theorem 16.75 (Removal of set-sized exceptions in several variables [22]; 16 for $m = 1$). *Let $F \in E(X_1, \dots, X_m)$ be defined and Λ -valued at every point of Λ^m outside a set $S \subseteq \Lambda^m$. Then F*

preserves Λ . Equivalently, the class of points of Λ^m at which F is undefined or not Λ -valued is empty or a proper class.

Proof. Let $F = P/Q \neq 0$ in lowest terms. If Q is not constant, choose j and ordinary integers $\mathbf{a}$ with $h_j(\mathbf{a}) \neq 0$ (Lemmas 16.68 and 16.69; only characteristic 0 is used, so this works over $\mathbf{No}[i]$). Put the supports of the coefficients and of all coordinates of all points of S into one set-sized G and take $b > G$, $t = \omega^b$. The point with j th coordinate t and other coordinates $\mathbf{a}$ is not in S , because the exponent b lies outside G , and there F is defined with a value outside $\mathcal{A}_{k,k} \supseteq \Lambda$ (Theorem 16.33), a contradiction. So F is a polynomial. If $F(\mathbf{a}) \notin \Lambda$ for some $\mathbf{a} \in \Lambda^m$, write $F(a_1 + X, a_2, \dots, a_m) = F(\mathbf{a}) + \sum_{n=1}^e c_n X^n$ and enlarge G by the supports of the coefficients of F , of $\mathbf{a}$ and of the points of S . For fresh t the value at $X = t$ differs from $F(\mathbf{a})$ by an element of Π_k (Corollary 16.32), which cannot move a nonmember of the additive group Λ into it, and $(a_1 + t, a_2, \dots, a_m) \notin S$, since its first coordinate has the new leading exponent b . $\square$

Theorem 16.75 and Theorem 16.51 are compatible for the reason given after Theorem 16.52: one prescribes good values on a set, the other assumes good values off a set. The two steps of the proof use different mechanisms (22). A denominator is detected by a negative Laurent block at a new scale. A polynomial failure is propagated by adding a positive-support block, which cannot alter an old negative tail or an old nonintegral constant coefficient; this step does not assume that the bad value is finite or that it has finitely many exponents.

Remark 16.76 (Specialization and composition [22]). The coefficientwise constant term of Theorem 16.7 respects composition: if $F \in \mathcal{N}_m(\mathbf{Oz})$ and $F_1, \dots, F_m \in \mathcal{N}_n(\mathbf{Oz})$, then $F(F_1, \dots, F_m) \in \mathcal{N}_n(\mathbf{Oz})$ and $\text{ct}_*(F(F_1, \dots, F_m)) = \text{ct}_*(F)(\text{ct}_*(F_1), \dots, \text{ct}_*(F_m))$, because all coefficients lie in the ring $\mathbb{Q} + \Pi$, on which ct is a ring homomorphism, and substitution is built from finitely many sums and products; the same holds for $\mathbf{Oz}[i]$. The decomposition $F = \text{ct}_*(F) + (F - \text{ct}_*(F))$ is an arithmetic classification, not a statement that the purely infinite part is negligible: a coefficient such as ω^α dominates every ordinary coefficient, and preservation places no condition on it beyond positive support (22).

Example 16.77 (Leading exponents are not enough [22]). If $0 < b < c$, the coefficient of $(1 + \omega^{-c})X$ has leading exponent 0, but the value $\omega^b + \omega^{b-c}$ at ω^b has a negative exponent. A scale chosen above leading exponents only does not give the conclusion of Corollary 16.32 for polynomials; G must contain the complete supports, remote tails included (compare the remark after Lemma 16.30).

Theorem 16.78 (Gaussian coordinate-degree grid [22]). *Let $F \in \mathbf{No}[i][Z_1, \dots, Z_m]$ with $\deg_{Z_j} F \leq d_j$, and $C_{d_j+1} = \{a + bi : a, b \in \{0, \dots, d_j\}\}$. Then $F(\mathbf{Oz}[i]^m) \subseteq \mathbf{Oz}[i]$ if and only if $F(\prod_j C_{d_j+1}) \subseteq \mathbf{Oz}[i]$; for a uniform bound d these are $(d+1)^{2m}$ tests.*

Proof. $H(\mathbf{U}, \mathbf{V}) = F(\mathbf{U} + i\mathbf{V})$ has degree at most d_j in each of U_j and V_j , and the tests are its values on the box $\prod_j \{0, \dots, d_j\}^2$. By (16.4) its Newton coefficients lie in $\mathbf{Oz}[i]$, and binomials of elements of $\mathbf{Oz}$ lie in $\mathbf{Oz}$ (Lemma 16.5), so $H(\mathbf{Oz}^{2m}) \subseteq \mathbf{Oz}[i]$; every element of $\mathbf{Oz}[i]$ is $x + iy$ with $x, y \in \mathbf{Oz}$. The converse is restriction. $\square$

This is the coordinate-degree twin of 18's simplex test (Proposition 16.64), which uses the total degree and the $\binom{2m+d}{2m}$ points $\mathbf{a} + i\mathbf{b}$ with $|\mathbf{a}| + |\mathbf{b}| \leq d$, a subset of C_{d+1}^m ; for polynomials of total degree at most d the simplex is the smaller test, and neither is claimed minimal. The ordinary box is not enough over $\mathbf{Oz}[i]$ (Remark 16.10).

Theorem 16.79 (Gaussian single-scale certificate [22]). *Let $F = P/Q \in \mathbf{No}[i](Z_1, \dots, Z_m)$ be in lowest terms with all coordinate degrees of P and Q at most $d \geq 1$, let G contain the supports of the real and imaginary parts of all coefficients, $b > G$ and $t = \omega^b$. Then F preserves $\mathbf{Oz}[i]$ if and only if it is defined and $\mathbf{Oz}[i]$ -valued on $C_{d+1}^m \cup \mathbf{R}_{m,d}(t)$, a set of at most $(d+1)^{2m} + m(2d^2 + 2d + 1)^{m-1}$ points. Every rational function preserving $\mathbf{Oz}[i]$ is a polynomial.*

Proof. Run the proof of Theorem 16.70 over K_G with $k = \mathbb{C}$: a nonconstant denominator gives a nonpolynomial one-variable restriction at ordinary integer values of the other coordinates, whose value at t has a negative block. So the remote tests force a constant denominator, and Theorem 16.78 applies. The last assertion is also Theorem 16.49. $\square$

The remote points need no Gaussian coordinates: ordinary integers already avoid a nonzero polynomial in characteristic 0, and only the arithmetic stage needs the Gaussian grid (22). Theorems 16.51 and 16.75 hold for $\mathbf{Oz}[i]$ as stated.

Theorem 16.80 (Tensor base change [22]; the decomposition is 10's, 16's and 18's). $\mathcal{N}_m^{\text{rat}}(\Lambda) = \mathcal{N}_m(\Lambda) = \text{Int}(D^m) \oplus \Pi_k[\mathbf{X}]$, and the multiplication map

$$\Lambda \otimes_D \text{Int}(D^m) \longrightarrow \mathcal{N}_m(\Lambda), \quad \lambda \otimes f \longmapsto \lambda f, \quad (16.29)$$

is a ring isomorphism, the tensor product being read through finite sums of pure tensors, or in a containing universe in which Λ is a set; it does not assert that Λ is a set in the original universe. The coefficientwise constant term is a split retraction onto $\text{Int}(D^m)$, and evaluation commutes with it.

Proof. The first equality is the polynomiality of preserving rational functions (Theorem 16.49; 22's Corollary 5.3 and Theorem 7.2), and the decomposition is Theorems 16.7 and 16.9 and Proposition 16.45; 22 proves it as 16 does, without a basis of $\text{Int}(D^m)$: values on a large ordinary box put the Newton coefficients, hence the monomial coefficients, in $F_0 + \Pi_k$; constant terms at points of D^m put the ordinary part in $\text{Int}(D^m)$; and $f(\mathbf{z} + \mathbf{u}) - f(\mathbf{z}) \in \Pi_k$ for $\mathbf{z} \in D^m$, $\mathbf{u} \in \Pi_k^m$. Put $M = \text{Int}(D^m)$. As D -modules $\Lambda = D \oplus \Pi_k$, so $\Lambda \otimes_D M \cong M \oplus (\Pi_k \otimes_D M)$. Since $D[\mathbf{X}] \subseteq M \subseteq F_0[\mathbf{X}]$ and every polynomial over F_0 has a common denominator in $D \setminus \{0\}$, localization gives $F_0 \otimes_D M \cong F_0[\mathbf{X}]$ (injective because M lies in the torsion-free module $F_0[\mathbf{X}]$); Π_k is an F_0 -vector space, so $\Pi_k \otimes_D M \cong \Pi_k \otimes_{F_0} (F_0 \otimes_D M) \cong \Pi_k[\mathbf{X}]$. Under these identifications the multiplication map is the direct-sum map of the decomposition. The assertions on the constant term follow from its multiplicativity on $F_0 + \Pi_k$ (Lemma 2.2). $\square$

Definition 16.81 ([22]). For a coordinate degree bound $\mathbf{d}$, a finite set $S \subseteq D^m$ is *universal* in that degree space if every $f \in F_0[\mathbf{X}]$ with $\deg_{X_j} f \leq d_j$ and $f(S) \subseteq D$ satisfies $f(D^m) \subseteq D$.

Theorem 16.82 (Lifting constant test sets [22]). *A finite set $S \subseteq D^m$ is universal in a coordinate degree space if and only if every $F \in E[\mathbf{X}]$ in that degree space with $F(S) \subseteq \Lambda$ satisfies $F(\Lambda^m) \subseteq \Lambda$.*

Proof. Let S be universal. Its evaluation matrix on the monomials of the degree space has full column rank over F_0 : otherwise a nonzero $h \in F_0[\mathbf{X}]$ in the space vanishes on S , Lemma 16.68 gives $\mathbf{c} \in \mathbb{Z}^m \subseteq D^m$ with $h(\mathbf{c}) \neq 0$, and $h/(2h(\mathbf{c}))$ is D -valued on S but takes the value $\frac{1}{2} \notin D$ at $\mathbf{c}$. Inverting a square invertible submatrix writes every coefficient of F as an F_0 -linear combination of values in $\Lambda = D \oplus \Pi_k$, so $F = f + g$ with $f \in F_0[\mathbf{X}]$ and $g \in \Pi_k[\mathbf{X}]$ in the same space. Constant terms at the points of S give $f(S) \subseteq D$, so $f \in \text{Int}(D^m)$, and F preserves Λ by Theorem 16.80. Conversely, if S tests Λ and $f \in F_0[\mathbf{X}]$ has $f(S) \subseteq D$, then $f(\Lambda^m) \subseteq \Lambda$ and $f(D^m) \subseteq \Lambda \cap F_0 = D$. $\square$

So among test points in the ordinary lattice D^m , minimal polynomial testing is the same problem before and after adjoining the purely infinite part (22). This turns the last assertion of [Proposition 16.64](#), the forward direction for $D = \mathbb{Z}[i]$ in one variable (18), into an equivalence in every number of variables, for $\mathbb{Z}$ and for $\mathbb{Z}[i]$. In one Gaussian variable the ordinary problem is the classical universal-set problem connected with simultaneous p -orderings (Byszewski, Frączyk and Szumowicz [7], with the work of Volkov and Petrov [40] discussed there); 22 does not determine the optimal sets, and supplies only the transfer and the grid of [Theorem 16.78](#). The theorem bears on [Question 18.14\(a\)](#), not on [Question 18.16\(c\)](#): its test points lie in D^m , and whether points of $\mathbf{Oz}[i]$ outside $\mathbb{Z}[i]^m$ can do better is not addressed.

Theorem 16.83 (Finite generators of image ideals [22]). *Let $F \in \mathcal{N}_m(\mathbf{Oz})$ have $\deg_{X_j} F \leq d_j$ and Newton coefficients $c_{\mathbf{k}}$ ([Theorem 16.6](#)), and let I_F be the ideal of $\mathbf{Oz}$ generated by $F(\mathbf{Oz}^m)$, the class of finite $\mathbf{Oz}$ -linear combinations of values. Then*

$$I_F = (c_{\mathbf{k}} : \mathbf{k} \leq \mathbf{d}) = \left(F(\boldsymbol{\mu}) : \boldsymbol{\mu} \in \prod_j \{0, \dots, d_j\} \right). \quad (16.30)$$

Hence for every ideal I of $\mathbf{Oz}$, $F(\mathbf{Oz}^m) \subseteq I$ if and only if the finitely many box values lie in I .

Proof. The Newton expansion writes each value $F(\mathbf{x})$ as an $\mathbf{Oz}$ -linear combination of the $c_{\mathbf{k}}$ ([Lemma 16.5](#)); (16.4) writes each $c_{\mathbf{k}}$ as an integral combination of box values; and box values are values. $\square$

A finite generating set avoids set-size questions about an ideal generated by a class, and no principal generator or greatest common divisor is assumed or inferred (22). The Gaussian version, with the box $\prod_j C_{d_j+1}$, follows in the same way from the $2m$ -variable polynomial H in the proof of [Theorem 16.78](#): the ideal of $\mathbf{Oz}[i]$ generated by all values on $\mathbf{Oz}[i]^m$ is generated by the finitely many box values.

Corollary 16.84 (Arithmetic birational rigidity [22]). *If $\Phi, \Psi : E^m \dashrightarrow E^m$ are rational maps inverse to each other as rational maps, and both are defined on Λ^m with values in Λ^m , then all coordinates of Φ and Ψ are polynomials in $\mathcal{N}_m(\Lambda)$, and Φ, Ψ are mutually inverse polynomial automorphisms of E^m .*

Proof. Each coordinate preserves Λ , so it is a polynomial ([Theorems 16.49](#) and [16.79](#)); the rational identities $\Psi \circ \Phi = \text{id}$ and $\Phi \circ \Psi = \text{id}$ are then polynomial identities. $\square$

The rational inverse is a hypothesis: nothing is asserted about a bijective polynomial map of Λ^m without one, no Jacobian-conjecture claim is made, and the coordinates need not have coefficients in Λ (22).

Example 16.85 ([22]). $\Phi(X, Y) = (X + \frac{1}{2}Y(Y-1), Y)$ and $\Psi(X, Y) = (X - \frac{1}{2}Y(Y-1), Y)$ are inverse polynomial maps that induce inverse bijections of $\mathbf{Oz}^2$, with the coefficient $\frac{1}{2} \notin \mathbf{Oz}$; the Gaussian analogue uses $Y(Y-1)/(1+i)$. So arithmetic polynomial maps form a larger class than maps with coordinates in $\mathbf{Oz}[X, Y]$. 16's shear $(x, y) \mapsto (x + y^2, y)$ (after [Theorem 16.43](#)) has coordinates in $\mathbf{Oz}[X, Y]$.

Three conditions on a polynomial are distinct (22): coefficients in Λ , values in Λ on Λ^m , and values in Λ at sampled points. The first implies the second but not conversely ($\binom{X}{2}$); for bounded degree the tests above make the second and third equivalent, while for rational functions no coefficient-independent set of samples does ([Theorem 16.51](#)). Coefficients in $\mathbf{Oz}$ for numerator and denominator do not make the denominator invertible at every omnific input ([Example 16.74](#)).

Proposition 16.86 (Fresh scales in a fixed Hahn workspace [22]). *Let Γ be a set-sized ordered abelian group, $G \subseteq \Gamma$ a subgroup containing the coefficient supports, and $\text{cvx}_\Gamma(G) = \{x \in \Gamma : |x| \leq g \text{ for some } g \in G\}$ the convex subgroup it generates. A positive $b \in \Gamma$ with $b > G$ exists if and only if $\text{cvx}_\Gamma(G) \neq \Gamma$. When it exists, the Laurent blocks, the detector and the one-point test (Theorems 16.31 and 16.33 and Corollary 16.32) and the certificates of Theorems 16.70 and 16.79 hold in $k((X^\Gamma))$, for $\mathcal{A}_{D,k}(\Gamma)$ and coefficients supported in G .*

Proof. If $b > G$, then $b \notin \text{cvx}_\Gamma(G)$. Conversely a positive $b \notin \text{cvx}_\Gamma(G)$ satisfies $b > |g|$ for all $g \in G$, since $b \leq |g|$ would put it in the convex subgroup. The remaining arguments use only supports, block separation, finite polynomial algebra and the split ring $D \oplus \Pi_k(\Gamma)$; the exponents $jb + g$ stay in Γ , and the countable union of the blocks is a legitimate support. $\square$

For an Archimedean Γ every nonzero G is cofinal, and no fresh exponent exists in Γ ; this is a failure of the method's hypothesis, not a theorem that every finite test fails there, and one may enlarge Γ by a lexicographic factor or work in **No** (22). Theorem 16.51 cannot hold for a set-sized ring, which is itself a set of tests; its strength comes from the unbounded supply of surreal scales, and Corollary 16.72 explains how set-sized computations coexist with it (22). Where no fresh exponent exists, rational collapse still holds by 18's discrete-gap argument, which uses none (Theorem 16.66 and Corollary 16.67); what 22's method loses there is the finite rational certificate, and whether another finite certificate exists is open (Question 18.14(c)) [merge].

Example 16.87 (One new coordinate exposes a denominator [22]). For $F(X, Y) = (X+Y)/(XY+1)$ over $\mathbb{Q}$, the leading coefficients in X are 1 and Y and $\text{Res}_X(X+Y, XY+1) = 1 - Y^2$, so $\mathbf{h}_X = Y(1 - Y^2)$. At $Y = 2$, $F(X, 2) = (X+2)/(2X+1) = \frac{1}{2} + \frac{3}{2(2X+1)}$, whose second term has a nonzero negative block at every fresh t . The resultant excludes exactly the values of Y at which the denominator loses degree or a cancellation occurs.

For every $u \in \Pi$, even of infinite support, $\binom{X}{3} - 4\binom{Y}{2} + uX^2Y$ preserves **Oz** with specialization $\binom{X}{3} - 4\binom{Y}{2}$, while $\omega^{-1}X$ fails at $X = 1$ and $\sqrt{2} + \omega \notin \mathbf{Oz}$: what matters is negative support or a nonintegral constant coefficient, not magnitude, and adding positive support cannot repair a nonintegral constant coefficient, which is the invariance behind the polynomial step of Theorem 16.75 (22; compare Example 16.8).

A construction of the certificate reduces P/Q over the coefficient field, collects every exponent of every coefficient support, forms the group they generate, chooses b above it, and tests exact values on the box and the remote grids. An effective version needs an implementation contract for equality, polynomial gcd, support bounds and membership in Λ , none of which can be presumed decidable for arbitrary notations of surreal numbers; when the coefficients can be inspected exactly, checking that the reduced denominator is constant and then the Newton coefficients may be cheaper than evaluating the grid. The theorem is an exact value certificate, not a complexity improvement over symbolic algebra. Once a nonzero resultant specialization is found it is also a witness generator: it names a failing point without numerical approximation, since the first nonzero negative Laurent coefficient contributes a whole block that cannot cancel (22).

Remark 16.88 (Other constant rings [merge]). The conclusion of Theorem 16.51 holds for $\mathcal{A}_{D,k}$ with every unital $D \subsetneq k$, in every number of variables. Take $c \in k \setminus D$. The only pole of $\omega^\beta/(X_1 - c)$ is $c \notin \mathcal{A}_{D,k}$; if $\beta > 0$ exceeds the absolute values of all exponents in the supports of the $(s_1 - c)^{-1}$, $s \in S$ (a set), the values on S lie in Π_k , while at a point with first coordinate $\omega^{2\beta}$ the value is the nonzero infinitesimal $\omega^{-\beta}(1 - c\omega^{-2\beta})^{-1}$. This is 16's proof with $\frac{1}{2}$ replaced by c ; for $D = k$ it does not apply.

22 summarizes its dependencies as follows: the certificate uses the fresh scale, the resultant specialization and the ordinary binomial test; the no-set-test theorem uses the fresh scale and a geometric inverse, not the resultant or the binomial machinery; the removal of exceptions combines denominator detection with invariance under positive-support additions; and the Gaussian versions change the constant arithmetic and keep the support mechanism. Its proofs need neither algebraic closedness of $\mathbf{No}[i]$, nor differentiation, nor a topology, and ω^γ is the normal-form monomial, not an analytic exponential.

17 Natural ordinal arithmetic is not a quotient of the omnific integers

This section is manuscript 11's. Let $\mathcal{G}_{\text{nat}}$ be the Grothendieck ring of the ordinals under Hessenberg's natural sum and natural product. Every ordinal has a finite Cantor normal form $\sum_j n_j \omega^{\gamma_j}$, each γ_j has its own finite Cantor normal form $\sum_\alpha m_{j,\alpha} \omega^\alpha$, and under natural multiplication $\omega^{\gamma_j} = \prod_\alpha (\omega^{\omega^\alpha})^{m_{j,\alpha}}$. Uniqueness of these forms gives the free commutative semiring $\mathbb{N}[T_\alpha : \alpha \in \mathbf{On}]$, and adjoining additive inverses gives

$$\mathcal{G}_{\text{nat}} \cong \mathbb{Z}[T_\alpha : \alpha \in \mathbf{On}], \quad (17.1)$$

with T_α represented by ω^{ω^α} under the natural embedding in $\mathbf{No}$. Natural operations, not the usual noncommutative ordinal operations, are essential. The description (17.1) appears in the question [18]; the argument above fixes the operations.

Theorem 17.1 (The ordinal quotient question [11, 13]). *Every unital ring map $h : \mathbf{Oz} \rightarrow \mathcal{G}_{\text{nat}}$ has image exactly the ordinary integral constants. In particular $\mathcal{G}_{\text{nat}}$ is not isomorphic to a quotient ring of $\mathbf{Oz}$. More generally, no ring map $\mathbf{Oz} \rightarrow \mathbb{Z}[X]$ with $X \neq \emptyset$ is surjective.*

Proof. First proof. $\mathbb{Z}[X]$ is residually set-sized by Theorem 5.12 (for any class X of indeterminates), so h kills Π and factors through ct ; its image is $\mathbb{Z}$, a proper subring since $T_0 \notin \mathbb{Z}$.

Second proof, specific to this target. The additive group of Π is divisible ($s/n \in \Pi$ for every ordinary $n \geq 1$), while no nonzero element of $\mathbb{Z}[T_\alpha]$ is divisible by every n (each of its finitely many integral coefficients would be). So every additive map from Π to $\mathbb{Z}[X]$ is zero, and a unital ring map from $\mathbf{Oz} = \mathbb{Z} + \Pi$ has image $\mathbb{Z}$. $\square$

The question was posted by Jesse Elliott as the final addition to MathOverflow question 188430, “Transcendence degree of the surreals over the subfield generated by the ordinals” (30 November 2014) [18]; the displayed answer, by Eric Wofsey, concerns the transcendence degree, a different question. The answer to the quotient question is negative. 11 gives the short additive proof precisely so that the answer is not mistaken for a difficult new factorization theorem: no claim is made that the elementary obstruction was unknown to specialists or absent from every other source. The first proof illustrates the general residual theorem, and the second isolates the weaker mechanism that suffices for this target. The obstruction is not a cardinal comparison, since both rings may be proper classes; it is their divisibility and representation properties (11).

Remark 17.2 (An independent answer [13]). Manuscript 13, dated a day after 11 and pinned to a repository commit that precedes 11's placement, answers the same addendum negatively with the same two proofs: its lemma that every unital map from $\mathbf{Oz}$ into a ring B with $\bigcap_n nB = 0$ factors through ct is the second proof above, and its route through the maps $q_F : \mathcal{G}_{\text{nat}} \rightarrow \mathbb{Z}[T_\alpha : \alpha \in F]$

(F finite) and the universal theorem is the first (its corollary for rings “separated by set-sized ring targets” is [Theorem 5.12](#)). 13 adds that the inclusion $\mathcal{G}_{\text{nat}} \subset \mathbf{Oz}$ has no unital ring retraction, which is immediate since every unital map has image $\mathbb{Z}$, and derives (17.1) from Cantor normal forms in the same way. 13 records that the question writes the ring as $K[\Omega]$, with K a Grothendieck construction; that notation was not rechecked for this merge. 13 does not claim to answer the original transcendence-degree question, and states that its priority as a first answer is not established.

18 Boundaries, non-claims and open questions

18.1 What the results do not say

The universal theorem is a statement about homomorphisms into set-sized rings and about set-sized modules. It is not an isomorphism $\mathbf{Oz} \cong \mathbb{Z}$; it does not classify class ideals, class modules, automorphisms or factorizations of $\mathbf{Oz}$; it concerns multiplicative maps and says nothing against additive or valuation-theoretic observations; and it fails verbatim for every fixed set-sized exponent group, whose identity map is a set-sized target. Size alone does not prove it (finite supports have proper-class many monomials and an augmentation), nor does divisibility alone (only targets of positive characteristic are handled). The set-sized thresholds are exact for the five constructions and the two fixed-group examples treated; no classification of exponent groups or support restrictions is claimed, and generator counts follow no general law stated here. The large-quotient results describe specified subalgebras of $\mathbf{Oz}/(1 + \omega^a)$ and conditional statements about primes; no existence theorem for nonarithmetic class primes is claimed. The later additions do not change this. The flat and projective dimensions of [Section 13](#) are those of set-sized finite-support cores, of coefficient-lattice modules and of the rings of [Section 12](#); they say nothing about the homological dimensions of the class ring $\mathbf{Oz}$, and a flat dimension computed for one subring of $\mathbf{Oz}$ bounds nothing for another ([Remark 13.16](#)). The finite-support cores are not invisible to set-sized rings, only to finite ones ([Proposition 13.2](#)). The fresh-scale theorems concern polynomials and rational functions with finitely many coefficients and use the whole surreal exponent class, or a fixed Hahn workspace in which the coefficient group generates a proper convex subgroup ([Proposition 16.86](#)). Nor do the third-batch additions: the relation counts and Tor groups of [Section 13.9](#) concern the lattice ideals $J_g(M)$ and constant matrices at one common scale, not arbitrary finitely generated ideals; the ordinary-output dichotomy is a one-variable statement; and the fibre theorem of [Section 15.8](#) gives no membership test for $\mathcal{N}$. The fourth-batch addition determines of $\text{Idem}(\mathcal{N}/\Pi\mathcal{N})$ that it is nontrivial, atomless and a proper class with independent families of every set size, not its isomorphism type, and its descent theorem is not a decision procedure; its fixed-workspace statements concern reduced fibres and assert neither radicality of $\Pi(G)\mathcal{N}(G)$ nor atomlessness of $\text{Idem}(\mathcal{C}(G))$ ([Section 15.9](#); status of batch 32: for cyclic G the seventh-batch addition decides both, the first negatively and the second positively, [Section 15.10](#)). Nor does the fifth-batch addition: its exact projective dimensions are those of the set-sized cores A_d and of specified modules over them, not global or weak global dimensions and not those of $\mathbf{Oz}$; its Tor statements over the class ring are read in two universes; and its reconstruction of an order needs the framed boundary data ([Section 13.10](#)). Nor does the sixth-batch addition: its certificates are relative to exact arithmetic, its rational bound and its Gaussian grid are not claimed optimal, its birational corollary assumes a rational inverse, and its fixed-workspace criterion says when its method applies, not that finite tests fail otherwise ([Section 16.10](#)). Nor does the seventh-batch addition: it concerns one cyclic exponent group, its root refinements and their bounded-denominator union, not arbitrary groups or support classes or the Hahn field over $\mathbb{Q}a$; its residue criterion is a criterion, not an algorithm; of the uncountable Boolean algebras it determines

atomlessness, cardinality and order density, not the isomorphism type; and its periodic resolution and infinite dimensions concern the coefficient fibre, not the arithmetic fibre or $\mathbf{Oz}$ (Section 15.10).

As an internal ring, $\mathbf{Oz}$ has all surreal scales available through its fraction field, a nonzero proper-class ideal Π , nonzero class-valued derivations and nonarithmetic prime quotients; as a source of representations on sets it carries no more scalar information than $\mathbb{Z}$. The gap is caused not by ignoring limits but by finite identities that become available once the source contains the appropriate Hahn sums (06). Adapting 06's list of hypotheses that must remain visible to all the constructions, the proofs use four different size restrictions, each for its own reason: individual supports are sets, so that a common positive gap exists; the full class of smaller scales supplies arbitrarily large collision families; in the set-sized models an uncountable bound admits the countable witnesses, and regularity (or cofinality, or the finite-coordinate invariant) bounds unions of fewer than the relevant number of supports; and cardinal arithmetic enters only in computing the size of the ring, which is the exact threshold. Dropping any of them needs a new argument: finite supports lack the witnesses (Section 9), full supports over a fixed set group need not have gaps (Remark 11.21), and singular support bounds may preserve field closure but not the simultaneous gap property.

18.2 Non-claims of the sources

Every limitation stated by a source survives. They are listed and numbered by source; several recur in the text where the corresponding result is proved.

03. (1) Not peer reviewed; no Lean declarations. (2) “Developed here” is relative to the inspected material; no certified priority. (3) Neither Conway’s general refinement problem nor the general factorization problems for omnific integers are settled. (4) No classification of exponent groups or support restrictions; the cardinal theorem is a realization theorem for its family. (5) cf $\kappa = \aleph_0$ is excluded. (6) Only the lower bound μ on the number of algebra generators of $\mathcal{N}_\kappa$ is proved. (7) No complete integrality test for mixed supports. (8) No descent equality $\mathcal{N} \cap F_\kappa^{03} = \mathcal{N}_\kappa$ (the equality holds by Corollary 15.59, a merge result from 20’s support projection). (9) No statement on proper-class-valued valuations. (10) Infinite set-sized images of the normalizations are not classified, and the absence of finite images is a general consequence of containing the real algebraic integers (19 has since classified the domain images, Theorem 15.46; images with zero divisors remain unclassified). (11) No class tensor products and no class prime-ideal theorem; all Tor is over set-sized rings. (12) The finite checks do not verify the cardinal or class arguments. (13) Restricted Hahn fields, truncation integer parts, complete-integral-closure terminology and the small-unit lemma are standard or elementary, with no priority claim; Kuhlmann is context only. (14) The full-class theorem, common divisors and clearing are credited to the companion manuscript (05). (15) Density produces no convergent set-indexed net: every set-generated intermediate algebra is uniformly discrete, so the dense normalization needs more than any set of generators. (16) $\frac{1}{2}\mathcal{N}[i]$ is not asserted to be a ring. (17) The flatness result is an application of standard module theory, not a new general flatness theorem. (18) Density does not impose continuity on abstract ring homomorphisms. (19) No claim about unrestricted surcomplex analytic continuation is made.

04. (1) Unrefereed; no Lean formalization; the universal theorems are candidate contributions without certified priority. (2) No classification of all class ideals, no class Zorn argument, no claim that proper class ideals extend to maximal ones. (3) The module equivalence covers set-sized modules only, not class modules, group representations of $\mathrm{GL}_n(\mathbf{Oz})$, or additive homomorphisms. (4) It does not say that $P(\mathrm{ct}(x)) = 0$ implies $P(x) = 0$. (5) It does not classify infinite points of

curves; the earlier project's curve questions are not declared solved. (6) Nothing is inferred about fraction fields of the bounded-support rings. (7) The fixed-workspace results are lower bounds only. (8) No named open conjecture is solved. (9) Additive coefficient maps escape the theorem and are not counterexamples. (10) The telescope has a direct precursor [23, Remark 3.4.4]; the constant term, finite quotients and common shifts come from the prior manuscript (02 of [46]). (11) The existence of standard part is classical and in the repository's infrastructure. (12) The support-threshold theorem differs from fixed-group completion or omitted-type questions. (13) The finite checks do not prove normal forms, infinite identities, class factorizations or novelty. (14) The quotient, module, derivation and localization results are a connected theorem package deduced from the universal theorems, not unrelated discoveries.

06. (1) Unrefereed, not Lean-verified; priority not established by the targeted searches. (2) $\text{pd } \mathbb{Z}$ is shown only to be at least 2. (3) Residue fields of nonarithmetic primes are not classified; the prime-reservoir proposition is a necessary condition, not an existence theorem. (4) The primality of $\omega^{\sqrt{2}} + \omega + 1$ is imported; no factorization, refinement, GCD or unique-factorization conjecture is used or claimed. (5) No automorphisms, class ideals or factorization patterns are described. (6) No elementarity claim for $A_\kappa^{06} \subseteq \mathbf{Oz}$; no compactness argument for a proper-class theory. (7) The Euler derivation is not asserted to be the Berarducci–Mantova derivation or compatible with exponentiation. (8) Real closedness of bounded-support Hahn fields is not a novelty claim, and regularity is not claimed necessary for field closure. (9) The idea of a universal set-sized quotient is not new to the repository. (10) The module correspondence is not a Morita equivalence of unrestricted module theories. (11) The invisible prime quotient does not say that a particular maximal class ideal fails to exist. (12) The polynomial-test theorem does not assert that a proper-class coordinate ring defines an ordinary scheme. (13) The finite-congruence observation is elementary and not a strong novelty claim. (14) Numerical tests and a successful build are not proofs; the formalization route was not executed.

07. (1) Conway's refinement conjecture is neither settled nor used (the Abramov development [1] is cited for its claim and caveats only); no historical priority; no referee; no Lean. (2) The Cantor algebra is a specified subalgebra; neither the idempotents nor the prime ideals of $\mathcal{Q}_a$ are classified, and $\mathcal{Q}_a$ is not claimed to be a function ring. (3) The statement on nonarithmetic primes is conditional; no existence is asserted. (4) Surreal self-similarity is credited to Bagayoko–van der Hoeven. (5) Θ_a is not claimed to preserve the ordinal embedding, birthdays, simplicity or the exponential; ϑ is not additive and Ψ not multiplicative. (6) Quotient, completion and module classifications are consequences, not separate major discoveries. (7) The small-image theorem does not say that no set-valued invariant sees positive exponents. (8) The module equivalence is not a homological-dimension statement over class modules. (9) The global theorem is not a theorem about small approximations of $\mathbf{Oz}$. (10) The finite checks do not verify the transfinite or ambient-support arguments. (11) The CRT isomorphisms at different stages need not be coherent; evaluation at a point of $\mathbb{Z}_2$ is not claimed to extend to $\mathcal{Q}_a$; no topology on $\mathcal{E}_a$ is used. (12) Class prime ideals are not classified, and no Zorn argument for a proper class of ideals is used. (13) No exponential function on surcomplex arguments is assumed.

08. (1) Candidate-original, not certified priority, not refereed, no Lean; the broad factorization questions for $\mathbf{Oz}$ are not addressed. (2) $\text{Frac}(A_\kappa^{08}) = F_\kappa^{08}$ is not claimed. (3) Residue-field isomorphism types are not classified. (4) No closure under derivations or exponentials, no initial-segment or birthday condition on the embedding. (5) The collision mechanism has published

precedents ([23, Proposition 8.2.1]); the constant retraction and finite reductions are not principal novelty claims. (6) Fixed Hahn workspaces lie outside the theorem (an Archimedean group has no small scales at all). (7) The non-integral-closure corollary is not a factorization theorem. (8) Class-sized modules are not classified; the theorem is not an isomorphism with $\mathbb{Z}$, and individual monomials can be represented faithfully in small rings. (9) The Hahn derivations D_η are not an exponential-compatible calculus. (10) No maximal-ideal theorem for proper-class rings; no assertion that proper class ideals lie in maximal ones. (11) The quantitative criterion gives sufficient conditions only. (12) The finite checks prove neither the class theorem, nor summability, nor real closedness. (13) The primality of $\omega^{\sqrt{2}} + \omega + 1$ is an imported published theorem.

09. (1) Novelty provisional after a targeted comparison; no referee; no proof-assistant verification. (2) No named factorization conjecture solved; primality imported. (3) The number of maximal ideals and the isomorphism types of the residue fields of $A_{\lambda,D}^{09}$ are not determined. (4) No assertion about the unrestricted (uncountable-support) ring over Γ_λ^{09} . (5) No class Zorn argument; the Jacobson statements concern set-sized rings. (6) $\mathbf{Oz}[i]$ is a subring of the surcomplex field, not the field. (7) The checks do not verify infinite supports, cardinal arithmetic or the existence of maximal ideals. (8) The geometric identity is elementary with precedents; profinite, derivation and flatness statements are consequences; the Gaussian 2-torsion computation is standard. (9) No global surcomplex exponential, sine, integration or topology is used; the arithmetic quotient is not an ordered quotient. (10) The rotation quotient is a conceptual predecessor, not imported. (11) No new exponential derivation is constructed and no uniqueness of the Berarducci–Mantova derivation is claimed. (12) $I/I^2 = 0$ is not a finite-presentation or smoothness claim.

10. (1) Unrefereed and not machine-checked; targeted review only; no named factorization conjecture solved. (2) For $r \geq 1$ the p -adic completion is not a profinite completion. (3) No classification of all class ideals, automorphisms or factorizations. (4) No transfer principle from $\mathbb{Z}$ to $\mathbf{Oz}$. (5) $\text{Int}(\mathbb{Z}[i]^r)$ is not replaced by a binomial basis. (6) The derivation D_h is not the Berarducci–Mantova derivation. (7) The universe-relative reading must carry the size restriction in its statement. (8) The rational classification is not a classification of arbitrary functions or of coordinatewise conjugation. (9) The congruence theorem does not make reduction modulo an infinite omnific integer an ordinary finite congruence, and does not classify congruence-preserving class functions. (10) Classical ingredients (Newton interpolation, binomial rings, the least-common-multiple pattern, p -adic approximation) are credited, not claimed. (11) The grid test assumes exact surreal input; no effective encoding is claimed. (12) The finite checks do not test class functions, support lemmas or priority.

11. (1) The main theorem is relative to one fixed universe, not to fragments viewed in a larger universe. (2) No classification of prime or maximal class ideals. (3) No class module of Kähler differentials. (4) The presentation theorem is not quantifier elimination or a reconstruction of class-valued solutions. (5) Sharpness of the coinitality theorem is stated under $\kappa^{<\kappa} = \kappa$ and $|K| \leq \kappa$ as explicit extra assumptions (the unconditional value is a merge result, [Proposition 11.23](#)). (6) Necessary closure conditions for intermediate rings are not given. (7) The ordinal answer is not claimed hard or unknown to specialists. (8) Bounded-support coefficient rings and lower-scale geometric division have precedents in [23, Section 8]. (9) Targeted novelty assessment only; the finite checks prove no infinite-support or class-size statement. (10) The rotation-group theme is acknowledged, not originated. (11) The only answer to the MathOverflow question concerns

transcendence degree. (12) No primality or refinement conjecture is newly settled; the ideals, class-valued representations and factorization theory of quotients invisible to set-sized rings are not determined, and ordinary factorization questions about **Oz** remain outside the conclusions.

12. (1) Nothing is claimed about the homological dimensions of the full proper-class ring **Oz**; all rings are set-sized and all embeddings use finite normal forms. (2) The weak global dimension of A_d is not computed; only the lower bound $d + 1$ is proved. (3) Projective dimensions are given only as bounds (the exact values in rank one are merge results; 21 has since determined them for every d , [Corollary 13.47](#) and [Theorem 13.48](#)). (4) Arbitrary non-simplicial cones, arbitrary ideals and all homological localizations of the cores are not classified. (5) The cores are not integer parts of **No**, not initial in the simplicity order, not preserved by all surreal automorphisms and not Hahn rings; the countable-rank ring still consists of finite sums. (6) $\hat{A}$ is the profinite completion, not a Hahn completion or a completion in a surreal-valued topology. (7) The $D + M$ framework (with the pullback literature, for instance [41]), filtered-colimit flatness, coherence criteria and the idempotent-kernel mechanism of formal étaleness are classical and credited; the proposed contributions are the Boolean Tor computation, its dimension consequences, the countable-rank realization and the ordered-cone comparison. (8) No named conjecture is claimed solved; the originality check was targeted, not an exhaustive priority audit. (9) Not refereed and not checked in Lean. (10) The finite audit uses $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$ on a five-point grid and proves nothing over $\mathbb{Z}$ or for all exponents. (11) The Gaussian version uses only the quadratic extension **No** $[i]$; no analytic continuation and no topology on a proper class is used. (12) “Nilpotent” is a test condition on algebras; the cores are domains.

13. (1) The theorems are relative to size; nothing is claimed about all maps out of a fixed set-sized Hahn ring, whose identity is a counterexample ($\mathbb{Z} + X\mathbb{R}[X]$ with evaluation at 1 is 13’s example). (2) No classification of class quotients, class residue objects or class ideals, and no maximal-ideal theorem or Zorn argument for class rings. (3) The ordinal result answers only the quotient addendum; the original transcendence-degree question has Wofsey’s answer; 13’s priority as a first answer is not established. (4) The integer divisibility of the purely infinite part is known (L’Innocente–Mantova); equation-only transfer along a retraction, filtered-colimit flatness and the algebra of idempotent quotient ideals are standard, and the homological and lifting statements are standard consequences, with no étaleness claimed under finite presentation. (5) The fiber theorem (printed in [46]) classifies only nonzero fibers of decomposable polynomials; there is no general classification of fibers, no Thue or Baker finiteness, and no enumeration of the ordinary solutions of its cubic. (6) The cardinal criterion is a necessary condition, not an existence theorem for detecting maps. (7) The statement on Hilbert’s tenth problem is a transfer of the classical theorem, with no complexity claim for Gaussian or number-field problems. (8) No claim about Conway’s refinement conjecture; no refinement, primality or unique-factorization property is used. (9) In the two-universe reading “not generated by a set” means not generated by a family from the smaller universe. (10) Unrefereed; no Lean; the finite checks verify identities only; the literature and repository searches were bounded, and the repository was neither audited nor built.

14. (1) Arbitrary finitely generated omnific ideals, with generators at several support levels, are not classified. (2) The rank-one modules with double dual B are not all listed. (3) No coefficient ring other than a principal ideal domain is treated. (4) The exact weak global dimension of A_0 is not claimed. (5) The ideal-class group under $\odot$ is not a subgroup of $\text{Pic}(A)$, and its identity is $L(\mathfrak{o})$, not A . (6) Full-class tensor products are explicit carriers only, and ordinary flat dimensions are asserted only over A_0 . (7) No novelty is attached to noncoherence; the $D + M$ context is credited

to Dobbs–Papick, and no valuation-ring hypothesis is used. (8) The number field spanned by a multiplier ring need not be totally real. (9) The unimodular-direction criterion concerns coordinate representatives, not all points of a projective scheme over a ring, and its example $[1 : \sqrt{2}]$ is not new. (10) The matrix theorem concerns constant matrices at one common monomial scale. (11) Constant-term arithmetic, flat idempotent support ideals and cardinal obstructions are credited to companion manuscripts and are not presented as new. (12) No Conway factorization or refinement claim; no peer review and no new Lean verification; priority is provisional; the finite checks are supplementary and certify no arbitrary support, class universal property, flatness statement or Eisenstein criterion in all degrees.

15. (1) The main theorem is for the full class only; the finite-support ring and fixed Hahn workspaces are counterexamples. (2) Its local cardinal criterion is sufficient only. (3) No classification of class prime or maximal ideals; its prime classification carries a set-sized residue hypothesis. (4) The primality of $\omega^{\sqrt{2}} + \omega + 1$ is imported from L’Innocente–Mantova and is not a new solution of Gonshor’s question. (5) No claim about Conway’s refinement problem; the repository `gaearon/conway-refinement` was noticed but neither built, audited nor relied on. (6) Only set-sized modules are classified; this is not a Morita statement. (7) Universe parameters must stay separate. (8) Geometric factorization and the congruence calculations are classical or elementary, the categorical deductions are not new techniques, and the rotation-group theorem is a precedent, not a dependency. (9) Blocking one extension of the finite-support augmentation is weaker than the theorem. (10) Embeddings of set-sized rings into $\mathbf{Oz}$ are not excluded, and $\mathbf{Oz}$ and $\mathbb{Z}$ are not claimed to have the same first-order theory. (11) Candidate originality rests on a bounded search; no referee; no Lean; the checks cover finite identities only.

16. (1) Rational-to-polynomial collapse is not new in principle (D-rings: Gunji–McQuillan, as recalled by Liu; related framework in [10]). (2) Surjectivity rigidity covers one variable and rational functions only; conjugation, multivariate triangular bijections, class functions, definable maps and formal power series are excluded, and not every bijection of $\mathbf{Oz}$ is claimed affine. (3) The fresh-scale arguments do not transfer to a fixed Hahn field without dominant exponents. (4) The certificates are relative to exact coefficient data, zero tests and a certificate of freshness; they are not a decision procedure for arbitrary surreal inputs. (5) No convergence, analytic continuation, surreal exponential, Berarducci–Mantova derivation, fine-topological completeness or factorization is used. (6) Newton interpolation and binomial bases are classical, and the Gaussian test is not claimed minimal. (7) The fiber theorem classifies the particular fresh fiber, not every fiber. (8) Priority rests on a bounded search; no referee; no Lean; the finite checks are not proofs.

17. (1) No Lean formalization and no peer review; no named factorization conjecture is resolved, and no factorization property is inferred from noncoherence. (2) The kernel theorem classifies constant matrices after one common monomial scaling, not matrices with several leading scales or positive tails; ideals not of the form $J_g(M)$ are not classified. (3) No upper bound on the weak global dimension of R_κ^{11} , no global dimension, and no projective dimension of Π_κ^{11} or $J_g(M)$ is claimed (the last two are asked); Dobbs’s valuation-pullback theorem does not apply because $K + \Pi_\kappa^{11}$ is not a valuation ring. (4) The power-growth law counts generators only; the lattices M^n need not stabilize when their ranks do, and it is not a reconstruction theorem. (5) Over the class ring, Hom means scalar codes, not a class of class maps; ordinary Tor, flat resolutions and weak dimensions are taken over set-sized models only, and no full-class Tor is inferred from the statement on modules killed by the tail. (6) Neither inversion closure of R_κ^{11} nor its equality with a Hahn field is asserted;

nothing is said about unrestricted Hahn series over an arbitrary divisible group, and the coinitality of the positive cone is not the cofinality of the group. (7) The appendix proof of the universal theorem is background, not a novelty candidate. (8) The $D + M$ noncoherence mechanism and the relation module M^{r-1} are classical (Dobbs, Dobbs–Papick); the pullback literature may contain antecedents of the Hom and tensor formulas; the audit of the repository and of the literature was targeted, not exhaustive. (9) The rank algorithm assumes an exact description of the coefficient span, not floating-point data. (10) The checks are finite arithmetic; kernel bases computed over $\mathbb{Q}$ are not integral lattice bases, and no proper-class argument, regularity statement, Hahn support or homological theorem is checked. (11) A field $\text{Frac}(D)M$ in the field test does not make M a ring. (12) The support-gap argument does not transfer to unrestricted Hahn series with real exponents.

18. (1) The dichotomy is one-variable: XY defeats any bound in several variables, only the linewise statement is proved, and how the exceptional cosets on different lines fit together is open. (2) The coefficient test is an algebraic certificate relative to exact coefficient operations, not an algorithm for arbitrary surreal representations. (3) Rational collapse is not new (D-rings: Gunji–McQuillan, recalled by Liu); the description of the integer-valued polynomials and the matrix and jet reductions are applications, and the all-size matrix test is a general algebraic observation. (4) Finite testing concerns polynomials of known degree only; no finite sample tests rational functions, and entire Hahn series and nonstandard degrees are excluded. (5) The Gaussian simplex test is not claimed minimal; the results of Volkov and Petrov are classical input. (6) Surjections are classified among rational maps of one variable only, not arbitrary bijections, automorphisms of the surreal field or polynomial automorphisms in several variables; injectivity alone is weaker. (7) Nothing is claimed about **No** or the finite surreals, where the discrete gap fails, and no factorization conjecture is addressed. (8) The checks cover finite-support models $\mathbb{Q}[H]$ and $\mathbb{Q}(i)[H]$ only and prove neither the cluster lemma, nor the proper-class reading, nor novelty. (9) Novelty rests on a bounded search; no referee; no Lean. (10) The sharp examples need an infinite element, and for a general discretely ordered ring finite testability of preservation must not be assumed.

19. (1) $\text{Idem}(\mathbb{B}_{\mathbb{R}})$ is not determined: its size, its atoms, whether it is trivial and whether it is a set (20 has since shown that it is nontrivial, atomless and a proper class, [Theorem 15.73](#); its isomorphism type is still not determined). (2) There is no membership criterion for $\mathcal{N}$ by normal forms (20 adds support descent and the one-scale rational case, [Theorems 15.57](#) and [15.60](#), but no general criterion). (3) $\mathbf{B}_{\bullet}$ is neither one finite product nor an unrestricted product of copies of $\mathbb{Z}$. (4) The real-case coefficient embedding depends on the choice of $\mathbf{v}$; no embedding of $\mathbb{C}$ into **No** is asserted. (5) No Bézout property of **Oz** or of the normalizations is claimed. (6) The characteristic- p fibres are neither reduced nor fields; the mod-2 splitting is a product of nonreduced rings. (7) Set-sized images with zero divisors are classified only up to the factorization through $\mathbf{B}_{\bullet}$; not all set-sized images are claimed countable; $\mathbf{B}_{\bullet}$ is not asserted to be a set (20 has since proved it a proper class), and no universal object in a category of set-sized rings is claimed. (8) The kernel detection uses an explicit foundation (class theory with global choice, no class Zorn lemma, no measurable cardinals), and its intersections are pointwise kernel characterizations, not sets of class maps. (9) In the finite-axiom form, v_T need not be infinitesimal, non-set-generation needs the full class, the small-image theorem needs a cardinal condition absent from the axioms, and nothing transfers to a fixed Hahn field, where the support $\{1/n\}$ blocks the monomial proof without being a counterexample (20’s reduced-fibre result, [Proposition 15.76](#), avoids the axioms and decides no radicality; 23 has since shown that radicality fails in the cyclic workspace, which is outside the axioms, [Remark 15.119](#)). (10) Integrality, absolute integral closure, Boolean powers and the Bézout

property of $\overline{\mathbb{Z}}$ are standard, and the small-image theorem is credited; priority is not established. (11) $\mathcal{N}$ itself is not absolutely integrally closed, lifted roots need not stay distinct modulo $\Pi\mathcal{N}$, and norm-one units of the fibre are not asserted to lift. (12) No named factorization conjecture is resolved; no referee; no Lean; the checks are finite identities and rational inequalities.

20. (1) The class isomorphism type of $\text{Idem}(\mathbf{B}_{\mathbb{R}})$ is not determined: atomlessness and free subalgebras of every set size do not make the class algebra free, complete or saturated for arbitrary Boolean types, and give no arbitrary infinite joins; no isomorphism $\mathbf{B}_{\mathbb{R}} \cong \mathbf{B}_{\mathbb{R}} \times \mathbf{B}_{\mathbb{R}}$ is claimed (the proved doubling is $\mathbf{B}_{\mathbb{C}} \cong \mathbf{B}_{\mathbb{R}} \times \mathbf{B}_{\mathbb{R}}$). (2) The fixed-workspace statements use $\sqrt{\Pi(G)\mathcal{N}(G)}$; $\Pi(G)\mathcal{N}(G)$ is not asserted radical in any fixed workspace, no atomlessness of $\text{Idem}(\mathbf{C}(G))$ for a fixed G is asserted, and the rank extension needs a new dominant coordinate, which a fixed workspace need not supply (23 has since decided both for every cyclic G , without a new coordinate: $\Pi(G)\mathcal{N}(G)$ is not radical and $\text{Idem}(\mathbf{C}(G))$ is atomless, [Theorem 15.86](#) and [Corollary 15.110](#)). (3) Support descent is not a decision algorithm; effective membership for general infinite supports is open, and the exact rational criterion concerns one monomial scale. (4) The matrix corollary computes no projective or global dimension and forms no category of class-sized modules; the higher projective dimensions of this report ([Question 18.3](#)) are not addressed. (5) $\sigma_{\pm}$ are specializations of a finite branch algebra, not evaluations of Laurent series at $T = 0$ or maps on the Laurent field; pr_G is linear, not a ring retraction; no evaluation map on all surreal numbers is used. (6) An idempotent residue does not say that its representative is idempotent; the quotient is not standard part. (7) The free algebra over an infinite index set is a directed union of finite products, not an unrestricted product. (8) The foundation is class theory with global choice; no set of all class homomorphisms or prime class ideals is formed, no set-sized Zorn argument is applied to a proper class, set-indexed families of class maps are single uniform class relations, and statements about all maps are pointwise. (9) The coefficient copy of $\overline{\mathbb{Z}}$ is not canonical. (10) Radicality, saturation, the Boolean power, integral doubling and domain-map detection are 19's and the small-image theorem is this report's, reproved and not claimed; general Boolean powers, multiquadratic extensions, the determinant trick, coefficient projection as a module map and ultrafilter methods are not new; it is not asserted that 19 was in the repository at 20's pin. (11) The questions resolved are those of 19, not a named conjecture; no Gonshor prime-element, refinement or other named factorization problem is solved. (12) The source comparison was targeted; historical priority is not established. (13) The finite checks verify polynomial identities, branch specializations, finite Boolean tables, finite support projections and initial Laurent coefficients, not class or set arguments, Hahn support theorems, arbitrary supports, integrality of arbitrary surreals or novelty. (14) No Lean formalization, no repository build, no peer review; Python and SymPy versions other than those recorded (3.13.5 and 1.14.0) were not tested.

21. (1) Unrefereed and AI-assisted; no Lean or other proof-assistant check; priority is not certified: the dimension question is answered relative to the inspected repository version, the literature search was targeted, and unintegrated companion manuscripts and historical sources were not exhaustively audited. (2) No major named factorization conjecture is claimed solved. (3) Arbitrary omnific ideals are not classified. (4) The value $d + 1$ concerns the specific set-sized core A_d , not $\text{pd } \mathbb{Z}$ over the class ring; no exact global or weak global dimension of $\mathbf{Oz}$ or of A_d is claimed, only the lower bounds $d + 2$ and $d + 1$. (5) The class rings are read under a two-universe convention, not as literal proper-class rings of ordinary set theory; the core theorems and all examples are ordinary set-sized mathematics. (6) The finite checks do not prove the infinite Ext nonvanishing, exactness in unbounded degree or the normal-form foundations, and replace no proof assistant. (7) Filtered colimits of flat modules, flat models of derived tensor products, countable telescopes (Positselski [28]) and the non-Noetherian

dimension methods for root-adjoining rings (Asgharzadeh [2]) are standard and not claimed. (8) The additive second-Tor formula is a dimension-shift consequence of the report's tensor calculation, not an independent discovery; the flat face resolution, $\text{fd}_{A_d} D = d$, the lattice syzygies and the tensor torsion are the report's and are credited. (9) The D -module decomposition of Tor_1 is not an A -module splitting. (10) The vanishing of Tor_2 is not Tor-independence ($\text{Tor}_1 \neq 0$). (11) The disjointness test creates no field extensions and does not decide linear disjointness without access to the arithmetic. (12) Reconstruction needs the boundary embedding and the distinguished 1; an abstract Tor_2 group, or an abstract derived equivalence, is not claimed to recover an order. (13) The DGA model is not claimed cofibrant or E_∞ -formal, and it determines no higher operations. (14) The minimal polynomial of an arbitrary element of degree at least three cannot be read from a two-generator Tor_2 . (15) $\text{Tor}_{>0}(D, E) = 0$ for every module E killed by the tail, despite $\text{pd } D = d + 1$; the nonvanishing uses the target A_d . (16) On the finite-support core the augmentation to the coefficient field exists, so “every set-sized target factors through ct ” is false there; the shadow theorem concerns the specified constant-term modules. (17) The resolutions control the specified face, lattice and quotient modules, not all ideals; the length-two flat resolution and the Tor calculations are asserted for $T_{\mathbb{Q}}$ and the class ring, not for A_d with $d \geq 2$, whose tail is not flat.

22. (1) AI-assisted and unrefereed, with complete written arguments; no Lean build, no Wolfram computation and no independent referee review. (2) No named longstanding conjecture is claimed settled; the finite-certificate and no-universal-set package is a proposed contribution, its priority is not certified, and its novelty statement is a qualified assessment of selected material (and is incomplete, Section 1.4). (3) The bare collapse of integer-valued rational functions of one variable has the D-ring antecedent (Gunji–McQuillan, through Liu's Proposition 1.12, the 1976 paper not read independently; the domain must be infinite), and the Newton basis and the binomial-ring viewpoint are classical (Elliott [17], Cahen–Chabert [8], the monograph not re-read in full); neither is claimed. (4) Normal forms, Hahn operations and $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ (L’Innocente–Mantova, Proposition 2.4.5) and the repository's splitting, constant-term and fraction background are inherited; no unreviewed geometric assertion of the repository is imported. (5) The certificates are relative to exact field operations and exact membership in $\mathbf{Oz}$ or $\mathbf{Oz}[i]$, not algorithms for arbitrary descriptions of surreal numbers; equality, gcd, support bounds and membership are not presumed decidable, and choosing b above an arbitrary support is a construction, not an effective operation. (6) Neither the rational bound $(d + 1)^m + m(2d^2 + 2d + 1)^{m-1}$ nor the Gaussian grid is claimed optimal, and the anisotropic refinement is available only after the presentation has been inspected. (7) The lower bound $(d + 1)^m$ concerns fixed point-value tests chosen independently of the coefficients, not algorithms that inspect the coefficients. (8) Point counts count exact evaluations, not bit operations or floating-point samples. (9) The no-set-test theorem needs the full proper class and fails for set-sized rings; the fixed-workspace extension needs an available fresh exponent, and its failure for Archimedean value groups is a failure of the method, not a proof that every finite test fails. (10) Arbitrary functions, entire Hahn series, surreal exponentials, infinite-dimensional operators and Diophantine varieties are not classified, nor are algebraic or semialgebraic maps ($|X|$ preserves $\mathbf{Oz}$ and is only piecewise polynomial). (11) The birational corollary assumes a rational inverse in both directions; no bijective polynomial map is claimed to have one, no Jacobian-conjecture claim is made, the coordinates need not have coefficients in Λ , and the automorphism group is not classified. (12) Image ideals are not assumed principal, and no greatest-common-divisor conclusion is drawn. (13) The tensor isomorphisms are read through finite sums of pure tensors or in a containing universe; they do not assert that $\mathbf{Oz}$ is a set. (14) Gaussian arithmetic is not replaced by a Gaussian binomial basis; the ordinary Gaussian universal-set problem (Byszewski–Frączyk–Szumowicz [7], of which only the

abstract and the bibliographic record were inspected; Volkov–Petrov [40]) is neither solved nor presented as newly discovered. (15) ω^γ is the normal-form monomial, not an analytic exponential; no algebraic closedness of $\mathbf{No}[z]$, differentiation or topology is used; ct is not a global standard part and is multiplicative only on nonnegative supports. (16) The 1,776 finite assertions (exact algebra and sampled lexicographic inequalities) verify no arbitrary support, infinite Hahn summability, cut above an arbitrary set, class-sized quantifier, proper-class alternative or originality; the lexicographic model illustrates the quantification over all exponents and does not replace it. (17) Only selected repository files were read, some responses were truncated, the repository was neither checked out nor built, and an empty search was not taken as evidence of absence. (18) Independent scrutiny may find simplifications, broader antecedents or errors.

23. (1) The answers concern the cyclic exponent groups $\mathbb{Z}a$ only, for the real pair and its Gaussian counterpart; they classify neither arbitrary ordered exponent groups nor support restrictions, and the proof does not classify the groups with a discrete convex component. (2) The Hahn field over $\mathbb{Q}a$ is open; the Puiseux result concerns bounded denominators only, is not an identification with that Hahn field, and does not say that the limit fibre is isomorphic to a reduced finite-stage fibre, whose image in it may grow. (3) The Boolean isomorphism types for $k = \mathbb{R}$ and $\mathbb{C}$ are not determined: the independent family does not make the algebra free, and neither completeness, nor infinite joins, nor a product representation of the ring, nor a free presentation follows from the count of $2^{2^{\aleph_0}}$ branches. (4) The weak global and global dimensions of the arithmetic fibre are open; the periodic resolution is exact only over the coefficient fibre, and no change-of-rings statement is made for the arithmetic fibre or for $\mathbf{Oz}$. (5) The residue criterion is a finite algebraic criterion for represented algebraic Laurent elements, not a decision procedure for arbitrary real coefficients or arbitrary surreal normal forms; substituting $T = 0$ in a singular equation need not separate branches. (6) The integer part of the Laurent field does not extend ct to the normalization; no single refinement removes all nilpotents; the ramification filtration is not a valuation. (7) The finite checks (18,040 assertions in 12 categories, most of them clipped-order compositions and threshold checks) count assertions, not theorems; they establish neither integral closure, nor the existence of finite places or prime extensions, nor arbitrary-support summability, nor the normal-form embedding, nor cardinalities, nor exactness of an infinite resolution. (8) AI-assisted and unrefereed; no Lean or other proof-assistant check, no repository build and no independent review; a compiled PDF is a document check. (9) Priority is not certified: the literature comparison was targeted, and the repository comparison covered selected sections of this report, not all reports or the history; nothing claims to settle a named factorization conjecture, uniqueness of surreal factorization, a global surcomplex analytic calculus or the general theory of surreal supports. (10) Integral dependence, normalization of curves, valuation domains, quadratic square classes and periodic resolutions are classical; the residue criterion is a proved pullback-style tool, not a claim to have invented pullback normalization; the period-two mechanism has antecedents in matrix factorizations; the Newton–Puiseux theorem, the lifting of idempotents across nil ideals and the uniqueness of the countable atomless Boolean algebra are classical and not claimed. (11) No continuum hypothesis and no regularity of $2^{\aleph_0}$ are used; all objects are sets, with ordinary choice and no class Zorn lemma, class module category or measurable cardinal; the repository’s descent theorem is compared, not used, and no unrefereed repository theorem is an input. (12) The questions concern precisely specified rings and do not assert that the cyclic calculation settles an unrestricted Hahn problem; divisibility of the exponent group alone is not a proof of radicality. (13) The formalization plan claims no implemented declaration and no repository build.

The merge. (1) The results marked [merge] are consequences of the sources' lemmas; they are no more refereed than the sources and are not formalized. (2) No general law is asserted for the generator counts. (3) For full-support Hahn rings over a prescribed set group, exact thresholds are computed only in [Examples 10.4](#) and [10.5](#); the general questions of 04 and 10 stay open. (4) The support-threshold theorem is stated for $k \in \{\mathbb{R}, \mathbb{C}\}$ as in 04 and 11, not for every set field. (5) No Gaussian analogue of 10's congruence-preservation criterion is given. (6) Nothing is claimed about the homological dimensions of the class ring $\mathbf{Oz}$; the weak global dimensions of the cores remain undetermined ($\mathrm{pd}_{A_d} D$ for $d \geq 2$, undetermined before the fifth batch, is $d + 1$ by 21's [Corollary 13.47](#)). (7) The finite-support cores of 12 and 14 are outside [Theorem 5.1](#): they have set-sized images detecting the tail. (8) [Proposition 13.12\(v\)](#) needs a countably generated tail; among the five constructions it settles only A_κ^{08} with $\mathrm{cf} \kappa = \aleph_0$. (9) The extensions of 17's results to $\kappa = \aleph_0$ and of its Tor formula to the cores say nothing about the class ring's homological dimensions, and $R_{\aleph_0}^{11}$, like the cores, is outside [Theorem 5.1](#). (10) The comparison of Gaussian tests concerns the sizes of two explicit certificates, not the minimal size of a test, and [Remark 15.52](#) by itself decides nothing about $\mathrm{Idem}(\mathbf{B}_{\mathbb{R}})$ (its candidates are decided in [Proposition 15.68](#)). (11) [Corollary 15.59](#) is a descent, not a membership test for $\mathcal{N}_\kappa$, and [Proposition 15.68](#) decides the idempotents $(1 + \bar{u}_H)/2$ only for $H = \iota_b(h)$ with $h \in TK_G[T]$, in particular for positive $H \in \Pi$ supported in one cyclic group; other H are not decided. (12) [Remark 15.75](#) concerns the splitting results only; the extension of branch maps and 19's detection still use global choice. (13) The extension of 21's lattice theorem to every unital D ([Theorem 13.48](#)) and [Remark 13.49](#) give no projective dimension for the five constructions or for R_κ^{11} with uncountable κ ; the remark shows only that (H2) cannot be dropped from [Proposition 13.12\(v\)](#). (14) The comparison of 22's and 16's rational certificates ([Remark 16.71](#)) concerns the sizes of two explicit certificates, not the least size of a test. (15) [Remark 16.88](#) says nothing for $D = k$, and the remark after [Proposition 16.86](#) decides nothing about finite certificates in a workspace without fresh exponents. (16) The comparisons of [Remark 15.119](#) and [Example 15.87](#) concern the real pair; [Remark 15.102](#) concerns the bounded-denominator Puiseux ring only and determines neither the isomorphism type nor the order density of its Boolean algebra; [Remark 15.114](#) rests on applying the proof of [Proposition 15.76](#) verbatim to the coefficient field $\overline{\mathbb{Q}} \cap \mathbb{R}$, and nothing is claimed for noncyclic groups or for the Hahn field over $\mathbb{Q}a$.

18.3 Open questions and their status

The sources pose the following questions, each as a question suggested by the proofs rather than as an established open problem. Several coincide and are merged.

Question 18.1 (Least detecting targets over a fixed exponent group; 03, 04, 07, 08, 09, 10, 11, 13, 15). For a set-sized ordered group Γ , a family of admissible supports and a specified element (or the whole positive-support ideal), what is the least cardinality of a ring target detecting it, and which intrinsic support conditions characterize it?

Status. Open in general. The field bound ([Proposition 10.3](#)) is a lower bound that is exact for the five constructions and for the two examples of [Section 10.2](#); there it answers 04's "optimal small targets" question for its own example, and for 10's question "when is the bound attained" it shows that the bound need not be attained (in the rational example, and for $A_{\aleph_1}^{03}$ if CH fails). 08's criterion and 11's coinitality theorem give sufficient conditions (07's "sharp set-sized analogues" and 09's "quantitative problem" are the same question). 13's question (how large a target must be to detect a prescribed series, and whether its divisibility bound is optimal) and 15's "optimal local cardinal bounds" are this question as well: their bounds are the monomial count of [Proposition 10.1\(ii\)](#),

which the field bound dominates; the threshold theorem ([Theorem 10.10](#)) answers them for the five constructions, and [Example 10.5](#) shows that the monomial bound need not be attained.

Question 18.2 (Cardinal arithmetic without the matching hypothesis; 06). For κ uncountable regular with $\kappa^{<\kappa} > \kappa$, what is the least size of a target detecting a nonzero element of Π_κ^{06} , and is there an integer part of size exactly κ with the same lower bound?

Status. Answered. The least size is $\kappa^{<\kappa}$ in ZFC ([Theorem 11.11](#), a merge result from 06's collision lemma); 08 constructs, for every infinite κ in ZFC, an integer part of size exactly κ with threshold κ ([Theorem 11.16](#)), without claiming its fraction field.

Question 18.3 (Higher projective dimension; 06). What is $\text{pd}_A \mathbb{Z}$ for the set-sized integer parts, and how does it depend on the cofinality of the principal-ideal chain?

Status. Partly answered. $\text{fd}_A D = 1$ and $\text{pd}_A D \geq 2$ are known for all five constructions. If the tail is countably generated, that is if the principal-ideal chain has countable cofinality, then $\text{pd}_A D = 2$ ([Proposition 13.12\(v\)](#), a merge result from 12's telescope); among the five constructions this is the case exactly for A_κ^{08} with $\text{cf } \kappa = \aleph_0$. For chains of uncountable cofinality (03, 06, 09, 11, and 08 with $\text{cf } \kappa > \aleph_0$) the question stays open. For comparison, the ordered-cone cores have $\text{pd } D = 2$ and the coordinate-cone cores $\text{pd}_{A_d} D = d + 1$ (21, [Corollary 13.47](#)), although their tails are countably generated: they fail (H2) ([Remark 13.49](#)); 17's countable lexicographic model has $\text{pd } D = 2$ and $\text{pd } L(M) = 2$ ([Remark 13.32](#)), and 17 asks the same question for its uncountable models ([Question 18.15](#)).

Question 18.4 (Nonarithmetic primes and their residue fields; 06, 07). Which prime or maximal class ideals of $\mathbf{Oz}$ have nonzero constant-term image and zero intersection with $\mathbb{Z}$, and which class fields occur as $\text{Frac}(\mathbf{Oz}/P)$ for primes $P \not\supseteq \Pi$? Can the embedded field $\mathcal{F}_a$ distinguish such primes?

Status. Open. Necessary conditions: [Corollary 14.10](#), [Proposition 14.12](#), and [Remark 14.13](#).

Question 18.5 (The ambient binomial quotient; 07). Which normal-form invariants describe the elements and ideals of $\mathcal{Q}_a$ beyond the Cantor algebra; which idempotents lie outside it; how are cyclotomic decompositions at different scales related?

Status. Open.

Question 18.6 (Residue fields of the set-sized models; 08, 09). Which characteristic-zero fields of the threshold cardinality occur as maximal residue fields of A_κ^{08} and of $A_{\lambda,D}^{09}$, and how many maximal ideals are there?

Status. Open; existence, cardinality and at least λ nonarithmetic maximal ideals (for 09) are proved.

Question 18.7 (Extra structure on the workspace; 08). Can prescribed residue-field behaviour be combined with closure under a surreal derivation or an exponential, or with an initial embedding, without destroying the finite-coordinate invariant or the cardinality?

Status. Open.

Question 18.8 (Intermediate rings; 04, 11, 15). Which rings between the finite-support ring and $D + \Pi_k$ (for instance, given by allowed families of support order types) still have set-sized reflection D ? Is enough closure under subordinate division also necessary?

Status. Open; [Theorem 9.1](#) settles the case of cardinal support bounds. 09's question on support bounds below a regular cardinal is addressed for the groups of 06 and 11 ([Sections 11.2](#) and [11.5](#)), where the support sizes of inverses and quotients are checked.

Question 18.9 (Beyond ring observations; 04). Which weaker multiplicative compatibility conditions still force factorization through the residue, and which structured families of set-sized local models can retain a specified family of scales?

Status. Open.

Question 18.10 (Invisible class domains and automorphisms; 10, 11, 15). What are the ideals, class-valued representations and factorization theory of proper-class quotient domains invisible to all set-sized rings? How much can an automorphism of $\mathbf{Oz}$ do inside Π while fixing ct ?

Status. Partly answered (batch 31). The first sentence is open. The second is answered in the report on omnific-preserving automorphisms [47] (`opa:as:cor:invisible`, from its automatic-strongness theorem `opa:as:thm:main` and its factorization `opa:as:thm:factor`, proved by that report's sources 12 and 13 and independently by its source 14). Every ring automorphism σ of $\mathbf{Oz}$ satisfies $\text{ct} \circ \sigma = \text{ct}$, and extends uniquely to a strong automorphism of $\mathbf{No}$ fixing $\mathbb{R}$ pointwise, of the form $M_{\chi, \tau} \circ u$. Here τ is an order automorphism of the exponents, χ a positive character and u a strong leading-term-fixing automorphism preserving $\mathbf{Oz}$, classified by that report's convex-support criterion `opa:thm:criterion`. On Π it therefore acts first by u , then by an exponent automorphism and a positive rescaling of monomials; every such composition occurs.

Question 18.11 (Normalization; 03). Give a membership test for $\mathcal{N}$ on mixed supports; describe the integral closure over a fixed finitely generated exponent subgroup; decide the descent $\mathcal{N} \cap F_{\kappa}^{03} = \mathcal{N}_{\kappa}$; find the algebra generation number of $\mathcal{N}_{\kappa}$ between $\text{cf } \kappa$ and $\kappa^{\aleph_0}$; find a class-sized replacement for the valuative criterion of integrality; describe the infinite set-sized images of $\mathcal{N}$.

Status. Partly answered. For the last clause, 19 classifies the set-sized *domain* images of $\mathcal{N}$ and $\mathcal{N}_{\mathbb{C}}$ (they are $\mathbb{Z}$ and $\mathbb{F}_p$, [Theorem 15.46](#)) and shows that every set-sized image factors through $\mathcal{N}/\Pi\mathcal{N} \cong \mathbb{Z}[\text{Idem}]$ ([Corollary 15.45](#) and [Theorem 15.38](#)); images with zero divisors are not classified. 20 shows that $\text{Idem}(\mathcal{N}/\Pi\mathcal{N})$ is atomless and a proper class and that no set-indexed family of set-sized images is jointly faithful ([Theorems 15.73](#) and [15.79](#)); its isomorphism type is open ([Question 18.18](#)). The descent clause is answered affirmatively: $\mathcal{N} \cap F_{\kappa}^{03} = \mathcal{N}_{\kappa}$ ([Corollary 15.59](#), a merge result from 20's support projection). The clause on a fixed exponent subgroup is partly answered: for every set-sized subgroup G , $\mathcal{N} \cap K_G$ is the integral closure $\mathcal{N}(G)$ of the omnific integers supported in G , so integrality witnesses at larger scales never help, and for $G = \mathbb{Z}a$ the rational one-scale slice is determined exactly ([Theorems 15.57](#) and [15.60](#)); a description of $\mathcal{N}(G)$ in general is open. *Status* (batch 32). For $G = \mathbb{Z}a$ the clause is answered by 23: $\mathcal{N}(\mathbb{Z}a)$ is the set of elements of $K_{\mathbb{Z}a}$ integral over $\mathbb{R}[\omega^a]$ whose residues at the finitely many places over $\omega^a = 0$ of each finite subextension are algebraic integers, and its $\mathbb{R}$ -linear span is the integral closure of $\mathbb{R}[\omega^a]$ ([Theorems 15.91](#) and [15.93](#); the same holds for the Gaussian pair with $\mathbb{C}$ in place of $\mathbb{R}$). This is a criterion, not an algorithm ([Question 18.17\(c\)](#)), and for noncyclic finitely generated G a description stays open. The membership test, the generation number and the valuative criterion stay open; 19 asks the same membership question, for supports in one finitely generated exponent subgroup first, allowing integrality witnesses at larger scales, and 20 asks for effective support-local certificates ([Question 18.17\(c\)](#)). 04's question on Diophantine fibers belongs to the sibling report [46], and so does 13's question on fibers of polynomials with nonlinear factors.

Question 18.12 (Homological dimensions of the cores; 12, 14, 21, merge). For $d \geq 2$, is $\text{pd}_{A_d} D$ equal to d or to $d + 1$? What are the weak global dimensions of A_d (at least $d + 1$) and of $A_0 = A_1$ (at least 2)? How do cores over non-simplicial cones, their arbitrary ideals and their homological localizations behave?

Status. Partly answered. The first clause is answered by 21: $\text{pd}_{A_d} D = d + 1$ for every $d \geq 1$ (Corollary 13.47); in rank one this is the merge value 2 of Proposition 13.12, and over every ordered-cone core $\text{pd} D = 2$. The weak global dimensions stay open, with lower bounds $d + 1$ and 2 (Section 13.5); 21 adds $\text{gl. dim } A_d \geq d + 2$ when $K \neq \text{Frac } D$ (Theorem 13.48) and asks whether the bounds $\text{gl. dim } A_d \geq d + 2$ and $\text{w. gl. dim } A_d \geq d + 1$ are sharp, which is merged here. Cores over non-simplicial cones, arbitrary ideals and homological localizations stay open.

Question 18.13 (Beyond one coefficient lattice; 14, 21). Replace the single coefficient lattice by a finite filtration of lattices indexed by support levels and determine which extension data survive multiplication; characterize all finitely generated rank-one modules with double dual B ; determine how coefficient rings that are not principal ideal domains change the tensor kernel.

Status. Open. 21 asks the first part again, for the derived tensor products of finitely generated omnific ideals whose generators lie at several independent boundary levels, and notes that its common-scale DGA (Theorem 13.50) does not cover them; its question is merged here.

Question 18.14 (Fresh scales; 16). (a) Can the factorial-residue Gaussian certificate be replaced by an optimal or near-optimal grid, and how does a regular basis of $\text{Int}(\mathbb{Z}[i])$ interact with the omnific splitting? (b) Under what hypotheses are the formal inverse branches of a polynomial map in several variables at independently fresh coordinates classified by finitely many forbidden support blocks (triangular automorphisms exclude a direct analogue of one-variable surjectivity rigidity)? (c) For a fixed set-sized Hahn subfield and its nonnegative-support ring, which coefficient subgroups admit enough dominant extensions inside the field to recover the image gap and the rational certificates?

Status. Partly answered. (a) is partly addressed by 18: its Gaussian simplex test is smaller than the factorial-residue certificate for $d \geq 3$, and every ordinary testing set for $\text{Int}(\mathbb{Z}[i])$, for instance the Volkov–Petrov sets of size $O(d)$ in one variable, also tests $\mathbf{Oz}[i]$ (Proposition 16.64); the optimal size is not determined. 22 reduces the constant-test part of (a) exactly to the ordinary problem: a finite subset of $\mathbb{Z}[i]^m$ tests preservation of $\mathbf{Oz}[i]$ in a coordinate degree space if and only if it tests $\text{Int}(\mathbb{Z}[i]^m)$ there (Theorem 16.82); its question on sharp Gaussian constant tests (how far known ordinary Gaussian universal sets improve the surcomplex certificate once combined with the remote denominator stage, distinguishing coordinate from total degree in several variables) is merged here, and 22 claims no optimality for its grid C_{d+1}^m . For (c), 18’s discrete-gap arguments, unlike the fresh-scale ones, do hold in fixed Hahn workspaces (Corollary 16.67); and 22 answers the part on rational certificates for its single-scale certificates: a coefficient subgroup G of the value group Γ has a fresh exponent in Γ exactly when the convex subgroup it generates is proper, and then the Laurent blocks, the detector and the certificates hold in the workspace (Proposition 16.86). The image gap in a fixed workspace, the corresponding condition for 16’s multi-exponent probes (Theorem 16.49), and the existence of other finite certificates when that convex subgroup is all of Γ stay open, as does (b).

Question 18.15 (Relations of lattice ideals; 17, 21). (a) Classify the kernels of omnific matrices with several monomial scales, and determine which coefficient-lattice data beyond q and ℓ they need. (b) Determine the projective dimensions of Π_κ^{11} and $J_g(M)$ over R_κ^{11} , and an upper bound for its weak global dimension. (c) How much of a finitely generated coefficient algebra do the product lattices, Hom modules and mixed multiplication kernels together reconstruct?

Status. Open, except (b) for $\kappa = \aleph_0$: $\text{pd } \Pi_{\aleph_0}^{11} = 1$ and $\text{pd } J_g(M) = 2$ for $r \geq 2$ (Remark 13.32, a merge result). For uncountable κ , (b) is Question 18.3 for $R_{\Gamma_{\kappa}, < \kappa}^{11}$; (a) is also 14's question (Question 18.13). (c) is partly addressed by 21: with the distinguished 1 and the boundary embedding, the self multiplication kernel $\mathcal{T}(\mathfrak{o}, \mathfrak{o})$ recovers the multiplication of an order (Theorem 13.56), and 17's Theorem 13.39 recovers the order from $\text{End}_A(J_\gamma(\mathfrak{o}))$; 21 asks which parts of these framed data can be recovered intrinsically from a category of omnific quotient modules with its tensor structure, which is merged into (c).

Question 18.16 (Ordinary-output loci; 18). (a) Which hypotheses on $f \in \mathcal{N}_n(\Lambda)$ exclude positive-dimensional fibres and make its ordinary-output locus a finite union of translated ordinary lattices, explaining both the linewise rigidity and the example XY ? (b) For a polynomial of degree d without a rich target, which patterns of the numbers $\#\mathcal{L}_{f,b} \in \{0, \dots, d\}$ occur as b varies? (c) Can Gaussian test points in $\mathbf{Oz}[i]$ that are not ordinary reduce the least number of test points below the ordinary minimum? (d) Can the cluster theorem be formalized for an abstract discretely ordered ring and specialized through the normal-form bridge?

Status. Open. The real analogue of (c) is settled by 16: for $\mathbf{Oz}^n$ no set of omnific points smaller than $|L|$ tests a lower-set space V_L (Theorem 16.46); for $\mathbf{Oz}[i]$ the lower bound $d + 1$ from the dimension is all that is known. 22's lifting theorem (Theorem 16.82) concerns test points in $\mathbb{Z}[i]^m$ only and does not bear on (c).

Question 18.17 (The arithmetic fibre of the normalization; 19). (a) Determine $\text{Idem}(\mathbf{B}_{\mathbb{R}})$: is it trivial, does it have atoms, can independent families of nontrivial idempotents be constructed, and is it a set? (b) For which fixed exponent groups and support classes is the extended ideal radical in the corresponding real integral closure, and when does real-rooted lifting give a Boolean-power fibre although set-family divisibility fails? (c) Given finitely many integral elements with monic witnesses, can a common Boolean partition and its algebraic-integer coefficients be extracted by a concrete symbolic procedure, including the equality tests in the quotient?

Status. Partly answered. (a) is answered by 20: $\text{Idem}(\mathbf{B}_{\mathbb{R}})$ is nontrivial and atomless, it is a proper class with no set-sized order-dense subset, the explicit family of Theorem 15.72 is independent and indexed by all ordinals, and free families of every set size exist over every set-sized subring (Theorems 15.66 and 15.71 to 15.73); the same holds for $\mathbf{B}_{\mathbb{C}}$ (Corollary 15.77). The earlier reduction of (a) to instances of the membership clause of Question 18.11 (Remark 15.52) is superseded: 20 needs no membership test, and the candidates of that remark are nontrivial (Proposition 15.68, a merge result). What remains of (a) is the isomorphism type (Question 18.18). (b) is partly answered: for every nonzero set-sized ordered group the reduced fibre $\mathbf{C}(G)$ is a Boolean power of $\mathbb{Z}$, and one dominant coordinate adds free Boolean families (Proposition 15.76 and Theorem 15.70); whether $\Pi(G)\mathcal{N}(G)$ is radical, and whether $\text{Idem}(\mathbf{C}(G))$ is atomless, in a fixed workspace was left open by 20, which asks the same; for cyclic groups it is now decided (batch 32, below). (c) is open; 20 asks it for algebraic Laurent elements given by equations and branch data, with complexity bounds, and settles the one-scale rational case (Theorem 15.60); 23 gives a criterion for cyclic groups (below). Theorem 15.42 holds throughout. 19's membership question is merged into Question 18.11.

Status (batch 32). (b) is decided for every nonzero cyclic group $G = \mathbb{Z}a$ by 23, in opposite directions: $\Pi(\mathbb{Z}a)\mathcal{N}(\mathbb{Z}a)$ is not radical, its fibre having nilpotents of every index (Theorem 15.86), and $\text{Idem}(\mathbf{C}(\mathbb{Z}a))$ is atomless of cardinality and order density $2^{\aleph_0}$ (Corollary 15.110 and Theorem 15.111), also for the Gaussian pair. For the bounded-denominator Puiseux support class over $\mathbb{Q}a$ the extended ideal is radical (Theorem 15.100), and there real-rooted lifting gives a Boolean-power fibre although set-family divisibility fails (Remark 15.102, a merge observation). For other groups and support

classes (b) stays open, and the Hahn field over $\mathbb{Q}a$ is [Question 18.21](#). 23 asks the same question (its [Question 10.2](#)): for a prescribed set-sized G , which properties of G and of the admitted supports determine radicality of $\Pi(G)\mathcal{N}(G)$, and can the one-variable ramification filtration be replaced by a finite-rank or cut-valued filtration when G has several Archimedean layers? 23's comments: the cyclic negative and the Puiseux positive results are two test cases that any criterion must distinguish; a next model is a lexicographic sum of two cyclic groups, where the coefficient block is a field with its own integer part and the residue-basis step of [Theorem 15.93](#) needs a replacement; the proof does not classify groups with a discrete convex component. (c) is addressed for cyclic groups by 23's residue criterion ([Theorem 15.91](#) and [Remark 15.92](#)), which is a criterion and not an algorithm. 23 asks (its [Question 10.7](#)), for coefficients in a number field or a computable presentation of $\overline{\mathbb{Q}}$, for an algorithm that takes a finite algebraic function field and a chosen Laurent branch at infinity, decides membership in $\mathcal{N}(\mathbb{Z}a)$, computes $\nu_T(f)$, and produces a monic certificate and the least refinement killing a nilpotent residue. 23's comments: the outputs are determined (normalize at the finite places, test the residues, compute the finitely many orders), and what remains is a representation and complexity problem; singular plane equations can hide branches, so normalization cannot in general be replaced by factoring one specialized polynomial; one would bound the certificate degree $dr \deg Q$ of [Remark 15.92](#) in the input degree and height and decide when smaller certificates exist.

Question 18.18 (The class Boolean algebra of the fibre; 20). Which further Boolean invariants determine $\text{Idem}(\mathbb{B}_{\mathbb{R}})$? Can its automorphisms, or its possible infinite joins, be characterized directly in terms of support-local arithmetic?

Status. Open. Known: it is atomless, a proper class with no set-sized order-dense subset, and relatively free over every set-sized subring in every set size ([Theorems 15.71](#) and [15.73](#)); this does not make it free, complete or saturated.

Question 18.19 (Finite certificates at new scales; 22). (a) What is the smallest coefficient-uniform remote test set for rational functions with given degree bounds and a prescribed fresh t ; can the bivariate count $5d^2 + 6d + 3$ of [Theorem 16.70](#) be improved with an explicit coefficient-independent choice of the ordinary coordinates? (b) Can total-degree bounds, Newton polytopes or prescribed monomial supports replace the rectangular degree space and give substantially smaller tests? (c) For which useful notations of surreal coefficients can a certified group bound and a symbolic fresh scale be produced effectively? (d) Given sign-sequence descriptions of a support bound, what ordinal birthday bounds hold for the witness t and for a bad input produced by the resultant method? (e) Can sufficient fresh-scale conditions for a bounded-degree family be expressed without naming normal-form supports, for instance in the pair **(No, Oz)**, possibly with predicates for convex subgroups? (f) Can the support-block and finite-certificate proofs be added as a small, independently auditable extension of the repository's Hahn layer?

Status. Open. 22's comments: in (a) the proof ignores the structure of the resultant and uses one bound in every direction, and an optimality claim must say whether the test set may depend on the coefficients or only on their support group and degrees; 16's probe certificate uses fewer values with $m + 1$ fresh exponents ([Remark 16.71](#)). In (b) the dimension lower bound holds for any finite-dimensional polynomial space with a rich input set, achieving it with integer-valued nodes is an arithmetic issue, and on the rational side sparse resultants must still exclude cancellation and degree loss; lower-set spaces, including total degree, are already tested optimally ([Theorem 16.46](#)), and 16's probes give total-degree rational certificates ([Corollary 16.50](#)). In (c) a first implementation would use finite normal forms with exponents in a computable ordered group and the fresh scale as an extra lexicographic generator, a second infinite supports with certified bounding data; coefficient equality,

support membership, support bounding and ring membership should be separate parts of the input contract, none following from the other three (compare 16's non-claim (4) and [Question 18.17\(c\)](#)). In (d) the construction of the fresh exponent as ω times a bound proves existence only; bounds should come from the representation used, with every multiplication and monomial operation accounted for. In (e) a merely transcendental test point is not enough, since its order relative to the coefficient scales is essential, and the certificate is not quantifier elimination for the pair. In (f) the milestones are the order embedding $(n, g) \mapsto nb + g$ on lexicographic blocks, reverse well-ordering and local finiteness of the Laurent image, multiplicativity, the nonnegative-support detector, the resultant bound and the grid lemma, and the explicit deceptive function, with the proper-class conclusions proved through an explicit small-set bounding interface; a finite-rank toy model would not establish the full claims ([Section 19.2](#)).

Question 18.20 (Arithmetic maps beyond affine polynomial space; 22). (a) Which single-valued algebraic or semialgebraic maps preserve $\mathbf{Oz}$, and do their branches admit analogous certificates? (b) Which affine varieties admit a counterpart of the constant-grid and remote-direction certificate for their omnific points? (c) How are the polynomial automorphisms whose two directions preserve $\mathbf{Oz}^m$ organized by their ordinary specializations and purely infinite parts? (d) How far do the finite certificates and the absence of set-sized tests extend to rings $D \oplus \Pi_k$ for other infinite subrings D of k , for instance rings of algebraic integers?

Status. Open, with (d) partly addressed. 22's comments: in (a) a global polynomial conclusion is false ($|X|$ preserves $\mathbf{Oz}$ and is only piecewise polynomial), and the target is a piecewise classification with explicit branch hypotheses and finite complexity bounds, controlling algebraic branches and their negative-support tails. In (b) the affine proof uses the Zariski density of an ordinary grid and free movement in one coordinate, neither automatic on a variety; a first theorem could fix an explicit dominant parametrization and track its exceptional locus, and varieties with very few ordinary integral points need a different mechanism. In (c) [Corollary 16.84](#) removes denominators but does not classify the group, which the triangular binomial shears of [Example 16.85](#) show to be larger than the group with coefficients in $\mathbf{Oz}$; a start is the subgroup generated by affine maps and triangular binomial shears, together with the kernel of specialization, and nothing is implied about all automorphisms in higher dimension. This is not the question on automorphisms of the ring $\mathbf{Oz}$ in [Question 18.10](#). In (d) the detector is insensitive to D , the polynomial step needs integer-valued polynomial theory over D , and the deceiver needs a constant outside D with a suitable cleared denominator; 16 already proves rational collapse, the transfer $\mathcal{N}_r(\mathcal{A}_{D,k}) = \text{Int}(D^r) \oplus \Pi_k[\mathbf{X}]$, the probe certificates and the one-variable removal of exceptions for every unital $D \subseteq k$ ([Theorems 16.42](#), [16.49](#) and [16.52](#) and [Proposition 16.45](#)), and the absence of set-sized tests holds for every $D \neq k$ ([Remark 16.88](#), a merge result); finite constant test sets for $\text{Int}(D^m)$ and the lifting theorem for general D remain open.

Question 18.21 (The full rational Hahn workspace; 23). Let $K_{\mathbb{Q}a} = k((X^{\mathbb{Q}a}))$, with arbitrary reverse well-ordered supports, let $\mathcal{A}(\mathbb{Q}a)$ be its omnific or Gaussian omnific integer part and $\Pi(\mathbb{Q}a)$ its positive-support ideal. Is $\Pi(\mathbb{Q}a) \text{IC}_{K_{\mathbb{Q}a}}(\mathcal{A}(\mathbb{Q}a))$ radical?

Status. Open. 23's comments: [Theorem 15.100](#) answers the bounded-denominator subcase affirmatively, and the obstruction to repeating its proof is exact: an element of $K_{\mathbb{Q}a}$, and the coefficients of its integrality equation, need not lie in a finite cyclic refinement. A first intermediate case would allow supports generated by finitely many rationally independent real exponents, with a restriction preventing positive supports from accumulating at zero. Divisibility of the exponent group alone is not a proof of radicality; the supports of the coefficients must be controlled as well.

For the real pair $K_{\mathbb{Q}a}$ is real closed, so [Theorem 15.51](#) would apply if its axiom (ii) held there; the support $\{a/n : n \geq 1\}$ blocks the monomial proof of (ii) (the paragraph after that theorem), and whether (ii) holds is not decided [merge].

Question 18.22 (The uncountable Boolean isomorphism type; 23). Determine the isomorphism types of $\text{Idem}(\mathbf{B}(\mathbb{Z}a))$ and $\text{Idem}(\mathbf{B}^{\text{co}}(\mathbb{Z}a))$ for $k = \mathbb{R}$ and $k = \mathbb{C}$. Are there invariants, beyond atomlessness, cardinality, order density and relative splitting, that distinguish them from familiar Boolean algebras?

Status. Open. 23's comments: the countable answer of [Theorem 15.118](#) is a baseline, and one would like to describe the Stone space directly from compatible places of finite function fields over $T = 0$. The explicit family gives $2^{2^{\aleph_0}}$ points on the coefficient side ([Corollary 15.112](#)), but neither a product representation of the ring nor a free presentation of the algebra follows from this count; countable joins, saturation and automorphisms should be asked separately, and relative splitting is weaker than all of them together. This is the fixed-cyclic counterpart of [Question 18.18](#), which concerns the proper-class algebra $\text{Idem}(\mathbf{B}_{\mathbb{R}})$; by [Remark 15.113](#) the idempotents of $\mathbf{B}(\mathbb{Z}a)$ and $\mathbf{C}(\mathbb{Z}a)$ are the same.

Question 18.23 (Arithmetic residue gluing; 23). Describe concretely the embedding $\mathbf{B}(\mathbb{Z}a) \hookrightarrow \mathbf{B}^{\text{co}}(\mathbb{Z}a)$ and the extension $0 \rightarrow \mathbf{J}/T\mathcal{B} \rightarrow \mathbf{B}(\mathbb{Z}a) \rightarrow \mathbf{C}(\mathbb{Z}a) \rightarrow 0$. How much of D can be recovered intrinsically from this extension and its multiplication, without naming the inclusion $D \subseteq k$?

Status. Open. 23's comments: the residue criterion says which elements of $\mathcal{B}$ lie in $\mathcal{N}(\mathbb{Z}a)$, and the task is to organize these conditions uniformly over all finite fields and all compatible branches. The strict inclusion of [Proposition 15.117](#) is an intrinsic arithmetic signal: multiplication by a nilpotent admits coefficients that cannot occur in the arithmetic fibre. A Boolean-power or locally constant description must keep this nonreduced information, not only the reduced data, for which see [Proposition 15.76](#) and [Remark 15.114](#).

Question 18.24 (Homological dimensions of the arithmetic fibre; 23). What are the weak global and global dimensions of $\mathbf{B}(\mathbb{Z}a)$? Can the kernels $\bar{z}_n^{n-j} \mathbf{B}^{\text{co}}(\mathbb{Z}a)$ be resolved over $\mathbf{B}(\mathbb{Z}a)$ so as to compute a nonzero sequence of higher Tor groups?

Status. Open. 23's comments: the infinite dimensions of $\mathbf{B}^{\text{co}}(\mathbb{Z}a)$ ([Theorem 15.116](#)) cannot be pulled back through the inclusion, since the alternating maps cease to be a resolution over $\mathbf{B}(\mathbb{Z}a)$ ([Proposition 15.117](#)); a direct approach would start from the $\mathbf{B}(\mathbb{Z}a)$ -module structure of $\mathbf{B}^{\text{co}}(\mathbb{Z}a)$, the modules $\bar{z}_n^r \mathbf{B}^{\text{co}}(\mathbb{Z}a)$ and their relation modules. This is not the question on projective dimensions of the constant module over the finite-support cores ([Question 18.12](#)); those rings, modules and maps are different.

Question 18.25 (The completed coefficient ring; 23). Describe $\widehat{\mathcal{B}} = \varprojlim_r \mathcal{B}/T^r \mathcal{B}$ intrinsically. Is there a useful representation by compatible completed local valuation rings, and which restricted-product or branch-compatibility conditions are needed?

Status. Open. 23's comments: by [Corollary 15.107](#) the arithmetic completion reduces to this problem together with the known first fibre. With $2^{2^{\aleph_0}}$ branches over zero, a naive identification with an unrestricted product of local completions needs proof and should not be assumed, and completion in the integer powers of T should be distinguished from completion for the filtration indexed by all positive rationals.

Question 18.26 (Refinement beyond monomial root maps; 23). For finite algebraic covers replacing $T = U^m$, which branchwise ramification data determine the kernel on arithmetic fibres? Is there an exact analogue of (15.30) when different branches have different ramification indices?

Status. Open. 23’s comments: the uniform multiplier m is what makes the threshold a single scalar, and a nonuniform cover needs the full tuple of branch orders, not only their minimum. Finite normalization and lying over give a concrete formulation, but compatibility with the arithmetic integer parts must be checked: a map of function fields alone need not induce the desired map of fibres.

Question 18.27 (A formal development; 23). Formalize the cyclic nonradicality theorem, the residue criterion and the refinement law in Lean, with an abstract algebraic-function-field layer separate from the embedding into the constructed surreal field.

Status. Open. 23’s plan: first the short argument for $\mathbf{z}_n$, with finite normalization and one valuation normalized by $v(T) = 1$; then the coefficient-span lemma and the exact ideal test; the Boolean splitting needs further finite-extension and ramification infrastructure, and the periodic resolution follows once the annihilator identity is available. No declaration has been implemented, and no repository build was run for 23 (Section 19.2).

19 Verification record and formalization routes

19.1 Finite checks

Each manuscript ships a program that checks finite identities with exact arithmetic. They are kept byte-identical in `code/` with their recorded outputs in `data/`; all were rerun for the seven merges on copies (Python 3.14.4, SymPy 1.14.0) and reproduce the recorded outputs, up to line endings, the recorded Python version, for 18 the recorded time stamp and for 23 its hash of the unshipped manuscript.

Source	What is checked	Checks	Needs
03	finite telescopes in a symbolic lattice, the unit series coefficients, lexicographic comparisons	2,362	stdlib
04	1,950 finite telescopes in $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$ with remainders and positive supports, 600 constant-term and 900 augmentation products, 3 support failures and 3 distinguishing examples	5,406	stdlib
06	geometric remainders and supports, constant terms, boundary cases, Euler derivation, real complex-structure blocks	17,586	stdlib
07	weighted geometric identities, binomial remainders, first idempotents, dyadic refinement maps	621	SymPy
08	telescopes, constant terms, Hahn Leibniz identities, binomial powers, root corrections	1,277	stdlib
09	telescopes, truncated comaximality, coefficient maps, finite-support evaluation, Gaussian derivation, gcd identities	597	stdlib
10	divided differences, Newton reconstruction, digit indicators and lifts, telescopes, rational differences	171,395	SymPy
11	finite division identities, positive quotient supports, augmentation and conjugation identities, Gaussian derivation	1,858	stdlib

Source	What is checked	Checks	Needs
12	homology of the Boolean tensor complex for $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$ on a five-point grid in ranks 1–4, with $d \circ d = 0$ (Theorem 13.7)	780	SymPy
13	51 finite geometric remainders and their positivity, a cubic norm determinant, a Laurent norm identity, a rank-deficient point, a constant-term product	6 groups	stdlib
14	ranks for $\alpha^d = 2$, homogenized Rees equations, quadratic torsion generators, $P^2 = 2\mathbb{Z}[\sqrt{10}]$ and the mod-5 obstruction, action associativity and tensor balancing	2,189	SymPy
15	finite telescopes and divisor truncations, constant-term products, finite-support evaluation, Gaussian matrix evaluation	820	stdlib
16	Newton reconstruction, lower-set interpolation matrices, Gaussian counterexample, 150 centered defects, 22 formal inverse branches	1,354	SymPy
17	coefficient multiplication matrices and kernels (quadratic, cubic, full orders, mixed, independent coefficients), the Pell matrix, fractional-linear identities, descent ranks q, ℓ	347	SymPy
18	Newton expansions on finite-support $\mathbb{Q}[H]$, $\mathbb{Q}(i)[H]$ models, centering formulas, the composition obstruction, rational differences, Lagrange leading coefficients, nilpotent matrices, Gaussian examples (20 categories)	499	SymPy
19	integral idempotent and doubling identities, spectral projectors, the Catalan expansion to degree 13, 216 endpoint signs of the lifting (24 polynomials, degrees 1–8)	253	SymPy
20	quadratic identities and branch specializations, simple-zero witnesses for 31 radicand products and all 16 four-branch assignments, initial Laurent coefficients to degree 15, Boolean atoms in 1–6 variables and the symmetric difference, 875 support-projection examples and one failure of multiplicativity	6,674	SymPy
21	Koszul squares and transition identities in dimensions 1–6, finite telescope identities and support barriers; integral multiplication-kernel bases, ranks and discriminants of $\mathbb{Z}[\sqrt{2}]$, $\mathbb{Z}[\sqrt{3}]$, $\mathbb{Z}[\sqrt[3]{2}]$, $\mathbb{Z}[\varphi]$, and the product rank for $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$	1,608	stdlib
22	Newton reconstruction in 1–3 variables, integral binomial values, grid determinants and integral inverses, resultant coordinate-degree bounds, good specializations (degrees, coprimality, nonzero bad-locus polynomial), first negative Laurent terms, the geometric remainder, Gaussian examples, fixed-divisor ideals, triangular inverses, sampled lexicographic block signs (22 categories, seed 23092026)	1,776	SymPy
23	binomial power identities of $\mathbf{z}_n$ for $n \leq 20$ to order 18 and the displayed coefficients of $\mathbf{z}_2$, refinement thresholds for $n \leq 100$, $m \leq 50$ and the table for $\mathbf{z}_{12}$, monic refinement certificates, the idempotent identity of $\mathbf{y}_c$ for $c = 1, 4, \dots, 1600$, finite Boolean word and partition models in $r \leq 8$ variables, truncated-polynomial annihilators and alternating compositions for $n \leq 40$, clipped refinement composition (12 categories)	18,040	stdlib

These checks catch sign, index and coefficient errors in finite special cases. None of them verifies

an infinite support, a class-size argument, a cardinal computation, the existence of a maximal ideal, an Ext computation, a primality statement, or novelty; those rest on the written proofs (all sources). The merge results were not machine-checked. For the second merge an independent exact computation, not shipped, rechecked [Theorem 13.7](#) in 1,626 cases (multidegree and subset F , including every proper F , over $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$ in ranks up to 3) and the formulas of [Corollary 13.23](#) for $d \leq 8$, $n \leq 12$; like the shipped checks it verifies finite cases only. For the third merge, independent exact computations, not shipped, rechecked the Pell matrix, the multiplication kernels and self-Tor endpoints for $r \leq 5$, the irreducibility of $X^r - (1 + i)$ over $\mathbb{Q}(i)$ for $r \leq 6$ and five $\mathrm{GL}_2(\mathbb{Z})$ cases (17); the composition obstruction of [Theorem 16.63](#) in 27 random cases and the formula for η_f (18); the idempotent identities, the Catalan expansion and 36 random cases of the lifting bound (19); and the sizes of the Gaussian certificates compared in [Proposition 16.64](#) for $n \leq 3$, $d \leq 5$. For the fourth merge an independent exact computation, not shipped, rechecked in 18 checks the certificate (15.24), the values of $\sigma_{\pm}$, the many-branch identity of [Theorem 15.70](#), the simple zeros used for the square classes, and the identities behind [Proposition 15.68](#). For the fifth merge an independent exact computation, not shipped, rechecked in 78 checks the converse formula of [Theorem 13.46](#) in 12 cases, the exponent identities behind it, the Koszul transition identity, the ranks of Tor_2 and Ind_2 and the discriminants of the four orders, and the product rank for $\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3})$; like 21's checks it proves no infinite Ext nonvanishing. For the sixth merge an independent exact computation, not shipped, rechecked in 101 checks the resultant and the partial fractions of [Example 16.87](#), the counts after [Theorem 16.70](#) and the comparison of [Remark 16.71](#) for $m \leq 4$, $d \leq 5$, the bound (16.26) in twelve random trivariate cases, the inverse shears of [Example 16.85](#), the two Gaussian denominators and the coefficients of [Example 16.74](#); it proves no statement about infinite supports. For the seventh merge 23's program was rerun on a copy: all 18,040 assertions pass in the same 12 categories, and the rewritten record differs from the shipped one only in the Python version and in its hash of the manuscript, which is `null` because the manuscript is not shipped beside the program (and, on Windows, in line endings). An independent exact computation, not shipped, rechecked in 2,642 checks (most of them the exponent inequality of axiom (ii) used in [Remark 15.102](#)) the relations (15.28) for $n \leq 7$ and (15.33), the refinement certificates of [Theorem 15.89](#), the Laurent coefficients and the full-class factorization of [Example 15.87](#), the residues of the Boolean words of [Theorem 15.111](#) modulo T for three parameters with the squarefreeness of their radicands, and the cancellation identity behind [Theorem 15.115](#); like 23's program it verifies no integral closure, place or cardinality.

19.2 Formalization routes

No statement of this report is formalized, and the repository has no omnific module. The sources propose compatible routes, recorded here as proposals only. First formalize the scaled-field collision lemma and the separation lemma, which need no surreals (06, 04, 07). Then the countable telescope in a set-sized Hahn workspace, with its two support inequalities, proved coefficientwise and not by convergence of partial sums (04, 06, 07, 08, 10). The set-sized criteria ([Theorems 10.7](#) and [10.8](#)) avoid proper-class syntax and are good first targets (08). The full-class theorem needs an explicit universe interface in which the target and the collision index are small and the exponent class supplies smaller scales below every small support; a universe-polymorphic theorem about an arbitrary field does not supply them (04, 08, 10, 11). For the constructions, the finite-coordinate invariant of 08 must be preserved through field generation, real closure and truncation, and the module bound at a cardinal must use cyclic quotients or the direct collision, not an endomorphism ring (08, 06). The homological part comes last, over

set-sized rings only (06). For 07's quotient, the support characterization of $\mathbf{Oz}[\omega^{-a}]$ and the exact binomial kernel should be kept separate from the finite Chinese-remainder algebra. The repository's small-normal-form modules (`SmallNormalForm.lean`, `SmallNormalFormTruncation.lean`, `SmallNormalFormEvaluation.lean`, `SignSequenceValuationBalls.lean`, see [50]) are possible prerequisites (04); they are not a certificate for any result here. The later manuscripts add three proposals. 12 would define its cores as subrings of a monoid algebra cut out by the constant-coefficient predicate, with the directed-union equality, exactness of the finite-support simplex complex and the degreewise classification of [Theorem 13.7](#) as the main obligations, and the surreal embedding last. 14 would start from an abstract domain $A = D \oplus P$ with P a k -linear ideal in which every element is a product of two, a hypothesis the finite-support model satisfies. 16 separates a finite part (simple-root recursion, centering, the leading-term argument, finite differences), a Hahn part (the block embedding) and an actual-surreal part (a common set-sized exponent workspace and a fresh exponent). The third batch adds three more. 17 separates a finite layer (saturated lattices, constant kernels, scalar multipliers and coefficient product matrices over a principal ideal domain), a support layer (the small-family gap and the union of principal ideals) and a homological layer over the set-sized models only. 18 would formalize the cluster theorem for an abstract discretely ordered ring and specialize it through the normal-form bridge. 19 would formalize the determinant contraction over any ring with a retraction, radicality under power divisibility, saturation, the root-forcing estimate, the spectral projectors and the doubling, and keep the class representation theorem and its ultrafilter foundation separate. 20 proposes its set-sized rank extension ([Theorem 15.70](#)) as a first target: Hahn coefficient projection, the finite determinant certificate for radical membership, iterated Hahn fields and the presentation and specialization of a multiquadratic algebra, all over set-sized ordered groups; the full-surreal theorem needs in addition the normal-form bridge, a bound above a set of exponents and universe management, and the class-ultrafilter representations should stay a separate foundational module. The fifth batch adds one. 21 would formalize its dimension theorem on the finite-support monoid algebra before any surreal embedding: the face ideals, the coefficientwise simplex contraction, the Koszul transitions, the mapping cone and the last Hom differential, with the nonvanishing lemma that a finitely supported element cannot have prescribed nonzero coefficients at infinitely many distinct exponents; its second part would start from an abstract ring $D \oplus \Pi$ with $k \oplus \Pi$, a common positive monomial factor and a flat tail, whose instantiation at the class rings is a separate obligation. The sixth batch adds one. 22 proposes a small extension of the repository's Hahn layer: the order embedding $(n, g) \mapsto nb + g$ on lexicographically ordered blocks, reverse well-ordering and local finiteness of the Laurent image, preservation of multiplication, the nonnegative-support detector, the resultant degree bound with the grid lemma, and the explicit deceptive function, with the proper-class conclusions proved through an explicit small-set bounding interface; a finite-rank toy model would not establish the full claims ([Question 18.19\(f\)](#)). The seventh batch adds one. 23 would first prove its $\mathbf{z}_n$ argument from finite normalization and one valuation, then the coefficient-span lemma and the exact ideal test, with an abstract algebraic-function-field layer separate from the embedding into the surreal field; the Boolean splitting needs further ramification infrastructure, and the periodic resolution follows from the annihilator identity ([Question 18.27](#)). None was executed.

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